



ExpressLane PEX 8724-BA/CA 24-Lane, 6-Port PCI Express Gen 3 Multi-Root Switch Data Book

Version 1.2

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Revision History

Version	Date	Description of Changes
1.0	March, 2012	Production Release, Silicon Revision BA.
1.1	June, 2012	Production Update, Silicon Revision BA / Production Release, Silicon Revision CA. - Added support for Silicon Revision CA. Updated register offset 108h and the JTAG ID. - Changed many register bits from RO to ROS. - Register offsets 700h[0], 704h[0], and 760h[19:17] – Changed to Factory Test Only. - Register offset 760h[16] – For Silicon Revision CA, re-purposed to enable ECAM access to all PEX 8724 registers. - Register offset F4Ch[28:24] – Added. - Applied miscellaneous corrections and enhancements throughout data book.
1.2	September, 2013	Production Update, Silicon Revisions BA and CA. The following list summarizes major changes incorporated in v1.2. Global/General - Updated latency values between symmetric ingress and egress Ports. - Changed 1- and 2-Bit Soft errors to 1-Bit ECC and 2-Bit errors, respectively. - Removed PRBS support. - Clarified GPIO11 signal availability note. Table 12-1 and Table 13-6 – Added “N/A” and footnote for offset 364h’s 4h value. - Applied miscellaneous corrections and enhancements throughout data book. Chapter 3 - I/O signals – Removed notes regarding input and output, and moved that information (“I” or “O”, respectively), into the signal’s Type column text. FATAL_ERR# – Added that FATAL_ERR# and/or PEX_INTA# outputs can signal the ECC error event. I2C_SCL1, I2C_SDA1, I2C_SCL0, and I2C_SDA0 – Changed pull-up resistor requirements and circuitry information. PEX_INTA# – Changed resistor requirement. PEX_PERST# – Added note with respect to Uncorrectable errors. PORT_GOODx# – Added that these signals can be re-assigned to any Port. STRAP_I2C_SMBUS_CFG_EN#, STRAP_I2C_SMBUS_EN, STRAP_MGMT_PORT_EN – Changed descriptions. STRAP_RESERVED signals – Corrected signal types, and the way in which these signals are listed. VSx_FATAL_ERR# – Added that ECC errors are reported on the VSx_FATAL_ERR# and/or VSx_PEX_INTA# outputs associated with the virtual switch that includes Port 0 of the Station in which the ECC error was detected. Clarified signal references within software de-assertion content. Continued...

Version	Date	Description of Changes
1.2	September, 2013	Production Update, Silicon Revisions BA and CA. (Cont.) Chapter 5 • “Virtual Switch Table” subsection – Updated. • “VSx_PERST#” subsection – Rewrote last paragraph. • “Reset and Clock Initialization Timing” subsection – Changed note. • “Initialization – Virtual Switch Mode” subsection – Added new content. • Table 5-4 – Corrected signal mapping information.” • Section 5.3.3 – Appended recommendation to note. Chapter 6 • “Serial EEPROM Load – Base Mode” subsection – Added that A30h[2] bit should be Set if NT mode is enabled. • “Serial EEPROM Load – Virtual Switch Mode” subsection – Added content regarding <i>NTx_PERST# Control</i> bit. • “Serial EEPROM Loading of NT Port Link Interface Registers – NT Mode” section – Added note that A30h[27] value can change when Virtual Switch mode and NT mode are both enabled. Chapter 7 • I²C Write and Read tables – Corrected Command Byte 3, Port 3, content to mention Station 0. • Figure 7-6 and Figure 7-7 – Corrected A and P symbols at end of Block Write. • Figure 7-9 – Corrected note to indicate msb. Chapter 9 • “General-Purpose Input/Output” section – Rewrote introduction. Chapter 10 • “I²C Write/Read to I/O Devices” and “Software Control of Hot Plug Output Signals” subsections – Added. • “Serial Hot Plug Controller” subsection – Changed second note. • “I²C I/O Expander Parts Selection and Pin Definition” subsection – Corrected quantity of possible 40-bit I/O Expanders. • I²C I/O Expander signal names – Prepended names with “SHP_”. • SHP_SLOTID[12:5] – Added new note and figure for when SHP_PORTID[4:0]=1_1111. • “Software Testing of Hot Plug Interrupts” subsection – Clarified content. Chapter 11 • Rewrote miscellaneous content. • Clarified wording used with respect to the various PM states. Chapter 12 • “Combined NT and Multiple Virtual Switches” subsection – Rewrote content. • “Port Migration – Moving a Port from One Virtual Switch to Another (VSx to VSy)” section – Clarified that traffic through VSx must be stopped prior to moving the Port. • Section 12.2.2.3 – Appended recommendation to note. Chapter 13 – Register Offsets • 58h and 5Ch – Clarified information related to bit 0. • 5Ch[3] – Corrected Type to be RO. • 104h and 108h – Removed <i>PCI Express Base r3.0</i>-related content. • 114h – Removed the Lane/Station part of the note. • 154h[0] – Corrected description. Continued...

Version	Date	Description of Changes
1.2	September, 2013	Production Update, Silicon Revisions BA and CA. (Cont.) Chapter 13 – Register Offsets (Cont.) • 1E8h[28:16] – Corrected default value. • 208h[30:24] – Corrected descriptions, affected Ports, and bit numbers used/Reserved. • 238h through 244h[31, 23:16] – Updated descriptions. • 258h – Corrected register map. • 26Ch[31:16] – Added information with respect to Address bits [7:0]. • 270h[31:0] – Changed Type to RO. • 364h – Corrected default values. • 3A8h – Added “Control” to register name. • 700h and 704h[25:23, 3:2] – Updated. • 700h and 710h – Clarified value of 1 descriptions. • 700h through 71Ch – Added Error Injection White Paper note preceding register map table. • 720h – Standardized naming of <i>Interrupt Output</i>-related bits. • 724h[14:13] – Added/expanded definitions. • 724h and 728h – Added second note. • 800h through 828h – Added Error Injection White Paper note preceding register map table. • 900h through 93Ch – Clarified introductory text to registers. • 9F0h[25:16, 8:0] – Changed Type to RO and added definition. • A00h through A08h – Removed references to x2. • A04h – Added “Atomic” to introductory description. • A04h[7:0] – Removed reference that credit is always available. • A30h[2] – Added note that bit should be Set if NT mode is enabled. • A30h[27] – Changed default value and added note that bit value can change when Virtual Switch mode and NT mode are both enabled. • BBCh – Updated related information. • F00h – Updated related information. • F20h – Updated related information. • F2Ch – Prepend “ACS” to register and bit names. • F30h – Changed to “All Ports.” • F60h[9, 3] – Changed bit 9 description and added a new note. Added support for bit 3. • F70h[29] – Changed to be Upstream only; Downstream is Reserved. • FB4h – Corrected Downstream Ports to point to register offset 10Ch. • FB8h – Added note with respect to Uncorrectable errors. • 22Ch, 764h[30], 77Ch, BD0h, F5Ch – Changed to Factory Test Only. • Silicon Revision CA only – 08h[7:0] – Updated to reflect Silicon Revision CA. – 74h[11:10] – Changed default to 10b (Disable ASPM L0s support). – 98h[21] – Changed Type from RW1C to RW1CS. – 204h[15:0] – Changed default to 1 (Enable EI inference mode, by default). – 220h[21:16] – Changed default to 1 (Enable EI inference on EIOS receipt). – 358h[7:6] – Changed from Reserved/RsvdP/No/00b to Factory Test Only/RWS/Yes/00b. – 570h[17] – Changed from Reserved/RsvdP/No/0 to Not used/RW/No/0. – 760h[16] – Changed note and removed NT-related information. – 760h[29] – Changed default to 1 (Disable Relaxed Ordering support for Completions with the <i>Relaxed Ordering</i> bit Set). – B7Ch[7:0] – Updated to reflect Silicon Revision CA. – BDCh Port x Signal Detect Level bits – Changed default to 011b (changed the signal detect threshold value). Continued...

Version	Date	Description of Changes
1.2	September, 2013	Production Update, Silicon Revisions BA and CA. (Cont.) Chapter 14 - Updated “Address Re-Map” subsection with respect to segment size. - Added “Ingress Support for Address Lookup Table” and its subsections. - Clarified wording used with respect to the various PM states. Chapter 15 58h and 5Ch – Clarified information related to bit 0. 74h[11:10] – Corrected default to 11b for Silicon Revision BA. D8h and E0h – Added RWS privilege note. 1E0h – Changed to Reserved. C60h – Corrected to indicate Link Interface. C6Ch through C88h – Clarified reset information. FB8h[16] – Added note. FBCh[14] and FC0h[14] – Added “Not applicable to switches.” Silicon Revision CA only 08h[7:0] – Updated to reflect Silicon Revision CA. 74h[11:10] – Changed default to 10b (Disable ASPM L0s support). Chapter 16 58h and 5Ch – Clarified information related to bit 0. 1E0h – Changed to Reserved. FB4h[31:20] – Corrected 138h’s structure name. FB8h through FC0h – Created NT Port Link Interface-specific copies of the registers. Updated register map to reflect the change. Silicon Revision CA only 08h[7:0] – Updated to reflect Silicon Revision CA. 74h[11:10] – Changed default to 10b (Disable ASPM L0s support). 98h[21] – Changed Type from RW1C to RW1CS. F70h[6:1] – Changed default from 00h to 02h. Chapter 17 Table 17-1 – Changed “Analog Loopback Slave Mode (LB_FEP)” description. “Analog Loopback Slave Mode (LB_FEP)” subsection – Added prerequisite bit Settings. Table 17-3 – Updated “Port <i>x</i> Test Pattern <i>x</i> Rate” field entry to match new description. “Port Good Status LEDs” section – Removed reference to PORT_GOOD11#. Chapter 18 Table 18-2 – Moved to appear under its correct section heading. - STRAP_RESERVED signals – Corrected how the signals are listed, and changed the pull/tie table reference. Table 18-6 – Updated signal group j and m information. Chapter 20 - Removed part ordering information for Silicon Revision BA.

Preface

The information in this data book is subject to change without warning. This PLX data book to be updated periodically as new information is made available.

Audience

This data book provides functional details of PLX Technology's ExpressLane PEX 8724-BA/CA 24-Lane, 6-Port PCI Express Gen 3 Multi-Root Switch, for hardware designers and software/firmware engineers. The information provided pertains to both Silicon Revisions (BA and CA), unless specified otherwise.

Supplemental Documentation

This data book assumes that the reader is familiar with the following documents:

- PLX Technology, Inc., www.plxtech.com
The [PLX PEX 8724 Toolbox](#) includes this data book and other supporting documentation, such as errata, white papers, and design and application notes.
- The Institute of Electrical and Electronics Engineers, Inc. (IEEE), www.ieee.org
 - *IEEE Standard 1149.1-1990, IEEE Standard Test Access Port and Boundary-Scan Architecture*
 - *IEEE Standard 1149.1a-1993, IEEE Standard Test Access Port and Boundary-Scan Architecture*
 - *IEEE Standard 1149.1-1994, Specifications for Vendor-Specific Extensions*
 - *IEEE Standard 1149.6-2003, IEEE Standard Test Access Port and Boundary-Scan Architecture Extensions*
- Intel Corporation, www.intel.com
 - [PHY Interface for the PCI Express Architecture, Version 2.00](#)
- NXP Semiconductors, ics.nxp.com
 - [The I2C-Bus Specification, Version 2.1](#)
- PCI Special Interest Group (PCI-SIG), www.pcisig.com
 - *PCI Local Bus Specification, Revision 3.0*
 - *PCI Bus Power Management Interface Specification, Revision 1.2*
 - *PCI Code and ID Assignment Specification, Revision 1.2*
 - *PCI to PCI Bridge Architecture Specification, Revision 1.2*
 - *PCI Express Base Specification, Revision 1.1*
 - *PCI Express Base Specification, Revision 2.0*
 - *PCI Express Base Specification, Revision 2.0 Errata*
 - *PCI Express Base Specification, Revision 2.1*
 - *PCI Express Base Specification, Revision 3.0*
 - *PCI Express Card Electromechanical Specification, Revision 2.0*
 - *PCI Express Mini Card Electromechanical Specification, Revision 1.1*
 - [PCI Express Architecture PCI Express Jitter and BER White Paper, Revision 1.0](#)

- Personal Computer Memory Card International Association (PCMCIA), www.pcmcia.org
 - *ExpressCard Standard Release 1.0*
- PXI System Alliance (PXI), www.pxisa.org
 - *PXI-5 PXI Express Hardware Specification, Revision 1.0*
- SBS Implementers Forum, smbus.org
 - *System Management Bus (SMBus) Specification, Version 2.0*

Note: In this data book, shortened titles are associated with the previously listed documents. The following table lists these abbreviations.

Abbreviation	Document
PCI r3.0	PCI Local Bus Specification, Revision 3.0
PCI Power Mgmt. r1.2	PCI Bus Power Management Interface Specification, Revision 1.2
PCI-to-PCI Bridge r1.2	PCI to PCI Bridge Architecture Specification, Revision 1.2
PCI Express Base r1.1	PCI Express Base Specification, Revision 1.1
PCI Express Base r2.0	PCI Express Base Specification, Revision 2.0
PCI Express Base r2.1	PCI Express Base Specification, Revision 2.1
PCI Express Base r3.0	PCI Express Base Specification, Revision 3.0
PCI ExpressCard CEM r2.0	PCI Express Card Electromechanical Specification, Revision 2.0
PCI ExpressCard Mini CEM r1.1	PCI Express Mini Card Electromechanical Specification, Revision 1.1
IEEE Standard 1149.1-1990	IEEE Standard Test Access Port and Boundary-Scan Architecture
IEEE Standard 1149.6-2003	IEEE Standard Test Access Port and Boundary-Scan Architecture Extensions
I ² C Bus v2.1 I2C Bus v2.1 ^a	The I ² C-Bus Specification, Version 2.1
SMBus v2.0	System Management Bus (SMBus) Specification, Version 2.0

- a. Due to formatting limitations, the specification name may appear without the superscripted “2” in its title.

Terms and Abbreviations

The following table lists common terms and abbreviations used in this data book. Most terms and abbreviations defined in the *PCI Express Base r3.0* are generally not included in this table.

Terms and Abbreviations	Definitions
8b/10b	Data-encoding scheme used on data transferred across a Link that is operating at either Gen 1 or Gen 2 Link speed (2.5 or 5.0 GT/s, respectively).
128b/130b	Data-encoding scheme used on data transferred across a Link that is operating at Gen 3 Link speed (8.0 GT/s).
ACK	Acknowledge Control Packet. A control packet used by a destination to acknowledge data packet receipt. A signal that acknowledges signal receipt.
AHB	Advanced High-Performance Bus.
APA	Autonomous Port Accumulation.
ARI	Alternative Routing-ID Interpretation.
ARP	Address Resolution Protocol.
ATT	Attenuator.
BAR	Base Address register.
Base Mode	PEX 8724 acts as a standard PCI Express switch, supporting one Host hierarchy (equivalent to Conventional PCI mode).
BER	Bit error rate.
BIST	Built-In Self Test.
CDR	Clock and Data Recovery.
CMU	Clock Management Unit.
CRC	Cyclic Redundancy Check.
CSR	Configuration Space register.
Data Beat	Single data transfer in a single clock period.
DFE	Decision Feedback Equalization. Optionally added to a Receiver to improve the signal.
DLL	Data Link Layer.
Downstream Device	Device that is connected to a Downstream Port.
Downstream Port	Port that is used to communicate with a device below it in the system hierarchy. A switch can have one or more Downstream Ports.
DRI	<i>Data Rate Identifier</i> field in Training Sets.
ECC	Error-Correcting Code.
ECRC	End-to-end Cyclic Redundancy Check.
EDB	TLP framing symbol used in place of an END symbol to indicate that the TLP is bad.
EIES	Electrical Idle Exit Sequence.
EIOS	Electrical Idle Ordered-Set.
Electrical Idle	Transmitter is in a High-Impedance state (+ and - are both at common mode voltage).
EP	Endpoint.

Terms and Abbreviations	Definitions
FC	Flow Control.
Field	Multiple register bits that are combined for a single function.
FTS	Fast Training Sequence.
Gen 1	<i>PCI Express Base r1.1</i> and below. Link transfer rate of 2.5 GT/s.
Gen 2	<i>PCI Express Base r2.1</i> . Link transfer rate of 5.0 GT/s.
Gen 3	<i>PCI Express Base r3.0</i> . Link transfer rate of 8.0 GT/s.
GPIO	General-Purpose Input/Output.
GPU	Graphics Processing Unit.
GT/s	Giga-Transfers per second.
INCH	Ingress Credit Handler.
ISR	Interrupt Service Routine.
Lane	Bidirectional pair of differential PCI Express I/O signals.
LCRC	Link Cyclic Redundancy Check.
LFSR	Linear Feedback Shift register.
Link	Active connection between two Ports.
Link Interface	Primary side of the NT Port, connects to external device pins.
Local	Reference to PCI Express attributes (<i>such as</i> credits) that belong to the PCI Express Station.
LTSSM	Link Training and Status State Machine.
LUT	Lookup Table.
LVDS	Low-Voltage Differential Signaling.
MPS	Maximum Payload Size.
MRL	Manually operated Retention Latch.
NACK	Negative Acknowledge. Used in the SMBus-related content.
NAK	Negative Acknowledge.
N_FTS	Number of Fast Training Sequences field in Training Sets.
NOP	No Operation.
NT	Non-Transparent. A bridging technique used in the PCI Express Switch to isolate Memory spaces by presenting the processor as an endpoint rather than another memory system. The PEX 8724 supports one NT Port.
OS	Ordered-Set.
P2P	Peer-to-Peer or PCI-to-PCI (as identified at point of use).
PCH	Intel Platform Controller Hub.
PCI Express Station	Functional unit that provides the PCI Express conforming system interface. Includes the Serializer/De-Serializer (SerDes) hardware interface modules and PCI Express interface, which provides the Physical Layer (PHY), Data Link Layer (DLL), and Transaction Layer (TL) logic.
PCS	Physical Coding Sublayer.
PEC	Packet Error Code.
PEX	PCI Express.
PHY	Physical Layer.

Terms and Abbreviations	Definitions
PIPE	PHY Interface for PCI Express architecture.
PLL	Phase-Locked Loop.
PM	Power Management.
PMA	Physical Media Attachment layer.
PME	Power Management Event.
PN	Port Number.
Port	Interface to a group of SerDes and supporting logic that is capable of creating a Link, for communication with another Port.
Port ID	Number, assigned in hardware, that associates a SerDes with a Port.
P-P	PCI-to-PCI.
QoS	Quality of Service.
RAS	Reliability, Availability, and Serviceability.
RoHS	Restrictions on the use of certain Hazardous Substances (RoHS) Directive.
RR	Round-Robin scheduling.
Rx	Receiver.
SATA	Serial ATA. Computer bus interface used for connecting Host Bus adapters to mass storage devices, <i>such as</i> hard disk and optical drives.
SerDes	Serializer/De-Serializer. A high-speed differential-signaling parallel-to-serial and serial-to-parallel conversion logic attached to Lane pads.
SMBus	System Management Bus.
SN	SerDes Number.
SPI	Serial Peripheral Interface.
SSC	Spread-Spectrum Clock.
Station	Logic block that implements the PCI Express function, bounded by the external pins of the differential Transceivers and the interface to the internal switch fabric.
Sticky Bits	Register bits in which the current values are unchanged by a Hot Reset, Link Down event or a Secondary Bus Reset, while the switch is powered. Sticky bits are reset to default values by a Fundamental Reset. HwInit, ROS, RW1CS, and RWS CSR types. (Refer to Table 13-4, “Register Types, Grouped by User Accessibility,” for CSR type definitions.)
Sticky State	Condition that causes a state machine to be stuck in a particular state, unable to make forward progress.

Terms and Abbreviations	Definitions
TC	Traffic Class.
TCB	Training Control Bits field in Training Sets.
TDM	Time Division Multiplex.
TL	Transaction Layer.
TLC	Transaction Layer Control. The module performing PCI Express Transaction Layer functions.
TLP	Transaction Layer Packet. PCI Express packet formation and organization.
Transparent	Refers to standard PCI Express Upstream-to-Downstream routing protocol.
TS	Tri-statable Output (High-Impedance when output is not driven). (Used in Chapter 3, “Signal Description,” in the Type column of the Signal tables.)
TS	Training Sequence.
TS1	Type 1 Training Sequence Ordered-Set.
TS2	Type 2 Training Sequence Ordered-Set.
Tx	Transceiver.
UDID	Unique Device Identifier.
UI	Unit Interval – 400 ps at 2.5 GT/s, 200 ps at 5.0 GT/s, 125 ps at 8.0 GT/s.
Upstream Device	Device that is connected to the Upstream Port(s).
Upstream Port	Port that is used to communicate with a device above it in the system hierarchy.
UTP	User Test Pattern.
VC	Virtual Channel. The PEX 8724 supports one Virtual Channel – VC0.
VC&T	Virtual Channel and Type.
Vector	Address and data.
Virtual Interface	Secondary side of the NT Port, connects to the internal virtual PCI Express interface.
Virtual Switch	Partition of a PCI Express switch that has its own virtual hierarchy. The PEX 8724 can support up to four virtual switches – VS0, VS1, VS2, and VS3. Each virtual switch is connected to its own Host.
Virtual Switch Mode	PEX 8724 supports up to four Hosts, creating up to four virtual switches within the PEX 8724 – each with its own virtual hierarchy.
VS	Virtual Switch.
WRR	Weighted Round-Robin scheduling.

Data Book Notations and Conventions

Notation / Convention	Description
Blue text	Indicates that the text is hyperlinked to its description elsewhere in the data book. Left-click the blue text to learn more about the hyperlinked information. This format is often used for register names, register bit and field names, register offsets, chapter and section titles, figures, and tables.
PEX_XXXn[x] PEX_XXXp[x]	When the signal name appears in all CAPS, with the primary Port description listed first, field [x] indicates the number associated with the signal balls/pads assigned to a specific SerDes module/Lane. The lowercase “n” (negative) or “p” (positive) suffix indicates the differential pair of signals, which are always used together.
# = Active-Low signals	Unless specified otherwise, Active-Low signals are identified by a “#” appended to the term (<i>for example</i> , PEX_PERST#).
Program/code samples	Monospace font (<i>program or code samples</i>) is used to identify code samples or programming references. These code samples are case-sensitive, unless specified otherwise.
Command/Status	Register names.
<i>Parity Error Detected</i>	Register parameter [bit or field] or control function.
Upper Base Address[31:16]	Specific Function in 32-bit register bounded by bits [31:16].
Number multipliers	k = 1,000 (10^3) is generally used with frequency response. K = 1,024 (2^{10}) is used for Memory size references. KB = 1,024 bytes. M = meg. = 1,000,000 when referring to frequency (decimal notation) = 1,048,576 when referring to Memory sizes (binary notation)
255d	d = Suffix that identifies decimal values.
1Fh	h = Suffix that identifies hex values. Each prefix term is equivalent to a 4-bit binary value (Nibble). Legal prefix terms are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
1010b	b = Suffix that identifies binary notation (<i>for example</i> , 01b, 010b, 1010b, and so forth). Not used with single-digit values of 0 nor 1.
0 through 9	Decimal numbers, or single binary numbers.
byte	Eight bits – abbreviated to “B” (<i>for example</i> , 4B = 4 bytes).
LSB	Least-Significant Byte.
lsb	Least-significant bit.
MSB	Most-Significant Byte.
msb	Most-significant bit.
DWord or DW	Double-Word (32 bits) is the primary register size in these devices.
QWord	Quad-Word (64 bits).
Reserved	Do not modify Reserved bits and words. Unless specified otherwise, these bits read as 0 and must be written as 0.
word	16 bits.

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16-54. FB8h Uncorrectable Error Status	852
16-55. FBCh Uncorrectable Error Mask	853
16-56. FC0h Uncorrectable Error Severity	854

1.1 Overview

This data book describes PLX Technology's ExpressLane™ PEX 8724, a fully non-blocking, low-latency, low-cost, and low-power 24-Lane, 6-Port PCI Express Gen 3 Multi-Root Switch. Conforming to the *PCI Express Base r3.0*, the PEX 8724 enables users to add high-bandwidth input/output (I/O) to various products, including servers, storage systems, and communications platforms. The PEX 8724's flexible hardware configuration and software programmability allows the switch to be tailored for a wide variety of application requirements.

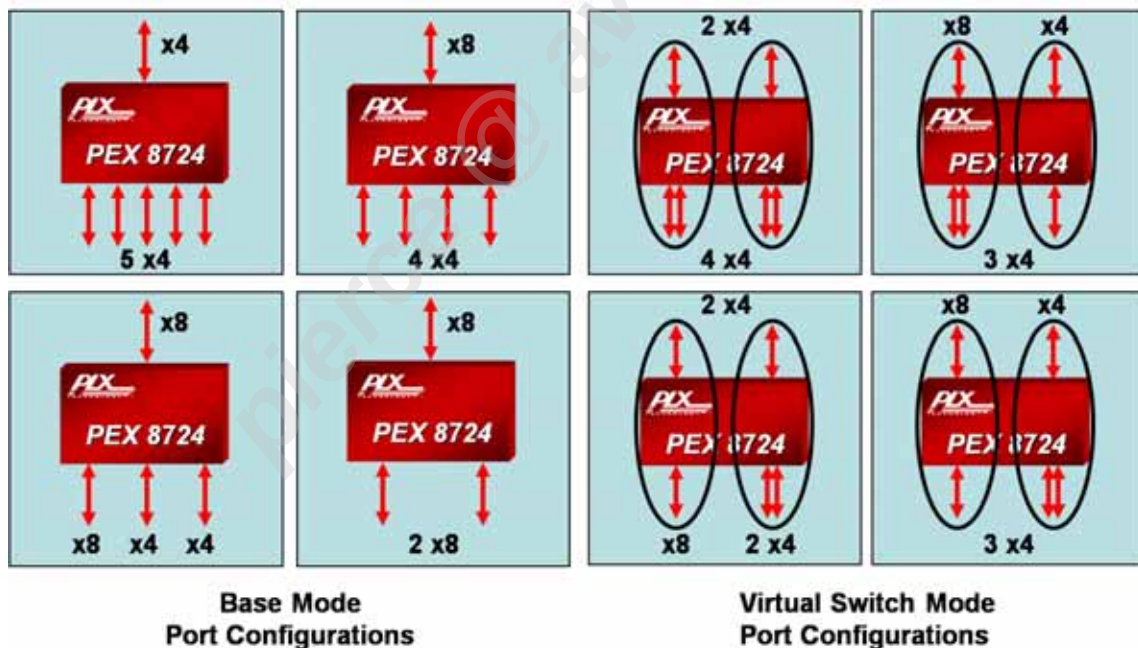
The PEX 8724 is well-suited for fan-in/out applications, as well as for applications requiring peer-to-peer communication. The PEX 8724 supports two functional modes – *Base* and *Virtual Switch*:

- **Base mode** – PEX 8724 acts as a standard PCI Express switch, supporting one Host hierarchy
- **Virtual Switch mode** – PEX 8724 supports up to four Hosts, creating up to four virtual switches within the PEX 8724, each with its own virtual hierarchy

The PEX 8724 supports multiple Port configuration options, to provide flexible solutions for optimal product design. [Figure 1-1](#) illustrates several of the possible PEX 8724 Port configurations, using differing Link widths, of x4, x8, or x16¹. When used with endpoints that require smaller Link widths, the PEX 8724 auto-negotiates down to support x1 and x2 Link width as well.

In Virtual Switch mode, the second Upstream Port, with no Downstream Ports assigned to it, can be used for failover. If the primary Host were to fail, the secondary Upstream Port would take over the Downstream Port assigned to the failing Upstream Port. The secondary Upstream Port need not initially have any Downstream Ports of its own.

Figure 1-1. Common Port Configurations



1. x16 is valid only for Port 0, Station 0.

1.2 Features

The PEX 8724 supports the following features:

- 6-Port PCI Express switch
 - 24 Lanes with integrated on-chip SerDes
 - Low-power SerDes (under 90 mW per Lane)
 - Relaxed Ordering
 - Port configuration
 - 6 independent Ports
 - Choice of Link width (quantity of Lanes) per unique Link/Port – x4, x8, or x16¹, depending upon Port configuration; x1 and x2 Link widths are also supported
 - Configurable with serial EEPROM, I²C, SMBus, and/or Host software
 - Designate any Port as the *Upstream Port* (Port 0 is recommended in Base mode)
- Multi-Root support
 - Up to four Upstream Ports supported
 - 1+1 Host Failover (one active and one backup)
 - N+1 Host Failover (N active and one backup)
 - Supported in both Transparent and Non-Transparent modes
- High Performance
 - 384 GT/s aggregate bandwidth (8.0 GT/s/Lane x 24 SerDes x 2 (full duplex))
 - Integrated 8.0 GT/s SerDes speed negotiation, for each Port
 - Full line rate on all Ports
 - Cut-Thru packet latency of less than 130 ns between symmetric (x8 to x8) ingress and egress Ports
 - Non-blocking Crossbar Switch interface supports TLP bandwidth capacity of each Link
 - Maximum Payload Size – 2,048 bytes
- *performancePAK*[™]
 - Read Pacing[™] (intelligent bandwidth allocation)
 - Multicast
 - Dynamic Buffer Pool Architecture for faster credit updates
- *visionPAK*[™]
 - Performance Monitoring
 - Per-Port Payload and Header Counters
 - Per-traffic type (Write, Read, Completion) Counters
 - PCI Express Packet Generator
 - Capable of saturating a x8 Gen 3 Link
 - SerDes Loopback
 - SerDes Eye Capture

1. x16 is valid only for Port 0, Station 0.

- Access Control Services (ACS) – Protection mechanisms for added data integrity in peer-to-peer transactions
- Alternative Routing-ID Interpretation (ARI) – Enables virtualized systems and/or highly integrated multi-function devices
- Quality of Service (QoS) support
 - All Ports support one, full-featured Virtual Channel, VCO
 - All Ports support eight Traffic Class (TC[7:0]) mapping, independently of the other Ports
 - Round-Robin (RR) and Weighted Round-Robin (WRR) Port arbitration
- Non-Transparent (NT) Bridging
 - Program any one Downstream Port as the *Upstream* NT Port
 - Enables Dual-Host, Dual-Fabric, Host-Failover applications
 - Moveable *Upstream Port*
 - Cross-link Port capability
- Reliability, Availability, Serviceability (RAS) features
 - PCI Express Standard Hot Plug Controller for three Transparent Downstream Ports, including optional usage models for Manually operated Retention Latch, by way of Manually operated Retention Latch (MRL) Sensor and Attention Button support
 - Serial Hot Plug, by way of I²C, for Hot Plug capability on up to five Transparent Downstream Ports
 - End-to-end Cyclic Redundancy Check (ECRC) and Poison bit support
 - Data path protection
 - Memory (RAM) error correction
 - Electromechanical Interlock supported with Power Enable output
 - Baseline and Advanced Error Reporting capability
 - Port (Link) Status bits available
 - Per-Port error diagnostics
 - Joint Test Action Group (JTAG) AC/DC boundary scan
- INTA# (PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) and FATAL ERROR (FATAL_ERR# (Base mode) or VSx_FATAL_ERR# (Virtual Switch mode)) (Conventional PCI SERR# equivalent) signal support
- 12 General-Purpose Input/Output (GPIO) signals
- Other PCI Express Capabilities
 - Lane reversal
 - Polarity reversal
 - Conventional PCI-compatible Link Power Management states – L0, L1¹, L2/L3 Ready, and L3 (Vaux is **not supported**)
 - Conventional PCI-compatible Device Power Management states – D0 and D3hot
 - Active State Power Management (ASPM) – L0s and L1
 - Dynamic speed (2.5, 5.0, or 8.0 GT/s (Gen 1, Gen 2, and Gen 3, respectively)) negotiation, for each Port
 - Dynamic Link width negotiation
- Out-of-Band Initialization options
 - Serial EEPROM
 - I²C and SMBus (7-bit Slave address with 100 Kbps)

1. To differentiate from ASPM L1, this method of L1 is referred to as PCI L1 in this data book.

- Testability – AC/DC JTAG support
- 19 x 19 mm², 324-ball Flip-Chip Ball Grid Array with Heat Spreader (HFC-BGA) package
- Typical power – 5.508W at 8.0 GT/s, with all Ports enabled
- Microsoft® Windows® Logo (WHQL)-compliant
- Compliant to the following specifications:
 - *PCI Local Bus Specification, Revision 3.0 (PCI r3.0)*
 - *PCI Bus Power Management Interface Specification, Revision 1.2 (PCI Power Mgmt. r1.2)*
 - *PCI Code and ID Assignment Specification, Revision 1.2*
 - *PCI to PCI Bridge Architecture Specification, Revision 1.2 (PCI-to-PCI Bridge r1.2)*
 - *PCI Express Base Specification, Revision 1.1 (PCI Express Base r1.1)*
 - *PCI Express Base Specification, Revision 2.0 (PCI Express Base r2.0)*
 - *PCI Express Base Specification, Revision 2.0 Errata*
 - *PCI Express Base Specification, Revision 2.1 (PCI Express Base r2.1)*
 - *PCI Express Base Specification, Revision 3.0 (PCI Express Base r3.0)*
 - *PCI Express Card Electromechanical Specification, Revision 2.0 (PCI ExpressCard CEM r2.0)*
 - *PCI Express Mini Card Electromechanical Specification, Revision 1.1 (PCI ExpressCard Mini CEM r1.1)*
 - *IEEE Standard 1149.1-1990, IEEE Standard Test Access Port and Boundary-Scan Architecture (IEEE Standard 1149.1-1990)*
 - *IEEE Standard 1149.1a-1993, IEEE Standard Test Access Port and Boundary-Scan Architecture*
 - *IEEE Standard 1149.1-1994, Specifications for Vendor-Specific Extensions*
 - *IEEE Standard 1149.6-2003, IEEE Standard Test Access Port and Boundary-Scan Architecture Extensions (IEEE Standard 1149.6-2003)*
 - *The I²C-Bus Specification, Version 2.1 (I²C Bus v2.1)*
 - *PHY Interface for the PCI Express Architecture, Version 2.00*
 - *System Management Bus Specification, Version 2.0 (SMBus v2.0)*



Chapter 2 Features and Applications

2.1 Flexible and Feature-Rich 24-Lane, 6-Port Switch

2.1.1 Highly Flexible Port Configurations

The PLX ExpressLane PEX 8724 PCI Express Gen 3 Multi-Root Switch offers a maximum of six configurable Ports (four per Station 0, and two per Station 1). To support specific bandwidth needs, Link widths can be individually configured as any power-of-two, from x1 to x8 or x16^{1,2}, through auto-negotiation, hardware strapping, an optional serial EEPROM, and/or the I²C Slave interface. Flexible buffer allocation, along with the PEX 8724's flexible packet flow control, maximizes throughput for applications where more traffic flows in the Downstream, rather than Upstream, direction.

The PEX 8724 architecture allows for the combining of multiple Lanes to flexible Port Link widths, as needed. *For example*, x16 Lanes can form a single x16 Port, two x8 Ports, or four x4 Ports.

Figure 1-1, “Common Port Configurations,” illustrates several of the possible PEX 8724 Port configurations in Conventional PCI mode (Base mode). The PEX 8724 can also be configured in Virtual Switch mode, where users can choose up to four Ports as Host/Upstream Ports, and assign a specific quantity of Downstream Ports to each Host. In this mode, a virtual switch is created for each Host Port and its associated Downstream Ports within the virtual switch. The traffic between a virtual switch's Ports is completely isolated from the traffic in other virtual switches. In addition to Base mode configurations, Figure 1-1 also illustrates common Port configurations in Virtual Switch mode, where each ellipse represents a virtual switch within the PEX 8724.

In Base mode, any one Port can be designated as, or dynamically changed to be, the Upstream Port (Port 0 is recommended). In Virtual Switch mode, any Port within the same virtual hierarchy can be designated as the Upstream Port for that particular hierarchy.

1. x16 is valid only for Port 0, Station 0.

2.

2.1.2 Non-Blocking Crossbar Switch Architecture

The Non-Blocking Crossbar Switch architecture is an on-chip interconnect switching fabric, which is built upon the existing PLX Switch Fabric Architecture technology. In addition to addressing concurrent multiple flows, the Crossbar Switch architecture incorporates functions required to support an efficient PCI Express switch fabric, including:

- Deadlock avoidance
- Priority preemption
- PCI Express Ordering rules
- Packet fair queuing
- Oldest first scheduling

The Crossbar Switch interconnect physical topology is that of a packet-based Crossbar Switch fabric (internal fabric), designed to simultaneously connect multiple on-chip Stations. The Crossbar Switch protocol is sufficiently flexible and robust to support a variety of embedded system requirements. The protocol is specifically designed to ease chip integration, by strongly enforcing Station boundaries and standardizing communication between Stations. The Crossbar Switch architecture basic features include:

- Multiple concurrent Data transfers
- Global ordering within the PEX 8724
- Three types of transactions – Posted, Non-Posted, and Completion (P, NP, and Cpl, respectively) – meet PCI and PCI Express Ordering and Deadlock Avoidance rules
- Optional weighting of source Ports, to support Source Port arbitration

2.1.3 Multi-Host Architecture – NT and Virtual Switch Modes

The PEX 8724 allows users to configure the switch as a single hierarchy of PCI-to-PCI bridges (Base mode), –or– as multiple hierarchies of PCI-to-PCI bridges, independent of one another (Virtual Switch mode). In Virtual Switch mode, the PEX 8724 can support up to four Hosts (one Host per virtual switch). This powerful architectural enhancement enables users to build PCI Express-based systems that support high-availability, failover, redundant, and clustered systems.

Note: When using the PEX 8724 in Virtual Switch mode, there is no data path between the virtual switch Hosts. In addition, a Host in one virtual switch cannot access nor be accessed by the endpoints associated in a different virtual switch (that is, Virtual Switch mode effectively splits the PEX 8724 to look like several independent switches).

2.1.3.1 Virtual Switch Mode (Multi-Host) and Failover Support

In Virtual Switch mode, the PEX 8724 can be configured with up to four Upstream Host Ports, each with its own dedicated Downstream Ports. The PEX 8724 can be configured for 1+1 or N+1 Host redundancy. The PEX 8724 allows the Hosts to communicate their status to one another, using special **Doorbell** registers.

In Failover mode, if a Host fails, the Host designated for failover disables the Upstream Port attached to the failing Host, then programs the Downstream Ports of that Host to its own domain. [Figure 2-1a](#) illustrates a two-Host system in Virtual Switch mode, with two virtual switches within the PEX 8724. [Figure 2-1b](#) illustrates Host 1 as being disabled after failing, and Host 2 having taken over all of Host 1's endpoints.

Figure 2-1. Virtual Switch Mode (Multi-Host) and Failover Support



Figure 2-2a. Multi-Host



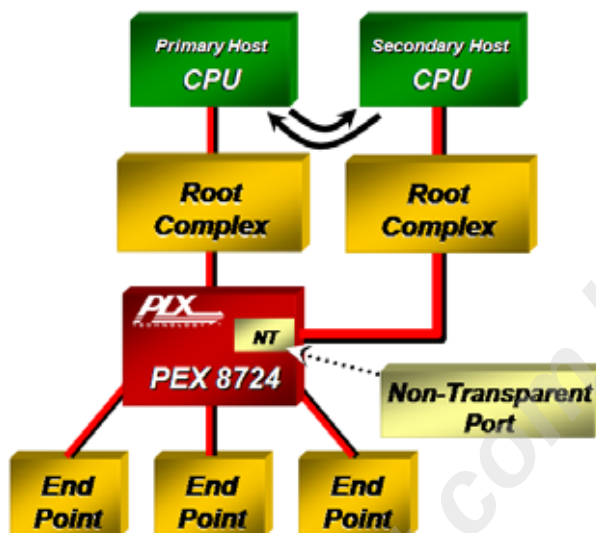
Figure 2-2b. Multi-Host Failover

2.1.3.2 Dual-Host and Failover Support – NT Mode

The PEX 8724 supports a single Non-Transparent (NT) Port (Figure 2-2), which enables the implementation of dual-Host systems for redundancy and Host failover capability. The NT Port allows systems to isolate Host memory domains, by presenting the processor subsystem as an endpoint, rather than as another memory system:

- **Base Address** registers (BARs) are used to translate addresses
- **Doorbell** registers are used to signal interrupts between the address domains
- **Scratchpad** registers are accessible from both address domains, to allow inter-processor communication

Figure 2-2. NT Port – NT Mode



2.1.4 Low Packet Latency and High Performance

The PEX 8724 architecture supports packet **Cut-Thru with a latency of less than 130 ns** between symmetric (**x8 to x8**) ingress and egress Ports. This, combined with large Packet memory, flexible common buffer/FC credit pool, and Non-Blocking Internal Switch architecture, provides full line rate on all Ports for performance-hungry applications, *such as* servers and switch fabrics. The low latency enables applications to achieve high throughput and performance. In addition to low latency, the PEX 8724 supports a Packet Payload size of up to 2,048 bytes, enabling users to achieve even higher throughput.

2.1.4.1 Data Payloads

The Data Payloads are variable length with a maximum of 2,048 bytes, as defined by the *Maximum Payload Size* field (available sizes are 128, 256, 512, 1,024, and 2,048, depending upon the quantity of enabled Ports, per Station). Read Requests **do not** include a Data Payload.

Note: Refer to the Port's **Device Control** register *Maximum Payload Size* field (offset 70h[7:5]) for Maximum Payload Size Port limitations.

2.1.4.2 Cut-Thru Mode

Cut-Thru mode can reduce latency, especially for longer packets, because the entire packet does not need to be stored before being forwarded. Instead, after the Header is decoded, the packet can be immediately forwarded. The PEX 8724 is designed to cut through TLPs, to and from every Port. By default, all Ports are enabled for Cut-Thru.

Cut-Thru mode is supported for the NT Port Link Interface, if the PEX 8724 is configured for NT mode.

Caution: *One of the drawbacks to using Cut-Thru mode is that the TLP is not known to be good until the last byte. If the TLP proves to be bad, the Cut-Thru packet must be discarded. If the TLP has already been forwarded to another device, that TLP will be framed with an EDB (End Data Bad), as opposed to the standard END.*

2.1.5 Virtual Channel and Traffic Classes

The PEX 8724 supports one Virtual Channel (VC0) and eight Traffic Classes (TC[7:0]). VC0 and TC0 are required by the *PCI Express Base r3.0*, and configured at device start-up.

2.1.6 Data Integrity

To enable designs that require **guaranteed error-free packets**, the PEX 8724 provides **End-to-end Cyclic Redundancy Check (ECRC)** protection and **Poison** bit support, as well as **Error-Correcting Code (ECC)** protection on the internal data paths and memory (RAM). ECC maintains packet integrity through the PEX 8724, by providing automatic correction of any 1-bit errors. These features are optional in the *PCI Express Base r3.0*; however, PLX provides them across its entire ExpressLane PCI Express Gen 2 and Gen 3 switch product lines.

2.1.7 Configuration Flexibility

The PEX 8724 provides several ways to configure its operations. *For example*, the PEX 8724 can be configured through Strapping signals, Host software, an optional serial EEPROM, and/or the I²C Slave interface. Additionally, the I²C Slave interface allows for easy debug during the Development phase, performance monitoring during the Operation phase, and driver and/or software upgrade.

2.1.8 Interoperability

The PEX 8724 is designed to be fully compliant with the *PCI Express Base r3.0*, and is backward-compatible to the *PCI Express Base r2.1*, *PCI Express Base r2.0*, *PCI Express Base r1.1*, and *PCI Express Base r1.0a*. Additionally, the switch supports **auto-negotiation**, **Lane reversal**, and **polarity reversal**, for maximum board design and board layout flexibility. Furthermore, the PEX 8724 is designed to be interoperable with many popular motherboards and server boards with PCI Express connections, and PCI Express endpoints (Ethernet, RAID Controllers), as well as PLX's family of PCI Express switches and bridges. All PLX ExpressLane devices undergo thorough interoperability testing at PLX's **Interoperability Lab** and compliance testing at the **PCI-SIG Compliance Workshop**, to ensure compatibility with PCI Express devices in the market.

2.1.9 Low Power with Granular SerDes Control

The PEX 8724 provides **low-power** capability that is fully compliant with the *PCI Express Base r3.0* and *PCI Power Mgmt. r1.2* Power Management (PM) specifications. Unused SerDes can be disabled, to further reduce power consumption.

The PEX 8724 supports **SerDes output software control**, to allow power and signal strength optimization within a system. The PLX SerDes implementation supports four power levels – *Off*, *Low*, *Typical*, and *High*. The SerDes block also supports **Loopback modes** and **Advanced Error Reporting**, which enables efficient system debug and management.

2.1.10 Dynamic Lane Reversal

The PEX 8724 supports dynamic Lane reversal during the Link training process. Lane reversal capability allows flexibility in determining board routing, so that PCI Express components can be connected without having to criss-cross wires. If the wiring of Lanes to a device is reversed (on both Transmitters and Receivers), only one of the two connected devices must support Lane reversal.

Either of the outside Lanes (Transmitter and Receiver pairs) of the PEX 8724 programmed Link width must be identified as being Lane 0. During Link training, both devices on the Link negotiate the Lane numbering. During the Link Training and Status State Machine (LTSSM)'s *Configuration* state, the Upstream device sends TS1 Ordered-Sets, in which each connected Lane is identified by a consecutive Lane Number, starting with Lane 0 corresponding to the physical Lane Number associated with the Port.

The Port reverses its Lane Numbers and attempts to re-train when any of the following conditions occur:

- No Receiver is detected on preferred Lane 0
- No valid Training Sets are received on preferred Lane 0 during the LTSSM's *Polling* state
- TS1 with a non-zero Lane Number Port is received on the Port's Lane 0

To confirm successful Lane Number negotiation, both devices exchange TS2 Ordered-Sets with identical Lane Numbers on each connected Lane.

2.1.11 Hot Plug for High Availability

Hot Plug capability allows users to replace hardware modules and perform maintenance, without having to power down the system. The PEX 8724 Hot Plug Capability and Advanced Error Reporting features make the switch suitable for High-Availability (HA) applications. The PEX 8724 supports both Parallel and Serial Hot Plug. Parallel Hot Plug is supported on three Transparent Downstream Ports, and/or Serial Hot Plug is supported on up to five Transparent Downstream Ports.

For further details, refer to [Chapter 10, "Hot Plug Support."](#)

2.1.12 Fully Compliant Power Management

In compliance with the *PCI Express Base r3.0* and *PCI Power Mgmt. r1.2* Power Management (PM) specifications, the PEX 8724 supports the following PM states:

- Conventional PCI-compatible Link PM states – L0, L1¹, L2/L3 Ready, and L3 (Vaux is **not supported**)
- Conventional PCI-compatible Device PM states – D0 and D3hot
- Active State Power Management (ASPM) – L0s and L1

For further details, refer to [Chapter 11, "Power Management."](#)

1. To differentiate from ASPM L1, this method of L1 is referred to as PCI L1 in this data book.

2.1.13 General-Purpose Input/Output Signals

The PEX 8724 contains 12 multiplexed General-Purpose Input/Output (GPIO) signals and associated registers, that can be programmed to function as GPIO, Link Status (PORT_GOOD) indicators, and/or Interrupt inputs. Default functionality is GPIO input; however, serial EEPROM, I²C/SMBus, and/or software can program the **GPIO** registers to define functionality for each I/O. Default functionality can also be modified by the logical value of the STRAP_TESTMODE[2:0] 3-state inputs, sampled at Fundamental Reset. Because typical designs implement PORT_GOOD functionality for enabled Ports, GPIO[6:0] are renamed as [PORT_GOOD\[6:0\]#](#).

For details, refer to the GPIO[11:7], PORT_GOOD[6:0]#, and STRAP_TESTMODE[2:0] signal descriptions in [Section 3.4, “Signal Descriptions.”](#)

Note: *The GPIO11 signal exists only in Virtual Switch mode, when four virtual switches are enabled. This signal does not exist in Base mode.*

Information related to how and when these multiplexed signals are used is provided in [Section 3.4.4.1, “STRAP_TESTMODE\[2:0\] 3-State Input Configurations.”](#)

2.1.14 **performancePAK**

Exclusive to PLX, *performancePAK* is a suite of unique and innovative performance features that enable PLX switches to be the highest-performing switches available in the market today. The *performancePAK* features consist of Read Pacing, Multicast, and Dynamic Buffer Pool.

2.1.14.1 **Read Pacing**

The Read Pacing feature allows users to throttle the quantity of Read Requests being made by Downstream devices. When a Downstream device requests several long Reads back-to-back, the Root Complex services the Read Requests from this Downstream Port in a sequential order. If this Port has a narrow Link and is therefore slow in receiving these Read packets from the Root Complex, other Downstream Ports may become starved, thus negatively impacting performance. This feature enhances performance by allowing for the adequate servicing of all Downstream devices, by intelligent handling of Read Requests.

For further details, refer to [Section 8.5, “Read Pacing.”](#)

2.1.14.2 **Multicast**

Multicast (MC) allows programs to concurrently write the same data to a group of multiple destinations. When Posted Memory Write and/or Address Routed Message TLPs entering the PEX 8724 are addressed to the MC Address range (*MC BARs*), the PEX 8724 automatically generates and transmits, if enabled, a copy of the original TLP (referred to as the *MC Copy TLP*) to the destination Ports. The MC Address space is divided into *MC Groups (MCG)*, defined by using *MC Base Address* and *MC Index Position*. Each PEX 8724 Port can elect to receive an MC Copy TLP by belonging to an *MCG*, by Setting the corresponding *MC Receive* bit. An MC TLP can be blocked using the *MC Block All* bit, if required. *MC Overlay Bar* can be used to replace the original MC TLP's address to a Unicast Address space, if the endpoint does not support MC.

For further details, refer to [Section 8.6, “Multicast.”](#)

2.1.14.3 **Dynamic Buffer Pool**

The PEX 8724 uses a dynamic buffer pool for FC management, which uses a common pool of FC Credits that is shared among other Ports within the same Station. This shared buffer pool is user-programmable, so FC credits can be allocated among the enabled Ports, as needed. Not only does this prevent wasted buffers and inappropriate buffer assignments, any un-allocated buffers remain in the common buffer pool, which can then be used by other Ports within the same Station, for faster FC credit updates.

2.1.15 visionPAK

Another PLX exclusive, *visionPAK* is a debug diagnostics suite of integrated hardware and software instruments that users can use to help bring their systems to market faster. *visionPAK* features consist of Performance Monitoring, Error Injection, SerDes Loopback, SerDes Eye Capture, and more.

2.1.15.1 Performance Monitoring

The PEX 8724's real-time performance monitoring allows users to literally “see” ingress and egress performance on each Port as traffic passes through the switch, using PLX's Software Development Kit (SDK). The monitoring is completely passive, and therefore, has no effect on overall system performance. Internal Counters provide extensive granularity down to traffic and packet type, and even allow for the filtering of traffic (*that is*, count only Memory Writes).

2.1.15.2 Error Injection

Using the PEX 8724's Error Injection feature, users can inject malformed packets and/or Fatal errors into their system, then evaluate the system's ability to detect and recover from such errors.

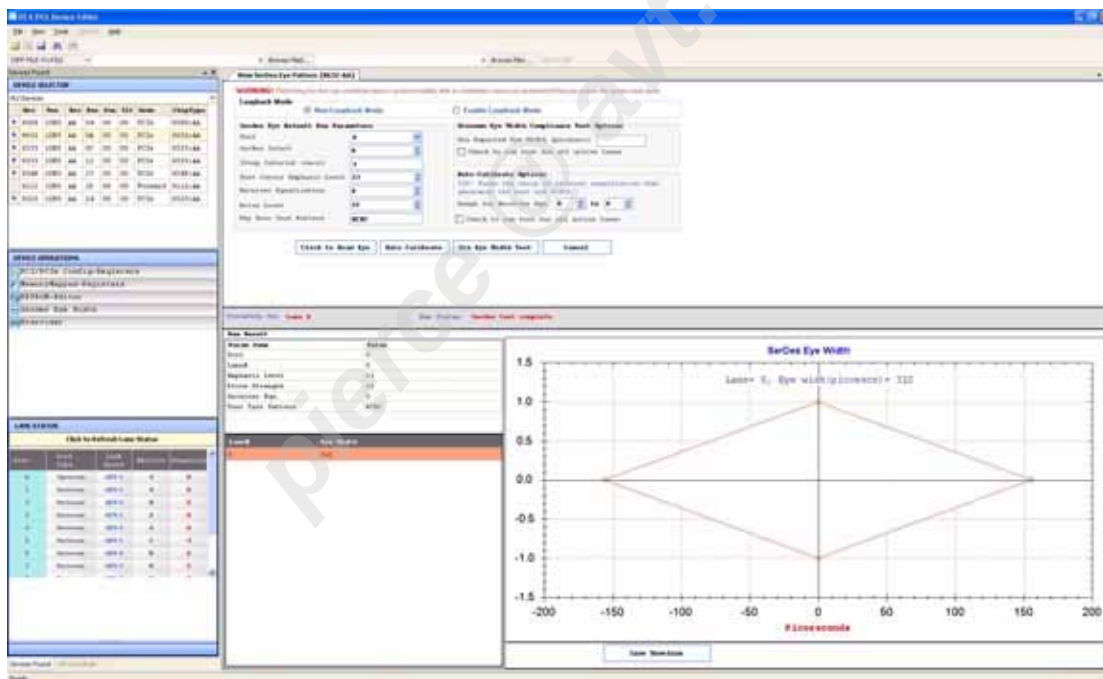
2.1.15.3 SerDes Loopback

The PEX 8724 supports External Tx, Recovered Clock, and Recovered Data Loopback modes.

2.1.15.4 SerDes Eye Capture

Users can evaluate their system's signal integrity at the Physical Layer (PHY), using the PEX 8724's SerDes Eye Capture feature. Using PLX's SDK, users can view the Receiver eye width of any Lane on the PEX 8724. Users can then modify SerDes Settings and see the impact on the Receiver eye. [Figure 2-3](#) presents a screenshot of the SDK's SerDes Eye Capture feature.

Figure 2-3. PLX SDK SerDes Eye Capture Feature



2.2 Applications

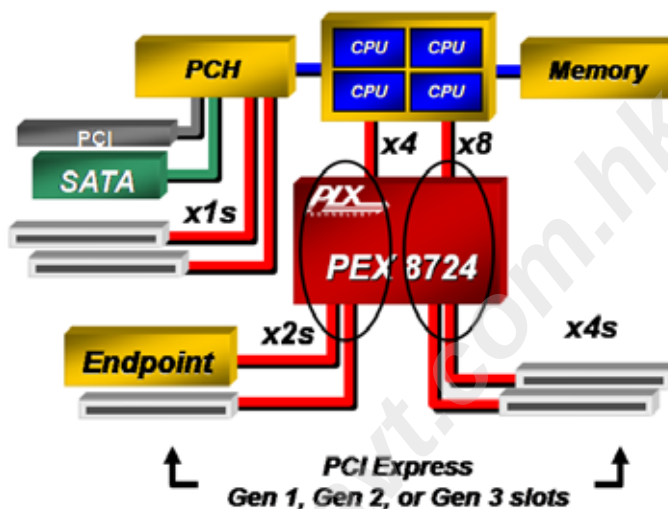
Suitable for **Host-centric** and **peer-to-peer traffic**, the PEX 8724 can be configured for a wide variety of form factors and applications.

2.2.1 Host-Centric Fan-Out

The PEX 8724, with its versatile symmetric or asymmetric Lane configuration capability, allows user-specific tuning to a variety of Host-centric applications.

Figure 2-4 illustrates a typical **server** design where, in a quad- or multi-processor system, users can assign endpoints/slots to CPU cores, to distribute the system load. The packets directed to different CPU cores go to different (user-assigned) PEX 8724 Upstream Port(s), providing better queuing and load-balancing capability, for higher performance. Conversely, the PEX 8724 can also be used in Base (Single Host) mode, to simply fan-out to endpoints.

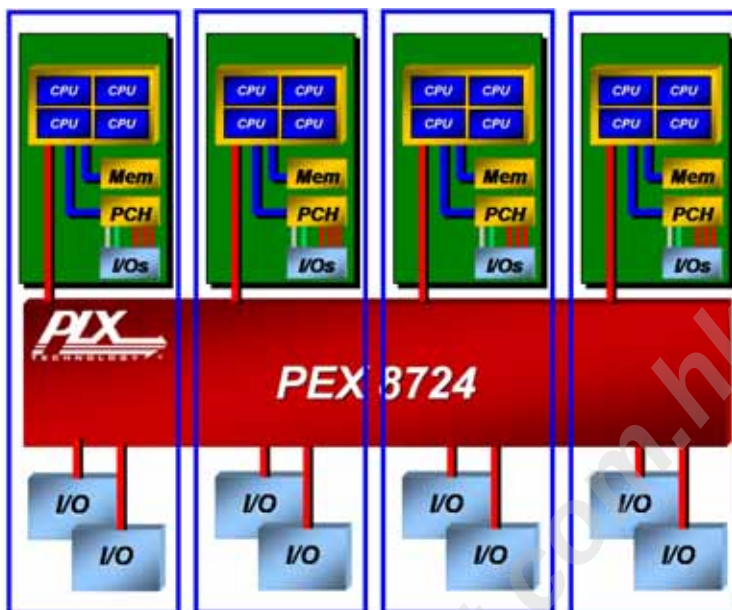
Figure 2-4. Host-Centric, Dual-Upstream



2.2.2 Multi-Host Systems

In Virtual Switch mode, with the ability to create and concurrently support up to four virtual switches/Hosts, the PEX 8724 allows each Host to fan-out to its respective endpoints. This reduces the quantity of switches required for fan-out, saving precious board space and power consumption. In [Figure 2-5](#), the PEX 8724 is being shared by four different servers (Hosts), with each server running its own applications (I/Os). The PEX 8724 assigns the endpoints to the appropriate Host, and isolates them from the other Hosts.

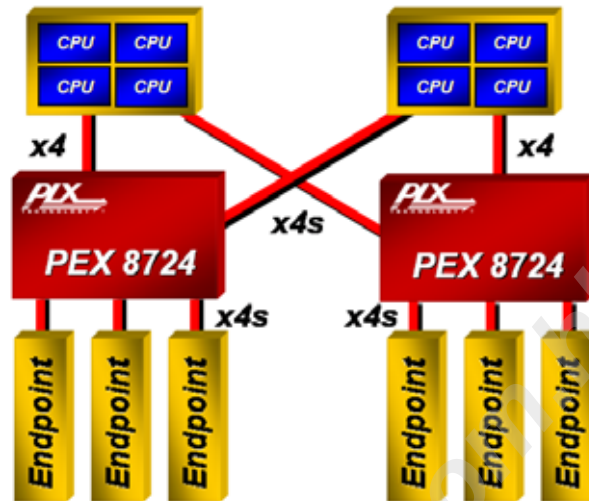
Figure 2-5. Multi-Host Systems



2.2.3 Host Failover

The PEX 8724 can also be used in applications where Host Failover is required. In the application illustrated in [Figure 2-6](#), two Hosts can be concurrently active and controlling their own domains, while exchanging status information through **Doorbell** registers and/or the I²C Slave interface. The devices can be programmed to trigger failover if the heartbeat information is not provided. In the event of a failure, the surviving device resets the endpoints connected to the failing CPU, then enumerates those endpoints within its own domain, without impacting the operation of endpoints already within its domain.

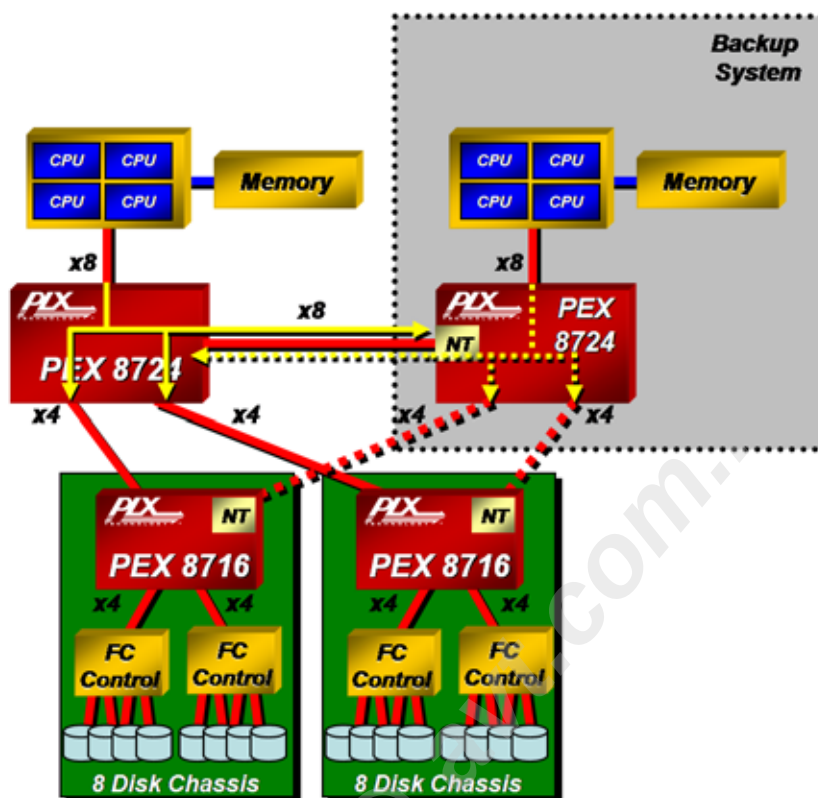
Figure 2-6. Host Failover



2.2.4 N+1 Failover in Storage Systems

The PEX 8724's Multi-Host feature can also be used to develop storage array clusters, where each Host manages a set of storage devices, independent of the other Hosts. Users can designate one of the Hosts as the Failover Host for all the other Hosts, while actively managing its own endpoints. The Failover Host communicates with the other Hosts for status/heartbeat information, and executes a Failover event if/when a Failover event is triggered. (Refer to [Figure 2-7.](#))

Figure 2-7. N+1 Failover



2.3 Software Usage Model

From the system model viewpoint, each PCI Express Port is a virtual PCI-to-PCI bridge, with its own set of PCI Express Configuration registers. The recommended Upstream Port in Base mode is Port 0; however, any Port can be configured as the Upstream Port through optional configuration, by way of a serial EEPROM, the I²C Slave interface, and/or Strapping signals. The BIOS or Host can configure the other Ports, by way of the Upstream Port, using Conventional PCI enumeration. In Virtual Switch mode, any Port within the same virtual hierarchy can be designated as the Upstream Port for that particular hierarchy.

2.3.1 System Configuration

The virtual PCI-to-PCI bridges within the PEX 8724 are compliant to the PCI and PCI Express system models. The Configuration Space registers (CSRs) in a virtual primary/secondary PCI-to-PCI bridge are accessible by Type 0 and Type 1 Configuration Requests, through the virtual primary bus interface (matching Bus Number, Device Number, and Function Number).

2.3.2 Interrupt Sources and Events

The PEX 8724 supports the INTx Interrupt Message type (compatible with *PCI r3.0* Interrupt signals) or Message Signaled Interrupts (MSIs), when enabled. The PEX 8724 generates interrupts/Messages for the following:

- Hot Plug- or Link State--triggered interrupt events
- PCI Express Hot Plug--triggered interrupt events
- Device-Specific NT-Link Port (NT mode only) or RAM ECC error events
- GPIO-generated events
- NT Doorbell-generated events (NT mode only)
- Management Port Doorbell events (VS Upstream Ports; Virtual Switch mode only)
- Management Link Status events (VS Upstream Ports; Virtual Switch mode only)
- Baseline and Advanced Error Reporting

Internally generated interrupts and interrupts forwarded from Downstream Ports are re-mapped and collapsed at the Upstream Port(s).

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Chapter 3 Signal Description

3.1 Introduction

This chapter describes the following signal-related information:

- [Signal Assignment Abbreviations](#)
- [Internal Pull-Up/Pull-Down Resistors](#)
- [Signal Descriptions](#)
- [Physical Layout](#)

3.2 Signal Assignment Abbreviations

Table 3-1 lists the abbreviations used within this chapter.

Table 3-1. Signal Assignment Abbreviations

Abbreviation	Description
#	Active-Low signal
3-state input	Associated with the Strapping signals. Can be Low (0), High-Impedance (Z; not connected/floating), or High (1).
A	Analog Input signal
APWR	Power (VDD09A) signals for SerDes Analog circuits
HCSLn ^a	High-Speed Current Steering Logic (HCSL) negative Clock inputs
HCSLp ^a	HCSL positive Clock inputs
CMLRn	Differential low-voltage, high-speed, CML negative Receiver inputs
CMLRp	Differential low-voltage, high-speed, CML positive Receiver inputs
CMLTn	Differential low-voltage, high-speed, CML negative Transmitter outputs
CMLTp	Differential low-voltage, high-speed, CML positive Transmitter outputs
CPWR	0.9V Power (VDD09) signals for low-voltage Core circuits
GND	Common Ground (VSS) for all circuits
I	Input
I/O	Bidirectional (Input or Output)
I/OPWR	1.8V Power (VDD18) signals for Input and Output interfaces
O	Output
OD	Open Drain output
PD	Weak internal pull-down resistor
PLL PWR	1.8V Power (VDD18A) signals for Phase-Locked Loop (PLL) circuits
PU	Weak internal pull-up resistor
SerDes	Serializer/De-Serializer differential low-voltage, high-speed, I/O signal pairs (negative and positive)
STRAP	Signals used for PEX 8724 configuration, operational mode Setting, and Factory Test ; these signals generally are not toggled at runtime
TS ^b	Tri-statable Output (High-Impedance when output is not driven)

a. For REFCLK inputs, HCSL source is recommended.

b. Used in the Signal tables, **Type** column. All other use of TS is for “Training Sequence.”

3.3 Internal Pull-Up/Pull-Down Resistors

The PEX 8724 contains I/O buffers that have weak internal pull-up or pull-down resistors, indicated in this chapter by PU or PD, respectively, in the signal description tables (**Type** column). If a signal with this notation is used and no board trace is connected to the signal, the internal resistor is usually sufficient to keep the signal from toggling. However, if a signal with this notation is not used, but is connected to a board trace and is not used nor driven by an external source at all times, the internal resistors might not be sufficiently strong, to hold the signal in the inactive state. In cases such as these, it is recommended that the signal be pulled or tied High to **VDD18** or Low to **VSS** (Ground), as appropriate, through a 3K Ω to 10K Ω resistor.

Table 3-2 lists the internal pull-up and pull-down resistor values.

Table 3-2. Internal Resistor Values

Internal Resistor	Minimum	Typical	Maximum	Units
PU	38K	57K	92K	Ω
PD	37K	53K	99K	Ω

3.4 Signal Descriptions

Note: If there is more than one ball per signal name that includes a numbered range, the locations are listed in the same sequence in which the range is listed, starting at the top row, from left to right. For example, *PEX_PERn11* is located at R13, *PEX_PERn10* is located at R12, and so forth.

If there is more than one ball per signal name that does not include a numbered range (such as *VDD09*), the locations are listed in ascending alphanumeric order.

The PEX 8724 signals are divided into the following groups:

- PCI Express Signals
- Hot Plug Signals
 - Parallel Hot Plug Signals
 - Serial Hot Plug Signals
- Serial EEPROM Signals
- Strapping Signals
 - STRAP_TESTMODE[2:0] 3-State Input Configurations
- JTAG Interface Signals
- I2C/SMBus Slave Interface Signals
- Device-Specific Signals
- External Resistor Signals
- Thermal Diode Signals
- No Connect Signals
- Power and Ground Signals

Each signal group is described in the sections that follow. The location and signal type, as well as whether the signal is multiplexed, is also included.

3.4.1 PCI Express Signals

Table 3-3 defines the PCI Express SerDes and Control signals.

Table 3-3. PCI Express Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PEX_PERn[15:0]	R18, R17, R15, R14, R13, R12, R11, R10, R9, R8, R7, R6, R5, R4, R2, R1	CMLRn	No	–
		Negative Half of PCI Express Receiver Differential Signal Pairs for Station 0 (16 Balls)		
PEX_PERn[31:24]	D12, D11, D10, D9, D8, D7, D5, D4	CMLRn	No	–
		Negative Half of PCI Express Receiver Differential Signal Pairs for Station 1 (8 Balls)		
PEX_PERp[15:0]	P18, P17, P15, P14, P13, P12, P11, P10, P9, P8, P7, P6, P5, P4, P2, P1	CMLRp	No	–
		Positive Half of PCI Express Receiver Differential Signal Pairs for Station 0 (16 Balls)		
PEX_PERp[31:24]	E12, E11, E10, E9, E8, E7, E5, E4	CMLRp	No	–
		Positive Half of PCI Express Receiver Differential Signal Pairs for Station 1 (8 Balls)		
PEX_PERST#	L3	I, PU	No	–
		PCI Express Reset Input Used to cause a Fundamental Reset. Base Mode Refer to Section 5.1.1, “Resets – Base Mode,” for further details. Virtual Switch Mode In Virtual Switch mode, PEX_PERST# assertion resets all virtual switches. Refer to Section 5.1.2, “Resets – Virtual Switch Mode,” for further details. Notes: The following Uncorrectable errors reported in the Uncorrectable Error Status register (offset FB8h) are fatal to the PEX 8724, and require a PEX_PERST# reset to recover: • Bit 17, Receiver Overflow Status • Bit 18, Malformed TLP Status • Bit 22, Uncorrectable Internal Error Status VSx_PERST# inputs, defined in Table 3-13 , are the Reset inputs for individual virtual switches.		
PEX_PETn[15:0]	V18, V17, V16, V15, V14, V13, V11, V10, V9, V8, V6, V5, V4, V3, V2, V1	CMLTn	No	–
		Negative Half of PCI Express Transmitter Differential Signal Pairs for Station 0 (16 Balls) PEX_PETn[15:0] must be AC-coupled. Use a 0.22 µF capacitor.		
PEX_PETn[31:24]	A12, A11, A9, A8, A7, A6, A5, A4	CMLTn	No	–
		Negative Half of PCI Express Transmitter Differential Signal Pairs for Station 1 (8 Balls) PEX_PETn[31:24] must be AC-coupled. Use a 0.22 µF capacitor.		

Table 3-3. PCI Express Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PEX_PETp[15:0]	U18, U17, U16, U15, U14, U13, U11, U10, U9, U8, U6, U5, U4, U3, U2, U1	CMLTp	No	–
		Positive Half of PCI Express Transmitter Differential Signal Pairs for Station 0 (16 Balls) PEX_PETp[15:0] must be AC-coupled. Use a 0.22 μ F capacitor.		
PEX_PETp[31:24]	B12, B11, B9, B8, B7, B6, B5, B4	CMLTp	No	–
		Positive Half of PCI Express Transmitter Differential Signal Pairs for Station 1 (8 Balls) PEX_PETp[31:24] must be AC-coupled. Use a 0.22 μ F capacitor.		
PEX_REFCLKn	M1	HCSLn	No	–
		Negative Half of 100-MHz PCI Express Constant Frequency Reference Clock Input Signal Pair An internal coupling capacitor is provided (an external capacitor is <i>not</i> needed).		
PEX_REFCLKp	M2	HCSLp	No	–
		Positive Half of 100-MHz PCI Express Constant Frequency Reference Clock Input Signal Pair An internal coupling capacitor is provided (an external capacitor is <i>not</i> needed).		

3.4.2 Hot Plug Signals

The PEX 8724 includes signals for both Parallel and Serial Hot Plug support.

Parallel Hot Plug can be implemented on any of three Transparent Downstream Ports, as selected by the **Parallel Hot Plug Control** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3A4h[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively).

Serial Hot Plug can be implemented on any Transparent Downstream Ports. If a Transparent Downstream Port is both Parallel- and Serial Hot Plug-capable, the Serial Hot Plug Controller is used, by default, unless the Port's **Power Management Hot Plug User Configuration** register *Serial Hot Plug Override Parallel Disable* bit (offset F70h[19]) is Set.

Hot Plug signals are enabled, configured, and accessed through the Transparent Downstream Port's **Slot Capability** and **Slot Status and Control** registers (Downstream Ports, offsets 7Ch and 80h, respectively). Also, each Port's **Power Management Hot Plug User Configuration** register provides additional Device-Specific configuration and control, for both Parallel and Serial Hot Plug implementations.

Both signal types are discussed in the sections that follow.

3.4.2.1 Parallel Hot Plug Signals

The PEX 8724 includes 10 signals, per Hot Plug-capable Port, that support the Parallel Hot Plug Controller (HP_), as defined in Table 3-4. These signals are active only for Hot Plug-capable Transparent Downstream Ports configured at start-up.

For further details regarding Hot Plug, refer to Chapter 10, “Hot Plug Support.”

Notes: All Parallel Hot Plug signals are I/O; however, their logical operation is either input or output, as described for each signal.

All Parallel Hot Plug signals are duplicated for each Hot Plug-capable Port, as A, B, and C signals, which map to any Transparent Downstream Port, as selected by the **Parallel Hot Plug Control** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3A4h[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively).

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_ATNLED_[C, B, A]#	F15, E1, B17	I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Attention LED Outputs (3 Balls) Active-Low Slot Control Logic outputs that are used to drive the Attention Indicator. Asserted Low to turn On (illuminate) the LED. Enabled when the Port's Slot Capability register <i>Attention Indicator Present</i> bit (Downstream Ports, offset 7Ch[3]) is Set and controlled by the Port's Slot Control register <i>Attention Indicator Control</i> field (Downstream Ports, offset 80h[7:6]). When software writes to the <i>Attention Indicator Control</i> field, a Command Completed interrupt can be generated to notify the Host that the command has been executed. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated to the Host, when the following conditions are met: - Port's Slot Capability register <i>Attention Indicator Present</i> bit (Downstream Ports, offset 7Ch[3]) is Set, and - Port's Slot Control register <i>Command Completed Interrupt Enable</i> bit is not masked (Downstream Ports, offset 80h[4], is Set), and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If HP_ATNLED_x# outputs are used, each requires an external current-limiting resistor. If HP_ATNLED_x# functionality is enabled but the outputs are not used, the signals can remain unconnected.		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_BUTTON_[C, B, A]#	B15, H2, E17	I/O as I, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Attention Button Inputs (3 Balls) Active-Low Slot Control Logic inputs that are connected directly to the Attention Button, with input assertion status latched in the Port's Slot Status register <i>Attention Button Pressed</i> bit (Downstream Ports, offset 80h[16]). Enabled when the Port's Slot Capability register <i>Attention Button Present</i> bit (Downstream Ports, offset 7Ch[0]) is Set. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated, to notify the Host of intended board insertion or removal, when the following conditions are met: - Port's HP_BUTTON_x# is not masked (Port's Slot Control register <i>Attention Button Pressed Enable</i> bit (Downstream Ports, offset 80h[0]) is Set), and - Port's Slot Capability register <i>Hot Plug Capable</i> bit (Downstream Ports, offset 7Ch[6]) is Set, and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If HP_BUTTON_x# functionality is enabled but the inputs are not used, the signals can remain unconnected. Note: HP_BUTTON_x# inputs are internally de-bounced, but must remain stable for at least 10 ms.		
HP_CLKEN_[C, B, A]#	D17, L18, G18	I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Reference Clock Enable Outputs (3 Balls) Active-Low Slot Control Logic outputs that control the connection of the external REFCLK to the slot. Enabled when the Port's Slot Capability register <i>Power Controller Present</i> bit (Downstream Ports, offset 7Ch[1]) is Set, and controlled by the Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10]). The time delay from HP_PWREN_x output assertion to HP_CLKEN_x# output assertion is programmable (through serial EEPROM load) from 128 to 512 ms, in the Port's Power Management Hot Plug User Configuration register <i>HPC Tpepv</i> field (Downstream Ports, offset F70h[4:3]). When this register field is programmed to 00b (default), HP_PWR_GOOD_x input assertion controls the time delay from HP_PWREN_x output assertion to HP_CLKEN_x# output assertion. If HP_CLKEN_x# functionality is enabled but the outputs are not used, the signals can remain unconnected.		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_MRL_[C, B, A]#	C14, F1, J18	I/O as I, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Manually Operated Retention Latch Sensor Inputs (3 Balls) Active-Low Slot Control Logic inputs that are connected directly to an optional Manually operated Retention Latch (MRL) Sensor that is logic Low when the latch is closed. Enabled when the Port's Slot Capability register <i>MRL Sensor Present</i> bit (Downstream Ports, offset 7Ch[2]) is Set. When enabled, HP_MRL_x# input assertion enables Hot Plug output sequencing to turn On the slot's power (HP_PWREN_x and HP_PWRLED_x#) and clock (HP_CLKEN_x#), and de-assert Reset (HP_PERST_x#) after reset, as illustrated in Figure 10-2, "Hot Plug Outputs when Power Controller Present and Power Controller Control Bits Are Cleared," or under software control. A change in the HP_MRL_x# input states is latched in the Port's Slot Status register <i>MRL Sensor Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[18]), and the state change can assert an interrupt to notify the Host of a change in the MRL Sensor state. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated, when the following conditions are met: - Port's HP_MRL_x# is not masked (Port's Slot Control register <i>MRL Sensor Changed Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[2]) is Set), and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If the associated Hot Plug-capable Transparent Downstream Port connects to a PCI Express board slot that does not implement an MRL Sensor, the HP_MRL_x# input is typically connected to the corresponding HP_PRST_x# input and a pull-up resistor, with the common node connected to the PRST2# signal(s) at the slot. If the associated Hot Plug-capable Transparent Downstream Port instead connects directly to a device, and Hot Plug signals (such as HP_PERST_x#) are used, pull the HP_MRL_x# input Low to VSS (Ground), to enable the automatic Hot Plug output sequencing following switch Reset (PEX_PERST# or Hot Reset) de-assertion. Otherwise, if Hot Plug signals are not used, HP_MRL_x# inputs can remain unconnected. When an HP_MRL_x# input is enabled and toggled High on a powered slot, REFCLK to the slot is automatically disconnected and the slot power is automatically turned Off, as illustrated in Figure 10-3, "Hot Plug Automatic Slot Power-Down Sequence." Note: HP_MRL_x# inputs are internally de-bounced, but must remain stable for at least 10 ms. HP_MRL_x#, if enabled, is not de-bounced when sampled immediately after reset.		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_PERST_[C, B, A]#	F14, J2, D18	I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Reset Outputs (3 Balls) Active-Low Slot Control Logic outputs that are used to reset the slot. When the Port's Slot Capability register <i>Power Controller Present</i> bit (Downstream Ports, offset 7Ch[1]) is Set, the HP_PERST_x# output states can be controlled by software, using the Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10]). If HP_PERST_x# functionality is enabled but the outputs are not used, the signals can remain unconnected.		
HP_PRSNT_[C, B, A]#	C15, G2, A17	I/O as I, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug PRSNT2# Inputs (3 Balls) Active-Low Slot Control Logic inputs that connect to the slot's PRSNT2# signal, which on the add-in board connects to the slot's PRSNT1# signal, which is typically grounded on the motherboard. Changes in the HP_PRSNT_x# input states are latched in the Port's Slot Status register <i>Presence Detect Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[19]), and the state changes can assert interrupts, to notify the Host of board presence or absence. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated, when the following conditions are met: - Port's HP_PRSNT_x# is not masked (Port's Slot Control register <i>Presence Detect Changed Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[3]) is Set), and Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If HP_PRSNT_x# functionality is enabled but the inputs are not used, the signals can remain unconnected. <i>Note: HP_PRSNT_x# inputs are internally de-bounced, but must remain stable for at least 10 ms.</i>		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_PWREN_[C, B, A]	D15, J3, C18	I/O as O, PD	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Power Enable Outputs (3 Balls) Active-High Slot Control Logic outputs that control the slot power state. When these outputs are High, power is enabled to the slot(s). Enabled when the Port's Slot Capability register <i>Power Controller Present</i> bit (Downstream Ports, offset 7Ch[1]) is Set. When software turns the slot's Power Controller On or Off (Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10])), a Command Completed interrupt can be generated, to notify the Host that the command has been executed. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated to the Host, when the following conditions are met: - Port's Slot Control register <i>Command Completed Interrupt Enable</i> bit is not masked (Downstream Ports, offset 80h[4], is Set), and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. When HP_MRL_x# are enabled (Port's Slot Capability register <i>MRL Sensor Present</i> bit (Downstream Ports, offset 7Ch[2]) is Set), HP_MRL_x# input assertion enables Hot Plug output sequencing to turn On the slot's power, by asserting HP_PWREN_x outputs after reset, as illustrated in Figure 10-2, "Hot Plug Outputs when Power Controller Present and Power Controller Control Bits Are Cleared," or under software control. If HP_PWREN_x functionality is enabled but the outputs are not used, the signals can remain unconnected. Note: HP_PWREN_x polarity is inverted with respect to HP_PWRENx# functionality in PLX ExpressLane Gen 1 switches.		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_PWRFLT_[C, B, A]#	E15, K3, H18	I/O as I, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Power Fault Inputs (3 Balls) Slot Control Logic inputs that, when asserted, indicate that the slot's external Power Controller detected a power fault on one or more supply rails. By default (with SPARE1=H when sampled at PEX_PERST# input de-assertion), HP_PWRFLT_x# are Active-Low; otherwise, if SPARE1=L, HP_PWRFLT_x are Active-High (however, the Serial Hot Plug SHP_PWRFLT# inputs from the external I²C I/O Expanders are always Active-Low). Enabled when the Port's Slot Capability register <i>Power Controller Present</i> bit (Downstream Ports, offset 7Ch[1]) is Set, and input assertion status is latched in the Port's Slot Status register <i>Power Fault Detected</i> bit (Downstream Ports, offset 80h[17]). An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated, to notify the Host of a power fault, when the following conditions are met: - Port's HP_PWRFLT_x# is not masked (Port's Slot Control register <i>Power Fault Detector Enable</i> bit (Downstream Ports, offset 80h[1]) is Set), and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If HP_PWRFLT_x# functionality is enabled but the inputs are not used, the signals can remain unconnected. <i>Note:</i> If HP_PWREN_x and HP_CLKEN_x# outputs are not used, HP_PWRFLT_x# can be used as General-Purpose inputs, with status reflected in the Port's Power Fault Detected bit, provided that the Port's Power Controller Present bit is Set.		

Table 3-4. Parallel Hot Plug Signals for Ports C, B, A^a (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
HP_PWR_GOOD_[C, B, A]	H15, D1, K18	I/O as I, PD	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Power Good Inputs (3 Balls) Active-High (default) inputs that, when enabled (default), cause the Slot Control Logic to delay HP_CLKEN_x# output assertion (to turn On REFCLK to the slot), until HP_PWR_GOOD_x inputs are asserted to indicate that the installed module's power supplies are active and stable. Signal polarity can be changed to Active-Low, by programming the serial EEPROM to Set the Port's Power Management Hot Plug User Configuration register <i>HP_PWR_GOOD_x Active-Low Enable</i> bit (Downstream Ports, offset F70h[6]). Polarity must not be changed by I²C, because that is too slow for initialization. HP_PWR_GOOD_x inputs are disabled when the Port's register's <i>HPC Tpepv</i> field (Downstream Ports, offset F70h[4:3]) is programmed to a value other than 00b, to cause HP_CLKEN_x# output assertion to follow HP_PWREN_x output assertion, by a fixed delay (128, 256, or 512 ms). If HP_PWR_GOOD_x functionality is enabled but the associated set of Hot Plug signals is not used, these inputs can remain unconnected.		
HP_PWRLED_[C, B, A]#	G15, H1, F18	I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Hot Plug Power LED Outputs (3 Balls) Active-Low Slot Control Logic outputs that are used to drive the Power Indicator. Asserted Low to turn On (illuminate) the LED. Enabled when the Port's Slot Capability register <i>Power Indicator Present</i> bit (Downstream Ports, offset 7Ch[4]) is Set, and controlled by the Port's Slot Control register <i>Power Indicator Control</i> field (Downstream Ports, offset 80h[9:8]). When software writes to the <i>Power Indicator Control</i> field, a Command Completed interrupt can be generated, to notify the Host that the command has been executed. An interrupt (INTx Message, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output, all mutually exclusive, on a per-Port basis) can be generated to the Host, when the following conditions are met: - Port's Slot Capability register <i>Power Indicator Present</i> bit (Downstream Ports, offset 7Ch[4]) is Set, and - Port's Slot Control register <i>Command Completed Interrupt Enable</i> bit is not masked (Downstream Ports, offset 80h[4], is Set), and - Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]) is Set. If HP_PWRLED_x# outputs are used, each requires an external current-limiting resistor. If HP_PWRLED_x# functionality is enabled but the outputs are not used, the signals can remain unconnected.		

- a. If Hot Plug outputs are used and HP_MRL_x# inputs are not used, pull the HP_MRL_x# inputs Low, so that Hot Plug outputs will properly sequence if the serial EEPROM is blank or missing. Default register values enable HP_MRL_x# inputs, which must then be asserted, to cause Hot Plug outputs to toggle (for example, to de-assert HP_PERST_x# outputs and assert HP_PWRLED_x# outputs).

3.4.2.2 Serial Hot Plug Signals

Transparent Downstream Ports can implement Hot Plug, by using external I²C I/O Expanders (one 16-pin Maxim MAX7311, NXP PCA9555, or TI PCA9555 per slot, –or– one 40-pin NXP PCA9698 per two slots). All Ports implementing Serial Hot Plug can concurrently use either type of I/O Expander (16- or 40-pin). The Serial Hot Plug Controller queries each I/O Expander for its Device ID. 40-Pin I/O Expanders implement Device ID, and 16-pin I/O Expanders do not. If the device responds to the PEX 8724's Device ID query, the Serial Hot Plug Controller assumes that the I/O Expander is a 40-pin device. The query can be disabled, by Setting the Port's **Power Management Hot Plug User Configuration** register *40-Pin I/O Expander Scan Disable* bit (offset F70h[17]).

Table 3-4 defines the signals that support Serial Hot Plug. Additionally, the PEX 8724 supports external Serial Hot Plug signals on the I²C I/O Expander. (Refer to Section 10.11.3, “I²C I/O Expander Parts Selection and Pin Definition.”)

Table 3-5. Serial Hot Plug Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
I2C_SCL1	K17	OD	No	–
		I²C Serial Clock Line for Serial Hot Plug Support I ² C Clock source. Used with the external I ² C I/O Expander, and must be bused to each I/O Expander's Clock (SCL) pin. In combination with I2C_SDA1, forms the PEX 8724 I ² C Master interface. Because the 1.8V signaling is less than the voltage defined by the I ² C Bus v2.1 (5V, 3.3V), external voltage level shifter circuitry might be required. I2C_SCL1 generally requires an external pull-up resistor. Alternatively, a voltage translation circuit can provide the current source.		
I2C_SDA1	L17	OD	No	–
		I²C Serial Data Line for Serial Hot Plug Support Transmits and receives I ² C data. Used with the external I ² C I/O Expander, and must be bused to each I/O Expander's Data (SDA) pin. In combination with I2C_SCL1, forms the PEX 8724 I ² C Master interface. Because the 1.8V signaling is less than the voltage defined by the I ² C Bus v2.1 (5V, 3.3V), external voltage level shifter circuitry might be required. I2C_SDA1 generally requires an external pull-up resistor. Alternatively, a voltage translation circuit can provide the current source.		

Table 3-5. Serial Hot Plug Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
SHPC_INT#	A15	I/O as I, PU	No	–
		Serial Hot Plug Controller Interrupt Input Active-Low interrupt input from external I²C I/O Expanders. Used only by Serial Hot Plug-capable Transparent Downstream Ports. The I/O Expander asserts its INT# output whenever any of its inputs change state, and de-asserts its INT# output when the corresponding Input Port Data register (that changed state) is read. When the SHPC_INT# Interrupt input (connected to the INT# output of all I/O Expanders) is asserted, the I²C Master interface begins reading the Input Port registers of all I/O Expanders, and copies the values to the appropriate bits in the Port's Slot Status register (Downstream Ports, offset 80h). The I²C Master interface halts the reading of I/O Expander registers when the SHPC_INT# input de-asserts. If used, SHPC_INT# input requires an external pull-up resistor; otherwise, this input can remain unconnected. Note: By default, SHPC_INT# input is internally de-bounced, but must remain stable for at least 10 ms. Internal de-bouncing can be disabled, by Setting the Port's Power Management Hot Plug User Configuration register Serial Hot Plug INTx De-Bounce Disable bit (offset F70h[18]).		

3.4.3 Serial EEPROM Signals

The PEX 8724 includes four signals for interfacing to a serial EEPROM, defined in [Table 3-6](#). The serial EEPROM is used to load register values immediately following Reset (**PEX_PERST#** or Hot Reset/DL_Down). Most registers can be programmed by serial EEPROM, as indicated in the register descriptions.

For information regarding serial EEPROM use, refer to [Chapter 6, “Serial EEPROM Controller.”](#)

Table 3-6. Serial EEPROM Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
EE_CS#	G16	I/O as O, PU	No	–
		Active-Low Serial EEPROM Chip Select Output		
EE_DI	H16	I/O as O, TS	No	–
		PEX 8724 Output to Serial EEPROM Data Input <i>Note: Always driven.</i>		
EE_DO	F17	I/O as I, PU	No	–
		PEX 8724 Input from Serial EEPROM Data Output Should be pulled High to VDD18 .		
EE_SK	E16	I/O as O, PU	No	–
		Serial EEPROM Clock Frequency Output Programmable, by way of the Serial EEPROM Clock Frequency register EepFreq[2:0] field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[2:0]), to the following: • 1 MHz (default) • 1.98 MHz • 5 MHz • 9.62 MHz • 12.5 MHz • 15.6 MHz • 17.9 MHz		

3.4.4 Strapping Signals

Note: *Table 4-1 indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.*

The PEX 8724 Strapping signals, described in [Table 3-7](#), define the configuration of Upstream Port and NT Port assignment, Link width, and various setup and test modes.

Many of the Strapping signals are I/O 3-state inputs, which can be Low (0), High-Impedance (Z; not connected/floating), or High (1).

Strapping signals that have 2-state inputs (rather than 3-state inputs; indicated simply as “inputs”) use internal pull-up and pull-down resistors to define the default configuration; if the PEX 8724 configuration must be changed from the default, external pull-up and/or pull-down resistors can be connected. External resistors are not required unless the Strapping signals:

- Must be changed from the default logic state, –or–
- Are connected to circuit traces (the internal resistors are relatively weak, and may not be sufficiently strong, to hold circuit traces to the default input states)

After a Fundamental Reset, the **Link Capability**, **Port Configuration**, and **VS0 Upstream** registers (offsets 74h, 300h, and 360h, respectively) capture Strapping signal status. Strapping signal Configuration data can be changed by writing new data to these registers from the serial EEPROM (if present) and/or Management Port. I²C/SMBus can also change Strapping signal Configuration data; however, the STRAP_I2C_SMBUS_CFG_EN# input should be pulled or tied Low to VSS (Ground), to prevent linkup and Host enumeration. Then, when I²C programming is complete, I²C/SMBus should lastly Set the **Configuration Release** register *Initiate Configuration* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3ACh[0]), to enable linkup and allow subsequent Host enumeration.

Table 3-7. Strapping Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States	
		Description			
STRAP_GEN1_GEN2	A16	I/O, 3-state input	No	–	
		PCI Express Data Rate Select Selects the maximum PCI Express data rate for all Ports, by globally changing the Maximum Link Speed and Target Link Speed (offsets 74h[3:0] and 98h[3:0], respectively) values, in all Ports. The value of this 3-state input is latched and reflected in the Strap Configuration register Gen 2 Mode Status and Gen 1 Mode Status bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[12:11], respectively). Note: Normally, this I/O should be pulled or tied High to VDD18 , to enable Gen 3 operation.			
		STRAP_GEN1_GEN2 Input State and Link Speed	Offset 74h[3:0] Value	Offset 98h[3:0] Value	Offset 46Ch[12:11] Value
		0 = Gen 1 (2.5 GT/s)	1h	1h	10b
		Z = Gen 2 (5.0 GT/s)	2h	2h	01b
		1 = Gen 3 (8.0 GT/s)	3h	3h	11b

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States												
		Description														
STRAP_I2C_SMBUS_CFG_EN#	K2	I/O as I, PU	No	–												
		I²C/SMBus Configuration Enable Enables or disables the option for I²C/SMBus to initialize register values prior to initial Link training. This option is similar to serial EEPROM initialization. While both serial EEPROM and I²C/SMBus can be used for device initialization, I²C/SMBus cannot replace serial EEPROM in programming the Clock Enable register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 314h). The value of this input is latched and its inverse value is reflected in the Strap Configuration register <i>I2C Configuration Enable Status</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[5]). Must be pulled or tied High to VDD18 or Low to VSS (Ground). L = Enables I²C/SMBus for initial device configuration. The serial EEPROM can also be used in this mode. After I²C and/or the serial EEPROM Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, offset 3ACh[0]), all Ports start Link training. H = Disables I²C/SMBus for initial device configuration. The <i>Initiate Configuration</i> bit is Set by hardware, immediately after the PEX 8724 comes out of reset. After the serial EEPROM load finishes, all Ports start Link training (assuming that the serial EEPROM does not Clear the <i>Initiate Configuration</i> bit). Note: The Delayed Linkup option might require software support to delay Host enumeration until I²C/SMBus completes switch initialization (by Setting the <i>Initiate Configuration</i> bit; otherwise system software could miss detection of the PCI Express hierarchy (precluding device enumeration)).														
STRAP_I2C_SMBUS_EN	D3	I/O, 3-state input	No	–												
		I²C/SMBus Protocol Select Selects the I²C or SMBus protocol, and defines the default SMBus Configuration register <i>SMBus Enable</i> and <i>ARP Disable</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[0 and 8], respectively) values, as listed in the subtable below. The value of this 3-state input is latched and reflected in the Strap Configuration register <i>SMBus Enable Status</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[6]). <table><tr><th rowspan="2">STRAP_I2C_SMBUS_EN Input State</th><th colspan="2">Offset 2C8h Values</th><th rowspan="2">Offset 46Ch[6] Value</th></tr><tr><th>Bit 0</th><th>Bit 8</th></tr><tr><td>0</td><td>0</td><td>–</td><td>1</td></tr><tr><td>Z, 1</td><td>1</td><td>1</td><td>0</td></tr></table> 0 = Enables I²C Slave protocol on the I2C_SCL0 and I2C_SDA0 2-wire bus Z, 1 = Enables SMBus without ARP, on the I2C_SCL0 and I2C_SDA0 2-wire bus			STRAP_I2C_SMBUS_EN Input State	Offset 2C8h Values		Offset 46Ch[6] Value	Bit 0	Bit 8	0	0	–	1	Z, 1	1
STRAP_I2C_SMBUS_EN Input State	Offset 2C8h Values		Offset 46Ch[6] Value													
	Bit 0	Bit 8														
0	0	–	1													
Z, 1	1	1	0													

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_MGMT_PORT_EN	H5	I/O, 3-state input	No	–
		Virtual Switch Mode Management Port Initialization Enables/Disables both the Virtual Switch mode Management Port, and the option for Management Port Host software to configure the PEX 8724 prior to releasing all other Ports to start Link training. The Management Port is selected by STRAP_UPSTRM_PORTSEL[2:0], –or– serial EEPROM configuring the Management Port Control register <i>Active Management Port Enable</i> and <i>Active Management Port</i> bits (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[5 and 4:0], respectively), in Virtual Switch mode. 0 = The Management Port is enabled, and the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, offset 3ACh[0]) is Set. After the serial EEPROM load finishes, all Ports start Link training (assuming that the serial EEPROM, or alternatively I²C/SMBus if the STRAP_I2C_SMBUS_CFG_EN# input is asserted Low, does not Clear the <i>Initiate Configuration</i> bit). Z = The Management Port is enabled, the <i>Initiate Configuration</i> bit is Cleared, and only the Management Port links up. After Management Port Host software programs register values and subsequently Sets the <i>Initiate Configuration</i> bit, or alternatively if the bit is Set by either the serial EEPROM, or I²C/SMBus if the STRAP_I2C_SMBUS_CFG_EN# input is asserted Low, all other Ports start Link training. 1 = Management Port is disabled, and the <i>Initiate Configuration</i> bit is Set. After the serial EEPROM load finishes, all Ports start Link training (assuming that the serial EEPROM, or alternatively I²C/SMBus if the STRAP_I2C_SMBUS_CFG_EN# input is asserted Low, does not Clear the <i>Initiate Configuration</i> bit). Note: Management Port software must not change the Port Configuration register (Port 0, accessible through the Management Port, offset 300h). When STRAP_MGMT_PORT_EN=Z, the Delayed Linkup option might require software support to delay Host enumeration until the Management Port Host and/or I²C/SMBus completes switch initialization (by Setting the <i>Initiate Configuration</i> bit); otherwise, system software could miss detection of the PCI Express hierarchy (precluding device enumeration).		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States	
Description					
STRAP_NT_UPSTRM_PORTSEL[2:0]	G17, D13, E13	I/O, 3-state input	No	—	
		Select Upstream NT Port (3 Balls)			
		Enable and select any Port to be the Upstream NT Port. The input values are sampled and latched upon PEX_PERST# input de-assertion, to define the default values of the VS0 Upstream register NT Enable and NT Port bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 360h[13 and 12:8], respectively).			
		Strap value of 111 disables NT mode; all other values enable NT mode.			
		The STRAP_NT_UPSTRM_PORTSEL[2:0] 3-state input can be overridden by the serial EEPROM value for the VS0 Upstream register. Because the NT Port value is Don't Care if the NT Enable bit is Cleared, NT mode can be disabled by simply Clearing these bits (that is, it is not necessary to program the NT Port field to the 1_1010b value listed in the table below).			
I²C/SMBus can enable and select a Port to be the Upstream NT Port, by writing to the VS0 Upstream register. To configure the PEX 8724 through I²C/SMBus, the STRAP_I2C_SMBUS_CFG_EN# input can be pulled or tied Low, to delay linkup and Host enumeration until I²C initialization is complete. After I²C initialization is complete, I²C/SMBus must then Set the Configuration Release register Initiate Configuration bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3ACh[0]).					
Refer to Section 14.8, “Port Programmability,” for further details.					
		STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Offset 360h Field [12:8] Value Bit 13 Value	Port Number	
		000	0_0000b	1	0
		00Z	0_0001b	1	1
		001	0_0010b	1	2
		0Z0	0_0011b	1	3
		Z00	0_1001b	1	9
		Z0Z	0_1010b	1	10
		111	1_1010b	0	NT mode is disabled
All other encodings are Reserved.					
STRAP_PLL_BYPASS#	A18	I, PU	No	—	
		Factory Test Only Must be pulled or tied High to VDD18.			
STRAP_PROBE_MODE#	L16	I, PU	No	—	
		Factory Test Only Must be pulled or tied High to VDD18.			

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_RESERVED16	K15	PWR	No	–
		Factory Test Only Reserved for future use Must be tied directly to VSS (Ground).		
STRAP_SERDES_MODE_EN#	H17	I, PU	No	–
		Factory Test Only Must be pulled or tied High to VDD18 .		
STRAP_STN0_PORTCFG[1:0]	M17, K16	I/O, 3-state input	No	–
		Select Port Configuration for Station 0 (Quantity of Enabled Ports (1, 2, 3, or 4), and Maximum Quantity of Lanes for Each Specific Port) (2 Balls) Defines the enabled Port Numbers and their Link widths, for Station 0. Programs the Port Configuration register <i>Port Configuration for Station 0</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[2:0]) default value. Notes: The default Maximum Payload Size supported is based upon the STRAP_STNx_PORTCFGx 3-state inputs. If the serial EEPROM or I ² C/SMBus changes the Port configuration (from the strapped value), it must also program the value of the Port's Device Capability register <i>Maximum Payload Size Supported</i> field (offset 6Ch[2:0]), accordingly (with the value dependent upon the Station having the most enabled Ports). In the values listed below, the first value is STRAP_STN0_PORTCFG[1:0]'s input state, and the second value is the Port Configuration Number, as listed in Table 4-1. 00 = 0 = Reserved 0Z = 1 = x16 01 = 2 = x8x8 Z0 = 3 = x8x4x4 ZZ = 4 = x4x4x4x4 Z1 = 5 = Reserved 10 = 6 = Reserved 1Z = 7 = Reserved 11 = 8 = Reserved		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_STN1_PORTCFG[1:0]	A1, A14	I/O, 3-state input	No	–
		Select Port Configuration for Station 1 (Quantity of Enabled Ports (1, 2, 3, or 4), and Maximum Quantity of Lanes for Each Specific Port) (2 Balls) Defines the enabled Port Numbers and their Link widths, for Station 1. Programs the Port Configuration register <i>Port Configuration for Station 1</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3]) default value. Notes: The default Maximum Payload Size supported is based upon the STRAP_STNx_PORTCFGx 3-state inputs. If the serial EEPROM or I²C/SMBus changes the Port configuration (from the strapped value), it must also program the value of the Port's Device Capability register <i>Maximum Payload Size Supported</i> field (offset 6Ch[2:0]), accordingly (with the value dependent upon the Station having the most enabled Ports). In the values listed below, the first value is STRAP_STN1_PORTCFG[1:0]'s input state, and the second value is the Port Configuration Number, as listed in Table 4-1. 00 = 0 = Reserved 0Z = 1 = Reserved 01 = 2 = x8 Z0 = 3 = x4x4 ZZ = 4 = Reserved Z1 = 5 = Reserved 10 = 6 = Reserved 1Z = 7 = Reserved 11 = 8 = Reserved		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_TESTMODE[2:0]	J1, K1, M18	I/O, 3-state input	No	Refer to Section 3.4.4.1 for details
		Test Mode Selects (3 Balls) The STRAP_TESTMODE[2:0] input states (first value listed below) indicate the test mode (second value listed below), which defines the default functionality of the multiplexed I/O signals. These signals can be used to configure Hot Plug, Virtual Switch mode, GPIOx, and/or PORT_GOODx# signal combinations. The value of this 3-state input is latched and reflected in the Strap Configuration register Strap Test Mode Status field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[28:24]). (Refer to Section 3.4.4.1 for details regarding the test modes, and the multiplexed I/O signals enabled within each mode. Refer also to PORT_GOOD[6:0]#, for details regarding PORT_GOODx# input and output functions, and VS_MODE, for input states that enable Virtual Switch mode.) Note: The GPIO11 signal exists only in Virtual Switch mode, when four virtual switches are enabled. This signal does not exist in Base mode. Base Mode 000 = 0 = Hot Plug A + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input 00Z = 1 = Hot Plug A + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input 001 = 2 = Hot Plug A + Hot Plug C + GPIO[10:0] Input 0Z0 = 3 = Hot Plug A + Hot Plug C + Hot Plug B + PORT_GOOD0# Virtual Switch Mode 0ZZ = 4 = VS[2:0] + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input 0Z1 = 5 = VS[2:0] + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input 010 = 6 = VS[2:0] + Hot Plug C + GPIO[10:0] Input 01Z = 7 = VS[2:0] + Hot Plug C + Hot Plug B + PORT_GOOD0# 011 = 8 = VS[2:0] + VS3 + PORT_GOOD[6:0]# + GPIO[11:7] Input Z00 = 9 = VS[2:0] + VS3 + SHP_PERST[6:0]# + GPIO[10:7] Input Z0Z = 10 = VS[2:0] + VS3 + GPIO[11:0] Input Z01 = 11 = VS[2:0] + VS3 + Hot Plug B + PORT_GOOD0# + GPIO1 Input All other encodings are Reserved.		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_UPSTRM_POR TSEL[2:0]	B16, F16, C16	I/O, 3-state input	No	–
		Base Mode Select Upstream Port (3 Balls) Select any Port as the VS0 Upstream Port. These inputs map to the VS0 Upstream register <i>VS0 Upstream Port</i> field (Port 0, offset 360h[4:0]). If I ² C/SMBus is used to configure the PEX 8724, the STRAP_I2C_SMBUS_CFG_EN# input can be pulled or tied Low to VSS (Ground), to delay linkup until I ² C/SMBus initialization is complete. After I ² C initialization is complete, I ² C/SMBus must then Set the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, offset 3ACh[0]). Refer to Section 14.8, “Port Programmability,” for further details.		
		STRAP_UPSTRM_PORTSEL[2:0] Input States	Offset 360h[4:0] Value	Port Number
		000	0_0000b	0
		00Z	0_0001b	1
		001	0_0010b	2
		0Z0	0_0011b	3
		011	0_1000b	8
		Z00	0_1001b	9
		Z0Z	0_1010b	10
		All other encodings are Reserved .		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
STRAP_UPSTRM_POR TSEL[2:0]	B16, F16, C16	I/O, 3-state input	No	–
		Virtual Switch Mode Select Management Port (3 Balls) If STRAP_MGMT_PORT_EN=0 or Z, the Management Port is enabled and these inputs select any virtual switch Upstream Port as the Management Port. These inputs map to the Management Port Control register Active Management Port field (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[4:0]). The Management Port Control register also controls the enabling or disabling of the Active Management Port, and designation and enabling of the Redundant Management Port (which is programmed by Management Port software, serial EEPROM, and/or I²C). The Active Management Port, the Redundant Management Port, and/or I²C/SMBus can promote the Redundant Management Port to be the Active Management Port, by programming the register. If STRAP_MGMT_PORT_EN=0, after a Fundamental Reset and the serial EEPROM load, only the Management Port links up. When the Management Port has completed its configuration of the virtual switches, Management Port software and/or serial EEPROM (but not I²C/SMBus) must Set the Configuration Release register Initiate Configuration bit (Port 0, accessible through the Management Port, offset 3ACh[0]), to release the hold that is preventing the remaining Links from coming up. If STRAP_MGMT_PORT_EN=Z, all Ports commence Link training after the serial EEPROM load completes, following a Fundamental Reset. If I²C is used to configure the PEX 8724, the STRAP_I2C_SMBUS_CFG_EN# input can be pulled or tied Low to VSS (Ground), to delay linkup until I²C initialization is complete. After I²C initialization is complete, I²C and/or the serial EEPROM must then Set the Initiate Configuration bit, to enable all Ports to begin Link training. Refer to the STRAP_UPSTRM_PORTSEL[2:0] Base mode description for the signal and register values associated with each Port Number. Note: Because the Virtual Switch Table registers (refer to Section 5.3.3.1, “Virtual Switch Table Registers”) exist in Port 0, Port 0 must not be held in Reset during setup of Virtual Switch mode. Therefore, for whichever virtual switch includes Port 0, its VSx_PERST# input must be de-asserted (High). Assigning Port 0 to be the Management Port might provide the simplest solution toward meeting this requirement.		

Table 3-7. Strapping Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States	
		Description			
VS_MODE	K18	I/O, 3-state input	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1	
		Virtual Switch Mode Virtual Switch Enable VS_MODE exists only when the STRAP_TESTMODE[2:0] 3-state inputs enable Virtual Switch mode. When used together, the VS_MODE and STRAP_TESTMODE[2:0] 3-state inputs, depending upon their input states, enable up to four virtual switches. The value of this 3-state input is latched and reflected in the Strap Configuration register VS Mode Status field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[15:13]). When the Test Mode is 4, 5, 6, or 7 (STRAP_TESTMODE[2:0] input state is 0ZZ, 0Z1, 010, or 01Z, respectively (refer to Table 3-9)), VS_MODE selects one, two, or three virtual switches. When the Test Mode is 8, 9, 10, or 11 (STRAP_TESTMODE[2:0] input state is 011, Z00, Z0Z, or Z01, respectively (refer to Table 3-10)), VS_MODE selects 4 virtual switches. The quantity of enabled virtual switches is reflected in the Virtual Switch Enable register VSx Enable bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 358h[3:0]). Table 12-1, “Virtual Switch Default Port Assignments – Virtual Switch Mode,” lists the default Port assignments for each virtual switch. Note: In Base mode, this signal is the HP_PWR_GOOD_A input.			
		STRAP_TEST MODE[2:0] Input States	VS_MODE Input State	# of Enabled Virtual Switches	VSx
		0ZZ, 0Z1, 010, or 01Z	0	1	VS0
			Z	2	VS0, VS1
			1	3	VS0, VS1, VS2
		011, Z00, Z0Z, or Z01	0	4	VS0, VS1, VS2, VS3

3.4.4.1 STRAP_TESTMODE[2:0] 3-State Input Configurations

The PEX 8724 has 31 multiplexed I/O signals, listed in [Table 3-8](#), [Table 3-9](#), and [Table 3-10](#). Thirty multiplexed signals are grouped into three sets of 10, with the PORT_GOOD0# signal being the 31st multiplexed signal.

The multiplexed signals include three sets (A, B, and C) of 10 Parallel Hot Plug signals, Virtual Switch signals (three per virtual switch, plus one strapping signal, VS_MODE), and 10 GPIOx signals that can function as PORT_GOODx#, SHP_PERST# (Reset) outputs (identified as SHP_PERSTx# in [Table 3-8](#), [Table 3-9](#), and [Table 3-10](#)), or GPIOx input.

Default functionality of each multiplexed signal is determined by the STRAP_TESTMODE[2:0] input states, which select one of 12 possible user configurations when sampled at Fundamental Reset. The three tables each include four columns (for a total of 12 columns), and each of these columns identifies one of the 12 possible configurations (only one of which can be selected at any time). Individual PORT_GOODx# and GPIOx signal functionality can be changed by serial EEPROM, I²C/SMBus, and/or software, to function as PORT_GOODx#, GPIOx inputs and/or outputs, and/or Interrupt inputs. PORT_GOODx# signals can be re-assigned to any Port.

Of the 31 multiplexed signals, none require external connection, with the exception of VS_MODE, which might require external termination, because the VS_MODE and STRAP_TESTMODE[2:0] 3-state inputs define the quantity of virtual switches.

If Hot Plug signals are to be used, the corresponding HP_MRL_x# input must be Low, to enable automatic signal sequencing to power up the slot. Because each HP_MRL_x# input has an internal pull-up resistor, external pull-down resistors (or connection to the corresponding PRSNT# input from the slot) can hold the HP_MRL_x# input Low.

Notes: The HP_Parallel Hot Plug signals are defined in [Table 3-4](#). The VSx, PORT_GOODx#, and GPIOx signals are defined in [Table 3-13](#). STRAP_TESTMODE[2:0] and VS_MODE are defined in [Table 3-7](#). Serial Hot Plug SHP_PERST# (SHP_PERSTx#) is defined in [Table 10-8](#).

Software, serial EEPROM and/or I²C/SMBus can change the functionality of the PORT_GOOD[6:0]#/SHP_PERST[6:0]#/GPIO[11:0] signals from the default configuration selected by the STRAP_TESTMODE[2:0] 3-state inputs. If Serial Hot Plug SHP_PERST# functionality is used, the STRAP_TESTMODE[2:0] 3-state inputs should be configured to enable that functionality as default (because the Downstream slots must be held in Reset at power-up).

The GPIO11 signal exists only in Virtual Switch mode, when four virtual switches are enabled. This signal does not exist in Base mode.

*PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the **PORT_GOOD Selector** x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) [6A0h](#) and [6A4h](#)).*

*[Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.*

Table 3-8. STRAP_TESTMODE[2:0] 3-State Input Configurations – Base Mode, Test Modes 0, 1, 2, 3

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Base Mode			
		0	1	2	3
		STRAP_TESTMODE[2:0] = 000	STRAP_TESTMODE[2:0] = 00Z	STRAP_TESTMODE[2:0] = 001	STRAP_TESTMODE[2:0] = 0Z0
		Hot Plug A + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input	Hot Plug A + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input	Hot Plug A + Hot Plug C + GPIO[10:0] Input	Hot Plug A + Hot Plug C + Hot Plug B + PORT_GOOD0#
B17	O, PU	HP_ATNLED_A#	HP_ATNLED_A#	HP_ATNLED_A#	HP_ATNLED_A#
E17	I, PU	HP_BUTTON_A#	HP_BUTTON_A#	HP_BUTTON_A#	HP_BUTTON_A#
G18	O, PU	HP_CLKEN_A#	HP_CLKEN_A#	HP_CLKEN_A#	HP_CLKEN_A#
J18	I, PU	HP_MRL_A#	HP_MRL_A#	HP_MRL_A#	HP_MRL_A#
D18	O, PU	HP_PERST_A#	HP_PERST_A#	HP_PERST_A#	HP_PERST_A#
A17	I, PU	HP_PRSNT_A#	HP_PRSNT_A#	HP_PRSNT_A#	HP_PRSNT_A#
K18	I, PD	HP_PWR_GOOD_A	HP_PWR_GOOD_A	HP_PWR_GOOD_A	HP_PWR_GOOD_A
C18	O, PD	HP_PWREN_A	HP_PWREN_A	HP_PWREN_A	HP_PWREN_A
H18	I, PU	HP_PWRFLT_A#	HP_PWRFLT_A#	HP_PWRFLT_A#	HP_PWRFLT_A#
F18	O, PU	HP_PWRLED_A#	HP_PWRLED_A#	HP_PWRLED_A#	HP_PWRLED_A#
F15	O, PU	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#
B15	I, PU	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#
D17	O, PU	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#
C14	I, PU	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#
F14	O, PU	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#
C15	I, PU	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#
H15	I, PD	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C
D15	O, PD	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C
E15	I, PU	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#
G15	O, PU	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

Table 3-9. STRAP_TESTMODE[2:0] 3-State Input Configurations – Virtual Switch Mode, Test Modes 4, 5, 6, 7

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Virtual Switch Mode; One to Three Virtual Switches, as Selected by VS_MODE Input State			
		4	5	6	7
		STRAP_TESTMODE[2:0] = 0ZZ	STRAP_TESTMODE[2:0] = 0Z1	STRAP_TESTMODE[2:0] = 010	STRAP_TESTMODE[2:0] = 01Z
		VS[2:0] + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input	VS[2:0] + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input	VS[2:0] + Hot Plug C + GPIO[10:0] Input	VS[2:0] + Hot Plug C + Hot Plug B + PORT_GOOD0#
B17	O, PU	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#
E17	I, PU	VS0_PERST#	VS0_PERST#	VS0_PERST#	VS0_PERST#
G18	O, PU	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#
J18	I, PU	VS1_PERST#	VS1_PERST#	VS1_PERST#	VS1_PERST#
D18	O, PU	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#
A17	I, PU	VS2_PERST#	VS2_PERST#	VS2_PERST#	VS2_PERST#
K18	I, 3-state	VS_MODE	VS_MODE	VS_MODE	VS_MODE
C18	O, PD	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#
H18	I, PU	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#
F18	O, PU	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#
F15	O, PU	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#
B15	I, PU	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#
D17	O, PU	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#
C14	I, PU	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#
F14	O, PU	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#
C15	I, PU	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#
H15	I/O, PD	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C
D15	O, PD	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C
E15	I, PU	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#
G15	O, PU	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

Table 3-10. STRAP_TESTMODE[2:0] 3-State Input Configurations – Virtual Switch Mode, Test Modes 8, 9, 10, 11

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Virtual Switch Mode; Four Virtual Switches, as Selected by VS_MODE Input State			
		8	9	10	11
		STRAP_TESTMODE[2:0] = 011	STRAP_TESTMODE[2:0] = Z00	STRAP_TESTMODE[2:0] = Z0Z	STRAP_TESTMODE[2:0] = Z01
		VS[2:0] + VS3 + PORT_GOOD[6:0]# + GPIO[11:7] Input	VS[2:0] + VS3 + SHP_PERST[6:0]# + GPIO[10:7] Input	VS[2:0] + VS3 + GPIO[11:0] Input	VS[2:0] + VS3 + Hot Plug B + PORT_GOOD0# + GPIO1 Input
B17	O, PU	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#
E17	I, PU	VS0_PERST#	VS0_PERST#	VS0_PERST#	VS0_PERST#
G18	O, PU	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#
J18	I, PU	VS1_PERST#	VS1_PERST#	VS1_PERST#	VS1_PERST#
D18	O, PU	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#
A17	I, PU	VS2_PERST#	VS2_PERST#	VS2_PERST#	VS2_PERST#
K18	I, 3-state	VS_MODE=0	VS_MODE=0	VS_MODE=0	VS_MODE=0
C18	O, PD	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#
H18	I, PU	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#
F18	O, PU	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#
F15	O, PU	VS3_PEX_INTA#	VS3_PEX_INTA#	VS3_PEX_INTA#	VS3_PEX_INTA#
B15	I, PU	VS3_PERST#	VS3_PERST#	VS3_PERST#	VS3_PERST#
D17	O, PU	RESERVED	RESERVED	RESERVED	RESERVED
C14	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
F14	O, PU	RESERVED	RESERVED	RESERVED	RESERVED
C15	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
H15	I/O, PD	GPIO11 input	GPIO11 input	GPIO11 input	GPIO11 input
D15	O, PD	RESERVED	RESERVED	RESERVED	RESERVED
E15	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
G15	O, PU	VS3_FATAL_ERR#	VS3_FATAL_ERR#	VS3_FATAL_ERR#	VS3_FATAL_ERR#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSTNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

3.4.5 JTAG Interface Signals

The PEX 8724 includes five signals for performing Joint Test Action Group (JTAG) boundary scan, defined in [Table 3-11](#). The JTAG interface is described in [Section 17.6, “JTAG Interface.”](#)

Table 3-11. JTAG Interface Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
JTAG_TCK	C13	I, PD	No	–
		JTAG Test Clock Input JTAG Test Access Port (TAP) Controller clock source. Frequency can be from 0 to 15 MHz.		
JTAG_TDI	A2	I, PD	No	–
		JTAG Test Data Input Serial input to the JTAG TAP Controller, for test instructions and data.		
JTAG_TDO	B14	I/O as O, TS	No	–
		JTAG Test Data Output Serial output from the JTAG TAP Controller test instructions and data. <i>Note: Driven when the TAP Controller is being used; otherwise, the output is High-Impedance.</i>		
JTAG_TMS	D14	I, PD	No	–
		JTAG Test Mode Select Input decoded by the JTAG TAP Controller, to control test operations.		
JTAG_TRST#	B1	I, PD	No	–
		JTAG Test Reset Active-Low input used to reset the Test Access Port. When JTAG functionality is not used, the JTAG_TRST# input should be driven Low, or pulled Low to VSS (Ground) through a 1.5K Ω resistor, to place the JTAG TAP Controller into the <i>Test-Logic-Reset</i> state, which disables the test logic and enables standard logic operation. Alternatively, if JTAG_TRST# input is High, the JTAG TAP Controller can be placed into the <i>Test-Logic-Reset</i> state by initializing the JTAG TAP Controller's Instruction register to contain the <i>IDCODE</i> instruction, or by holding the JTAG_TMS input High for at least five rising edges of the JTAG_TCK input.		

3.4.6 I²C/SMBus Slave Interface Signals

Table 3-12 defines the three I²C/SMBus Slave interface signals. The I²C/SMBus Slave interface provides access to the PEX 8724 registers. Most registers that can be programmed by serial EEPROM (including registers that are Read Only to PCI Express), can be also programmed by I²C/SMBus. However, I²C/SMBus cannot replace the serial EEPROM programming of some registers, *such as* SerDes- and Hot Plug-related registers that (generally) must be configured prior to automatic link training, immediately following Reset. I²C/SMBus can also be used to program the serial EEPROM.

The I²C/SMBus Slave interface used with PLX software is a powerful debugging tool that enables register access, regardless of whether the PCI Express Link is Up. Figure 3-1 illustrates the suggested Header pin layout on 0.1-inch centers, for direct connection to the Total Phase Aardvark I²C-USB converter (as used with PLX Rapid Development Kits (RDKs)).

For further details, refer to Chapter 7, “I²C/SMBus Slave Interface Operation.”

Figure 3-1. Suggested I²C Header Pin Layout on 0.1-Inch Centers

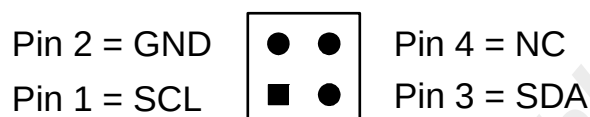


Table 3-12. I²C/SMBus Slave Interface Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States								
		Description										
I2C_ADDR0	C3	I/O, 3-state input	No	—								
		I²C Slave/SMBus Device Address Bit Used to define the two least significant bits of the PEX 8724 7-bit I ² C Slave/SMBus Device address, which is reflected in both the I2C Configuration register <i>Slave Address</i> and SMBus Configuration register <i>SMBus Device Address</i> fields (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 294h[6:0] and 2C8h[7:1], respectively). If I ² C or SMBus configuration is used, the I2C_ADDR0 3-state input should be strapped to a unique address, to avoid an address conflict with any other I ² C/SMBus devices (on the same I ² C Bus/SMBus segment). The three possible default I ² C Slave/SMBus Device Address values are listed below.										
		<table><thead><tr><th>I2C_ADDR0 Input State</th><th>I²C Slave/SMBus Device Address</th></tr></thead><tbody><tr><td>0</td><td>0111_000b (default)</td></tr><tr><td>Z</td><td>0111_001b</td></tr><tr><td>1</td><td>0111_010b</td></tr></tbody></table>			I2C_ADDR0 Input State	I ² C Slave/SMBus Device Address	0	0111_000b (default)	Z	0111_001b	1	0111_010b
		I2C_ADDR0 Input State	I ² C Slave/SMBus Device Address									
0	0111_000b (default)											
Z	0111_001b											
1	0111_010b											

Table 3-12. I²C/SMBus Slave Interface Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
I2C_SCL0	H3	OD	No	—
		I²C/SMBus Serial Clock Line I ² C/SMBus Clock line. Data on the I ² C Bus can be transferred at rates of up to 100 kbit/s (Standard mode). Because the 1.8V signaling is less than the voltage defined by the I ² C Bus v2.1 (5V, 3.3V), external voltage level shifter circuitry might be required. I2C_SCL0 generally requires an external pull-up resistor. Alternatively, a voltage translation circuit can provide the current source. <i>Note: The PEX 8724 I²C/SMBus Slave Interface can stretch the Low period of the I²C/SMBus clock while a concurrent in-band Request that also targets PEX 8724 registers is being processed.</i>		
I2C_SDA0	D2	OD	No	—
		I²C/SMBus Serial Data Line Transmits and receives I ² C/SMBus data during I ² C/SMBus accesses to PEX 8724 registers. Because the 1.8V signaling is less than the voltage defined by the I ² C Bus v2.1 (5V, 3.3V), external voltage level shifter circuitry might be required. I2C_SDA0 generally requires an external pull-up resistor. Alternatively, a voltage translation circuit can provide the current source.		

3.4.7 Device-Specific Signals

Table 3-13 defines the Device-Specific signals – signals that are unique to the PEX 8724.

Table 3-13. Device-Specific Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
FATAL_ERR#	E14	I/O as O, TS	No	–
		Base Mode Fatal Error Output FATAL_ERR# is asserted Low when a Fatal error is detected in the PEX 8724 and the following conditions are met (all the same conditions that are required to send a Fatal Error Message to the Host). For <i>PCI Express Base r3.0</i> errors: - Specific error is defined as <i>Fatal</i> in the Uncorrectable Error Severity register (offset FC0h), and - Reporting of the specific error condition is enabled, not masked by the Uncorrectable Error Mask register's (offset FBCh) corresponding <i>Interrupt Mask</i> bit, and Device Control register <i>Fatal Error Reporting Enable</i> bit (offset 70h[2]) or PCI Command register <i>SERR# Enable</i> bit (offset 04h[8]) is Set Device-Specific (RAM ECC) errors can be reported as a PCI Express Uncorrectable Error, and/or by interrupt (INTx, MSI, or PEX_INTA#) signaling, by the FATAL_ERR# output: - FATAL_ERR# and/or PEX_INTA# output can signal the error event. - Reporting of the specific error condition in the Device-Specific Error Status x Mask x register bit(s), if not masked in their corresponding Device-Specific Error Mask x register bit(s) (Port 0 and/or 8, offsets 700h, 708h, 710h, and/or 718h (Status) and offsets 704h, 70Ch, 714h, and/or 71Ch (Mask)). - Uncorrectable Internal error in the Uncorrectable Error Status register <i>Uncorrectable Internal Error Status</i> bit (Port 0 and/or 8, offset FB8h[22]) is defined as <i>Fatal</i> in the Uncorrectable Error Severity register <i>Uncorrectable Internal Error Severity</i> bit (Port 0 and/or 8, offset FC0h[22]), and - Reporting of Uncorrectable Internal errors is enabled (not masked) by the Uncorrectable Error Mask register <i>Uncorrectable Internal Error Mask</i> bit (Port 0 and/or 8, offset FBCh[22]), and Device Control register <i>Fatal Error Reporting Enable</i> bit (offset 70h[2]) or PCI Command register <i>SERR# Enable</i> bit (offset 04h[8]) is Set The Device Status register <i>Fatal Error Detected</i> bit (offset 70h[18]) is Set, and the specific error is flagged in the Uncorrectable Error Status register (offset FB8h). Continued...		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
FATAL_ERR#	E14	I/O as O, TS	No	–
		Base Mode Fatal Error Output (Cont.) Software can be used to de-assert FATAL_ERR#, with the following procedure: 1. Set the Port's Error Check Disable register <i>Software Force Error Enable</i> bit (Transparent Downstream Ports, offset 720h[2]). 2. Write 5h to bits [11:8] of the Uncorrectable Error Status register (offset FB8h), in addition to Clearing <i>Status</i> bits that are Set, using a Read-Modify-Write operation. When the Port's <i>Software Force Error Enable</i> bit is Set, writing 1 to an <i>Error Status</i> bit that is Cleared causes that bit to be Set, and generates the corresponding ERR_FATAL or ERR_NONFATAL Message, if enabled and not masked. Therefore, software should not, <i>for example</i>, write all 1s to this register while the Port's <i>Software Force Error Enable</i> bit is Set. 3. Clear the Port's <i>Software Force Error Enable</i> bit. If FATAL_ERR# output is not used, this signal can remain unconnected. Notes: Always driven. Device-Specific (RAM ECC) errors are reported in the lowest numbered enabled Port within the Station.		
GPIO[11:7]	H15, H1, K3, J3, D1,	GPIO[8:7] – I/O, PD GPIO[11:9] – I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		General-Purpose I/Os (5 Balls) GPIO[11:7] provide GPIO functionality, determined according to the GPIO registers associated with each GPIO (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 600h through 63Ch, as well as Virtual Switch mode – Port 0, accessible through the Management Port, offsets 64Ch through 65Ch and A34h through A54h). GPIO interrupts are described in Section 9.5, “General-Purpose Input/Output.” Note: GPIOx signals are always mapped to Port 0 registers in Base mode. In Virtual Switch mode, these GPIO signals can be re-assigned to VS Upstream Ports (in the Port 0 registers located at offsets 64Ch through 65Ch).		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PEX_INTA#	C2	OD	No	—
		Base Mode Interrupt Output PEX_INTA# Interrupt output is enabled if: - INTx Messages are enabled (Port's PCI Command register <i>Interrupt Disable</i> bit, offset 04h[10], is Cleared), and MSIs are disabled (Port's MSI Control register <i>MSI Enable</i> bit, offset 48h[16], is Cleared) - PEX_INTA# output is enabled (Error Check Disable register <i>Enable Ball Interrupt Output(s) for x Interrupt</i> bit(s) (offset 720h[7, 6, 5, and/or 4]; refer to the register description, for Port associations) are Set) The three interrupt mechanisms, listed below, are mutually exclusive modes of operation, on a per-Port basis, for all interrupt sources: - Conventional PCI INTx Message generation - Native MSI transaction generation - Device-Specific PEX_INTA# assertion PEX_INTA# assertion (Low) indicates that one or more of the following events and/or errors (if not masked) were detected: - Hot Plug or Link State events - PCI Express Hot Plug events - General-Purpose Input Interrupt events - Device-Specific NT-Link Port errors and events - NT-Virtual Doorbell events - NT-Link Doorbell events Refer to Section 9.1.1, “Interrupt Sources or Events,” for details. If used, PEX_INTA# output requires a pull-up resistor; otherwise, this output can remain unconnected.		
PEX_NT_PERST#	E18	I, PU	No	—
		Active-Low Input Used to Reset the NT Port Link Interface in NT Mode If PEX_NT_PERST# input is not used, the signal can either be pulled High, or left unconnected because the internal pull-up resistor is sufficiently strong, to hold PEX_NT_PERST# High.		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) PORT_GOODx# signals can function as PORT_GOODx# outputs, Serial Hot Plug Reset outputs (SHP_PERSTx#), or General-Purpose input/output (I/O). Default functionality for all PORT_GOODx# signals is selected according to the STRAP_TESTMODE[2:0] input states (sampled at Fundamental Reset (PEX_PERST#) de-assertion), and the functionality of each signal can be programmed by the serial EEPROM, I²C/SMBus, and/or software. PORT_GOODx# signals can be re-assigned to any Port. If Serial Hot Plug functionality is required, the STRAP_TESTMODE[2:0] inputs should select that as the default functionality (STRAP_TESTMODE[2:0] = 00Z, 0Z1, or Z00). These GPIO signals can function independently, as General-Purpose inputs (default functionality), General-Purpose outputs, or Interrupt inputs (that can trigger interrupt outputs using MSI Writes, INTx Messages, PEX_INTA#, or VSx_PEX_INTA# signaling). Notes: Table 4-1 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The GPIO11 signal exists only in Virtual Switch mode, when four virtual switches are enabled. This signal does not exist in Base mode. PORT_GOOD[6:0]# signals that are enabled to function as PORT_GOODx# outputs correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals that are enabled to function as PORT_GOODx# outputs can be re-assigned to any Port, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). However, if Serial Hot Plug Reset functionality (SHP_PERSTx#) functionality is enabled, the PORT_GOODx# signals cannot be re-assigned to a different Port. Instead, the PORT_GOODx# signals are assigned to Serial Hot Plug Ports, in increasing order, starting with PORT_GOOD0#: • PORT_GOOD0# is assigned to the first Serial Hot Plug Port (which is selected by the SHP_PORTID[4:0] inputs on the I/O Expander mapped to I²C/SMBus address 40h). • PORT_GOOD1# is assigned to the second Serial Hot Plug Port. If a 40-bit I/O Expander is used (for two Serial Hot Plug Ports), this Port assignment is determined by the value of the second set of PORTID inputs (PORTID1[4:0]). If a 16-bit I/O Expander is used, the Port is selected by the PORTID[4:0] inputs on the I/O Expander mapped to I²C/SMBus address 41h. • Port assignment continues, as described above, with each sequential I/O Expander I²C/SMBus address. PORT_GOODx# signals that are enabled to function as GPIO are always mapped to Port 0 registers in Base mode. In Virtual Switch mode, these GPIO signals can be re-assigned to VS Upstream Ports (in the Port 0 registers located at offsets 64Ch through 65Ch).		

Continued...

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) (Cont.) PORT_GOOD Output Function For PORT_GOODx# signals that are: - Assigned to enabled Ports, according to the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h), and - Configured for PORT_GOOD output functionality by the STRAP_TESTMODE[2:0] input states, sampled at Fundamental Reset (refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1 for details), –or– - Configured for PORT_GOOD output functionality by subsequent programming (by serial EEPROM, I²C, and/or software) of the appropriate GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and/or 604h) value(s) The output states are not directly available from a single register (due to encoded, possibly blinking output); however, software can determine LANE_GOOD status (Physical Layer Link status for each Lane) from the Station 0/1 Lane Status register <i>Lane x Up Status</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 330h[31:24 and 15:0]). Software can also determine the Port's Maximum Link Width and Maximum Link Speed, from the Port's Link Capability register (offset 74h[9:4 and 3:0], respectively), as well as the Port's Negotiated Link Width and Current Link Speed, from the Port's Link Status register (offset 78h[25:20 and 19:16], respectively). If PORT_GOOD output functionality is enabled, but some Ports are not enabled due to the STRAP_STNx_PORTCFGx input states, the PORT_GOODx# signals associated with non-enabled Ports function as GPIOx signals. LED behavior when connected to PORT_GOODx# signals: Off – Link is Down Blinking, 512 ms On, 512 ms Off (1 Hz) – Link is Up, 2.5 GT/s, any negotiated width Blinking, 256 ms On, 256 ms Off (2 Hz) – Link is Up, 5.0 GT/s, any negotiated width On – Link is Up, 8.0 GT/s, any negotiated width The PORT_GOODx#/GPIOx I/Os operate from the VDD18 supply, and maximum input voltage is 2.5V. Because LED forward voltage drop is typically greater than 1.8V, the LED voltage source will likely be higher than 2.5V. In this case, if the LED is connected directly to the PORT_GOODx# input, the input voltage level (while the LED is turned Off) would exceed the specification. Therefore, a PORT_GOODx# signal typically cannot drive an LED directly; however, it can be used to gate a transistor that switches the LED On and Off. For further details regarding LED behavior, refer to Section 17.7, “Port Good Status LEDs.”		

Continued...

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) (Cont.) Serial Hot Plug Reset The I²C I/O Expander provides a serialized Reset output; however, a parallel output is recommended because: - I²C/SMBus is relatively slow. In the case of a power fault (SHP_PWRFLT#=0), software should assert Reset to the Downstream slot, as quickly as possible. - If PEX_PERST# input to the PEX 8724 is asserted, Downstream devices should also be reset; however, the PEX 8724 I²C/SMBus Controller is in a Reset state, and cannot generate the I²C/SMBus command to assert the I/O Expander Reset output. Reset to the Downstream slot can be implemented with the output from a 2-input AND gate, that has PORT_GOODx# (SHP_PERSTx#) and PEX_PERST# as inputs. Continued...		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) (Cont.) General-Purpose Inputs For PORT_GOODx# signals that are configured as General-Purpose inputs (by the STRAP_TESTMODE[2:0] input states, sampled at Fundamental Reset; refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1 for details), or by subsequent programming (by serial EEPROM, I²C, and/or software) of the appropriate GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and/or 604h) value(s), input states are reflected in the GPIO 0_11 Input Data register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 61Ch). Inputs can be internally de-bounced, by Setting the corresponding GPIO 0_11 Input De-Bounce register bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 614h). De-bouncing is disabled, by default; if de-bouncing is enabled, an input must be stable for approximately 1.3 ms to be latched. General-Purpose Outputs For PORT_GOODx# signals that are configured as General-Purpose outputs in the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and/or 604h), output states are controlled by the corresponding GPIO 0_11 Output Data register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 624h) bit values. Interrupt Inputs For PORT_GOODx# signals that are configured as Interrupt inputs in the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s), input states are reflected in the GPIO 0_11 Interrupt Status register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 634h). Inputs can be internally de-bounced, by Setting the corresponding GPIO 0_11 Input De-Bounce register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 614h) bits. De-bouncing is disabled, by default; if de-bouncing is enabled, an input must be stable for approximately 1.3 ms to be latched. Interrupt polarity (Active-High or Active-Low) is individually programmable in the GPIO 0_11 Interrupt Polarity register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 62Ch). Interrupt generation can be selectively masked in the GPIO 0_11 Interrupt Mask register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 63Ch). If a GPIO 0_11 Interrupt Mask register bit is Set, a read of the corresponding GPIO 0_11 Interrupt Status register <i>Interrupt Status</i> bit returns a value of 0. Continued...		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) (Cont.) GPIO Function – Virtual Switch Mode <i>Note:</i> If DLL state change interrupts must be routed to the Management Port, use the Virtual Switch Device-Specific registers (Port 0, accessible through the Management Port, offsets 900h through 90Ch). If DLL state change interrupts must be routed to the Upstream Port(s), use the PCI Express Capability registers. In Virtual Switch mode, PORT_GOODx# signals that are enabled for GPIO functionality are associated to specific VS Upstream Ports, by one of two mechanisms: 1. If STRAP_MGMT_PORT_EN=Z, to enable the Management Port and initially Clear the Configuration Release register Initiate Configuration bit (Port 0, accessible through the Management Port, offset 3ACh[0]) (to delay linkup on all other Ports until Management Port software, serial EEPROM, and/or I²C/SMBus releases the hold by Setting the Initiate Configuration bit), all PORT_GOODx# signals are initially assigned to the Management Port. In this case, the PORT_GOODx# signals are not associated to corresponding Ports in the virtual switches, until Management Port software, the serial EEPROM, and/or I²C/SMBus Sets the Virtual Switch GPIO Update register VS GPIOs Update bit (Port 0, accessible through the Management Port, offset 64Ch[0]), to complete the assignment of individual PORT_GOODx# signals to specific virtual switches (after software, the serial EEPROM, and/or I²C/SMBus configures functionality in the GPIO registers (Port 0, accessible through the Management Port, offsets 600h, 604h, 614h, 61Ch, and 624h)), and assigns the signals to virtual switches by programming the VSx GPIO_PG 0_11 Assignment register(s) (Port 0, accessible through the Management Port, offset(s) 650h through 65Ch). Therefore, in this mode, the PORT_GOODx# signals do not reflect Link status for virtual switch Ports, until Management Port software, the serial EEPROM, and/or I²C/SMBus completes PORT_GOODx# initialization, with one exception – the PORT_GOODx# signal that corresponds to the Management Port reflects the Management Port Link status, provided that PORT_GOOD output functionality is enabled (either as default with STRAP_TESTMODE[2:0] 3-state inputs (sampled at PEX_PERST# de-assertion; refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1 for details), or by Management Port software, the serial EEPROM, and/or I²C/SMBus programming the corresponding GPIO 0_x Direction Control register(s) (Port 0, accessible through the Management Port, offset(s) 600h and/or 604h)).		

Continued...

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
PORT_GOOD[6:0]#	G2, J2, F1, L18, H2, E1, C1	I/O, PU	Yes Refer to Section 3.4.4.1	Refer to Section 3.4.4.1 for details
		Active-Low PCI Express Port Linkup Status Indicator Outputs –or– Programmable General-Purpose I/Os (7 Balls) (Cont.) GPIO Function – Virtual Switch Mode (Cont.) 2. If STRAP_MGMT_PORT_EN=1 or 0 to either (respectively) disable the Management Port, or enable the Management Port and Set the Configuration Release register Initiate Configuration bit (Port 0, accessible through the Management Port, offset 3ACh[0]) to allow all Ports to begin Link training after the serial EEPROM download completes immediately following Fundamental Reset (assuming the serial EEPROM does not Clear the Initiate Configuration bit), the PORT_GOODx# signals are individually associated to corresponding Ports. Therefore, the PORT_GOODx# signals reflect Link status of all enabled Ports, provided that PORT_GOOD output functionality is enabled (either as default with STRAP_TESTMODE[2:0] 3-state inputs (sampled at PEX_PERST# de-assertion; refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1 for details), or by Management Port software, the serial EEPROM, and/or I²C/SMBus programming the GPIO 0_x Direction Control register(s) (Port 0, accessible through the Management Port, offset(s) 600h and/or 604h)). To assign individual PORT_GOODx# signals to specific virtual switch Upstream Ports, Management Port software, the serial EEPROM, and/or I²C can program the associated VSx GPIO_PG 0_11 Assignment register(s) (Port 0, accessible through the Management Port, offset(s) 650h through 65Ch). Individual PORT_GOODx# signal assignments must be mutually exclusive among the virtual switches. The 12 bits in each VSx GPIO_PG 0_11 Assignment register correspond to the 12 PORT_GOODx# and GPIOx signals. (Refer to the registers for signal-to-bit mapping.) Setting a bit in one of the four registers assigns the signal to the virtual switch indicated by the register name. Because any PORT_GOODx# signal must not be assigned to more than one virtual switch, each of the 12 bits can be Set exclusively in only one of the four registers. A maximum of 12 PORT_GOODx# and GPIOx signals can be assigned to any one virtual switch. After the VSx GPIO_PG 0_11 Assignment register(s) are programmed, the actual assignments do not take effect, until Management Port software, the serial EEPROM, and/or I²C/SMBus Sets the Virtual Switch GPIO Update register VS GPIOs Update bit (Port 0, accessible through the Management Port, offset 64Ch[0]). After the PORT_GOODx# signals are assigned to virtual switches, each virtual switch can then configure individual PORT_GOODx# signal functionality, by programming the Virtual Switch GPIO_PG 0_x Direction Control register(s) (VS Upstream Port(s), offset(s) A34h and/or A38h). In these registers, the PORT_GOODx# signals are virtualized, and the bit numbers do not correspond to specific signal names. Software can determine how many and which PORT_GOODx# and GPIOx signals are assigned to a unique virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register Number of GPIO_PGs Available field (VS Upstream Port(s), offset A3Ch[3:0]). Virtual switch software does not determine which of the actual PORT_GOODx# and GPIOx signals are assigned to the virtual switch. The bit numbering is relative within the context of each virtual switch.		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
VS3_FATAL_ERR# VS2_FATAL_ERR# VS1_FATAL_ERR# VS0_FATAL_ERR#	G15, C18, H18, F18	VS3_FATAL_ERR# – I/O as O, PU VS2_FATAL_ERR# – I/O as O, PD VS1_FATAL_ERR# – I/O as O, PU VS0_FATAL_ERR# – I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Virtual Switch Mode Fatal Error Outputs (4 Balls) VSx_FATAL_ERR# are used only in Virtual Switch mode (one per virtual switch – Virtual Switches 3 through 0, respectively). VSx_FATAL_ERR# are asserted Low when a Fatal error is detected in the PEX 8724 and the following conditions are met (all the same conditions that are required to send a Fatal Error Message to the Host): For <i>PCI Express Base r3.0</i> errors: - Specific error is defined as <i>Fatal</i> in the Uncorrectable Error Severity register (offset FC0h), and - Reporting of the specific error condition is enabled, not masked by the Uncorrectable Error Mask register's (offset FBCh) corresponding <i>Interrupt Mask</i> bit, and Device Control register <i>Fatal Error Reporting Enable</i> bit (offset 70h[2]) –or– PCI Command register <i>SERR# Enable</i> bit (offset 04h[8]) is Set Device-Specific (RAM ECC) errors can be reported as a PCI Express Uncorrectable Error, and/or by interrupt (INTx, MSI, or PEX_INTA#) signaling, by a VSx_FATAL_ERR# output: - The error event can be signaled by the VSx_FATAL_ERR# and/or VSx_PEX_INTA# outputs that are associated with the virtual switch that includes Port 0 of the Station in which the ECC error occurred. - Reporting of the specific error condition in the Device-Specific Error Status x register bit(s), if not masked in their corresponding Device-Specific Error Mask x register bit(s) (Port 0 and/or 8, offsets 700h, 708h, 710h, and/or 718h (Status) and offsets 704h, 70Ch, 714h, and/or 71Ch (Mask)). - Uncorrectable Internal error in the Uncorrectable Error Status register <i>Uncorrectable Internal Error Status</i> bit (Port 0 and/or 8, offset FB8h[22]) is defined as <i>Fatal</i> in the Uncorrectable Error Severity register <i>Uncorrectable Internal Error Severity</i> bit (Port 0 and/or 8, offset FC0h[22]), and - Reporting of Uncorrectable Internal errors is enabled (not masked) by the Uncorrectable Error Mask register <i>Uncorrectable Internal Error Mask</i> bit (Port 0 and/or 8, offset FBCh[22]), and Device Control register <i>Fatal Error Reporting Enable</i> bit (offset 70h[2]) or PCI Command register <i>SERR# Enable</i> bit (offset 04h[8]) is Set The Device Status register <i>Fatal Error Detected</i> bit (offset 70h[18]) is Set, and the specific error is flagged in the Uncorrectable Error Status register (offset FB8h). Continued...		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
VS3_FATAL_ERR# VS2_FATAL_ERR# VS1_FATAL_ERR# VS0_FATAL_ERR#	G15, C18, H18, F18	VS3_FATAL_ERR# – I/O as O, PU VS2_FATAL_ERR# – I/O as O, PD VS1_FATAL_ERR# – I/O as O, PU VS0_FATAL_ERR# – I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Virtual Switch Mode Fatal Error Outputs (4 Balls) (Cont.) Software can be used to de-assert VSx_FATAL_ERR# outputs, with the following procedure: 1. Set the Port's Error Check Disable register <i>Software Force Error Enable</i> bit (Transparent Downstream Ports, offset 720h[2]). 2. Write 5h to bits [11:8] of the Uncorrectable Error Status register (offset FB8h), in addition to Clearing <i>Status</i> bits that are Set, using a Read-Modify-Write operation. When the Port's <i>Software Force Error Enable</i> bit is Set, writing 1 to an <i>Error Status</i> bit that is Cleared causes that bit to be Set, and generates the corresponding ERR_FATAL or ERR_NONFATAL Message, if enabled and not masked. Therefore, software should not, for example, write all 1s to this register while the Port's <i>Software Force Error Enable</i> bit is Set. 3. Clear the Port's <i>Software Force Error Enable</i> bit. If VSx_FATAL_ERR# outputs are not used, these signals can remain unconnected. <i>Notes:</i> Always driven. Device-Specific (RAM ECC) errors are reported in the lowest numbered enabled Port within the Station.		
VS3_PERST# VS2_PERST# VS1_PERST# VS0_PERST#	B15, A17, J18, E17	I/O as I, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Virtual Switch Mode Fundamental Reset (4 Balls) VSx_PERST# inputs are used only in Virtual Switch mode, to cause a Fundamental Reset (PERST#; one per virtual switch – Virtual Switches 3 through 0, respectively). (Refer to Section 5.1.2, “Resets – Virtual Switch Mode,” for further details.) <i>Notes:</i> PEX_PERST# input, defined in Table 3-3, is the Reset input used in Base mode. Because the Virtual Switch Table registers (refer to Section 5.3.3.1, “Virtual Switch Table Registers”) exist in Port 0, Port 0 must not be held in Reset during setup of Virtual Switch mode. Therefore, for whichever virtual switch includes Port 0, its VSx_PERST# input must be de-asserted (High). Assigning Port 0 to be the Management Port might provide the simplest solution toward meeting this requirement.		

Table 3-13. Device-Specific Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
VS3_PEX_INTA# VS2_PEX_INTA# VS1_PEX_INTA# VS0_PEX_INTA#	F15, D18, G18, B17	I/O as O, PU	Yes Refer to Section 3.4.4.1	Refer to STRAP_TESTMODE[2:0] and Section 3.4.4.1
		Virtual Switch Mode Interrupt Outputs (4 Balls) VSx_PEX_INTA# are used only in Virtual Switch mode (one per virtual switch – Virtual Switches 3 through 0, respectively). VSx_PEX_INTA# Interrupt output is enabled if: - INTx Messages are enabled (Port's PCI Command register <i>Interrupt Disable</i> bit, offset 04h[10], is Cleared), and MSIs are disabled (Port's MSI Control register <i>MSI Enable</i> bit, offset 48h[16], is Cleared) - VSx_PEX_INTA# output (Error Check Disable register <i>Enable Ball Interrupt Output(s) for x Interrupt</i> bit(s) (offset 720h[9, 8, 7, 6, 5, and/or 4]; refer to the register description, for Port associations) are Set) The three interrupt mechanisms, listed below, are mutually exclusive modes of operation, on a per-Port basis, for all interrupt sources: - Conventional PCI INTx Message generation - Native MSI transaction generation - Device-Specific VSx_PEX_INTA# assertion VSx_PEX_INTA# assertion (Low) indicates that one or more of the following events and/or errors (if not masked) were detected: - Hot Plug or Link State events - PCI Express Hot Plug events - General-Purpose Input Interrupt events - Device-Specific NT-Link Port errors and events - NT-Virtual Doorbell events - NT-Link Doorbell events - Management Port Doorbell-Generated interrupts - Management Link Status events Refer to Section 9.1.1, “Interrupt Sources or Events,” for details. If used, VSx_PEX_INTA# outputs require an external pull-up resistor; otherwise, these outputs can remain unconnected.		

3.4.8 External Resistor Signals

Table 3-14. External Resistor Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
REXT_A[7:6, 3:0]	F5, F3, N16, M15, M4, N3	A	No	–
		External Bias Resistor Signals (6 Balls) One REXT_Ax signal is provided for each SerDes quad, per Station. An external resistor (3.00KΩ, 1%) must be connected between each REXT_Ax signal and VSS (Ground). Each resistor should be placed close to the PEX 8724, for minimal noise injection. Bypass capacitors must not be connected to these signals. <i>Notes: A 3.01KΩ, 0.1% resistor can be used, because that value is within the 3.00KΩ, $\pm 1\%$ specification.</i>		
REXT_B[7:6, 3:0]	F6, F4, M16, L15, L4, M3	A	No	–
		External Bias Resistor Signals (6 Balls) One REXT_Bx signal is provided for each SerDes quad, per Station. An external resistor (3.00KΩ, 1%) must be connected between each REXT_Bx signal and VSS (Ground). Each resistor should be placed close to the PEX 8724, for minimal noise injection. Bypass capacitors must not be connected to these signals. <i>Notes: A 3.01KΩ, 0.1% resistor can be used, because that value is within the 3.00KΩ, $\pm 1\%$ specification.</i>		
REXT_CMU	L5	A	No	–
		Clock Management Unit for SerDes An external resistor (3.00KΩ, 1%) must be attached between REXT_CMU and VSS (Ground). <i>Notes: A 3.01KΩ, 0.1% resistor can be used, because that value is within the 3.00KΩ, $\pm 1\%$ specification.</i>		

3.4.9 Thermal Diode Signals

The PEX 8724 includes a thermal-sensing diode, to facilitate the measurement of on-die device junction temperature. On the package level, the thermal diode inputs listed in [Table 3-15](#) connect to anode and cathode terminals of a diode-connected transistor (PNP) implemented on the die.

For further details, refer to [Section 17.8, “Thermal Sensor Diode.”](#)

Table 3-15. Thermal Diode Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
THERMAL_DIODE _n	J15	I	No	—
		Thermal Diode Negative THERMAL_DIODE _n can be connected to the negative terminal of a Temperature Sensor IC, to monitor the approximate temperature of the die. <i>Note: Subtract 24°C from the temperature value reported by the Temperature Sensor IC, to determine the correct die temperature.</i>		
THERMAL_DIODE _p	J16	I	No	—
		Thermal Diode Positive THERMAL_DIODE _p can be connected to the positive terminal of a Temperature Sensor IC, to monitor the approximate temperature of the die. <i>Note: Subtract 24°C from the temperature value reported by the Temperature Sensor IC, to determine the correct die temperature.</i>		

3.4.10 No Connect Signals

Caution: Do not connect these signals to board electrical paths.
These signals are internally connected to the device.

Table 3-16. No Connect Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
N/C	A10, B10, D6, E6, J4, L1, L2, P3, P16, R3, R16, U7, U12, V7, V12	Reserved	No	–
		No Connect (15 Balls) Do not connect these signals to board electrical paths.		
SPARE3	E2	I/O, PD	No	–
		Spare 3 Reserved for future use Note: Normally, this I/O should be pulled Low to VSS (Ground). Optionally, this I/O can remain unconnected, because the internal pull-down resistor holds the input Low.		
SPARE2	K4	I/O as I, PU	No	–
		Compatibility Enable for Non-Compliant Gen 1 Endpoints When SPARE2 is Low and the Link training sequence fails during either the LTSSM <i>Polling</i> or <i>Configuration</i> state, the next time the LTSSM exits the <i>Detect</i> state, TS Ordered-Sets advertise only the Gen 1 data rate, and no Autonomous Change support. When SPARE2 is High, the Data Rate Identifier symbol in the TS Ordered-Sets always advertises support for both the Gen 3 data rate and Autonomous Change. Notes: This feature should be enabled only if a non-compliant Gen 1 device will not linkup if these Data Rate Identifier bits are Set. Normally, this input should be pulled High to VDD18 . Optionally, this input can remain unconnected, because the internal pull-up resistor holds the input High.		

Table 3-16. No Connect Signals (Cont.)

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
SPARE1	H4	I/O as I, PU	No	–
		HP_PWREN_x Polarity Inversion The SPARE1 input state is not mapped to a register bit (and therefore HP_PWREN_x polarity is not programmable by software, serial EEPROM, nor I²C/SMBus). An external resistor is not required unless: • HP_PWREN_x outputs must be Active-Low, –or– • SPARE1 is connected to a circuit trace (the internal resistor is relatively weak, and might not be sufficiently strong, to hold the SPARE1 input High) 0 = HP_PWREN_x outputs are Active-Low 1 = HP_PWREN_x outputs are Active-High (default, by virtue of the internal pull-up resistor) Notes: Serial Hot Plug SHP_PWREN outputs (from connected I²C I/O Expanders) are always Active-High. This input is sampled at the moment PEX_PERST# input is de-asserted. Normally, this input should be pulled High to VDD18. Optionally, this input can remain unconnected, because the internal pull-up resistor holds the input High.		
SPARE0	B2	I/O, PU	No	–
		Spare 0 Reserved for future use Note: Normally, this I/O should be pulled High to VDD18. Optionally, this I/O can remain unconnected, because the internal pull-up resistor holds the input High.		

3.4.11 Power and Ground Signals

In general, each power signal should connect to one de-coupling capacitor. Use a mix of values, *such as* 0.1 and 0.01 μ F.

Due to VDD09 in-rush current at the moment **PEX_PERST#** input is de-asserted, bulk capacitance is needed, to prevent the voltage rail from falling below minimum voltage requirements. *For example*, the PEX 8724 RDK uses three 680 μ F capacitors for this. This particular capacitance value is not required; however, it is best to use multiple capacitors in parallel. Inductance must be Low.

Table 3-17. Power and Ground Signals

Signal Name	Location	Type	Multiplexed Y/N	STRAP_TESTMODE[2:0] Input States
		Description		
VDD09	G7, G11, G13, H6, H8, H10, H12, H14, J5, J7, J9, J11, J13, K6, K8, K10, K12, K14, L7, L9, L11, L13, M6, M8	CPWR	No	–
		0.9V -5%, +5.55% Power for Core and SerDes Digital Logic (24 Balls)		
VDD09A	F7, F8, G5, G9, M5, M9, M10, M14, N7, N8, N12, N13	APWR	No	–
		0.9V -5%, +5.55% Power for SerDes Analog Circuits (12 Balls)		
VDD18	D16, E3, F13, N9, N14	I/OPWR	No	–
		1.8V -5%, +10% Power for I/O Logic Functions (5 Balls)		
VDD18A	G8, M7, M12	PLL PWR	No	–
		1.8V -5%, +10% Power for PLL Circuits (3 Balls)		
VSS	A3, A13, B3, B13, C4, C5, C6, C7, C8, C9, C10, C11, C12, C17, F9, F10, F11, F12, G3, G6, G10, G12, G14, H7, H9, H11, H13, J6, J8, J10, J12, J14, J17, K5, K7, K9, K11, K13, L6, L8, L10, L12, L14, M11, M13, N1, N2, N4, N5, N6, N10, N11, N15, N17, N18, T1, T2, T3, T4, T5, T6, T7, T8, T9, T10, T11, T12, T13, T14, T15, T16, T17, T18	GND	No	–
		Ground Connections (73 Balls)		

3.5 Physical Layout

Figure 3-2. Physical Ball Assignment (See-Through Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
A	STRAP_ST N1_PORTC FG1	JTAG_TDI	VSS	PEX_PETn 24	PEX_PETn 25	PEX_PETn 26	PEX_PETn 27	PEX_PETn 28	PEX_PETn 29	N/C	PEX_PETn 30	PEX_PETn 31	VSS	STRAP_ST N1_PORTC FG0	SHPC_INT#	STRAP_GE N1_GEN2	HP_PRST A#_VS2_P ERST#	STRAP_PL L_BYPASS #	A
B	JTAG_TRS T#	SPARE0	VSS	PEX_PETp 24	PEX_PETp 25	PEX_PETp 26	PEX_PETp 27	PEX_PETp 28	PEX_PETp 29	N/C	PEX_PETp 30	PEX_PETp 31	VSS	JTAG_TDO	HP_BUTTO N_C#_VS3 PERST#	STRAP_UP STRM_POR TSEL2	HP_ATNLE D_A#_VS0 PEX_INTA#	STRAP_RE SERVED2	B
C	PORT_GO OD0#	PEX_INTA#	I2C_ADDR0	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	JTAG_TCK	HP_MRL_C #_RESERV ED	HP_PRST C#_RESE RVED	STRAP_UP STRM_POR TSEL0	VSS	HP_PWRE N_A#_VS2_F ATAL_ERR #	C
D	HP_PWR_ GOOD_B_G PIO7	I2C_SDA0	STRAP_I2C SMBUS_E N	PEX_PERn 24	PEX_PERn 25	N/C	PEX_PERn 26	PEX_PERn 27	PEX_PERn 28	PEX_PERn 29	PEX_PERn 30	PEX_PERn 31	STRAP_NT UPSTRM_ PORTSEL1	JTAG_TMS	HP_PWRE N_C#_RESE RVED	VDD18	HP_CLKEN _C#_RESE RVED	HP_PRST A#_VS2_P EX_INTA#	D
E	HP_ATNLE D_B#_POR T_GOOD1#	SPARE3	VDD18	PEX_PERp 24	PEX_PERp 25	N/C	PEX_PERp 26	PEX_PERp 27	PEX_PERp 28	PEX_PERp 29	PEX_PERp 30	PEX_PERp 31	STRAP_NT UPSTRM_ PORTSEL0	FATAL_ER R#	HP_PWRFL T_C#_RES ERVED	EE_SK	HP_BUTTO N_A#_VS0 PERST#	PEX_NT_P ERST#	E
F	HP_MRL_B #_PORT_C OOD4#	STRAP_RE SERVED3	REXT_A6	REXT_B6	REXT_A7	REXT_B7	VDD09A	VDD09A	VSS	VSS	VSS	VSS	VDD18	HP_PRST C#_RESE RVED	HP_ATNLE D_C#_VS3 PEX_INTA#	STRAP_UP STRM_POR TSEL1	EE_DO	HP_PWRLE D_A#_VS0 FATAL_ER R#	F
G	STRAP_RE SERVED1	HP_PRST _B#_PORT _GOOD6#	VSS	STRAP_RE SERVED4	VDD09A	VSS	VDD09	VDD18A	VDD09A	VSS	VDD09	VSS	VDD09	VSS	HP_PWRLE D_C#_VS3 FATAL_ER R#	EE_CS#	STRAP_NT UPSTRM_ PORTSEL2	HP_CLKEN A#_VS1_P EX_INTA#	G
H	HP_PWRLE D_B#_GPIO 10	HP_BUTTO N_B#_POR T_GOOD2#	I2C_SCL0	SPARE1	STRAP_MG MT_PORT_ EN	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	HP_PWR_ GOOD_C_ GPIO11	EE_DI	STRAP_SE RDES_MO DE_EN#	HP_PWRFL T_A#_VS1_ FATAL_ER R#	H
J	STRAP_TE STMODE2	HP_PRST _B#_PORT _GOOD5#	HP_PWRE N_B#_GPIO8	N/C	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	THERMAL DIODEn	THERMAL_ DIODEp	VSS	HP_MRL_A #_VS1_P ERST#	J
K	STRAP_TE STMODE1	STRAP_I2C SMBUS_C FG_EN#	HP_PWRFL T_B#_GPIO 9	SPARE2	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	STRAP_RE SERVED16	STRAP_ST N0_PORTC FG0	I2C_SCL1	HP_PWR_ GOOD_A_V S_MODE	K
L	N/C	N/C	PEX_PERS T#	REXT_B1	REXT_CMU	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	VDD09	VSS	REXT_B2	STRAP_PR OBE_MOD E#	I2C_SDA1	HP_CLKEN B#_PORT _GOOD3#	L
M	PEX_REFC LKp	PEX_REFC LKp	REXT_B0	REXT_A1	VDD09A	VDD09	VDD18A	VDD09	VDD09A	VDD09A	VSS	VDD18A	VSS	VDD09A	REXT_A2	REXT_B3	STRAP_ST N0_PORTC FG1	STRAP_TE STMODE0	M
N	VSS	VSS	REXT_A0	VSS	VSS	VSS	VDD09A	VDD09A	VDD18	VSS	VSS	VDD09A	VDD09A	VDD18	VSS	REXT_A3	VSS	VSS	N
P	PEX_PERp 0	PEX_PERp 1	N/C	PEX_PERp 2	PEX_PERp 3	PEX_PERp 4	PEX_PERp 5	PEX_PERp 6	PEX_PERp 7	PEX_PERp 8	PEX_PERp 9	PEX_PERp 10	PEX_PERp 11	PEX_PERp 12	PEX_PERp 13	N/C	PEX_PERp 14	PEX_PERp 15	P
R	PEX_PERn 0	PEX_PERn 1	N/C	PEX_PERn 2	PEX_PERn 3	PEX_PERn 4	PEX_PERn 5	PEX_PERn 6	PEX_PERn 7	PEX_PERn 8	PEX_PERn 9	PEX_PERn 10	PEX_PERn 11	PEX_PERn 12	PEX_PERn 13	N/C	PEX_PERn 14	PEX_PERn 15	R
T	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	T
U	PEX_PETp 0	PEX_PETp 1	PEX_PETp 2	PEX_PETp 3	PEX_PETp 4	PEX_PETp 5	N/C	PEX_PETp 6	PEX_PETp 7	PEX_PETp 8	PEX_PETp 9	N/C	PEX_PETp 10	PEX_PETp 11	PEX_PETp 12	PEX_PETp 13	PEX_PETp 14	PEX_PETp 15	U
V	PEX_PETn 0	PEX_PETn 1	PEX_PETn 2	PEX_PETn 3	PEX_PETn 4	PEX_PETn 5	N/C	PEX_PETn 6	PEX_PETn 7	PEX_PETn 8	PEX_PETn 9	N/C	PEX_PETn 10	PEX_PETn 11	PEX_PETn 12	PEX_PETn 13	PEX_PETn 14	PEX_PETn 15	V
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	



Chapter 4 Functional Overview

4.1 Introduction

The PEX 8724 is designed with a flexible, modular architecture. This chapter provides an overview of the PEX 8724's major functions:

- [Modes of Operation](#)
- [Software Architecture](#)
- [Hardware Architecture](#)
- [PCI Express Station Functional Description](#)
- [Physical Layer](#)
- [Transaction Layer](#)

Subsequent chapters go into further detail.

4.2 Modes of Operation

The PEX 8724 supports and implements two modes of operation – [Base Mode](#) and [Virtual Switch Mode](#).

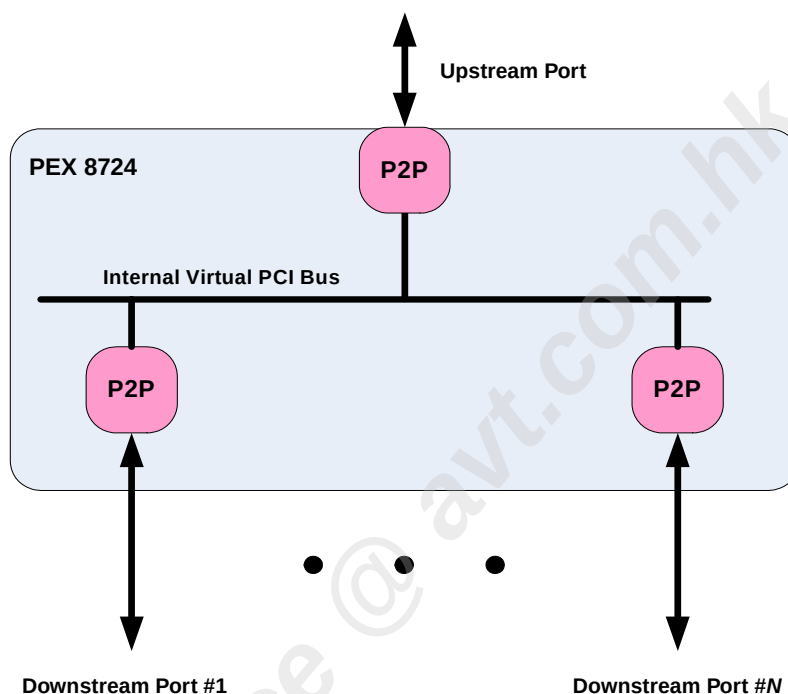
4.2.1 Base Mode

The PEX 8724 is a 24-Lane, 6-Port switch. [Figure 4-1](#) illustrates the PEX 8724 in Base mode, from a software point of view. In Base mode, the PEX 8724 supports one Upstream Port and up to five Downstream Ports.

The PEX 8724 also implements a single NT Port, which can optionally be enabled and configured to be any Port within the PEX 8724. NT mode is useful in supporting high-availability systems and failover.

Note: The PCI-to-PCI (P2P) blocks in [Figure 4-1](#) are a logical representation of how a Port presents itself to software. Each P2P block is a Type 1 PCI function, which is a PCI-to-PCI bridge.

Figure 4-1. Software Point-of-View – Base Mode



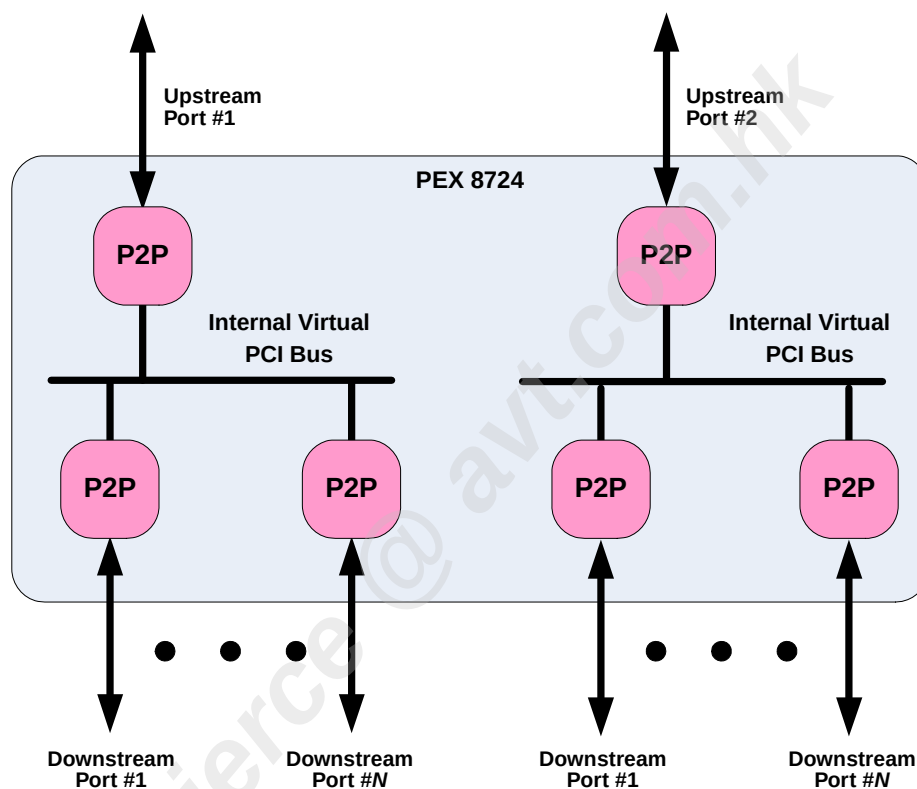
4.2.2 Virtual Switch Mode

In Virtual Switch mode, the PEX 8724 can be partitioned in up to four independent virtual switches, using one piece of silicon. The switches are still separate – there is no path between them. Each virtual switch is part of an independent PCI Express hierarchy, and the different virtual switches do not share Ports. Although the virtual switches share the same hardware infrastructure, they enforce the security between them so the traffic from one virtual switch does not leak into other virtual switches. Additionally, by re-programming the virtual switch to which a Port belongs, it is possible to support various failover and load balancing uses. A virtual switch can span across both Stations, and a Station can be shared by multiple virtual switches.

For example, the PEX 8724 can be partitioned into two virtual switches, as illustrated in Figure 4-2. This enables the PEX 8724 to be used in applications such as two-way machines, where one Root Complex is attached to one Upstream Port, and a second Root Complex is attached to a second Upstream Port.

Refer to [Chapter 12, “Virtual Switch Mode,”](#) for further details.

Figure 4-2. Example of Using Two Virtual Switches – Virtual Switch Mode



4.3 Software Architecture

4.3.1 PCI-Compatible Software Model

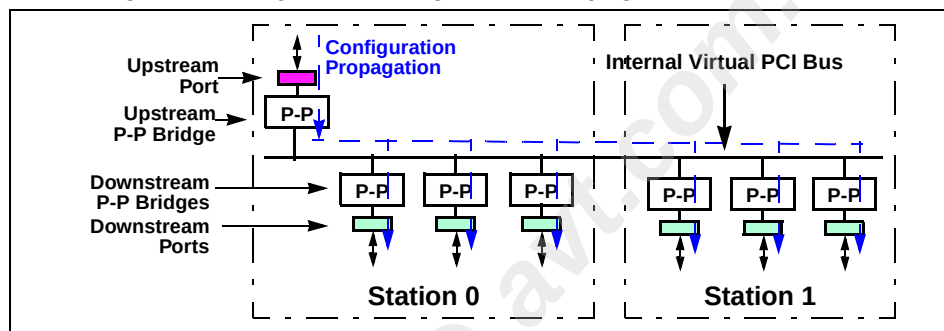
The PEX 8724 can be thought of as a hierarchy of PCI-to-PCI bridges, with one Upstream PCI-to-PCI bridge and one or more Downstream PCI-to-PCI bridges connected by an internal Virtual PCI Bus. (Refer to [Figure 4-3](#), which represents the PEX 8724 in Base mode.) PCI-to-PCI bridges are compliant with the PCI and PCI Express system models, allowing software written for Conventional PCI devices to work on the latest PCI Express switches.

The Upstream Port has the lowest Bus Number in a PCI Express switch. From the point of view of the PCI-to-PCI bridge, there is a:

- Primary Bus Number (the Upstream Bus Number),
- Secondary Bus Number (the Downstream Bus Number), and
- Subordinate Bus Number (highest Bus Number below the bridge).

The Secondary and Subordinate Bus Numbers form a range, within which all Downstream devices must fit. This can cause problems for systems that might allow hot-insertion of additional devices, unless planned for in advance, and a gap is created within in the Bus Number range, to allow for new device(s). Typically, system BIOS scans the PCI Bus and assigns Bus Numbers; therefore, support for hot insertion can depend upon the BIOS used.

Figure 4-3. System Configuration Propagation – Base Mode



Note: [Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

The Configuration Space registers (CSRs) in the Upstream PCI-to-PCI bridge are accessible by Type 0 Configuration Requests that target the Upstream bus interface. The Upstream Port(s) capture(s) the Type 0 Configuration Write Target Bus Number and Device Number. The Upstream Port(s) then use(s) this Captured Bus Number and Captured Device Number, as part of the Requester ID and Completer ID for the Requests and Completions generated by the Upstream Port(s).

The CSRs in the Downstream Port PCI-to-PCI bridges are accessible by Type 1 Configuration Requests received at the Upstream Port(s) that target the internal virtual PCI Bus, by having a Bus Number value that matches the Upstream bridge's Secondary Bus Number value. Each Downstream bridge is associated with a unique Device Number, as explained in [Section 4.4.1.2](#).

The CSRs of Downstream devices are hit in two ways. If the Configuration Request matches the PEX 8724 Downstream Port Secondary Bus Number, the PEX 8724 converts the Type 1 Configuration Request into a Type 0 Configuration Request. However, if the Bus Number does **not** match the Secondary Bus Number, but falls within the Subordinate Bus Number range, the Type 1 Configuration Request is forwarded out of the PEX 8724, unchanged. A Type 1 Configuration Request that targets a Bus Number that is not within range is invalid, and is terminated by the PEX 8724 Upstream Port(s) as an Unsupported Request (UR).

After all PCI devices have been located and assigned Bus and Device Numbers, software can assign a Memory map and I/O map. In the PEX 8724, each Downstream bridge has its own Base and Limit. Software will Set the Upstream PCI-to-PCI bridge Base and Limit to include all Downstream Base and Limit ranges. As a result, all Downstream devices are contiguous in the Memory map. Requests (Memory or I/O) go Downstream if they fall within a bridge's Base and Limit range. Alternatively, Requests (Memory or I/O) go Upstream if they do not target anything within the Upstream bridge's Base and Limit range.

Completions are routed by the Bus Number established in the Configuration registers. If the Bus Number is in the Secondary or Subordinate range, the packet goes Downstream; otherwise, the packet goes Upstream.

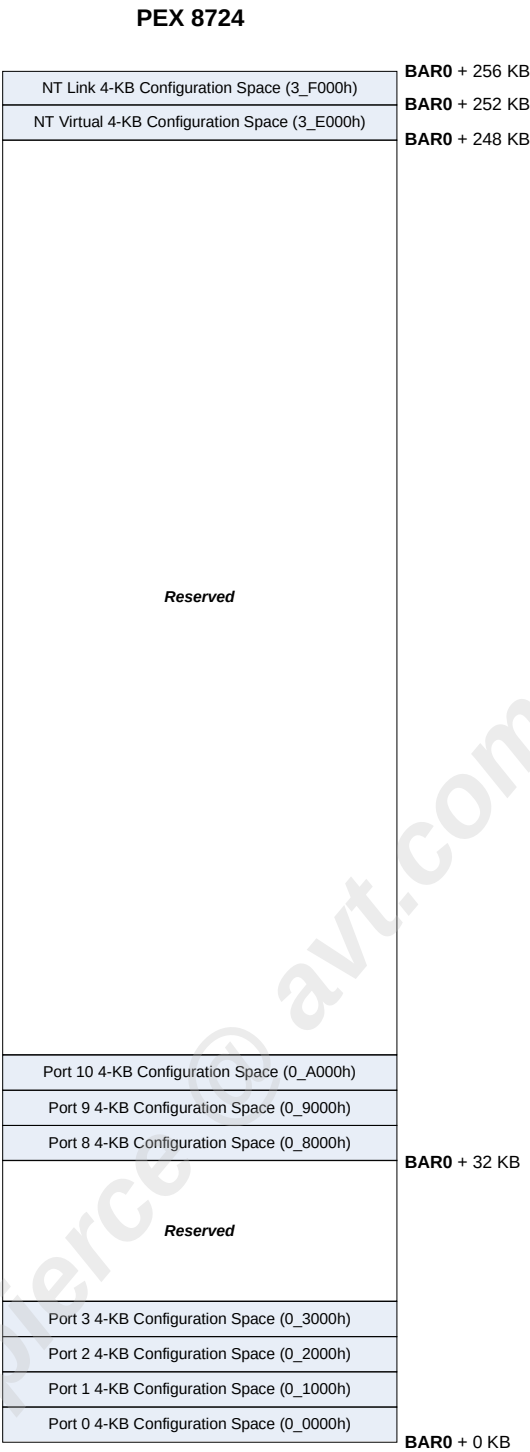
PLX has enhanced the basic PCI-to-PCI bridge with PCI Express-defined capabilities, Device-Specific capabilities, and additional functions.

4.3.2 PCI Express Memory-Mapped Configuration Space

The Configuration space is also mapped into Memory space that is seen by **Base Address Register 0 (BAR0)** of the PCI-to-PCI bridges, which is termed *Memory-Mapped Configuration space*. All PEX 8724 Ports, including the NT Port, fit into a single, flattened Memory space, as illustrated in [Figure 4-4](#).

Each Port Configuration space occupies 4 KB of the CSR Memory map. Port 0 is located at offset 0 KB, Port 1 is located at offset 4 KB, and so forth, for all enabled Ports, up to 256 KB. 4-KB register blocks that correspond to non-enabled Ports are **Reserved**.

Figure 4-4. PCI Express Memory-Mapped Configuration Space



Note: Table 4-1 indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

4.3.3 NT Bridge – NT Mode

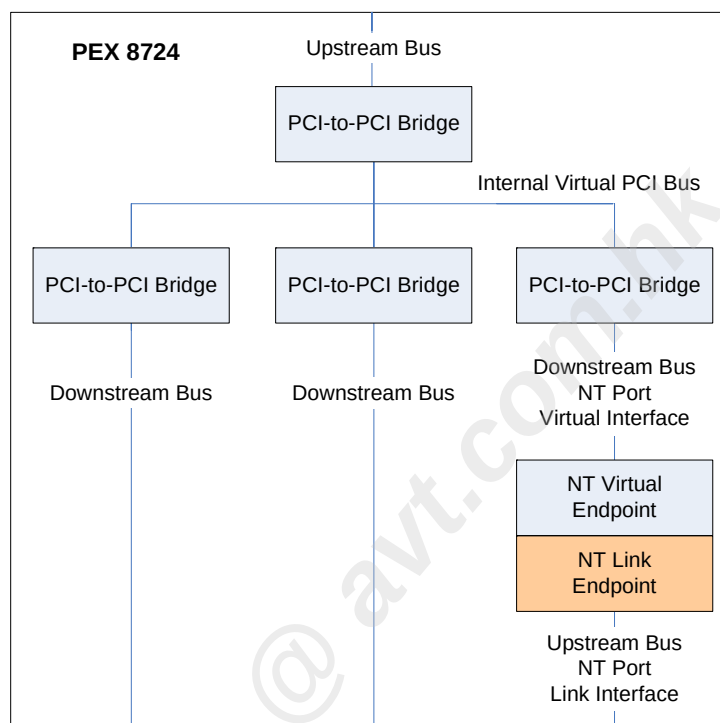
An NT bridge (NTB) is comprised of two back-to-back Type 0 endpoints. One of the endpoints is “found” by the Host that is enumerating the PCI Express Link – that is the *Link side endpoint*, or *NT Port Link Interface*. The other endpoint is “found” by the Host enumerating the internal virtual PCI Bus – that is the *Virtual side endpoint*, or *NT Port Virtual Interface*.

The NTB connects two different Host domains. Each NT endpoint has up to four Base Address registers, **BAR2** through **BAR5**, that can point to a window within the other Host’s Address space.

The PEX 8724’s NT endpoints show up to software as if they are separate devices Downstream of a PCI-to-PCI bridge, as illustrated in Figure 4-5.

For further details, refer to [Chapter 14, “Non-Transparent Bridging – NT Mode.”](#)

Figure 4-5. NT Bridge – NT Mode

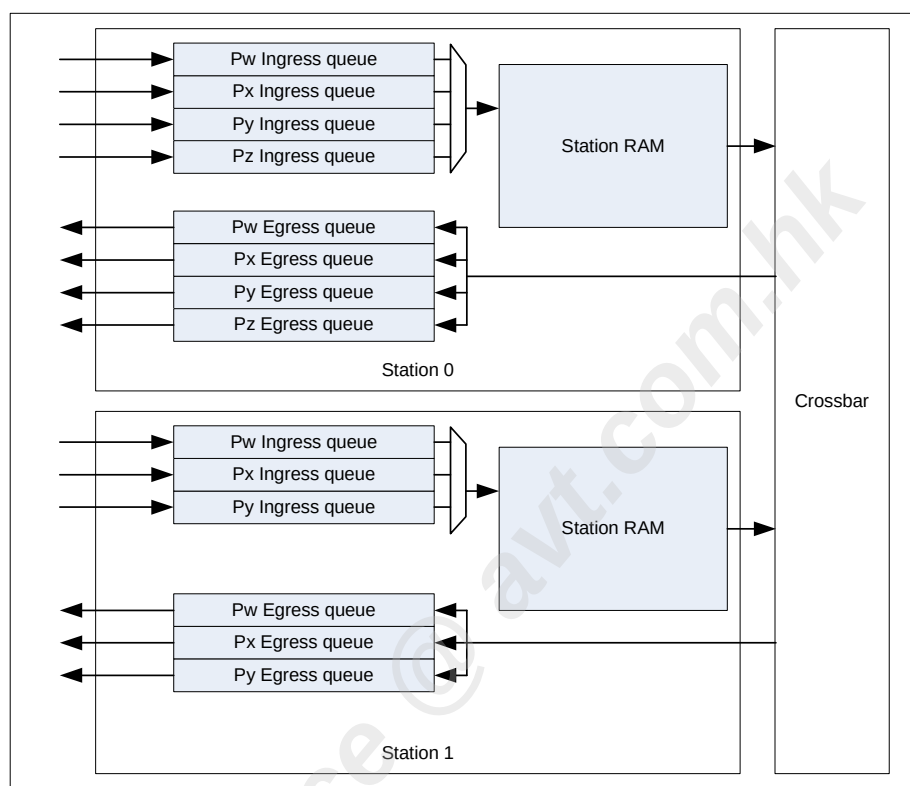


4.4 Hardware Architecture

The PEX 8724 has 24 PCI Express Lanes, implemented as 16 Lanes across Station 0 and eight Lanes across Station 1 (refer to [Section 4.4.1.1](#)). The Stations are connected to one another by the internal non-blocking fabric. [Figure 4-6](#) illustrates a block diagram of the PEX 8724.

Station 0 can have from one to four Ports associated with it (Ports 0 through 3). Station 1 can have one or two Ports associated with it (Ports 9 and 10). Station 1 also has a third Port, Port 8, which is a **Software Only** Port. The quantity of enabled Ports, and Link widths associated with those Ports, are configurable, on a Station-by-Station basis. Each Port has its own Ingress and Egress queue. Ports within a Station share the same Station RAM, but are otherwise completely independent of one another. Each Port implements the *PCI Express Base r3.0* Physical, Data Link, and Transaction Layers (PHY, DLL, and TL, respectively).

Figure 4-6. PEX 8724 Block Diagram



Notes: *Px* represents the first Port Number within the Station. *Py* represents the last enabled Port Number within the Station.

[Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the *STRAP_STNx_PORTCFGx* input states and/or serial EEPROM programming.

4.4.1 Station and Port Functions

4.4.1.1 Port Configurations

[Table 4-1](#) defines the PEX 8724 Port configurations, as well as the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Lanes are assigned to each enabled Port, in sequence.

The Upstream and Downstream Ports' maximum Link widths are initially defined by the STRAP_STNx_PORTCFGx 3-state inputs, which can be pulled or tied Low (0) to VSS (Ground), High-Impedance (Z; not connected/floating), or High (1) to VDD18. The serial EEPROM option can be used to re-configure the Ports, with the options defined in [Table 4-1](#), by programming the **Port Configuration** register *Port Configuration for Station x* field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]). Serial EEPROM configuration occurs following a Fundamental Reset, and overrides the configuration defined by the STRAP_STNx_PORTCFGx 3-state inputs at that time. Port configuration can also be changed through the I²C Slave interface, if programmed prior to the I²C/SMBus Write sequence that Sets the **Configuration Release** register *Initiate Configuration* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3ACh[0]) when STRAP_I2C_SMBUS_CFG_EN# is strapped Low. The final Link width can be automatically negotiated down from the programmed width, through Link-width negotiation for linkup to a device with fewer Lanes. The narrowest Port on one end of the Link determines the maximum Link width. Additionally, if a connection is broken on one of the Lanes, the training sequence removes the broken Lane and negotiates to a narrower Link width. *For example*, the x16 Port can negotiate down to x8, x4, x2, or x1.

If the Port cannot train to x1 (Lane 0 is broken), the Port reverses its Lanes and attempts to retrain. *For example*, a x8 Port that cannot train to x8 attempts to negotiate down to x4, x2, or x1; if x1 linkup fails, the Port reverses its Lanes and re-attempts linkup negotiation. Either the lowest Lane (Lane 0) or highest Lane (if Lanes are reversed) of the programmed Link width must connect to the other device's Lane 0.

Each Port can run independently at Gen 1, Gen 2, or Gen 3 (2.5, 5.0, or 8.0 GT/s, respectively) Link speed.

Each Station's Port configurations are independent of the other Station's Port configurations. Ports that are not configured or enabled are invisible to software.

Note: Hot Plug Port assignment is discussed in [Section 10.9, "Default Parallel Hot Plug Ports."](#)
Virtual Switch Port assignment is discussed in [Section 12.1.1, "Default Virtual Switch Port Assignments."](#)

Table 4-1. Port Configurations

# ^a	Port Configuration Strapping Input States STRAP_STN0_PORTCFG[1:0] ^b	Port Configuration Register Value Port 0, Offset 300h[2:0] ^c	Station 0 [Lanes/SerDes]/Port			
			Port 0 ^d	Port 1	Port 2	Port 3
1	0Z	001b	x16 [0-15]			
2	01	010b	x8 [0-7]	x8 [8-15]		
3	Z0	011b	x8 [0-7]	x4 [8-11]	x4 [12-15]	
4	ZZ	100b	x4 [0-3]	x4 [4-7]	x4 [8-11]	x4 [12-15]
# ^a	Port Configuration Strapping Input States STRAP_STN1_PORTCFG[1:0] ^b	Port Configuration Register Value Port 0, Offset 300h[5:3] ^c	Station 1 [Lanes/SerDes]/Port			
			Port 8 ^{d, e}	Port 9	Port 10	–
2	01	010b	Software Only	x8 [24-31]		
3	Z0	011b	Software Only	x4 [24-27]	x4 [28-31]	

- a. Port Configuration Number, which is the decimal equivalent of the **Port Configuration** register Port Configuration for Station x field(s)' (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]) binary value.
- b. “Z” means “floating.” Z is used for the 3-state inputs that have three functions, depending upon whether the input is pulled or tied High to **VDD18**, pulled or tied Low to **VSS** (Ground), or left floating (not connected). Each one of these three input states (1, 0, or Z, respectively) results in a different function for that input.
- c. **Port Configuration** register Port Configuration for Station x field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]).
- d. Ports 0 and 8 are also “Station Ports” within their respective Stations. Station Ports are used to control the Station-specific registers.
- e. Port 8 is a **Software Only** Station Port. The PEX 8724 Ports that connect to PCI Express Transmitters and Receivers are Ports 0, 1, 2, 3, 9, and 10, and each Port has its own set of registers. An additional set of Port 8 registers is visible to software, although the Port is not connected to external signals. The Port 8 registers include:
- Other Device-Specific registers.
 - Station-specific registers, which control all Ports in Station 1 (Ports 9 and 10), and are identical to the Port 0 registers that control all Ports in Station 0 (Ports 0, 1, 2, and 3). The Station-specific registers include Physical Layer registers and Device-Specific Error registers.

4.4.1.2 Port Numbering

Available Port Numbers that can be used by the PEX 8724 are as follows (refer to [Table 4-1](#) and [Figure 4-7](#)):

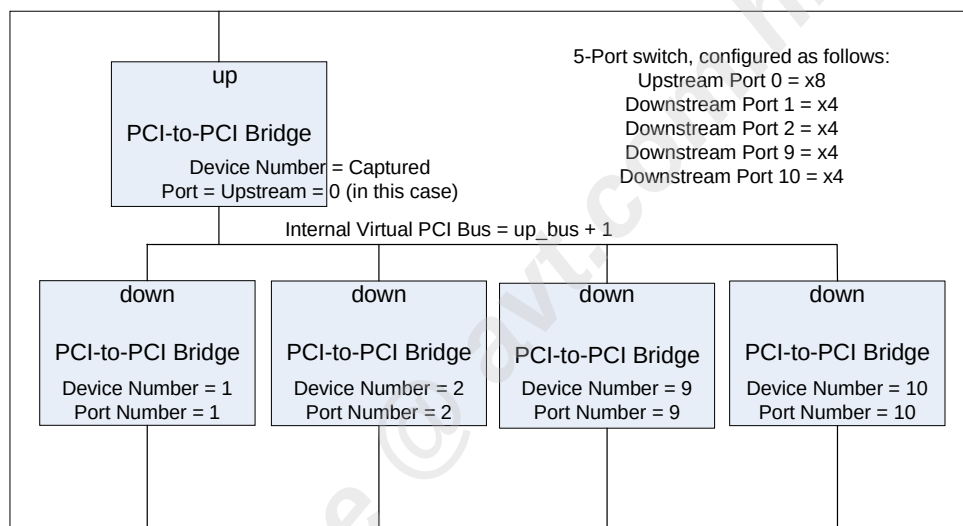
- **Station 0** – Ports 0, 1, 2, and 3
- **Station 1** – Ports 8, 9, and 10

All Downstream Device Numbers match their corresponding Port Number. *For example*, if Port 9 is a Downstream Port, the PCI-to-PCI bridge associated with that Port is Device Number 9.

Any Port can be configured as, or dynamically changed to be, the Upstream Port (Port 0 is recommended in Base mode, described later in this chapter). The PCI-to-PCI bridge implemented on the Upstream Port does not assume a Device Number – it accepts the Device Number assigned by the Upstream device. Generally, the Upstream device assigns Device Number 0, according to the *PCI Express Base r3.0*.

[Figure 4-7](#) illustrates a sample Base mode configuration, with the PEX 8724 configured as a 5-Port switch with Upstream x8 on Port 0, and Downstream Ports as follows – Ports 1, 2, 9, and 10 (x4). The Device Numbers for the PCI-to-PCI bridges implemented on the Downstream Ports match the Port Numbers.

Figure 4-7. Sample Port and Device Numbering for a 5-Port Configuration – Base Mode



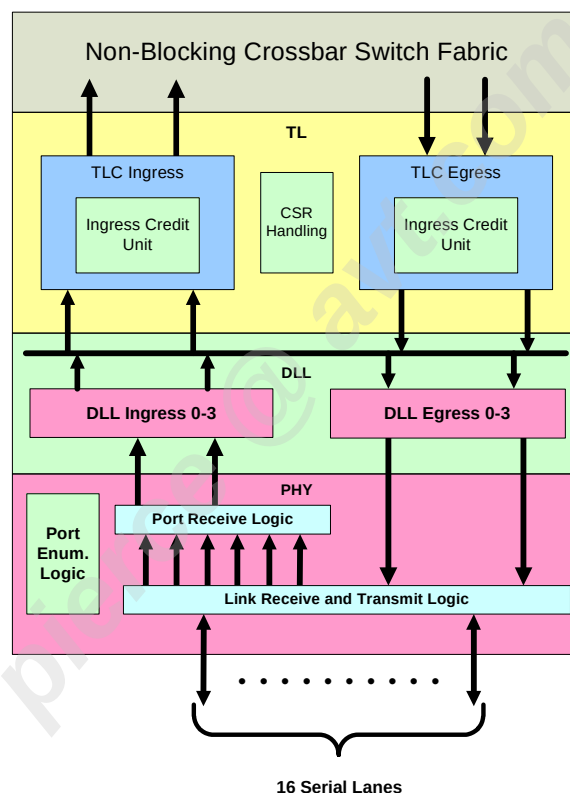
4.5 PCI Express Station Functional Description

The PEX 8724 groups SerDes together into a Station (16 SerDes in Station 0, and 8 SerDes in Station 1), as listed in [Table 4-1](#). Station 0 can have one to four Ports, and Station 1 can have one to two Ports, depending upon the Station's Port configuration, with each Port having its own SerDes assignments.

Each Station implements the PCI Express PHY and DLL functions for each of its Ports, and aggregates traffic from these Ports onto a transaction-based, non-blocking internal crossbar fabric. The PCI Express Station also performs many TL functions, while the packet queuing and ordering aspects of this layer are handled by the Crossbar Switch Control blocks. During system initialization, software initiates Configuration Requests that set up the PCI Express interfaces, Device Numbers, and Address maps across the various Ports. These maps are used to direct traffic between Ports during standard system operation. Traffic flow between Ports within the same Station, or Ports on different Stations, is supported through the internal crossbar fabric.

At the top level, each Station has a layered organization consisting of the PHY, DLL, and TL blocks, as illustrated in [Figure 4-8](#). The PHY and DLL blocks have Port-specific data paths (one per PCI Express Port) that operate independently of one another. The Transaction Layer Control (TLC) ingress section of the TL block aggregates traffic for all ingress Ports within the Station, then sends the traffic to the internal crossbar fabric. The TLC egress section of the TL block accepts packets, by way of the internal crossbar fabric, from all ingress Ports, and schedules them to be sent out the appropriate egress Port.

Figure 4-8. PCI Express Station Block Diagram (Station 0)



4.6 Physical Layer

The Physical Layer (PHY) converts information received from the DLL into an appropriate serialized format and transmits it across the PCI Express Link. The PHY also receives the serialized input from the Serializer/De-Serializer (SerDes), converts it to parallel data (internal Data Bus), then writes it to the Transaction Layer Control (TLC) Ingress buffer.

The PHY includes all circuitry for PCI Express Link interface operation, including:

- Driver and input buffers
- Parallel-to-serial and serial-to-parallel conversion
- Phase-Locked Loops (PLLs) and clock circuitry
- Impedance matching circuitry
- Interface initialization and maintenance functions
- Programmable pre-emphasis
- Decision Feedback Equalization (DFE)
- Constant Time Linear Equalization (CTLE)

4.6.1 Physical Layer Features

- 24 SerDes, implemented across two Stations, as 16 Lanes (Station 0) and 8 Lanes (Station 1) (refer to [Section 4.4.1.1](#))
- Up to four Ports in Station 0, and two Ports in Station 1
- Each Port can belong to the same or different virtual switch hierarchy, in any combination
- Multiple Upstream Port support in Virtual Switch mode – Up to four Upstream Ports, per instance
- Hardware Link training and initialization
- Hardware detection of polarity inversion
- Hardware detection of Lane reversal
- User-configurable Link widths, per Port (refer to [Table 4-1](#))
- Supported Link widths – x4, x8, or x16¹, depending upon Port configuration; x1 and x2 Link widths are also supported for auto-negotiation
- Supported Link speeds – 2.5, 5.0, and 8.0 GT/s
- Modified Compliance Pattern support with Receiver Error Counters, per Lane
- Hardware insertion of Sequence Number, STP, SDP, END, and EDB symbols
- Hardware Autonomous Speed Control supported
- Dynamic Link speed control supported
- Dynamic Link width supported
- Data scrambling and 8b/10b (Gen 1 and Gen 2 Link speeds) or 128b/130b (Gen 3 Link speed) encode/decode
- Receiver Error checking (Elastic buffer over/underflow, disparity and symbol encoding, Gen 3 framing)
- Modified Compliance Pattern support with Receiver Error Counters, per Lane
- Supported Link Power Management (PM) states
 - Conventional PCI-compatible Link Power Management states – L0, L1², L2/L3 Ready, and L3 (Vaux is **not supported**)
 - Active State Power Management (ASPM) – L0s and L1
- Upstream Port(s) can operate as Link Negotiation Master (Upstream cross-link)
- Downstream Port(s) can operate as Link Negotiation Slave (Downstream cross-link)
- Checks and removes Data Link Layer Packet (DLLP) framing symbols/tokens
- Checks and removes DLLP Link Cyclic Redundancy Check (LCRC)

1. x16 is valid only for Port 0, Station 0.

2. To differentiate from ASPM L1, this method of L1 is referred to as PCI L1 in this data book.

4.6.2 Physical Layer Status and Command Registers

The PHY operating conditions are defined in:

- [Section 13.16.4, “Device-Specific Registers – Physical Layer \(Offsets 200h – 25Ch\)”](#)
- [Section 13.18.2, “Device-Specific Registers – Physical Layer \(Offsets B80h – C8Ch\)”](#)

The Link side Host can track the Link operating status and re-configure Link parameters, by way of these registers.

4.6.3 Hardware Link Interface Configuration

The PHY can include up to 16 integrated SerDes modules on Station 0, and up to eight on Station 1. The SerDes modules are distributed among four SerDes quads (Quads 0, 1, 2, and 3) and provide the PCI Express hardware interface Lanes. The SerDes modules also provide all physical communication controls and functions required by the *PCI Express Base r3.0*, as well as the Links (clustered into Ports) that connect the PEX 8724 to other PCI Express devices.

The quantity of Ports, quantity of Lanes per Port, and SerDes connected to those Ports, are configurable, as defined in [Table 4-1](#). Initial Port configuration is determined by Strapped inputs, serial EEPROM, or auto Link-width negotiation. After the Ports are configured using the auto-negotiation process, the Link widths can narrow or widen (by combining multiple adjacent Lanes within the Station).

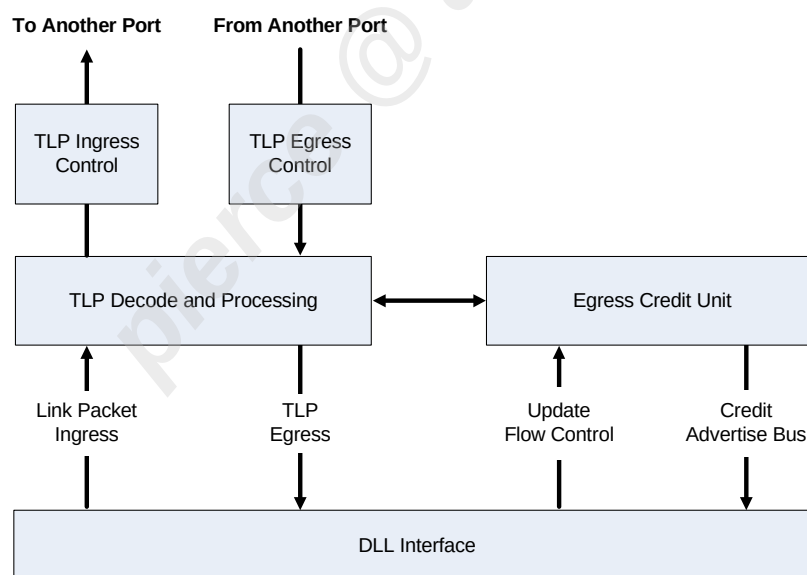
4.7 Transaction Layer

Transaction Layer (TL) functions include:

- Decoding and checking rules for the incoming TLP
- Memory-Mapped CSR access
- Checking incoming packets for malformed or unsupported packets
- Data Poisoning and end-to-end data integrity detection
- End-to-end Cyclic Redundancy Check (ECRC) of incoming packets
- Error logging and reporting for incoming packets
- TLP dispatching
- Write control to the packet RAM and packet Link List RAM
- Destination lookup and TC-VC mapping
- Credit-based scheduling
- Pipelined full Split Transaction protocol
- PCI/PCI-X-compatible ordering
- Interrupt handling (INTx or Message Signaled Interrupts (MSIs))
- Power Management (PM) support
- Hot Plug and PCI Express Hot Plug event support
- Link State event support
- QoS support
- Ordering
- Ingress and Egress credit management

The hardware functions provided by the PEX 8724 to implement the *PCI Express Base r3.0* TL requirements are illustrated in [Figure 4-9](#). The blocks provide a combination of ingress and egress control, as well as the data management at each stage within the flow sequence.

Figure 4-9. TL Controller



4.7.1 Locked Transactions

The PEX 8724 forwards Locked transactions that are destined across the switch (that is, the PEX 8724 cannot be a target of a Locked Request; however, it can pass along the Request). Locked transactions that target PEX 8724 Memory space are terminated with an Unsupported Request (UR) error.

While the PEX 8724 understands Locked transactions, it does not lock the resources. This is consistent with limitations for Locked transaction use, as outlined in the *PCI r3.0*, Appendix F, “Exclusive Accesses,” and prevents potential deadlock, as well as serious performance degradation, that could occur with Locked transaction use.

4.7.2 Relaxed Ordering

The PEX 8724 supports Relaxed Ordering for Completions. By default, if the RO attribute is Set within a Completion, that Completion can bypass Posted transactions, if Posted TLPs are blocked at the egress Port (due to insufficient Posted credits from the connected device). This behavior can be disabled, by Setting the **Station-Based Control** register *No Special Treatment for Relaxed Ordering Traffic* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 760h[29]), in each Station.

4.7.3 TL Transmit/Egress Protocol – End-to-End Cyclic Redundancy Check

End-to-End Cyclic Redundancy Check (ECRC) is an optional 32-bit field appended to the end of the outgoing packet. ECRC is calculated over the entire packet, starting with the Header and including the Data Payload, except for the *EP* bit and bit 0 of the *Type* field, which are always considered to be a value of 1 for ECRC calculations. The *ECRC* field is transmitted, unchanged, as it moves through the fabric to the Completer device. The PEX 8724 checks the ECRC on all incoming TLPs, if enabled (**Advanced Error Capabilities and Control** register *ECRC Check Enable* bit (offset FCCh[8]) is Set, in each Port), and can optionally report detected errors. (When the ECRC is detected, the **Uncorrectable Error Status** register *ECRC Error Status* bit (offset FB8h[19]) can be used to log ECRC errors.)

Additionally, the PEX 8724 can optionally append ECRC to the end of internally generated TLPs, such as Interrupt and Error Messages, if enabled (**Advanced Error Capabilities and Control** register *ECRC Generation Enable* bit (offset FCCh[6]) is Set, in each Port).

4.7.4 TL Receive/Ingress Protocol

The ingress side TL collects and stores inbound TLP traffic in the packet RAM. The incoming TLP is checked for LCRC error, valid type field, length matching the Header *Transfer Size* field, and other TLP-specific errors defined by the *PCI Express Base r3.0*.

Header and Data Payload information is forwarded to the Source Scheduler, to be routed across the internal fabric, to the egress Port. When ECRC errors are detected, the packet is discarded.

4.7.5 Flow Control Credit Initialization

The initial quantity of VC0 Flow Control (FC) credits is advertised as programmed for each type of Header and Payload. After VC0 FC initialization is complete, the FC credits received are transferred to the TL egress. The TL ingress must schedule an UpdateFC Data Link Layer Packet (DLLP) for transmission, to increase the quantity of advertised credits.

4.7.6 Flow Control Protocol

The PEX 8724 implements FC protocol that ensures that the switch:

- Does not transmit a TLP over a Link to a remote Receiver, unless the receiving device has sufficient VC Buffer space to accommodate the packet
- Generates FC credit updates to the remote Transmitter, to replace credits used to send TLPs to the PEX 8724

This FC is automatically managed by the hardware, and is transparent to software. Software is used only to enable additional Buffer space, to supplement the initial default buffer assignment.

The initial default FC credits, which are enabled after Link training, allow TLP traffic immediately after Link training completes. The Configuration transactions are the first transactions to use the default VC credits, to set up the initial device operating modes and capabilities.

The TL Ingress Credit Unit transmits DLLPs (referred to as *FC packets*) that update the FC to the remote Transmitter device, on a periodic basis. The DLLPs contain FC credit information that updates the Transmitter regarding the amount of available Buffer space in the PEX 8724.

The TL Egress Credit Unit receives DLLPs from the remote device, indicating the amount of Buffer space available in the remote Receiver. The unit uses this credit information to schedule the sending of TLPs to the remote device.



Chapter 5 Reset and Initialization

5.1 Reset

This section describes the resets supported in Base mode and Virtual Switch mode.

Reset is a mechanism that returns a device to its initial state. Reset is propagated from Upstream to Downstream. Hardware or software mechanisms can trigger three different levels of reset – Fundamental, Hot, or Secondary Bus (each is described in the sections that follow). The re-initialized states following a reset vary, depending upon the reset type.

For details regarding reset in NT Mode, refer to [Section 14.1.5, “NT Port Reset.”](#)

5.1.1 Resets – Base Mode

[Table 5-1](#) summarizes each type of reset in Base mode.

Table 5-1. Reset Summary – Base Mode

PCI Express Definition	Reset Source	Impact to Different Internal Components (upon De-Assertion)	Impact to Internal Registers
Fundamental Reset - Cold Reset - Warm Reset	PEX_PERST# input assertion	- Initializes everything - Serial EEPROM contents are loaded - HwInit types are evaluated	All registers are initialized
Hot Reset	- TS Ordered-Set <i>Hot Reset</i> bit is Set, at the Upstream Port - Upstream Port enters the <i>DL_Down</i> state	- Initializes all Station Ports - Initializes internal credits and queues - Selectively reloads serial EEPROM contents	All registers, except: - Port Configuration registers - All Sticky bits not affected by Hot Reset (HwInit, ROS, RW1CS, RWS)
Secondary Bus Reset	Downstream Port's Bridge Control register Secondary Bus Reset bit (offset 3Ch[22]) is Set	- Downstream Port PHY generates a Hot Reset - Downstream Port Data Link Layer (DLL) is down - Downstream Port Transaction Layer (TL) is initialized, exhibits <i>DL_Down</i> behavior, and TLP Requests to that Port are dropped - Upstream Port and Downstream Ports drain traffic, corresponding to the <i>DL_Down</i> condition on the Downstream Port, and initialize credits corresponding to that Downstream Port	Does not affect registers (other than to initialize credits)
	Upstream Port's Bridge Control register Secondary Bus Reset bit (offset 3Ch[22]) is Set	- All Downstream Ports propagate a Hot Reset - DLL of each Downstream Port is down - TL of each Downstream Port is initialized, exhibits <i>DL_Down</i> behavior, and drops TLP Requests that target Downstream Ports - Upstream Port TL exhibits <i>DL_Up</i> behavior	Initializes Downstream Ports registers to default values

5.1.1.1 Fundamental Reset – Base Mode

Fundamental Reset is a hardware mechanism defined by the *PCI Express Base r3.0*, Section 6.6. Fundamental Reset input, through the **PEX_PERST#** signal, resets all Port states and Configuration registers to default conditions.

5.1.1.2 Hot Reset – Base Mode

Hot Reset is an in-band Reset that propagates from an Upstream PCI Express Link to all its Transparent Downstream Ports, through the Physical Layer (PHY) mechanism. The PHY mechanism communicates a reset to Downstream devices through a training sequence (TS1/TS2 Ordered-Set, in which the *Hot Reset Training Control Bit* is Set). Hot Reset is also referred to as a *Soft Reset*.

A Hot Reset initializes all Ports, resets registers that are not defined as Sticky, and resets the serial EEPROM logic to reload registers from serial EEPROM, if present. Hot Reset does not reset the Clock logic, and can be caused by any of the following:

- Upstream Port PHY receives two consecutive TS1 Ordered-Sets in which the *Hot Reset Training Control bit* is Set. Hot Reset is generated from an Upstream device, *such as* by Setting its **Bridge Control** register *Secondary Bus Reset* bit (offset 3Ch[22]).
- Upstream Port unexpectedly enters the *DL_Down* state.
Exception – If the Upstream Port Link is in the L2/L3 Ready Link PM state and the Link goes down, the Downstream Ports do *not* generate Hot Reset.
- Upstream Port PHY enters either the *Loopback* or *Disabled* state, upon receiving two consecutive TS1 or TS2 Ordered-Sets in which either the *Loopback* or *Disable Link Training Control bit* is Set, respectively. An Upstream device can generate the *Disable Link* sequence, by Setting its **Link Control** register *Link Disable* bit (offset 78h[4]).

5.1.1.3 Secondary Bus Reset – Base Mode

Any virtual Upstream or Downstream PCI-to-PCI bridge within the PEX 8724 can reset its Downstream hierarchy, by Setting the **Bridge Control** register *Secondary Bus Reset* bit (offset 3Ch[22]).

When the *Secondary Bus Reset* bit is Set on the Upstream Port, all the Downstream Ports are initialized to their default states, as defined by the *PCI Express Base r3.0*. Each of the Transparent Downstream Ports generates an in-band Hot Reset onto its Downstream Links (the NT Port Link Interface does *not* generate Hot Reset). In addition, writable registers defined by the *PCI Express Base r3.0*, in all Downstream Ports, are initialized to default values (Upstream Port registers are not reset, and the serial EEPROM does not reload registers).

When the *Secondary Bus Reset* bit is Set on a Downstream Port, that Port is reset to its default state as defined by the *PCI Express Base r3.0*, and generates an in-band Hot Reset onto its Downstream Link. The registers of that Downstream Port are not affected.

5.1.1.4 Register Bits that Affect Hot Reset – Base Mode

Setting the **Bridge Control** register *Secondary Bus Reset* bit (offset 3Ch[22]) generates a Hot Reset to Downstream Ports and Downstream devices.

5.1.2 Resets – Virtual Switch Mode

The *PCI Express Base r3.0* discusses a Conventional Reset (Cold, Warm, and Hot). For Virtual Switch mode, the above cases are slightly extended (from how they function in Base mode), as discussed in the sections that follow.

5.1.2.1 PEX_PERST# Input Fundamental Reset – Virtual Switch Mode

PEX_PERST# input is used as the Fundamental (Cold or Warm) Reset for the entire PEX 8724. This reset affects the virtual switches, and impacts all on-chip components – Stations and their Ports, Serial EEPROM Controller, Clock logic, so forth. All registers and states are initialized.

Use of the **PEX_PERST#** input is the only way to Clear Fatal errors detected in the PEX 8724.

After **PEX_PERST#** input de-assertion, the PEX 8724 can be initialized by way of serial EEPROM, I²C, and/or the Management Port.

5.1.2.2 VSx_PERST# Input Fundamental Reset – Virtual Switch Mode

There is a virtual Fundamental Reset, per virtual switch. This input attempts to mimic the **PEX_PERST#** input, but on a virtual-switch basis. The Port's **Port Reset** register *Reset Port x Vector* bit (Port 0, accessible through the Management Port, offset 3A0h[10:8, 3:0]) can also be used to generate the equivalent effect of the **VSx_PERST#** inputs.

The **VSx_PERST#** inputs reset all PCI-to-PCI bridges in the virtual switch hierarchy, from the Upstream PCI-to-PCI bridge, down to the Downstream PCI-to-PCI bridges and Downstream Ports owned by the virtual switch that caused the **VSx_PERST#** input to assert.

Because some of the PEX 8724 data structures might be shared across virtual switches, **VSx_PERST#** inputs cannot reset the entire switch to a clean state. If there are any Fatal errors in any virtual switch, a **PEX_PERST#** input assertion is required.

VSx_PERST# assertion resets Sticky register bits in the associated virtual switch. The inputs do not function as Hot Resets.

5.1.2.3 Inband Reset (TS1 Ordered-Set) or Upstream Port DL_Down (Hot Reset) – Virtual Switch Mode

The *PCI Express Base r3.0* defines an inband reset (with a TS1 Ordered-Set) or the Upstream Port going down as a Hot Reset. These work exactly as indicated by the *PCI Express Base r3.0*. The reset is, by default, propagated Downstream to all Ports in the virtual switch whose Upstream Port receives the Hot Reset.

5.1.2.4 Secondary Bus Reset (Soft Reset) – Virtual Switch Mode

Every PCI-to-PCI bridge has a *Secondary Bus Reset* bit that resets the entire Downstream hierarchy. The Upstream PCI-to-PCI bridge's Secondary Bus Reset resets all Downstream PCI-to-PCI bridges owned by the virtual switch. A Downstream PCI-to-PCI bridge's Secondary Bus Reset sends a Hot Reset training set across the Link.

5.1.2.5 Reset Propagation Prevention – Virtual Switch Mode

It is possible to prevent the propagation and effect of the various conventional resets (besides the Cold Reset). By Setting the virtual switch's **Virtual Switch Debug** register *Upstream Port and NT-Link Port DL_Down Reset Propagation Disable* bit (VS Upstream Port(s), offset A30h[4]), Hot Resets are no longer propagated. The default value is that the bit is Cleared, and the resets are propagated.

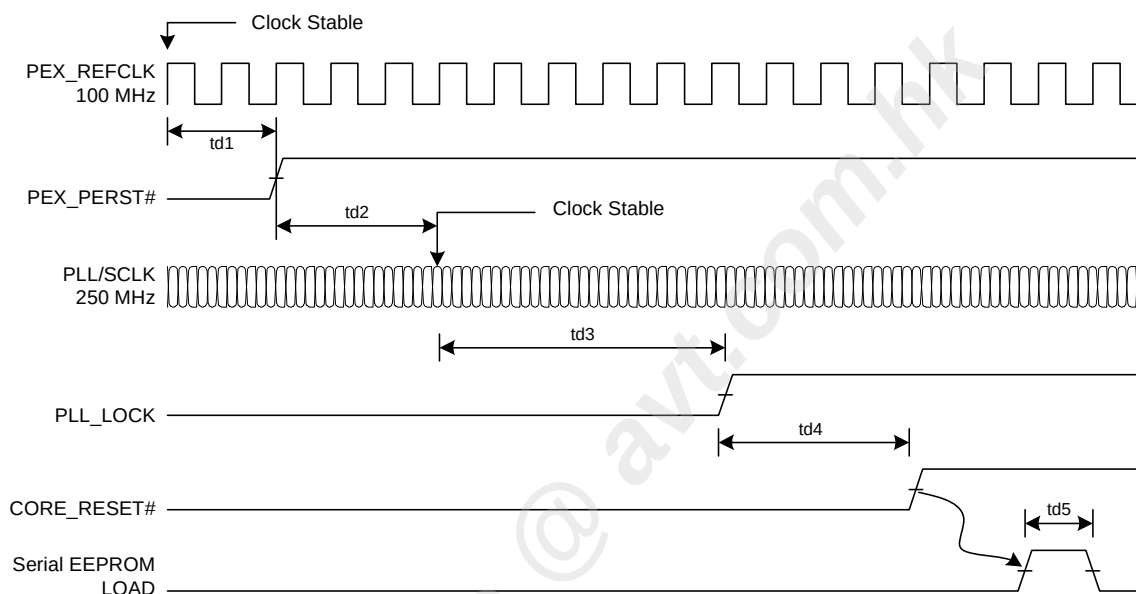
5.1.3 Reset and Clock Initialization Timing

Note: Serial Hot Plug scan and I²C I/O Expander initialization start after serial EEPROM load is complete.

Table 5-2. Reset and Clock Initialization Timing

Symbol	Description	Typical Delay
td1	REFCLK stable to PEX_Reset release time	100 μ s
td2	PEX_Reset release to Reset de-bounce	1.32 ms
td3	Reset de-bounce to Phase-Locked Loop (PLL) Lock	105 μ s
td4	Reset de-bounce to Core Reset release	2.63 ms
td5	Serial EEPROM load time with no serial EEPROM present	17 μ s

Figure 5-1. Reset and Clock Initialization Timing



5.2 Initialization – Base Mode

The PEX 8724 initialization process starts upon exit from a Fundamental Reset. The serial EEPROM and/or I²C can be used to program initial register values, prior to BIOS/OS enumeration. If the [STRAP_I2C_SMBUS_CFG_EN#](#) input is Low, linkup of all Ports is delayed, until I²C software Sets the **Configuration Release** register *Initiate Configuration* bit (Port 0, offset [3ACh\[0\]](#)).

Serial EEPROM download operates much faster than I²C access. I²C is relatively slow. Consequently, I²C initialization might not complete prior to the first BIOS/OS Configuration access, unless the system is designed to delay BIOS/OS Configuration access until the PCI Express subsystem is ready. If I²C is used to program the initial register values, and the PHY or DLL register values that affect SerDes parameters or Link initialization need to be changed, those registers must be programmed by serial EEPROM, so that the values are loaded prior to initial Link training.

5.2.1 Serial EEPROM Load Time

Serial EEPROM initialization loads only the Configuration register data that is specifically programmed into the serial EEPROM. Registers that are not included in the serial EEPROM data are initialized to default register values.

Each register entry in the serial EEPROM consists of two Address bytes and four Data bytes (refer to [Section 6.4, “Serial EEPROM Data Format”](#)); therefore, each register entry (6 bytes, or 48 bits) requires 48 serial EEPROM clocks to download. Thus, at the serial EEPROM clock default frequency of 1 MHz, after initial overhead to read the **Serial EEPROM Status** register (Port 0, offset [260h](#)) (16 serial EEPROM clocks, or 16 μ s), plus another 40 serial EEPROM clocks (40 μ s) to begin reading the register data, each register entry in the serial EEPROM requires 48 μ s to download. A serial EEPROM containing 50 register entries (typical configuration, assuming the serial EEPROM is programmed only with non-default register values) and clocked at 1 MHz, takes approximately 5.2 ms to load ((16 + 40 + 48) x 50 μ s = 5,200 μ s).

To reduce the serial EEPROM initialization time, the first register entry in the serial EEPROM can increase the clock frequency, by programming the **Serial EEPROM Clock Frequency** register (Port 0, offset [268h](#)), to a value of 2h (5 MHz), or 3h (9.62 MHz), if the serial EEPROM supports the higher frequency at the serial EEPROM supply voltage (typically 1.8 to 2.5V). At 5-MHz clocking, the serial EEPROM load time for 50 register entries can be reduced to approximately 575 μ s. Because the *PCI Express Base r3.0* allows a 20-ms budget for system hardware initialization, the default 1-MHz serial EEPROM clock is often sufficient when the quantity of Ports and registers programmed by serial EEPROM is relatively small.

If the NT Port Expansion ROM (stored within the serial EEPROM) is used, the serial EEPROM clock frequency ([EE_SK](#)) must be at least 5 MHz. For further details, refer to [Chapter 6, “Serial EEPROM Controller.”](#)

5.2.2 I²C Load Time

Initialization using I²C is slower than serial EEPROM initialization, because the I²C Slave interface operates at a lower clock frequency (100 KHz maximum) and the quantity of bits per Register access is increased (because the Device address is included in the bit stream). Writing one register using 100-KHz clocking takes approximately 830 μ s (83 clock periods).

For further details, refer to [Section 7.2, “I2C Slave Interface.”](#)

5.3 Initialization – Virtual Switch Mode

To enable Virtual Switch mode, the **VS_MODE** and **STRAP_TESTMODE[2:0]** input states at power-up must not be Low. The serial EEPROM, I²C/SMBus, and/or software cannot override the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state inputs to change between Base mode and Virtual Switch mode. However, in Virtual Switch mode, the serial EEPROM, I²C/SMBus, and/or software can change the **Virtual Switch Table**, including the quantity of enabled virtual switches. (Refer to [Section 5.3.3](#))

If a design must support both Base mode and Virtual Switch mode, strap the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state inputs to a non-Low state, to enable Virtual Switch mode. If Base mode is needed, the serial EEPROM, I²C/SMBus, and/or software can then assign all Ports to VS0 (with no Ports assigned to other virtual switches).

In addition to the Strapping signals, there are several ways to initialize the PEX 8724 when it is in Virtual Switch mode, by way of serial EEPROM, I²C/SMBus, and/or the Management Port. One of these three agents must initialize the Virtual Switch Table. The serial EEPROM and I²C/SMBus can effectively program all registers, whereas Management Port software generally cannot modify Read-Only (RO) registers. The serial EEPROM (programmable by I²C/SMBus and/or software) requires the least software support.

The PEX 8724 sequences these methods, in the following order:

1. Serial EEPROM
2. I²C/SMBus
3. Management Port

Note: It is possible to use more than one method to initialize the PEX 8724.

*If using a serial EEPROM and/or I²C, it programs the necessary Virtual Switch registers. If using a Virtual Switch Management Port, enabled by **STRAP_MGMT_PORT_EN**=0 or Z, the Management Port Sets the required Virtual Switch registers. If **STRAP_MGMT_PORT_EN**=Z to initially allow only the Management Port to link up, the Virtual Switch Management Port releases the remainder of the PEX 8724, by Setting the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset 3ACh[0]).*

Strapping inputs are used to determine whether Virtual Switch mode uses I²C, or the Management Port, to initialize:

- If **STRAP_I2C_SMBUS_CFG_EN#**=L, I²C is used to configure the PEX 8724. When I²C has finished, it must write to the *Initiate Configuration* bit, to release the hold that is preventing linkup.
- If **STRAP_I2C_SMBUS_CFG_EN#**=H, I²C is not used to configure the PEX 8724.
- The **STRAP_MGMT_PORT_EN** 3-state input is used to control two bringup options:
 - If **STRAP_MGMT_PORT_EN**=0 after the serial EEPROM is loaded, all Ports come up at the same time.
 - If **STRAP_MGMT_PORT_EN**=Z (after the serial EEPROM is loaded (or does not exist)), the Management Port comes up first to configure the PEX 8724. When the Management Port has completed its configuration, it must Set the *Initiate Configuration* bit, to release the hold that is preventing linkup.

Note: Because the Virtual Switch Table registers (refer to [Section 5.3.3.1](#)) exist in Port 0, Port 0 must not be held in Reset during setup of Virtual Switch mode. Therefore, for whichever virtual switch includes Port 0, its **VSx_PERST#** input must be de-asserted (High). Assigning Port 0 to be the Management Port might provide the simplest solution toward meeting this requirement.

The Active Management Port is defined by the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset **354h**). This register also defines an optional Redundant Management Port that can be used to back up the Active Management Port.

The RWS Setting in the **Management Port Control** register indicates that the bits are readable, writable, and Sticky.

Note: Sticky means that, as long as power is not removed, the bits are not changed with Hot Reset. Note however, that if a Hot Reset causes a serial EEPROM and/or I²C load, these bits can be overwritten from their pre-reset value.

The Active Management Port can read or write all the Virtual Switch Table registers. (Refer to [Section 5.3.3.1](#).) No other Ports are allowed to do so.

The quantity of enabled virtual switches is reflected in the **Virtual Switch Enable** register **VSx Enable** bit(s) (Port 0, accessible through the Management Port, offset **358h**). This register is initially configured by the **STRAP_TESTMODE[2:0]** and **VS_MODE** 3-state inputs.

The Upstream Port for each virtual switch can also be Set by the Management Port, using the **VSx Upstream** register(s) (Port 0, accessible through the Management Port, offsets **360h** through **36Ch**), per virtual switch.

The Management Port configures which Downstream Port is owned by which Upstream Port, by writing a **VSx Port Vector** register (Port 0, accessible through the Management Port, offsets **380h** through **38Ch**), per virtual switch. Each Downstream Port can be owned by a single Upstream Port only. If software sets two or more owners for a Downstream Port, behavior is not deterministic. The NT endpoints are considered to be separate Ports for this virtual switch ownership.

After the ownership of all Ports is defined, the **Configuration Release** register *Initiate Configuration* bit (Port 0, accessible through the Management Port, offset **3ACh[0]**) is Set, allowing the Links to come up.

There can be a Redundant Management Port as well. The Redundant Management Port can promote itself to be the new Management Port, if the Management Port Host fails. The implementation of Redundant Management Port promotion to Management Port is application-specific.

After the *Initiate Configuration* bit is Set, each virtual switch Host enumerates its hierarchy, as if it had its own independent switch. Each hierarchy is independent and there is no order in which the hierarchies must be initialized. Each Host finds only the PCI-to-PCI bridges that it owns. Each Host has its own bus numbering scheme, which applies to the Upstream Port, through to the internal virtual PCI Bus and down to any Downstream buses. The same is true for memory addressing – each virtual switch has a completely independent Address map, both in 32- and 64-bit space.

5.3.1 Management Port Policies

All Device-Specific registers can be accessed from the Active Management Port, serial EEPROM, and/or I²C.

The Redundant Management Port can access only the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset **354h**). Each Upstream Port can access the PCI-SIG-defined registers of each Port, within its own virtual switch.

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5.3.2 Active and Redundant Management Ports

The **Management Port Control** register *Active Management Port* field (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[4:0]) defines which Virtual Switch Upstream Port is assigned as the Active Management Port. In Virtual Switch mode, there is no Management Port when the following conditions are met:

- STRAP_MGMT_PORT_EN 3-state input is pulled or tied High (1) to VDD18, and
- **Management Port Control** register *Active Management Port Enable* bit (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[5]) is Cleared, and
- **Configuration Release** register *Initiate Configuration* bit (Port 0, accessible through the Management Port, offset 3ACh[0]) is Set.

The **Management Port Control** register *Redundant Management Port* field (field [12:8]) defines which Virtual Switch Upstream Port is assigned as the Redundant Management Port. The Redundant Management Port provides a Failover capability, should the Management Port Host fail. Software can demote the Management Port and promote the Redundant Management Port to be the new Management Port, by programming the **Management Port Control** register. (Refer to [Section 5.3.6.5](#).)

This register can be accessed by the Management-capable (Active and Redundant) Ports, Strapping signals, and/or the I²C Bus. The **Reserved** register bits return zeros (0) during Reads. Writes to **Reserved** register bits do not affect the register.

The register's RWS fields/bits are represented in [Table 5-3](#). (For complete details, refer to the register offset 354h description provided in [Section 13.16.7](#), "Device-Specific Registers – Port Configuration and Lane Status (Offsets 300h – 46Ch).")

Table 5-3. Management Port Control Register (Port 0, accessible through the Management Port and Redundant Management Port, Offset 354h)

Bit(s) ^a	Description
4:0	Active Management Port Indicates the Port Number of the Active Management Port. The value of this field is latched in, upon reset de-assertion, from the STRAP_UPSTRM_PORTSEL[2:0] input states.
5	Active Management Port Enable Enables the Active Management Port. The value of this bit is latched in, upon reset de-assertion, from the STRAP_MGMT_PORT_EN input state. 0 = Management Port is disabled (STRAP_MGMT_PORT_EN=1) 1 = Management Port is enabled (STRAP_MGMT_PORT_EN=0 or Z)
6	Active Management Port EEPROM Load on Hot Reset for Chip and Station Registers Enable <i>Valid only for the Management Port.</i> After the Management Port receives a Hot Reset or DL_Down condition, the serial EEPROM reloads registers, as described below. 0 = Serial EEPROM reloads Management Port Port-specific registers (default) 1 = Serial EEPROM reloads: • Chip-specific registers (might affect all virtual switches), • Station-specific registers (might affect all virtual switches), and, • Management Port Port-specific registers
12:8	Redundant Management Port Indicates the Redundant Management Port's Port Number.
13	Redundant Management Port Enable Enables the Redundant Management Port.

a. Bits not identified in [Table 5-3](#) are **Reserved** or **Factory Test Only**.

5.3.3 Virtual Switch Table

Note: In Virtual Switch mode, the Virtual Switch Table registers include the **VSx Port Vector** and **VSx Upstream** registers (Port 0, accessible through the Management Port, offsets **380h** through **38Ch**, and **360h** through **36Ch**, respectively). These two sets of registers define which Ports are associated to each virtual switch, and which Port is the Upstream Port for each virtual switch, respectively. These registers must be initialized by one (or more) of the following agents:

- Serial EEPROM.
- I²C/SMBus, provided that the **STRAP_I2C_SMBUS_CFG_EN#** input is Low (to delay linkup until I²C/SMBus Sets the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset **3ACh[0]**). This option might require software support, to delay Host enumeration until I²C/SMBus Sets the **Initiate Configuration** bit after programming the PEX 8724 Configuration registers.
- Management Port software, provided that **STRAP_MGMT_PORT_EN** is left floating (to enable the Management Port, and delay linkup of all other Ports until software and/or I²C/SMBus Sets the **Initiate Configuration** bit).

Use of serial EEPROM for the initialization might be the best choice for most applications, because it is the simplest solution. Therefore, for Virtual Switch mode applications, the serial EEPROM is required to initialize the Virtual Switch Table registers, unless I²C/SMBus and/or Management Port software can perform this task.

The PEX 8724 supports up to four virtual switches. The Virtual Switch Table defines the Port Numbers and associated Virtual Switch Numbers and Upstream Port of each virtual switch. Each table entry must be updated as a single atomic operation. Because Chip-specific registers exist in Port 0, the Management Port, if enabled, is recommended to be assigned to Port 0.

Additional information related to the table is discussed in the sections that follow.

Note: In Virtual Switch mode, the Virtual Switch Table registers include the **VSx Port Vector** and **VSx Upstream** registers (Port 0, accessible through the Management Port, offsets **380h** through **38Ch**, and **360h** through **36Ch**, respectively). These two sets of registers define which Ports are associated to each virtual switch, and which Port is the Upstream Port for each Virtual Switch, respectively. These registers must be initialized by one (or more) of the following agents:

- Serial EEPROM.
- I²C/SMBus, provided that the **STRAP_I2C_SMBUS_CFG_EN#** input is Low (to delay linkup until I²C/SMBus Sets the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset **3ACh[0]**). This option might require software support, to delay Host enumeration until I²C/SMBus Sets the **Initiate Configuration** bit after programming the PEX 8724 Configuration registers.
- Management Port software, provided that **STRAP_MGMT_PORT_EN** is left floating (to enable the Management Port, and delay linkup of all other Ports until software and/or I²C/SMBus Sets the **Initiate Configuration** bit).

Use of serial EEPROM for the initialization might be the best choice for most applications, because it is the simplest solution. Therefore, for Virtual Switch mode applications, the serial EEPROM is required to initialize the Virtual Switch Table registers, unless I²C/SMBus and/or Management Port software can perform this task.

5.3.3.1 Virtual Switch Table Registers

The Virtual Switch Table consists of three registers, listed in Table 5-4. The first two can be accessed in any order, and the third (**Virtual Switch Enable**) must be accessed last.

Table 5-4. Virtual Switch Table Registers^a

Offset(s)	Register	Description
358h[3:0]	Virtual Switch Enable	The register's <i>VSx Enable</i> bits are used to enable or disable virtual switches within the system. There is one bit, per virtual switch (VS0 through VS3). Setting a bit enables the corresponding virtual switch. Clearing a bit disables the corresponding virtual switch. If a <i>VSx Enable</i> bit is Set, another virtual switch is being established with the new set of Upstream and Downstream Ports. One Write to this register can disable the previous virtual switch, and enable a new virtual switch. This ensures that Virtual Switch Enable and Disable can be implemented with a single Write to a register. The STRAP_TESTMODE[2:0], STRAP_UPSTRM_PORTSEL[2:0], and VS_MODE 3-state inputs map to this register.
360h – 36Ch	VSx Upstream	These registers define the Upstream Port of each virtual switch. There is one register, per virtual switch (VS0 through VS3). The registers' <i>VSx Upstream Port</i> bits [4:0] define which Port is the singular Upstream Port, within the corresponding virtual switch. A virtual switch must include a single, unique Upstream Port.
380h – 38Ch	VSx Port Vector	These registers define the Upstream and Downstream Ports associated with each virtual switch. There is one register, per virtual switch (VS0 through VS3), and each register has one bit, per Port. Any Port can be assigned to a virtual switch. Setting a Port's bit in a VSx Port Vector register assigns that Port to the virtual switch associated with the register. A single Port can be assigned to only one virtual switch at any time. Therefore, each bit (in the range listed above) must be Set in only one of the four VSx Port Vector registers, at any time. A Downstream Port can be re-assigned to a different virtual switch, by Clearing the Port's bit in one VSx Port Vector register, and Setting the same bit in another VSx Port Vector register. The STRAP_TESTMODE[2:0] and VS_MODE 3-state inputs map to these registers.

a. All registers listed in this table are located in Port 0, accessible through the Management Port.

5.3.3.2 Virtual Switch Table Programming Sequence

The following describes the programming sequence for the Virtual Switch Table, by the Management Port Host.

To program the Virtual Switch Table:

1. Strap the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state inputs to enable Virtual Switch mode and the quantity of virtual switches needed.

Note: When used together, the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state inputs, depending upon their input states, enable up to four virtual switches. The quantity of enabled virtual switches is reflected in the **Virtual Switch Enable** register **VSx Enable** bit(s) (Port 0, accessible through the Management Port, offset 358h[3:0]). (Refer to the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state input descriptions, in Table 3-7, “Strapping Signals,” and Section 3.4.4.1, “STRAP_TESTMODE[2:0] 3-State Input Configurations,” for details.)

2. Strap the **STRAP_MGMT_PORT_EN** 3-state input to 0, to enable the Active Management Port and allow it to linkup, while delaying linkup of all other Ports. The Management Port Host must Set the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset 3ACh[0]), to allow the remaining Ports to linkup.

Note: When the Active Management Port is enabled and the **VS_MODE** and **STRAP_TESTMODE[2:0]** 3-state inputs are strapped to enable one active virtual switch (**VS0**), all enabled Ports are assigned to **VS0**.

3. Strap the **STRAP_UPSTRM_PORTSEL[2:0]** 3-state inputs to the Active Management Port Number. The **VS0 Upstream** register **VS0 Upstream Port** field (Port 0, accessible through the Management Port, offset 360h[4:0]) defaults to the Port Number defined as the **VS0 Upstream Port** by these inputs.
4. Adjust all **VSx Upstream** and **VSx Port Vector** registers (Port 0, accessible through the Management Port, offsets 360h through 36Ch, and 380h through 38Ch, respectively) to the new table values, ensuring that the following conditions are met:
 - a. The Management Port must remain in its original virtual switch.
 - b. A Port can appear in only one **VSx Port Vector** register. The exception is when the Management Port is being migrated from one virtual switch to another. If this is the case, the Management Port can appear in two **VSx Port Vector** registers – the original enabled virtual switch, and the newly disabled virtual switch.
 - c. The VS Upstream Port defined in the **VSx Upstream** register must be included in the corresponding **VSx Port Vector** register. If the Management Port is migrating to a different virtual switch, the new virtual switch’s **VSx Upstream** register must be programmed to the Management Port Number.
5. **If the Management Port is migrating to a different virtual switch** – Enable all other needed virtual switches, using the **Virtual Switch Enable** register **VSx Enable** bit(s) (Port 0, accessible through the Management Port, offset 358h[3:0], as appropriate), and disable the original virtual switch that contains the Management Port, in the same Write operation. If the original virtual switch must remain disabled, skip to step 7.

If the Management Port is not migrating to a different virtual switch – Enable all other needed virtual switches, using the **VSx Enable** bit(s), and skip to step 8.

6. Remove the Management Port from the original Management Port virtual switch (**VSx Port Vector** register(s)), and update the **VSx Upstream** register(s) to the Management Port's new Port Number.
7. To enable all other virtual switches that need to be enabled, write to their corresponding **Virtual Switch Enable** register *VSx Enable* bit(s).
8. Set the **Configuration Release** register *Initiate Configuration* bit, to allow all Ports to begin the linkup process.

5.3.4 Port Activity Vector

The **Egress Control and Status** register *Port Activity* bit (offset F30h[31]) is implemented for each Port, and collectively, this same bit in all Ports forms the Port Activity Vector. The *Port Activity* bit is used to indicate that for this Port, there are no pending transactions inside the virtual switch. When the bit is Set, the Port's queues are empty. This bit is polled by software, to decommission and re-assign the Port to another virtual switch.

The PEX 8724 waits 20 ms after reset to the virtual switch, then indicates that the Port's queues are empty and Sets the bit. If the Port is known to be inactive, the user can write to the Virtual Switch Table, to switch over in less than 20 ms, without looking at this bit. Reset the *Port Activity* bit after writing the Virtual Switch Table entry for this Port.

The Virtual Switch Table is described in [Section 5.3.3](#).

5.3.5 Link-Related Registers

Link-related registers are used when a Management Port is used to manage the PEX 8724. These registers provide information, and optionally generate an interrupt to the Management Port Host when the PEX 8724's Ports change the Links' status. In standard operation, the Port's **Link Status** register (offset 78h) provides the Link status information to the respective Virtual Switch Host, and Transparent Downstream Ports can optionally generate a **Data Link Layer State Changed** interrupt when Downstream Port linkup is lost or regained. When a Management Port is connected to manage the PEX 8724, it also must be informed of the status of all Ports.

Table 5-5. Link-Related Registers^a

Offset	Register	Description
3A0h	Port Reset	When driven with a value of 1, this register holds the Port in reset, including the Port PCI-to-PCI bridge and the hierarchy below it. A value of 0 indicates to un-reset the PCI-to-PCI bridge and the hierarchy below it. There is one bit, per Port.
900h	Switch Link Up	When the Port's Link state transitions from down to up, the Port's bit is Set. The Port's bit is Cleared, by using software to explicitly write 1 to the bit. The register has one bit, per Port.
904h	Switch Link Down	When the Port's Link state transitions from up to down, the Port's bit is Set. The Port's bit is Cleared, by using software to explicitly write 1 to the bit. The register has one bit, per Port.
908h	Switch Link Event Mask	If the <i>Mask</i> bit is Set, the corresponding <i>Up</i> or <i>Down</i> bit (located in register offsets 900h and 904h, respectively) transition does not generate an interrupt to the Management Port. If not masked, the bit transition generates an interrupt to the Active Management Port. The register has one <i>Mask</i> bit, per Port.
90Ch	Switch Link Status	This Read-Only register indicates Link status. The register has one <i>Status</i> bit, per Port.

a. All registers listed in this table are located in Port 0, accessible through the Management Port.

5.3.6 Reconfiguration of Virtual Switches

Virtual switches and their Ports can be re-configured for a variety of reasons, including failures or insertion or removal of cards on the Downstream or Upstream Ports.

5.3.6.1 Graceful De-Allocation of Downstream Port

To gracefully de-allocate a Downstream Port from a virtual switch, higher-level software ensures that new Requests are not initiated by the Port and its associated I/O endpoint. Higher-level software reads the Port activity register, and checks the Port's **Egress Control and Status** register *Port Activity* bit (offset F30h[31]), to determine whether the Port's queues are empty. All pending Requests are serviced. After the Port is in an Idle state, with no pending Requests (*Port Activity* bit is Cleared), the Port can be assigned to another virtual switch, by programming the Virtual Switch Table. (Refer to Section 5.3.3.)

The operation sequence is as follows:

1. Management Port software writes in the **Bridge Control** register (offset 3Ch) or Hot Reset to the source virtual switch. (This can also be done by asserting a VSx_PERST# input or Setting the corresponding bit.)
2. Management Port software re-programs the Virtual Switch Table, to add this Downstream Port to the new virtual switch.
3. Hot Plug Controller on the Upstream Port of the new virtual switch generates a **Presence Detect Changed** interrupt, if not masked. The Port's **Power Management Hot Plug User Configuration** register *Upstream Hot Plug Enable* bit (Upstream Port(s), offset F70h[14]) must be Set.
4. BIOS running on a Root Complex attached to the Upstream Port of the new virtual switch enumerates the devices. Whether the enumeration is partial or full, is dependent upon the software.
5. The driver is invoked on a Root Complex, and starts communicating with the endpoint connected to this Downstream Port.

Definitions:

- **BIOS** – Basic Input Output Software
- **Driver software** – Software running on the Host
- **Management Port software** – Software running on the Management Port
- **New virtual switch** – Location to which the Downstream Port is being re-allocated/assigned
- **Old virtual switch** – Location from which the Downstream Port is being de-allocated

5.3.6.2 Surprise Removal of Downstream Device

When an endpoint connected to a virtual switch Downstream Port is removed without any indication to software, the Link goes down and an interrupt is generated to the virtual switch's Host, as well as to the Management Port Host. If a Hot Plug Controller is implemented on the Downstream Port where the Surprise Down event occurs, an Uncorrectable Error Message is sent to the Upstream Host.

The operation sequence is as follows:

1. Management Port is connected to a Management Port.
2. Link Down interrupt is sent to the Management Port software running on the Management Port, and a **Data Link Layer State Changed** interrupt is sent to the Virtual Switch Host.
3. Host software Clears the Port's **Slot Status** register **Data Link Layer State Changed** bit (Downstream Ports, offset 80h[24]) interrupt, and determines that the Link is Down (Port's **Link Status** register **Data Link Layer Link Active** bit (Downstream Ports, offset 78h[29]) is Cleared).
4. Host software marks the device as unreachable.
5. Interrupt Service Routine (ISR) of Management Port software reads the Device-Specific **Switch Link Up**, **Switch Link Down**, and **Switch Link Status** registers (Port 0, accessible through the Management Port, offsets 900h, 904h, and 90Ch, respectively) (Management Port's PCI-to-PCI bridge's *MSI Enable* bit is used to generate an INTx interrupt or Message Signaled Interrupt (MSI) to the Management Port).
6. ISR of Management Port software determines that the Link is Down, then Clears the interrupt.
7. ISR of Management Port software again reads the Device-Specific **Switch Link Up**, **Switch Link Down**, and **Switch Link Status** registers, and determines that the interrupt has been Cleared and the status is Link Down.
8. ISR leaves the routine, marking the Downstream Port Number as "down".
9. Management Port software reports the Downstream Port as "not in use" to the software's upper layer.
10. Based upon policy, this Downstream Port can be assigned to the Management Port, or remain with the virtual switch to which it belongs.

If there is no Management Port, the virtual switches set up by the serial EEPROM and Upstream Ports manage their own virtual hierarchies.

5.3.6.3 Graceful De-Allocation of Upstream Port

When an Upstream Port must be de-commissioned for scheduled maintenance, the following actions are taken:

1. Downstream Port's traffic is stopped, by disabling all endpoints it owns.
2. Software waits until the Upstream Port's **Egress Control and Status** register **Port Activity** bit (offset **F30h**[31]) indicates a value of 1.
3. Management Port software notifies the administrator that the virtual switch Upstream Port is idle.
4. Administrator can remove the Upstream Port device.
5. Hot Reset is sent (as a result of item 4) to all Downstream Ports that belong to this Root Port's hierarchy.
6. Based upon policy, Downstream Ports can be assigned to another Root Port's hierarchy, or remain as is for future use by the same Upstream Port.

5.3.6.4 Surprise Removal of Upstream Port

An Upstream Port can be removed from the Root Port's hierarchy, without notifying the Management Port software. Although this is not advisable, software and hardware ensure that the removal does not bring down shared resources, and that the system gracefully recovers from this event.

The following actions are taken after the Upstream Port is detected as having been removed:

1. Link goes down for this Upstream Port.
2. Management Port software is interrupted.
3. ISR reads the Device-Specific **Switch Link Up**, **Switch Link Down**, and **Switch Link Status** registers (Port 0, accessible through the Management Port, offsets **900h**, **904h**, and **90Ch**, respectively) (Management Port's PCI-to-PCI bridge's **MSI Enable** bit is used to generate an INTx interrupt or MSI to the Management Port).
4. ISR determines that the Port's **Switch Link Down** register **Port x Link Down** Interrupt bit (Port 0, accessible through the Management Port, offset **904h**[10:8, 3:0]) is Set.
5. DL_Down event on the Upstream Port generates a Hot Reset within its virtual hierarchy.
6. All Downstream Ports that belong to this virtual hierarchy are reset, and propagate a Hot Reset Downstream.
7. Based upon policy, Management Port software programs the Virtual Switch Table (refer to [Section 5.3.3](#)), then adds the Downstream Ports to its own virtual switch.

5.3.6.5 Management-Capable Port Switch Over

Typically, a system design implements a heartbeat mechanism between the Active and Redundant Management Ports. When the Redundant Management Port Host detects that the Active Management Port must be switched over, the Redundant Management Port Host writes to the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset **354h**), removes the existing Active Management Port, and promotes itself to be the Active Management Port Host.

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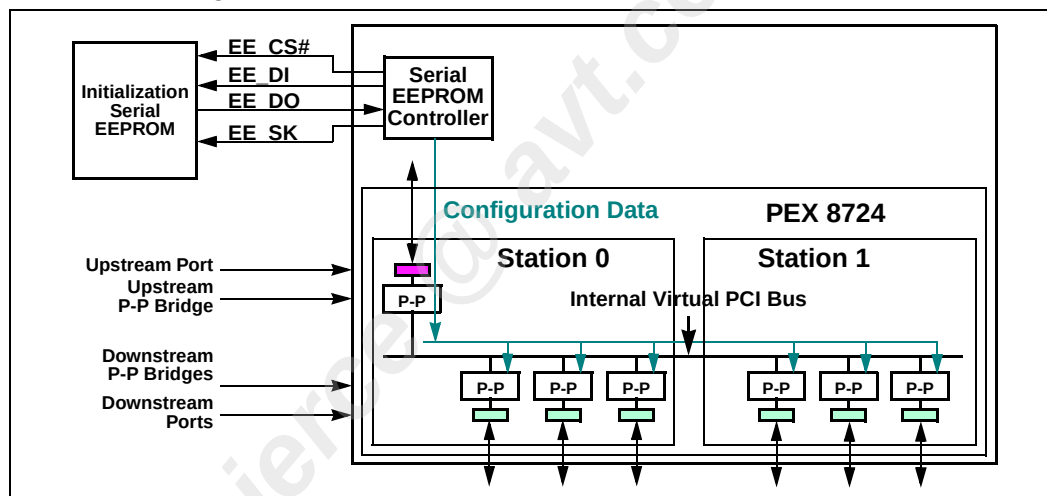
6.1 Overview

The PEX 8724 provides a Serial EEPROM Controller and interface to Serial Peripheral Interface (SPI)-compatible serial EEPROMs. (Refer to [Figure 6-1](#), which represents the PEX 8724 in Base mode.) This interface consists of a Chip Select, Clock, Serial Data In, and Serial Data Out signals, and operates at a programmable frequency of up to 17.9 MHz. The PEX 8724 supports serial EEPROMs that use 1-, 2-, or 3-byte addressing; the PEX 8724 automatically determines the appropriate addressing mode.

The controller provides access to non-volatile memory. This external memory can be used for three different purposes:

- The serial EEPROM can be used to store register data, for switch configuration and initialization. When a serial EEPROM device is connected to the PEX 8724, immediately after reset, the Serial EEPROM Controller reads data from the serial EEPROM that is used to update the PEX 8724 register default values.
- System or application data can be stored into, and read from, the serial EEPROM, by software, I²C and/or SMBus, initiating random-access Read or Write Requests to the serial EEPROM.
- In NT mode, the serial EEPROM can provide up to 32 KB of Expansion ROM, for the NT Port Link (default) or Virtual Interface. When software reads the NT Port Expansion ROM (starting at the NT Port Expansion ROM Base Address), the PEX 8724 reads from the serial EEPROM, to return the requested ROM image.

Figure 6-1. Serial EEPROM Connections – Base Mode



Note: [Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

6.2 Features

- Detection of whether a serial EEPROM is present/not present
- Supports high-speed serial EEPROMs with Serial Peripheral Interface (SPI) interface
- Non-volatile storage for register default values loaded during Power-On Reset
- Cyclic Redundancy Check (CRC) compared on each download, to ensure data integrity prior to loading registers
- 4-byte Write/Read access to the serial EEPROM, through the Upstream Port (Base mode), the Management Port (Virtual Switch mode), and/or I²C/SMBus
- Serial EEPROM data format allows for loading registers by Station/Port/Address location
- Required serial EEPROM size is dependent upon the quantity of registers being changed
- Automatic support for 1-, 2-, or 3-byte-addressable serial EEPROMs
- Manual override for quantity of serial EEPROM Address bytes
- Programmable serial EEPROM clock frequency
- Programmable serial EEPROM clock-to-chip select timings
- Supports one NT Port Expansion ROM (**not supported** for 1-byte address serial EEPROMs), for the NT Port Link (default) or Virtual Interface

6.3 Serial EEPROM Load

6.3.1 Serial EEPROM Load – Base Mode

The Serial EEPROM Controller performs a serial EEPROM download when the following conditions are met:

- Serial EEPROM is present^a, and
- Validation signature (first byte read from the serial EEPROM) value is 5Ah, and
- One of the following events occurs:
 - PEX_PERST# is returned High, following a Fundamental Reset (such as a Cold or Warm Reset to the entire PEX 8724)
 - Hot Reset is received at the Upstream Port (downloading upon this event can be optionally disabled, by Setting the **Virtual Switch Debug** register *Disable Serial EEPROM Load on Hot Reset* and/or *Upstream Hot Reset Control* bit (Upstream Port, offset A30h[3 and/or 2], respectively))
 - Upstream Port exits a DL_Down state (downloading upon this event can be optionally disabled, by Setting the **Virtual Switch Debug** register *Upstream Port and NT-Link Port DL_Down Reset Propagation Disable* and/or *Upstream Hot Reset Control* bit (Upstream Port, offset A30h[4 and/or 2], respectively))

Serial EEPROM reload upon receiving a Soft Reset can also be disabled, by Setting the *Upstream Hot Reset Control* bit. Setting this bit effectively converts Hot Reset severity to the lower severity of a Secondary Bus Reset. The *Upstream Hot Reset Control* bit should be Set if NT mode is enabled.

*a. The serial EEPROM is considered to be present if it returns a non-zero value in response to an initial Read Status command of its Status register. This value is copied to the **Status Data from Serial EEPROM** register bits (Port 0, offset 260h[31:24]). Serial EEPROM presence is reported in the register's *EepPrsnt[1:0]* field (field [17:16]); a value of 01b or 11b indicates that the serial EEPROM is present.*

6.3.2 Serial EEPROM Load – Virtual Switch Mode

Serial EEPROM functionality in Virtual Switch mode is identical to that in Base mode, with the following exceptions:

- Only the Management Port, I²C, and/or SMBus can access the serial EEPROM. A PCI Express Master can access the serial EEPROM, only if that virtual switch Upstream Port is designated as the Management Port (as reflected in the **Management Port Control** register *Active Management Port* field (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[4:0])).
- When the serial EEPROM reloads registers following a Reset, it reloads only the Ports that are associated with that virtual switch. Determination of which Ports are associated to a particular virtual switch is dependent upon the **VSx Port Vector** register (Port 0, accessible through the Management Port, offset(s) 380h through 38Ch) value(s), initially programmed to the configuration defined by the enabled virtual switches, as listed in Table 12-1, “Virtual Switch Default Port Assignments – Virtual Switch Mode,”
- By default, Chip- and Station-specific registers are not reloaded from serial EEPROM following a Virtual Switch Fundamental Reset (VSx_PERST# inputs are de-asserted), a Port’s **Port Reset** register *Reset Port x Vector* bit (Port 0, accessible through the Management Port, offset 3A0h[10:8, 3:0]) being toggled from 1 to 0, and/or a Soft Reset (Hot Reset or Upstream Port *DL_Down* state).

A Chip- and Station-specific register reload from serial EEPROM can be enabled by the Management Port, I²C, and/or SMBus Setting the **Management Port Control** register *Active Management Port EEPROM Load on Hot Reset for Chip and Station Registers Enable* bit (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[6]).

The Serial EEPROM Controller performs a serial EEPROM download when the following conditions are met:

- Serial EEPROM is present^a, and
- Validation signature (first byte read from the serial EEPROM) value is 5Ah, and
- One of the following events occurs:
 - PEX_PERST# input is returned High, to exit a Fundamental Reset (such as a Cold or Warm Reset to the entire chip)
 - VSx_PERST# inputs are returned High, to exit a Virtual Switch Fundamental Reset (such as a Cold or Warm Reset to a single virtual switch)
 - VSx_PERST# and NT_PERST# Control and Status register PEX_NT_PERST# Control bit (Port 0, accessible through the Management Port, offset 3A8h[24]) corresponding to one or more enabled virtual switches that are Set and then Cleared, to release the virtual switches from Fundamental Reset
 - VSx_PERST# and NT_PERST# Control and Status register VSx_PERST# Control bit(s) (Port 0, accessible through the Management Port, offset 3A8h[11:8]) corresponding to one or more enabled virtual switches that are Set and then Cleared, to release the virtual switches from Fundamental Reset
 - Hot Reset is received at a virtual switch Upstream Port (downloading upon this event can be optionally disabled, by Setting the Virtual Switch Debug register Disable Serial EEPROM Load on Hot Reset bit (VS Upstream Port(s), offset A30h[3]))
 - Virtual switch Upstream Port exits a DL_Down state (downloading upon this event can be optionally disabled, by Setting the Virtual Switch Debug register Upstream Port and NT-Link Port DL_Down Reset Propagation Disable bit (VS Upstream Port(s), offset A30h[4]))

Serial EEPROM reload upon receiving a Soft Reset can also be disabled, by Setting the Virtual Switch Debug register Upstream Hot Reset Control bit (VS Upstream Port(s), offset A30h[2]). Setting this bit effectively converts Hot Reset severity to the lower severity of a Secondary Bus Reset.

a. The serial EEPROM is considered to be present if it returns a non-zero value in response to an initial Read Status command of its Status register. This value is copied to the Status Data from Serial EEPROM register bits (Port 0, accessible through the Management Port, offset 260h[31:24]). Serial EEPROM presence is reported in the register's EepPrsnt[1:0] field (field [17:16]); a value of 01b or 11b indicates that the serial EEPROM is present.

6.4 Serial EEPROM Data Format

The data in the serial EEPROM is stored in the format defined in [Table 6-1](#). The [Validation Signature](#) byte is located in the first address. The Serial EEPROM Controller reads this byte to determine whether a valid serial EEPROM image exists versus a blank image. REG_BYTE_COUNT[15:0] contains the quantity of bytes of serial EEPROM data to be loaded. It is equal to the quantity of registers to be loaded times 6 (6 serial EEPROM bytes, per register). If the REG_BYTE_COUNT[15:0] value is not a multiple of 6, the last incomplete register entry is ignored.

For the remaining register-related locations, data is written into a 2-byte address that represents the Configuration register offset and Port Number, and the 4 bytes following are the data loaded for that Configuration register. Only Configuration register data specifically programmed into the serial EEPROM is loaded after the PEX 8724 exits reset.

[Table 6-2](#) defines the Configuration register Address format (REGADDR[15:0] from [Table 6-1](#)):

- Bits [9:0] represent bits [11:2] of the Register address
- Bits [15:10] represent the Port Number of the register selected to be programmed by serial EEPROM

Because the PEX 8724 Serial EEPROM Controller always accesses 4 bytes of serial EEPROM data (for DWord-aligned Register addresses), register offsets are stored in the serial EEPROM as DWord address values.

To determine the 2-byte serial EEPROM value that represents the PEX 8724 Port and register offset, shift the register offset 2 bits to the right (divide by 4), then OR the resulting value with the appropriate Port Identifier value from [Table 6-2](#).

For example, to load Port 1 register offset 7Ch, shift the address to the right by 2 bits (this becomes 1Fh), then concatenate 0000_01b. The resulting DWord address in the serial EEPROM is 0000_0100_0001_1111b (041Fh).

Table 6-1. Serial EEPROM Data Format

Location	Value	Description
0h	5Ah	Validation Signature
1h	–	Bit 7 = CRC checking enable ^a
2h	REG BYTE COUNT (LSB)	Configuration register Byte Count (LSB)
3h	REG BYTE COUNT (MSB)	Configuration register Byte Count (MSB)
4h	REGADDR (LSB)	1 st Configuration Register Address (LSB)
5h	REGADDR (MSB)	1 st Configuration Register Address (MSB)
6h	REGDATA (Byte 0)	1 st Configuration Register Data (Byte 0)
7h	REGDATA (Byte 1)	1 st Configuration Register Data (Byte 1)
8h	REGDATA (Byte 2)	1 st Configuration Register Data (Byte 2)
9h	REGDATA (Byte 3)	1 st Configuration Register Data (Byte 3)
Ah	REGADDR (LSB)	2 nd Configuration Register Address (LSB)
Bh	REGADDR (MSB)	2 nd Configuration Register Address (MSB)
Ch	REGDATA (Byte 0)	2 nd Configuration Register Data (Byte 0)
Dh	REGDATA (Byte 1)	2 nd Configuration Register Data (Byte 1)
Eh	REGDATA (Byte 2)	2 nd Configuration Register Data (Byte 2)
Fh	REGDATA (Byte 3)	2 nd Configuration Register Data (Byte 3)
...	...	...
End of Serial EEPROM data (BYTE COUNT +3)	REGDATA (Byte 3)	Last Configuration Register Data (Byte 3)
BYTE COUNT +4	Stored CRC Byte 0	Optional, used when CRC is enabled ^a
BYTE COUNT +5	Stored CRC Byte 1	
BYTE COUNT +6	Stored CRC Byte 2	
BYTE COUNT +7	Stored CRC Byte 3	

- a. There is an optional CRC checking at the end of serial EEPROM auto load. Bit 7 of the second byte (after the 5Ah Validation Signature) determines whether CRC checking is to be performed. The CRC value is calculated for serial EEPROM addresses REG BYTE COUNT (LSB) (that is, 2h) through BYTE COUNT +3. If the “REG BYTE COUNT (LSB)” value is not a multiple of 4, two additional bytes (of value 0h) are included in the calculation, so that CRC polynomial is always applied to DWord data.

Serial EEPROM CRC errors do **not** inhibit serial EEPROM auto load of registers.

The calculated CRC value is always placed into the **Serial EEPROM CRC Value** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 270h) after serial EEPROM auto load. If CRC checking is enabled, and the stored 32-bit CRC value read from the end of the serial EEPROM content does **not** match the calculated value:

1. The **Serial EEPROM Status** register **EepCRCErr** bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h[19]) is Set, and,
2. An Uncorrectable Internal error is triggered at Port 0 of Station 0 (**Uncorrectable Error Status** register **Uncorrectable Internal Error Status** bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset FB8h[22]) is Set). The severity for Internal errors is Fatal, by default (**Uncorrectable Error Severity** register **Uncorrectable Internal Error Severity** bit (Port 0, offset FC0h[22]) is Set).

Table 6-2. Configuration Register Address Format

Port Number ^{a, b}	REGADDR Bits [15:10] Value ^{c, d}	Port Identifier
Port 0	0000_00b	0000h
Port 1	0000_01b	0400h
Port 2	0000_10b	0800h
Port 3	0000_11b	0C00h
Port 8	0010_00b	2000h
Port 9	0010_01b	2400h
Port 10	0010_10b	2800h
Virtual Switch Mode Station-Specific Registers		
Station 0, Port 0	1100_00b	C000h
Station 1, Port 8	1100_01b	C400h
NT Mode		
NT Port Link Interface	1110_00b	E000h
NT Port Virtual Interface	1110_01b	E400h
NT PCI-to-PCI ^e	0XXX_XXb	XX00h

- a. Refer to [Table 4-1](#) for the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The table also indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.
- b. Station Port 8 is **Software Only**.
- c. Encodings not listed are **Reserved**.
- d. REGADDR bits [15:13] indicate the Station Number. REGADDR bits [12:10] indicate the relative Port Number (0, 1, 2, or 3) within the Station.
- e. Use the values for the Station Number and Port Number for the Port that is configured as the NT Port in the **VS0 Upstream** register **NT Port** field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 360h[12:8]).

6.4.1 Serial EEPROM Cyclic Redundancy Check

A Cyclic Redundancy Check (CRC) value is always calculated for serial EEPROM data stored in serial EEPROM addresses 2h through BYTE_COUNT +3. If the value of “REG BYTE COUNT plus 2” is not a multiple of 4, the final data is still put into a CRC calculation, with unread byte(s) of the final DWord equal to 00h. The NT Port Expansion ROM code (if present in the serial EEPROM content) is not included in the CRC calculation. The constant value used for CRC calculation is DB71_0641h. The calculated CRC value is always placed into the **Serial EEPROM CRC Value** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 270h), after serial EEPROM auto load.

There is an optional CRC checking at the end of serial EEPROM auto load. If bit 7 of the second byte (address 01h, following the **Validation Signature**) is Set, CRC checking is enabled. When CRC checking is performed, the calculated CRC value is compared to a stored CRC value read from serial EEPROM addresses [BYTE_COUNT+4] through [BYTE_COUNT+7]. If the calculated CRC value does not match the stored CRC value, the **Serial EEPROM Status** register *EepCRCErr* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h[19]) is Set, and an Uncorrectable Internal Error is triggered in Port 0 of Station 0 (**Uncorrectable Error Status** register *Uncorrectable Internal Error Status* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset FB8h[22])). The serial EEPROM loading of registers is not aborted.

6.4.1.1 Programming the Serial EEPROM CRC Value

To add CRC to existing serial EEPROM code, PLX software can be used to automatically calculate the value and program that value to the correct location in the serial EEPROM.

Alternatively, the PEX 8724 can calculate the CRC value. This procedure requires a programmed serial EEPROM that contains an odd quantity of register entries. A serial EEPROM that has an even quantity of entries can be converted to an odd quantity, *such as* by duplicating an existing entry.

The procedure to add CRC for an odd quantity of entries is as follows:

1. Boot up the PEX 8724 with a programmed serial EEPROM that has an odd quantity of register entries.
2. Read the **Serial EEPROM CRC Value** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 270h) to obtain the 32-bit calculated CRC value, and append that value to the existing serial EEPROM code (in serial EEPROM offsets BYTE_COUNT+4 through BYTE_COUNT+7).
3. Enable CRC checking, by changing the first two bytes in the serial EEPROM from 5A_00 to 5A_80.
4. Re-boot and verify that the **Serial EEPROM Status** register *EepCRCErr* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h[19]) is not Set.

Note: PLX software does not require an odd number of EEPROM entries, to program the CRC value.

6.5 Serial EEPROM Initialization

After the device Reset is de-asserted, the PEX 8724 determines whether a serial EEPROM is present. The serial EEPROM is considered to be present if it returns a value other than FFh in response to an initial Read Status command of its **Status** register, in which case this value is copied to the **Status Data from Serial EEPROM** register bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h[31:24]). Serial EEPROM presence is reported in the **Serial EEPROM Status** register *EepPrsnt[1:0]* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h[17:16]); a value of 01b or 11b indicates that the serial EEPROM is present.

If a serial EEPROM is detected, the first byte (**Validation Signature**) is read. If a value of 5Ah is read, it is assumed that the serial EEPROM is programmed for the PEX 8724. The serial EEPROM address width is determined while the first byte is read. If the first byte's value is not 5Ah, the serial EEPROM is blank or programmed with invalid data. In this case, no more data is read from the serial EEPROM, and the **Serial EEPROM Status** register *EepAddrWidth* field (offset 260h[23:22]) reports a value of 00b (undetermined width).

If the *EepAddrWidth* field reports a value of 00b, any subsequent accesses to the serial EEPROM (through the PEX 8724 Serial EEPROM registers) default to a serial EEPROM address width of 1 byte, unless the **Serial EEPROM Status** register *EepAddrWidth Override* bit (offset 260h[21]) is Set. The *EepAddrWidth* field is usually Read-Only (RO); however, it is writable if the *EepAddrWidth Override* bit is Set (both can be programmed by a single Write instruction).

If the serial EEPROM contains valid data, the REG_BYTE_COUNT values in Bytes 2 and 3 determine the quantity of serial EEPROM locations that contain Configuration register addresses and data. Each Configuration register entry consists of 2 bytes of register Address and 4 bytes of register Write data. The REG_BYTE_COUNT must be a multiple of 6.

The **EE_SK** output clock frequency is determined by the **Serial EEPROM Clock Frequency** register *EepFreq[2:0]* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[2:0]). The default clock frequency is 1 MHz. At this clock rate, it takes approximately 48 μ s per DWORD during Configuration register initialization. For faster loading of large serial EEPROMs that support a faster clock, the first Configuration register load from the serial EEPROM could be to the **Serial EEPROM Clock Frequency** register.

There is an optional CRC checking at the end of serial EEPROM auto load. Bit 7 of the second byte (after the 5Ah **Validation Signature**) determines whether CRC checking is to be performed. The CRC value is calculated for serial EEPROM addresses 2h through BYTE COUNT +3. The calculated CRC value is always placed into the **Serial EEPROM CRC Value** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 270h) after serial EEPROM auto load. If CRC checking is enabled, and the stored 32-bit CRC value read from the end the serial EEPROM content does not match the calculated value, an Uncorrectable Internal error is triggered at Port 0 of Station 0.

6.6 Serial EEPROM Registers

The Serial EEPROM register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 260h through 270h) parameters defined in Section 13.16.5, “Device-Specific Registers – Serial EEPROM (Offsets 260h – 270h),” can be changed, using the serial EEPROM. It is recommended that the first entry in the serial EEPROM be used to change the **Serial EEPROM Clock Frequency** register (offset 268h) value, to increase the clock frequency, and thereby reduce the time needed for the remainder of the serial EEPROM load. When the NT Port Expansion ROM feature is used, the serial EEPROM clock frequency must be 5 MHz or higher. At the last serial EEPROM entry, the **Serial EEPROM Status and Control** register (offset 260h) can be programmed to issue a Write Status Register (WRSR) command, to enable the Write Protection feature(s) within the serial EEPROM data, if needed.

6.7 Serial EEPROM Random Write/Read Access

To access the serial EEPROM, a PCI Express, I²C, or SMBus Master uses the following registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port):

- **Serial EEPROM Status and Control** (offset 260h)
- **Serial EEPROM Buffer** (offset 264h)
- **Serial EEPROM 3rd Address Byte** (offset 26Ch)

Note: To help streamline the text in the following subsections, the specific Port location/access of each register offset is not repeated – only the offset location is mentioned.

The Master can only access the serial EEPROM on a DWord basis (4 bytes aligned to one DWord address).

6.7.1 Writing to Serial EEPROM

To write a DWord to the serial EEPROM:

1. If the 3rd Address byte (Address bits [23:16]) is needed (when the **Serial EEPROM Status** register *EepAddrWidth* field bits (offset 260h[23:22]) are both Set), write the value to the **Serial EEPROM 3rd Address Byte** register *Serial EEPROM 3rd Address Byte* field (offset 26Ch[7:0]).
2. Write the 32-bit data into the **Serial EEPROM Buffer** register (offset 264h).
3. Issue a Write Enable instruction to the serial EEPROM (Command = 110b, Set Write Enable Latch), by writing the value 0000_C000h into the **Serial EEPROM Status and Control** register (offset 260h).
4. Calculate and write the combined Address and Command value to write into the **Serial EEPROM Control** register (offset 260h), by combining the serial EEPROM 3-bit Write Data instruction (value 010b) as the *EepCmd[2:0]* field [15:13], together with the serial EEPROM address. Serial EEPROM Address bits [14:2] must be programmed into the register's *EepBlkAddr* field [12:0], and serial EEPROM Address bit 15 must be programmed into the **Serial EEPROM Status** register *EepBlkAddr Upper Bit* bit (*that is*, Set offset 260h[20] if the serial EEPROM address is in the upper 32 KB of any 64-KB address block within the serial EEPROM). The data in the **Serial EEPROM Buffer** register is written to the serial EEPROM when the **Serial EEPROM Status and Control** register is written.
5. The serial EEPROM Write operation is complete when a subsequent read of the **Serial EEPROM Status** register *EepCmdStatus* bit (offset 260h[18]) returns a value of 0. At this time, another serial EEPROM access can be started.

Because each PEX 8724 Port and Register address value (REGADDR; refer to [Section 6.5](#)), and its corresponding Data value (REGDATA), require 6 bytes of serial EEPROM memory, and the PEX 8724 serial EEPROM interface accesses 4 bytes at a time, two serial EEPROM Writes may be needed to store each set of REGADDR (one word) and REGDATA (1 DWord) entries into the serial EEPROM. To avoid overwriting a word of another set of 6-byte REGADDR and REGDATA values, one of the two Serial EEPROM Writes might need to be a Read-Modify-Write type of operation (preserving one word read from the serial EEPROM, and writing the value back along with a new word value).

6.7.2 Reading from Serial EEPROM

To read a DWord from the serial EEPROM:

1. If the 3rd Address byte (Address bits [23:16]) is needed (when the **Serial EEPROM Status** register *EepAddrWidth* field bits (offset 260h[23:22]) are both Set), write the value to the **Serial EEPROM 3rd Address Byte** register *Serial EEPROM 3rd Address Byte* field (offset 26Ch[7:0]).
2. Calculate the combined Address and Command value to write into the **Serial EEPROM Control** register (offset 260h), by combining the serial EEPROM 3-bit Read Data instruction (value 011b) as bits [15:13], together with the serial EEPROM address. Serial EEPROM Address bits [14:2] must be programmed into **Serial EEPROM Control** register bits [12:0], and serial EEPROM Address bit 15 must be programmed into **Serial EEPROM Status** register bit 20 (*that is*, Set bit 20 if the serial EEPROM address is in the upper 32 KB of any 64-KB address block within the serial EEPROM).
3. Poll the **Serial EEPROM Status** register until the *EepCmdStatus* bit (offset 260h[18]) is Cleared, which signals that the transaction is complete.
4. Read the four bytes of serial EEPROM data from the **Serial EEPROM Buffer** register (offset 264h).

For example, to read the first DWord in the serial EEPROM, write the value 0000_6000h to Port 0, register offset 260h, and then read Port 0, register offset 264h.

6.7.3 Programming a Blank Serial EEPROM

The PEX 8724 supports 1-, 2-, or 3-byte serial EEPROM addressing. 8-Kbit to 512-Kbit SPI EEPROMs use 2-byte addressing. The PEX 8724 requires that the first byte in the serial EEPROM must be the value 5Ah (ASCII Z), as a **Validation Signature**.

The 2nd and 3rd bytes contain the quantity of bytes within the serial EEPROM image, beginning with the first register entry at serial EEPROM address 04h. If this Byte Count value exceeds the actual quantity of register entries times 6 (*for example*, if the first DWord is programmed to the value 5A00_FFFFh), the system could hang. To simplify programming of a blank EEPROM (*such as* in a typical production build), the serial EEPROM could be pre-programmed with the first DWord, 0000_005Ah.

A 2-byte address serial EEPROM that is blank (or corrupted) can be programmed according to the following procedure (when the PEX 8724 is in 1-Byte Address mode).

To program a blank serial EEPROM:

1. Write the value 0000_005Ah into the **Serial EEPROM Buffer** register at address [Upstream Port BAR0 + 264h].
2. Issue a Write Enable instruction (Command = 110b, Set Write Enable Latch, and enable 2-byte addressing, by writing the value 00A0_C000h into the **Serial EEPROM Status and Control** register (offset 260h).
3. Copy this data value to serial EEPROM location 0, by writing the value 00A0_4000h into the **Serial EEPROM Status and Control** register. At this point, the first four bytes in the serial EEPROM now contain the value 0000_005Ah.
4. Reboot the system, to reset the PEX 8724 so that it re-detects the serial EEPROM.

6.8 Serial EEPROM Loading of NT Port Link Interface Registers – NT Mode

The **Virtual Switch Debug** register *Load Only EEPROM NT-Link on Hot Reset* and *Inhibit EEPROM NT-Link Load on Hot Reset* bits (Upstream Port(s), offset A30h[27:26], respectively) control whether the serial EEPROM is to load registers following a Soft Reset (Hot Reset or DL_Down) to the Upstream Port or NT Port Link Interface, as defined in Table 6-3.

Note: In Virtual Switch mode, with NT mode enabled, the default value of the *Load Only EEPROM NT-Link on Hot Reset* bit changes to 1 in a virtual switch that implements the NT Port; the default value of the bit is 0 in a virtual switch that does not implement the NT Port.

Table 6-3. Serial EEPROM Loading of NT Port Link Interface Registers (Upstream Port(s), Offset A30h[27:26] Values) – NT Mode

Bit 27 Value	Bit 26 Value	Action
0	0	Load all registers from the serial EEPROM.
0	1	Load all registers, except the NT Port Link Interface registers, from the serial EEPROM.
1	0	Load only NT Port Link Interface registers from the serial EEPROM.
1	1	Disable serial EEPROM loading of all registers.

6.9 NT Port Expansion ROM – NT Mode

The PEX 8724 NT Port Virtual and Link Interfaces support Expansion ROM, as defined in the *PCI r3.0*. The NT Port Expansion ROM can be implemented for either interface, but not both simultaneously. The Expansion ROM image is stored in the serial EEPROM, and its size can be either 16 KB (default, bit is Cleared) or 32 KB (maximum), based upon the **Serial EEPROM Clock Frequency** register *Expansion ROM Size* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[16]) value. When the NT Port Expansion ROM feature is used, the serial EEPROM clock frequency must be 5 MHz or higher.

By default, the NT Port Expansion ROM is enabled on the NT Port Link Interface; however, it can be enabled instead for the NT Port Virtual Interface, by Setting the **Ingress Chip Control** register *NT Port Expansion ROM Virtual Side* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[0]). The **NT Port Expansion ROM Base Address** register (BAR) must be enabled, by Setting the register's *Expansion ROM Enable* bit, in either the NT Port Virtual Interface (offset 30h[0]) or NT Port Link Interface (offset 30h[0]).

The starting address (Base address) of the NT Port Expansion ROM within the serial EEPROM is programmed in the **Serial EEPROM 3rd Address Byte** register *Expansion ROM Base Address* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 26Ch[31:16]), which is mapped to serial EEPROM Byte Address bits [23:8].

The default serial EEPROM Base Address value is as follows:

- **16-KB NT Port Expansion ROM (*Expansion ROM Size* bit is Cleared)** – The value is 0020h, which corresponds to serial EEPROM Byte address 2000h (8 KB). The serial EEPROM size must be at least 32 KB.
- **32-KB NT Port Expansion ROM (*Expansion ROM Size* bit is Set)** – The value is 0040h, which corresponds to serial EEPROM Byte address 4000h (16 KB). The serial EEPROM size must be at least 64 KB.

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Chapter 7 I²C/SMBus Slave Interface Operation

7.1 Introduction

This chapter discusses the [I2C Slave Interface](#) and [SMBus Slave Interface](#).

7.2 I²C Slave Interface

7.2.1 I²C Support Overview

Note: This section applies to the I²C Slave interface, which uses the [I2C_ADDR0](#), [I2C_SCL0](#), and [I2C_SDA0](#) signals for PEX 8724 register access by an I²C Master. The [I2C_SCL1](#) and [I2C_SDA1](#) signals form the PEX 8724 I²C Master interface, which is used only for Serial Hot Plug operation. (Refer to [Section 10.11](#), “Serial Hot Plug Controller.”)

Inter-Integrated Circuit (I²C) is a bus used to connect Integrated Circuits (ICs). Multiple ICs can be connected to an I²C Bus, and I²C devices that have I²C mastering capability can initiate a Data transfer. I²C is used for Data transfers between ICs at relatively low rates (100 Kbps), and is used in a variety of applications. For further details regarding I²C Buses, refer to the [I2C Bus, v2.1](#).

The PEX 8724 is an I²C Slave. Slave operations allow the PEX 8724 Configuration registers to be read from or written to by an I²C Master, external from the device. I²C is a sideband mechanism that allows the device Configuration registers to be programmed, read from, or written to, independent of the PCI Express Upstream Link.

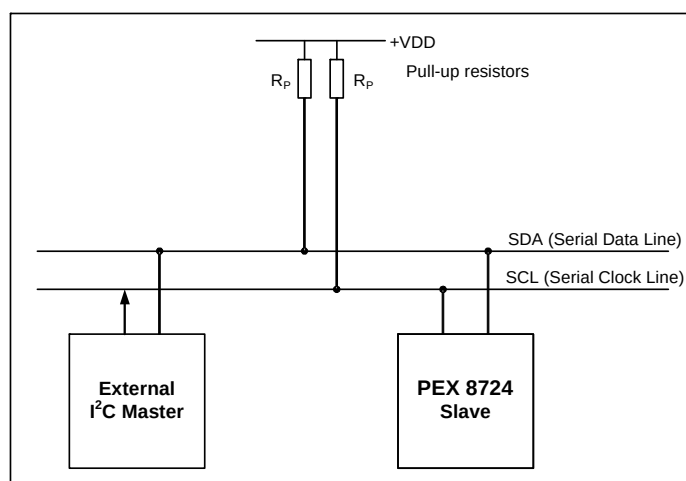
With I²C, users have the option of accessing all PEX 8724 registers through the I²C Slave interface. I²C provides an alternative to using a serial EEPROM, except the I²C/SMBus cannot access the registers (in time) prior to initial Link training (unless linkup is delayed by strapping the [STRAP_I2C_SMBUS_CFG_EN#](#) input Low, or by Clearing the [Configuration Release](#) register [Initiate Configuration](#) bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset [3ACh\[0\]](#)) by serial EEPROM); therefore, the serial EEPROM is recommended for complete programmability options. I²C can also be used for debugging, such as if the PEX 8724 Upstream Port(s) fail(s) to linkup.

Accordingly, it is recommended that both I²C/SMBus access, and the serial EEPROM (or at least its footprint), be included in designs.

The I2C_SCL0 and I2C_SDA0 signals can be brought out to a 2x2 pin header on the board, to allow PLX software (for example, running on a laptop computer) to access the PEX 8724 registers, using an Aardvark USB-I²C adapter connected to this header. (Refer to the [PEX 8724 RDK Hardware Reference Manual](#) for the header pin design.)

Figure 7-1 provides a block diagram that illustrates how standard devices connect to the I²C Bus.

Figure 7-1. Standard Devices to I²C Bus Connection Block Diagram



7.2.2 I²C Addressing – Slave Mode Access

To access the PEX 8724 Configuration registers through the I²C Slave interface, the PEX 8724 I²C Slave address must be configured.

The PEX 8724 supports a 7-bit I²C Slave address, defined by the `I2C_ADDR0` 3-state input, which can be strapped Low (0), left floating (Z), or High (1), to select a different Slave address – `0111_000b`, `0111_001b`, or `0111_010b`, respectively.

Up to three PEX 8724 devices can share the same I²C Bus segment without conflict, provided that each PEX 8724 has its `I2C_ADDR0` 3-state input strapped to a unique state. More than three PEX 8724 devices can share the I²C Bus, however, if the upper Address bits are programmed in the serial EEPROM.

The Slave address can also be programmed by the serial EEPROM (if present; recommended, if the default address must be changed), or by a Memory Write, in the **I2C Configuration** register *Slave Address* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset `294h[6:0]`). The bits of field [6:0] correspond to Address Byte bits [7:1], with bit 0 of the byte indicating a Write (0) or Read (1).

Note: *The I²C Slave address must not be changed by an I²C Write command.*

7.2.3 I²C Command Format

An I²C transfer starts as a packet with Address Phase bytes, followed by four Command Phase bytes, and one or more Data Phase bytes. The I²C packet Address Phase Byte format is illustrated in [Figure 7-2a](#). The I²C Command portion must include 4 bytes of data that contain the following:

- I²C Transfer type (Read/Write)
- PCI Express Configuration Register address
- PEX 8724 Port Number being accessed
- Byte Enable(s) of the register data being accessed

When the I²C Master is writing to the PEX 8724, the I²C Master must transmit the Data bytes to be written to that register within the same packet that contains the Command bytes. [Table 7-2](#) describes each I²C Command byte for Write access. [Figure 7-2b](#) illustrates the Command phase portion of an I²C Write packet.

When the I²C Master is reading from the PEX 8724, the I²C Master must separately transmit a Command Phase packet and Data Phase packet. [Table 7-6](#) describes each I²C Command byte for Read access. [Figure 7-4b](#) illustrates the Command phase portion of an I²C Read packet.

Each I²C packet must contain 4 bytes of data. Pad unused packet Data bytes with zeros (0) to meet this requirement.

7.2.4 I²C Register Write Access

The PEX 8724 Configuration registers can be read from and written to, based upon I²C register Read and Write operations, respectively. An I²C Write packet consists of Address Phase bytes and Command Phase bytes, followed by one to four additional I²C Data bytes. [Table 7-1](#) defines mapping of the I²C Data bytes to the Configuration register Data bytes. [Figure 7-2c](#) illustrates the I²C Data byte format.

The I²C packet starts with the *S* (START condition) bit. Data bytes are separated by the *A* (Acknowledge Control Packet (ACK)) or *N* (Negative Acknowledge (NAK)) bit. The packet ends with the *P* (STOP condition) bit.

If the Master generates an invalid command, the targeted PEX 8724 register is not modified.

The PEX 8724 considers the 1st Data byte of the 4-byte Data phase, following the four Command bytes in the Command phase, as register Byte 3 (bits [31:24]). The next three Data bytes access register Bytes 2 through 0, respectively. Four Data bytes are required, regardless of the Byte Enable Settings in the Command phase. The Master can then generate either a STOP condition (to finish the transfer) or a repeated START condition (to start a new transfer). If the I²C Master sends more than the four Data bytes (violating PEX 8724 protocol), the PEX 8724 returns a NAK for the extra Data byte(s). (For further details regarding I²C protocol, refer to the [I2C Bus, v2.1](#).)

[Table 7-2](#) describes each I²C Command byte for Write access. In the packet described in [Figure 7-2](#), Command Bytes 0 through 3 for Writes follow the format specified in [Table 7-2](#).

Table 7-1. I²C Register Write Access

I ² C Data Byte Order	PCI Express Configuration Register Bytes
0	Written to register Byte 3
1	Written to register Byte 2
2	Written to register Byte 1
3	Written to register Byte 0

Table 7-2. I²C Command Format for Write Access

Field (Byte) On Bus	Bit(s)	Value/Description
Command Byte 1	7:3	Reserved Should be Cleared.
	2:0	Command 011b = Write register Do not use other encodings for Writes.
Command Byte 2	7:6	Reserved Should be Cleared.
	5:4	Access Mode 00b = Transparent Ports 01b = NT Port Link Interface 10b = NT Port Virtual Interface 11b = Reserved
	3:2	Station Select (when <i>Access Mode</i> = 00b) 00b = Station 0 01b = Station 1 10b, 11b = Reserved
	1:0	Port Select, Bits [2:1]

Table 7-2. I²C Command Format for Write Access (Cont.)

Field (Byte) On Bus	Bit(s)	Value/Description										
Command Byte 3	7	Port Select, Bit 0 When <i>Access Mode</i> = 00b, <i>Port Select</i> [2:0] selects the Port to access, as indicated by <i>Station Select (when Access Mode = 00b)</i> . 000b = Port 0 of the Station 001b = Port 1 of the Station 010b = Port 2 of the Station 011b = Port 3 of the Station (Station 0 Only) Notes: If the Port Number indicated is not enabled within the Station, the associated encoding is Reserved . <i>Port 8 is Software Only.</i> When <i>Access Mode</i> = 01b or 10b, <i>Port Select</i> [2:0] selects the NT Port. 000b = NT Port 001b = Reserved All other encodings are Reserved .										
	6	Reserved Should be Cleared.										
	5:2	Byte Enables <table><thead><tr><th>Bit</th><th>Description</th></tr></thead><tbody><tr><td>2</td><td>Byte Enable for Byte 0 (PEX 8724 register bits [7:0])</td></tr><tr><td>3</td><td>Byte Enable for Byte 1 (PEX 8724 register bits [15:8])</td></tr><tr><td>4</td><td>Byte Enable for Byte 2 (PEX 8724 register bits [23:16])</td></tr><tr><td>5</td><td>Byte Enable for Byte 3 (PEX 8724 register bits [31:24])</td></tr></tbody></table> 0 = Corresponding PEX 8724 register byte will not be modified 1 = Corresponding PEX 8724 register byte will be modified All 16 combinations are valid values.	Bit	Description	2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])	3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])	4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])	5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])
	Bit	Description										
2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])											
3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])											
4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])											
5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])											
1:0	PEX 8724 Register Address, Bits [11:10]											
Command Byte 4	7:0	PEX 8724 Register Address [9:2] Note: All register addresses are DWord-aligned. Therefore, Address bits [1:0] are implicitly Cleared, and then internally incremented for successive I ² C byte Writes.										

Figure 7-2. I²C Write Packet**Figure 7-2a I²C Write Packet Address Phase Bytes**

1 st Cycle			
START	7 6 5 4 3 2 1	0	ACK/NAK
S	Slave Address[7:1]	Read/Write Bit 0 = Write	A

Figure 7-2b I²C Write Packet Command Phase Bytes

Command Cycle							
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK
Command Byte 0	A	Command Byte 1	A	Command Byte 2	A	Command Byte 3	A

Figure 7-2c I²C Write Packet Data Phase Bytes

Write Cycle								
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Data Byte 0 (to selected register Byte 3)	A	Data Byte 1 (to selected register Byte 2)	A	Data Byte 2 (to selected register Byte 1)	A	Data Byte 3 (to selected register Byte 0)	A	P

7.2.4.1 I²C Register Write Example

The following tables illustrate a sample I²C packet for writing the PEX 8724 **MSI Upper Address** register (offset **50h**) for Port 1 (Port 1 of Station 0) with data 1234_5678h.

Note: The PEX 8724 has a default I²C Slave address value of **0111_000b**, **0111_001b**, or **0111_010b**, according to whether the **I2C_ADDR0** 3-state input is strapped Low (0), left floating (Z), or High (1), respectively. The actual address is reflected in the **I2C Configuration** register **Slave Address** field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset **294h**[6:0]).

The following example assumes the I²C Slave address is 0111_000b (I2C_ADDR0=0). The byte sequence on the I²C Bus, as listed in the following tables and figures, occurs after the START and before the STOP bits, by which the I²C Master frames the transfer.

Table 7-3. I²C Register Write Access Example – 1st Cycle

Phase	Value	Value/Description
Address	70h	Bits [7:1] for PEX 8724 I²C Slave Address (0111_000b) Last bit (bit 0) for Write = 0.

Table 7-4. I²C Register Write Access Example – Command Cycle

Byte	Value	Value/Description
0	03h	[7:3] Reserved Should be Cleared. [2:0] Command 011b = Write register
1	00h	[7:6] Reserved Should be Cleared. [5:4] Access Mode [3:2] Station Select [1:0] Port Select, Bits [2:1]
2	BCh	7 Port Select, Bit 0 6 Reserved Should be Cleared. [5:2] Byte Enables All active. [1:0] PEX 8724 Register Address, Bits [11:10]
3	14h	[7:0] PEX 8724 Register Address [9:2]

Table 7-5. I²C Register Write Access Example – Write Cycle

Byte	Value	Description
0	12h	Data to Write for Byte 3
1	34h	Data to Write for Byte 2
2	56h	Data to Write for Byte 1
3	78h	Data to Write for Byte 0

Figure 7-3. I²C Write Command Packet Example
Figure 7-3a I²C Write Packet Address Phase Bytes

1 st Cycle			
START	7 6 5 4 3 2 1	0	ACK/NAK
S	Slave Address[7:1]	Read/Write Bit 0 0 = Write	A

Figure 7-3b I²C Write Packet Command Phase Bytes

Command Cycle							
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK
Command Byte 0 0000_0011b	A	Command Byte 1 0000_0000b	A	Command Byte 2 1011_1100b	A	Command Byte 3 0001_0100b	A

Figure 7-3c I²C Write Packet Data Phase Bytes

Write Cycle								
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Data Byte 0 0001_0010b	A	Data Byte 1 0011_0100b	A	Data Byte 2 0101_0110b	A	Data Byte 3 0111_1000b	A	P

7.2.5 I²C Register Read Access

When the I²C Master attempts to read a PEX 8724 register, two packets are transmitted. The 1st packet consists of Address and Command Phase bytes to the Slave. The 2nd packet consists of Address and Data Phase bytes.

According to the [I2C Bus, v2.1](#), a Read cycle is triggered when the Read/Write bit (bit 0) of the 1st cycle is Set. The Command phase reads the requested register content into the internal buffer. When the I²C Read access occurs, the internal buffer value is transferred on to the I²C Bus, starting from Byte 3 (bits [31:24]), followed by the subsequent bytes, with Byte 0 (bits [7:0]) being transferred last. If the I²C Master requests more than four bytes, the PEX 8724 re-transmits the same byte sequence, starting from Byte 3 of the internal buffer.

The 1st and 2nd I²C Read packets (illustrated in [Figure 7-4](#) and [Figure 7-5](#), respectively) perform the following functions:

- **1st packet** – Selects the register to read
- **2nd packet** – Reads the register (sample 2nd packet provided is for a 7-bit PEX 8724 I²C Slave address)

Although two packets are shown for the I²C Read, the I²C Master can merge the two packets together into a single packet, by not generating the STOP at the end of the first packet (Master does not relinquish the bus) and generating REPEAT START.

[Table 7-6](#) describes each I²C Command byte for Read access. In the packet described in [Figure 7-4](#), Command Bytes 0 through 3 for Reads follow the format specified in [Table 7-6](#).

Table 7-6. I²C Command Format for Read Access

Field (Byte) On Bus	Bit(s)	Value/Description
Command Byte 1	7:3	Reserved Should be Cleared.
	2:0	Command 100b = Read register Do not use other encodings for Reads.
Command Byte 2	7:6	Reserved Should be Cleared.
	5:4	Access Mode 00b = Transparent Ports 01b = NT Port Link Interface 10b = NT Port Virtual Interface 11b = Reserved
	3:2	Station Select (when Access Mode = 00b) 00b = Station 0 01b = Station 1 10b, 11b = Reserved
	1:0	Port Select, Bits [2:1]

Table 7-6. I²C Command Format for Read Access (Cont.)

Field (Byte) On Bus	Bit(s)	Value/Description										
Command Byte 3	7	Port Select, Bit 0 When <i>Access Mode</i> = 00b, <i>Port Select</i> [2:0] selects the Port to access, as indicated by <i>Station Select (when Access Mode = 00b)</i> . 000b = Port 0 of the Station 001b = Port 1 of the Station 010b = Port 2 of the Station 011b = Port 3 of the Station (Station 0 Only) Notes: If the Port Number indicated is not enabled within the Station, the associated encoding is Reserved . <i>Port 8 is Software Only.</i> When <i>Access Mode</i> = 01b or 10b, <i>Port Select</i> [2:0] selects the NT Port. 000b = NT Port 001b = Reserved All other encodings are Reserved .										
	6	Reserved Should be Cleared.										
	5:2	Byte Enables <table><thead><tr><th>Bit</th><th>Description</th></tr></thead><tbody><tr><td>2</td><td>Byte Enable for Byte 0 (PEX 8724 register bits [7:0])</td></tr><tr><td>3</td><td>Byte Enable for Byte 1 (PEX 8724 register bits [15:8])</td></tr><tr><td>4</td><td>Byte Enable for Byte 2 (PEX 8724 register bits [23:16])</td></tr><tr><td>5</td><td>Byte Enable for Byte 3 (PEX 8724 register bits [31:24])</td></tr></tbody></table> 0 = Corresponding PEX 8724 register byte will not be modified 1 = Corresponding PEX 8724 register byte will be modified All 16 combinations are valid values.	Bit	Description	2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])	3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])	4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])	5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])
	Bit	Description										
2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])											
3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])											
4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])											
5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])											
1:0	PEX 8724 Register Address, Bits [11:10]											
Command Byte 4	7:0	PEX 8724 Register Address [9:2] Note: All register addresses are DWord-aligned. Therefore, Address bits [1:0] are implicitly Cleared, for the I ² C Read.										

Figure 7-4. I²C Read Command Packet (1st Packet)**Figure 7-4a I²C Read Command Packet Address Phase Bytes**

1 st Cycle			
START	7 6 5 4 3 2 1	0	ACK/NAK
S	Slave Address[7:1]	Read/Write Bit 0 = Write	A

Figure 7-4b I²C Read Command Packet Command Phase Bytes

Command Cycle								
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Command Byte 0	A	Command Byte 1	A	Command Byte 2	A	Command Byte 3	A	P

Figure 7-5. I²C Read Data Packet (2nd Packet)**Figure 7-5a I²C Read Data Packet Address Phase Bytes**

1 st Cycle			
START	7 6 5 4 3 2 1	0	ACK/NAK
S	Slave Address[7:1]	Read/Write Bit, 1 = Read	A

Figure 7-5b I²C Read Data Packet Data Phase Bytes

Read Cycle								
7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Register Byte 3	A	Register Byte 2	A	Register Byte 1	A	Register Byte 0	A	P

7.2.5.1 I²C Register Read Address Example – Phase and Command Packet

The following is a sample I²C packet for reading the PEX 8724 **MSI Upper Address** [63:32] register (offset **50h**) in Port 1 (Port 1 of Station 0), assuming the register value is ABCD_EF01h.

Note: The PEX 8724 has a default I²C Slave address value of **0111_000b**, **0111_001b**, or **0111_010b**, according to whether the **I2C_ADDR0** 3-state input is strapped Low (0), left floating (Z), or High (1), respectively. The actual address is reflected in the **I2C Configuration** register **Slave Address** field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset **294h**[6:0]).

The following example assumes the I²C Slave address is 0111_000b (I2C_ADDR0=0). The byte sequence on the I²C Bus, as listed in the following tables, occurs after the START and before the STOP bits, by which the I²C Master frames the transfer.

Table 7-7. I²C Register Read Access Example – 1st Packet

Phase	Value	Value/Description
Address	70h	Bits [7:1] for PEX 8724 I²C Slave Address (0111_000b) Last bit (bit 0) for Write = 0.

Table 7-8. I²C Register Read Access Example – Command Cycle

Byte	Value	Value/Description
0	04h	[7:3] Reserved Should be Cleared. [2:0] Command 100b = Read register
1	00h	[7:6] Reserved Should be Cleared. [5:4] Access Mode [3:2] Station Select [1:0] Port Select, Bits [2:1]
2	BCh	7 Port Select, Bit 0 6 Reserved Should be Cleared. [5:2] Byte Enables All active. [1:0] PEX 8724 Register Address, Bits [11:10]
3	14h	[7:0] PEX 8724 Register Address [9:2]

7.3 SMBus Slave Interface

7.3.1 SMBus Features

- Compliant to the *SMBus v2.0*
- Supports the SMBus Slave function only
- PEX 8724 internal registers can be read and written, through the SMBus Slave interface
- Supports Address Resolution Protocol (ARP-capable)
- [I2C_ADDR0](#) input state, serial EEPROM, software, and/or ARP define the SMBus Device address
- Supports Block Read, Block Write, and Block Write - Block Read Process Call commands to access the registers
- Supports Packet Error Checking
- 10 to 100 KHz Bus operation frequency range

7.3.2 SMBus Operation

Based upon I²C's principles of operation, SMBus is a two-wire bus used for communication between IC components and the remainder of the system. Electrically, I²C and SMBus devices are compatible, and both protocol devices can co-exist on the same bus. Multiple devices, both Masters and Slaves, can be connected to an SMBus segment. PCI Express cards have two optional SMBus pins defined on the connector – SMCLK and SMDAT.

The PEX 8724 implements an *SMBus v2.0*-compliant Slave device, and is used to read and write PEX 8724 registers, through SMBus commands. The PEX 8724 SMBus uses the same SDA Data and SCL Clock lines that are used for I²C. Additionally, the SMBus Device Address is the same value as the I²C Slave address, used by the I²C protocol (default addresses [0111_000b](#), [0111_001b](#), or [0111_010b](#), according to whether the [I2C_ADDR0](#) 3-state input is strapped Low (0), left floating (Z), or High (1), respectively). At any time, either the I²C or SMBus feature is enabled, dependent upon the **SMBus Configuration** register *SMBus Enable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset [2C8h\[0\]](#)) state, which is latched (at Fundamental Reset) according to the [STRAP_I2C_SMBUS_EN](#) 3-state input, to one of the following protocols:

- 0 = I²C (*SMBus Enable* bit is Cleared)
- Z, 1 = SMBus with ARP (*SMBus Enable* bit is Set)

Software can toggle the *SMBus Enable* bit to change between I²C and SMBus functionality.

The PEX 8724 SMBus logic also supports the commands that are required to support ARP. ARP is a feature specific to *SMBus v2.0*, through which an SMBus ARP Master can dynamically assign a unique address to each of the SMBus Targets residing on the same bus. Although ARP is an optional feature of the *SMBus v2.0*, PCI and PCI Express cards are required to support ARP. The ARP feature is enabled when the **SMBus Configuration** register *ARP Disable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset [2C8h\[8\]](#)) is Cleared; this bit is initially latched (at Fundamental Reset) according to the [STRAP_I2C_SMBUS_EN](#) input state, as described above.

The PEX 8724 also supports Packet Error Checking and Packet Error Code (PEC) generation, as explained in the *SMBus v2.0*. The *SMBus v2.0* optional feature, *Notify ARP Master* (which requires Master capability on the SMBus) is **not** supported.

7.3.3 Supported SMBus Commands

For register access, the SMBus logic supports three commands:

- Block Write (command BEh) is used to write the registers
- Block Write (command BAh), followed by Block Read (command BDh), can be used to read the registers
- Block Write - Block Read Process Call (commands BAh, CDh) can also be used to read registers

Unsupported SMBus commands – Quick Command, Send Byte, Receive Byte, Write Byte, Write Word, Read Byte, Read Word, and Process Call – are NACKed.

7.3.3.1 SMBus Block Write

The Block Write command is used to write to the PEX 8724 registers. General SMBus Block Writes are illustrated in [Figure 7-6](#) and [Figure 7-7](#). The sequence of Bytes include the following, in the sequence listed:

- 7-bit address,
- Command Code that indicates it is Block Write,
- *Byte Count* field with a value of 8h that indicates 4 bytes to set up the register to write (Port Number, register address, Command Byte Enable, and so forth), followed by
- 4 bytes of data to be written into the register

[Figure 7-8](#) explains the elements used in [Figure 7-6](#) and [Figure 7-7](#), and [Figure 7-9](#) indicates the Data Bytes written.

Figure 7-6. SMBus Block Write Command Format, to Write to a PEX 8724 Register without PEC

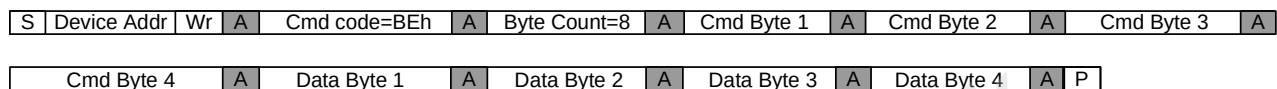


Figure 7-7. SMBus Block Write Command Format, to Write to a PEX 8724 Register with PEC

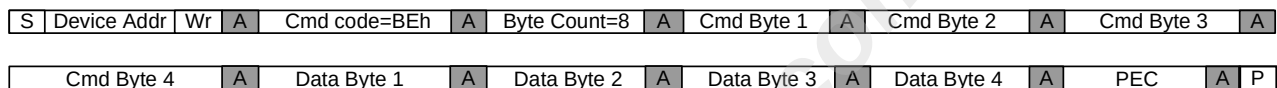


Figure 7-8. SMBus Packet Protocol Diagram Element Key

- S -> START condition
P -> STOP condition
A -> Acknowledge (this bit position may be 0 for an ACK or 1 for a NACK)
□ -> Master to Slave
■ -> Slave to Master

Figure 7-9. SMBus Block Write Bytes, as Written to Register

31:24	23:16	15:8	7:0
Data Byte 1	Data Byte 2	Data Byte 3	Data Byte 4

Note: Within each byte, the most significant bit (msb) is transmitted first.

Table 7-9 provides a description of bytes for an SMBus Block Configuration Space register (CSR) Write.

Block Write transactions that are received with incorrect byte Settings are NACKed, starting from the wrong byte Setting, and including subsequent bytes in the packet. *For example*, if the Byte Count value is not 8, the PEX 8724 NACKs the byte corresponding to the Byte Count value, as well as any Data bytes following within the same packet.

The byte after Data Byte 4, if present, is taken as the PEC byte, and if present, the PEC is checked. If a packet fails Packet Error Checking, the PEX 8724 drops the packet (ignores the Write), and returns NACK for the PEC byte, to the SMBus Master. Packet Error Checking can be disabled, by Setting the **SMBus Configuration** register *PEC Check Disable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[9]). The Byte Count value, by definition, does not include the PEC byte.

Table 7-9. Bytes for Block CSR Write on SMBus

Field (Byte) On Bus	Bit(s)	Value/Description
Command Code	7:0	BEh for Block Write.
Byte Count	7:0	08h = 8 bytes to follow (4 Command and 4 Data bytes). The PEC byte is not counted.
Command Byte 1	7:3	Reserved Should be Cleared.
	2:0	Command 011b = Write register Do not use other encodings for Writes.
Command Byte 2	7:6	Reserved Should be Cleared.
	5:4	Access Mode 00b = Transparent Ports 01b = NT Port Link Interface 10b = NT Port Virtual Interface 11b = Reserved
	3:2	Station Select (when Access Mode = 00b) 00b = Station 0 01b = Station 1 10b, 11b = Reserved
	1:0	Port Select, Bits [2:1]

Table 7-9. Bytes for Block CSR Write on SMBus (Cont.)

Field (Byte) On Bus	Bit(s)	Value/Description										
Command Byte 3	7	Port Select, Bit 0 When <i>Access Mode</i> = 00b, <i>Port Select</i> [2:0] selects the Port to access, as indicated by <i>Station Select (when Access Mode = 00b)</i> . 000b = Port 0 of the Station 001b = Port 1 of the Station 010b = Port 2 of the Station 011b = Port 3 of the Station (Station 0 Only) <i>Notes: If the Port Number indicated is not enabled within the Station, the associated encoding is Reserved.</i> <i>Port 8 is Software Only.</i> When <i>Access Mode</i> = 01b or 10b, <i>Port Select</i> [2:0] selects the NT Port. 000b = NT Port 001b = Reserved All other encodings are Reserved .										
	6	Reserved Should be Cleared.										
	5:2	Byte Enables <table><tr><th>Bit</th><th>Description</th></tr><tr><td>2</td><td>Byte Enable for Byte 0 (PEX 8724 register bits [7:0])</td></tr><tr><td>3</td><td>Byte Enable for Byte 1 (PEX 8724 register bits [15:8])</td></tr><tr><td>4</td><td>Byte Enable for Byte 2 (PEX 8724 register bits [23:16])</td></tr><tr><td>5</td><td>Byte Enable for Byte 3 (PEX 8724 register bits [31:24])</td></tr></table> 0 = Corresponding PEX 8724 register byte will not be modified 1 = Corresponding PEX 8724 register byte will be modified All 16 combinations are valid values.	Bit	Description	2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])	3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])	4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])	5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])
	Bit	Description										
2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])											
3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])											
4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])											
5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])											
1:0	PEX 8724 Register Address, Bits [11:10]											
Command Byte 4	7:0	PEX 8724 Register Address [9:2] <i>Note: All register addresses are DWord-aligned. Therefore, Address bits [1:0] are implicitly Cleared, and then internally incremented for successive SMBus byte Writes.</i>										

7.3.3.2 SMBus Block Read

A Block Read command is used to read PEX 8724 registers. Similar to register Reads using I²C, an SMBus Write sequence must first be performed to select the register to read, followed by an SMBus Read of the corresponding register. There are two ways a PEX 8724 register can be read:

- Use a Block Write, followed by a Block Read. The Block Write sets up the parameters including Port Number, register address and Byte Enables, and the Block Read performs the actual Read operation.
- Use a Block Write - Block Read Process Call. This command is defined by the *SMBus v2.0*, and performs a Block Write and Block Read, using a two-part message. The Block Write portion of the message sets up the register to be read, and then a repeated START followed by the Block Read portion of the message returns the register data specified by the Block Write.

Note: There is no STOP condition before the repeated START condition.

CSR Read, Using SMBus Block Write, followed by SMBus Block Read

A general SMBus Block Write and Block Read sequence is illustrated in Figure 7-10.

Table 7-10 describes the Byte definitions for a Block Write bus protocol, to prepare for a subsequent Block Read of the PEX 8724 register.

The PEX 8724 always NACKs any incorrect command sequences, starting with the wrong Byte. Upon receiving the Block Read command (listed in Table 7-11), the PEX 8724 returns a PEC to the Master if, after the 4th byte of register data, the Master still requests one more Byte. As a Slave, the PEX 8724 recognizes the end of the Master's Read cycle, by observing the Master's NACK response for the last Data Byte transmitted by the PEX 8724.

Figure 7-10. SMBus Block Write to Set up Read, and Resulting Read that Returns CSR Value

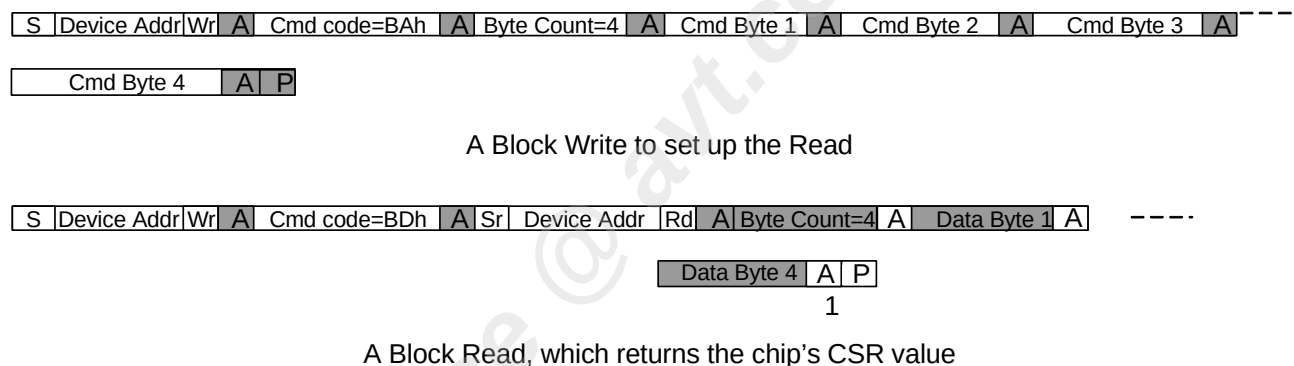


Table 7-10. SMBus Block Read Bytes

Field (Byte) On Bus	Bit(s)	Value/Description
Command Code	7:0	BAh, to set up the Read, using Block Writes.
Byte Count	7:0	04h = 4 Command bytes.
Command Byte 1	7:3	Reserved Should be Cleared.
	2:0	Command 100b = Read register Do not use other encodings for Reads.
Command Byte 2	7:6	Reserved Should be Cleared.
	5:4	Access Mode 00b = Transparent Ports 01b = NT Port Link Interface 10b = NT Port Virtual Interface 11b = Reserved
	3:2	Station Select (when <i>Access Mode</i> = 00b) 00b = Station 0 01b = Station 1 10b, 11b = Reserved
	1:0	Port Select, Bits [2:1]

Table 7-10. SMBus Block Read Bytes (Cont.)

Field (Byte) On Bus	Bit(s)	Value/Description										
Command Byte 3	7	Port Select, Bit 0 When <i>Access Mode</i> = 00b, <i>Port Select</i> [2:0] selects the Port to access, as indicated by <i>Station Select (when Access Mode = 00b)</i> . 000b = Port 0 of the Station 001b = Port 1 of the Station 010b = Port 2 of the Station 011b = Port 3 of the Station (Station 0 Only) Notes: If the Port Number indicated is not enabled within the Station, the associated encoding is Reserved . <i>Port 8 is Software Only.</i> When <i>Access Mode</i> = 01b or 10b, <i>Port Select</i> [2:0] selects the NT Port. 000b = NT Port 001b = Reserved All other encodings are Reserved .										
	6	Reserved Should be Cleared.										
	5:2	Byte Enables <table><tr><th>Bit</th><th>Description</th></tr><tr><td>2</td><td>Byte Enable for Byte 0 (PEX 8724 register bits [7:0])</td></tr><tr><td>3</td><td>Byte Enable for Byte 1 (PEX 8724 register bits [15:8])</td></tr><tr><td>4</td><td>Byte Enable for Byte 2 (PEX 8724 register bits [23:16])</td></tr><tr><td>5</td><td>Byte Enable for Byte 3 (PEX 8724 register bits [31:24])</td></tr></table> 0 = Corresponding PEX 8724 register byte will not be modified 1 = Corresponding PEX 8724 register byte will be modified All 16 combinations are valid values.	Bit	Description	2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])	3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])	4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])	5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])
	Bit	Description										
2	Byte Enable for Byte 0 (PEX 8724 register bits [7:0])											
3	Byte Enable for Byte 1 (PEX 8724 register bits [15:8])											
4	Byte Enable for Byte 2 (PEX 8724 register bits [23:16])											
5	Byte Enable for Byte 3 (PEX 8724 register bits [31:24])											
1:0	PEX 8724 Register Address, Bits [11:10]											
Command Byte 4	7:0	PEX 8724 Register Address [9:2] Note: All register addresses are DWord-aligned. Therefore, Address bits [1:0] are implicitly Cleared, for the SMBus Read.										

Table 7-11. Command Format for SMBus Block Read

Field (Byte) On Bus	Bit(s)	Value/Description
Cmd Code	7:0	CDh, for Block Read (Process Call Rea D).

7.3.3.3 CSR Read, Using SMBus Block Write - Block Read Process Call

A general SMBus Block Write - Block Read Process Call sequence is illustrated in [Figure 7-11](#). Alternatively, a general SMBus Block Write - Block Read Process Call with PEC sequence is illustrated in [Figure 7-12](#).

Using this command, the register to be read can be set up and read back with one SMBus cycle (a transaction with a START and ending in STOP). There is no STOP condition before the repeated START condition.

[Table 7-11](#) lists the Command format for Block Read.

Figure 7-11. CSR Read Operation Using SMBus Block Write - Block Read Process Call

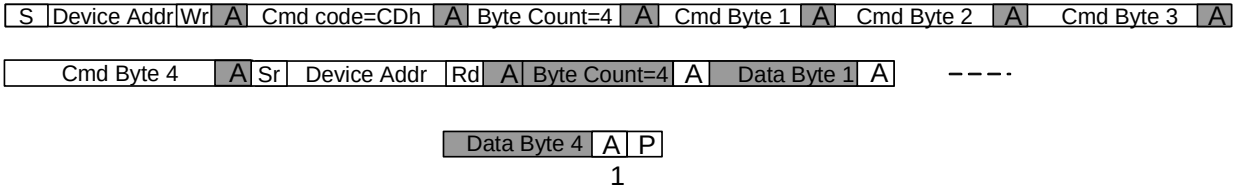
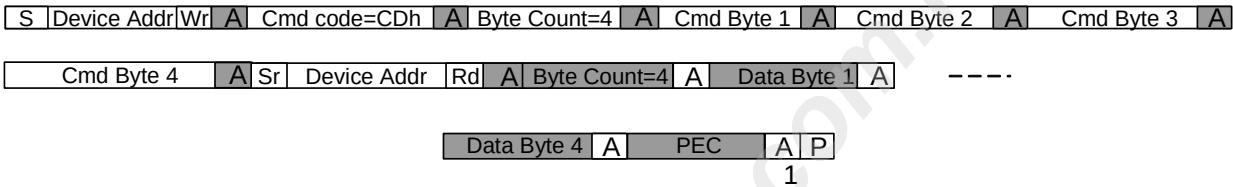


Figure 7-12. CSR Read Operation Using SMBus Block Write - Block Read Process Call with PEC



7.3.4 SMBus Device Address Resolution Protocol

Address Resolution Protocol (ARP) is a protocol by which SMBus devices that implement an assignable SMBus Device address feature are enumerated and dynamically assigned non-conflicting Device addresses, rather than using a fixed Device address. Although optional in the *SMBus v2.0*, it is mandatory per the *PCI r3.0* for add-in boards, to support ARP. This feature avoids conflicts with addresses used by other devices on a motherboard. ARP also allows multiple devices of the same type to co-exist on the same bus segment, without address conflicts.

To support this feature, a Slave device must implement a unique 128-bit ID, called *Unique Device Identifier (UDID)*. The fields of this ID are provided in [Figure 7-13](#). All ARP commands use the default Device Address, 1100_001b. There are also two flags that the SMBus devices must implement to support the ARP process:

- **Address Resolved flag (AR)** – A flag bit or device internal state that indicates whether the ARP Master has resolved the device's SMBus Device address
- **Address Valid flag (AV)** – A flag bit or device internal state that indicates whether the device's SMBus Device address is valid

The process of assigning an SMBus Device address starts with the ARP Master issuing a Reset Device or Prepare to ARP command, using the default Device address. This Clears the AR flag in the Slave device (both flags are Cleared by a Reset Device command). The Master then issues a general Get UDID command. This causes all devices that support ARP to start driving their UDID onto the serial bus. A Target that loses the SMBus arbitration, backs off. Arbitration loss means that a device keeps the SMDAT line floating and it detects 0 driven by another device on the bus. Slave devices that lose arbitration issue NACK in response to further Bytes transmitted on the bus. After the ARP Master finishes the Get UDID sequence, it issues a Set Address command to the Slave device, using the Slave's UDID. All Slave devices on the bus monitor the UDID that is transmitted by the ARP Master, but only the particular device that has the matching UDID adopts the new SMBus Device address, and Sets its own AV and AR flags. After the Slave device sets its AR flag, that device no longer responds to a general Get UDID command, which allows other devices to participate in the ARP process. All ARP commands require PEC checking and generation.

7.3.4.1 SMBus UDID

The 128-bit UDID is comprised of the following fields, as illustrated in [Figure 7-13](#) (not to scale). Each UDID field and its default value implemented in the PEX 8724 and meaning are explained in the tables that follow.

Figure 7-13. 128-Bit SMBus UDID

8 bits	8 bits	16 bits	16 bits	16 bits	16 bits	16 bits	32 bits
127:120	119:112	111:96	95:80	79:64	63:48	47:32	31:0
Device Capability	Version/Revision	Vendor ID	Device ID	Interface	Subsystem Vendor ID	Subsystem Device ID	Vendor-Specific ID

Table 7-12. SMBus Vendor-Specific ID [31:0]

Field	Name	Default Value	Description
31:0	Vendor-Specific ID	Depends upon the I2C_ADDR0 input state, which provides the following ID values: 0 = 7000_0000h Z = B000_0000h 1 = D000_0000h	The Vendor-Specific ID is used to provide a unique ID for functionally equivalent devices. This is for devices that would otherwise return identical UDIDs for the purpose of dynamic address assignment. The I2C_ADDR0 3-state input produces three unique Vendor-Specific ID values, for a maximum of three SMBus-enabled PEX 8724 switches to co-exist on the same SMBus segment.

Table 7-13. SMBus Subsystem Device ID [47:32]

Field	Name	Default Value	Description
15:0	Subsystem Device ID	8724h	PLX part number for the PEX 8724.

Table 7-14. SMBus Subsystem Vendor ID [63:48]

Field	Name	Default Value	Description
15:0	Subsystem Vendor ID	10B5h	PLX Vendor ID.

Table 7-15. SMBus Interface [79:64]

Field	Name	Default Value	Description
3:0	SMBus Version	0100b	<i>SMBus v2.0.</i>
15:4	Reserved	000h	Supported protocols.

Table 7-16. SMBus Device ID [95:80]

Field	Name	Default Value	Description
15:0	Device ID	8724h	PEX 8724 default Device ID value.

Table 7-17. SMBus Vendor ID [111:96]

Field	Name	Default Value	Description
15:0	Vendor ID	10B5h	PLX Vendor ID.

Table 7-18. SMBus Version/Revision [119:112]

Field	Name	Default Value	Description
2:0	Silicon Revision ID	001b	PEX 8724, Silicon Revisions BA and CA.
5:3	UDID Version	001b	UDID version defined for the <i>SMBus v2.0.</i>
7:6	Reserved	00b	

Table 7-19. SMBus Device Capability [127:120]

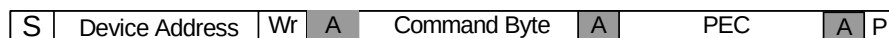
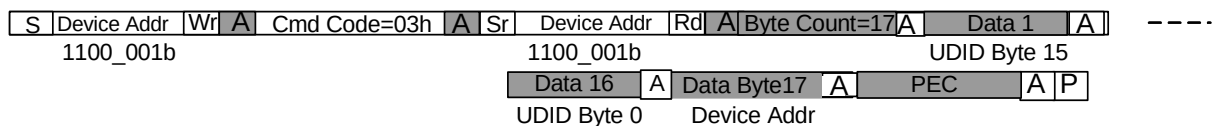
Field	Name	Default Value	Description
0	PEC Supported	1	By default, PEC generation and checking are enabled.
5:1	Reserved	00_000b	
7:6	Address Type	10b	The PEX 8724 SMBus Address Type is implemented as dynamic and volatile. 00b = Fixed address 01b = Dynamic and persistent 10b = Dynamic and volatile (default) 11b = Random number device

7.3.4.2 Supported SMBus ARP Commands

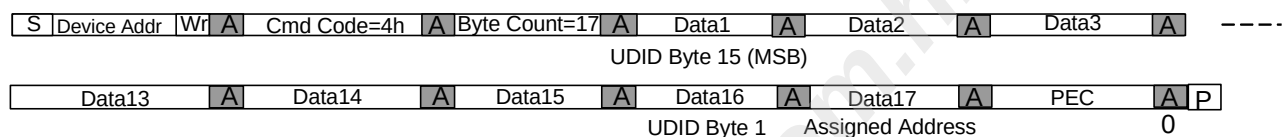
The PEX 8724 supports all ARP Slave commands. The Notify ARP Master command, which requires Master functionality, is **not** supported. Table 7-20 explains the PEX 8724 response to each received ARP command.

Table 7-20. Supported SMBus ARP Commands, Format, and Actions

ARP Command	SMBus Command Format	SMBus Device Address	Command Code	Action
Prepare to ARP (Only General)	Send Byte (Refer to Figure 7-14)	SMBus default Device Address 1100_001b	01h	Clear the <i>AR Flag</i> and prepare for the ARP process. <i>AV Flag</i> will have no change.
Reset Device (General)	Send Byte (Refer to Figure 7-14)	SMBus default Device Address 1100_001b	02h	Clear the <i>AR Flag</i> and <i>AV Flag</i> .
Reset Device (Directed)	Send Byte (Refer to Figure 7-14)	SMBus default Device Address 1100_001b	Target Device Address[7:1] + 0	If the <i>AV Flag</i> is Set, Set ACK and Clear the <i>AR Flag</i> and <i>AV Flag</i> ; else, NACK/REJECT.
Get UDID (General)	Block Read (Refer to Figure 7-15)	SMBus default Device Address 1100_001b	03h	Respond only if the <i>AR Flag</i> is Cleared; else, NACK/REJECT. <i>AR Flag</i> and <i>AV Flag</i> are not changed. Address returned is all ones (1), if the <i>AV Flag</i> is Cleared.
Get UDID (Directed)	Block Read	SMBus default Device Address 1100_001b	Target Device Address[7:1] + 1	<i>AR Flag</i> and <i>AV Flag</i> are not changed. ACK if <i>AV Flag</i> =1; else, NACK/REJECT. Data Byte 17 returned will be the SMBus Device address.
Assign Address ARP	Block Write (Refer to Figure 7-16)	SMBus default Device Address 1100_001b	04h	Always ACK and Set the <i>AR Flag</i> and <i>AV Flag</i> , if the UDID matches.

Figure 7-14. Prepare SMBus ARP Command and SMBus Reset Device Command Format**Figure 7-15. Get SMBus UDID Command Format (General Get UDID Command with PEC)**

Note: If the **SMBus Configuration** register **AR Flag** bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[11]) is Cleared, the device returns the register's **SMBus Device Address** field (offset 2C8h[7:1]) as 1111_111b; otherwise, it returns the device SMBus Device address. Bit 0 (lsb) in the Data Byte 17 field should be 1.

Figure 7-16. Assign SMBus Device Address ARP Command Format

Note: Bit 0 (lsb) of the Data 17 field is ignored in the Assign Address command field.

7.3.5 SMBus PEC Handling

The PEX 8724 supports the optional SMBus v2.0 PEC generation and checking feature. This feature is required for the ARP process; however, it is optional for standard data transfer operation. The PEX 8724 supports PEC Cyclic Redundancy Check (CRC) generation and checking during ARP, as well as during Read/Write transfers to the PEX 8724 registers. The CRC polynomial used for PEC calculation is:

$$C(x) = x^8 + x^2 + x + 1$$

An 8-bit parallel CRC is implemented. The PEC calculation does not include ACK, NACK, START, STOP, nor repeated START bits. An SMBus Master can determine whether a Slave device supports PEC, from the UDID value returned by the Slave device, in response to a Get UDID command.

As a Slave device, the PEX 8724 checks the PEC, if the Master transmits the additional PEC byte and the PEX 8724 PEC checking feature is enabled (default). PEC checking can be disabled, by Setting the **SMBus Configuration** register *PEC Check Disable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[9]).

Additionally, when PEC is enabled, packets received with an incorrect PEC value are dropped. If PEC checking is disabled and a received PEC byte value is incorrect, the PEX 8724 accepts the packet. During a register Read, if the Master requests the additional PEC byte, the PEX 8724 generates and transmits the PEC byte after the register data.

7.3.6 Addressing PEX 8724 SMBus Slave

By default, the PEX 8724 supports ARP when **STRAP_I2C_SMBUS_EN**=1, and expects the ARP Master to define the PEX 8724 SMBus Device address. If ARP is disabled (**STRAP_I2C_SMBUS_EN**=0, and the default **SMBus Configuration** register *ARP Disable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[8]) is Set), the default **SMBus Configuration** register *SMBus Device Address* bits [7:1] are 0111_000b, 0111_001b, or 0111_010b, according to whether the **I2C_ADDR0** 3-state input is strapped Low (0), left floating (Z), or High (1), respectively (Address bits [7:1] are 0111_000b, with Address bit [1:0] values loaded from **I2C_ADDR0**). The input state allows a maximum of three PEX 8724 SMBus Slaves to co-exist without address conflict on the SMBus, using SMBus Device Address byte values of 70h, 72h, and 74h. The **I2C_ADDR0** 3-state input is loaded immediately after Fundamental Reset, and any subsequent change of input state does not affect functionality.

If the register's *UDID Address Type* field [13:12] is programmed as fixed address (both bits are Cleared) without disabling ARP, the PEX 8724 still participates in ARP, but does not Set the Device address after ARP successfully completes.

The SMBus Slave Address can be changed at any time, by using software to write to the register's *SMBus Device Address* field. ARP can also be enabled or disabled at runtime, by writing to the register's *ARP Disable* bit. If ARP is disabled by software after initially being enabled, the default address (70h) is not used for subsequent transactions. In this case, software must program a Device address into the *SMBus Device Address* field. When software writes the Device address, it must also Set the register's *AR Flag* and *AV Flag* bits (bits [11:10], respectively), to indicate that the address is valid and resolved.

Whenever software changes the register's *AV Flag*, *ARP Disable*, and/or *SMBus Device Address* values, software must also Set the register's *SMBus Parameter Reload* bit 15. Writes to this register bit take effect only when the register's *SMBus Command In-Progress* bit 28 is Cleared, which indicates that the PEX 8724 SMBus interface is in the Idle state.

7.3.7 SMBus Timeout

Unlike I²C, where the Slave or Master can indefinitely hold the I2C_SCL0 line Low, SMBus has a timeout condition. No device is allowed to hold the I2C_SCL0 line Low for more than 25 ms. When the PEX 8724, as a Slave-Transmitter, detects that it has pulled the I2C_SCL0 line Low for more than 25 ms, the PEX 8724 releases I2C_SCL0, and the logic returns to its default state and waits for another START condition. This can also occur when the Master pulls the I2C_SCL0 line Low for more than 25 ms during any single Clock Low interval within a transfer in progress, or during the ACK phase if the Master pulls the I2C_SCL0 line Low to process a task. Generally, the PEX 8724 pulls the I2C_SCL0 line Low if SMBus access to registers is delayed by internal arbitration for register access.

7.4 Switching between SMBus and I²C Bus Protocols

The PEX 8724's I²C implementation allows switching between the SMBus and I²C protocols, by toggling the **SMBus Configuration** register *SMBus Enable* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 2C8h[0]).

When operating in SMBus mode, Clearing this bit, using the SMBus Block Write protocol, enables I²C protocol for subsequent register accesses. This SMBus Block Write can be transmitted from an SMBus or I²C Master, provided that the Block Write Byte sequence conforms to the sequence explained in [Section 7.3.3.1](#). In I²C mode, writing 1 to the *SMBus Enable* bit turns On the SMBus protocol, immediately after the Write operation is complete.

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Chapter 8 Performance Features

8.1 Introduction

This chapter discusses guidelines for programming on-chip registers, to boost performance beyond that provided by the general-purpose default values, specifically:

- [DLLP Policies](#)
- [Latency](#)
- [Queuing Options](#)
- [Read Pacing](#)
- [Multicast](#)

8.2 DLLP Policies

Data Link Layer Packet (DLLP) rates can vary from 0 to 2 or more DLLPs/TLP. The PEX 8724 allows programming to affect the DLLP rate. An increase in DLLPs reduces the total TLP throughput. Therefore, for designs that require high performance, it would be beneficial to minimize DLLP rates. Transmitting fewer DLLPs, however, can result in credit starvation or Replay buffer overflow, which can have a detrimental effect on TLP bandwidth. Care must be taken when changing the default PEX 8724 DLLP transmission rate.

Typically, TLPs have higher transmission priority on the wire than DLLPs. The PEX 8724, however, allows DLLPs to have higher priority under certain conditions, meaning that DLLPs can transmit before starting a new TLP. The decision to transmit a DLLP ahead of a TLP is referred to as *DLLP policy*.

The PEX 8724 can be programmed to alter its default DLLP policies, to emphasize improved TLP throughput, faster acknowledgement, more credit, or simplest behavior. (Refer to [Section 8.2.1](#).) The PEX 8724 default policies are designed to achieve optimal performance for most applications. Programmable choices for a DLLP policy, however, allow for further optimization.

8.2.1 ACK DLLP Policy

An *ACK DLLP* is a response indicating to the TLP Transmitter that the Receiver received a “good” copy of the TLP, meaning that it acknowledged receipt of the TLP. The simplest policy is to send 1 ACK for every received TLP, resulting in a 1 DLLP/TLP rate for ACK alone. What an ACK means to the TLP Transmitter is that the TLP Transmitter can remove any stored copy of that TLP, because it is unnecessary to resend the TLP. ACK DLLPs can be combined, so that one ACK DLLP can serve to acknowledge multiple TLPs. This collapsing of ACKs is the basis of the ACK DLLP policy choices. Less-frequent, more-collapsed ACKs have the least impact on TLP transmit bandwidth, meaning that less-frequent ACKs result in less than 1 DLLP/TLP.

The PEX 8724 ACK policy consists of two parts – a Timer and TLP Counter. The default ACK Timer policy/value varies according to the negotiated Link width, operating Link speed, and Maximum Packet Size, as recommended in the *PCI Express Base r1.1*, *PCI Express Base r2.1*, or *PCI Express Base r3.0*.

The ACK Transmission Latency Timer loads the appropriate value when a TLP is received and known to be good, meaning a few clocks after the END framing symbol is received. The Timer counts down, as follows:

- **2.5 GT/s** – Each symbol time, once every 4 ns
- **5.0 GT/s** – Each symbol time, once every 2 ns
- **8.0 GT/s** – Each block time, twice every 2 ns

When the Timer reaches 0, an ACK DLLP takes higher priority over new TLPs (*that is*, an ACK DLLP is transmitted before a new TLP is started). The ACK DLLP transmitted acknowledges all TLPs, up to the most recently arrived good TLP.

The TLP Counter counts down on each TLP arrival until it reaches 0, then schedules a high-priority ACK DLLP. The default initialization value for the TLP Counter is 16, meaning a high-priority ACK is scheduled upon the arrival of 16 TLPs. The **Ingress Port-Based Control** register *ACK TLP Counter Timeout* field (Base mode – All Ports; Virtual Switch mode – VS Upstream Port, offset **F48h[1:0]**) value controls the ACK TLP Counter, as follows:

- **00b** – Allows 16 TLPs before a high-priority ACK (default)
- **01b** – Allows 8 TLPs before a high-priority ACK
- **10b** – Allows 4 TLPs before a high-priority ACK
- **11b** – Disables the Counter

Either the Latency Timer or TLP Counter mechanism can cause a high-priority ACK DLLP to be scheduled, and the first one to do so re-initializes both mechanisms to their starting parameters. *For example*, the time for 16 TLPs can be less than the ACK Timer above, in which case an ACK is sent earlier. The TLP Counter is useful for any system with a large programmed MPS (resulting in a large Timer value) that is capable of sending short TLPs (*such as* 12-byte Memory Reads). Rather than require the Transmitter to possibly save 100+ small TLPs, it need only save 16 TLPs, plus whatever else arrives during the round-trip time.

If there is no TLP traffic being transmitted (*that is*, the Transmit Link is idle), an ACK DLLP can be transmitted immediately, before the Latency Timer expires. This is an opportunistic, low-priority ACK because it does not contend with a TLP in transmission. When an opportunistic, low-priority ACK is transmitted, both the Latency Timer and TLP Counter re-initialize, waiting for a new TLP to arrive to begin counting again.

The PEX 8724 allows a programmable override of the default *Ack_Latency_Timer* value, on a per-Port basis, by programming the **ACK Transmission Latency Limit** register *ACK Transmission Latency Limit* field (offset **FA8h[11:0]**). The value in this register is loaded when a new TLP arrives and a high-priority ACK DLLP is attempted when the Timer reaches 0. For fastest ACK response, this Timer can be programmed to 000h, resulting in one DLLP ACK transmitted immediately per each TLP received. For less impact on Transmit TLP bandwidth, a larger value can be programmed, resulting in less-frequent ACKs.

In general, a slower ACK response aids the TLP Transmit stream, and does not impact the Receive TLP stream. Every PCI Express device contains storage (Retry buffer) for storing TLPs while waiting for ACKs. The amount of Retry buffer storage a device contains is vendor-dependent. The quantity of TLPs that the PEX 8724 can store depends upon the type and size of TLPs received. The PEX 8724 holds TLPs in the Retry buffer while waiting for an ACK. At some point, if the Retry buffer storage fills, no new TLPs can be sent until a new received ACK frees up space. In this case, the ACK can become a performance bottleneck.

8.2.2 UpdateFC DLLP Policy

An *UpdateFC DLLP* is transmitted in response to a received TLP. The UpdateFC DLLP replenishes the connected device with additional credit, to allow the Transmitter to transmit more TLPs of that type. Each TLP that arrives consumes credit, and eventually, a stream of TLPs consume all the available credit, unless an UpdateFC DLLP provides additional credit. However, if the connected device has sufficient credit to transmit more TLPs, it is not necessary to transmit UpdateFC DLLPs to it. The UpdateFC policy determines how and when to transmit an UpdateFC DLLP.

There are two parts to the UpdateFC policy – frequency of transmitting the updates and credit amount. This section discusses only the frequency.

If the PEX 8724 is not transmitting TLPs (*that is*, the Transmit Link is idle), and credit to replenish the credit used becomes available, the PEX 8724 immediately transmits an UpdateFC DLLP to the connected device. This is an opportunistic, low-priority UpdateFC DLLP.

However, if the PEX 8724 is busy transmitting TLPs to the connected device, the PEX 8724 does not transmit an UpdateFC DLLP until a programmed threshold is crossed. The PEX 8724 provides four threshold options – 100%, 75% (default), 50%, and 25%. Whenever the remaining credit drops below the programmed threshold, an UpdateFC DLLP is given high priority, meaning that the UpdateFC DLLP is transmitted before a new TLP is started. There is a separate threshold for Header and Payload credits, for each TLP type – Posted, Non-Posted, and Completion – located in the **Ingress Credit Handler (INCH) Threshold** registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets [A00h](#) through [A08h](#)).

8.2.3 Unidirectional DLLP Policies

For unidirectional traffic, the PEX 8724 DLLP policies allow the most-frequent DLLPs, because DLLPs do not interfere with TLPs. (DLLPs flow in the opposite direction of TLPs.)

The PEX 8724 can transmit a DLLP ACK almost immediately upon receiving and verifying a TLP. A faster ACK results in fast Transmitter de-allocation of the TLP, and can therefore allow a shallow TLP Replay buffer. The default values can be overwritten, to increase or decrease the ACK DLLP rate. For unidirectional traffic, a small number *such as* 1, is recommended.

The number programmed into the **ACK Transmission Latency Limit** register *ACK Transmission Latency Limit* field (offset [FA8h](#)[11:0]) Sets the ACK Transmission Latency Timer, to count the quantity of symbol times after receiving a TLP, before transmitting an ACK.

Similar to the ACK programmability, the PEX 8724 can immediately transmit an UpdateFC after receiving only the TLP Header. By transmitting an UpdateFC earlier, the total credit advertised can be minimized. By programming fewer credits and having a fast UpdateFC policy, the system does not run out of credits and the PEX 8724 does not waste Buffer space on reservations that do not arrive. The following are the recommended Settings:

- Set the UpdateFC policy for unidirectional traffic to 100%
- Set the credits to be sufficient to allow 3 to 4 TLPs

8.3 Latency

Latency is the length of time it takes to proceed from one event to another. Latency can be measured in several different ways, but perhaps the most common measurement for a switch is *Start TLP-to-Start TLP (STP-to-STP) latency*. [Figure 8-1](#) illustrates an STP-to-STP Latency Measurement. When the Egress Start TLP symbol is transmitted out of a switch before the Ingress Port End symbol arrives, the transfer is termed *Cut-Thru*. If an Egress Port queue is not already established, the PEX 8724 always cuts the packet through. The PEX 8724 has the same latency, regardless of whether the traffic is Upstream or peer-to-peer.

As expected with the PEX 8724 Cut-Thru architecture, STP-to-STP latency is basically constant for all Payload sizes. A faster Link can receive the Header for decode faster, with a slightly lower latency. There will generally be a constant latency for any ingress width to the same egress width, or any ingress width to a smaller egress width, operating at the same Link speed.

For cases in which the egress Port has a higher bandwidth than the ingress Port, then a fraction of the packet, given by the following formula:

$$F = (E - I) / E$$

must be buffered (to prevent under-run in the middle of the TLP), before the TLP can be forwarded to the egress Link.

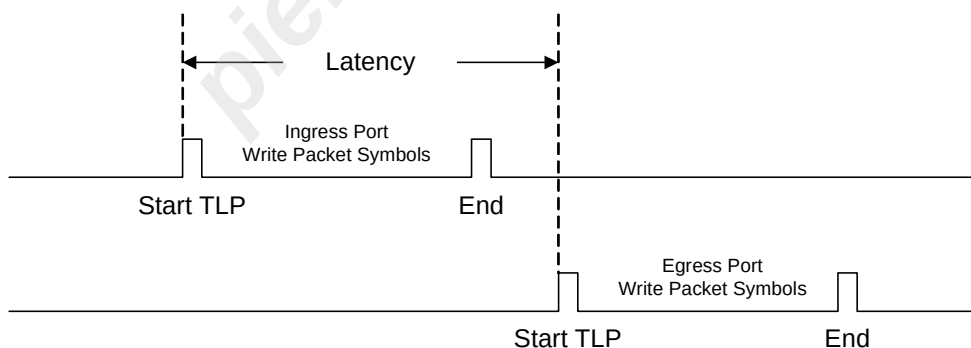
where

- *F* is the fraction sum
- *E* is egress bandwidth
- *I* is ingress bandwidth

For example, with an ingress x4 Port and egress x16 Port, $(16 - 4) / 16 = 3/4$ of the packet must be buffered before it can be forwarded. Moving in the opposite direction, with an ingress x16 Port and egress x4 Port, the ingress Port has more bandwidth than the egress Port; therefore, buffering is not needed, and the packet can immediately be cut through.

Latency is important when a Downstream device is reading Main memory, Upstream, because the reader has a finite number of Reads that it can send. Often in this case, the Upstream Port is wider than the Downstream Port and, as a result, latency through the PEX 8724 is optimal in both directions. The Memory Read Request that is transmitting Upstream is quite short (12 to 20 bytes); therefore, the buffering required for going from narrower to wider Link widths does not significantly increase latency. The resultant Completion with Data (CplD), with a Payload returning Downstream, is a significantly larger packet; however, the wider to narrower Link width rule applies, and therefore no buffering is needed before forwarding the packet.

Figure 8-1. Start TLP-to-Start TLP Latency Measurement



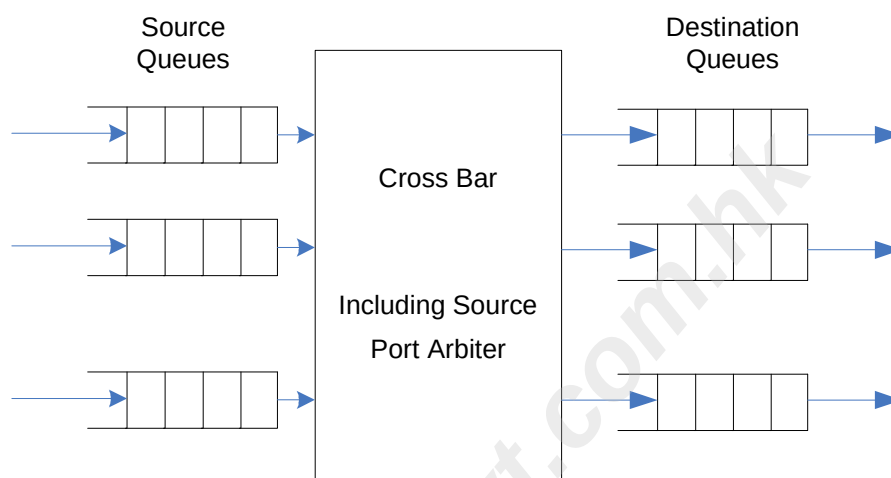
8.4 Queuing Options

On-chip queuing does not exist in balanced bandwidth scenarios, where the total ingress bandwidth is less than or equal to the egress bandwidth. In the common case, where the total ingress bandwidth is greater than the egress bandwidth, queues develop on the PEX 8724. The PEX 8724 provides two alternatives, as to where to locate such queuing (refer to [Figure 8-2](#)):

- **Destination queue** – Associated with a single destination Port. All the TLPs in a Destination queue will egress out the same Port.
- **Source queue** – Associated with a single ingress Port. All the TLPs in a Source queue come from the same Port.

Each queue is discussed in the sections that follow.

Figure 8-2. On-Chip Queuing



8.4.1 Destination Queuing

Note: For the queuing examples provided in this section, “Port 1” indicates “first Port,” not the Port physically identified as Port 1.

The default behavior is for all queues to develop at the destination Port. If TLPs are arriving from four sources to a common destination Port, the TLPs are scheduled according to First-In, First-Out (FIFO). The crossbar can forward a TLP every 4 ns, to each Destination queue; therefore, it is unlikely that a Source queue can develop or last very long.

A Destination queue develops whenever the *ingress rate* – the sum of all ingress Ports targeting a destination Port – exceeds the egress rate. A Destination queue might also develop in a credit-starved situation, where there is no credit available to forward TLPs.

For example, if TLPs arriving from four sources all go to a common destination Port, the TLPs are scheduled, based upon the order in which they arrive at the Destination queue FIFO^a. If all four flows are equally active, the TLPs naturally interleave as 1, 2, 3, 4, 1, 2, 3, 4. If three of the Ports, however, have a head start before the fourth Port turns On, the output can be 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4. In this case, all the new Port (fourth Port) TLPs must wait for the earlier Port 1, 2, 3 traffic to be transferred before the fourth Port TLPs can be transferred. Therefore, the latency for the fourth Port traffic to travel through the PEX 8724 can widely vary, based upon the traffic passing through the switch.

a. Conventional PCI Strong Ordering rules can override the FIFO. Conventional PCI requires Posted TLPs to be able to pass Non-Posted and Completion TLPs, to avoid deadlock.

8.4.2 Source Queuing

Caution: *Source Queuing and [Read Pacing](#) should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors.*

Note: *For the queuing examples provided in this section, “Port 1” indicates “first Port,” not the Port physically identified as Port 1.*

Source queuing can be enabled for applications that require deterministic bounded latency for traffic that flows to a single Port.

Source queuing limits the Destination queue depth by applying an upper and lower threshold. When the Destination queue reaches the programmed upper threshold, any subsequent TLPs targeting that destination Port are not forwarded, but instead are queued up in a per-Source Port-based queue. The Source Port queue does not begin to forward TLPs again (to any Port), until its first entry’s destination Port’s queue drops to a programmed lower threshold.

Note: *A Source Port queue that cannot forward to a Destination queue blocks all subsequent TLPs arriving on that same Source Port, even if those subsequent TLPs target different destination Ports.*

The **Port Egress TLP Threshold** register (offset [F38h](#)) controls the upper and lower queue thresholds. [Table 8-1](#) summarizes the register bit Settings. The Port Lower TLP Counter is the quantity of TLPs that the Destination queue must reach after becoming saturated, before re-enabling TLP forwarding. The Port Upper TLP Counter is the quantity of TLPs that can be queued in the Destination queue.

In the Destination queue example provided in [Section 8.4.1](#), the early arriving Port 1, 2, 3 TLPs stalled the fourth Port’s TLP for an indeterminate length of time. By programming, with source queuing enabled and a Destination Port Lower TLP Counter programmed to 1 and Port Upper TLP Counter programmed to 3 (TLPs), the worst case is that the fourth Port must wait for three TLPs (1, 2, 3) before getting its first turn. With these Settings, the example TLP output would be 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4. The *turn to be forwarded* refers to a Port Arbitration wait, described in [Section 8.4.3](#).

To help prevent idles on the destination Link, increase the Port Lower TLP Counter to a value somewhat higher than 1. The minimum value should be 2; however, 3 is recommended. Also, the separation of the two counters should be at least 2, for efficiency.

Table 8-1. Port Egress TLP Threshold Register Port Lower and Upper TLP Counters (Offset F38h)

Bit(s) ^a	Name	Description
11:0	Port Lower TLP Counter	When Source Scheduling is disabled due to Threshold, it is re-enabled when the Port TLP Counter goes below this Threshold value.
27:16	Port Upper TLP Counter	When the Port TLP Counter is greater than or equal to this value, the Source Scheduler disables TLP scheduling to this egress Port.

a. Bits not identified in [Table 8-1](#) are **Reserved**.

8.4.3 Port Arbitration

Port Arbitration refers to the arbitration at an egress Port of a switch between other ingress Ports. In the PEX 8724, a different Port Arbiter connects all the Source queues to each Destination queue. The PEX 8724 supports, on a per-Port basis, either a hardware-fixed Round-Robin (RR) arbitration scheme or a 64-slot Weighted Round Robin (WRR) arbitration scheme, the assignment of which is a bit complicated and will be discussed at the end of this section.

The Port Arbiter has the most effect when source queuing is enabled (refer to [Section 8.4.2](#) for details), and TLPs from multiple ingress Ports are enqueued such that the Port Arbiter has choices about which TLP to forward. Every egress Port has its own independent Port Arbiter state machine.

Port arbitration in the PEX 8724 makes decisions on a per-TLP basis. As a consequence, the throughput may not look fair for two otherwise equal Ports if the average TLP size of each is different. *For example*, if two ingress Ports, PortA and PortB, target the same egress Port and both have the same weight, but PortA has short TLPs while PortB has long TLPs, PortB will appear to get a higher throughput. The seeming unfairness can be compensated for, by using the WRR Arbiter, and by adjusting the weights of the Ports to be in a ratio similar to the ratio of TLP size.

The RR Port Arbiter scheme decides which ingress Port to forward to a single egress Port, with all ingress Ports having the same weight. All Downstream Ports, by default, use an RR Port Arbiter.

The WRR Port Arbiter scheme also decides which ingress Port to forward to a single egress Port; however, the ingress Ports may have different weights. The initial weights are based upon the negotiated Link width. With 64 slots, the resolution of the weights goes down to a x1 Gen 3 Link (8 GT/s). Therefore, a x1 Gen 3 Link would have 1 slot, while a x8 Gen 3 Link would have 8 slots, and so forth. A x4 Gen 1 Link, x2 Gen 2 Link, and x1 Gen 3 Link all have the same weight, because each has the same throughput capability of approximately 8 GT/s.

Software can change the default WRR weights to any value. The WRR Port Arbiter uses the *PCI Express Base r3.0*-defined **Port Arbitration Capability** structure, as reflected in the Port's **VC0 Resource Capability** register *Port Arbitration Capability* field (offset 158h[2:0]), with the Port's **VC0 Resource Control** register *Port Arbitration Select* field (offset 15Ch[19:17]) programmed to 010b.

Software performs two steps to change the WRR weights.

In the first step, the weights are programmed in the **Port Arbitration Table Phase x** registers (offsets 178h through 1B4h). (Refer to the *PCI Express Base r3.0*, as well as [Section 13.15.1](#), “WRR Port Arbitration Table Registers (Offsets 178h – 1BCh),” for further details.)

The final step requires a write to the Port's **VC0 Resource Control** register *Load Port Arbitration Table* bit (offset 15Ch[16]). When written, the weights programmed in the **Port Arbitration Table Phase x** registers in the first step are transferred to the WRR arbitration logic and take effect. Before writing the *Load Port Arbitration Table* bit, the **Port Arbitration Table Phase x** registers have no effect.

The selection of which egress Port gets a WRR Port Arbiter is limited, due to hardware constraints. The main constraint is that only one Port within a Port group can use the WRR capability. The Port groups are as follows:

- **Group 0** – Ports 0, 2
- **Group 1** – Ports 1, 3
- **Group 2** – Port 10
- **Group 3** – Port 9

Each Port group has one WRR that is always enabled. As previously mentioned, the WRR defaults to weights based upon the negotiated Link width of each ingress Port. The decision algorithm looks at whether any Port is Upstream, in which case that Port “wins” use of the WRR capability. If there is no Upstream Port, then the first Port in the group gets WRR capability. Thus, a Downstream Port can use only the first Port within a group (Port 0, 1, 9, or 10).

In Virtual Switch mode, where there can be multiple Upstream Ports within the same group, the last Port within the group wins use of the WRR capability. *For example*, if the Upstream Port is Port , it uses the Group WRR Port Arbiter, relegating (enabled) Port to an RR Port Arbiter. The only way for each Upstream Port to get a WRR Port Arbiter is to choose Upstream Ports that belong to different Port groups.

8.4.4 Port Bandwidth Allocation

For applications that need to allocate a fixed bandwidth to each Port, the PEX 8724 can help enforce the relative bandwidth ratio between Ports in a congested scenario. By combining source queuing, Port Arbitration, and initial credit, as well as some knowledge of average Payload size, many combinations of Port bandwidth allocation are possible. For further assistance, contact PLX Technical Support.

8.5 Read Pacing

Note: Source Queuing and Read Pacing should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors.

The Read Pacing feature is supported on all Ports. The Read Pacing Configuration registers, however, are implemented as follows – Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port.

PCI Express has a weakness concerning the number of outstanding bytes requested by Reads. It is possible that a single device can overwhelm the system with a relatively small number of large Read Requests, thereby impacting the performance of other connected devices, by filling the ingress transaction queue in the Root Complex.

The Root Complex must handle the transactions in the order in which they are posted. Transactions posted from less aggressive reading devices, which may be more sensitive to latency, suffer performance reductions due to the unfairly weighted path (head of line blocking) in the transaction queue that the large reads represent.

Read Pacing attempts to apply some rules to Memory Read Requests, so that no one Port can overwhelm a system. There are two aspects to the PEX 8724's Read Pacing capability:

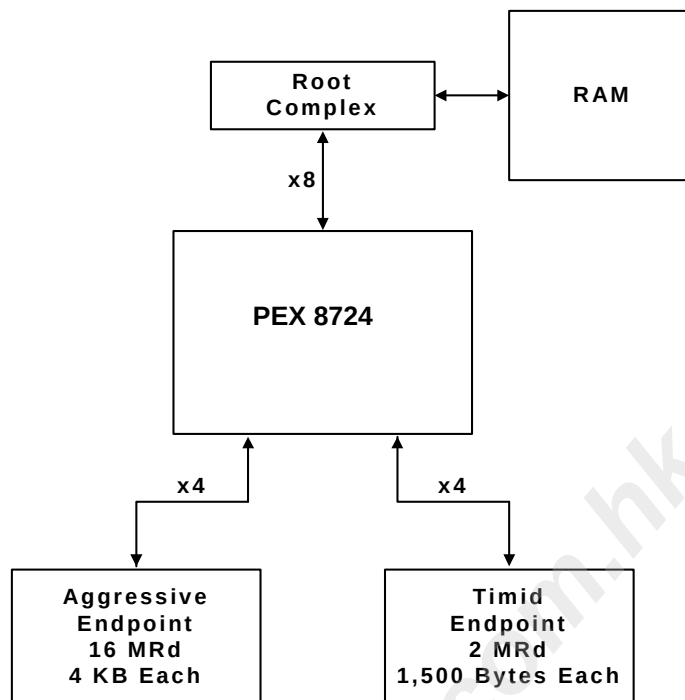
- Read spacing
- Read threshold

The following sections provide examples and further information regarding Read Pacing.

8.5.1 Read Pacing Example

Figure 8-3 illustrates an example of a system that benefits from Read Pacing.

Figure 8-3. Read Pacing Example (Station 0)



In a typical Host-centric application, endpoints have Direct Memory Access (DMA) engines that write to and read from Main memory. A performance bottleneck can occur during the Read to Main memory, through the Root Complex. For the example illustrated in [Figure 8-3](#), the aggressive endpoint sends many large (16, 4-KB) Memory Read Requests, while another endpoint, or Timid Endpoint (TEP), sends only two 1,500-byte Memory Read Requests. The TEP then waits for a response before sending additional Read Requests^a.

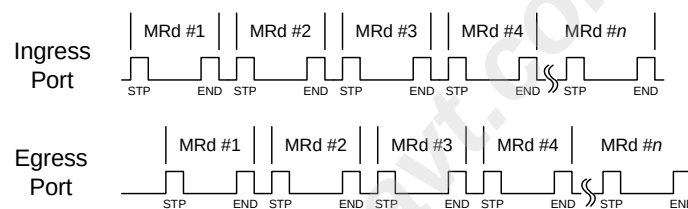
If either endpoint is running by itself, neither sees a problem. However, if both endpoints are concurrently active, the aggressive endpoint dominates the Root Complex Memory Controller. In addition, due to the bandwidth mismatch, Completions can queue up in the PEX 8724, creating too many Completions for the switch to store at one time. As a result, the PEX 8724 backpressures the Root Complex for Completions. The Root Complex can only forward Completions to the PEX 8724 at the aggressive endpoint's rate, which is significantly less than the Root Complex could otherwise handle.

The net impact is not to the aggressive endpoint, because there are a sufficient number of Completions queued up in the PEX 8724 to keep it busy. In fact, the aggressive endpoint experiences better performance with a switch, than connected directly to the Root Complex^b. Rather, the TEP experiences lower performance results. Its Memory Read Requests wait in line behind multiple aggressive endpoint Requests, and the Root Complex can drain Requests only at the same rate of the PEX 8724, not at the Upstream Link's capacity.

[Figure 8-4](#) illustrates how a PCI Express switch, without Read Pacing, forwards MRds.

Read Pacing solves the performance loss seen by the TEP, while improving the aggressive endpoint's performance. The following sections provide examples of the way in which the PEX 8724 functions when Read Pacing is enabled, and Read Spreading is enabled or disabled.

Figure 8-4. Read Pacing Off (Disabled)



a. This is based upon an actual setup in a third-party lab. Fibre Channel endpoints can easily send 16, 4-KB MRd at a time, while Gigabit Ethernet endpoints might send only one or two 1,500-byte endpoints at a time.

b. Without a switch, when the Root Complex has something else to do, the aggressive endpoint loses its data stream. With a switch, the buffering of multiple Completions hides the fact that the Root Complex is multitasking.

8.5.2 Read Spacing (Spreading) Logic

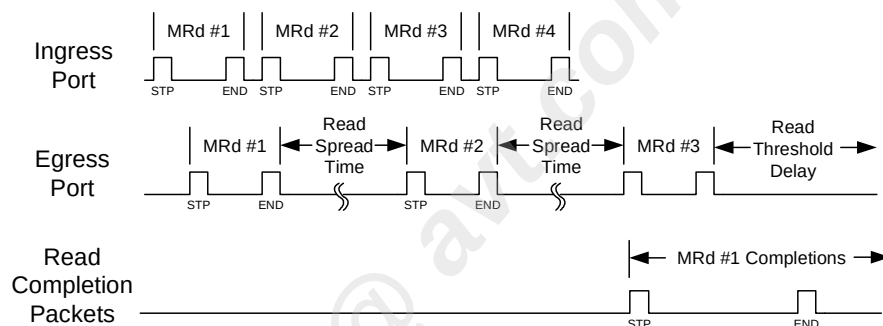
Read Spacing (also referred to as *Read Spreading*) spreads out Read Requests. The PEX 8724 Read Spacing logic looks at the Read Request size and the endpoint's bandwidth, to determine how often to forward subsequent Read Requests. *For example*, Read Requests arriving on a x4 Link can only sink data at a x4 rate. If a x4 endpoint submits multiple Read Requests to a x8 Link, the Read Spacing logic does not forward the subsequent Read Requests until the endpoint has sufficient time to sink a portion of the Completion data from the previous Read Requests.

Initially, a queue of Completions must build up to hide the time that it takes for the data to return. As a result, Reads are forwarded at 2x the endpoint's bandwidth. This 2x rate is maintained until a threshold of outstanding Read data is reached, at which time Reads are forwarded at 1x the endpoint's bandwidth.

Read Pacing must be enabled for Read Spreading to be enabled. That is, for a Port to have Read Spreading enabled, the Port's **Read Pacing Control** register *Port x Read Pacing Disable* and *Port x Memory Read Spreading Disable* bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 1D0h[1:0 and 17:16], respectively) must both be Cleared.

Figure 8-5 illustrates the way in which the PEX 8724 forwards Read Requests when the Read Pacing- and Read Spreading-related bits are enabled. (Refer to Section 8.5.5 for additional register/bit information.) The PEX 8724 continues to spread and forward the Read Requests, until the amount of Completion data for which it is waiting exceeds the value programmed in the **Read Pacing Threshold x** register(s) for that Link width (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 1D4h and 1D8h).

Figure 8-5. Read Pacing On (Enabled) and Read Spreading On (Enabled)

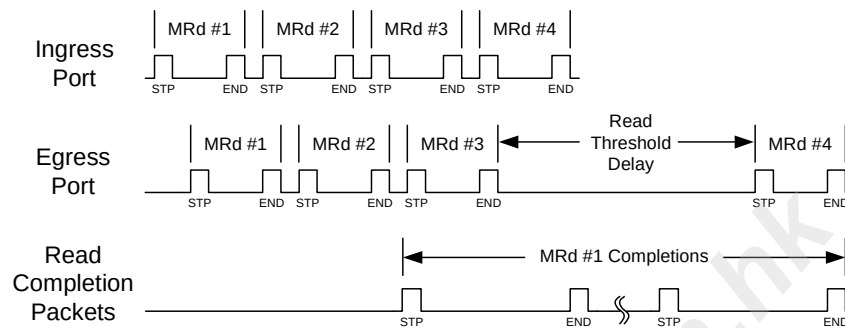


8.5.3 Read Threshold

The Read threshold is the maximum number of outstanding DWords (1 DWord = 4 bytes) that the endpoint Port requested to be read, but were not yet returned as Completion data. The threshold is related to the PEX 8724's buffering capacity – all outstanding Read data ought to be able to be buffered in the switch, to remain out of the way of other Completions for other endpoint's Read Requests.

After a Port reaches its Read threshold, subsequent Read Requests from that Port queue up in the PEX 8724, waiting for Completion data to reduce the outstanding count to below the threshold. If an overabundant number of Read Requests queue in the PEX 8724, no additional Read credit is allocated, which backpressures the Read Requester. Figure 8-6 illustrates the way in which the PEX 8724 forwards Read Requests when its Read Spacing logic is enabled and Read Spreading logic is disabled.

Figure 8-6. Read Pacing On (Enabled) and Read Spreading Off (Disabled)



8.5.4 Read Pacing Benefits

When Read Pacing logic is enabled, the PEX 8724 provides the follow benefits:

- Maximum Read latency that an endpoint may experience can be dramatically reduced.
By reducing the amount of queued Read Requests, and therefore pending Read Completion data at the Root Complex, new Read Requests from Ports that do not have pending Read Requests can be serviced with a predictable and/or reasonable amount of latency.
- Timid endpoint bandwidth is dramatically increased in busy applications.
Because queues of pending Read Requests in the Root Complex are limited, and congestion caused by a large amount of Completion data intended for a high-bandwidth, needy Port (or Ports) is avoided, the bandwidth needs of endpoints with smaller bandwidth requirements are met (*that is*, the endpoints are not starved).
- PEX 8724's Read Pacing Threshold logic allows all busy Ports to be equally serviced in congested scenarios, regardless of their individual Read requesting behavior.

For example, all Ports might simultaneously request data, some aggressively and some timidly. While unable to quickly drain their queued Completions, the Ports' Read Pacing Threshold logic forwards the additional Read Requests to the Root Complex, equally and fairly, while ensuring Completion data is available for each Port, when the Port is ready to accept it.

8.5.5 Enabling Read Pacing and Read Spreading

Read Pacing is disabled, by default. To enable Read Pacing, the Port's **Read Pacing Control** register *Port x Read Pacing Disable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 1D0h[1:0]) must be Cleared. Each bit corresponds to a range of Ports. A value of 0 enables Read Pacing for the Ports associated with the bit, whereas a value of 1 (default) disables Read Pacing for the Ports associated with the bit.

The Port's **Read Pacing Control** register *Port x Memory Read Spreading Disable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 1D0h[17:16]) is used to enable or disable Read Spreading. Each bit corresponds to a range of Ports. A value of 1 disables Read Spreading for the Ports associated with the bit. Read Spreading is enabled, by default (value of 0); however, it is overridden by the Port's *Port x Read Pacing Disable* bit, by default.

Both sets of Read Spreading and Pacing Control register bits are represented in Table 8-2. (For complete details, refer to the register offset 1D0h description provided in Section 13.16.1, “Device-Specific Registers – Read Pacing (Offsets 1D0h – 1D8h).”) Figure 8-4 through Figure 8-6 illustrate what occurs when the bits are enabled or disabled.

The Read Pacing thresholds are Set, based upon the Source Port's programmed Link width. The **Read Pacing Threshold 1** register controls the threshold values for x16 and x8 Link width threshold values, and the **Read Pacing Threshold 2** register controls the x4 and x2 Link width threshold values (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 1D4h and 1D8h, respectively). The thresholds are in DWords. Narrower Link widths have lower thresholds, because they must buffer smaller quantities.

Table 8-2. Read Pacing Control Register Read Pacing and Memory Read Spreading Disable Bits (Offset 1D0h)
(Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port)

Bit(s)	Description	Default
1:0	Port x Read Pacing Disable 0 = Read Pacing is enabled for this Port 1 = Read Pacing is disabled for this Port	11b
17:16	Port x Memory Read Spreading Disable 0 = Memory Read Spreading is enabled for this Port 1 = Memory Read Spreading is disabled for this Port	00b

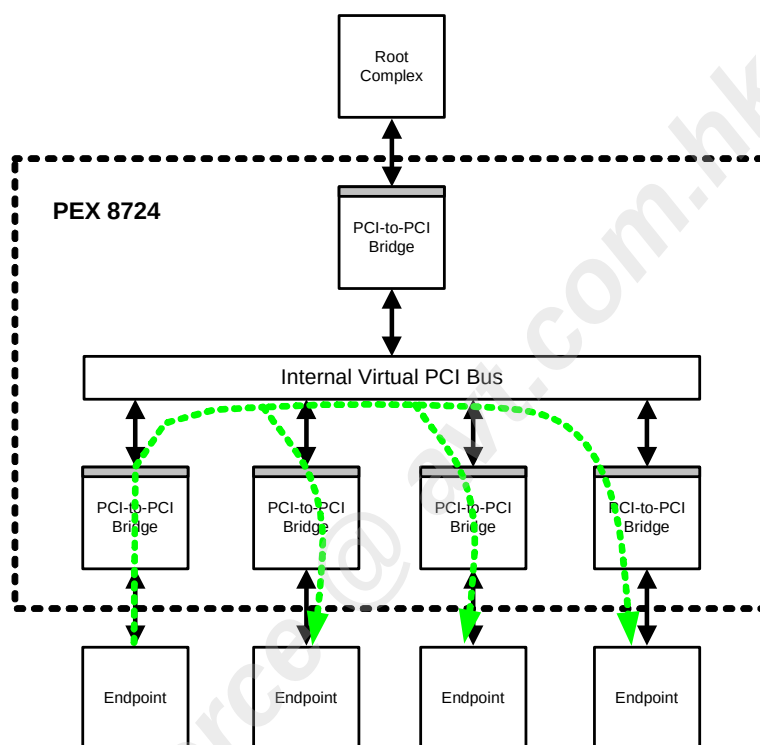
8.6 Multicast

This section describes the functions and registers of the Multicast (MC) feature.

Multicast allows software to concurrently write the same data to a group of multiple destinations. When a Posted Memory Write TLP entering the PEX 8724 is addressed to the MC Address Range (*MC BARs*), and Multicast is enabled, the PEX 8724 automatically generates and transmits a copy of the original TLP (referred to as the *MC Copy TLP*) to one or more destination Port(s). The MC Address Space is divided into *MC Groups (MCG)*, defined by using *MC Base Address* and *MC Index Position*. Each PEX 8724 Port can elect to receive an MC Copy TLP by belonging to an *MCG*, by Setting the corresponding *MC Receive* bit. An MC TLP can be blocked by using *MC Block All*, if required. *MC Overlay Bar* can be used to replace the original MC TLP's address to a Unicast address space, if the endpoint does not support MC.

Figure 8-7 illustrates an example of a Multicast operation on a peer-to-peer Write Request. An endpoint's Write Request is transmitted as copies to additional peer destination endpoints.

Figure 8-7. Peer-to-Peer Multicast



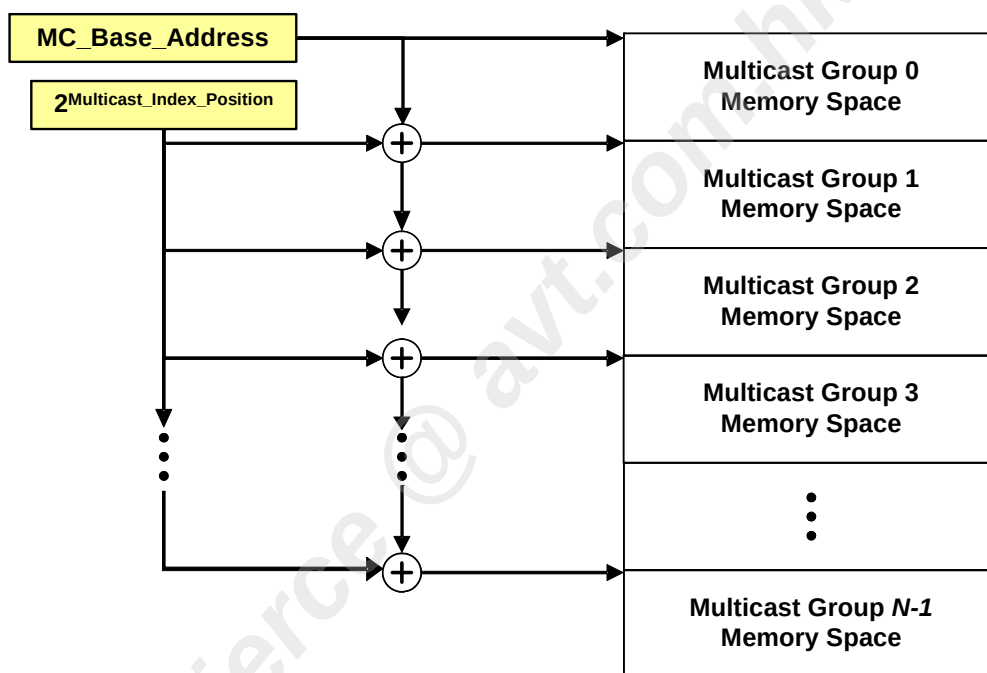
8.6.1 Multicast Extended Capability

MC functionality is controlled by the **Multicast Extended Capability** structure. Multiple copies of this structure are required – one for each Port that supports MC. To provide implementation efficiencies, certain fields within each of the MC Capability structures within a component must be programmed the same. Results are indeterminate if this is not the case. The fields and registers that must be configured with the same values include MC_Enable, MC_Num_Group, MC_Base_Address, and MC_Index_Position. These same fields in an endpoint's **Multicast Extended Capability** structure must match those configured within a **Multicast Extended Capability** structure of the Port Upstream from the endpoint, or in which the Root Complex integrated endpoint is integrated.

8.6.2 Multicast Address Range Segmentation

The **Multicast Extended Capability** structure defines an MC Address range, the segmentation of that range into a number, N , of equally sized MC windows, and the association of each MC window with an MCG, as illustrated in Figure 8-8. Each function that supports MC within a component implements a **Multicast Extended Capability** structure that provides routing directions and permission checking for each MCG for TLPs that pass through. The MCG is a field of up to 6 bits in width, which is embedded in the address, beginning at the MC_Index_Position.

Figure 8-8. Multicast Address Range Segmentation



8.6.3 Multicast TLP Processing

A TLP is processed as an MC TLP if an MC Hit occurs when all of the following conditions are true:

- MC_Enable is Set
- TLP is a Memory Write (Posted Request)
- TLP Address $\geq$ MC_Base_Address
- TLP Address $< (MC_Base_Address + (MC_Index_Position^2) \times (MC_Num_Group + 1))$

While processing the TLP, each PEX 8724 ingress Port uses values of MC_Enable, MC_Base_Address, MC_Index_Position, and MC_Num_Group from its registers. The software is required to identically configure all these fields in all Ports. If this is not the case, results are indeterminate.

If an MC Hit occurs, standard address routing rules do not apply. Instead, the TLP is processed by first extracting the MCG from the address in the TLP, using the ingress Port's values for MC_Base_Address and MC_Index_Position. Specifically:

$$MCG = ((Address_{TLP} - MC_Base_Address) \gg MC_Index_Position) \& 3Fh$$

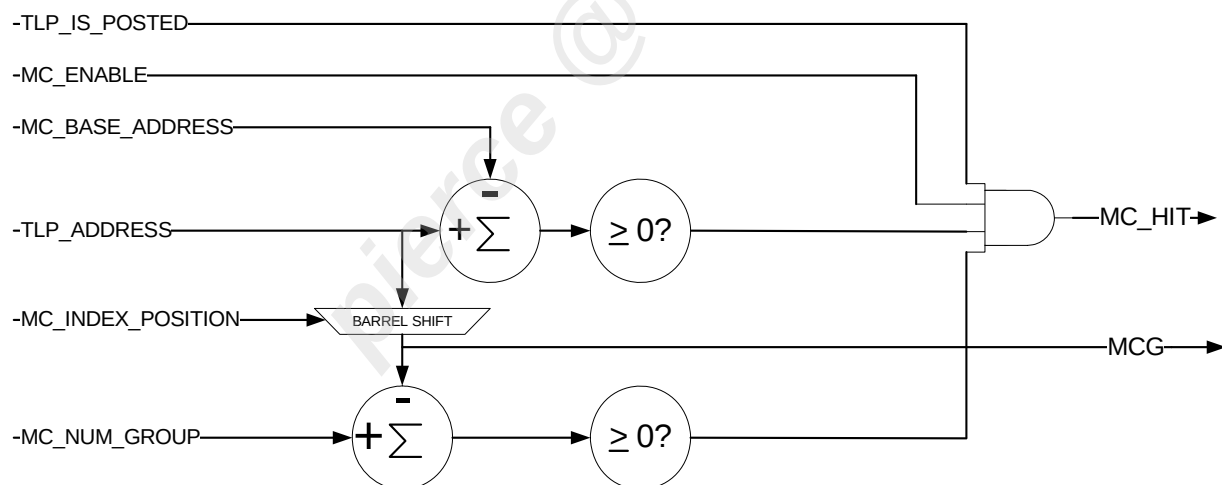
Next, the PEX 8724 checks the MC_Block_All and MC_Block_Untranslated bits corresponding to the extracted MCG, using the MC_Block_All and MC_Block_Untranslated registers associated with the ingress Port. If the MC_Block_All bit corresponding to the extracted MCG is Set, the TLP is handled as an MC Blocked TLP. If the MC_Block_Untranslated bit corresponding to the extracted MCG is Set, and the TLP contains an Untranslated Address, the TLP is also handled as an MC Blocked TLP.

If the TLP is not blocked in the PEX 8724, it is forwarded out of all the Ports, with the exception of its ingress Port, whose MC_Receive bit corresponding to the extracted MCG is Set. If no Ports forward the TLP, the TLP is silently dropped.

Figure 8-9 illustrates Multicast processes TLPs.

Note: To prevent loops, it is prohibited for a PEX 8724 Port to forward a TLP back out through its ingress Port, regardless of whether it is specified to do so, by the Port's MC_Receive register.

Figure 8-9. Multicast TLP Processing



An MC Hit suspends standard address routing, including default Upstream routing in the PEX 8724. When an MC Hit occurs, the TLP is forwarded out only through those egress Ports whose *MC_Receive* bit associated with the MCG extracted from the address in the TLP is Set. The PEX 8724 attempts to simultaneously broadcast the packets. However, if the Egress queues are not empty, the packet remains queued for each egress Port, because ordering must be maintained. Therefore, the MC packets could be transmitted at different times, depending upon the traffic flow.

If the address in the TLP does not decode to a Downstream Port using standard address decode, the TLP is copied to the Upstream Port, if specified to do so, by the Upstream Port's **MC_Receive** register.

If the address in a Non-Posted Memory Request hits in an MC window, no MC Hit occurs, and the TLP is processed as a Unicast.

If an MC Hit occurs, the only ACS access control that can apply is ACS Source Validation. In particular, neither ACS re-direction, nor the Port's **ACS Egress Control Vector** register *ACS Peer-to-Peer Port x Control* bit (Downstream Ports, offset *F2Ch*[10:8, 3:0]), affects operations during an MC Hit.

8.6.4 Multicast Ordering

No new ordering rules are defined for processing MC TLPs. All MC TLPs are Posted Requests and follow Posted Request ordering rules. MC TLPs are ordered per standard ordering rules, relative to other TLPs in a component's ingress stream, through the point of replication. Once copied into an egress stream, an MC TLP follows the same ordering as other Posted Requests in the stream.

8.6.5 Multicast Extended Capability Structure Field Updates

Certain fields of the **Multicast Extended Capability** structure can be changed at any time. Others cannot be changed with predictable results, unless the *MC_Enable* bit is Cleared in every component function. The latter group includes *MC_Base_Address* and *MC_Index_Position*. Fields that software can change at any time include *MC_Enable*, *MC_Num_Group*, *MC_Receive*, *MC_Block_All*, and *MC_Block_Untranslated*. Updates to these fields must be ordered.

For example, TLPs *A* and *B* arrive in that order, at the same ingress Port, and in the same TC. If *A* uses *X* for one of these fields, then *B* must use the same value or a newer value.

8.6.6 MC Blocked TLP Processing

When a TLP is blocked by the *MC_Block_All* or *MC_Block_Untranslated* mechanisms, the TLP is dropped. The ingress Port blocking the TLP serves as the Completer. The ingress Port logs and signals this MC Blocked TLP. In addition, the ingress Port Sets the **PCI Status** register *Signaled Target Abort* bit (offset *04h*[27]) or **Secondary Status** register *Signaled Target Abort* bit (offset *1Ch*[27]), as appropriate. If the error occurs with a TLP received by an ingress Port, the error is reported by that ingress Port.

8.6.7 MC_Overlay and ECRC Re-Generation

The MC Overlay mechanism is provided to allow a single BAR in an endpoint that does not contain a **Multicast Extended Capability** structure to be used for both MC and Unicast TLP reception. Software can configure the MC_Overlay mechanism to affect this, by Setting the MC_Overlay_BAR in a Downstream Port of the PEX 8724, so that the MC Address range, or a portion thereof, is re-mapped (overlaid) onto the Memory Space range accepted by the endpoint's BAR. At the Upstream Port, the mechanism can be used to overlay a portion of the MC Address range onto a Memory Space range associated with Host memory.

When enabled, the overlay operation specifies that MC TLP Address bits, whose bit numbers are greater than or equal to the MC_Overlay_Size field, be replaced by the corresponding MC_Overlay_BAR bits.

That is:

```
If (MC_Overlay_Size = 0)
Then:
    Egress_TLP_Addr = Ingress_TLP_Addr;
Otherwise:
    Egress_TLP_Addr =
        {MC_Overlay_BAR[63:MC_Overlay_Size], Ingress_TLP_Addr[MC_Overlay_Size-1:0]};
```

If the TLP with the modified address contains the optional ECRC, the unmodified ECRC will almost certainly indicate an error. The action to be taken if a TLP containing an ECRC is MC copied to an egress Port that has MC_Overlay enabled, are outlined in [Table 8-3](#). If MC_Overlay is not enabled, the TLP is forwarded, unmodified. If MC_Overlay is enabled and the TLP has no ECRC, the modified TLP, with its address replaced as specified in the previous paragraph, is forwarded. If the TLP has an ECRC but ECRC re-generation is not enabled, then the modified TLP is forwarded with its ECRC dropped and the *TD* bit in the header Cleared, to indicate that no ECRC is attached. If the TLP has an ECRC and ECRC re-generation is enabled, an ECRC is performed before the TLP is forwarded. If the ECRC passes, the TLP is forwarded with a re-generated ECRC. If the ECRC check fails, the TLP is forwarded with an inverted re-generated ECRC.

Table 8-3. ECRC Rules for MC_Overlay^a

MC_Overlay Enabled	TLP with ECRC	ECRC Re-Generation Supported	Action if ECRC Check Passes	Action if ECRC Check Fails
No	X	X	Forward TLP unmodified.	
Yes	No	X	Forward modified TLP.	
Yes	Yes	Yes	Forward modified TLP with re-generated ECRC.	Forward modified TLP with inverted re-generated ECRC.

a. "X" is "Don't Care."

8.6.7.1 Multicast to Endpoints without Multicast Extended Capability

An endpoint function that does not contain a **Multicast Extended Capability** structure cannot distinguish MC TLPs from Unicast TLPs. It is possible to take advantage of this, to use such endpoints as MC targets. The Port Upstream from the device can be configured to overlap at least part of the MC Address range, or the MC_Overlay mechanism can be used.

8.6.7.2 Congestion Avoidance

The use of MC increases the output Link use of switches to a degree proportional to both the size of the MC groups used and the fraction of MC traffic to total traffic, which can increase the risk of congestion, as well as skewing of individual Multicast TLP instance arrival times. To mitigate this risk, design components that are intended to serve as MC targets to consume MC TLPs at wire speed. Components intended to serve as MC sources should consider adding a rate-limiting mechanism.

8.6.8 Multicast with NT – NT Mode

The following subsections describe Multicast behavior in NT mode.

8.6.8.1 NT Multicast from NT Port Virtual-to-Link Interface Direction (NT Target) – NT Mode

When a PEX 8724 Transparent Port receives a TLP that is a Multicast Hit, the Multicast TLP that is routed to the egress Ports belongs to the extracted Multicast Group. If a virtual side NT endpoint is behind one of the Multicast Targeted Ports, the ingress Port routes the Multicast TLP through the virtual Downstream Port to the NT Port Virtual Interface, if all the following conditions are true:

- Virtual Downstream Port **Command** register *Memory Enable* bit is Set
- If Multicast overlay is not enabled, the Multicast TLP address hits the NT Port Virtual Interface BARs (**BAR2** to **BAR5**, offsets **18h** through **24h**, respectively)
- If Multicast overlay is enabled, the Multicast overlay translated address hits the NT Port Virtual Interface BARs
- NT Port Virtual Interface **PCI Command** register *Memory Access Enable* bit (offset **04h**[1]) is Set
- NT Port Link Interface **PCI Command** register *Bus Master Enable* bit (offset **04h**[2]) is Set
- Multicast TLP Requester ID hits the NT Port Virtual Interface Requester ID Translation LUT

If any one of these conditions is not met, the Multicast TLP to the NT Port is handled as an *Unsupported Request (UR)*. If a Multicast TLP's Traffic Class (TC) value does not map to the Port's **VC0 Resource Control** register *TC/VC Map* bits (offset **15Ch**[7:1 and 0]), the Multicast TLP is handled as a *Malformed TLP*.

The NT Port receives the Multicast TLP with either an un-translated address or Multicast overlay translated address (based upon the virtual Downstream Port MC_Overlay_Enable (**Multicast BAR0** and **Multicast BAR0** registers, offsets **E28h** and **E2Ch**, respectively). The NT Port does the Address and Requester ID translation, as defined in the NT Port Virtual Interface **Memory BARx Address Translation** registers (offsets **C3Ch** through **C48h**) and **Requester ID Translation LUT** registers (addresses **D94h** through **DD0h**). The translated Multicast TLP is sent out from the NT Port.

8.6.8.2 NT Multicast from NT Port Link-to-Virtual Interface Direction (NT Source) – NT Mode

When the NT Port Link Interface receives a Memory Write TLP, it performs the following checks, before qualifying the TLP as a Multicast Hit:

- TLP hits the NT Port Link Interface BARs (**BAR2** to **BAR5**, offsets **18h** through **24h**, respectively).
- NT Port Link Interface **PCI Command** register *Memory Access Enable* bit (offset **04h[1]**) is Set.
- NT Port Virtual Interface **PCI Command** register *Bus Master Enable* bit (offset **04h[2]**) is Set.
- Virtual Downstream Port **PCI Command** register *Bus Master Enable* bit is Set.
- Received TLP Requester ID hits the NT Port Link Interface Requester ID Translation LUT registers.
- NT Port Link Interface performs Address translation and Requester ID translation, as defined in the NT Port Link Interface **Memory BARx Address Translation** registers (offsets **C3Ch** through **C48h**) and **Requester ID Translation LUT** registers (addresses **DB4h** through **DF0h**). The translated address hits the Multicast BAR in the virtual Downstream's Port **Multicast Extended Capability** structure and satisfies all other Multicast Hit conditions.

When the NT Port Link Interface receives a TLP that is a Multicast Hit, and the Multicast destination Port(s) **PCI Command** register *Memory Access Enable* or *Bus Master Enable* bit (based upon the traffic direction) is not Set, the Multicast TLP is handled as a UR for the corresponding destination Port(s).

When the NT Port Link Interface receives a TLP that is a Multicast Hit, and TLP TC value does not map to the enabled Virtual Channel, the Multicast TLP is handled as a Malformed TLP for the corresponding destination Port(s).

A Multicast TLP received at the NT Port Link Interface is forwarded with address translation and Requester ID translation to the Multicast destination Port(s). Multicast destination Port(s) transmit the TLP, either un-modified or with MC_Overlay address translation (based upon MC_Overlay_Enable).

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Chapter 9 Interrupts

9.1 Interrupt Support

The PEX 8724 supports the PCI Express interrupt model, which uses two mechanisms:

- INTx Interrupt Message-type emulation (compatible with the *PCI r3.0*-defined Interrupt signals)
- Message Signaled Interrupt (MSI), when enabled

For Conventional PCI compatibility, the PCI INTx emulation mechanism is used to signal interrupts to the System Interrupt Controller. This mechanism is compatible with existing PCI software, provides the same level of service as the corresponding PCI interrupt signaling mechanism, and is independent of System Interrupt Controller specifics. The PCI INTx emulation mechanism virtualizes PCI physical Interrupt signals, by using an in-band signaling mechanism, for the assertion and de-assertion of INTx interrupt signals.

In addition to PCI INTx-compatible interrupt emulation, the PEX 8724 supports the MSI mechanism. The PCI Express MSI mechanism is compatible with the MSI Capability defined in the *PCI r3.0*.

INTx and MSIs are mutually exclusive, on a per-Port basis; either can be enabled in a system (depending upon which interrupt type the system software supports), but never concurrently within the same domain. (Refer to the **PCI Command** register *Interrupt Disable* bit, offset 04h[10], and **MSI Control** register *MSI Enable* bit, offset 48h[16], respectively.) The PEX 8724 does not convert received INTx Messages to MSI Messages.

The PEX 8724's external Interrupt output, **PEX_INTA#** (Base mode) or **VSx_PEX_INTA#** (Virtual Switch mode), indicates the assertion and/or de-assertion of the internally generated INTx signal:

- **Non-Hot Plug-triggered interrupts** – PEX_INTA# or VSx_PEX_INTA# assertion is controlled by the following **Error Check Disable** register bits:
 - *Enable Ball Interrupt Output for GPIO-Generated Interrupts* (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 720h[6])
 - **NT mode only**
 - *Enable Ball Interrupt Output for NT-Virtual Doorbell-Generated Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[7])
 - *Enable Ball Interrupt Output for Device-Specific NT-Link Port Event-Triggered Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[5])
 - **Virtual Switch mode only**
 - *Enable Ball Interrupt Output for Management Link Status Event-Generated Interrupts* bit (Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[9])
 - *Enable Ball Interrupt Output for Management Port Doorbell-Generated Interrupts* bit (Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[8])

When any of these bits are Set, Device-Specific errors trigger PEX_INTA# or VSx_PEX_INTA# assertion; however, PEX_INTA# or VSx_PEX_INTA# assertion and INTx Message generation are mutually exclusive, on a per-Port basis.

- **Hot Plug or Link State-triggered INTx events** – PEX_INTA# or VSx_PEX_INTA# assertion is controlled by the **Error Check Disable** register *Enable Ball Interrupt Output for Hot Plug or Link State Event-Triggered Interrupts* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 720h[4]). When this bit is Set, Hot Plug or Link State events trigger PEX_INTA# or VSx_PEX_INTA# assertion; however, an INTx Message is not generated in this case. PEX_INTA# or VSx_PEX_INTA# assertion and INTx Message generation for Hot Plug or Link State cases are mutually exclusive, on a per-Port basis.

The NT Port Virtual and Link Interfaces can each independently support the interrupt mechanism (INTx or MSI) used in their respective domains. (Refer to [Section 14.5, “NT Port Interrupts,”](#) for details.)

9.1.1 Interrupt Sources or Events

The PEX 8724 internally generated interrupt/Message sources include:

- For Hot Plug-capable Ports
 - [Presence Detect Changed](#) (logical OR of PRSNT# (HP_PRSNT_# or Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input), and SerDes Receiver Detect^a on Lane(s) associated with that Port)
 - [Attention Button Pressed](#)
 - [Power Fault Detected](#)
 - [MRL Sensor Changed](#)
 - [Command Completed](#)
 - [Data Link Layer State Changed](#)
 - [Link Bandwidth Management Status](#)
 - [Link Autonomous Bandwidth Status](#)
- For non-Hot Plug-capable Downstream Ports
 - [Presence Detect Changed](#) (SerDes Receiver Detect^a on Lane(s) associated with that Port)
 - [Data Link Layer State Changed](#)
- Device-Specific NT-Link events (also refer to [Section 14.5, “NT Port Interrupts”](#))
 - NT-Link Port Correctable error (NT Port Link Interface, offset FC4h)
 - NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h)
 - NT-Link Port Data Link Layer State change
 - NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message
- General-Purpose Input/Output (GPIO) events
- Non-Transparent (NT) Doorbell events (also refer to [Section 14.5, “NT Port Interrupts”](#))
- Virtual Switch mode
 - Link Status management
 - Doorbells

The PEX 8724 externally generated interrupt/Message sources include INTx Messages from Downstream devices.

[Table 9-1](#) lists the interrupt sources.

a. The SerDes Receiver Detect mechanism is comprised of the [Physical Layer Receiver Detect Status](#) register [Receiver Detected on Lane x](#) bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset [200h\[31:16\]](#)) or Serial Hot Plug [SHP_PRSNT#](#) (SHP_PRSNTx#) (from external I²C I/O Expander) input for the Port.

Table 9-1. Interrupt Sources

Event/Error	Description
Base and Virtual Switch Modes	
Hot Plug or Link State events	Port's Slot Status register (offset 80h): <i>Presence Detect Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, bit 19) is Set <i>Data Link Layer State Changed</i> bit (Downstream Ports, bit 24) is Set
PCI Express Hot Plug events	The master control of Hot Plug interrupts is the Port's Slot Control register <i>Hot Plug Interrupt Enable</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5]). There are six sources of Hot Plug interrupts. Each Hot Plug source has its own <i>Enable</i> bit in the Port's Slot Control register: - Downstream Ports <i>Attention Button Pressed</i> (bit 16) <i>Power Fault Detected</i> (bit 17) <i>Command Completed</i> (bit 20) <i>Data Link Layer State Changed</i> (bit 24) - Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports <i>MRL Sensor Changed</i> (bit 18) <i>Presence Detect Changed</i> (bit 19) The interrupt status of each Hot Plug source is provided by the Port's Slot Status register (Downstream Ports, offset 80h). Note: <i>Presence</i> (Port's Slot Status register <i>Presence Detect State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[22])) is determined by the logical OR of: <i>SerDes Receiver Detect</i> (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), and <i>HP_PRSENT_x#</i>, –or– <i>Serial Hot Plug SHP_PRSENT#</i> (<i>SHP_PRSENTx#</i>) (from external I²C I/O Expander) input for the Port
General-Purpose Input Interrupt events	External interrupt from any of the PORT_GOODx# and GPIOx signals that are configured as an Interrupt input in the GPIO x_y Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and 604h).
Device-Specific Error conditions	Device-Specific (RAM ECC) errors indicated by the Device-Specific Error Status x register bit(s), if not masked in their corresponding Device-Specific Error Mask x register bit(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 700h, 708h, 710h, and/or 718h (Status) and offsets 704h, 70Ch, 714h, and/or 71Ch (Mask)). Note: <i>Device-Specific (RAM ECC) errors can be signaled by interrupt, and/or as an Uncorrectable Internal error that is fatal</i> (Uncorrectable Error Status register <i>Uncorrectable Internal Error Status</i> and Uncorrectable Error Severity register <i>Uncorrectable Internal Error Severity</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets FB8h[22] and FC0h[22], respectively, are Set).

Table 9-1. Interrupt Sources (Cont.)

Event/Error	Description
NT Mode Only	
Device-Specific NT-Link Port errors and events	- NT Port Link Interface Correctable errors reported by the NT Port Virtual Interface (to the Host of the Upstream Port domain), collectively flagged in the Link Error Status Virtual register <i>Link Side Correctable Error Status</i> bit (NT Port Virtual Interface, offset FE0h[0]), with specific errors indicated in the Correctable Error Status register (NT Port Link Interface, offset FC4h), if not masked both globally in the Link Error Mask Virtual register <i>Link Side Correctable Error Mask</i> bit (NT Port Virtual Interface, offset FE4h[0]), nor individually in the Correctable Error Mask register (NT Port Link Interface, offset FC8h). - NT Port Link Interface Uncorrectable errors reported by the NT Port Virtual Interface (to the Host of the Upstream Port domain), collectively flagged in the Link Error Status Virtual register <i>Link Side Uncorrectable Error Status</i> bit (NT Port Virtual Interface, offset FE0h[1]), with specific errors indicated in the Uncorrectable Error Status register (NT Port Link Interface, offset FB8h), if not masked both globally in the Link Error Mask Virtual register <i>Link Side Uncorrectable Error Mask</i> bit (NT Port Virtual Interface, offset FE4h[1]), nor individually in the Uncorrectable Error Mask register (NT Port Link Interface, offset FBCh). - NT Port Link Interface State change – Interrupt to the NT Port Virtual Interface Host, if enabled (not masked) by the Link Error Mask Virtual register <i>Link Side DL Active Change Mask</i> bit (NT Port Virtual Interface, offset FE4h[2]). - Link Side Uncorrectable Error Message Drop interrupt to the NT Port Virtual Interface Host, if enabled (not masked) by the Link Error Mask Virtual register <i>Link Side Uncorrectable Error Message Drop Mask</i> bit (NT Port Virtual Interface, offset FE4h[3]). This feature supports applications using back-to-back NT Ports, where an Uncorrectable Error Message received (and properly dropped) by the NT Port Link Interface can trigger an interrupt to the NT Port Virtual Interface Host.
NT-Virtual Doorbell events	NT Virtual Interface IRQ Set/Clear register (offsets C4Ch[15:0] and/or C50h[15:0]) bit is Set while the corresponding NT Virtual Interface IRQ Set/Clear register (offsets C54h[15:0] and/or C58h[15:0]) bit is Cleared.
NT-Link Doorbell events	NT Link Interface IRQ Set/Clear register (offsets C5Ch[15:0] and/or C60h[15:0]) bit is Set while the corresponding NT Link Interface IRQ Set/Clear register (offsets C64h[15:0] and/or C68h[15:0]) bit is Cleared.

Table 9-1. Interrupt Sources (Cont.)

Event/Error	Description
Virtual Switch Mode Only	
Management Port Doorbell-Generated interrupts	- Writing 1 to the VS Upstream to Management Upstream Doorbell Request register (VS Upstream Port(s) and Management Port, offset 910h[3:0]) signals an interrupt (unless it is masked by the VS Upstream to Management Upstream Doorbell Mask register (VS Upstream Port(s) and Management Port, offset 914h[3:0])) to the Management Port - Writing 1 to the Management Upstream to VS Upstream Doorbell Request register (VS Upstream Port(s) and Management Port, offset 928h[3:0]) signals an interrupt (unless it is masked by the Management Upstream to VS Upstream Doorbell Mask register (VS Upstream Port(s) and Management Port, offset 92Ch[3:0])) to the Management Port Note: If a virtual switch Upstream Port is also the Management Port, that Upstream Port cannot send Doorbell interrupts to itself.
Management Link Status events	- A Link transition from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state Sets the bit that corresponds to the Port Number (Switch Link Up register (Port 0, accessible through the Management Port, offset 900h[10:8, 3:0]), and generates an interrupt (unless it is masked by the Switch Link Event Mask register (Port 0, accessible through the Management Port, offset 908h[10:8, 3:0])) to the Management Port - A Link transition from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state Sets the bit that corresponds to the Port Number (Switch Link Down register (Port 0, accessible through the Management Port, offset 904h[10:8, 3:0]), and generates an interrupt (unless it is masked by the Switch Link Event Mask register (Port 0, accessible through the Management Port, offset 908h[10:8, 3:0])) to the Management Port

9.1.2 Interrupt Handling

The PEX 8724 provides an Interrupt Generation module with each Port. The module reads the Request for interrupts from different sources, then generates an MSI or PCI-compatible Assert_INTx/Deassert_INTx Interrupt Message. MSIs support a PCI Express edge-triggered interrupt, whereas Assert_INTx and Deassert_INTx Message transactions emulate PCI level-triggered interrupt signaling. The System Interrupt Controller functions include:

- Sensing Interrupt events
- Signaling the interrupt, by way of the INTx mechanism, and Setting the *Interrupt Status* bit
- Signaling the interrupt, by way of the MSI mechanism
- Handling INTx-type Interrupt Messages from Downstream devices

9.1.2.1 Interrupt Handling – Base Mode

Base mode supports INTx or MSIs generated by the PEX 8724, as per the *PCI Express Base r3.0*.

One INTA# Interrupt output, [PEX_INTA#](#), is implemented.

9.1.2.2 Interrupt Handling – Virtual Switch Mode

Virtual Switch mode supports INTx or MSIs generated by the PEX 8724, as per the *PCI Express Base r3.0*, within the respective hierarchy.

One INTA# Interrupt output, VSx_PEX_INTA#, is implemented, per virtual switch.

9.2 INTx Emulation Support

The PEX 8724 supports PCI INTx emulation, to signal interrupts to the System Interrupt Controller. This mechanism is compatible with existing PCI software. PCI INTx emulation virtualizes PCI physical Interrupt signals, by using the in-band signaling mechanism.

PCI **Interrupt** registers (defined in the *PCI r3.0*) are supported. The Port's *PCI r3.0* **PCI Command** register *Interrupt Disable* and **PCI Status** register *Interrupt Status* bits are also supported (offset 04h[10 and 19], respectively).

Although the *PCI Express Base r3.0* provides INTA#, INTB#, INTC#, and INTD# for INTx signaling, the PEX 8724 uses only INTA# for internal Interrupt Message generation, because it is a single-function device. However, incoming Messages from Downstream devices can be of INTA#, INTB#, INTC#, or INTD# type. Internally generated INTA# Messages from the Downstream Port are also re-mapped and collapsed at the Upstream Port(s), according to the Downstream Port's Device Number, with its own Device Number and Received Device Number from the Downstream device.

When an interrupt is requested, the Port's **PCI Status** register *Interrupt Status* bit is Set. If INTx interrupts are enabled (Port's **PCI Command** register *Interrupt Disable* and **MSI Control** register *MSI Enable* bits, offsets 04h[10] and 48h[16], are both Cleared, respectively), an Assert_INTx Message is generated and transmitted Upstream to indicate the Port interrupt status. For each interrupt event, there is a corresponding *Interrupt Mask* bit; an Interrupt Message can be generated only when the corresponding *Interrupt Mask* bit is Cleared. Software reads and Clears the event and *Interrupt Status* bit after servicing the interrupt.

A Port de-asserts INTx, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode), interrupts, in response to one or more of the following conditions:

- Port's **PCI Command** register *Interrupt Disable* bit (offset 04h[10]) is Set
- Corresponding *Interrupt Mask* bit is Set
- Upstream Port Link goes down (DL_Down condition), or receives a Hot Reset (unless Hot Reset/DL_Down Reset is disabled, by Setting the **Virtual Switch Debug** register *Upstream Port and NT-Link Port DL_Down Reset Propagation Disable* bit (Upstream Port(s), offset A30h[4]))
- Software Clears the corresponding *Interrupt Status* bit

9.2.1 INTx-Type Interrupt Message Re-Mapping and Collapsing

The Upstream Port(s) re-map(s) and collapse(s) the INTx *virtual wires* received at a Downstream Port, based upon the Downstream Port's Device Number and Received INTx Message Requester ID Device Number, and generate(s) a new Interrupt Message, according to the mapping defined in [Table 9-2](#).

Each virtual PCI-to-PCI bridge of a Downstream Port specifies the Port Number associated with the INTx (Interrupt) Messages received or generated, and forwards the Interrupt Messages Upstream.

A Downstream Port transmits an Assert_INTA/Deassert_INTA Message to the Upstream Port(s), due to:

- Hot Plug- or Link State-triggered interrupt events
- PCI Express Hot Plug-triggered interrupt events
- Device-Specific NT-Link Port (NT mode only) or RAM ECC error events
- GPIO-generated events
- NT Doorbell-generated events (NT mode only)
- Management Port Doorbell events (VS Upstream Ports; Virtual Switch mode only)
- Management Link Status events (VS Upstream Ports; Virtual Switch mode only)

Internally generated INTx Messages always originate as type INTA Messages, because the PEX 8724 is a single-function device. Internally generated Interrupt INTA Messages from Downstream Ports are re-mapped at the Upstream Port(s) to INTA, INTB, INTC, or INTD Messages, according to the mapping defined in [Table 9-2](#).

INTx Messages from Downstream devices and from internally generated Interrupt Messages are ORed together to generate INTA, INTB, INTC, or INTD level-sensitive signals, and edge-detection circuitry in the Upstream Port(s) generates the Assert_INTx and Deassert_INTx Messages. The Upstream Port(s) then forward(s) the new Messages Upstream, by way of its (their) Link.

Table 9-2. Downstream/Upstream Port INTx Interrupt Message Mapping

Device Number ^{a, b}	At Downstream Port	By Upstream Port(s)
0, 8	INTA	INTA
	INTB	INTB
	INTC	INTC
	INTD	INTD
1, 9	INTA	INTB
	INTB	INTC
	INTC	INTD
	INTD	INTA
2, 10	INTA	INTC
	INTB	INTD
	INTC	INTA
	INTD	INTB
3	INTA	INTD
	INTB	INTA
	INTC	INTB
	INTD	INTC

- a. Device 8 is associated with **Software Only** Port 8.
- b. [Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

9.2.1.1 Interrupt Re-Mapping and Collapsing – NT Mode

In NT mode, an NT Port Virtual Interface-generated interrupt is treated like an external event to the PCI-to-PCI bridge immediately Upstream, for tracking purposes. In this mode, when the Upstream Port(s) receive(s) an INTx Message from the NT Port Virtual Interface, the Upstream Port(s)' re-mapping-collapsing logic performs double Swizzling, one based upon the NT Port Virtual Interface's Captured Device Number, and another based upon the virtual Downstream Port (PCI-to-PCI) Device Number.

If software asserts a [Secondary Bus Reset](#) to this PCI-to-PCI bridge, the PCI-to-PCI bridge de-asserts the NT Port Virtual Interface interrupt.

9.3 MSI Support

One of the interrupt schemes supported by the PEX 8724 is the MSI mechanism, which is required for PCI Express devices. The MSI method uses Memory Write transactions to deliver interrupts. MSIs are edge-triggered interrupts.

Note: *MSIs and INTx are mutually exclusive, on a per-Port basis. The mechanisms that generate these types of interrupts **cannot** be concurrently enabled.*

9.3.1 MSI Operation

Note: *The descriptions in this section are applicable only to Transparent Ports. For the NT Port, refer to:*

- [Section 15.7, “NT Port Virtual Interface MSI Capability Registers \(Offsets 48h – 64h\)”](#)
- [Section 16.7, “NT Port Link Interface MSI Capability Registers \(Offsets 48h – 64h\)”](#)

At configuration time, system software traverses the function Capability list. If a **Capability ID** of 05h is found, the function implements MSIs. System software reads the **MSI Capability** structure registers, to determine function capabilities.

The Port’s **MSI Control** register *Multiple Message Capable* field (offset 48h[19:17]) default value is 011b, which indicates that the PEX 8724 can request up to eight MSI Vectors (Address and Data). When the Port’s register’s *Multiple Message Enable* field (offset 48h[22:20]) is Cleared (default), only one Vector is allocated, and therefore, the PEX 8724 can generate only one Vector for all errors or events. When system software writes a non-zero value to the *Multiple Message Enable* field, multiple-Vector support is enabled (the quantity of Vectors supported is dependent upon the value). [Table 9-3](#) lists the supported MSI Vector types.

System software initializes the Port’s MSI Address registers (offsets 4Ch and 50h) and **MSI Data** register (offset 54h), with a system-specified Vector. After system software enables the MSI function (by Setting the Port’s **MSI Control** register *MSI Enable* bit, offset 48h[16]), when an Interrupt event occurs, the Interrupt Generation module generates a DWord Memory Write to the address specified by the Port’s **MSI Address** (lower 32 bits of the *Message Address* field) and **MSI Upper Address** (upper 32 bits of the *Message Address* field) register contents (offsets 4Ch and 50h, respectively). The single DWord Payload includes zero (0) for the upper two bytes, and the lower two bytes are taken from the Port’s **MSI Data** register. The Port’s **MSI Control** register *Multiple Message Enable* field (offset 48h[22:20]) can be programmed to a value of 000b, 001b, 010b, or 011b. When programmed to 011b, the lower three bits of Message data are changed to indicate the general type of interrupt event that occurred. (Refer to [Table 9-3](#).)

Table 9-3. Supported MSI Vector Types

Vector Type	Modes	
	Base	Virtual Switch
Power Management-, or Hot Plug- or Link State-triggered interrupt events	✓	✓
Device-Specific NT-Port Link or RAM ECC error events	✓	✓
GPIO interrupts	✓	✓
Management Link Status events		✓
Management Port Doorbell interrupts		✓

The quantity of MSI Vectors generated is dependent upon the quantity enabled:

- If **one** MSI Vector is enabled (default), all six interrupt events are combined into a single Vector
- If **two** MSI Vectors are allocated, the individual Vectors indicate:
 - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events, GPIO-generated events, Virtual Switch Doorbell events, and Management Port Link Status events
 - Vector[1] Device-specific NT-Link Port events or RAM ECC error events
- If **four** MSI Vectors are allocated, the individual Vectors indicate:
 - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events
 - Vector[1] Device-specific NT-Link Port events or RAM ECC error events
 - Vector[2] GPIO-generated events
 - Vector[3] Virtual Switch Doorbell events and Management Port Link Status events
- If **eight** MSI Vectors are allocated (up to six Vectors are used; the upper two Vectors (having the highest values of Message Data bits [2:0]) are **not** used), the individual Vectors indicate:
 - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events
 - Vector[1] Device-specific NT-Link Port events or RAM ECC error events
 - Vector[2] GPIO-generated events
 - Vector[3] **Reserved**
 - Vector[4] Virtual Switch Doorbell events
 - Vector[5] Management Port Link Status events

If a non-masked Interrupt event occurs before system software Sets the Port's **MSI Control** register *MSI Enable* bit, normally (but unlike Conventional PCI interrupts, which are level-triggered), an MSI packet is sent immediately after software Sets the *MSI Enable* bit, to notify the system of the prior event. Alternatively, MSIs for prior events can be disabled, on a per-Port basis, by Setting the **Error Check Disable** register *Disable Sending MSI if MSI Is Enabled after Interrupt Status Set* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 720h[10]).

When the error or event that caused the interrupt is serviced, the PEX 8724 can generate a new MSI Memory Write as a result of new events. Because MSIs are edge-triggered events, the Port's **MSI Mask** register *Interrupt Mask* bits (offset 58h[5:4, 2:0]) are provided, to use for masking the events. A new MSI can be generated only after the *Interrupt Status* bits are serviced. System software should mask these bits when an MSI event is being processed.

The Port's **MSI Control** register *MSI 64-Bit Address Capable* bit is enabled (offset 48h[23] is Set), by default. If the serial EEPROM and/or I²C/SMBus Clears the bit, the **MSI Capability** structure is reduced by 1 DWord (*that is*, register offsets 54h, 58h, and 5Ch change to 50h, 54h, and 58h, respectively).

9.3.1.1 Message Signaled Interrupts – NT Mode

If NT mode is enabled, NT Port Virtual Interface MSI TLPs are not generated if the **PCI Command** register *Bus Master Enable* bit (offset 04h[2]) is Cleared in the Upstream Port(s), NT Port Virtual Interface, and Virtual Downstream PCI-to-PCI bridge.

9.3.2 MSI Capability Registers

For details, refer to [Section 13.9, “Message Signaled Interrupt Capability Registers \(Offsets 48h – 64h\).”](#)

9.4 PEX_INTA# or VSx_PEX_INTA# Interrupts

PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) Interrupt output is enabled when the following conditions are met:

- INTx Messages are enabled (Port's **PCI Command** register *Interrupt Disable* bit, offset 04h[10], is Cleared) and MSIs are disabled (Port's **MSI Control** register *MSI Enable* bit, offset 48h[16], is Cleared)
- PEX_INTA# or VSx_PEX_INTA# output is enabled for the following errors and events, when the **Error Check Disable** register bit associated with that error or event is Set:
 - *Enable Ball Interrupt Output for GPIO-Generated Interrupts* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 720h[6])
 - *Enable Ball Interrupt Output for Hot Plug or Link State Event-Triggered Interrupts* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 720h[4])
 - **NT mode only**
 - *Enable Ball Interrupt Output for NT-Virtual Doorbell-Generated Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[7])
 - *Enable Ball Interrupt Output for Device-Specific NT-Link Port Event-Triggered Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[5])
 - **Virtual Switch mode only**
 - *Enable Ball Interrupt Output for Management Link Status Event-Generated Interrupts* bit (Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[9])
 - *Enable Ball Interrupt Output for Management Port Doorbell-Generated Interrupts* bit (Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[8])

The three interrupt mechanisms, listed below, are mutually exclusive modes of operation, on a per-Port basis, for all interrupt sources:

- Conventional PCI INTx Message generation
- Native MSI transaction generation
- Device-Specific PEX_INTA# or VSx_PEX_INTA# assertion

PEX_INTA# or VSx_PEX_INTA# assertion (Low) indicates that the PEX 8724 detected one or more of the events and/or errors (if not masked) listed in [Table 9-1](#).

Note: PEX_INTA# or VSx_PEX_INTA# assertion and INTx messaging are mutually exclusive for a given interrupt event. When MSIs are enabled (Port's *MSI Enable* bit is Set), both INTx and PEX_INTA# or VSx_PEX_INTA# are disabled for PEX 8724 internally generated interrupts. The forwarding of external INTx Messages received from a Downstream Port to an Upstream Port is always enabled.

9.5 General-Purpose Input/Output

The PEX 8724 contains 12 multiplexed General-Purpose Input/Output (GPIO) signals. The signals can be used as GPIO, PORT_GOOD, or inputs to generate interrupts (MSI, INTx Message, or [PEX_INTA#](#) (Base mode) or [VSx_PEX_INTA#](#) (Virtual Switch mode)), or Serial Hot Plug [SHP_PERST#](#) pins. The signals/pins can default to the PORT_GOOD output function or Serial Hot Plug SHP_PERST# function, either through register programming and/or the [STRAP_TESTMODE\[2:0\]](#) input states.

In Base mode, the Virtual Switch **GPIO** registers (offsets [64Ch](#) through [65Ch](#) and [A34h](#) through [A54h](#)) are *not* used. In Virtual Switch mode, the Management Port and/or I²C has access to all **GPIO** registers (Chip-specific (offsets [600h](#) through [63Ch](#)), Management Port (offsets [64Ch](#) through [65Ch](#)), and Virtual Switch (offsets [A34h](#) through [A54h](#)) registers). Each virtual switch can access only its own **Virtual Switch** registers.

The 12 GPIO signals are divided among the virtual switches, according to the enabled Ports that are assigned to each virtual switch. The default Settings that determine which signals are assigned to each virtual switch are based upon the Virtual Switch Table registers (Virtual Switch mode – Port 0, accessible through the Management Port, offsets [354h](#) through [38Ch](#)). (Refer to [Section 5.3.3](#), “Virtual Switch Table.”)

The **Virtual Switch GPIO_PG 0_11 Availability** register *Number of GPIO_PGs Available* field (VS Upstream Port(s), offset [A3Ch\[3:0\]](#)) can be read by the Virtual Switch Host, to determine how many GPIO_PG signals are available to that virtual switch. The Management Port can then adjust the quantity of GPIO_PG signals assigned to each virtual switch. A maximum of 12 GPIO_PG signals can be assigned to any virtual switch. A single GPIO_PG signal cannot be assigned to more than one virtual switch.

If the [STRAP_TESTMODE\[2:0\]](#) 3-state inputs call for the Serial Hot Plug SHP_PERST# function by default, two GPIO_SHP pins are assigned to each virtual switch, by default. It is up to the Management Port to make any further assignments of the GPIO_SHP pins to the virtual switches.

The logic that controls the GPIO signal function is driven from the **GPIO** Chip-specific registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets [600h](#) through [63Ch](#)). When a GPIO signal is assigned to a virtual switch, the GPIO information from the **VS GPIO** registers is multiplexed into the corresponding Chip-specific **GPIO** registers. As a result, the **VS GPIO** registers now controls the data in the Chip-specific **GPIO** registers, for the GPIOs assigned to it, which leaves the Chip-specific **GPIO** registers to act as Read-Only registers for the GPIOs that are assigned to virtual switches.

Table 9-4 lists the registers used for GPIO functionality.

Table 9-4. Registers Used for GPIO Functionality

Offset	Register
Chip-Specific Registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)	
600h	GPIO 0_9 Direction Control
604h	GPIO 10_11 Direction Control
614h	GPIO 0_11 Input De-Bounce
61Ch	GPIO 0_11 Input Data
624h	GPIO 0_11 Output Data
62Ch	GPIO 0_11 Interrupt Polarity
634h	GPIO 0_11 Interrupt Status
63Ch	GPIO 0_11 Interrupt Mask
Virtual Switch Management Port Registers (Virtual Switch mode – Port 0, accessible through the Management Port)	
64Ch	Virtual Switch GPIO Update
650h	VS0 GPIO_PG 0_11 Assignment
654h	VS1 GPIO_PG 0_11 Assignment
658h	VS2 GPIO_PG 0_11 Assignment
65Ch	VS3 GPIO_PG 0_11 Assignment
Virtual Switch Registers (Virtual Switch mode – VS Upstream Port(s))	
A34h	Virtual Switch GPIO_PG 0_9 Direction Control
A38h	Virtual Switch GPIO_PG 10_11 Direction Control
A3Ch	Virtual Switch GPIO_PG 0_11 Availability
A40h	Virtual Switch GPIO_PG 0_11 Input De-Bounce
A44h	Virtual Switch GPIO_PG 0_11 Input Data
A48h	Virtual Switch GPIO_PG 0_11 Output Data
A4Ch	Virtual Switch GPIO_PG 0_11 Interrupt Polarity
A50h	Virtual Switch GPIO_PG 0_11 Interrupt Status
A54h	Virtual Switch GPIO_PG 0_11 Interrupt Mask

9.6 Management Port Interrupts – Virtual Switch Mode

The CPU connected to the Management Port receives interrupts for the following two events:

- Switch Ports Link Status
- Doorbell

Both are described in the sections that follow.

9.6.1 Switch Port Link Status Events – Virtual Switch Mode

The Management Port has four registers that provide Link status to the Management CPU (located in Port 0, accessible through the Management Port):

- **Switch Link Up** register (offset 900h)
- **Switch Link Down** register (offset 904h)
- **Switch Link Event Mask** register (offset 908h)
- **Switch Link Status** register (offset 90Ch)

In all four registers, each non-**Reserved** bit corresponds to one Port (for example, bit 0 corresponds to Port 0, bit 1 corresponds to Port 1, and so forth).

When a Port Link goes to an active state (`DL_ACTIVE=1`) from a down state, it Sets the Port's **Switch Link Up** register *Port x Link Up* bit (offset 900h[10:8, 3:0]), regardless of the Port's **Switch Link Event Mask** register *Port x Link Event Mask* bit (offset 908h[10:8, 3:0]) value. If the interrupt is not masked, the Management Port interrupt handler signals an Assert_INTA Message to the Management CPU if Conventional PCI interrupts (INTx) are enabled and MSIs are disabled. If instead MSIs are enabled, the Management Port generates an MSI Message instead of a Conventional PCI Assert_INTA Message. When the Interrupt Service Routine. (ISR) services this interrupt, it writes 1 to the Port's **Switch Link Up** register *Port x Link Up* bit, to Clear this event. This Clear event or Interrupt Mask event generates a Deassert_INTA Message to the Management CPU, if all Interrupt events in the Port's **Switch Link Up** and **Switch Link Down** registers are Cleared, provided Conventional PCI interrupts are enabled and MSIs are disabled.

When a Port Link goes to a down state (`DL_ACTIVE=0`) from an active state, it Sets the Port's **Switch Link Down** register *Port x Link Down* bit (offset 904h[10:8, 3:0]), regardless of the Port's **Switch Link Event Mask** register *Port x Link Event Mask* bit (offset 908h[10:8, 3:0]) value. If the interrupt is not masked, the Management Port interrupt handler signals an Assert_INTA Message to the Management CPU, if Conventional PCI interrupts (INTx) are enabled and MSIs are disabled. If instead MSIs are enabled, the Management Port generates an MSI Message instead of a Conventional PCI Assert_INTA Message. When the ISR services this interrupt, it writes 1 to the Port's **Switch Link Down** register *Port x Link Down* bit, to Clear this event. This Clear event or Interrupt Mask event generates a Deassert_INTA Message to the Management CPU, if all Interrupt events in the Port's **Switch Link Up** and **Switch Link Down** registers are Cleared, provided Conventional PCI interrupts are enabled and MSIs are disabled.

9.6.1.1 Special Handling for Race Conditions – Virtual Switch Mode

If multiple DL_ACTIVE and DL_INACTIVE events occur before the ISR is able to service the Interrupt event, a race condition exists with event ordering. The Management Port implements the **Switch Link Status** register (Port 0, accessible through the Management Port, offset 90Ch) for this purpose. If the Port's bits are Set in both the **Switch Link Up** and **Switch Link Down** registers (Port 0, accessible through the Management Port, offsets 900h and 904h, respectively), the ISR looks at the **Switch Link Status** register, to determine the order of these two events:

- If a **Switch Link Status** register bit is Cleared, the first event is DL_ACTIVE and the latest event is DL_INACTIVE
- If a bit in the **Switch Link Status** register is Set, the first event is DL_INACTIVE and the latest event is DL_ACTIVE

9.6.2 Doorbell Interrupts – Virtual Switch Mode

In Virtual Switch mode, each virtual switch Upstream Port (other than the active Management Port) implements **Doorbell** and **Scratchpad** registers for communication (located in the VS Upstream Port(s) and Management Port):

- Virtual switch to Management CPU direction
 - **VS Upstream to Management Upstream Doorbell Request** register (offset 910h)
 - **VS Upstream to Management Upstream Doorbell Mask** register (offset 914h)
 - **VS Upstream to Management Upstream Scratchpad 1** register (offset 918h)
 - **VS Upstream to Management Upstream Scratchpad 2** register (offset 91Ch)
 - **VS Upstream to Management Upstream Scratchpad 3** register (offset 920h)
 - **VS Upstream to Management Upstream Scratchpad 4** register (offset 924h)
- Management CPU to virtual switch direction
 - **Management Upstream to VS Upstream Doorbell Request** register (offset 928h)
 - **Management Upstream to VS Upstream Doorbell Mask** register (offset 92Ch)
 - **Management Upstream to VS Upstream Scratchpad 1** register (offset 930h)
 - **Management Upstream to VS Upstream Scratchpad 2** register (offset 934h)
 - **Management Upstream to VS Upstream Scratchpad 3** register (offset 938h)
 - **Management Upstream to VS Upstream Scratchpad 4** register (offset 93Ch)

Software uses these registers to establish communication between the active Management CPU and the other CPUs. There is no in-band communication mechanism between CPUs outside the active Management Port domain.

A Fundamental Reset resets all the registers listed above.

A Hot Reset to the Management Port domain Clears the **VS Upstream to Management Upstream Doorbell Request**, **VS Upstream to Management Upstream Doorbell Mask**, and **VS Upstream to Management Upstream Scratchpad x** registers in the Virtual Switch to Management CPU direction.

A Hot Reset to virtual switches other than the Management Port domain Clears the **Management Upstream to VS Upstream Doorbell Request**, **Management Upstream to VS Upstream Doorbell Mask**, and **Management Upstream to VS Upstream Scratchpad x** registers in the Management CPU to Virtual Switch direction.

Software writing to either **Doorbell Request** register generates an interrupt if the corresponding **Doorbell Mask** register bit is not masked, and interrupt signaling is enabled.



Chapter 10 Hot Plug Support

10.1 Introduction

Note: In this chapter, unless stated otherwise, “Hot Plug Controller” references both the Parallel Hot Plug Controllers and Serial Hot Plug Controller.

Hot Plug capability allows board insertion and removal from a running system, without adversely affecting the system. Boards are typically inserted or removed to repair faulty boards or re-configure the system without system down time. Hot Plug capability allows systems to isolate faulty boards in the event of a failure.

The PEX 8724 includes one Parallel Hot Plug Controller per Hot Plug-capable Transparent Downstream Port and one Serial Hot Plug Controller per PEX 8724, as well as signals for both Parallel and Serial Hot Plug support. The quantity of Parallel Hot Plug Ports depends upon the [STRAP_TESTMODE\[2:0\]](#) and [VS_MODE](#) input states, and can be up to three (A, B, and C). All Transparent Downstream Ports can support Serial Hot Plug.

Parallel Hot Plug can be implemented on any Transparent Downstream Port, as selected by the **Parallel Hot Plug Control** register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset [3A4h](#)[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively).

Serial Hot Plug can be implemented on any Transparent Downstream Port. If a Transparent Downstream Port is both Parallel- and Serial Hot Plug-capable, the Serial Hot Plug Controller is used, by default, unless the Port’s **Power Management Hot Plug User Configuration** register [Serial Hot Plug Override Parallel Disable](#) bit (Downstream Ports, offset [F70h](#)[19]) is Set.

Hot Plug signals are enabled, configured, and accessed through the Port’s **Slot Capability** and **Slot Status and Control** registers (Downstream Ports, offsets [7Ch](#) and [80h](#), respectively). Also, each Port’s **Power Management Hot Plug User Configuration** register provides additional Device-Specific configuration and control, for both Parallel and Serial Hot Plug implementations.

Normally, Hot Plug cannot be implemented on PEX 8724 Upstream Port(s). However, as an option, Presence Detect and its corresponding interrupt can be enabled for an Upstream Port, by Setting the Port’s **Power Management Hot Plug User Configuration** register [Upstream Hot Plug Enable](#) bit (Upstream Port(s), offset [F70h](#)[14]).

10.2 Hot Plug Features

The following are the PEX 8724 Hot Plug features:

- Hot Plug features are supported on all Transparent Downstream Ports.
- Zero to three Parallel Hot Plug Ports (one Parallel Hot Plug Port, per Station), depends upon the [STRAP_TESTMODE\[2:0\]](#) and [VS_MODE](#) input states.
- Any three Transparent Downstream Ports can be programmed as a Parallel Hot Plug Port.
- Additional Hot Plug signals for all other Transparent Downstream Ports (that are not being used as a Parallel Hot Plug Port, up to five) are implemented with external I²C I/O Expanders, which alert the PEX 8724 through the [SHPC_INT#](#) input, that inputs have toggled, and the PEX 8724 internal Serial Hot Plug Controller and registers automatically communicate with and control the I/O Expanders, using the PEX 8724 I²C Master interface ([I2C_SCL1](#) and [I2C_SDA1](#) lines).
- Insertion and removal of PCI Express boards, without removing system power.
- Board Present and Manually operated Retention Latch (MRL) signals are implemented. Presence Detect is accomplished through either an in-band SerDes Receiver Detect mechanism ([Physical Layer Receiver Detect Status](#) register [Receiver Detected on Lane x](#) bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset [200h\[31:16\]](#))) –or– by using the [HP_PRSENT_x#](#) inputs.
- Power Indicator and Attention Indicator Output signals are controlled.
- Attention Button is monitored.
- Power Fault detection and Faulty board isolation.
- Power switch for controlling Downstream device power. In Parallel Hot Plug Ports, polarity of this output is defined by the [SPARE1](#) input sampled at [PEX_PERST#](#) input de-assertion. In Serial Hot Plug Ports, this output is Active-High.
- Generates Power Management Event (PME) for Hot Plug events in sleeping systems (D3hot Device Power Management (PM) state).
- Electromechanical Interlock Control feature available on Serial Hot Plug-capable Ports.
- Hot Plug interrupts can be sent in-band using [INTx](#) or MSI Messages, or signaled externally using [PEX_INTA#](#) (Base mode) or [VSx_PEX_INTA#](#) (Virtual Switch mode).

10.3 Hot Plug Elements

Table 10-1 summarizes the Hot Plug elements required for PCI Express Hot Plug implementation. For specific platform requirements, refer to the PCI Express Form Factor specifications.

Table 10-1. Required Hot Plug Elements for PCI Express Implementation

Element	Purpose
Attention Button	To request Hot Plug operations. Implemented on the PEX 8724's Hot Plug-capable Transparent Downstream Ports.
Attention Indicator	Implemented on the PEX 8724's Hot Plug-capable Transparent Downstream Ports. LED functions: • Off – Standard operation. • On – Operational Problem at this slot. • Blinking – Slot is being identified at user's request. Blinking frequency is 1 Hz. 50% duty cycle.
Power Indicator	Implemented on the PEX 8724's Hot Plug-capable Transparent Downstream Ports. LED functions: • Off – Slot is powered Off. Board insertion or removal is permitted. • On – Board insertion or removal is not permitted. • Blinking – Slot is in the process of powering Up or Down. Blinking frequency is 2 Hz. 50% duty cycle.
MRL	Manually-operated Retention Latch, that holds add-in boards in place.
MRL Sensor	Reports the position of a slot's MRL to the Port. A logic Low indicates that the latch is closed.
Electromechanical Interlock	Prevents removal of adapter from slot.

10.4 Hot Plug Signals

10.4.1 Hot Plug Port External Signals

The on-chip signals for Parallel Hot Plug Controller support are defined in [Section 3.4.2.1, “Parallel Hot Plug Signals.”](#)

The on-chip signals for Serial Hot Plug Controller support are defined in [Section 3.4.2.2, “Serial Hot Plug Signals.”](#) In addition to the set of on-chip Serial Hot Plug signals, the PEX 8724 supports Serial Hot Plug signals to and from the I²C I/O Expander, which are used with Serial Hot Plug-capable Transparent Downstream Ports. (Refer to [Section 10.11.3.](#))

10.4.2 Hot Plug Output States for Disabled Hot Plug Slots

When a Hot Plug slot is disabled, the Hot Plug outputs for that Port are in the logic states defined in [Table 10-2.](#)

Table 10-2. Hot Plug Outputs for Disabled Hot Plug Slot

Output Signal	Logic	Comments
HP_ATNLED_x#	High	Attention LED is turned Off
HP_CLKEN_x#	High	Reference Clock is not driven to the slot
HP_PERST_x#	Low	Slot remains in reset
HP_PWREN_x	Low	Power Controller is turned Off
HP_PWRLED_x#	High	Power LED is turned Off

10.4.3 Software Control of Hot Plug Output Signals

For Hot Plug operation, typically the Hot Plug output signal states can be determined and controlled through the Port's **Slot Control** register (Downstream Ports, offset 80h). Toggling the Slot Control register can generate a Command Complete interrupt.

The Hot Plug Controller additionally includes a Device-Specific mechanism for directly reading and writing Hot Plug Output signal states. If software Sets the Port's **Power Management Hot Plug User Configuration** register *HPC I/O Reload* bit (offset F70h[20]), the Hot Plug Output signals can be directly controlled by this same register Write, to the register's *HPC Output Reload Value* field [26:21], without generating a Command Complete interrupt. The mapping of these register bits to Hot Plug signals is listed in Table 10-3.

Table 10-3. Power Management Hot Plug User Configuration Register *HPC Output Reload Value* Field (Offset F70h[26:21]) Mapping when Bit 20 Is Set

Bit	Hot Plug Output Signal
21	HP_PWRLED_x# or Serial Hot Plug <i>SHP_PWRLED#</i> (SHP_PWRLEDx#)
22	HP_ATNLED_x# or Serial Hot Plug <i>SHP_ATNLED#</i> (SHP_ATNLEDx#)
23	HP_PWREN_x or Serial Hot Plug <i>SHP_PWREN</i> (SHP_PWRENx)
24	HP_CLKEN_x# or Serial Hot Plug <i>SHP_REFCLKEN#</i> (SHP_REFCLKENx#)
25	HP_PERST_x# or Serial Hot Plug <i>SHP_PERST#</i> (SHP_PERSTx#)
26	Serial Hot Plug <i>SHP_INTERLOCK</i> (SHP_INTERLOCKx)

10.5 Hot Plug Registers

All Transparent Downstream Ports and Stations include identical sets of Hot Plug registers, and all Hot Plug Ports use the identical register sets, regardless of whether Hot Plug is implemented using the PEX 8724 Hot Plug signals, or Serial Hot Plug signals on the external I²C I/O Expanders. Therefore, other than initial configuration (typically programmed by serial EEPROM), whether Hot Plug functionality for a Port is implemented using a Parallel Hot Plug Controller or Serial Hot Plug Controller (with external I²C I/O Expander) is effectively transparent to software.

The PCI Express Hot Plug Configuration, Capability, Command, Status, and Event registers are described in [Section 13.10, “PCI Express Capability Registers \(Offsets 68h – A0h\).”](#)

Device-Specific Hot Plug configuration features are programmable in the **Power Management Hot Plug User Configuration** register (offset F70h) of each Station and Transparent Downstream Port.

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10.6 Hot Plug Interrupts

Refer to [Chapter 9, “Interrupts,”](#) for interrupt details.

10.6.1 Software Testing of Hot Plug Interrupts

Hot Plug interrupts can be generated by software (such as for testing Interrupt Handler software), without having to toggle Hot Plug signals to trigger interrupts. The Downstream Port **Slot Status** register bits at offset 80h[24:16] are usually Read Only (RO; as required by the *PCI Express Base r3.0*); however, if the Port's **Power Management Hot Plug User Configuration** register *Software-Controlled Hot Plug Enable* bit (offset F70h[12]) is Set, the Port's Hot Plug input signals (from either Parallel or Serial Hot Plug sources for BUTTON#, PWRFLT#, MRL#, PRSNT#, and INTERLOCK#) are disconnected from the pins, and software is given control of the corresponding *Status* register bits.

When this feature is enabled, the following **Slot Status** register (Downstream Ports, offset 80h) *Status* bits for the Port can be Set by software writing 1 to these bit positions, but only if the *Status* bit value is already 0:

- For BUTTON#, *Attention Button Pressed* bit (bit 16)
- For PWRFLT#, *Power Fault Detected* bit (bit 17)
- For INTERLOCK#, *Electromechanical Interlock Status* bit (bit 23)
- *Command Completed* bit (bit 20)

If the bit is already Set, writing 1 Clears the *Status* bit (same as in standard operation).

Software **cannot** write 1 to directly change the Port's **Slot Status** register *MRL Sensor Changed* and *Presence Detect Changed* bits. However, software can write to the following **Slot Status** register bits, if the software Write to the corresponding *Status* bits changed the original value:

- Writes to the *MRL Sensor State* bit (bit 21) change the register's *MRL Sensor Changed* bit (bit 18)
- Writes to the *Presence Detect State* bit (bit 22) change the register's *Presence Detect Changed* bit (bit 19)

During software testing, the Port's **Power Management Hot Plug User Configuration** register *Software-Controlled Power Good* bit (offset F70h[13]) replaces the HP_PWR_GOOD_x inputs.

If software Sets the Port's **Slot Status** register *Power Fault Detected* bit (Downstream Ports, offset 80h[17]) when the bit is Cleared, the Port's HP_PERST_x# or Serial Hot Plug *SHP_PERST#* (*SHP_PERSTx#*) output is asserted for Parallel Hot Plug (**Power Management Hot Plug User Configuration** register *Port Is Serial Hot Plug Port* bit, offset F70h[15], is Cleared) or Serial Hot Plug (*Port Is Serial Hot Plug Port* bit is Set), respectively.

If software changes the MRL# state to 1 (open), that Port executes a Power Down sequence on its slot.

10.7 Hot Plug and Virtual Switch Mode Failover

In Virtual Switch mode, a Downstream Port belonging to a failed virtual switch can be moved to another virtual switch that is up and running. A Transparent Downstream Port that supports Parallel Hot Plug continues to support Parallel Hot Plug when that Port is re-assigned to another virtual switch. Likewise, a Downstream Port that does **not** support Parallel Hot Plug remains a non-Hot Plug Port when that Port is re-assigned to another virtual switch. The PEX 8724 does **not** change the **Parallel Hot Plug Control** register (Port 0, accessible through the Management Port, offset 3A4h[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively) value when the Port is re-assigned to a different virtual switch.

10.7.1 Virtual Switch Mode Failover and Upstream Port Hot Plug Functionality

To inform software of the addition of a bridge in failover while the PEX 8724 is in Virtual Switch mode, the Upstream Port Hot Plug (Device-Specific) functionality can be enabled, by Setting the Port's **Power Management Hot Plug User Configuration** register *Upstream Hot Plug Enable* bit (Upstream Port(s), offset F70h[14]). This feature enables the MRL Sensor and Presence Detect-related register bits, and generates the corresponding Hot Plug interrupts, using the Upstream Port(s)' **Slot Status and Control** register (offset 80h, which is usually **Reserved** in Upstream Ports).

The Hot Plug-related bits that can be enabled for the Upstream Port in the **Slot Status and Control** register are:

- *MRL Sensor Changed Enable* (bit 2)
- *Presence Detect Changed Enable* (bit 3)
- *Hot Plug Interrupt Enable* (bit 5)
- *MRL Sensor Changed* (bit 18)
- *Presence Detect Changed* (bit 19)
- *MRL Sensor State* (bit 21)
- *Presence Detect State* (bit 22)

If the *Upstream Hot Plug Enable* bit is Set, software can indirectly program the *MRL Sensor State* and *Presence Detect State* bit values, by writing to the Port's **Power Management Hot Plug User Configuration** register *Software MRL State Value* and *Software Present Detect State Value* bits (Upstream Port(s), offset F70h[28:27], respectively). The *MRL Sensor Changed* and *Presence Detect Changed* bits are updated accordingly, and Hot Plug interrupts can be generated (if enabled), to indicate the addition of a bridge behind the Upstream Port.

10.8 Hot Plug Controller Slot Power-Up/Down Sequence

If a Hot Plug-capable Transparent Downstream Port is enabled, the Port's Hot Plug Controller can power-up or power-down the slot. This section describes how this process occurs.

10.8.1 Slot Power-Up Sequence

If a Hot Plug-capable Transparent Downstream Port is connected to a slot, its associated Hot Plug Controller can power up that slot, with or without an external serial EEPROM. Hot Plug Controller sequencing is determined by the states of the following bits:

- Port's **Slot Capability** register *Power Controller Present* bit (Downstream Ports, offset 7Ch[1])
- Port's **Slot Capability** register *MRL Sensor Present* bit (Downstream Ports, offset 7Ch[2])
- Port's **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10])

and the HP_MRL_x# input states, if the *MRL Sensor Present* bit is Set. Hot Plug-configurable features are programmable only by the serial EEPROM and/or I²C.

10.8.1.1 Configuring Slot Power-Up Sequence Features with Serial EEPROM

An external serial EEPROM can be used to configure the Hot Plug Controller and Hot Plug outputs. Features can be changed by using the registers defined in [Table 10-4](#). The Hot Plug Controller outputs remain in the default state described in [Table 10-2](#), before the serial EEPROM image is loaded into the device.

After the serial EEPROM image is loaded, the Hot Plug Controller starts a power-up sequence on each slot that has the Port's **Slot Capability** register *Power Controller Present* bit (Downstream Ports, offset 7Ch[1]) Set and **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10]) Cleared.

Table 10-4. Configuring Slot Power-Up Sequence Features with Serial EEPROM

Register Bit(s) ^a	Hot Plug Controller and Hot Plug Output Signal Configurable Features
<i>Power Controller Present</i> Slot Capability register, offset 7Ch[1]	Reserved for the Upstream Port(s) and NT Port. The <i>Power Controller Present</i> bit enables or disables the Hot Plug Controller on the Hot Plug-capable Transparent Downstream Ports. If this bit is Cleared, the Hot Plug Controller is disabled for that slot and a power-up sequence is not executed. The slot remains in the disabled state, as defined in Table 10-2 . If this bit is Set, the Hot Plug Controller powers up the slot when the MRL is closed and the Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10]) is Cleared. Otherwise, if the Port's Slot Capability register <i>MRL Sensor Present</i> bit (Downstream Ports, offset 7Ch[2]) is Cleared, the MRL's position has no effect on powering up the slot.
<i>MRL Sensor Present</i> Slot Capability register, offset 7Ch[2]	Reserved for the Upstream Port(s) and NT Port. If this bit is Cleared, the MRL position is "Don't Care," and has no effect on powering up the slot. If this bit is Set, the PEX 8724 senses whether the MRL is open or closed for a slot. The MRL should be Low for power-on for that slot.
<i>Attention Indicator Present</i> Slot Capability register, offset 7Ch[3]	Reserved for the Upstream Port(s) and NT Port. When Set, this bit controls whether the HP_ATNLED_x# output for the slot drives out Active-Low. Otherwise, this output is not functional on the slot.
<i>Power Indicator Present</i> Slot Capability register, offset 7Ch[4]	Reserved for the Upstream Port(s) and NT Port. When Set, this bit controls whether the HP_PWRLLED_x# output for the slot drives out Active-Low. Otherwise, this output is not functional on the slot.
<i>HPC Tpepv</i> Power Management Hot Plug User Configuration register, offset F70h[4:3]	Functionality associated with this field is enabled only on Downstream Ports. This field indicates the delay from when an HP_PWREN_x output is asserted High, to when power is valid at a slot. (Refer to Section 10.8.1.2) 00b = Feature is disabled, and HP_PWR_GOOD_x inputs are used for Power Valid 01b = 128 ms 10b = 256 ms 11b = 512 ms
<i>HP_PWR_GOOD_x Active-Low Enable</i> Power Management Hot Plug User Configuration register, offset F70h[6]	Functionality associated with this bit is enabled only on Downstream Ports. Controls the HP_PWR_GOOD_x input polarity. (Refer to Section 10.8.1.2) 0 = HP_PWR_GOOD_x inputs are Active-High 1 = HP_PWR_GOOD_x inputs are Active-Low

a. All register bits listed in this table are located in the Downstream Ports.

10.8.1.2 Slot Power-Up Sequencing when *Power Controller Present* Bit Is Set

By default, when the MRL is open, the Port's **Slot Capability** register *MRL Sensor Present* and *Power Controller Present* bits (Downstream Ports, offset 7Ch[2:1], respectively) and **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10]) are Set on a Hot Plug-capable Downstream Port. When the serial EEPROM is not present, present but blank, or programmed with default register values, the Hot Plug Controller is initially powered up, the Port's **PCI Express Capability** register *Slot Implemented* bit (Downstream Ports, offset 68h[24]) is Set, and the PEX 8724 is in the following state:

1. Hot Plug Controller is enabled for Hot Plug-capable Transparent Downstream Port.
2. Slots associated with Hot Plug-capable Transparent Downstream Ports are enabled to be powered up.
3. Attention and Power LEDs (HP_ATNLED_x# and HP_PWRLED_x# outputs, respectively) are High on the slot chassis.

Immediately after the PEX 8724 exits Reset (PEX_PERST# and/or VSx_PERST# input goes High), if the Hot Plug-capable Transparent Downstream Port's *MRL Sensor Present* bit is Set (default), the HP_MRL_x# input for that slot is sampled. If HP_MRL_x# input is enabled and asserted (value of 0), the device Clears the *Power Controller Control* bit, to enable slot power-up. If the *Power Controller Control* bit is not Cleared, either by initially enabling it (default) and asserting HP_MRL_x#, or by programming both the *MRL Sensor Present* and *Power Controller Control* bit values to 0 in the serial EEPROM, the Downstream slot is not powered up and remains in the disabled state, as defined in Table 10-2.

If a slot's *Power Controller Present* bit is Set, and the *Power Controller Control* bit is Cleared (either by initially enabling and asserting HP_MRL_x#, or by programming the *MRL Sensor Present* and *Power Controller Control* bits to 0 in the serial EEPROM), the slot starts power-up sequencing with HP_PWREN_x and HP_PWRLED_x# assertion, following PEX_PERST# and/or VSx_PERST# input de-assertion and serial EEPROM initialization. The serial EEPROM initialization delay is determined by the following:

- Serial EEPROM clock (EE_SK) frequency, programmable through the **Serial EEPROM Clock Frequency** register *EepFreq[2:0]* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[2:0])
- Quantity of registers that are programmed to be initialized by the serial EEPROM

The power-up sequence is as follows:

1. The Hot Plug Controllers drive the HP_PWRLED_x# outputs Low, to turn On the Power Indicators, and drives the HP_PWREN_x outputs High, to turn On the external Power Controllers.
2. After the HP_PWR_GOOD_x inputs are sampled asserted High or T_{pepv} delay following HP_PWREN_x output assertion, power to the slots are valid and the Hot Plug Controllers drive the HP_CLKEN_x# outputs Low, to turn On the Reference Clock (PEX_REFCLKp/n) to the slots.

The T_{pepv} time delay is specified by programming the Port's **Power Management Hot Plug User Configuration** register *HPC Tpepv* field (Downstream Ports, offset F70h[4:3]) to a non-zero value. Values of 01b, 10b, or 11b program the delay to 128, 256, or 512 ms, respectively. The default value, 00b, disables the feature, and uses the HP_PWR_GOOD_x inputs instead.

3. After the 100-ms T_{pvperl} time delay following HP_CLKEN_x# output assertion, the Hot Plug Controllers de-assert the HP_PERST_x# outputs, to release slot reset.

Consideration should be given to the combination of the serial EEPROM clock (*EE_SK*) frequency (programmable through the **Serial EEPROM Clock Frequency** register *EepFreq[2:0]* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[2:0])), along with the quantity of registers to be initialized by serial EEPROM, as well as any delay for cascaded resets through multiple devices, and allow sufficient margin for devices to be ready for Host enumeration.

Figure 10-1 illustrates the timing sequence with the Port's **Slot Capability** register *Power Controller Present* bit (Downstream Ports, offset 7Ch[1]) Set. This timing sequence occurs at system power-up, or when a slot is being powered up by the user, using software control.

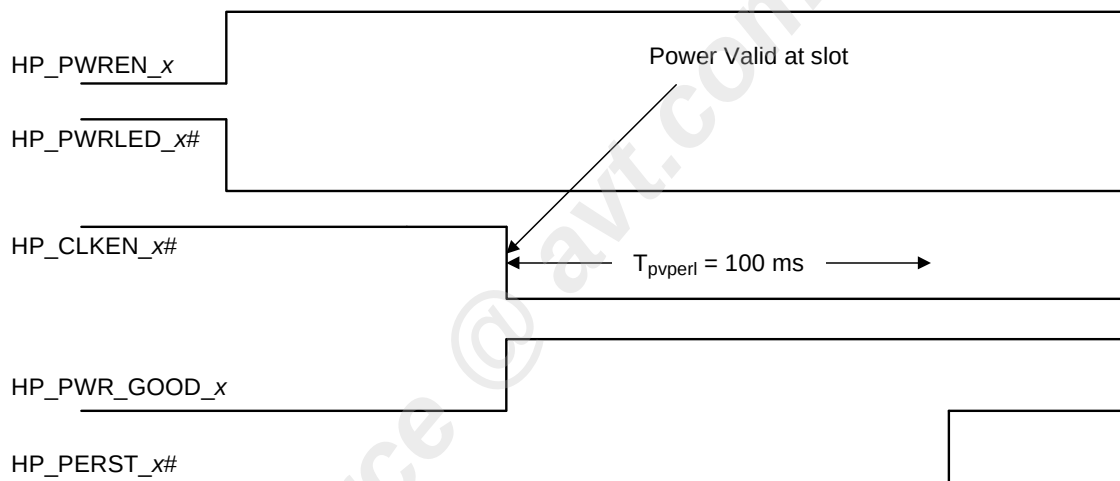
If the HP_MRL_x# inputs are enabled, but not asserted to power-up the slots immediately after reset, the HP_MRL_x# inputs can be asserted at runtime to start the slot power-up sequence, provided that the Port's **Slot Capability** register *MRL Sensor Present* and *Power Controller Present* bits (Downstream Ports, offset 7Ch[2:1], respectively) are both Set (either by default values when the serial EEPROM is not present or blank, or by programming the serial EEPROM to Set these bits), and the Port's **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10]) is Cleared (by the programmed serial EEPROM and/or software).

Power-up sequencing at runtime is controlled by software Clearing the *Power Controller Control* bit, in response to an interrupt caused by one or both of the following:

- HP_MRL_x# input assertion (if an MRL Sensor is present, and the Port's **Slot Control** register *Hot Plug Interrupt Enable* and *MRL Sensor Changed Enable* bits (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[5 and 2], respectively) are both Set)
- User pressing the Attention Button, if enabled (Port's register's *Attention Button Pressed Enable* (Downstream Ports, offset 80h[0]) and *Hot Plug Interrupt Enable* bits must both be Set)

HP_MRL_x# and HP_BUTTON_x# input assertion and de-assertion at runtime are not latched until the 10-ms de-bounce ensures that their state changes are stable.

Figure 10-1. Slot Power-Up Timing when *Power Controller Present* Bit Is Set



Note: HP_PWRLED_x# is **not** asserted if the serial EEPROM and/or I²C Slave interface Clears the Port's **Slot Capability** register *Power Indicator Present* bit (Downstream Ports, offset 7Ch[4]).

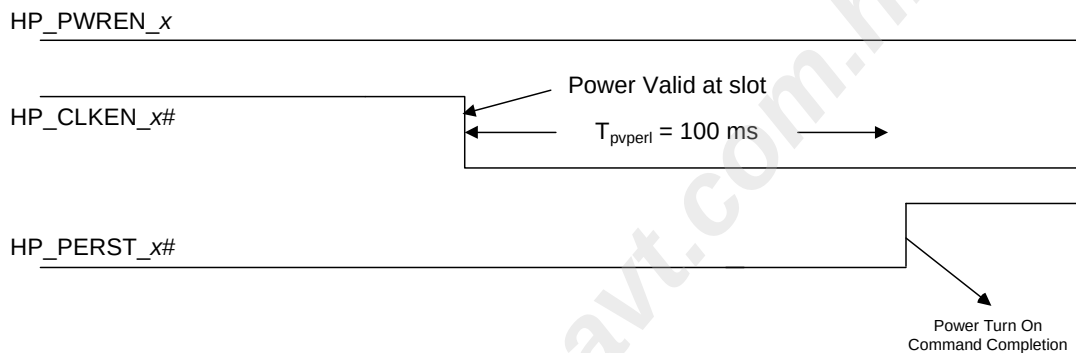
10.8.1.3 HP_PERST_x# (Reset) and HP_PWRLED_x# Output Power-Up Sequencing when *Power Controller Present* Bit Is Cleared

The HP_PERST_x# and HP_PWRLED_x# outputs can be used without enabling the Hot Plug Power Controllers (HP_PWREN_x and HP_CLKEN_x# outputs and HP_PWRFLT_x# inputs). *For example*, the HP_PERST_x# outputs can be used to reset an on-board Downstream device.

If the Port's **Slot Capability** register *Power Controller Present* bit (Downstream Ports, offset 7Ch[1]) and **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10]) are both Cleared by the serial EEPROM, the HP_PERST_x# outputs are de-asserted (High) and the HP_PWRLED_x# outputs are asserted (Low), after the Root Complex PERST# input is de-asserted, as illustrated in Figure 10-2. However, the HP_PWRLED_x# outputs are **not** asserted, if the serial EEPROM also Cleared the Port's **Slot Capability** register *Power Indicator Present* bit (Downstream Ports, offset 7Ch[4]).

If the serial EEPROM is initially blank, causing register default values to be loaded, the HP_PERST_x# outputs are asserted and the HP_PWRLED_x# outputs are **not** asserted, unless the HP_MRL_x# inputs are Low. Therefore, if the HP_PERST_x# and/or HP_PWRLED_x# outputs are used (and an MRL is **not** used), pull the HP_MRL_x# inputs Low, to allow the outputs to toggle, regardless of whether the serial EEPROM is blank.

Figure 10-2. Hot Plug Outputs when *Power Controller Present* and *Power Controller Control* Bits Are Cleared



Note: HP_PWRLED_x# are **not** asserted if the serial EEPROM and/or I²C Slave interface Clears the Port's Power Indicator Present bit.

10.8.1.4 Disabling Power-Up Hot Plug Output Sequencing

If the serial EEPROM is used to Set the Port's **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10]), the HP_PWRLED_x# and HP_CLKEN_x# outputs remain High, and the HP_PERST_x# and HP_PWREN_x outputs remain Low. The HP_PWRLED_x# and HP_CLKEN_x# outputs also remain High, if the HP_MRL_x# inputs are not asserted in the default Hot Plug power-up sequencing described in [Section 10.8.1.2](#).

10.8.2 Slot Power-Down Sequence

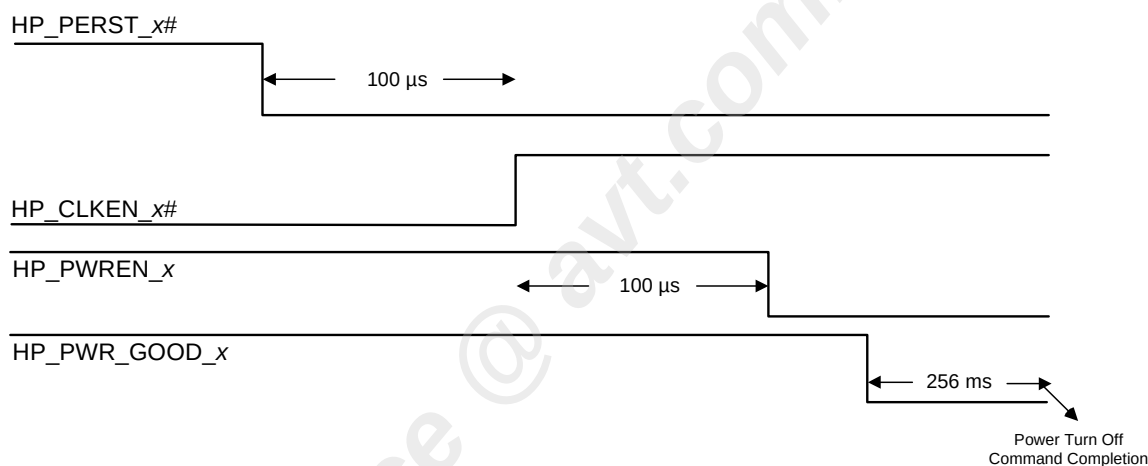
Slots can be powered down by either:

- De-asserting the HP_MRL_x# input, by manually opening the closed MRL, –or–
- Software Setting the Port's **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset 80h[10])

[Figure 10-3](#) illustrates the following power-down timing sequence for either event:

1. HP_PERST_x# output to the Port is asserted.
2. HP_CLKEN_x# output is de-asserted to the slot, 100 μ s after HP_PERST_x# output is asserted.
3. HP_PWREN_x output is de-asserted to the slot, 100 μ s after HP_CLKEN_x# output is de-asserted.

Figure 10-3. Hot Plug Automatic Slot Power-Down Sequence



10.9 Default Parallel Hot Plug Ports

Default assignment of Hot Plug Controllers to Transparent Downstream Ports depends upon the test mode and Port configuration. By default, the test mode is defined by the [STRAP_TESTMODE\[2:0\]](#) 3-state inputs, and Port configuration is defined by the [STRAP_STNx_PORTCFGx](#) 3-state inputs, all of which are sampled and latched immediately following Fundamental Reset (at [PEX_PERST#](#) de-assertion); however, the following agents can subsequently override the Strapping inputs:

- Serial EEPROM
- I²C/SMBus, if [STRAP_I2C_SMBUS_CFG_EN#](#)=L, to delay linkup until the **Configuration Release** register *Initiate Configuration* bit (Port 0, offset 3ACh[0]) is Set
- Management Port software (in Virtual Switch mode), if [STRAP_MGMT_PORT_EN](#)=Z, to delay linkup until the *Initiate Configuration* bit is Set

These three agents can also override the Strapping inputs in assigning Hot Plug Controllers to Transparent Downstream Ports, by programming the **Parallel Hot Plug Control** register (Port 0, offset 3A4h[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively). However, if the Hot Plug Port assignment must be changed (from the default assignment determined by the Strapping inputs, as indicated in the **Parallel Hot Plug Control** register), use of the serial EEPROM is recommended, rather than I²C/SMBus or Management Port software, to configure the Hot Plug Slot signals as quickly as possible. By default, Port 1 of each Station (Ports 1 and 9, if they are Transparent Downstream Ports), has highest priority for default assignment as a Parallel Hot Plug Port.

Hot Plug Controller availability is as follows:

- Hot Plug Controller A is available in Test modes [0 through 3].
- Hot Plug Controller B is available in Test modes [3, 7, and 11].
- Hot Plug Controller C is available in Test modes [0 through 7].
- Hot Plug Controllers are not available in Test Modes [8, 9, and 10].

For Test modes [0 through 6], Hot Plug Controllers A, B, and C are assigned in the following order:

1. Hot Plug Controller C,
2. Hot Plug Controller A, if the Test mode is [0, 1, 2, or 3],
3. Hot Plug Controller B, if the Test mode is [3].

For Test modes [7 and 11], the Hot Plug Controller assignment order is:

1. Hot Plug Controller B,
2. Hot Plug Controller C, if the Test mode is [7].

Transparent Downstream Ports have priority for default assignment of Hot Plug Controllers, in the following order – 1, 9, 0, 8 (Port 8 is a **Software Only** Port), 2, 10, and 3.

Note: [Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the [STRAP_STNx_PORTCFGx](#) input states and/or serial EEPROM programming.

[Table 10-5](#), [Table 10-6](#), and [Table 10-7](#) lists the [STRAP_TESTMODE\[2:0\]](#) configurations, which includes which Hot Plug Controllers are enabled for each of the 12 Test mode configurations. For further details, refer to [Section 3.4.4.1](#), “[STRAP_TESTMODE\[2:0\]](#) 3-State Input Configurations.”

Table 10-5. STRAP_TESTMODE[2:0] 3-State Input Configurations – Base Mode, Test Modes 0, 1, 2, 3

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Base Mode			
		0	1	2	3
		STRAP_TESTMODE[2:0] = 000	STRAP_TESTMODE[2:0] = 00Z	STRAP_TESTMODE[2:0] = 001	STRAP_TESTMODE[2:0] = 0Z0
		Hot Plug A + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input	Hot Plug A + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input	Hot Plug A + Hot Plug C + GPIO[10:0] Input	Hot Plug A + Hot Plug C + Hot Plug B + PORT_GOOD0#
B17	O, PU	HP_ATNLED_A#	HP_ATNLED_A#	HP_ATNLED_A#	HP_ATNLED_A#
E17	I, PU	HP_BUTTON_A#	HP_BUTTON_A#	HP_BUTTON_A#	HP_BUTTON_A#
G18	O, PU	HP_CLKEN_A#	HP_CLKEN_A#	HP_CLKEN_A#	HP_CLKEN_A#
J18	I, PU	HP_MRL_A#	HP_MRL_A#	HP_MRL_A#	HP_MRL_A#
D18	O, PU	HP_PERST_A#	HP_PERST_A#	HP_PERST_A#	HP_PERST_A#
A17	I, PU	HP_PRSNT_A#	HP_PRSNT_A#	HP_PRSNT_A#	HP_PRSNT_A#
K18	I, PD	HP_PWR_GOOD_A	HP_PWR_GOOD_A	HP_PWR_GOOD_A	HP_PWR_GOOD_A
C18	O, PD	HP_PWREN_A	HP_PWREN_A	HP_PWREN_A	HP_PWREN_A
H18	I, PU	HP_PWRFLT_A#	HP_PWRFLT_A#	HP_PWRFLT_A#	HP_PWRFLT_A#
F18	O, PU	HP_PWRLED_A#	HP_PWRLED_A#	HP_PWRLED_A#	HP_PWRLED_A#
F15	O, PU	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#
B15	I, PU	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#
D17	O, PU	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#
C14	I, PU	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#
F14	O, PU	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#
C15	I, PU	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#
H15	I, PD	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C
D15	O, PD	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C
E15	I, PU	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#
G15	O, PU	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

Table 10-6. STRAP_TESTMODE[2:0] 3-State Input Configurations – Virtual Switch Mode, Test Modes 4, 5, 6, 7

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Virtual Switch Mode; One to Three Virtual Switches, as Selected by VS_MODE Input State			
		4	5	6	7
		STRAP_TESTMODE[2:0] = 0ZZ	STRAP_TESTMODE[2:0] = 0Z1	STRAP_TESTMODE[2:0] = 010	STRAP_TESTMODE[2:0] = 01Z
		VS[2:0] + Hot Plug C + PORT_GOOD[6:0]# + GPIO[10:7] Input	VS[2:0] + Hot Plug C + SHP_PERST[6:0]# + GPIO[10:7] Input	VS[2:0] + Hot Plug C + GPIO[10:0] Input	VS[2:0] + Hot Plug C + Hot Plug B + PORT_GOOD0#
B17	O, PU	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#
E17	I, PU	VS0_PERST#	VS0_PERST#	VS0_PERST#	VS0_PERST#
G18	O, PU	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#
J18	I, PU	VS1_PERST#	VS1_PERST#	VS1_PERST#	VS1_PERST#
D18	O, PU	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#
A17	I, PU	VS2_PERST#	VS2_PERST#	VS2_PERST#	VS2_PERST#
K18	I, 3-state	VS_MODE	VS_MODE	VS_MODE	VS_MODE
C18	O, PD	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#
H18	I, PU	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#
F18	O, PU	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#
F15	O, PU	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#	HP_ATNLED_C#
B15	I, PU	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#	HP_BUTTON_C#
D17	O, PU	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#	HP_CLKEN_C#
C14	I, PU	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#	HP_MRL_C#
F14	O, PU	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#	HP_PERST_C#
C15	I, PU	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#	HP_PRSNT_C#
H15	I/O, PD	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C	HP_PWR_GOOD_C
D15	O, PD	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C	HP_PWREN_C
E15	I, PU	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#	HP_PWRFLT_C#
G15	O, PU	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#	HP_PWRLED_C#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

Table 10-7. STRAP_TESTMODE[2:0] 3-State Input Configurations – Virtual Switch Mode, Test Modes 8, 9, 10, 11

Location	Type	Multiplexed I/O Signals Used, by Test Mode – Virtual Switch Mode; Four Virtual Switches, as Selected by VS_MODE Input State			
		8	9	10	11
		STRAP_TESTMODE[2:0] = 011	STRAP_TESTMODE[2:0] = Z00	STRAP_TESTMODE[2:0] = Z0Z	STRAP_TESTMODE[2:0] = Z01
		VS[2:0] + VS3 + PORT_GOOD[6:0]# + GPIO[11:7] Input	VS[2:0] + VS3 + SHP_PERST[6:0]# + GPIO[10:7] Input	VS[2:0] + VS3 + GPIO[11:0] Input	VS[2:0] + VS3 + Hot Plug B + PORT_GOOD0# + GPIO1 Input
B17	O, PU	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#	VS0_PEX_INTA#
E17	I, PU	VS0_PERST#	VS0_PERST#	VS0_PERST#	VS0_PERST#
G18	O, PU	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#	VS1_PEX_INTA#
J18	I, PU	VS1_PERST#	VS1_PERST#	VS1_PERST#	VS1_PERST#
D18	O, PU	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#	VS2_PEX_INTA#
A17	I, PU	VS2_PERST#	VS2_PERST#	VS2_PERST#	VS2_PERST#
K18	I, 3-state	VS_MODE=0	VS_MODE=0	VS_MODE=0	VS_MODE=0
C18	O, PD	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#	VS2_FATAL_ERR#
H18	I, PU	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#	VS1_FATAL_ERR#
F18	O, PU	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#	VS0_FATAL_ERR#
F15	O, PU	VS3_PEX_INTA#	VS3_PEX_INTA#	VS3_PEX_INTA#	VS3_PEX_INTA#
B15	I, PU	VS3_PERST#	VS3_PERST#	VS3_PERST#	VS3_PERST#
D17	O, PU	RESERVED	RESERVED	RESERVED	RESERVED
C14	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
F14	O, PU	RESERVED	RESERVED	RESERVED	RESERVED
C15	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
H15	I/O, PD	GPIO11 input	GPIO11 input	GPIO11 input	GPIO11 input
D15	O, PD	RESERVED	RESERVED	RESERVED	RESERVED
E15	I, PU	RESERVED	RESERVED	RESERVED	RESERVED
G15	O, PU	VS3_FATAL_ERR#	VS3_FATAL_ERR#	VS3_FATAL_ERR#	VS3_FATAL_ERR#
E1	I/O, PU	PORT_GOOD1#	SHP_PERST1#	GPIO1 input	HP_ATNLED_B#
H2	I/O, PU	PORT_GOOD2#	SHP_PERST2#	GPIO2 input	HP_BUTTON_B#
L18	I/O, PU	PORT_GOOD3#	SHP_PERST3#	GPIO3 input	HP_CLKEN_B#
F1	I/O, PU	PORT_GOOD4#	SHP_PERST4#	GPIO4 input	HP_MRL_B#
J2	I/O, PU	PORT_GOOD5#	SHP_PERST5#	GPIO5 input	HP_PERST_B#
G2	I/O, PU	PORT_GOOD6#	SHP_PERST6#	GPIO6 input	HP_PRSTNT_B#
D1	I/O, PD	GPIO7 input	GPIO7 input	GPIO7 input	HP_PWR_GOOD_B
J3	I/O, PD	GPIO8 input	GPIO8 input	GPIO8 input	HP_PWREN_B
K3	I/O, PU	GPIO9 input	GPIO9 input	GPIO9 input	HP_PWRFLT_B#
H1	I/O, PU	GPIO10 input	GPIO10 input	GPIO10 input	HP_PWRLED_B#
C1	I/O, PU	PORT_GOOD0#	SHP_PERST0#	GPIO0 input	PORT_GOOD0#

10.10 HP_PWREN_x Output Polarity Control

HP_PWREN_x output polarity is configured by the SPARE1 input:

- **SPARE1 input is pulled or tied High to VDD18, or left unconnected (and therefore High due to the internal pull-up resistor)** – HP_PWREN_x outputs are Active-High. Therefore, when the slot power is turned On (Port's Slot Control register Power Controller Control bit (Downstream Ports, offset 80h[10]) is Cleared), the HP_PWREN_x outputs are driven High.
- **SPARE1 input is pulled or tied Low to VSS (Ground)** – HP_PWREN_x outputs are Active-Low. Therefore, when the slot power is turned On, the HP_PWREN_x outputs are driven Low.

HP_PWREN_x output polarity **cannot** be changed by serial EEPROM, I²C/SMBus, nor software. Polarity of Serial Hot Plug SHP_PWREN output(s) is **not** programmable.

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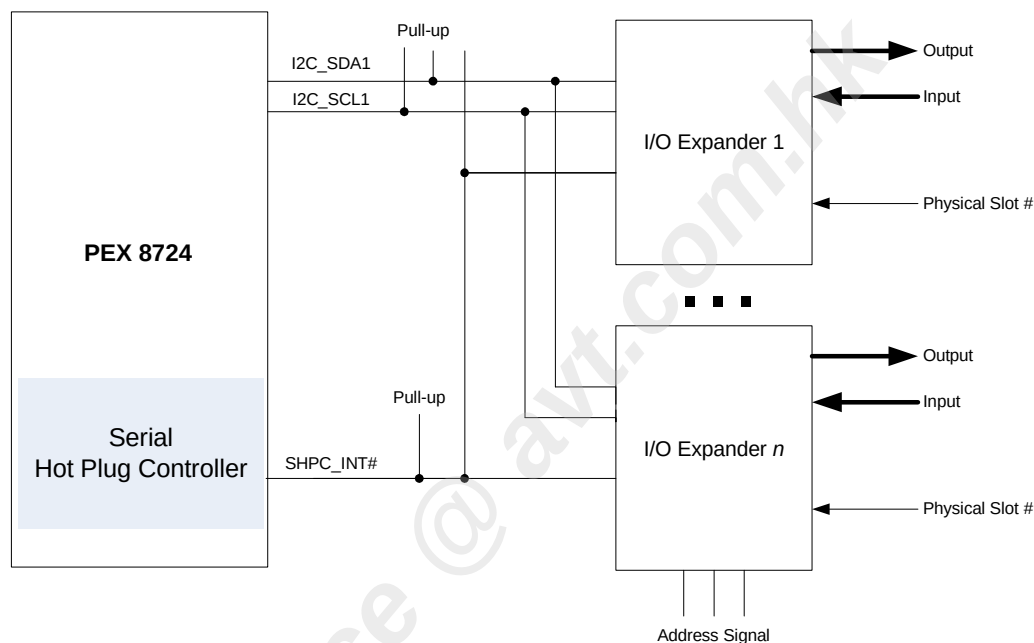
10.11 Serial Hot Plug Controller

Notes: The I²C Master interface is described in this section. The Master capabilities are limited to the Serial Hot Plug Controller.

Serial Hot Plug scan and I²C I/O Expander initialization start after serial EEPROM load is complete.

Using I/O Expander ICs sitting on an I²C Bus, the PEX 8724 has the option of Hot Plug capability on all its Transparent Downstream Ports. Figure 10-4 illustrates the internal Serial Hot Plug Controller interface. The Serial Hot Plug Controller controls the output Ports on the I/O Expanders and retrieves the Port status, such as device connect status, Power Fault, and MRL Sensor position, from all I/O Expanders. When there is an input change to an I/O Expander, an INTx interrupt from an I/O Expander goes Low and the PEX 8724 reads the I/O Expander. When an I/O Expander output Port requires updating with a new value, the PEX 8724 writes to the I/O Expander through the I²C Bus.

Figure 10-4. Serial Hot Plug Interface Diagram



10.11.1 Hot Plug Operations by way of I²C I/O Expander

When software issues a Slot Power On command, the Serial Hot Plug Controller issues an I²C Write to the I²C I/O Expander, to assert the [SHP_PWREN](#) output on the I/O Expander, and thereby turn On the power. After the Write is complete, either the [SHP_PWR_GOOD](#) input is sampled asserted or the T_{pepv} time has elapsed, the Serial Hot Plug Controller issues another Write to the I/O Expander, to assert its [SHP_REFCLKEN#](#) output, and thereby turn On the Reference Clock (REFCLK) at the slot.

Note: The T_{pepv} value is used when the [SHP_PWR_GOOD](#) input is not used, as indicated by the Port's [Power Management Hot Plug User Configuration](#) register [HPC Tpepv](#) field (Downstream Ports, offset [F70h\[4:3\]](#)) not being Cleared (Cleared is the default).

After the Serial Hot Plug [SHP_REFCLKEN#](#) output is asserted, the Serial Hot Plug Controller waits 100 ms, then issues another Write, to de-assert the Serial Hot Plug [SHP_PERST#](#) output to the slot. If there are commands, such as Attention or Power LED changes along with the Power Control command, the Serial Hot Plug Controller includes the LED output value change, along with these Writes, to change the LED status. The same procedure applies to other commands, such as Port Power Off. After the Serial Hot Plug Controller completes all Write operations, it Sets the Port's [Slot Status](#) register [Command Completed](#) bit (Downstream Ports, offset [80h\[20\]](#)). When another command is issued before the current command completes, the results are undefined. The I²C clock ([I2C_SCL1](#)) frequency is approximately 250 kHz. Read and Write cycles on the I²C Bus are illustrated in the figures provided in [Section 10.11.2](#).

10.11.2 I²C Write/Read to I/O Devices

10.11.2.1 I²C Write Access to I/O Devices

An I²C Write cycle on the I²C Bus would look as illustrated in [Figure 10-5](#). (The number of the Data byte to be written can vary from 0 to 8.)

Figure 10-5. I²C Write Access to I/O Devices

START	Address + W		ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK
S	Slave Address[6:0]	0	A	Command Byte	A

7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Data Byte 0	A	Data Byte 1	A	Data Byte 2	A	Data Byte 3	A	P

10.11.2.2 I²C Read Access from I/O Devices

An I²C Read cycle on the I²C Bus would look as illustrated in [Figure 10-6](#). (The number of the Data byte to be read can vary from 0 to 8.)

Figure 10-6. I²C Read Access from I/O Devices

START	Address + W		ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	START	Address + R	
S	Slave Address[6:0]	0	A	Command Byte	A	S	Slave Address[6:0]	1

7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	7 6 5 4 3 2 1 0	ACK/NAK	STOP
Data Byte 0	A	Data Byte 1	A	Data Byte 2	A	Data Byte 3	A	P

10.11.3 I²C I/O Expander Parts Selection and Pin Definition

Two types of I²C I/O Expanders can be used for Serial Hot Plug:

- **16-bit device** – For the 16-bit device, the 7-bit I²C Address must be 010X_XXXb; a Maxim MAX7311, NXP PCA9555, or TI PCA9555 is recommended. I/O Expander addresses must begin with the lowest address (0100_000b), and increment sequentially (corresponding to increasing PEX 8724 Port Numbers), for each device. For MAX7311 (which supports more than the eight addresses provided by the other 16-bit devices), the 7-bit I²C addresses can be in two ranges, 010X_XXXb and 1010_XXXb. A 16-bit device (and its 16 I/O pins) supports one Serial Hot Plug Port. (Refer to [Figure 10-7](#).)

For further details, refer to the manufacturer's data sheets for the Maxim MAX7311, NXP PCA9555, or TI PCA9555.

- **40-bit device** – For the 40-bit device, the 7-bit I²C Address must be within two ranges, 0100_XXXb and 1010_XXXb; an NXP PCA9698 is recommended. I/O Expander addresses must begin with the lowest address, and increment sequentially (corresponding to increasing PEX 8724 Port Numbers), for each device. The lower 32 I/O pins are used for two Ports. A 40-bit device can support two sets of Serial Hot Plug pins. The two sets are indicated with suffix PX and PY, in [Figure 10-9](#).

For further details, refer to the manufacturer's data sheet for the NXP PCA9698.

To provide Hot Plug services on all Transparent Downstream Ports, the PEX 8724 can concurrently support (refer to the [Power Management Hot Plug User Configuration](#) register *40-Pin I/O Expander Scan Disable* bit, offset F70h[17]):

- Up to six, 16-bit I/O Expanders (for up to five Transparent Downstream Ports, plus one for optional Slot ID load), –or–
- Up to three, 40-bit I/O Expanders (for up to five Transparent Downstream Ports), –or–
- Mix of 16- and 40-bit I/O Expanders

The NXP PCA9555 and TI PCA9555 devices have only eight programmable I²C addresses; therefore, the maximum quantity of Serial Hot Plug-capable Ports with these I/O Expanders is eight. However, the Maxim MAX7311 can have up to 64 programmable Slave addresses; therefore, if using the MAX7311 I/O Expanders, all Transparent Downstream Ports can be made Hot Plug-capable.

[Table 10-8](#) defines the external I²C I/O Expander pins used for Serial Hot Plug, in location order.

Table 10-8. External I²C I/O Expander Pin Definitions used for Serial Hot Plug, by Location

Signal Name	Direction	Description	Location	
			16-Bit Device ^a	40-Bit Device ^b
SHP_PWRLED#	O	Serial Hot Plug Power LED Output Same function as HP_PWRLED_x#.	IO0_0 or P00	IO0_0 IO2_0
SHP_ATNLED#	O	Serial Hot Plug Attention LED Output Same function as HP_ATNLED_x#.	IO0_1 or P01	IO0_1 IO2_1
SHP_PWREN	O	Serial Hot Plug Power Enable Output Same function as HP_PWREN_x.	IO0_2 or P02	IO0_2 IO2_2
SHP_REFCLKEN#	O	Serial Hot Plug Reference Clock Enable Output Same function as HP_CLKEN_x#.	IO0_3 or P03	IO0_3 IO2_3

Table 10-8. External I²C I/O Expander Pin Definitions used for Serial Hot Plug, by Location (Cont.)

Signal Name	Direction	Description	Location	
			16-Bit Device ^a	40-Bit Device ^b
SHP_PERST#	O	Serial Hot Plug Reset Output Same function as HP_PERST_x#. <i>Note: SHP_PERSTx# outputs are recommended over I²C I/O Expander PERST# output, to avoid delay from using I²C to toggle reset.</i>	IO0_4 or P04	IO0_4 IO2_4
SHP_INTERLOCK	O	Serial Hot Plug Electromechanical Interlock Output Control Used to physically lock the adapter in place, until software releases it. The signal default is 0. The current state of the Electromechanical Interlock is reflected in the Port's Slot Status register <i>Electromechanical Interlock Status</i> bit (Downstream Serial Hot Plug-Enabled Ports, offset 80h[23]). This output can be toggled by writing 1 to the Port's Slot Control register <i>Electromechanical Interlock Control</i> bit (Downstream Serial Hot Plug-Enabled Ports, offset 80h[11]). A Write of 0 has no effect. SHP_INTERLOCK is enabled when the Port's Slot Capability register <i>Electromechanical Interlock Present</i> bit (Downstream Serial Hot Plug-Enabled Ports, offset 7Ch[17]) is Set (default for Serial Hot Plug-capable Transparent Downstream Ports).	IO0_5 or P05	IO0_5 IO2_5
SHP_PORTID[4:0]	I	Serial Hot Plug Port ID Straps <i>Note: Table 4-1 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.</i> 0_0000b = Port 0 0_0001b = Port 1 0_0010b = Port 2 0_0011b = Port 3 0_1000b = Port 8 0_1001b = Port 9 0_1010b = Port 10 Configures to which Downstream Port this I ² C I/O Expander maps. 1_1111b is valid only for 16-bit I/O Expanders, for loading the Port's Slot Capability register <i>Physical Slot Number</i> field (Downstream Ports, offset 7Ch[31:19]) from the I/O Expander SHP_SLOTID[12:5] inputs. All other encodings are Reserved .	{IO1_2:0, IO0_7:6} or {P1[2:0], P0[7:6]}	{IO1_2:0, IO0_7:6} {IO3_2:0, IO2_7:6}

Table 10-8. External I²C I/O Expander Pin Definitions used for Serial Hot Plug, by Location (Cont.)

Signal Name	Direction	Description	Location	
			16-Bit Device ^a	40-Bit Device ^b
SHP_PRSNT#	I	Serial Hot Plug PRSNT2# Input Same function as HP_PRSNT_x#.	IO1_3 or P13	IO1_3 IO3_3
SHP_MRL#	I	Serial Hot Plug Manually Operated Retention Latch Sensor Input Same function as HP_MRL_x#.	IO1_4 or P14	IO1_4 IO3_4
SHP_BUTTON#	I	Serial Hot Plug Attention Button Input Same function as HP_BUTTON_x#.	IO1_5 or P15	IO1_5 IO3_5
SHP_PWRFLT#	I	Serial Hot Plug Power Fault Input Same function as HP_PWRFLT_x#.	IO1_6 or P16	IO1_6 IO3_6
SHP_PWR_GOOD	I	Hot Plug Power Good Input Same function as HP_PWR_GOOD_x.	IO1_7 or P17	IO1_7 IO3_7
SHP_SLOTID[12:5]	I	Hot Plug Slot ID Defines the value of the upper eight bits of the Port's Slot Capability register <i>Physical Slot Number</i> field (bits [31:24] of Downstream Ports, offset 7Ch[31:19]; the lower five bits ([23:19]) are automatically programmed to be equal to the Serial Hot Plug-capable Port's Port Number). The 40-bit I²C I/O Expander has provision for two sets of SHP_SLOTID[12:5] inputs, for two Serial Hot Plug-capable Ports. A 16-bit I²C I/O Expander can be used: • For Serial Hot Plug functionality, –or– • Exclusively for SHP_SLOTID[12:5] input, with that 8-bit value applied globally to all Hot Plug-capable Ports (Parallel and Serial) Note: If <i>SHP_PORTID[4:0]</i> = 1_1111 to load <i>SlotID[31:24]</i> in all Hot Plug Ports, the I²C I/O Expander pinout is re-defined, as illustrated in <i>Figure 10-8</i>.	If SHP_PORTID[4:0] =1_1111: {IO0_5:0, IO1_7:6} or {P0[5:0], P1[7:6]}	IO4_7:0

a. **16-bit device** – Refer to *Figure 10-7*, –or– *Figure 10-8* (*SHP_PORTID[4:0]* = 1_1111), for pinout.

b. **40-bit device** – Refer to *Figure 10-9* for pinout.

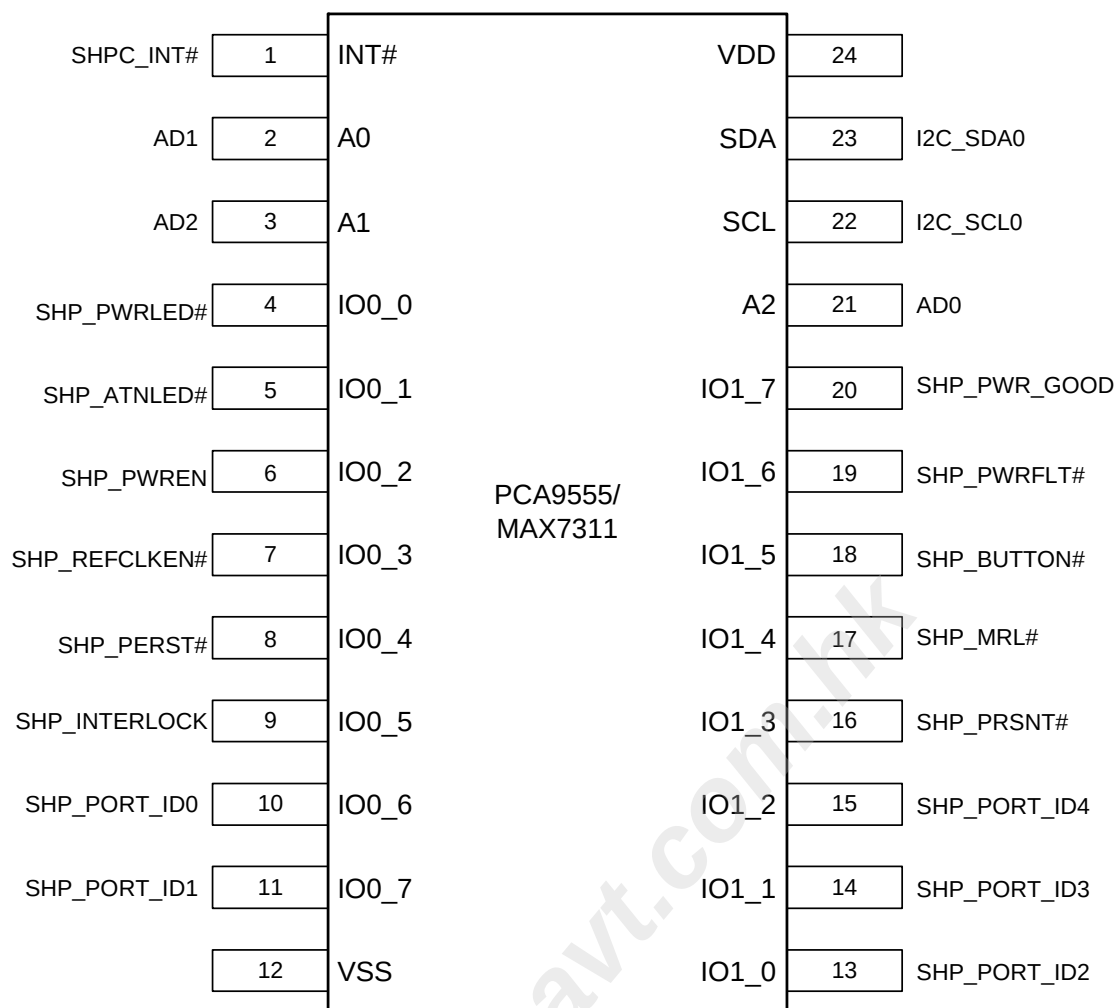
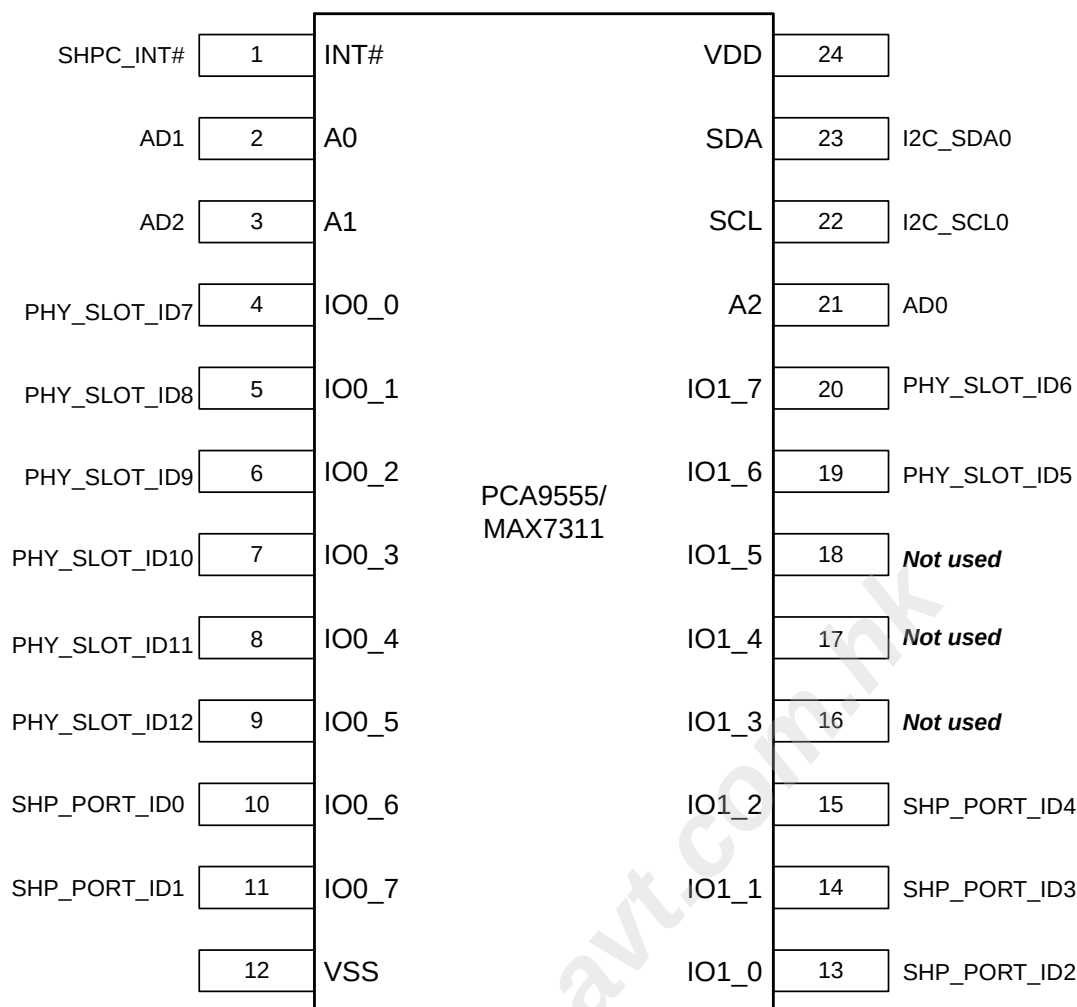
Figure 10-7. 16-Bit I²C I/O Expander Pinout

Figure 10-8. 16-Bit I²C I/O Expander Pinout when SHP_PORTID[4:0]=1_1111

Note: If $SHP_PORTID[4:0]=1_1111$ to load SlotID[31:24] in all Hot Plug Ports, the I²C I/O Expander pinout is re-defined, as illustrated in [Figure 10-8](#).

Figure 10-9. 40-Bit I²C I/O Expander Pinout

I2C_SDA1	1	SDA	RESET#	56	
I2C_SCL1	2	SCL	INT#	55	SHPC_INT#
SHP_PWRLED# (PX)	3	IO0_0	IO4_7	54	SHP_SLOTID12
SHP_ATNLED# (PX)	4	IO0_1	IO4_6	53	SHP_SLOTID11
SHP_PWREN (PX)	5	IO0_2	IO4_5	52	SHP_SLOTID10
	6	VSS	VSS	51	
SHP_REFCLKEN# (PX)	7	IO0_3	IO4_4	50	SHP_SLOTID9
SHP_PERST# (PX)	8	IO0_4	IO4_3	49	SHP_SLOTID8
SHP_INTERLOCK (PX)	9	IO0_5	IO4_2	48	SHP_SLOTID7
SHP_PORTID0 (PX)	10	IO0_6	IO4_1	47	SHP_SLOTID6
	11	VSS	VDD	46	
SHP_PORTID1 (PX)	12	IO0_7	IO4_0	45	SHP_SLOTID5
SHP_PORTID2 (PX)	13	IO1_0	IO3_7	44	SHP_PWR_GOOD (PY)
SHP_PORTID3 (PX)	14	IO1_1	IO3_6	43	SHP_PWRFLT# (PY)
SHP_PORTID4 (PX)	15	IO1_2	IO3_5	42	SHP_BUTTON# (PY)
SHP_PRSNT# (PX)	16	IO1_3	IO3_4	41	SHP_MRLN# (PY)
SHP_MRLN# (PX)	17	IO1_4	IO3_3	40	SHP_PRSNT# (PY)
	18	VDD	VSS	39	
SHP_BUTTON# (PX)	19	IO1_5	IO3_2	38	SHP_PORTID4 (PY)
SHP_PWRFLT# (PX)	20	IO1_6	IO3_1	37	SHP_PORTID3 (PY)
SHP_PWR_GOOD (PX)	21	IO1_7	IO3_0	36	SHP_PORTID2 (PY)
SHP_PWRLED# (PY)	22	IO2_0	IO2_7	35	SHP_PORTID1 (PY)
	23	VSS	VSS	34	
SHP_ATNLED# (PY)	24	IO2_1	IO2_6	33	SHP_PORTID0 (PY)
SHP_PWREN (PY)	25	IO2_2	IO2_5	32	SHP_INTERLOCK (PY)
SHP_REFCLKEN# (PY)	26	IO2_3	IO2_4	31	SHP_PERST# (PY)
	27	AD0	OE	30	
	28	AD1	AD2	29	

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10.11.4 Serial Hot Plug Port Enumeration, Assignment, and Initialization

Serial Hot Plug can be implemented using either 16- or 40-bit I²C I/O Expanders, or a combination of both. The PEX 8724 Serial Hot Plug Controller has the intelligence to discover and differentiate between the 16- and 40-bit I/O Expanders, and assigns the I/O Expanders to a corresponding Transparent Downstream Port. The 16-bit I/O Expanders can support one Serial Hot Plug Port, and 40-bit I/O Expanders can support one or two Serial Hot Plug Ports.

After [PEX_PERST#](#) and/or [VSx_PERST#](#) input de-asserts and the serial EEPROM (if present) load completes, the Serial Hot Plug Controller scans the I²C Bus for I/O Expanders, starting with Device address 0100_000b. If the Controller receives an Acknowledge (ACK) from the I/O Expander, it performs a Device ID code Read from that I/O Expander, to detect the presence of a 40-bit I/O Expander (40-bit device scan is enabled, by default). This scan can be disabled, by Setting the Transparent Downstream Port's **Power Management Hot Plug User Configuration** register *40-Pin I/O Expander Scan Disable* bit (offset [F70h\[17\]](#)). If the Device ID code Read fails, that I/O Expander is identified as a 16-bit I/O Expander, and the Serial Hot Plug [SHP_PORTID\[4:0\]](#) value on the 16-bit I/O Expander determines which PEX 8724 Port is associated with the I/O Expander. If the Device ID code Read fetches the correct Device ID, that I/O Expander is identified as being 40-bit capable. The two PEX 8724 Serial Hot Plug Ports that correspond to the two sets of Serial Hot Plug pins in a 40-bit I/O Expander are determined from two unique [SHP_PORTID\[4:0\]](#) values. (Refer to [Table 10-8](#) for valid [SHP_PORTID\[4:0\]](#) values.)

The Serial Hot Plug Controller logic uses the I²C Master interface to program the I/O Expander's I/O Configuration registers, and Sets the initial Hot Plug state for successfully scanned I/O Expanders. (Refer to [Table 10-2](#) for the initial states of the Hot Plug outputs.) On an I/O Expander scan, if the Serial Hot Plug Controller receives a Negative Acknowledge (NAK), the Hot Plug Controller stops scanning for I²C I/O Expanders. After the I²C scan is complete, the Serial Hot Plug Controller starts the slot power-on sequence for Ports in which the following is true:

- Port's [SHP_MRL#](#) input is sampled Low (Port's **Slot Status** register *MRL Sensor State* bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset [80h\[21\]](#)) is Cleared), and
- Port's **Slot Control** register *Power Controller Control* bit (Downstream Ports, offset [80h\[10\]](#)) is not Set by the serial EEPROM load

The I²C address for the I/O Expanders must be contiguous, with the first I/O Expander's I²C address programmed to 0100_000b, the next programmed to 0100_001b, and so forth.

10.11.5 I²C I/O Expander Interrupt Processing

The I/O Expander Interrupt outputs (INT#) must all be connected together in an Open Drain (OD) manner, to the PEX 8724's SHPC_INT# input. When an I/O Expander input state changes on any of the I/O Expanders, the SHPC_INT# input is asserted Low. The Serial Hot Plug Controller, through the I²C Bus, scans the I/O Expanders, starting with address 40h, until the SHPC_INT# input de-asserts High (as a result of the I²C Read), signaling which I/O Expander asserted the interrupt (INT#). SHPC_INT# is internally de-bounced for 10 ms; therefore, if SHPC_INT# asserts Low for less than 10 ms, its assertion is ignored. The 10 ms de-bounce on the SHPC_INT# input can be disabled, by Setting the Port's **Power Management Hot Plug User Configuration** register *Serial Hot Plug INTx De-Bounce Disable* bit (offset F70h[18]). SHPC_INT# assertion due to an I/O Expander input state change can cause a corresponding PEX 8724 **Slot Status** register bit to be Set, and cause the Port to send an INTx Message to the Root Complex, if the corresponding interrupt is enabled.

10.11.6 Serial Hot Plug-Capable Port Command Completion

For slot power On or Off commands from software to turn power On or Off to a specific Port, the Port's **Slot Status** register *Command Completed Interrupt Enable* bit (Downstream Ports, offset 80h[4]) is Set after the sequence of I²C Master Write operations to the I²C I/O Expander, that perform the power On or Off sequence, have completed.

10.11.7 Physical Slot Number Loading from I²C I/O Expander

The Port's **Slot Capability** register *Physical Slot Number* field (Downstream Ports, offset 7Ch[31:19]) in Serial Hot Plug Ports must be assigned a unique, non-zero Port identifier within the system. The register's MSB [31:24] can be loaded from the SHP_SLOTID[12:5] input Settings on the I²C I/O Expander. The 40-bit I/O Expander has SHP_SLOTID[12:5] inputs implemented along with two sets of Serial Hot Plug pins. The SHP_SLOTID[12:5] inputs define the *Physical Slot Number* field in both Ports identified by the SHP_PORTID[4:0] (PX) and SHP_PORTID[4:0] (PY) inputs. LSB [4:0] (bits [23:19]) of the Physical Slot Number are loaded with the Port's Port Number. The combination of SHP_SLOTID[12:5] and Port Number forms a unique Physical Slot Number value, for each Port.

For 16-bit I/O Expanders, the SHP_PORTID[4:0] input value enables one of two possible implementations:

- If the SHP_PORTID[4:0] value matches a valid Downstream Port Number, the I/O Expander provides a set of Hot Plug pins for that designated Port, –or–
- If SHP_PORTID[4:0]=1_1111, this single 16-bit I/O Expander globally loads the set of {IO0_5:0, IO1_7:6} or {P0[5:0], P1[7:6]} as the upper 8 bits of a 13-bit Physical Slot Number, into all Hot Plug Ports. The lower five bits in each Port are initialized to the Port's Port Number, to provide a unique Physical Slot Number value for each Hot Plug-capable Port.

The 16-bit I/O Expander {IO1_5:3} or {P1[5:3]} pins are not used for SHP_SLOTID[12:5] load. This I/O Expander itself is **not** associated to any Port, does **not** provide Hot Plug signaling, and does **not** enable Hot Plug.

Note: 16-bit I/O Expanders that have Serial Hot Plug SHP_PORTID[4:0] programmed as 1_1111 **cannot** be used for Serial Hot Plug operation, because that value is used only for the Physical Slot Number field, as explained in Table 10-8. An alternative is to use the serial EEPROM to program the **Slot Capability** register.

10.12 Hot Plug Board Insertion and Removal Process

Table 10-9 defines the board insertion procedure supported by the PEX 8724. Table 10-10 defines the board removal procedure. Both processes apply to Parallel and Serial Hot Plug-capable Transparent Downstream Ports.

Table 10-9. Hot Plug Board Insertion Process

Operator / Action	Hot Plug Controller	Software
A. Places board in slot.	1. Sets the Port's Slot Status register <i>Presence Detect State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[22]).	Clears the Port's <i>Presence Detect Changed</i> bit.
	2. Sets the Port's Slot Status register <i>Presence Detect Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[19]).	
	3. Generates Interrupt Message due to <i>Presence Detect Changed</i> event, if enabled.	
	4. Transmits an Interrupt de-assertion Message, if enabled.	
B. Locks MRL.	5. Clears the Port's Slot Status register <i>MRL Sensor State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[21]).	Clears the Port's <i>MRL Sensor Changed</i> bit.
	6. Sets the Port's Slot Status register <i>MRL Sensor Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[18]).	
	7. Generates an Interrupt Message due to <i>MRL Sensor Changed</i> event, if enabled.	
	8. Transmits an Interrupt de-assertion Message, if enabled.	
C. Presses Attention Button.	9. Sets the Port's Slot Status register <i>Attention Button Pressed</i> bit (Downstream Ports, offset 80h[16]).	Clears the Port's <i>Attention Button Pressed</i> bit.
	10. Generates an Interrupt Message due to <i>Attention Button Pressed</i> event, if enabled.	
	11. Transmits an Interrupt de-assertion Message, if enabled.	Programs the Port's Slot Control register <i>Power Indicator Control</i> field (Downstream Ports, offset 80h[9:8]) value to 10b, to blink the Power Indicator LED, which indicates that the board is being powered up.
Continued...		

Table 10-9. Hot Plug Board Insertion Process (Cont.)

Operator / Action	Hot Plug Controller	Software
D. Power Indicator blinks.	12. Blinks the Power Indicator LED. 13. Sets the Port's Slot Status register <i>Command Completed</i> bit (Downstream Ports, offset 80h[20]). 14. Generates an Interrupt Message due to Power Indicator Blink command Completion, if enabled.	Clears the Port's <i>Command Completed</i> bit.
	15. Transmits an Interrupt de-assertion Message, if enabled.	Clears the Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10]), to turn On power to the Port.
	16. Slot is powered up. 17. After the HP_PWR_GOOD_x inputs are sampled asserted High or T _{pepv} delay, Sets the Port's <i>Command Completed</i> bit. 18. Generates an Interrupt Message due to Power Turn On command Completion, if enabled.	Clears the Port's <i>Command Completed</i> bit.
	19. Transmits an Interrupt de-assertion Message, if enabled.	Programs the Port's Slot Control register <i>Power Indicator Control</i> field (Downstream Ports, offset 80h[9:8]) value to 01b, to turn On the Power Indicator LED, which indicates that the slot is fully powered On.
E. Power Indicator On.	20. Turns On the Power Indicator LED. 21. Transmits an Interrupt assertion Message due to Power Indicator Turn On command Completion, if enabled.	Clears the Port's <i>Command Completed</i> bit.
	22. Transmits an Interrupt de-assertion Message, if enabled. 23. After the Data Link Layer is up, Sets the Port's Slot Status register <i>Data Link Layer State Changed</i> bit (Downstream Ports, offset 80h[24]), and transmits the corresponding interrupt, if enabled.	Software can now read the Port's Link Status register <i>Data Link Layer Link Active</i> bit (Downstream Ports, offset 78h[29]). A value of 1 in this bit indicates that the board is ready to be used. Clears the Port's <i>Data Link Layer State Changed</i> bit and interrupt, if enabled.
	24. Transmits an Interrupt de-assertion Message, if enabled.	

Table 10-10. Hot Plug Board Removal Process

Operator / Action	Hot Plug Controller	Software
A. Presses Attention Button.	1. Sets the Port's Slot Status register <i>Attention Button Pressed</i> bit (Downstream Ports, offset 80h[16]). 2. Generates an Interrupt Message due to Attention Button pressed, if enabled.	Clears the Port's <i>Attention Button Pressed</i> bit.
	3. Transmits an Interrupt de-assertion Message, if enabled.	Programs the Port's Slot Control register <i>Power Indicator Control</i> field (Downstream Ports, offset 80h[9:8]) value to 10b, to blink the Power Indicator LED, which indicates that the board is being powered down.
B. Power Indicator blinks.	4. Blinks the Power Indicator LED. 5. Sets the Port's Slot Status register <i>Command Completed</i> bit (Downstream Ports, offset 80h[20]). 6. Generates an Interrupt Message due to Power Indicator Blink command Completion, if enabled.	Clears the Port's <i>Command Completed</i> bit.
	7. Transmits an Interrupt de-assertion Message, if enabled.	Sets the Port's Slot Control register <i>Power Controller Control</i> bit (Downstream Ports, offset 80h[10]), to turn Off power to the Port.
C. Power Indicator Off.	8. Slot is powered Off. 9. Sets the Port's Slot Status register <i>Data Link Layer State Changed</i> bit (Downstream Ports, offset 80h[24]), and transmits an interrupt, if enabled. 10. After a 256-ms delay from HP_PWR_GOOD_x sampled de-asserted (if the HP_PWR_GOOD_x inputs are enabled through the Port's Power Management Hot Plug User Configuration register <i>HPC Tpepv</i> field (Downstream Ports, offset F70h[4:3], is Cleared)), Sets the Port's <i>Command Completed</i> bit. 11. Generates an Interrupt Message due to Power Turn Off command Completion, if enabled.	Clears the Port's <i>Data Link Layer State Changed</i> bit and interrupt. Clears the Port's <i>Command Completed</i> bit. Programs the Port's <i>Power Indicator Control</i> field value to 11b, to turn Off the Power Indicator LED, which indicates that the slot is fully powered Off and the board can be removed.
D. Power Indicator Off, board ready to be removed.	12. Turns Off the Power Indicator LED. 13. Sets the Port's <i>Command Completed</i> bit, due to Power Indicator Off command Completion.	Clears the Port's <i>Command Completed</i> bit.
	14. Transmits an Interrupt de-assertion Message, if enabled.	
Continued...		

Table 10-10. Hot Plug Board Removal Process (Cont.)

Operator / Action	Hot Plug Controller	Software
E. Unlocks MRL.	15. Sets the Port's Slot Status register <i>MRL Sensor State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[21]). 16. Sets the Port's Slot Status register <i>MRL Sensor Changed</i> bit (Downstream Ports, offset 80h[18]). 17. Generates an Interrupt Message due to MRL Sensor state change, if enabled.	Clears the Port's <i>MRL Sensor Changed</i> bit.
	18. Transmits an Interrupt de-assertion Message, if enabled.	
F. Removes board from slot.	19. Clears the Port's Slot Status register <i>Presence Detect State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[22]). 20. Sets the Port's Slot Status register <i>Presence Detect Changed</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[19]). 21. Generates an Interrupt Message due to Presence Detect change, if enabled.	Clears the Port's <i>Presence Detect Changed</i> bit.
	22. Transmits an Interrupt de-assertion Message, if enabled.	

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11.1 Overview

The PEX 8724 Power Management (PM) features provide the following services:

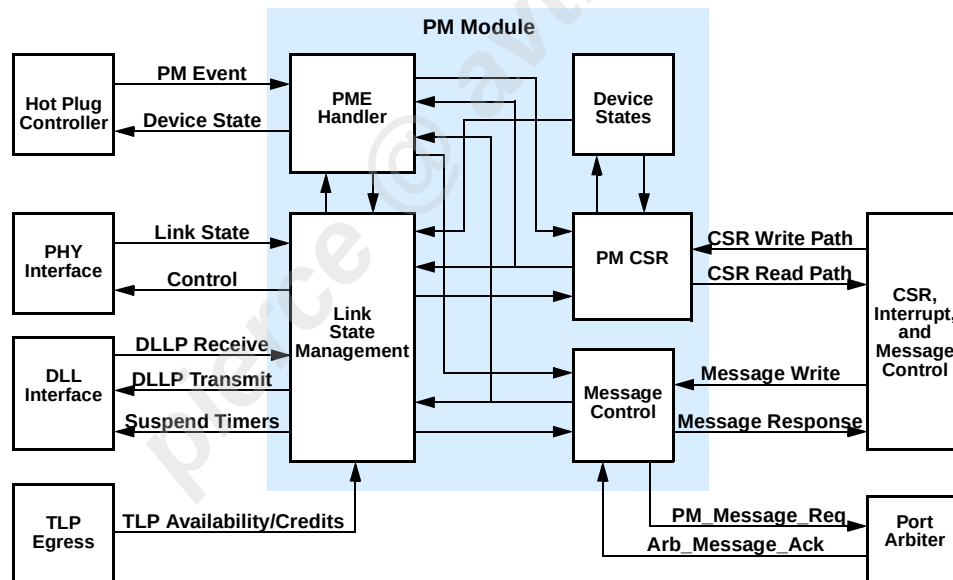
- Mechanisms to identify PM capabilities
- Ability to transition into certain PM states
- Notification of the current PM state of each Port
- Support for the option to wake up the system upon a specific event

The PEX 8724 supports hardware-autonomous PM and software-driven Device PM states. The switch also supports the L0s and L1 Link PM states in hardware-autonomous Active State Power Management (ASPM), as well as the L1, L2/L3 Ready, and L3 Link PM states in Conventional PCI-compatible PM. The D0, D3hot, and D3cold Device PM states are also supported in Conventional PCI-compatible PM. Because the PEX 8724 does **not support** Vaux, Power Management Event (PME) generation from the D3cold Device PM state is **not supported**.

The PM module interfaces with a Physical Layer (PHY) electrical sub-block, to transition the Link state into a low-power state, when the module receives a Power State Change Request from a Downstream component, or an internal event forces the Link state entry into low-power states in hardware-autonomous ASPM mode. PCI Express Link states are not directly visible to Conventional PCI Bus driver software; however, they are derived from the PM state of the components residing on those Links.

Figure 11-1 provides a functional block diagram of the PEX 8724 PM module.

Figure 11-1. PM Module Functional Block Diagram



Note: The Hot Plug Controller is available only on Hot Plug-capable Transparent Downstream Ports.

11.2 Power Management Features

- *PCI Express Base r3.0-compliant*
- *PCI Power Mgmt. r1.2-compliant*
- Link PM states
 - Conventional PCI-compatible Link PM states – L1^a, L2/L3 Ready, and L3 (Vaux is **not supported**)
 - Active State Power Management (ASPM) – L0s and L1
- Device PM states
 - Conventional PCI-compatible Device PM states – D0 (D0 Uninitialized and D0 Active) and D3 (D3hot and D3cold)
- Power Management Event (PME) support from the D3hot Device PM state
- PME due to Hot Plug and/or PCI Express Hot Plug events
- Forwards PME_Turn_Off Broadcast Messages
- Implements Gen 3-specific Control and Status register and associated interrupts
- Supports ASPM L0s, ASPM L1, PCI L1, and L2/L3 Ready Link PM states in NT mode, as well as Virtual Switch mode (ASPM L0s must be enabled through the optional serial EEPROM and/or I²C)

a. To differentiate from ASPM L1, this method of L1 is referred to as PCI L1 in this data book.

11.3 Power Management Capability

11.3.1 Device Power Management States

The PEX 8724 supports the D0 and D3hot Device PM states. The D1 and D2 Device PM states, which are optional in the *PCI Express Base r3.0*, are **not supported**.

The D3hot Device PM state can be entered from the D0 Uninitialized or D0 Active Device PM states, when system software programs the Port's **PCI Power Management Status and Control** register *Power State* field (offset 44h[1:0]) to 11b. The D0 Uninitialized Device PM state can be entered from the D3hot Device PM state when system software Clears the Port's *Power State* field.

11.3.1.1 D0 Device Power Management State

A PCI Express device must be placed into the D0 Device PM state before the device can be used. When the device enters the D0 Device PM state following a Fundamental Reset, the device is in the D0 Uninitialized Device PM state. A subsequent Reset forces the device into the D0 Uninitialized Device PM state.

Other than the Fundamental Reset that forces the device into the D0 Uninitialized Device PM state, all other PM state transitions are explicitly software-controlled.

A device enters the D0 Active Device PM state when system software Sets any combination of the **PCI Command** register *Bus Master Enable*, *Memory Access Enable*, and/or *I/O Access Enable* bits (offset 04h[2, 1, and/or 0], respectively).

11.3.1.2 D3hot Device Power Management State

In the D3hot Device PM state, Hot Plug and/or Link State operations can cause the PEX 8724 to generate a PME.

Only Type 0 Configuration accesses are allowed in the D3hot Device PM state. Memory and I/O transactions result in an Unsupported Request (UR). Completions flowing in either direction are not affected.

Type 1 transactions flowing toward a PEX 8724 Port in the D3hot Device PM state are terminated as URs. Type 0 Configuration transactions successfully complete. When an Upstream Port is programmed to the D3hot Device PM state, the Port initiates PCI L1 Link PM state entry.

11.3.2 Link Power Management States

PEX 8724 components hold their Upstream and Downstream Links in the L0 Link PM state during standard device operation (D0 Active Device PM state). ASPM defines a mechanism for components in the D0 Device PM state, to reduce Link power by placing their Links into a low-power state and instructing the other end of the Link to do likewise. This capability allows hardware-autonomous, dynamic Link power reduction beyond what is achievable by software-only-controlled PM. [Table 11-1](#) defines the relationship between a component's Power state and the Upstream Link. [Table 11-2](#) defines the relationship between Link PM states and power-saving actions.

The PCI L1 and L2/L3 Ready Link PM states are controlled by software programming the PEX 8724 into the D3hot Device PM state, and subsequently causing the Root Complex to broadcast the PME_Turn_Off Message to the Downstream hierarchy.

Table 11-1. Relationship between Component Power State and Upstream Link

Downstream Component Device PM State	Permissible Upstream Component Device PM State	Permissible Interconnect Link PM State
D0	D0	L0, and optionally ASPM L0s and L1 (if ASPM is supported and enabled).
D3hot	D0-to-D3hot	PCI L1 and L2/L3 Ready.
D3cold (no Vaux)	D0-to-D3cold	L3 (off). Zero power.

Table 11-2. Relationship between Link PM States and Power-Saving Actions

Link PM State	Power-Saving Actions
Tx ASPM L0s	PHY Tx Lanes are in a High-Impedance state.
Rx ASPM L0s	PHY Rx Lanes are in a low-power state.
ASPM or PCI L1	PHY Tx and Rx Lanes are in a low-power state. FC timers are suspended.
L2/L3 Ready	PHY Tx and Rx Lanes are in a low-power state. FC timers are suspended.
L3 (D3cold)	Component is fully powered Off.

11.3.2.1 Active State Power Management – L0s Link Power Management State Transition

The L0s Link PM state is allowable only in Active State Link Power Management (ASPM). Each device reports its support to ASPM L0s Link PM state entry through bit 10 of the Port's **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10])^a. Software enables ASPM L0s Link PM state entry, by writing 1 to bit 0 of the Port's **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]). The ASPM L0s Link PM state is optimized for short entry and exit latencies, while providing substantial power savings. If a device has the ASPM L0s Link PM state enabled, the device's unused Transmitter Links must be brought into the ASPM L0s Link PM state.

Link Layer Timers are **not** affected by a transition to the L0 Link PM state. When in the ASPM L0s Link PM state, Links return to the L0 Link PM state, regardless of whether there are any TLP traffic situations to transmit Flow Control (FC) updates, which occurs every 30 μ s minimum.

Rx ASPM L0s Link Power Management State Entry

The PEX 8724 allows its Rx Lanes to enter the ASPM L0s Link PM state, regardless of whether its Tx is enabled for ASPM L0s.

Rx ASPM L0s Link Power Management State Exit

The endpoint Transmitter at the opposite end of a PCI Express Link transmits the Fast Training Sequence (FTS) and returns to the L0 Link PM state. If the PEX 8724 Receiver achieves symbol and bit lock, the Receiver settles into the L0 Link PM state. Otherwise, the Receiver enters the *Recovery* state.

Tx L0s Link Power Management State Entry Conditions

L0s Link Power Management State Entry Conditions – Upstream Port(s)

The following four conditions must be met for Upstream Port(s) to enter the ASPM L0s Link PM state:

1. All PEX 8724 Downstream Port Rx Lanes are in the ASPM L0s (or deeper) Link PM state. If the PEX 8724 Downstream Port Link is in a *Link Down* state, this condition is also satisfied.
2. No pending TLPs (including Retry TLPs) or credits available to transmit pending TLPs.
3. No pending DLLPs in the DLL.
4. Enabled for entering the ASPM L0s Link PM state:
 - Bit 10 of the Port's **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10]) is Set
 - Bit 0 of the Port's **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]) is Set

The PM module directs the Upstream Link to enter the ASPM L0s Link PM state when Conditions 1, 2, and 3 are met for a specified period of time and Condition 4 is satisfied. The idle time can be selected, as either 1 μ s (default) or 4 μ s, in the Port's **Power Management Hot Plug User Configuration** register *L0s Entry Idle Count* bit (offset F70h[0]). If these conditions are met for the specified time period, PM directs the PHY to the ASPM L0s Link PM state.

^a. The ASPM L0s Link PM state is **not** supported, by default. To enable the ASPM L0s Link PM state, use the serial EEPROM and/or I²C to write 1 to bit 0 of the Port's Active State Power Management (ASPM) field.

L0s Link Power Management State Entry Conditions – Downstream Ports

The following four conditions must be met for Downstream Ports to enter the ASPM L0s Link PM state:

1. All PEX 8724 Upstream Port Rx Lanes are in the ASPM L0s Link PM state.
2. No pending TLPs (including Retry TLPs) or there are no credits available to transmit pending TLPs.
3. No pending DLLPs.
4. Enabled for entering the ASPM L0s Link PM state:
 - Bit 10 of the Port's **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10]) is Set
 - Bit 0 of the Port's **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]) is Set

The PM module directs the Downstream Link to enter the ASPM L0s Link PM state when Conditions 1, 2, and 3 are met for a specified period of time and Condition 4 is satisfied. The idle time can be selected, as either 1 μ s (default) or 4 μ s, in the Port's **Power Management Hot Plug User Configuration** register *L0s Entry Idle Count* bit (offset F70h[0]).

L0s Link Power Management State Exit Conditions

Components from either end of a PCI Express Link can initiate exit from the ASPM L0s Link PM state.

L0s Link Power Management State Exit Conditions – Upstream Port(s)

One or more of the following conditions must be met for Upstream Port(s) to exit the ASPM L0s Link PM state:

- If there are pending DLLPs or TLPs to transmit, –or– any of the Downstream Port Rx Lanes exit the ASPM L0s Link PM state
- Upstream Port ASPM L1 Link PM state entry conditions are met
- Upstream Port is placed into the D3hot Device PM state
- Link is Set to retrain, as a result of a DLLP timeout or Replay Number rollover

L0s Link Power Management State Exit Conditions – Downstream Ports

One or more of the following conditions must be met for a Downstream Port to exit the ASPM L0s Link PM state and enter the L0 Link PM state:

- If there are pending DLLPs or TLPs to transmit, –or– any of the Upstream Port Rx Lanes exit the ASPM L0s Link PM state
- Port's **Link Control** register *Retrain Link* or *Link Disable* bit (Downstream Ports, offset 78h[5:4], respectively) is Set
- If a PCI L1 Link PM State Entry Request or ASPM L1 Link PM State Request is received from the Downstream direction

The PM module directs the PHY to change from the ASPM L0s Link PM state to the operational L0 Link PM state, when any of the conditions listed above are met. The PHY transmits the FTS, then enters the L0 Link PM state.

11.3.2.2 L1 Link Power Management State

L1 Link Power Management State Entry

The L1 Link PM state is a deeper power-saving state than the ASPM L0s Link PM state, with both Tx and Rx Lanes settling into Electrical Idle after L1 Link PM state negotiation is complete.

A PCI Express Upstream Link can enter the L1 Link PM state when either of the following conditions are met:

- Upstream Port(s) is (are) placed into the D3hot Device PM state
- Upstream Port(s) is (are) in the D0 Active Device PM state, Idle conditions (explained below) are satisfied, and bit 1 of the Port's **Link Control** register *Active State Power Management (ASPM)* field is programmed for ASPM L1 Link PM state support (bit 1 of offset 78h[1:0] is Set)

Downstream Port Links can enter the L1 Link PM state when either of the following conditions are met:

- Downstream device is in the D0 Device PM state and initiates L1 Link PM state entry (bit 1 of the Port's *Active State Power Management (ASPM)* field is Set)
- Downstream device is placed into a D3hot Device PM state, and initiates L1 Link PM state entry

Entry into the PCI L1 Link PM state is always initiated by an Upstream Port or Downstream device. PCI L1 Link PM state entry is also possible when an Upstream Port is placed into a state other than the D0 Device PM state. This transition is required. Downstream Ports never initiate PCI L1 Link PM state entry.

The ASPM L1 Link PM state is optional, and software-controlled by writing 1 to bit 1 of the *Active State Power Management (ASPM)* field. Conditions for entering the ASPM L1 Link PM state are implementation-specific.

A Port cannot directly transition from the L0s to L1 Link PM state while in ASPM – the Port transitions from L0s -> L0 -> L1, instead. After the Port enters the ASPM L1 Link PM state, FC Update Timers are suspended.

PCI L1 Link PM state entry protocol uses a PM_Enter_L1 DLLP. In contrast, ASPM L1 Link PM state entry protocol uses a PM_Active_State_Request_L1 DLLP. Downstream Ports never NAK the PM_Enter_L1 DLLP received from a Downstream device; however, Downstream Ports can NAK a PM_Active_State_Request_L1 DLLP, if there is a packet to transmit.

ASPM L1 State Entry Conditions – Upstream Port(s)

If an Upstream Port is in the D0 Device PM state, and is enabled for ASPM L1 Link PM state entry (bit 1 of the Port's **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]) is Set), the Upstream Port requests ASPM L1 Link PM state entry if the following conditions are met for 2 to 3 μ s:

- All Downstream Port Links are in the L1 (or deeper) Link PM state, or the Link is Down
- No pending TLPs to transmit, including TLPs waiting for DLL ACK/NAK (in the Retry buffer)
- No pending DLLPs, except for FC DLLPs

Also, the Upstream Port should have the maximum FC credits replenished for each packet type. The Upstream Port suspends transmission of new TLPs.

The Upstream Port continuously transmits a PM_Active_State_Request_L1 DLLP, until it sees a PM_Request_Ack DLLP or PM_Active_State_Nak TLP Message. The PM_Active_State_Request_L1 DLLPs are transmitted with no more than a 4-symbol time gap on the Upstream Link.

After the Upstream Port receives a PM_Request_Ack DLLP, it directs the PHY to enter the L1 Link PM state. The Link settles into the L1 Link PM state when an Electrical Idle Ordered-Set (EIOS) is transmitted, and Electrical Idle is received. After the Upstream Port receives a PM_Active_State_Nak TLP Message, the Link enters the ASPM L0s Link PM state as soon as possible, if conditions are met.

The PEX 8724 Upstream Port, when enabled for the ASPM L0s and L1 Link PM states, and is in the ASPM L0s Link PM state for more than 2 μ s and the ASPM L1 Link PM state entry conditions listed above are met, exits the ASPM L0s Link PM state and negotiates for ASPM L1 Link PM state entry.

Downstream Port Accepts Request from Downstream Device PM_Active_State_Request_L1 DLLP

A Downstream Port accepts a Request from the Downstream device PM_Active_State_Request_L1 DLLP and suspends transmission of new TLPs, when the following three conditions are met:

- Downstream Port supports, and is enabled for, ASPM L1 Link PM state entry:
 - Bit 11 of the Port's **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10]), and
 - Bit 1 of the Port's **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]) is Set
- No TLPs are scheduled for transmission (including sent TLPs that are waiting for an ACK/NAK DLLP)
- No pending DLLPs, other than FC DLLPs

The Downstream Port continuously transmits a PM_Request_Ack DLLP, with no more than a 4-symbol time gap between DLLPs, until the Rx Lanes enter Electrical Idle, which indicates that the other side of the Link has received an ACK. After the Downstream Port receives Electrical Idle, PM directs the PHY to enter the L1 Link PM state.

If the conditions for going to the ASPM L1 Link PM state are not satisfied, the Downstream Port transmits a PM_Active_State_Nak Message and remains in the L0 Link PM state.

Upstream Port Requests PCI L1 Link Power Management State Entry

An Upstream Port requests PCI L1 Link PM state entry, when the following six conditions are met:

- Software places the Upstream Port into the D3hot Device PM state.
- Upstream Port suspends the scheduling of new TLPs, after it has sent the Completion of the TLP that enabled the D3hot Device PM state.
- Upstream Port waits until all previously transmitted TLPs are acknowledged. During this time, the Upstream Port receives TLPs and DLLPs, and transmits ACK/NAK and FC DLLPs.
- Upstream Port waits for a sufficient quantity of FC credits, for all TLP packet types.
- Upstream Port waits for all Retry TLPs to complete.
- After the Upstream Port transmits all pending TLPs, it transmits a PM_Enter_L1 DLLP Upstream, with no more than a 4-symbol time gap.

When the Upstream Port receives a PM_Request_Ack DLLP, it directs the Link to enter the PCI L1 Link PM state.

When in the PCI L1 Link PM state, if any Receiver within the Port detects exit from Electrical Idle, or a TLP must be transmitted, the Upstream Port does the following:

1. Exits the PCI L1 Link PM state.
2. Transmits pending TLP(s), if required.
3. Re-enters the PCI L1 Link PM state after a 16- μ s delay.

The Upstream Port must detect a continuous 16 μ s of the PHY L0 Link PM state for the next state entry.

Downstream Port Accepts Request from Downstream Device PM_Enter_L1 DLLP

A Downstream Port accepts a Request from the Downstream device PM_Enter_L1 DLLP, if the following three conditions are met:

- No TLPs are scheduled for transmission
- TLP transmissions are suspended
- Downstream Port received a Link Layer ACK for all transmitted TLPs, and gathered sufficient maximum packet size credits for all packet types

The Downstream Port continuously transmits PM_Enter_Ack DLLPs, with no more than a 4-symbol time gap between subsequent DLLPs. When the Downstream Port “sees” the Rx Lanes enter Electrical Idle, the Port directs the PHY to enter the L1 Link PM state.

L1 Link Power Management State Exit Conditions

Devices on either end of a PCI Express Link can initiate exit from the L1 Link PM state. Unlike the L1 Link PM state entry protocol, in which both ends of the Link must negotiate for the resultant Link state, L1 Link PM state exit does **not** require negotiation.

PCI L1 Link Power Management State Exit Conditions – Upstream Port(s)

One or more of the following conditions must be met for Upstream Port(s) to exit the PCI L1 Link PM state:

- Any Upstream Port Rx Lane exits Electrical Idle, –or–
- TLPs are pending transmission. (This occurs if a PME Message transmission is pending, due to Hot Plug.)
- Downstream Port exited the PCI L1 Link PM state as the result of a Downstream Port Rx Lane exiting Electrical Idle while in the PCI L1 Link PM state.

PCI L1 Power Management State Exit Conditions – Downstream Ports

The following conditions must be met for a Downstream Port to exit the PCI L1 Link PM state:

- Any Downstream Rx Lanes exit Electrical Idle
- TLPs are pending transmission on the Link

ASPM L1 Link Power Management State Exit Conditions – Upstream Port(s)

The following conditions must be met for Upstream Port(s) to exit the ASPM L1 Link PM state:

- Any Downstream Port exits the ASPM L1 Link PM state
- TLPs are pending transmission
- Any Upstream Port Rx Lane exits Electrical Idle
- Upstream Port is placed into the D3hot Device PM state

ASPM L1 Link Power Management State Exit Conditions – Downstream Ports

The following conditions must be met for a Downstream Port to exit the ASPM L1 Link PM state:

- Any Downstream Rx Lanes exit Electrical Idle.
- TLPs are pending transmission on the Link.
- Upstream Port starts to exit the ASPM L1 Link PM state. Downstream Ports, which are in the ASPM L1 Link PM state as the result of a Downstream device being placed into the D3hot Device PM state, are not affected by this transition.

L1 Link Power Management State Entry Negotiation Interruption

Per the *PCI Express Base r3.0*, when L1 Link PM state entry negotiation is interrupted, the state machine on both ends of the PCI Express Link should return to an Idle state. A Port can be in any of the states described in the sections that follow, when L1 Link PM state entry negotiation is interrupted, usually as the result of a trip through the *Recovery* state.

L1 Link Power Management State Entry Negotiation Interruption – Upstream Port(s)

What occurs after L1 Link PM state entry negotiation is interrupted on an Upstream Port is dependent upon the following:

- **Started requesting ASPM L1 Link PM state entry, using PM_Active_State_L1_Request DLLP** – When the Tx or Rx *Recovery* state occurs, the Port stops sending Requests, returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **Waiting for a PM_Request_Ack DLLP** – When the Tx or Rx *Recovery* state occurs, the Port returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **After receiving a PM_Request_Ack DLLP, directs the PHY to the L1 Link PM state, and the PHY has not reached the L1 Link PM state and the PHY enters the Tx or Rx Recovery state** – The Port returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **After the PHY transmits an EIOS and settles in the Tx L1 Link PM state, but the Rx has not yet reached the L1 Link PM state, and an Rx Recovery state occurs** – The Port directs the PHY Tx to exit the L1 Link PM state.

L1 Link Power Management State Entry Negotiation Interruption – Downstream Ports

What occurs after L1 Link PM state entry negotiation is interrupted on a Downstream Port is dependent upon the following:

- **Transmits PM_Request_Ack DLLPs for an L1 Link PM State Entry Request** – When the Tx or Rx *Recovery* state occurs, the Port returns to an Idle state, then waits 10 μ s before validating another L1 Link PM State Entry Request from the Downstream device, –or– the Rx should enter the ASPM L0s Link PM state, to validate the next L1 Link PM State Entry Request.
- **Waits for Electrical Idle from the other side of the PCI Express Link** – When the Tx or Rx *Recovery* state occurs, the Port returns to an Idle state, then waits 10 μ s before validating another L1 Link PM State Entry Request from the Downstream device, –or– the Rx should enter the ASPM L0s Link PM state, to validate the next L1 Link PM State Entry Request.
- **Rx is in the L1 Link PM State and directs the PHY to enter the L1 Link PM State** – If the PHY enters the Tx *Recovery* state because the Port's **Link Control** register *Retrain Link* bit (Downstream Ports, offset 78h[5]) is Set, the Port returns to an Idle state, then waits 10 μ s before validating another L1 Link PM State Entry Request from the Downstream device, –or– the Rx should enter the ASPM L0s Link PM state, to validate the next L1 Link PM State Entry Request.

11.3.2.3 L2/L3 Ready Link Power Management State Entry

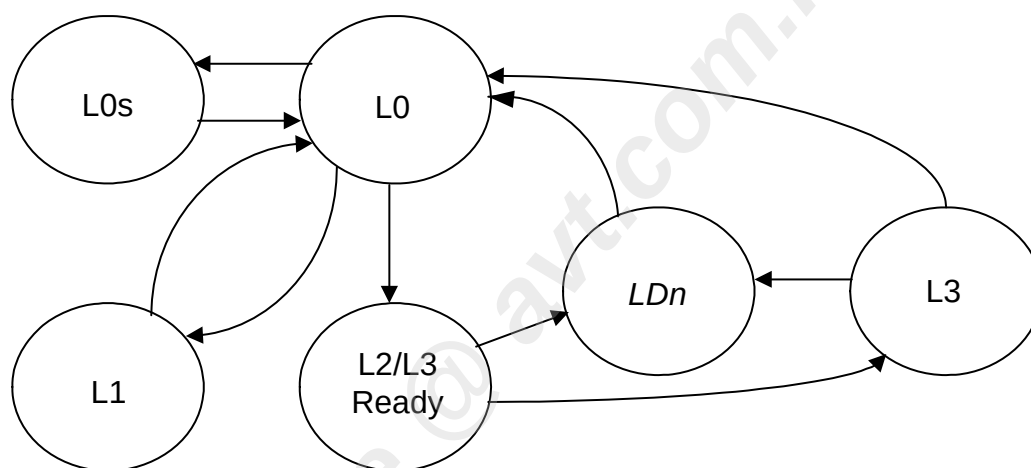
The L2/L3 Ready and L3 Link PM states are the staging points for removing main power. Support for the L2/L3 Ready Link PM state transition protocol is required.

The L2/L3 Ready Link PM state is related to the Device PM state transitions – it is the state that a given Link enters when the platform is preparing to enter its system sleep or shutdown state. Following completion of the L2/L3 Ready Link PM state transition protocol, the Link enters the L2/L3 Ready Link PM state. After main power is removed, the Link settles into the L3 Link PM state. The time for the L2/L3 Ready Link PM state entry transition is indicated by completion of the PME_Turn_Off/PME_TO_Ack handshake sequence. Any actions on the part of the Downstream component necessary to ready itself for power loss must be completed before initiating transition to the L2/L3 Ready Link PM state. After all preparations for loss of power and clock are complete, the Downstream component initiates L2/L3 Ready Link PM state entry, by transmitting the PM_Enter_L23 DLLP Upstream.

Exit from the L2/L3 Ready Link PM state, back to the L0 Link PM state, can be initiated only by an Upstream transaction that targets the Downstream component (Upstream Electrical Idle exit) in the same manner that an Upstream transaction would trigger transition from the L1 to L0 Link PM states. The transition of the L2/L3 Ready Link PM state back to the L0 Link PM state is always through the *Link Down (LDn)* state. This transition can also result from **PEX_PERST#** input assertion (Fundamental Reset).

Figure 11-2 illustrates the Link PM state transitions.

Figure 11-2. Link Power Management State Transitions



PME_TO_Ack TLP Flow from Downstream Device to Upstream Port(s)

Because all previously transmitted packets must first be transmitted Upstream toward the Root Complex, the PEX 8724 performs the following tasks before transmitting the PME_TO_Ack packet:

1. When a PME_TO_Ack Message is received from a Downstream device, the Message is routed to the Station's Upstream Egress module.
2. The Egress module decodes the PME_TO_Ack TLP, then informs the Upstream Port's PM module that it received the PME_TO_Ack Message.
3. When the Upstream Port receives the PME_TO_Ack Message from all enabled Downstream Ports, it transmits the Message Upstream toward the Root Complex. This PME_TO_Ack Message flushes all previously transmitted PM_PME Messages, due to PEX 8724 internal events.

L2/L3 Ready Link Power Management State Entry Sequence – Upstream Port(s)

System software places all Downstream devices into the D3hot Device PM state, before broadcasting the PME_Turn_Off Message. The Root Complex then transmits the PME_Turn_Off Message to an Upstream Port. The Upstream Port then unicasts the PME_Turn_Off Message to all Downstream Ports, which reports an active Data Link Layer (DLL). The Downstream Ports receive a PME_Turn_Off Message from Upstream, then transmit the Message to the Downstream device(s).

1. Upstream Port waits for a PME_TO_Ack Message, from all Downstream devices.
2. Upstream Port waits for all pending TLP transmission to complete.
3. After all Downstream Ports receive the PME_TO_Ack, the PEX 8724 transmits the PME_TO_Ack Upstream.
4. PEX 8724 transmits the PM_Enter_L23 DLLP Upstream, after all Downstream Ports are in the L2/L3 Ready Link PM state.
5. When in the L2/L3 Ready Link PM state, if **PEX_PERST#** input is de-asserted (main power is removed), the Link settles into the L3 Link PM state.

Note: When in the L2/L3 Ready Link PM state, a Receiver exiting Electrical Idle also causes exit from the L2/L3 Ready Link PM state. The LTSSM enters the Link Down (LDn) state, then the PEX 8724 generates an internal Soft Reset, which also brings all Downstream Ports to the LDn state, from the L2/L3 Ready Link PM state.

After an Upstream Port is in the L2/L3 Ready Link PM state, Rx Electrical Idle exit causes the Link to transition to the LDn state, then subsequently enter the Detect state.

L2/L3 Ready Link Power Management State Entry Sequence – Downstream Ports

When a Downstream Port is entering the L2/L3 Ready Link PM state, the PEX 8724 performs the following tasks:

1. Downstream Port receives a PME_Turn_Off Message from an Upstream Port.
2. Downstream Port suspends transmission of new PM_PME Messages (upon global power-off).
3. Downstream Port transmits PME_Turn_Off Message to the Downstream device, after Clearing all pending TLPs.
4. Downstream Port waits for a PM_Enter_L23 DLLP from the Downstream device. When the PEX 8724 receives the PM_Enter_L23 DLLP, the switch transmits a PM_Request_Ack DLLP, then waits for the Rx to enter the L2/L3 Ready Link PM state. After the Rx enters Electrical Idle, the Tx enters the L2/L3 Ready Link PM state. The Upstream scoreboard is updated through vendor-defined Messages.
5. When in the L2/L3 Ready Link PM state, if [PEX_PERST#](#) input is de-asserted (main power is removed), the Link settles into the L3 Link PM state.

11.4 Power Management Tracking

Note: *NT Port Link Interface entry and exit to Link PM states are not dependent upon the Transparent Upstream nor Downstream Port power states or traffic. They are solely dependent upon the NT Port Link Interface's traffic conditions.*

Upstream Port logic tracks the status of each Downstream and Upstream Port Link, to derive the following conditions:

- Upstream Port(s) enter(s) the ASPM L0s Link PM state when all enabled Downstream Receivers are in the ASPM L0s (or deeper) Link PM state, or in a *Link Down* state.
- Upstream Port(s) enter(s) the ASPM L1 Link PM state, only when all Downstream Ports are in the ASPM L1 (or deeper) Link PM state, or the Link is Down.
- When a Downstream Port is in the ASPM L1 Link PM state and an ASPM L1 Link PM state exit is occurring in the Downstream Port, the Upstream Port(s) exit(s) the L1 Link PM state.
- When the Upstream Port(s) is (are) in the ASPM L1 Link PM state and an ASPM L1 Link PM state exit is occurring due to Rx Electrical Idle exit, the Downstream Ports exit the L1 Link PM state.
- When a PME_TO_Ack Message is received only on all active (not in *Link Down*) Downstream Ports, a PME_TO_Ack Message is issued toward the Upstream Port(s). The NT Port Virtual Interface is marked as being in the *DL_Down* state.
- When all Downstream Ports are in the L2/L3 Ready Link PM or *Link Down* state, the Upstream Port(s) transmit(s) PM_Enter_L23 DLLPs toward the Root Complex.

11.5 Power Management Event Handler

PM_PME Messages are Posted TLPs that inform the PM software which agent within the PCI Express hierarchy has requested a PM-state change. PM_PME Messages are always routed Upstream toward the Root Complex.

PCI Express components are permitted to wake the system from any supported PM state, through a PME Request.

When a Transparent Downstream Port is in the D3hot Device PM state, the following Hot Plug and/or PCI Express Hot Plug events cause the Port's **PCI Power Management Status and Control** register *PME Status* bit (offset 44h[15]) to be Set:

- Hot Plug-capable Ports
 - *Presence Detect Changed* (logical OR of PRSNT# (HP_PRSNT_x# or Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input), and SerDes Receiver Detect^a on Lane(s) associated with that Port)
 - *Attention Button Pressed*
 - *Power Fault Detected*
 - *MRL Sensor Changed*
 - *Command Completed*
 - *Data Link Layer State Changed*
 - *Link Bandwidth Management Status*
 - *Link Autonomous Bandwidth Status*
- Non-Hot Plug-capable Downstream Ports
 - *Presence Detect Changed* (SerDes Receiver Detect^a on Lane(s) associated with that Port)
 - *Data Link Layer State Changed*

This causes the Downstream Port to generate a PM_PME Message, if the Port's **PCI Power Management Status and Control** register *PME Enable* bit (offset 44h[8]) is Set.

The PEX 8724 also forwards PM_PME Messages from Downstream Endpoint devices, Upstream toward the Root Complex. There is no *Status* bit that indicates forwarding of PM_PME Messages.

a. The SerDes Receiver Detect mechanism is comprised of the **Physical Layer Receiver Detect Status** register *Receiver Detected on Lane x* bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16]) or Serial Hot Plug *SHP_PRSNT#* (SHP_PRSNTx#) (from external I²C I/O Expander) input for the Port.

11.6 Power Management – Virtual Switch Mode

In Virtual Switch mode, the PEX 8724 can have up to four Upstream Ports. Downstream Ports are assigned to different Upstream Ports, depending upon the Virtual Switch Table. (Refer to [Section 5.3.3, “Virtual Switch Table.”](#)) Each virtual switch works as an independent switch for deciding entry into, or exit from, different PM states.

ASPM L0s and L1 Link PM state entry/exit works the same way as defined in the previous sections. However, each virtual switch Upstream Port monitors Link status only for its own Downstream Ports, for ASPM L0s and L1 Link PM state entry/exit. Similarly, each virtual switch’s Downstream Ports monitor Link status for their own Upstream Port, for deciding entry/exit from the ASPM L0s and L1 Link PM states.

A virtual switch Upstream Port that is programmed into the D3hot Device PM state (Port’s **PCI Power Management Status and Control** register *Power State* field bits, offset 44h[1:0], are both Set) requests the PCI L1 Link PM state, and finally settles into the PCI L1 Link PM state after the Upstream Port returns a Completion for this Configuration Write and PCI L1 Link PM state negotiation successfully completes.

A PM_Turn_Off Message received at a virtual switch Upstream Port is broadcast only to the Downstream Ports assigned to that virtual switch. When a PME_TO_Ack Message is subsequently received from all Downstream Ports assigned to the virtual switch, a single PME_TO_Ack Message is sent Upstream of that virtual switch.

If a virtual switch has only an Upstream Port and no Downstream Ports associated with it (*such as* the case of a Management Port), the virtual switch behaves like an endpoint for entry/exit to the ASPM L0s, ASPM L1, PCI L1, and L2/L3 Ready Link PM states. Entry/exit to different PM states depends only upon the idle/traffic conditions on that Upstream Port.

When a new Downstream Port is added to a virtual switch due to re-configuration, future entry into/exit from different power states on the virtual switch’s Upstream Port also depends upon the newly added Downstream Port’s power states. After a Downstream Port is removed from a virtual switch, that Port’s Link states are not taken into consideration for future entry/exit into power states.

Interrupts and PM_PME Messages generated on a virtual switch’s Downstream Ports are routed to its Upstream Port.

12.1 Multiple Virtual Switches

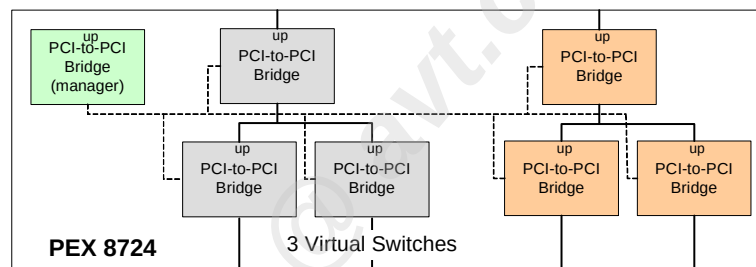
The PEX 8724 implements multiple virtual PCI Express switches, up to four total. Each virtual switch has its own Upstream Port and zero or more Downstream Ports. From a software point of view, each virtual switch consists of one Upstream PCI-to-PCI (P2P or P-P) bridge and zero to N Downstream PCI-to-PCI bridges. Each virtual switch has its own Reset, Interrupt, and Error signals. While the virtual switches share the same physical switch, traffic from one virtual switch cannot migrate to another virtual switch.

Figure 12-1 illustrates a partitioning of three virtual switches in a single physical switch – a Management Port hierarchy and two regular PCI Express 3-Port switches. In this example, the right three PCI-to-PCI bridges are owned by the right-most Upstream Port, and the middle three PCI-to-PCI bridges are owned by the middle Upstream Port. In this scenario, a third Upstream Port is used for a Management Port, and this Upstream Port has access to all other hierarchies, as indicated by the dashed line. The Management Port configures the virtual switches so that each Port – Upstream or Downstream – is placed in the proper configuration. Each Port can be assigned to only one virtual switch at a time. The Management Port can own Downstream Ports, although this mode is not shown in Figure 12-1.

Host-to-Host messages across virtual switch boundaries can be sent by the Management Port's **Scratchpad** and **Mailbox** registers.

Base mode can be emulated in Virtual Switch mode, by enabling two virtual switches, with all remaining Ports assigned to VS0.

Figure 12-1. Multiple Virtual Switches in a Single Physical Switch



12.1.1 Default Virtual Switch Port Assignments

Note: Default Port assignments must be changed, by using the serial EEPROM, I²C/SMBus, and/or the Management Port, prior to Setting the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset 3ACh[0]).

The Management Port, if enabled (STRAP_MGMT_PORT_EN=0 or Z) is assigned by the STRAP_UPSTRM_PORTSEL[2:0] 3-state inputs, or as programmed in the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h). The value must correspond to a virtual switch Upstream Port.

Table 12-1 lists the default virtual switch Port assignments, which are dependent upon the quantity of enabled virtual switches. The Port Numbers referenced correspond to those listed in Table 4-1, “Port Configurations.”

Table 12-1. Virtual Switch Default Port Assignments – Virtual Switch Mode

# of Enabled Virtual Switches	Test Modes	STRAP_TESTMODE[2:0] Input States	VS_MODE Input State	Virtual Switch Enable Register (Offset 358h) Value	VSx Upstream Register Offset and Value (Upstream Port Number) ^{a, b}	VSx Downstream Ports ^a	VSx Vector Register Offset
1	4	0ZZ	0	1h	VS0 (360h) = 0h (0)	1, 2, 3, 8, 9, 10	VS0 (380h)
	5	0Z1					
	6	010					
	7	01Z					
2	4	0ZZ	Z	3h	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A)	VS0 = 1, 2, 3, 10 VS1 = 8, 9	VS0 (380h) VS1 (384h)
	5	0Z1					
	6	010					
	7	01Z					
3	4	0ZZ	1	7h	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A) VS2 (368h) = Ah (10)	VS0 = 1, 2, 3 VS1 = 8, 9 VS2 = None	VS0 (380h) VS1 (384h) VS2 (388h)
	5	0Z1					
	6	010					
	7	01Z					
4	8	011	0	Fh	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A) VS2 (368h) = Ah (10) VS3 (36Ch) = 10h	VS0 = 1, 2, 3 VS1 = 8, 9 VS2 = None VS3 = None	VS0 (380h) VS1 (384h) VS2 (388h) VS3 (38Ch)
	9	Z00					
	10	Z0Z					
	11	Z01					

a. Table 4-1, “Port Configurations,” indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

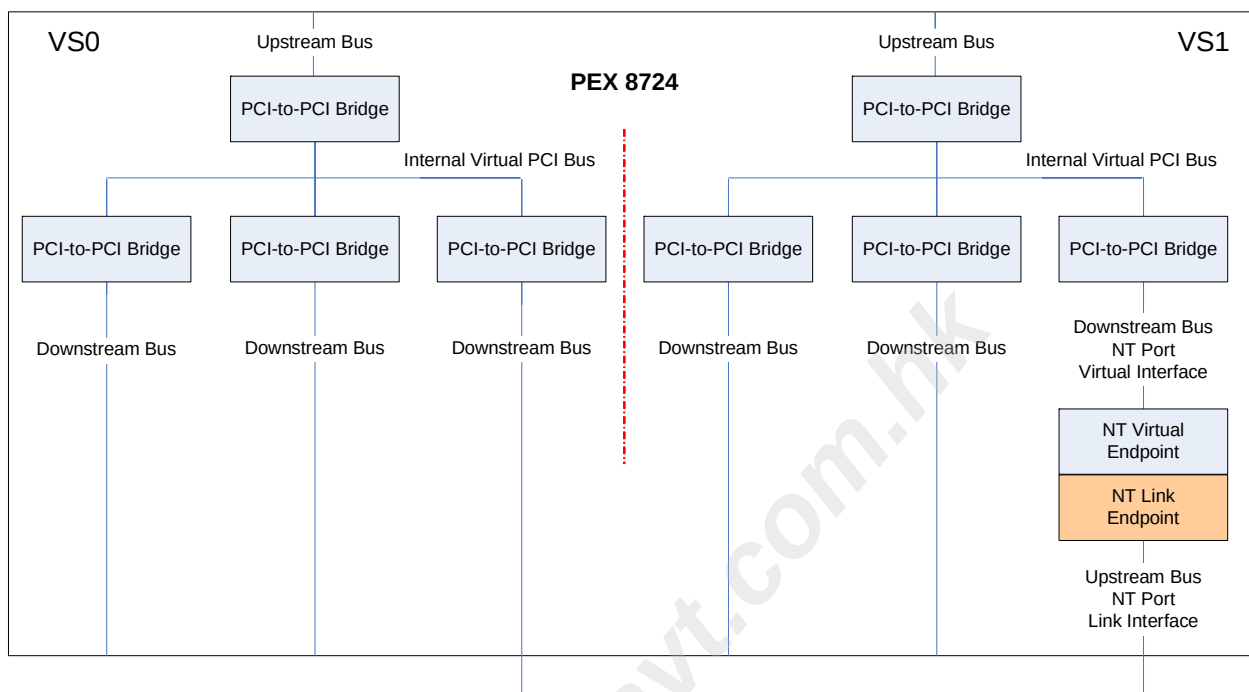
b. In Virtual Switch mode, the VS1 Upstream register (Port 0, accessible through the Management Port, offset 364h) default value (4h) must be changed by serial EEPROM (or by I²C/SMBus, provided that STRAP_I2C_SMBUS_CFG_EN# is Low), to a valid Port Number.

12.1.2 Combined NT and Multiple Virtual Switches

The PEX 8724 supports the combination of a Non-Transparent bridge (NTB) and multiple virtual switches. The NT Port Virtual Interface is enumerated by only one of the virtual switches, as illustrated in Figure 12-2, for a sample configuration when using the VS0 and VS1 virtual switches.

In this example, a Downstream Port in VS0 can be connected to the VS1 NT Port Link Interface, so that the VS0 and VS1 Hosts can communicate with one another, while having their own independent hierarchies.

Figure 12-2. Sample Configuration when NT Is Used in Combination with Multiple Virtual Switches



12.2 Management Port

The PEX 8724 supports configuration and management through various options, described in the sections that follow. In Virtual Switch mode, the PEX 8724's registers are accessible by the designated Active Management Port, serial EEPROM, and/or I²C Slave interface. The PEX 8724 can be configured by serial EEPROM and/or Strapping signals, without restrictions. With the PEX 8724, any two PCI Express Ports can be configured as Management Ports, with one designated as the primary Management Port, and the other as the Redundant Management Port.

The Management Port can be used to:

- Configure virtual switches
- Move Ports from one virtual switch to another
- Monitor all virtual switch Links
- Configure PEX 8724-wide registers (*such as* the Physical Layer (PHY) registers)
- Access the serial EEPROM – In Virtual Switch mode, in-band access to the serial EEPROM is restricted to the Management Port, as designated by the **Management Port Control** register *Active Management Port* field (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[4:0])

The Redundant Management Port can be promoted to become the new Active Management Port, by software (through the Redundant Management Port, Management Port, and/or I²C/SMBus) copying the register's *Redundant Management Port* field [12:8] value to the *Active Management Port* field. (Refer to [Section 12.5.2](#) for further details.)

12.2.1 Out-of-Band Interfaces

This section briefly describes the out-of-band interfaces supported by the PEX 8724.

12.2.1.1 Unused PCI Express Port – Management-Capable Port

Any Port can be designated as the Management Port and connected to a small service processor. This configuration is used in Virtual Switch mode.

12.2.1.2 Strapping Signals

The Strapping signals are used to load default configurations. Refer to [Section 3.4.4, “Strapping Signals,”](#) for details.

12.2.1.3 Serial EEPROM

An on-board serial EEPROM can be used to override the Strapping signals and configure the PEX 8724. Refer to [Chapter 6, “Serial EEPROM Controller,”](#) for details.

12.2.1.4 I²C Bus/SMBus

The PEX 8724 supports both the I²C Bus and System Management Bus (SMBus) interfaces for configuring the registers. However, if the **STRAP_I2C_SMBUS_CFG_EN#** input is enabled to delay linkup until the **Configuration Release** register *Initiate Configuration* bit (Port 0, accessible through the Management Port, offset 3ACh[0]) is Set, this register Write can only be performed by I²C (not SMBus).

Refer to [Chapter 7, “I2C/SMBus Slave Interface Operation,”](#) for details.

12.2.2 In-Band Interface

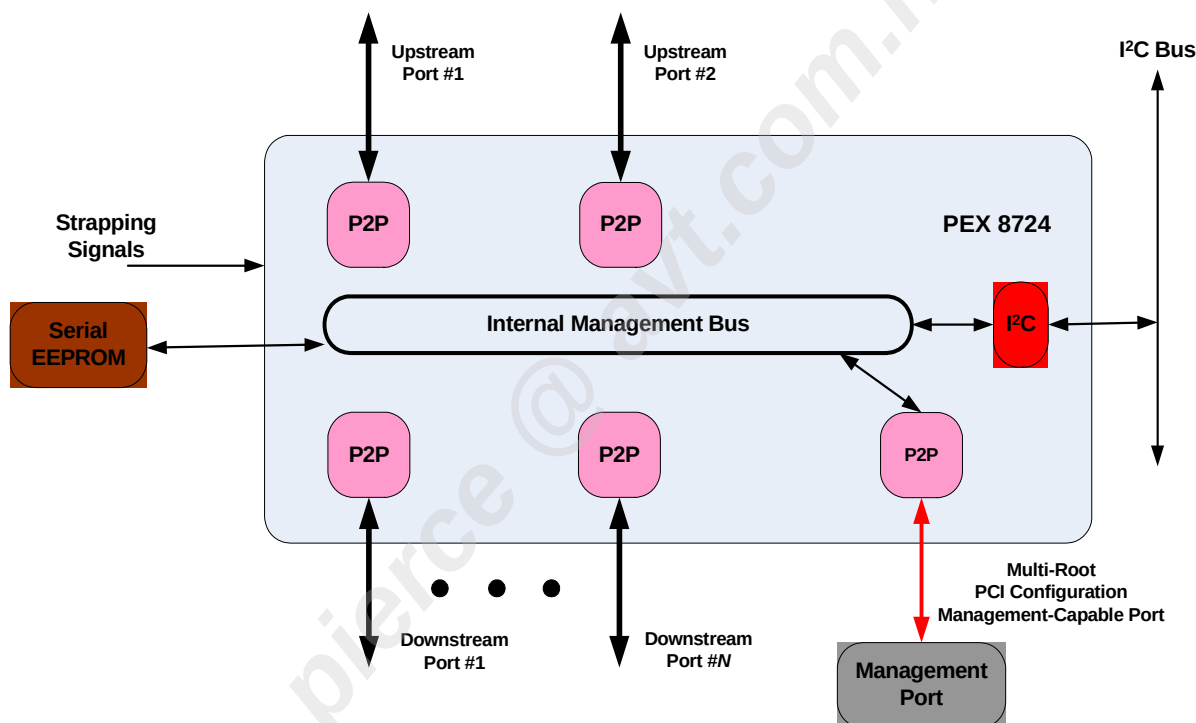
An in-band interface is typically used to manage PCI Express fabric within a single domain hierarchy, where the virtual switch's Upstream Port manages the PCI Express switch. In the case of Multi-Root switches, the shared resources are being modified by a trusted resource, *such as* a designated PCI Express Port. PLX Multi-Root switches support this mode of operation, in which one of the switch's Ports can be used for in-band operations, where the CPU can send data (regular traffic), as well as control data to control the PCI Express switch fabric's behavior. Additionally, a second Port can be designated as a Redundant Management Port. If the primary Port fails, the Redundant Management Port takes over the responsibility of managing the PCI Express switch fabric.

12.2.2.1 Configuration and Management

Figure 12-3 illustrates various options for accessing the Management Bus and internal registers in Virtual Switch mode, using two virtual switches. Each Upstream Port manages its own partition; however, another Port or interface is designated as the Management Port, with responsibility for defining and modifying partitions, as required. The PEX 8724 can also be configured by Strapping inputs, serial EEPROM, and/or I²C.

Note: The PCI-to-PCI (P2P) blocks in Figure 12-3 are a logical representation of how a Port presents itself to software.

Figure 12-3. Configuration and Management – Virtual Switch Mode



Note: Table 12-1, “Virtual Switch Default Port Assignments – Virtual Switch Mode,” lists the default virtual switch Port assignments. Table 4-1 indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

12.2.2.2 In-Band Management Port

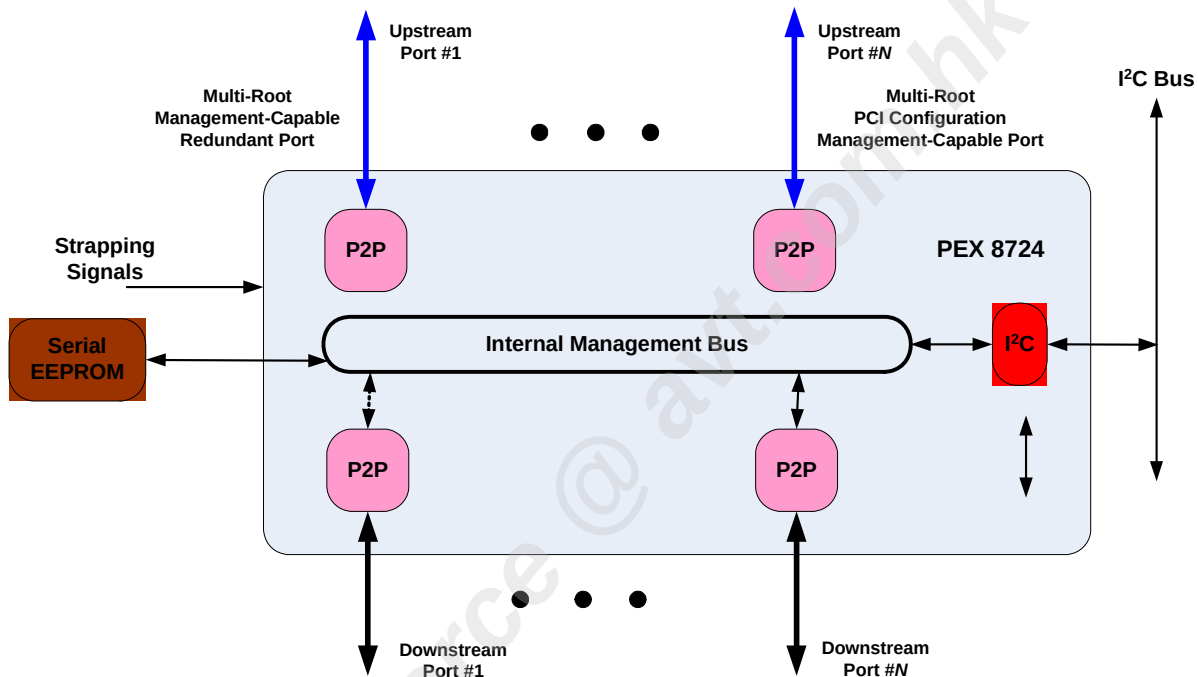
In Virtual Switch mode, an in-band PCI Express Port can be used as a Management Port and have access to all registers. (Refer to [Figure 12-4](#).)

The Redundant Management Port can access only the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h), to promote itself to be the Management Port if the Active Management Port Host fails, using Memory-Mapped access to offset 354h within its own **BAR0/1** register map.

Notes: All Configuration Space registers (CSRs) requiring access by the Management Port are memory-mapped, using the Management Port's **Base Address 0** and **Base Address 1** registers (**BAR0** and **BAR1**, respectively), located in the Type 1 CSR headers. The In-Band Management Port uses Memory-Mapped transactions, to access all registers in all virtual switches. Because Ports within a Station can be assigned to different virtual switches, Management Port Writes to Station-specific registers can affect multiple virtual switches.

The PCI-to-PCI (P2P) blocks in [Figure 12-4](#) are a logical representation of how a Port presents itself to software.

Figure 12-4. Configuration and Management of In-Band Management – Virtual Switch Mode



Note: [Table 12-1, “Virtual Switch Default Port Assignments – Virtual Switch Mode,”](#) lists the default virtual switch Port assignments. [Table 4-1](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

12.2.2.3 Management Ports and Restriction

The shared registers listed in [Table 12-2](#), and all PCI-to-PCI registers, are accessed by the Active Management Port. The Redundant Management Port can access the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset [354h](#)), to promote itself to be the Management Port if the Active Management Port Host fails.

In Virtual Switch mode, the [STRAP_UPSTRM_PORTSEL\[2:0\]](#) 3-state inputs define which Port is the Active Management Port, when the Management Port is enabled ([STRAP_MGMT_PORT_EN](#)=0 or Z).

The Management Port can also be enabled and designated by serial EEPROM and/or I²C writing an appropriate value into the register's [Active Management Port Enable](#) and [Active Management Port](#) bits (bits [\[5 and 4:0\]](#), respectively). As previously mentioned, the Redundant Management Port can be promoted to become the new Active Management Port, by software (through the Redundant Management Port, Management Port, and/or I²C/SMBus) copying the register's [Redundant Management Port](#) field [\[12:8\]](#) value to the [Active Management Port](#) field.

Note: In Virtual Switch mode, the Virtual Switch Table registers include the **VSx Port Vector** and **VSx Upstream** registers (Port 0, accessible through the Management Port, offsets [380h](#) through [38Ch](#), and [360h](#) through [36Ch](#), respectively). These two sets of registers define which Ports are associated to each virtual switch, and which Port is the Upstream Port for each virtual switch, respectively. These registers must be initialized by one (or more) of the following agents:

- Serial EEPROM.
- I²C/SMBus, provided that the [STRAP_I2C_SMBUS_CFG_EN#](#) input is Low (to delay linkup until I²C/SMBus Sets the [Configuration Release](#) register [Initiate Configuration](#) bit (Port 0, accessible through the Management Port, offset [3ACh\[0\]](#)). This option might require software support, to delay Host enumeration until I²C/SMBus Sets the [Initiate Configuration](#) bit after programming the PEX 8724 Configuration registers.
- Management Port software, provided that [STRAP_MGMT_PORT_EN](#) is left floating (to enable the Management Port, and delay linkup of all other Ports until software and/or I²C/SMBus Sets the [Initiate Configuration](#) bit).

Use of serial EEPROM for the initialization might be the best choice for most applications, because it is the simplest solution. Therefore, for Virtual Switch mode applications, the serial EEPROM is required to initialize the Virtual Switch Table registers, unless I²C/SMBus and/or Management Port software can perform this task.

Table 12-2. Virtual Switch Management Registers (accessible from Primary Management Port)^a

Offset(s)	Register	Description ^b
354h	Management Port Control	This register contains bits that enable and indicate the Port Number of the Active and Redundant Management Ports.
358h[3:0]	Virtual Switch Enable	The register's <i>VSx Enable</i> bits are used to enable or disable virtual switches within the system. There is one bit, per virtual switch (VS0 through VS3). ^c
360h – 36Ch	VSx Upstream	These registers define the Upstream Port of each virtual switch. There is one register, per virtual switch (VS0 through VS3). ^c
380h – 38Ch	VSx Port Vector	These registers define the Upstream and Downstream Ports associated with each virtual switch. There is one register, per virtual switch (VS0 through VS3). Each register has one bit, per Port. ^c
3A0h	Port Reset	When driven with a value of 1, this register holds the Port in reset, including the Port PCI-to-PCI bridge and the hierarchy below it. A value of 0 indicates to un-reset the PCI-to-PCI bridge and the hierarchy below it. There is one bit, per Port.
900h	Switch Link Up	When the Port's Link state transitions from down to up, the Port's bit is Set. The Port's bit is Cleared, by using software to explicitly write 1 to the bit. The register has one bit, per Port.
904h	Switch Link Down	When the Port's Link state transitions from up to down, the Port's bit is Set. The Port's bit is Cleared, by using software to explicitly write 1 to the bit. The register has one bit, per Port.
908h	Switch Link Event Mask	If the <i>Mask</i> bit is Set, the corresponding <i>Up</i> or <i>Down</i> bit (located in register offsets 900h and 904h, respectively) transition does not generate an interrupt to the Management Port. If not masked, the bit transition generates an interrupt to the Active Management Port. The register has one <i>Mask</i> bit, per Port.
90Ch	Switch Link Status	This Read-Only register indicates Link status. The register has one <i>Status</i> bit, per Port.

- a. All registers listed in this table are located in Port 0, accessible through the Management Port.
- b. For more complete descriptions, refer to the individual registers listed.
- c. Additional information is also provided in [Table 5-4, “Virtual Switch Table Registers.”](#)

12.3 Virtual Switch Reset and Initialization

12.3.1 Virtual Switch Reset

For details regarding the Virtual Switch reset, refer to [Section 5.1.2, “Resets – Virtual Switch Mode.”](#)

12.3.2 Virtual Switch Initialization

For details regarding Virtual Switch Initialization, refer to [Section 5.3, “Initialization – Virtual Switch Mode.”](#)

12.3.3 Virtual Switch Table Programming Sequence

For details regarding the Virtual Switch Table Programming Sequence, refer to [Section 5.3.3.2, “Virtual Switch Table Programming Sequence.”](#)

12.4 Port Migration – Moving a Port from One Virtual Switch to Another (VSx to VSy)

A Downstream Port can be moved from one virtual switch (VSx) to another (VSy), by programming both virtual switch's **VSx Port Vector** registers (Port 0, accessible through the Management Port, offsets [380h](#) through [38Ch](#)). (The reason for the Port to be moved, and the coordination with the Management Port, are beyond the scope of this data book.) Prior to moving the Port, traffic through VSx must be stopped.

First, the **VSx Port Vector** register for VSx must **remove** the Port. This can be done by Clearing the bit(s) in the active Port Vector for the Port(s) that is (are) moving **out** of VSx.

Second, the **VSx Port Vector** register for VSy must **add** the Port. This can be done by Setting the bit(s) in the active Port Vector for the Port(s) that just left VSx, that is (are) moving **to** VSy.

12.5 Failover in Virtual Switch Mode

This section discusses the various types of failover in Virtual Switch mode:

- [Virtual Switch Host Failover](#)
- [Active Management Port Failover](#)
- [Virtual Switch Host to NT Host Failover](#)

Another type of failover in NT mode, that is related to Virtual Switch mode, is described in [Section 14.9.3, “NT Failover for Virtual Switch Operation.”](#)

12.5.1 Virtual Switch Host Failover

A planned or unplanned event can cause a Host in one virtual switch to no longer be active. In either case, the first step is to make the Management Host aware that another Host is no longer available, and to quiesce the traffic in Ports owned by the no-longer-responding Host. After that, both planned or unplanned failover follow the same steps to complete the failover.

12.5.2 Active Management Port Failover

The Active Management Port has a backup (Redundant) Management Port. In the event that the Active Management Port fails, the Redundant Management Port can promote itself to be the Active Management Port while simultaneously demoting the previous Active Management Port to a backup status. The Redundant Management Port writes the **Management Port Control** register (Port 0, accessible through the Management Port and Redundant Management Port, offset [354h](#)) with the following sequence:

1. Initially, the **Management Port Control** register status is:
 - [Active Management Port](#) field [4:0] holds the Port Number of the Active Management Port
 - [Redundant Management Port](#) field [12:8] holds the Port Number of the Redundant Management Port
 - [Active Management Port Enable](#) and [Redundant Management Port Enable](#) bits (bits [5 and 13], respectively)
2. The Redundant Management Port determines that the Active Management Port is no longer active.
3. The Redundant Management Port writes to the **Management Port Control** register:
 - Its own Management Port Number, to the *Active Management Port* field
 - The Port Number of the previous Active Management Port, to the *Redundant Management Port* field
 - Holds the *Active Management Port Enable* and *Redundant Management Port Enable* bits Set
4. Failover is then complete. The new Active Management Port should scan the **VS Upstream to Management Upstream Doorbell Request** and **Management Upstream to VS Upstream Doorbell Request** registers (VS Upstream Port(s) and Management Port, offsets [910h](#) and [928h](#), respectively), to determine whether there are any active interrupts to service, and might inform other Hosts that a new Active Management Port has been promoted.

12.5.3 Virtual Switch Host to NT Host Failover

The NT Port can be held as a backup for one or more Hosts in different virtual switches. Suppose a Host within a virtual switch fails – the NT Host can replace it.

To do this, the Management Port must first move the NT Port to the failed virtual switch. (The reason is beyond the scope of this data book, but one reason, *for example*, might be because it stopped receiving heartbeats from an Upstream virtual switch.) This is done by writing the NT Port's **VSx Port Vector** register(s) (Port 0, accessible through the Management Port, offset(s) 380h through 38Ch), to remove the NT Port from the original virtual switch hierarchy, and then writing the **VSx Port Vector** register(s) again, to add the NT Port to the new virtual switch.

After the NT Port is in the failed virtual switch, the NT Port can perform a standard NT failover sequence. In brief, the NT Port upgrades itself to be the Upstream Port, it might reset the Downstream Ports, and takes over handling of Ports within that hierarchy.

12.6 Performance

The virtual switch is built upon a non-blocking, peer-to-peer switch. The virtual switch assignment merely restricts which Ports can communicate with one another. Performance for each virtual switch should be equivalent to an ideal, single-hierarchy switch.

There is one exception, with respect to the Data buffers. The Data buffers are set up on a Station-basis (16 (Station 0) or 8 (Station 1) neighboring Lanes). If two or more Ports within a Station belong to different virtual switches, the common pool shared by all Ports associated with the Station's Lanes can be unfairly used by one virtual switch, to the detriment of another. To prevent this from occurring, the initial credits and Port pool should be adjusted (Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the Management Port, offset [9FCh](#)), so as to reduce the common pool to zero (0).

12.7 Host-to-Host Communication

In Base mode, the PEX 8724 is usually managed by I²C or the Root Port. In contrast, for bladed systems, Root Complexes are not typically trusted entities, and are therefore managed by a dedicated Management Port connected through an out-of-band mechanism (*such as* I²C or a dedicated PCI Express Port).

A virtual switch sends Doorbell interrupts and **Scratchpad** register data to the Management Port, using either Configuration Requests, or Memory Requests to the offset within its own **BAR0/1** register map.

The Management Port sends Doorbell interrupts and **Scratchpad** register data to individual virtual switches, by Memory-Mapped access to the offset within the virtual switch Upstream Port registers (rather than the offset within the Management Port, which would generate an interrupt to itself). *For example*, if Port 1 is a non-Management virtual switch Upstream Port, the Management Port can generate a Doorbell interrupt to it, by writing to the **Management Upstream to VS Upstream Doorbell Request** register (VS Upstream Port(s) and Management Port, offset [928h](#)).

Every virtual switch Upstream Port has the following **Scratchpad** registers and four corresponding **Doorbell** registers (located in the VS Upstream Port(s) and Management Port):

- Virtual switch to Management CPU direction
 - **VS Upstream to Management Upstream Doorbell Request** register (offset [910h](#))
 - **VS Upstream to Management Upstream Doorbell Mask** register (offset [914h](#))
 - **VS Upstream to Management Upstream Scratchpad 1** register (offset [918h](#))
 - **VS Upstream to Management Upstream Scratchpad 2** register (offset [91Ch](#))
 - **VS Upstream to Management Upstream Scratchpad 3** register (offset [920h](#))
 - **VS Upstream to Management Upstream Scratchpad 4** register (offset [924h](#))
- Management CPU to virtual switch direction
 - **Management Upstream to VS Upstream Doorbell Request** register (offset [928h](#))
 - **Management Upstream to VS Upstream Doorbell Mask** register (offset [92Ch](#))
 - **Management Upstream to VS Upstream Scratchpad 1** register (offset [930h](#))
 - **Management Upstream to VS Upstream Scratchpad 2** register (offset [934h](#))
 - **Management Upstream to VS Upstream Scratchpad 3** register (offset [938h](#))
 - **Management Upstream to VS Upstream Scratchpad 4** register (offset [93Ch](#))

Virtual switches cannot send messages directly to another virtual switch; instead, a virtual switch Upstream Port must first send the message to the Management Port, and the Management Port can then send the message to the other virtual switch.

Refer to [Section 9.6.2, “Doorbell Interrupts – Virtual Switch Mode,”](#) for further details.



Chapter 13 Transparent Port Registers

13.1 Introduction

This chapter defines the Transparent Port registers. Each Port has its own Configuration, Capability, Control, and Status register space. The register mapping is the same for each Port. (Refer to [Table 13-1](#).) This chapter also presents the programmable registers and the order in which they appear in the register map. Register descriptions, when applicable, include details regarding their use and meaning in the Upstream Port(s) and Downstream Ports. (Refer to [Table 13-3](#).) Other registers are defined in:

- [Chapter 15, “NT Port Virtual Interface Registers – NT Mode”](#)
- [Chapter 16, “NT Port Link Interface Registers – NT Mode”](#)

All Transparent Port registers can be accessed by Memory Requests. Registers outside the following Device-Specific ranges can also be accessed by Configuration Requests:

- 200h through AFCh
- B0Ch through B6Ch
- B80h through C30h

For reliable operation, software should not write to register offsets that are identified as **Reserved**.

For further details regarding register names and descriptions, refer to the following specifications:

- *PCI r3.0*
- *PCI Power Mgmt. r1.2*
- *PCI-to-PCI Bridge r1.2*
- *PCI Express Base r3.0*

13.2 Type 1 Port Register Map

Table 13-1 defines the Transparent mode Type 1 Port register mapping.

Table 13-1. Type 1 Port Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
PCI-Compatible Type 1 Configuration Header Registers (Offsets 00h – 3Ch)																																00h
																																...
																																34h
																																...
																																3Ch
																Next Capability Pointer (48h)								Capability ID (01h)								40h
PCI Power Management Capability Registers (Offsets 40h – 44h)																																44h
																Next Capability Pointer (68h)								Capability ID (05h)								48h
Message Signaled Interrupt Capability Registers (Offsets 48h – 64h)																																...
																																64h
																Next Capability Pointer (A4h)								Capability ID (10h)								68h
PCI Express Capability Registers (Offsets 68h – A0h)																																...
																																A0h
																Next Capability Pointer (00h)								SSID/SSVID Capability ID (0Dh)								A4h
Subsystem ID and Subsystem Vendor ID Capability Registers (Offsets A4h – FCh)																																...
																																FCh
Next Capability Offset (FB4h)								1h								PCI Express Extended Capability ID (0003h)																100h
Device Serial Number Extended Capability Registers (Offsets 100h – 108h)																																...
																																108h
Next Capability Offset (148h)								1h								PCI Express Extended Capability ID (0019h)																10Ch
Secondary PCI Express Extended Capability Registers (Offsets 10Ch – 134h)																																...
																																134h
Next Capability Offset (10Ch)								1h								PCI Express Extended Capability ID (0004h)																138h
Power Budget Extended Capability Registers (Offsets 138h – 144h)																																...
																																144h
Next Capability Offset (E00h)								1h								PCI Express Extended Capability ID (0002h)																148h
Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)																																...
																																1BCh
Device-Specific Registers (Offsets 1C0h – A74h)																																1C0h
																																...
																																A74h
Reserved																																A78h – AFCh

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset (B70h)																1h				PCI Express Extended Capability ID (0018h)																B00h
Latency Tolerance Reporting Extended Capability Registers (Offsets B00h – B08h)																																...				
																																B08h				
Reserved																																B0Ch – B6Ch				
Next Capability Offset 2 (000h)																1h				PCI Express Extended Capability ID 2 (000Bh)																B70h
Device-Specific Registers (Offsets B70h – DFCh)																																...				
																																DFCh				
Next Capability Offset (B00h or F24h)																1h				PCI Express Extended Capability ID (0012h)																E00h
Multicast Extended Capability Registers (Offsets E00h – E2Ch)																																...				
																																E2Ch				
Reserved																																E30h – EFCh				
Device-Specific Registers – Failover and Port Cut-Thru (Offsets F00h – F20h)																																F00h				
																																...				
																																F20h				
Next Capability Offset (B70h)																1h				PCI Express Extended Capability ID (000Dh)																F24h
ACS Extended Capability Registers (Offsets F24h – F2Ch)																																...				
																																F2Ch				
Device-Specific Registers (Offsets F30h – FB0h)																																F30h				
																																...				
																																FB0h				
Next Capability Offset (138h or 10Ch)																1h				PCI Express Extended Capability ID (0001h)																FB4h
Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)																																...				
																																FDCh				
Reserved																																FE0h – FFCh				

13.3 Port Register Configuration and Map

The PEX 8724 Port registers are configured similarly – not all the same. Port 0 of Station 0 and Port 8 of Station 1 include more Device-Specific registers than the other Ports. Port 0 also contains registers that are used to set up and control the PEX 8724, as well as a serial EEPROM interface, I²C Slave interface, and SMBus Slave interface logic and control. The Port registers contain setup and control information specific to the Station and its Port(s). [Table 13-2](#) defines the Port register configuration and map.

Note: In Virtual Switch mode, Chip-specific registers are accessible only to the Management Port, serial EEPROM, I²C, and SMBus.

Table 13-2. Port Register Configuration and Map

Register Types	Station 0, Port 0	Station 1, Port 8 ^a	Station 0, Ports 1, 2, 3 Station 1, Ports 9, 10 ^b
PCI-Compatible Type 1 Configuration Header Registers (Offsets 00h – 3Ch)	00h – 0Ch, 18h – 3Ch	00h – 0Ch, 18h – 3Ch	00h – 0Ch, 18h – 3Ch
	Upstream Port(s) 10h, 14h		
PCI Power Management Capability Registers (Offsets 40h – 44h)	40h – 44h	40h – 44h	40h – 44h
Message Signaled Interrupt Capability Registers (Offsets 48h – 64h)	48h – 64h	48h – 64h	48h – 64h
PCI Express Capability Registers (Offsets 68h – A0h)	68h – 78h, 84h – A0h	68h – 78h, 84h – A0h	68h – 78h, 84h – A0h
	Downstream Ports 7Ch, 80h		
Subsystem ID and Subsystem Vendor ID Capability Registers (Offsets A4h – FCh)	A4h – FCh	A4h – FCh	A4h – FCh
Device Serial Number Extended Capability Registers (Offsets 100h – 108h)	100h – 108h	100h – 108h	100h – 108h
Secondary PCI Express Extended Capability Registers (Offsets 10Ch – 134h)	10Ch – 134h	10Ch – 134h	10Ch – 134h
Power Budget Extended Capability Registers (Offsets 138h – 144h)	Upstream Port(s) 138h – 144h		
Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)	148h – 1BCh	148h – 1BCh	148h – 174h
WRR Port Arbitration Table Registers (Offsets 178h – 1BCh)	178h – 1BCh	178h – 1BCh	
Device-Specific Registers (Offsets 1C0h – A74h)			
Device-Specific Registers – Read Pacing (Offsets 1D0h – 1D8h)	1D0h – 1D8h	1D0h – 1D8h	
Device-Specific Registers – Captured Bus and Device Numbers (Offsets 1DCh – 1E4h)	1DCh – 1E4h	1DCh – 1E0h	1DCh – 1E0h
Device-Specific Registers – Read Pacing (Offset 1E8h)	1E8h		
Device-Specific Registers – Physical Layer (Offsets 200h – 25Ch)	200h – 25Ch	200h – 25Ch	
Device-Specific Registers – Serial EEPROM (Offsets 260h – 270h)	260h – 270h		
Device-Specific Registers – I2C and SMBus Slave Interfaces (Offsets 290h – 2FCh)	290h – 2FCh		

Table 13-2. Port Register Configuration and Map (Cont.)

Register Types	Station 0, Port 0	Station 1, Port 8 ^a	Station 0, Ports 1, 2, 3 Station 1, Ports 9, 10 ^b
Device-Specific Registers – Port Configuration and Lane Status (Offsets 300h – 46Ch)	300h – 46Ch		
Device-Specific Registers – Exerciser (Offsets 56Ch – 570h)	56Ch – 570h		
Device-Specific Registers – General-Purpose Input/Output (Offsets 600h – 68Ch)	600h – 68Ch		
Device-Specific Registers – PORT_GOOD Selector (Offsets 6A0h – 6A8h)	6A0h – 6A8h		
Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch)	700h – 75Ch	700h – 75Ch	724h, 728h
Device-Specific Registers – Control (Offsets 760h – 774h)	760h – 774h	760h – 774h	
Device-Specific Registers – Error (Offsets 778h – 8FCh)	778h – 8FCh	778h – 8FCh	
Device-Specific Registers – Virtual Switch (Offsets 900h – 9E8h), Virtual Switch Mode	900h – 90Ch, 940h – 9E8h		
	VS Upstream Port(s) and Management Port 910h – 93Ch		
Device-Specific Registers – Ingress Credit Handler (Offsets 9ECh – A2Ch)	9ECh – A2Ch	9ECh – A2Ch	9FCh – A2Ch (Base mode)
	VS Upstream Port(s) 9FCh – A2Ch (Virtual Switch mode)		
Device-Specific Registers – Virtual Switch Debug and GPIO Status and Control (Offsets A30h – A74h)	Upstream Port(s) A30h		
	VS Upstream Port(s) A34h – A74h (Virtual Switch mode)		
Latency Tolerance Reporting Extended Capability Registers (Offsets B00h – B08h)	Upstream Port(s) B00h – B08h		
Device-Specific Registers (Offsets B70h – DFCh)			
Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch)	B70h – B7Ch	B70h – B7Ch	B70h – B7Ch
Device-Specific Registers – Physical Layer (Offsets B80h – C8Ch)	B80h – C8Ch	B80h – C8Ch	
Device-Specific Registers – Requester ID Read Back (Offset C90h)	C90h	C90h	C90h
Multicast Extended Capability Registers (Offsets E00h – E2Ch)	E00h – E2Ch	E00h – E2Ch	E00h – E2Ch
Device-Specific Registers – Failover and Port Cut-Thru (Offsets F00h – F20h)	Upstream Port(s) F00h – F20h		
ACS Extended Capability Registers (Offsets F24h – F2Ch)	Downstream Ports F24h – F2Ch		

Table 13-2. Port Register Configuration and Map (Cont.)

Register Types	Station 0, Port 0	Station 1, Port 8 ^a	Station 0, Ports 1, 2, 3 Station 1, Ports 9, 10 ^b
Device-Specific Registers (Offsets F30h – FB0h)			
Device-Specific Registers – Egress Control (Offsets F30h – F40h)	F30h – F40h	F30h – F40h	F30h – F40h
Device-Specific Registers – Ingress Control and Port Enable (Offsets F44h – F6Ch)	F44h, F5Ch (Base mode)	F44h, F5Ch (Base mode)	F44h, F5Ch (Base mode)
	Upstream Port(s) F4Ch – F58h, F60h VS Upstream Port(s) F48h, F5Ch (Virtual Switch mode)		
Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h)	F70h – FB0h	F70h – FB0h	F70h – FB0h
Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)	FB4h – FDCh	FB4h – FDCh	FB4h – FDCh

- a. Port 8 is a **Software Only Station Port**. Refer to [Table 13-5](#) for further details.
- b. Some Ports are **Software Only**, or used only within specific Port configurations. Refer to [Table 13-5](#) for details.

13.4 Register Access

Each PEX 8724 Port implements a 4-KB Configuration Space. The lower 256 bytes (offsets 00h through FFh) comprise the PCI-compatible Configuration Space, and the upper 960 DWords (offsets 100h through FFFh) comprise the PCI Express Extended Configuration Space. The PEX 8724 supports six mechanisms for accessing the Transparent Mode registers:

- [PCI r3.0-Compatible Configuration Mechanism](#)
- [PCI Express Enhanced Configuration Access Mechanism](#)
- [Device-Specific Memory-Mapped Configuration Mechanism](#)
- I²C Slave Interface (refer to [Section 7.2, “I2C Slave Interface”](#))
- SMBus Slave Interface (refer to [Section 7.3, “SMBus Slave Interface”](#))
- Serial Peripheral Interface (SPI) Bus (refer to [Chapter 6, “Serial EEPROM Controller”](#))

The sideband register access mechanisms (serial EEPROM, I²C, and/or SMBus) can modify Read-Only (RO) register values.

Each Port captures the Bus Number and Device Number on every Type 0 Configuration Write, as required by the *PCI r3.0*. Therefore, following a Fundamental Reset, software must initially perform a Configuration Write to each Port (using either the [PCI r3.0-Compatible Configuration Mechanism](#) or [PCI Express Enhanced Configuration Access Mechanism](#)), to allow each Port to capture its designated Bus Number and Device Number. The initial access to each Port, *for example*, could be a Configuration Write Request to a RO register, such as the **Device ID / Vendor ID** register (offset 00h).

Note: An option is provided to allow the serial EEPROM and/or I²C/SMBus to initialize the Port's **Captured Bus and Device Numbers** registers (offset 1DCh), instead of the required initial Configuration Write to each Port; however, this option is not recommended.

13.4.1 PCI r3.0-Compatible Configuration Mechanism

The *PCI r3.0*-Compatible Configuration mechanism provides standard access to the PCI Express Configuration Space within each Port. PCI Express Configuration Request transactions use the following Addressing fields:

- *Bus Number*
- *Device Number*
- *Function Number*
- *Extended Register Number and Register Number*

Notes: When an Alternative Routing-ID Interpretation (ARI) device is targeted, and the Downstream Port immediately above it is enabled for ARI Forwarding, the Device Number (in associated Routing IDs, Requester IDs, and Completer IDs) is implied to be 0, and the traditional 5-bit Device Number and 3-bit Function Number fields are interpreted as a single 8-bit Function Number field.

For ARI devices, discovery and enumeration of Extended Functions require ARI-aware software.

Together, the *Extended Register Number* and *Register Number* fields specify the Configuration Space address of the register being accessed (concatenated such that the *Extended Register Number* forms the more-significant bits). For PCI-compatible Configuration Requests (that target the first 256 bytes of a Function's Configuration Space, the *Extended Register Address* field must be all zeros (0). The PCI Express Extended Configuration Space (offsets 100h through FFFh) can be accessed by the [PCI Express Enhanced Configuration Access Mechanism](#) (ECAM), except for the register ranges noted in [Section 13.4.2](#), or [Device-Specific Memory-Mapped Configuration Mechanism](#), which can access all registers.

The *PCIr3.0*-Compatible Configuration mechanism uses PCI Type 0 and Type 1 Configuration transactions to access the PEX 8724 Configuration registers. Each Port can convert a Type 1 Configuration Request (destined to a Downstream Port or device) to a Type 0 Configuration Request (targeting the next Downstream Port or device), as described below.

The PEX 8724 decodes all Type 1 Configuration accesses received on its Upstream Port(s), when any of the following conditions are met:

- If the Bus Number in the Configuration access is not within the Upstream Port(s)' Secondary Bus Number and Subordinate Bus Number range, the Upstream Port(s) respond with an Unsupported Request (UR).
- Specified Bus Number in the Configuration access is the PEX 8724 internal virtual PCI Bus Number, the PEX 8724 automatically converts the Type 1 Configuration access into the appropriate Type 0 Configuration access for the specified device.
 - If the specified device corresponds to the PCI-to-PCI bridge in one of the PEX 8724 Downstream Ports, the PEX 8724 processes the Read or Write Request to the specified Downstream Port register specified in the original Type 1 Configuration access.
 - If the specified Device Number does not correspond to any of the PEX 8724 Downstream Port Device Numbers, the PEX 8724 responds with a UR.
- If the specified Bus Number in the Type 1 Configuration access is not the PEX 8724 internal virtual PCI Bus Number, but is the Bus Number of one of the PEX 8724 Downstream Port secondary/subordinate buses, the PEX 8724 passes the Configuration access on to the PCI Express Link attached to that PEX 8724 Downstream Port.
 - If the specified Bus Number is the Downstream Port Secondary Bus Number, and the specified Device Number is 0, the PEX 8724 converts the Type 1 Configuration access to a Type 0 Configuration access before passing it on.
 - If the specified Device Number is not 0, the Downstream Port drops the Transaction Layer Packet (TLP) and generates a UR.
- If the specified Bus Number is not the Downstream Port Secondary Bus Number, the PEX 8724 passes along the Type 1 Configuration access, without change.

13.4.2 PCI Express Enhanced Configuration Access Mechanism

The PCI Express Enhanced Configuration Access mechanism (ECAM) is implemented on all PCI Express PCs and on systems that do not implement a processor-specific firmware interface to the Configuration Space. The mechanism provides a Memory-Mapped Address space in the Root Complex, through which the Root Complex translates a Memory access into one or more Configuration Requests. Device drivers normally use an application programming interface (API) provided by the Operating System (OS), to use this mechanism.

The mechanism can be used to access PEX 8724 registers that are defined by Specifications. This mechanism typically **cannot** access the Device-Specific registers (which are instead accessible by the Device-Specific Memory-Mapped Configuration Mechanism) in the following offset ranges, per Port:

- 200h through AFCh
- B0Ch through B6Ch
- B80h through C30h

Silicon Revision CA provides an option to enable ECAM access to all PEX 8724 registers, however, by Setting the **Station-Based Control** register *Device-Specific Register Access by Extended Configuration Request Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 760h[16]), in the Station(s) that contain(s) an Upstream Port or NT Port.

13.4.3 Device-Specific Memory-Mapped Configuration Mechanism

The Device-Specific Memory-Mapped Configuration mechanism provides a method to access the Configuration registers of all Ports, within a single 256-KB Memory map, as listed in [Table 13-3](#). The registers of each Port are contained within a 4-KB range. The PEX 8724 supports a maximum of up to six concurrently active Ports.

This mechanism follows the *PCI Express Base r3.0* Configuration Request Routing rules, which do not allow the propagation of Configuration Requests from Downstream-to-Upstream, nor peer-to-peer. By default, if any PEX 8724 Downstream Port receives a Memory Request from a Downstream device targeting the PEX 8724 Configuration registers, the Port:

- Responds to a Memory Read Request with a Completion indicating an Unsupported Request (UR) Uncorrectable error
- For Memory Writes:
 - Silently discards a Memory Write Request (default, in compliance with the *PCI Express Base r2.1*), –or–
 - If the **Error Check Disable** register *Software Force Error Enable* bit (Transparent Downstream Ports, offset 720h[2]) is Set, the Port responds with a UR

In Memory Requests that target PEX 8724 registers, the Payload Length indicated within the Memory Request Header must be 1 DWord. Lengths greater than 1 DWord result in a Completer Abort error.

To use this mechanism, program the Upstream Port(s) Type 1 Configuration Space **Base Address 0** and **Base Address 1** registers (**BAR0** and **BAR1**, offsets 10h and 14h, respectively), which are typically enumerated at boot time by BIOS or the OS software. After the PEX 8724 Upstream Port(s) BARs are enumerated, Port 0 registers can be accessed with Memory Reads from and Writes to the first 4 KB (0000h to 0FFFh), Port 1 registers can be accessed with Memory Reads from and Writes to the second 4 KB (1000h to 1FFFh), and so forth. Within each of these 4-KB windows, individual registers are located at the DWord offsets indicated in [Table 13-1](#).

In Virtual Switch mode, each virtual switch Upstream Port has its own BAR0/1 register for Memory Request access to all Port registers (excluding Chip- and Station-specific registers) within its Virtual Switch hierarchy. The Port mapping listed in [Table 13-3](#) applies to all virtual switches; however, only the Management Port mapping can access the Chip- and Station-specific registers, as well as the registers of all virtual switches. The 4-KB Address blocks corresponding to Ports that exist in a different virtual switch **cannot** be accessed with the **BAR0/1** mapping of an Upstream Port that is not the designated Management Port. Such attempted accesses are handled as No Operation (NOP) (Writes are ignored, Reads return a value of 0h).

Table 13-3. Register Offsets from Upstream Port BAR0/1 Base Address

Port Number ^a	Internal Register 4-KB Memory Space Range	Location Range
Port 0	0_0000h to 0_0FFFh	0 to 4 KB
Port 1	0_1000h to 0_1FFFh	4 to 8 KB
Port 2	0_2000h to 0_2FFFh	8 to 12 KB
Port 3	0_3000h to 0_3FFFh	12 to 16 KB
Port 8	0_8000h to 0_8FFFh	32 to 36 KB
Port 9	0_9000h to 0_9FFFh	36 to 40 KB
Port 10	0_A000h to 0_AFFFh	40 to 44 KB

a. Port 8 is **Software Only**.

13.5 Register Descriptions

The remainder of this chapter details the PEX 8724 registers, including:

- Bit/field names
- Description of register functions for the PEX 8724 Upstream Port(s), Downstream Ports, and virtual switches
- Type (such as RW or HwInit; refer to [Table 13-4](#) for Type descriptions)
- Whether the power-on/reset value can be modified, by way of the PEX 8724 serial EEPROM and/or I²C/SMBus Initialization feature
- Default power-on/reset value

Table 13-4. Register Types, Grouped by User Accessibility

Type	Description
HwInit	Hardware-Initialized Refers to the PEX 8724 Hardware-Initialization mechanism or PEX 8724 Serial EEPROM and/or I ² C register Initialization features. RO after initialization and can only be reset with a Fundamental Reset. HwInit register bits are not modified by a Soft Reset.
RO	Read-Only Read-Only and cannot be altered by software. Permitted to be initialized by the PEX 8724 Hardware-Initialization mechanism or PEX 8724 serial EEPROM and/or I ² C/SMBus register Initialization features.
ROS	Read-Only, Sticky Same as RO, except that bits are neither initialized nor modified by a Soft Reset.
RsvdP	Reserved and Preserved Reserved for future RW implementations. Registers are RO and must return a value of 0 when read. Software must preserve value read for Writes to bits.
RsvdZ	Reserved and Zero Reserved for future RW1C implementations. Registers are RO and must return a value of 0 when read. Software must use 0 for Writes to bits.
RW	Read-Write Read/Write and permitted to be Set or Cleared by software to the needed state.
RW1C	Write 1 to Clear Status (Transparent mode) Indicates status when read. A status bit Set by the system (to indicate status) is Cleared, by writing 1 to that bit. Writing 0 has no effect. Read-Write, Clear Interrupt (NT mode) In the NT registers, RW1C is used for the Doorbell interrupts, to indicate that a value of 1 Clears the interrupt.
RW1CS	Write 1 to Clear, Sticky Same as RW1C, except that bits are neither initialized nor modified by a Soft Reset.
RW1S	Read-Write, Set Interrupt (NT mode) In the NT registers, RW1S is used for the Doorbell interrupts, to indicate that a value of 1 Sets the interrupt.
RWS	Read-Write, Sticky Same as RW, except that bits are Set or Cleared by software to the needed state. Bits are neither initialized nor modified by a Soft Reset.
RZ	Software Read Zero Software Read always returns a value of 0; however, software is allowed to write this register.

13.6 Port Configurations and Station/Station Port/Port/SerDes Module/Lane Relationship

This section discusses the PEX 8724 Port configurations, in Base mode and Virtual Switch mode, as well as information related to Stations, Station Ports, Ports, SerDes modules, and Lanes.

In this chapter, the term “SerDes quad” or “quad” refers to assembling SerDes modules into groups of four contiguous Lanes for testing purposes.

13.6.1 Port Configurations

Table 13-5 defines the PEX 8724 Port configurations, as well as the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. For further details regarding Port configuration, refer to Section 4.4.1.1, “Port Configurations.”

Table 13-5. Port Configurations

# ^a	Port Configuration Strapping Input States STRAP_STN0_PORTCFG[1:0] ^b	Port Configuration Register Value Port 0, Offset 300h[2:0] ^c	Station 0 [Lanes/SerDes]/Port			
			Port 0 ^d	Port 1	Port 2	Port 3
1	0Z	001b	x16 [0-15]			
2	01	010b	x8 [0-7]	x8 [8-15]		
3	Z0	011b	x8 [0-7]	x4 [8-11]	x4 [12-15]	
4	ZZ	100b	x4 [0-3]	x4 [4-7]	x4 [8-11]	x4 [12-15]
# ^a	Port Configuration Strapping Input States STRAP_STN1_PORTCFG[1:0] ^b	Port Configuration Register Value Port 0, Offset 300h[5:3] ^c	Station 1 [Lanes/SerDes]/Port			
			Port 8 ^{d,e}	Port 9	Port 10	–
2	01	010b	Software Only	x8 [24-31]		
3	Z0	011b	Software Only	x4 [24-27]	x4 [28-31]	

- Port Configuration Number, which is the decimal equivalent of the **Port Configuration** register Port Configuration for Station x field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]) binary value.
- “Z” means “floating.” Z is used for the 3-state inputs that have three functions, depending upon whether the input is pulled or tied High to **VDD18**, pulled or tied Low to **VSS** (Ground), or left floating (not connected). Each one of these three input states (1, 0, or Z, respectively) results in a different function for that input.
- Port Configuration** register Port Configuration for Station x field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]).
- Ports 0 and 8 are also “Station Ports” within their respective Stations. Station Ports are used to control the Station-specific registers.

- e. *Port 8 is a **Software Only** Station Port. The PEX 8724 Ports that connect to PCI Express Transmitters and Receivers are Ports 0, 1, 2, 3, 9, and 10, and each Port has its own set of registers. An additional set of Port 8 registers is visible to software, although the Port is not connected to external signals. The Port 8 registers include:*
- *Other Device-Specific registers.*
 - *Station-specific registers, which control all Ports in Station 1 (Ports 9 and 10), and are identical to the Port 0 registers that control all Ports in Station 0 (Ports 0, 1, 2, and 3). The Station-specific registers include Physical Layer registers and Device-Specific Error registers.*

13.6.2 Virtual Switch Port Assignments – Virtual Switch Mode

Note: Default Port assignments must be changed, by using the serial EEPROM, I²C/SMBus, and/or the Management Port, prior to Setting the **Configuration Release** register **Initiate Configuration** bit (Port 0, accessible through the Management Port, offset **3ACh[0]**).

The Strapping signals define the quantity of enabled virtual switches (up to four, based upon the **STRAP_TESTMODE[2:0]** and **VS_MODE** input states), the Upstream and Downstream Ports, Port size, and Management Port. The Management Port and/or I²C/SMBus Slave interface can re-configure the virtual switches and dynamically re-assign Downstream Ports belonging to one virtual switch, to any other virtual switch, by programming the **VSx Port Vector** register(s) (Port 0, accessible through the Management Port, offset(s) **380h** through **38Ch**).

Each Port can be assigned to only one virtual switch; therefore, a Port must be removed from the Virtual Switch *x* Vector to which it is assigned, prior to adding that Port to a different Virtual Switch *x* Vector. Following dynamic re-assignment of a Downstream Port from one virtual switch to another virtual switch, that Port should be reset by Management Port software and/or the I²C/SMBus Slave interface by Setting, and then Clearing, the Port's **Port Reset** register **Reset Port *x* Vector** bit(s) (Port 0, accessible through the Management Port, offset **3A0h[10:8, 3:0]**).

The Management Port, serial EEPROM, and/or I²C/SMBus Slave interface can configure any single Port within a virtual switch as that virtual switch's Upstream Port, by programming the **VSx Upstream** register(s) (Port 0, accessible through the Management Port, offset(s) **360h** through **36Ch**).

Table 13-6 lists the default virtual switch Port assignments, which are dependent upon the quantity of enabled virtual switches. The Port Numbers referenced correspond to those listed in **Table 13-5**.

For further details, refer to **Chapter 12, "Virtual Switch Mode."**

Table 13-6. Virtual Switch Default Port Assignments – Virtual Switch Mode

# of Enabled Virtual Switches	Test Modes	STRAP_TESTMODE[2:0] Input States	VS_MODE Input State	Virtual Switch Enable Register (Offset 358h) Value	VSx Upstream Register Offset and Value (Upstream Port Number) ^{a, b}	VSx Downstream Ports ^a	VSx Vector Register Offset
1	4	0ZZ	0	1h	VS0 (360h) = 0h (0)	1, 2, 3, 8, 9, 10	VS0 (380h)
	5	0Z1					
	6	010					
	7	01Z					
2	4	0ZZ	Z	3h	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A)	VS0 = 1, 2, 3, 10 VS1 = 8, 9	VS0 (380h) VS1 (384h)
	5	0Z1					
	6	010					
	7	01Z					
3	4	0ZZ	1	7h	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A) VS2 (368h) = Ah (10)	VS0 = 1, 2, 3 VS1 = 8, 9 VS2 = None	VS0 (380h) VS1 (384h) VS2 (388h)
	5	0Z1					
	6	010					
	7	01Z					
4	8	011	0	Fh	VS0 (360h) = 0h (0) VS1 (364h) = 4h (N/A) VS2 (368h) = Ah (10) VS3 (36Ch) = 10h (N/A)	VS0 = 1, 2, 3 VS1 = 8, 9 VS2 = None VS3 = None	VS0 (380h) VS1 (384h) VS2 (388h) VS3 (38Ch)
	9	Z00					
	10	Z0Z					
	11	Z01					

- a. *Table 13-5* indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.
- b. In Virtual Switch mode, the **VS1 Upstream** register (Port 0, accessible through the Management Port, offset 364h) default value (4h) must be changed by serial EEPROM (or by I²C/SMBus, provided that STRAP_I2C_SMBUS_CFG_EN# is Low), to a valid Port Number.

13.7 PCI-Compatible Type 1 Configuration Header Registers (Offsets 00h – 3Ch)

This section details the PCI-Compatible Type 1 Configuration Header registers. [Table 13-7](#) defines the register map.

Table 13-7. PCI-Compatible Type 1 Configuration Header Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Device ID																Vendor ID																00h
PCI Status																PCI Command																04h
PCI Class Code																		PCI Revision ID														08h
PCI BIST (<i>Not Supported</i>)								PCI Header Type								Master Latency Timer (<i>Not Supported</i>)								Cache Line Size								0Ch
Base Address 0																																10h
Base Address 1																																14h
Secondary Latency Timer (<i>Not Supported</i>)								Subordinate Bus Number								Secondary Bus Number								Primary Bus Number								18h
Secondary Status								<i>Not Supported/Reserved</i>								I/O Limit								I/O Base								1Ch
Memory Limit																Memory Base																20h
Prefetchable Memory Limit																Prefetchable Memory Base																24h
Prefetchable Memory Upper Base Address																																28h
Prefetchable Memory Upper Limit Address																																2Ch
I/O Limit Upper 16 Bits																I/O Base Upper 16 Bits																30h
<i>Reserved</i>																								Capability Pointer (40h)								34h
Expansion ROM Base Address (<i>Reserved</i>)																																38h
<i>Not Supported/Reserved</i>								Bridge Control								PCI Interrupt Pin								PCI Interrupt Line								3Ch

Register 13-1. 00h PCI Configuration ID (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Vendor ID Identifies the device manufacturer. Defaults to the PCI-SIG-issued Vendor ID of PLX (10B5h), if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	10B5h
31:16	Device ID Identifies the particular device. Defaults to the PLX part number for the PEX 8724, if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	8724h

**Register 13-2. 04h PCI Command/Status
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Command				
0	I/O Access Enable 0 = PEX 8724 ignores I/O Space accesses on the Port's primary interface 1 = PEX 8724 responds to I/O Space accesses on the Port's primary interface	RW	Yes	0
1	Memory Access Enable 0 = PEX 8724 ignores Memory Space accesses on the Port's primary interface 1 = PEX 8724 responds to Memory Space accesses on the Port's primary interface	RW	Yes	0
2	Bus Master Enable Controls PEX 8724 Memory and I/O Request forwarding Upstream. Does not affect Message (including INTx Interrupt Messages) forwarding nor Completions traveling Upstream or Downstream. 0 = PEX 8724 handles Memory and I/O Requests received on the Port's Downstream/secondary interface as Unsupported Requests (URs); for Non-Posted Requests, the PEX 8724 returns a Completion with UR Completion status. Because MSI Messages are in-band Memory Writes, disables MSI Messages as well. 1 = PEX 8724 forwards Memory and I/O Requests Upstream.	RW	Yes	0
3	Special Cycle Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
4	Memory Write and Invalidate Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
5	VGA Palette Snoop <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
6	Parity Error Response Enable Controls bit 24 (<i>Master Data Parity Error Detected</i>).	RW	Yes	0
7	IDSEL Stepping/Wait Cycle Control <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0

Register 13-2. 04h PCI Command/Status
(All Ports) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	SERR# Enable Controls bit 30 (<i>Signaled System Error</i>). 1 = Enables reporting of Fatal and Non-Fatal errors detected by the device to the Root Complex, and, enables primary interface forwarding of ERR_FATAL and ERR_NONFATAL Messages from Downstream Ports and devices when the Port's Bridge Control register <i>SERR# Enable</i> bit (offset 3Ch[17]) is Set	RW	Yes	0
9	Fast Back-to-Back Transactions Enable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
10	Interrupt Disable 0 = Port is enabled to generate INTx Interrupt Messages and assert <i>PEX_INTA#</i> (Base mode) or <i>VSx_PEX_INTA#</i> (Virtual Switch mode) output 1 = Port is prevented from generating INTx Interrupt Messages and asserting <i>PEX_INTA#</i> (Base mode) or <i>VSx_PEX_INTA#</i> (Virtual Switch mode) output	RW	Yes	0
15:11	Reserved	RsvdP	No	0-0h
PCI Status				
18:16	Reserved	RsvdP	No	000b
19	Interrupt Status 0 = No pending INTx Interrupt Message 1 = INTx Interrupt Message is pending internally to the Port, –or– <i>PEX_INTA#</i> (Base mode) or <i>VSx_PEX_INTA#</i> (Virtual Switch mode) (if enabled) is asserted	RO	No	0
20	Capability List Capability function is supported. Set, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	1
21	66 MHz Capable Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
22	Reserved	RsvdP	No	0
23	Fast Back-to-Back Transactions Capable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0

Register 13-2. 04h PCI Command/Status
(All Ports) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Master Data Parity Error Detected If bit 6 (<i>Parity Error Response Enable</i>) is Set, the Port Sets this bit when the Port: - Forwards the poisoned Transaction Layer Packet (TLP) Write Request from the secondary to the primary interface, –or– - Receives a Completion marked as poisoned on the primary interface If the <i>Parity Error Response Enable</i> bit is Cleared, the PEX 8724 never Sets this bit. This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
26:25	DEVSEL# Timing Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00b
27	Signaled Target Abort The Upstream Port(s) Set this bit when any one of the following conditions are met: - Upstream Port receives a Memory Request targeting a PEX 8724 register, and the Payload Length (indicated within the Memory Request Header) is greater than 1 DWord - Upstream Port receives a Memory Request targeting a PEX 8724 register address within a non-existent Port - Transparent Downstream Port Sets this bit if it detects an Access Control Services (ACS) violation This error is reported by the Uncorrectable Error Status register <i>Completer Abort Status</i> bit (offset FB8h[15]), which is mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
28	Received Target Abort Reserved	RsvdP	No	0
29	Received Master Abort Reserved	RsvdP	No	0
30	Signaled System Error If bit 8 (<i>SERR# Enable</i>) is Set, the Port Sets this bit when it transmits or forwards an ERR_FATAL or ERR_NONFATAL Message Upstream. This error is natively reported by the Device Status register <i>Fatal Error Detected</i> and <i>Non-Fatal Error Detected</i> bits (offset 70h[18:17], respectively), which are mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
31	Detected Parity Error This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility. 1 = Port received a Poisoned TLP on its primary side, regardless of the bit 6 (<i>Parity Error Response Enable</i>) state	RW1C	Yes	0

**Register 13-3. 08h PCI Class Code and Revision ID
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
PCI Revision ID					
7:0	Revision ID Unless overwritten by the serial EEPROM, returns the Silicon Revision (BAh or CAh), the PLX-assigned Revision ID for this version of the PEX 8724. The PEX 8724 Serial EEPROM register Initialization capability is used to replace the PLX Revision ID with another Revision ID.	RO	Yes	BAh	CAh
PCI Class Code				060400h	
15:8	Register-Level Programming Interface The PEX 8724 Ports support the <i>PCI-to-PCI Bridge r1.2</i> requirements, but not subtractive decoding, on their Upstream interface.	RO	Yes	00h	
23:16	Sub-Class Code PCI-to-PCI bridge.	RO	Yes	04h	
31:24	Base Class Code Bridge device.	RO	Yes	06h	

**Register 13-4. 0Ch Miscellaneous Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Cache Line Size				
7:0	Cache Line Size System Cache Line Size. Implemented as a RW field for Conventional PCI compatibility purposes and does not impact PEX 8724 functionality.	RW	Yes	00h
Master Latency Timer				
15:8	Master Latency Timer <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00h
PCI Header Type				
22:16	Configuration Layout Type The Port's Configuration Space Header adheres to the Type 1 PCI-to-PCI Bridge Configuration Space layout defined by the <i>PCI-to-PCI Bridge r1.2</i> .	RO	No	01h
23	Multi-Function Device 0 = Single-function device 1 = Indicates multiple (up to eight) functions (logical devices), each containing its own, individually addressable Configuration Space, 256 DWords in size	RO	No	0
PCI BIST				
31:24	PCI BIST <i>Not supported</i> Built-In Self-Test (BIST) Pass or Fail.	RsvdP	No	00h

**Register 13-5. 10h Base Address 0
(Upstream Port(s))**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
0	Memory Space Indicator 0 = Base Address register maps the PEX 8724 Configuration registers into Memory space <i>Note: The Upstream Port(s) are hardwired to 0.</i>	Upstream	RO	No	0
	Reserved	Downstream	RsvdP	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are Reserved .	Upstream	RO	Yes	00b
	Reserved	Downstream	RsvdP	No	00b
3	Prefetchable 0 = Base Address register maps the PEX 8724 Configuration registers into Non-Prefetchable Memory space <i>Note: The Upstream Port(s) are hardwired to 0.</i>	Upstream	RO	Yes	0
	Reserved	Downstream	RsvdP	No	0
17:4	Reserved		RsvdP	No	0-0h
31:18	Base Address 0 Base Address (BAR0) for the Device-Specific Memory-Mapped Configuration mechanism.	Upstream	RW	Yes	0-0h
	Reserved	Downstream	RsvdP	No	0-0h

**Register 13-6. 14h Base Address 1
(Upstream Port(s))**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Base Address 1 For 64-bit addressing, Base Address 1 (BAR1) extends Base Address 0 (BAR0) to provide the upper 32 Address bits when the Base Address 0 register <i>Memory Map Type</i> field (Upstream Port(s), offset 10h[2:1]) is programmed to 10b.	Upstream	RW	Yes	0000_0000h
	RO when the Base Address 0 register is not enabled as a 64-bit BAR (<i>Memory Map Type</i> field (Upstream Port(s), offset 10h[2:1]) is not programmed to 10b).		RO	Yes	0000_0000h
	Reserved	Downstream	RsvdP	No	0000_0000h

**Register 13-7. 18h Bus Number
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Primary Bus Number Primary Bus Number of this PCI-to-PCI bridge. Records the Bus Number of the PCI Bus segment to which the primary interface of this Port is connected. Programmed by Configuration software.	RW	Yes	00h
15:8	Secondary Bus Number Secondary Bus Number of this PCI-to-PCI bridge. Records the Bus Number of the PCI Bus segment that is the secondary interface of this Port. Programmed by Configuration software.	RW	Yes	00h
23:16	Subordinate Bus Number Subordinate Bus Number of this PCI-to-PCI bridge. Records the Bus Number of the highest numbered PCI Bus segment that is subordinate to this Port. Programmed by Configuration software.	RW	Yes	00h
31:24	Secondary Latency Timer Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00h

**Register 13-8. 1Ch Secondary Status, I/O Limit, and I/O Base
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: If I/O Address decoding is enabled (PCI Command register I/O Access Enable bit, offset 04h[0], is Set), the Port forwards I/O transactions from the Port's primary interface to the Port's secondary interface (Downstream) if an I/O address is within the range defined by the I/O Base and I/O Limit registers when the Base is less than or equal to the Limit. Conversely, the Port forwards I/O transactions from the Port's secondary interface to the Port's primary interface (Upstream) if an I/O address is outside this Address range. If the Port does not implement an I/O Address range, the Port forwards all I/O transactions on the Port's secondary interface Upstream, to the Port's primary interface.				
I/O Base				
3:0	I/O Base Addressing Capability 1h = 32-bit I/O Address decoding is supported All other encodings are Reserved .	RO	Yes	1h
7:4	I/O_BAR[15:12] I/O Base Address[15:12]. The Ports use their I/O Base and I/O Limit registers to determine the address range of I/O transactions to forward from one interface to the other. I/O Base Address[15:12] bits specify the Port's I/O Base Address[15:12]. The PEX 8724 assumes I/O Base Address[11:0]=000h. For 16-bit I/O addressing, the PEX 8724 assumes Address[31:16]=0000h. For 32-bit addressing, the PEX 8724 decodes Address[31:0], and uses the I/O Upper Base and Limit Address register I/O Base Upper 16 Bits and I/O Limit Upper 16 Bits fields (offset 30h[15:0 and 31:16], respectively).	RW	Yes	Fh
I/O Limit				
11:8	I/O Limit Addressing Capability 1h = 32-bit I/O Address decoding is supported All other encodings are Reserved .	RO	Yes	1h
15:12	I/O_Limit[15:12] I/O Limit Address[15:12]. The Ports use their I/O Base and I/O Limit registers to determine the Address range of I/O transactions to forward from one interface to the other. I/O Limit Address[15:12] bits specify the Port's I/O Limit Address[15:12]. The PEX 8724 assumes I/O Limit Address[11:0]=FFFh. For 16-bit I/O addressing, the PEX 8724 decodes Address bits [15:0] and assumes I/O Limit Address[31:16]=0000h. For 32-bit addressing, the PEX 8724 decodes Address bits [31:0], and uses the I/O Upper Base and Limit Address register I/O Base Upper 16 Bits and I/O Limit Upper 16 Bits fields (offset 30h[15:0 and 31:16], respectively).	RW	Yes	0h

**Register 13-8. 1Ch Secondary Status, I/O Limit, and I/O Base
(All Ports) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Secondary Status				
20:16	Reserved	RsvdP	No	0-0h
21	66 MHz Capable Not supported 0 = Not enabled, because PCI Express does not support 66 MHz	RsvdP	No	0
22	Reserved	RsvdP	No	0
23	Fast Back-to-Back Transactions Capable Reserved Not enabled, because PCI Express does not support this function.	RsvdP	No	0
24	Master Data Parity Error If the Bridge Control register <i>Parity Error Response Enable</i> bit (offset 3Ch[16]) is Set, the Port Sets this bit when transmitting or receiving a TLP on its Downstream side, and when either of the following two conditions occur: - Port receives Completion marked poisoned - Port forwards poisoned TLP Write Request If the <i>Parity Error Response Enable</i> bit is Cleared, the PEX 8724 never Sets this bit. These errors are reported by the Port's Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), and mirrored to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
26:25	DEVSEL# Timing Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00b
27	Signaled Target Abort Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
28	Received Target Abort Cleared, as required by the <i>PCI Express Base r3.0</i> , because the PEX 8724 never initiates a Request itself.	RsvdP	No	0
29	Received Master Abort Cleared, as required by the <i>PCI Express Base r3.0</i> , because the PEX 8724 never initiates a Request itself.	RsvdP	No	0
30	Received System Error 1 = Downstream Port received an ERR_FATAL or ERR_NONFATAL Message on its secondary interface from a Downstream device	RW1C	Yes	0
31	Detected Parity Error 1 = Downstream Port received a poisoned TLP from a Downstream device (Set regardless of the Bridge Control register <i>Parity Error Response Enable</i> bit (offset 3Ch[16]) state)	RW1C	Yes	0

**Register 13-9. 20h Memory Base and Limit
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: The Port forwards Memory transactions from its primary interface to its secondary interface (Downstream) if a Memory address is within the range defined by the Memory Base and Memory Limit registers (when the Base is less than or equal to the Limit). Conversely, the Port forwards Memory transactions from its secondary interface to its primary interface (Upstream) if a Memory address is outside this Address range (provided that the address is not within the range defined by the Prefetchable Memory Base (offsets 28h + 24h[15:0]) and Prefetchable Memory Limit (offsets 2Ch + 24h[31:16]) registers).				
Memory Base				
3:0	Reserved	RsvdP	No	0h
15:4	MEM_BAR[31:20] Memory Base Address[31:20]. Specifies the Port's Non-Prefetchable Memory Base Address[31:20]. The PEX 8724 assumes Memory Base Address[19:0]=0_0000h.	RW	Yes	FFFh
Memory Limit				
19:16	Reserved	RsvdP	No	0h
31:20	MEM_Limit[31:20] Memory Limit Address[31:20]. Specifies the Port's Non-Prefetchable Memory Limit Address[31:20]. The PEX 8724 assumes Memory Limit Address[19:0]=F_FFFFh.	RW	Yes	000h

Register 13-10. 24h Prefetchable Memory Base and Limit (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: The Port forwards Memory transactions from its primary interface to its secondary interface (Downstream) if a Memory address is within the range defined by the Prefetchable Memory Base (offsets 28h + 24h[15:0]) and Prefetchable Memory Limit (offsets 2Ch + 24h[31:16]) registers (when the Base is less than or equal to the Limit). Conversely, the Port forwards Memory transactions from its secondary interface to its primary interface (Upstream) if a Memory address is outside this Address range (provided that the address is not within the range defined by the Memory Base and Memory Limit registers (offset 20h).				
Prefetchable Memory Base				
0	Prefetchable Memory Base Capability 0 = Port supports 32-bit Prefetchable Memory Addressing 1 = Port defaults to 64-bit Prefetchable Memory Addressing support, as required by the <i>PCI Express Base r3.0</i> Note: If the application needs 32-bit only Prefetchable space, the serial EEPROM and/or I ² C must Clear both this bit and bit 16 (Prefetchable Memory Limit Capability).	RO	Yes	1
3:1	Reserved	RsvdP	No	000b
15:4	PMEM_BAR[31:20] Prefetchable Memory Base Address[31:20]. Specifies the Port's Prefetchable Memory Base Address[31:20]. The PEX 8724 assumes Prefetchable Memory Base Address[19:0]=0_0000h.	RW	Yes	FFFh
Prefetchable Memory Limit				
16	Prefetchable Memory Limit Capability 0 = Port supports 32-bit Prefetchable Memory Addressing 1 = Port defaults to 64-bit Prefetchable Memory Addressing support, as required by the <i>PCI Express Base r3.0</i>	RO	Yes	1
19:17	Reserved	RsvdP	No	000b
31:20	PMEM_Limit[31:20] Prefetchable Memory Limit Address[31:20]. Specifies the Port's Prefetchable Memory Limit Address[31:20]. The PEX 8724 assumes Prefetchable Memory Limit Address[19:0]=F_FFFFh.	RW	Yes	000h

**Register 13-11. 28h Prefetchable Memory Upper Base Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
31:0	PBUP[63:32] Prefetchable Memory Base Address[63:32]. The PEX 8724 uses this register for Prefetchable Memory Upper Base Address[63:32].	Offset 24h[0]=1	RW	Yes	0000_0000h
	When the Prefetchable Memory Base register <i>Prefetchable Memory Base Capability</i> field indicates 32-bit addressing, this register is RO and returns a value of 0000_0000h.	Offset 24h[0]=0	RO	No	0000_0000h

**Register 13-12. 2Ch Prefetchable Memory Upper Limit Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
31:0	PLIMUP[63:32] Prefetchable Memory Limit Address[63:32]. The PEX 8724 uses this register for Prefetchable Memory Upper Limit Address[63:32]. When the Prefetchable Memory Limit register <i>Prefetchable Memory Limit Capability</i> field indicates 32-bit addressing, this register is RO and returns a value of 0000_0000h.	Offset 24h[16]=1	RW	Yes	0000_0000h
	<i>Note: The serial EEPROM must not write a non-zero value into this register when the RO attribute is Set for this register.</i>	Offset 24h[16]=0	RO	No	0000_0000h

**Register 13-13. 30h I/O Upper Base and Limit Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
I/O Base Upper 16 Bits					
15:0	I/O Base Upper 16 Bits The PEX 8724 uses this register for I/O Base Address[31:16]. When the I/O Base register <i>I/O Base Addressing Capability</i> field indicates 16-bit addressing, this register is RO and returns a value of 0000h.	Offset 1Ch[3:0]=1h	RW	Yes	0000h
		Offset 1Ch[3:0]=0h	RO	No	0000h
I/O Limit Upper 16 Bits					
31:16	I/O Limit Upper 16 Bits The PEX 8724 uses this register for I/O Limit Address[31:16]. When the I/O Limit register <i>I/O Limit Addressing Capability</i> field indicates 16-bit addressing, this register is RO and returns a value of 0000h. Note: The serial EEPROM must not write a non-zero value into this register when the RO attribute is Set for this register.	Offset 1Ch[11:8]=1h	RW	Yes	0000h
		Offset 1Ch[11:8]=0h	RO	No	0000h

**Register 13-14. 34h Capability Pointer
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability Pointer Default 40h points to the PCI Power Management Capability structure.	RO	Yes	40h
31:8	Reserved	RsvdP	No	0000_00h

**Register 13-15. 38h Expansion ROM Base Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Expansion ROM Base Address Reserved	RsvdP	No	0000_0000h

**Register 13-16. 3Ch Bridge Control and PCI Interrupt Signal
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Interrupt Signal				
7:0	PCI Interrupt Line Interrupt line routing value. The PEX 8724 does not use this register; however, the register is included for operating system and device driver use.	RW	Yes	00h
15:8	PCI Interrupt Pin Identifies the Conventional PCI Interrupt Message(s) that the device (or device function) uses. Only value 00h or 01h is allowed in the PEX 8724. 00h = Device does not use Conventional PCI Interrupt Message(s) 01h, 02h, 03h, and 04h = Maps to Conventional PCI Interrupt Messages for INTA#, INTB#, INTC#, and INTD#, respectively	RO	Yes	01h
Bridge Control				
16	Parity Error Response Enable Controls the response to Poisoned TLPs. 0 = Disables the Port's Secondary Status register <i>Master Data Parity Error</i> bit (offset 1Ch[24]) 1 = Enables the Port's Secondary Status register <i>Master Data Parity Error</i> bit (offset 1Ch[24])	RW	Yes	0
17	SERR# Enable Controls forwarding of ERR_COR, ERR_FATAL, and ERR_NONFATAL from the secondary interface to the primary interface. When Set, and the Port's PCI Command register <i>SERR# Enable</i> bit (offset 04h[8]) is also Set, enables the Port's PCI Status register <i>Signaled System Error</i> bit (offset 04h[30]).	RW	Yes	0
18	ISA Enable Modifies the PEX 8724's response to Conventional PCI ISA I/O addresses enabled by the Port's I/O Base and I/O Limit registers (offset 1Ch[7:0 and 15:8], respectively) and located in the first 64 KB of the PCI I/O Address space (0000_0000h to 0000_FFFh). 0 = Port's I/O Base and I/O Limit registers, and I/O Base Upper 16 Bits and I/O Limit Upper 16 Bits registers (offset 30h[15:0 and 31:16], respectively), define the Port's I/O Address range 1 = If the Port's I/O Address range (defined by the Port's I/O Base , I/O Limit , I/O Base Upper 16 Bits , and I/O Limit Upper 16 Bits registers) includes I/O addresses below 64 KB, the upper 768 I/O addresses within each 1-KB block below 64 KB are excluded from the Port's I/O Address range Note: Refer also to the Ingress Control register <i>Disable VGA BIOS Memory Access Decoding</i> bit (Upstream Port(s), offset F60h[28]).	RW	Yes	0

**Register 13-16. 3Ch Bridge Control and PCI Interrupt Signal
(All Ports) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19	VGA Enable Modifies the bridge response to VGA-compatible addresses. When Set, the bridge positively decodes and forwards the following addresses on the primary interface to the secondary interface (and, conversely, blocks forwarding of these addresses from the secondary interface to the primary interface): - Memory addresses within the range 000A_0000h to 000B_FFFFh - I/O addresses in the first 64 KB of the I/O Address space (AD[31:16] is 0000h), where AD[9:0] is within the ranges 3B0h to 3BBh and 3C0h to 3DFh (inclusive of ISA address aliases – AD[15:10] is not decoded) Additionally, when Set, forwarding of these addresses is independent of the: - Memory and I/O Address ranges defined by the bridge I/O Base, I/O Limit, Memory Base, Memory Limit, Prefetchable Memory Base, and Prefetchable Memory Limit registers - Bit 18 (<i>ISA Enable</i>) Setting VGA address forwarding is qualified by the Port's PCI Command register <i>Memory Access Enable</i> and <i>I/O Access Enable</i> bits (offset 04h[1:0], respectively). The default state of this bit after reset must be 0. 0 = Do not forward VGA-compatible Memory and I/O addresses from the primary to the secondary interface (addresses defined above) unless they are enabled for forwarding by the defined Memory and I/O Address ranges 1 = Forward VGA-compatible Memory and I/O addresses (addresses defined above) from the primary interface to the secondary interface (when the <i>Memory Access Enable</i> and <i>I/O Access Enable</i> bits are Set), independent of the Memory and I/O Address ranges and independent of the <i>ISA Enable</i> bit Notes: When Set in an egress Port, the Port is configured as a non-Cut-Thru path. (Refer to Section 2.1.4.2, “Cut-Thru Mode,” for further details.) Refer also to the Ingress Control register <i>Disable VGA BIOS Memory Access Decoding</i> bit (Upstream Port(s), offset F60h[28]). Conventional PCI VGA support – To avoid potential I/O address conflicts, if the <i>VGA Enable</i> bit is Set in an Upstream Port and a Downstream Port, Set the Port's PCI Command register <i>I/O Access Enable</i> bit (offset 04h[0]) in the remaining Downstream Ports, unless those Downstream Ports are configured to use default 32-bit address decoding and their I/O Address range is programmed above 1_0000h.	RW	Yes	0

**Register 13-16. 3Ch Bridge Control and PCI Interrupt Signal
(All Ports) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
20	VGA 16-Bit Decode Enable Used only when bit 19 (<i>VGA Enable</i>) is also Set, enabling VGA I/O decoding and forwarding by the bridge. Status after reset is 0. Enables system configuration software to select between 10- and 16-bit I/O address decoding, for VGA I/O register accesses forwarded from the primary interface to the secondary interface. 0 = Execute 10-bit address decodes on VGA I/O accesses 1 = Execute 16-bit address decodes on VGA I/O accesses <i>Note:</i> Refer also to the <i>Ingress Control</i> register <i>Disable VGA BIOS Memory Access Decoding</i> bit (Upstream Port(s), offset F60h[28]).	RW	Yes	0
21	Master Abort Mode <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
22	Secondary Bus Reset 1 = Causes a Hot Reset on the Port's Downstream Link	RW	Yes	0
23	Fast Back-to-Back Transactions Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
24	Primary Discard Timer <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
25	Secondary Discard Timer <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
26	Discard Timer Status <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
27	Discard Timer SERR# Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
31:28	Reserved	RsvdP	No	0h

13.8 PCI Power Management Capability Registers (Offsets 40h – 44h)

This section details the PCI Power Management Capability registers. [Table 13-8](#) defines the register map.

Table 13-8. PCI Power Management Capability Register Map (All Ports)

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0	Offsets
PCI Power Management Capability		Next Capability Pointer (48h)	Capability ID (01h)	40h
PCI Power Management Data	PCI Power Management Control/Status Bridge Extensions (<i>Reserved</i>)	PCI Power Management Status and Control		44h

Register 13-17. 40h PCI Power Management Capability (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability ID Program to 01h, to indicate that the Capability structure is the PCI Power Management Capability structure.	RO	Yes	01h
15:8	Next Capability Pointer Default 48h points to the MSI Capability structure.	RO	Yes	48h
18:16	Version Default 011b indicates compliance with the <i>PCI Power Mgmt. r1.2</i> .	RO	Yes	011b
19	PME Clock Power Management Event (PME) clock. Does not apply to PCI Express. Returns a value of 0.	RsvdP	No	0
20	Reserved	RsvdP	No	0
21	Device-Specific Initialization 0 = Device-Specific Initialization is <i>not</i> required	RO	Yes	0
24:22	AUX Current The PEX 8724 does <i>not support</i> PME generation from the D3cold Device PM state; therefore, the serial EEPROM value for this field should be 000b.	RO	Yes	000b
25	D1 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D1 Device PM state	RsvdP	No	0
26	D2 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D2 Device PM state	RsvdP	No	0
31:27	PME Support Bits [31, 30, and 27] must be Set, to indicate that the PEX 8724 will forward PME Messages, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	19h

**Register 13-18. 44h PCI Power Management Status and Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Status and Control				
1:0	Power State Used to determine the Port's current Device PM state, and to program the Port into a new Device PM state. 00b = D0 01b = D1 – <i>Not supported</i> 10b = D2 – <i>Not supported</i> 11b = D3hot If software attempts to write an unsupported state to this field, the Write operation completes normally; however, the data is discarded and no state change occurs.	RW	Yes	00b
2	Reserved	RsvdP	No	0
3	No Soft Reset 1 = Devices transitioning from the D3hot to D0 Device PM state, because of Power State commands, do not perform an internal reset	ROS	Yes	1
7:4	Reserved	RsvdP	No	0h
8	PME Enable 0 = Disables PME generation by the Port ^a 1 = Enables PME generation by the Port	RWS	No	0
12:9	Data Select Initially writable by serial EEPROM and/or I ² C only ^b . This Configuration Space register (CSR) access privilege changes to RW after a Serial EEPROM and/or I ² C Write occurs to this register. Selects the Power Number at a particular state (for field [14:13] (<i>Data Scale</i>)), which is reflected in the Port's PCI Power Management Data register <i>Data</i> field (offset 44h[31:24]). 0h = D0 Device PM state power consumed 3h = D3hot Device PM state power consumed 4h = D0 Device PM state power dissipated 7h = D3hot Device PM state power dissipated All other encodings are Reserved .	RO	Yes	0h
14:13	Data Scale Writable by serial EEPROM and/or I ² C only ^b . Indicates the scaling factor to be used when interpreting the PCI Power Management Data register <i>Data</i> field (offset 44h [31:24]) value. The value and meaning of the <i>Data Scale</i> field varies, depending upon which data value is selected by field [12:9] (<i>Data Select</i>). There are four internal <i>Data Scale</i> fields (one each, per <i>Data Select</i> values 0h, 3h, 4h, and 7h), per Port. For other <i>Data Select</i> values, the <i>Data Scale</i> value returned is 0h.	RO	Yes	00b
15	PME Status 0 = PME is not generated by the Port ^a 1 = PME is being generated by the Port	RW1CS	No	0

**Register 13-18. 44h PCI Power Management Status and Control
(All Ports) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Control/Status Bridge Extensions				
21:16	Reserved	RsvdP	No	0-0h
22	B2/B3 Support Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
23	Bus Power/Clock Control Enable Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
PCI Power Management Data				
31:24	Data Writable by serial EEPROM and/or I ² C only ^b . There are four supported <i>Data Select</i> values (0h, 3h, 4h, and 7h), per Port. For other <i>Data Select</i> values, the <i>Data</i> value returned is 00h. Selected by the Port's PCI Power Management Status and Control register <i>Data Select</i> field (offset 44h[12:9]).	RO	Yes	00h

- a. Because the PEX 8724 does not consume auxiliary power, this bit is not Sticky, and is always Cleared at power-on reset.
- b. With no serial EEPROM nor previous I²C programming, Reads return a value of 00h for the **PCI Power Management Status and Control** register *Data Scale* and **PCI Power Management Data** register *Data* fields (for all *Data Selects*).

13.9 Message Signaled Interrupt Capability Registers (Offsets 48h – 64h)

This section details the Message Signaled Interrupt (MSI) Capability registers. [Table 13-9](#) defines the register map.

Table 13-9. MSI Capability Register Map (All Ports)^a

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0	Offsets
MSI Control		Next Capability Pointer (68h)	Capability ID (05h)	48h
MSI Address				4Ch
MSI Upper Address				50h
Reserved		MSI Data		54h
Reserved			MSI Mask	58h
MSI Status				5Ch
Reserved				60h – 64h

- a. Offsets 54h, 58h, and 5Ch change to 50h, 54h, and 58h, respectively, when the Port's **MSI Control** register **MSI 64-Bit Address Capable** bit (offset 48h[23]) is Cleared.

**Register 13-19. 48h MSI Capability
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
MSI Capability Header				
7:0	Capability ID Program to 05h, as required by the <i>PCI r3.0</i> .	RO	Yes	05h
15:8	Next Capability Pointer Program to 68h, to point to the PCI Express Capability structure.	RO	Yes	68h
MSI Control				
16	MSI Enable 0 = MSIs for the Port are disabled 1 = MSIs for the Port are enabled, and INTx Interrupt Messages and PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output assertion are disabled	RW	Yes	0
19:17	Multiple Message Capable 000b = Port can request only one MSI Vector 001b = Port can request two MSI Vectors 010b = Port can request four MSI Vectors 011b = Port can request eight MSI Vectors All other encodings are Reserved .	RO	Yes	011b
22:20	Multiple Message Enable 000b = Port is allocated one MSI Vector, by default. 001b = Port is allocated two MSI Vectors. 010b = Port is allocated four MSI Vectors. 011b = Port is allocated eight MSI Vectors. Up to six MSI Vectors are used; the upper two MSI Vectors (having the highest values of Message Data bits [2:0]) are not used. All other encodings are Reserved . Note: Should not be programmed with a value larger than that of field [19:17] (Multiple Message Capable). If the value of this field is larger than that of field [19:17], the Multiple Message Capable value takes effect.	RW	Yes	000b
23	MSI 64-Bit Address Capable 0 = PEX 8724 is capable of generating MSI 32-bit addresses (Port's MSI Address register, offset 4Ch, is the Message address) 1 = PEX 8724 is capable of generating MSI 64-bit addresses (Port's MSI Address register, offset 4Ch, is the lower 32 bits of the Message address, and MSI Upper Address register, offset 50h, is the upper 32 bits of the Message address)	RO	Yes	1
24	Per Vector Masking Capable 0 = PEX 8724 does not have Per Vector Masking capability 1 = PEX 8724 has Per Vector Masking capability	RO	Yes	1
31:25	Reserved	RsvdP	No	0-0h

**Register 13-20. 4Ch MSI Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	Reserved	RsvdP	No	00b
31:2	Message Address <i>Note:</i> If the Port's MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Set (default), refer to the Port's MSI Upper Address register for the MSI Upper Address (offset 50h).	RW	Yes	0-0h

**Register 13-21. 50h MSI Upper Address
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Message Upper Address Valid/used only when the Port's MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Set. MSI Write transaction upper address[63:32]. <i>Note:</i> Refer to the MSI Address register for the MSI Lower Address (offset 4Ch).	RW	Yes	0000_0000h

**Register 13-22. 54h MSI Data
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
<i>Note:</i> The offset for this register changes from 54h, to 50h, when the Port's MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared.				
15:0	Message Data MSI Write transaction TLP payload.	RW	Yes	0000h
31:16	Reserved	RsvdP	No	0000h

**Register 13-23. 58h MSI Mask
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Port interrupt sources are grouped into six categories – Hot Plug or Link State-/PM-triggered interrupt events, Device-Specific NT-Link Port or RAM ECC error events, GPIO-generated interrupts, Virtual Switch mode Doorbell events, and Virtual Switch mode Link Status events. The quantity of MSI Vectors that are generated is determined by the Port's MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (offset 48h[19:17 and 22:20], respectively): • If one MSI Vector is enabled (default), all six interrupt events are combined into a single Vector • If two MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Hot Plug or Link State-/PM-triggered interrupt events, GPIO-generated events, Virtual Switch Doorbell events, and Management Port Link Status events – Vector[1] Device-specific NT-Link Port events or RAM ECC error events • If four MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Hot Plug or Link State-/PM-triggered interrupt events – Vector[1] Device-specific NT-Link Port events or RAM ECC error events – Vector[2] GPIO-generated events – Vector[3] Virtual Switch Doorbell events and Management Port Link Status events • If eight MSI Vectors are allocated (up to six Vectors are used; the upper two Vectors (having the highest values of Message Data bits [2:0]) are not used), the individual Vectors indicate: – Vector[0] Hot Plug or Link State-/PM-triggered interrupt events – Vector[1] Device-specific NT-Link Port events or RAM ECC error events – Vector[2] GPIO-generated events – Vector[3] Reserved – Vector[4] Virtual Switch Doorbell events – Vector[5] Management Port Link Status events Notes: The offset for this register changes from 58h, to 54h, when the Port's MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be used to mask the Port's respective MSI Status register bits (offset 5Ch).					
0	MSI Mask for Hot Plug or Link State Events MSI mask for Power Management- or Hot Plug or Link State-triggered interrupt events.	All Ports, when offset 48h[22:20]=010b or 011b	RW	Yes	0
	MSI Mask for Shared Interrupt Sources MSI mask for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	All Ports, when offset 48h[22:20]≤001b	RW	Yes	0

Register 13-23. 58h MSI Mask
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
1	Transparent Mode MSI Mask for RAM ECC Error Events	Base Mode 0, 8, when offset 48h[22:20]=001b, 010b or 011b Virtual Switch Mode 0, 8, accessible through the Management Port, when offset 48h[22:20]=001b, 010b or 011b	RW	Yes	0
	NT Mode MSI Mask for Device-Specific NT-Link Port Events MSI mask for Device-Specific NT-Link Port event-generated interrupts. Implemented only for the NT Port Virtual Interface. Enables MSIs for the following NT-Link Port events, defined in the Link Error Status Virtual and Link Error Mask Virtual registers (NT Port Virtual Interface, offsets FE0h and FE4h, respectively): - NT-Link Port Correctable error (NT Port Link Interface, offset FC4h) - NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h) - NT-Link Port Data Link Layer State change - NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message	NT Port Virtual Interface, when offset 48h[22:20]=001b, 010b or 011b	RW	Yes	0
	Reserved	Otherwise, when offset 48h[22:20]=000b	RsvdP	No	0
2	Base Mode MSI Mask for GPIO-Generated Interrupts	0, when offset 48h[22:20]=010b or 011b	RW	Yes	0
	Virtual Switch Mode MSI Mask for GPIO-Generated Interrupts	VS Upstream, when offset 48h[22:20]=010b or 011b	RW	Yes	0
	Reserved	Otherwise, when offset 48h[22:20]≤001b	RsvdP	No	0
3	Reserved		RW	Yes	0

**Register 13-23. 58h MSI Mask
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode MSI Mask for Management Port Doorbell-Generated Interrupts	VS Upstream and Management, when offset 48h[22:20]=011b	RW	Yes	0
	<i>Reserved</i>	Otherwise, when offset 48h[22:20]≤010b	RsvdP	No	0
5	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode MSI Mask for Management Link Status Event	Management, when offset 48h[22:20]=011b	RW	Yes	0
	<i>Reserved</i>	Otherwise, when offset 48h[22:20]≤010b	RsvdP	No	0
7:6	<i>Reserved</i>		RW	Yes	00b
31:8	<i>Reserved</i>		RsvdP	No	0000_00h

**Register 13-24. 5Ch MSI Status
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Port interrupt sources are grouped into six categories – Hot Plug or Link State-/PM-triggered interrupt events, Device-Specific NT-Link Port or RAM ECC error events, GPIO-generated interrupts, Virtual Switch mode Doorbell events, and Virtual Switch mode Link Status events. The quantity of MSI Vectors that are generated is determined by the Port's MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (offset 48h[19:17 and 22:20], respectively): - If one MSI Vector is enabled (default), all six interrupt events are combined into a single Vector - If two MSI Vectors are allocated, the individual Vectors indicate: - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events, GPIO-generated events, Virtual Switch Doorbell events, and Management Port Link Status events - Vector[1] Device-specific NT-Link Port events or RAM ECC error events - If four MSI Vectors are allocated, the individual Vectors indicate: - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events - Vector[1] Device-specific NT-Link Port events or RAM ECC error events - Vector[2] GPIO-generated events - Vector[3] Virtual Switch Doorbell events and Management Port Link Status events - If eight MSI Vectors are allocated (up to six Vectors are used; the upper two Vectors (having the highest values of Message Data bits [2:0]) are not used), the individual Vectors indicate: - Vector[0] Hot Plug or Link State-/PM-triggered interrupt events - Vector[1] Device-specific NT-Link Port events or RAM ECC error events - Vector[2] GPIO-generated events - Vector[3] Reserved - Vector[4] Virtual Switch Doorbell events - Vector[5] Management Port Link Status events Notes: The offset for this register changes from 5Ch, to 58h, when the Port's MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be masked by the Port's respective MSI Mask register bits (offset 58h).					
0	MSI Pending Status for Hot Plug or Link State Events MSI pending status for Power Management- or Hot Plug or Link State-triggered interrupt events.	All Ports, when offset 48h[22:20]=010b or 011b	RO	No	0
	MSI Pending Status for Shared Interrupt Sources MSI pending status for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	All Ports, when offset 48h[22:20]≤001b	RsvdP	No	0

**Register 13-24. 5Ch MSI Status
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
1	Transparent Mode MSI Pending Status for RAM ECC Error Events	Base Mode 0, 8, when offset 48h[22:20]=001b, 010b or 011b Virtual Switch Mode 0, 8, accessible through the Management Port, when offset 48h[22:20]=001b, 010b or 011b	RO	No	0
	NT Mode MSI Pending Status for Device-Specific NT-Link Port Events Implemented only for the NT Port Virtual Interface. Enables MSIs for the following NT-Link Port events, defined in the Link Error Status Virtual and Link Error Mask Virtual registers (NT Port Virtual Interface, offsets FE0h and FE4h, respectively): - NT-Link Port Correctable error (NT Port Link Interface, offset FC4h) - NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h) - NT-Link Port Data Link Layer State change - NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message	NT Port Virtual Interface, when offset 48h[22:20]=001b, 010b or 011b	RO	No	0
	Reserved	Otherwise, when offset 48h[22:20]=000b	RsvdP	No	0
2	Base Mode MSI Pending Status for GPIO-Generated Interrupts	0, when offset 48h[22:20]=010b or 011b	RO	No	0
	Virtual Switch Mode MSI Pending Status for GPIO-Generated Interrupts	VS Upstream, when offset 48h[22:20]=010b or 011b	RO	No	0
	Reserved	Otherwise, when offset 48h[22:20]≤001b	RsvdP	No	0
3	Reserved		RO	Yes	0

**Register 13-24. 5Ch MSI Status
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode MSI Pending Status for Management Port Doorbell-Generated Interrupts	VS Upstream and Management, when offset 48h[22:20]=011b	RO	No	0
	<i>Reserved</i>	Otherwise, when offset 48h[22:20]≤010b	RsvdP	No	0
5	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode MSI Pending Status for Management Link Status Event	Management, when offset 48h[22:20]=011b	RO	No	0
	<i>Reserved</i>	Otherwise, when offset 48h[22:20]≥010b	RsvdP	No	0
31:6	<i>Reserved</i>		RsvdP	No	0-0h

13.10 PCI Express Capability Registers (Offsets 68h – A0h)

This section details the PCI Express Capability registers. Hot Plug Capability, Command, Status, and Events are included in these registers. [Table 13-10](#) defines the register map.

Table 13-10. PCI Express Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
PCI Express Capability																Next Capability Pointer (A4h)										Capability ID (10h)						68h
Device Capability																																6Ch
Device Status																Reserved/Not Supported										Device Control						70h
Link Capability																																74h
Link Status																Link Control																78h
Reserved (Upstream) Slot Capability (Downstream)																																7Ch
Slot Status (Upstream (Virtual Switch Mode Only) and Downstream)																Slot Control (Upstream (Virtual Switch Mode Only) and Downstream)																80h
Reserved																																84h – 88h
Device Capability 2																																8Ch
Device Status 2 (Reserved)																Device Control 2																90h
Link Capability 2																																94h
Link Status 2																Link Control 2																98h
Reserved																																9Ch – A0h

Register 13-25. 68h PCI Express Capability List and Capability (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
PCI Express Capability List					
7:0	Capability ID Program to 10h, as required by the <i>PCI Express Base r3.0</i> .		RO	Yes	10h
15:8	Next Capability Pointer Program to A4h, to point to the Subsystem Capability structure.		RO	Yes	A4h
PCI Express Capability					
19:16	Capability Version The PEX 8724 Ports program this field to 2h, as required by the <i>PCI Express Base r3.0</i> .		RO	Yes	2h
23:20	Device/Port Type Set at reset, as required by the <i>PCI Express Base r3.0</i> .	Upstream	RO	Yes	5h
		Downstream	RO	Yes	6h
24	Slot Implemented 0 = Disables or connects to an Upstream Port	Upstream	RsvdP	No	0
	0 = Disables or connects to an integrated component 1 = Downstream Port connects to a slot, as opposed to being connected to an integrated component or being disabled Note: The PEX 8724 serial EEPROM register Initialization capability, and/or I ² C/SMBus when <i>STRAP_I2C_SMBUS_CFG_EN#</i> =L, can be used to Clear this bit, indicating that the Downstream Port connects to an integrated component or is disabled.	Downstream	ROS	Yes	1
29:25	Interrupt Message Number The serial EEPROM writes 00_000b, because the Base Message and MSI Messages are the same.		RO	Yes	00_000b
31:30	Reserved		RsvdP	No	00b

**Register 13-26. 6Ch Device Capability
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
2:0	Maximum Payload Size Supported <i>Note:</i> The default Maximum Payload Size supported is based upon the STRAP_STNx_PORTCFGx 3-state inputs. If the serial EEPROM or I ² C/SMBus changes the Port configuration (from the strapped value), it must also program the value of this field, accordingly (with the value dependent upon the Station having the most enabled Ports). The value for all Ports is determined by the Station having the most enabled Ports: - One or two enabled Ports = 100b - Three enabled Ports = 011b - Four enabled Ports = 010b 000b = Port supports a 128-byte maximum payload 001b = Port supports a 256-byte maximum payload 010b = Port supports a 512-byte maximum payload 011b = Port supports a 1,024-byte maximum payload 100b = Port supports a 2,048-byte maximum payload No other encodings are supported.		HwInit	Yes	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
4:3	Phantom Functions Supported <i>Not supported</i>		RO	Yes	00b
5	Extended Tag Field Supported 0 = Maximum <i>Tag</i> field is 5 bits 1 = Maximum <i>Tag</i> field is 8 bits		RO	Yes	0
8:6	Endpoint L0s Acceptable Latency <i>Not supported</i> Because the PEX 8724 is a switch and not an endpoint, the PEX 8724 does not support this feature. 000b = Disables the capability		RO	Yes	000b
11:9	Endpoint L1 Acceptable Latency <i>Not supported</i> Because the PEX 8724 is a switch and not an endpoint, the PEX 8724 does not support this feature. 000b = Disables the capability		RO	Yes	000b
14:12	Reserved , as required by the <i>PCI Express Base r3.0</i> .		RsvdP	No	000b
15	Role-Based Error Reporting		RO	Yes	1

**Register 13-26. 6Ch Device Capability
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
17:16	Reserved		RsvdP	No	00b
25:18	Captured Slot Power Limit Value For Upstream Port(s), the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [27:26] (<i>Captured Slot Power Limit Scale</i>).	Upstream	RO	Yes	00h
	Not valid	Downstream	RsvdP	No	00h
27:26	Captured Slot Power Limit Scale For Upstream Port(s), the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [25:18] (<i>Captured Slot Power Limit Value</i>). 00b = 1.0 01b = 0.1 10b = 0.01 11b = 0.001	Upstream	RO	Yes	00b
	Not valid	Downstream	RsvdP	No	00b
31:28	Reserved		RsvdP	No	0h

**Register 13-27. 70h Device Status and Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Control				
0	Correctable Error Reporting Enable 0 = Disables 1 = Enables the Port to report Correctable errors	RW	Yes	0
1	Non-Fatal Error Reporting Enable 0 = Disables 1 = Enables the Port to report Non-Fatal errors	RW	Yes	0
2	Fatal Error Reporting Enable 0 = Disables 1 = Enables the Port to report Fatal errors	RW	Yes	0
3	Unsupported Request Reporting Enable 0 = Disables 1 = Enables the Port to report UR errors	RW	Yes	0
4	Enable Relaxed Ordering <i>Not supported</i> The value of this bit has no effect upon internal logic.	RW	Yes	1
7:5	Maximum Payload Size Software can change this field to configure the Ports to support other Payload Sizes; however, software cannot change this field to a value larger than that indicated by the Device Capability register <i>Maximum Payload Size Supported</i> field (offset 6Ch[2:0]). 000b = Port supports a 128-byte maximum payload 001b = Port supports a 256-byte maximum payload 010b = Port supports a 512-byte maximum payload 011b = Port supports a 1,024-byte maximum payload 100b = Port supports a 2,048-byte maximum payload No other encodings are supported.	RW	Yes	000b
8	Extended Tag Field Enable <i>Not supported</i>	RsvdP	No	0
9	Phantom Functions Enable <i>Not supported</i>	RsvdP	No	0
10	AUX Power PM Enable <i>Not supported</i>	RsvdP	No	0
11	Enable No Snoop <i>Not supported</i>	RW	Yes	1
14:12	Max Read Request Size <i>Not supported</i>	RsvdP	No	000b
15	Reserved Hardwired to 0, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0

**Register 13-27. 70h Device Status and Control
(All Ports) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Status				
16	Correctable Error Detected 0 = Port did not detect a Correctable error 1 = Port detected a Correctable error, regardless of the bit 0 (<i>Correctable Error Reporting Enable</i>) state	RW1C	Yes	0
17	Non-Fatal Error Detected 0 = Port did not detect a Non-Fatal error 1 = Port detected a Non-Fatal error, regardless of the bit 1 (<i>Non-Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
18	Fatal Error Detected 0 = Port did not detect a Fatal error 1 = Port detected a Fatal error, regardless of the bit 2 (<i>Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
19	Unsupported Request Detected 0 = Port did not detect a UR 1 = Port detected a UR, regardless of the bit 3 (<i>Unsupported Request Reporting Enable</i>) state	RW1C	Yes	0
20	AUX Power Detected <i>Not supported</i>	RsvdP	No	0
21	Transactions Pending <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
31:22	Reserved	RsvdP	No	0-0h

**Register 13-28. 74h Link Capability
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
Note: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STN _x _PORTCFG _x input states and/or serial EEPROM programming. The table also lists the relationship between the Ports and Lanes (Link widths).						
3:0	Maximum Link Speed Indicates the Port's maximum Link speed. The encoding is the binary value of the bit location in the Port's Link Capability 2 register <i>Supported Link Speeds Vector</i> field (offset 94h[7:1]) that corresponds to the maximum Link speed. <i>For example</i> , the default value 3h indicates that the maximum Link speed is that which corresponds to bit 3 in the <i>Supported Link Speeds Vector</i> field, which is 8.0 GT/s. The default value for all Ports can be globally changed to 2h (Gen 2 maximum speed) or 1h (Gen 1 maximum speed), according to the STRAP_GEN1_GEN2 input state (Z = 2h (Gen 2), or 0 = 1h (Gen 1)). 1h = Gen 1 (2.5 GT/s) 2h = Gen 2 (5.0 GT/s) 3h = Gen 3 (8.0 GT/s)		RO	Yes	3h	
9:4	Maximum Link Width The PEX 8724 maximum Link width is x16 = 01_0000b. Actual maximum Link width is defined by the STRAP_STN _x _PORTCFG _x 3-state inputs. 00_0000b = Reserved 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are not supported .		ROS	No	Defined by STRAP_STN _x _PORTCFG _x input states, or programmed by serial EEPROM value for the Port Configuration -related registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 300h through 308h)	
11:10	Active State Power Management (ASPM) Support Active State Link PM support. Indicates the level of ASPM supported by the Port. 00b = No ASPM support 01b = ASPM L0s Link PM state is supported 10b = ASPM L1 Link PM state is supported 11b = ASPM L0s and L1 Link PM states are supported		RO	Yes	11b	10b

**Register 13-28. 74h Link Capability
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
14:12	L0s Exit Latency Indicates the ASPM L0s Link PM state exit latency for the Port's PCI Express Link, when the state is supported and enabled: • Bit 10 of field [11:10] (Port's <i>Active State Power Management (ASPM) Support</i> field) is Set • Bit 0 of the Port's Link Control register <i>Active State Power Management (ASPM)</i> field (offset 78h[1:0]) is Set The value reported indicates the length of time this Port requires to complete the ASPM L0s to L0 Link PM state transition. If the ASPM L0s Link PM state is not supported (bit 10 of the Port's <i>Active State Power Management (ASPM) Support</i> field is Cleared), the value of this field is undefined; however, for the recommended value, refer to the Implementation Note in the <i>PCI Express Base r3.0</i>, Section 5.4.1.1, "Potential Issues with Legacy Software When L0s is Not Supported." 000b = Less than 64 ns 001b = 64 ns to less than 128 ns 010b = 128 ns to less than 256 ns 011b = 256 ns to less than 512 ns 100b = 512 ns to less than 1 μs 101b = 1 μs to less than 2 μs 110b = 2 to 4 μs (default) 111b = More than 4 μs		RO	No	110b	

**Register 13-28. 74h Link Capability
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
17:15	L1 Exit Latency Indicates the ASPM L1 Link PM state exit latency for the Port's PCI Express Link. The value reported indicates the length of time that this Port requires to complete the ASPM L1 to L0 Link PM state transition. If the L0 Link PM state is not supported, the value of this field is undefined. 000b = Less than 1 μ s 001b = 1 μ s to less than 2 μ s (5.0 GT/s) 010b = 2 μ s to less than 4 μ s (default, 2.5 GT/s) 011b = 4 μ s to less than 8 μ s 100b = 8 μ s to less than 16 μ s 101b = 16 μ s to less than 32 μ s 110b = 32 to 64 μ s 111b = More than 64 μ s		RO	Yes	010b	
18	Clock Power Management Capable <i>Not supported</i>	Upstream	RO	Yes	0	
	<i>Not supported</i> Must be hardwired to 0.	Downstream	RO	No	0	
19	Reserved Must be hardwired to 0, for Upstream Port(s) and components that do not support this optional capability.	Upstream	RsvdP	No	0	
	Surprise Down Error Reporting Capable Must be Set if the component supports the optional capability of detecting and reporting a Surprise Down error condition. If this bit is Cleared, the Port's Uncorrectable Error Status register <i>Surprise Down Error Status</i> bit (Downstream Ports, offset FB8h[5]) is disabled.	Downstream	RO	Yes	1	

**Register 13-28. 74h Link Capability
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
20	Reserved	Upstream	RsvdP	No	0	
	Data Link Layer Link Active Reporting Capable	Downstream	RO	Yes	1	
21	Reserved Hardwired to 0, as required by the <i>PCI Express Base r3.0</i> .	Upstream	RsvdP	No	0	
	Link Bandwidth Notification Capability 1 = Indicates support for the Link Bandwidth Notification status and interrupt mechanisms	Downstream	RO	Yes	1	
22	ASPM Optionality Compliance		HwInit	Yes	1	
23	Reserved		RsvdP	No	0	
31:24	Port Number The Port Number is defined by 3-state input Strapping options. Note: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.</i> Station 0, STRAP_STN0_PORTCFG[1:0] – Ports 0, 1, 2, 3 Station 1, STRAP_STN1_PORTCFG[1:0] – Ports 8, 9, 10		ROS	Yes	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register for Station x field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])	

**Register 13-29. 78h Link Status and Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Link Control					
1:0	Active State Power Management (ASPM) 00b = Disable ^a 01b = Enables only ASPM L0s Link PM state entry 10b = Enables only ASPM L1 Link PM state entry 11b = Enables both ASPM L0s and L1 Link PM state entries		RW	Yes	00b
2	Reserved		RsvdP	No	0
3	Read Request Return Parameter Control Read Request Return Parameter “R” control. Read Completion Boundary (RCB). Cleared, as required by the <i>PCI Express Base r3.0</i> .		RO	Yes	0

Register 13-29. 78h Link Status and Control
(All Ports) (Cont.)

Bit(s)	Description		Ports	Type	Serial EEPROM and I ² C	Default
4	Not valid	Offset F70h[29]=0	Upstream	RsvdP	No	0
	Link Disable Functionality is enabled when the Port's Power Management Hot Plug User Configuration register <i>PHY Upstream Port Control Write Enable</i> bit (offset F70h[29]) is Set. 1 = Places the Upstream Port(s)' Link(s) into the <i>Disabled</i> Link Training state	Offset F70h[29]=1	Upstream	RW	Yes	0
	1 = Places the Downstream Port's Link into the <i>Disabled</i> Link Training state		Downstream	RW	Yes	0
5	Not valid Always read as 0.	Offset F70h[29]=0	Upstream	RsvdP	No	0
	Retrain Link Functionality is enabled when the Port's Power Management Hot Plug User Configuration register <i>PHY Upstream Port Control Write Enable</i> bit (offset F70h[29]) is Set. Always returns a value of 0 when read; however, software is allowed to write this register. Writing 1 to this bit causes Upstream Port(s) to initiate retraining of its (their) PCI Express Link.	Offset F70h[29]=1	Upstream	RZ	Yes	0
	Always returns a value of 0 when read; however, software is allowed to write this register. Writing 1 to this bit causes the Downstream Port to initiate retraining of its PCI Express Link.		Downstream	RZ	Yes	0
6	Common Clock Configuration 0 = Port and the device at the other end of the Port's PCI Express Link use an asynchronous Reference Clock source 1 = Port and the device at the other end of the Port's PCI Express Link use a common (synchronous) Reference Clock source (constant phase relationship)			RW	Yes	0
7	Extended Sync Setting this bit causes the Port to transmit: • 4,096 FTS Ordered-Sets in the ASPM L0s Link PM state, • Followed by a single SKIP Ordered-Set prior to entering the L0 Link PM state, • Finally, transmission of 1,024 TS1 Ordered-Sets in the <i>Recovery</i> state.			RW	Yes	0

**Register 13-29. 78h Link Status and Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
8	Clock Power Management Enable <i>Not supported</i>		RsvdP	No	0
9	Hardware-Autonomous Width Disable <i>Reserved</i>		RsvdP	No	0
10	<i>Reserved</i>	Upstream	RsvdP	No	0
	Link Bandwidth Management Interrupt Enable 0 = Disables interrupt generation 1 = Enables generation of an interrupt, to indicate that the Port's Link Status register <i>Link Bandwidth Management Status</i> bit (Downstream Ports, offset 78h[30]) has been Set	Downstream	RW	Yes	0
11	<i>Reserved</i>	Upstream	RsvdP	No	0
	Link Autonomous Bandwidth Interrupt Enable 0 = Disables interrupt generation 1 = Enables generation of an interrupt, to indicate that the Port's Link Status register <i>Link Autonomous Bandwidth Status</i> bit (Downstream Ports, offset 78h[31]) has been Set	Downstream	RW	Yes	0
15:12	<i>Reserved</i>		RsvdP	No	0h
Link Status					
19:16	Current Link Speed Indicates the negotiated Link speed of the Port's PCI Express Link. The encoding is the binary value of the bit location in the Port's Link Capability 2 register <i>Supported Link Speeds Vector</i> field (offset 94h[7:1]) that corresponds to the current Link speed. <i>For example</i> , a value of 1h indicates that the current Link speed is that which corresponds to bit 1 in the <i>Supported Link Speeds Vector</i> field, which is 2.5 GT/s. If the Link is not up, the value of this field is undefined.		RO	No	1h
25:20	Negotiated Link Width Dependent upon the physical Port configuration. Link width is determined by the negotiated value with the attached Lane/Port. If the Link is not up, the value of this field is undefined. 00_0000b = Link is Down (default) 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are <i>not supported</i> .		RO	No	00_0000b
26	<i>Reserved</i>		RsvdP	No	0
27	<i>Reserved</i>	Upstream	RsvdP	No	0
	Link Training 1 = Downstream Port requested Link training, and the Link training is in-progress or about to start	Downstream	RO	No	0

Register 13-29. 78h Link Status and Control
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
28	Slot Clock Configuration 0 = PEX 8724 uses an independent clock 1 = PEX 8724 uses the same physical Reference Clock that the platform provides on the connector		HwInit	Yes	0
29	Reserved	Upstream	RsvdP	No	0
	Data Link Layer Link Active When Set, and the Port's Link Capability register <i>Data Link Layer Link Active Reporting Capable</i> bit (Downstream Ports, offset 74h[20]) is also Set, indicates the following: - Data Link Layer (DLL) is in the <i>DL_Active</i> state - Link is operational - Flow Control (FC) Initialization has successfully completed	Downstream	RO	Yes	0
30	Reserved	Upstream	RsvdP	No	0
	Link Bandwidth Management Status Set by hardware to indicate that either of the following has occurred, without the Port transitioning through <i>DL_Down</i> status: - Link retraining has completed following a Write of 1 to the Port's Link Control register <i>Retrain Link</i> bit (Downstream Ports, offset 78h[5]) - Hardware has changed Link speed or width, to attempt to correct unreliable Link operation, either through an Link Training and Status State Machine (LTSSM) timeout or higher-level process	Downstream	RW1C	Yes	0
31	Reserved	Upstream	RsvdP	No	0
	Link Autonomous Bandwidth Status Set by hardware to indicate that hardware has autonomously changed Link speed or width, without the Port transitioning through <i>DL_Down</i> status, for reasons other than to attempt to correct unreliable Link operation.	Downstream	RW1C	Yes	0

- a. The Port Receiver must be capable of entering the ASPM L0s Link PM state, regardless of whether the state is disabled.

**Register 13-30. 7Ch Slot Capability
(Downstream Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: For bits [6, 4:0], the default values are shown to be 1 for Downstream Ports, which is true only if the Port is Parallel or Serial Hot Plug-capable; otherwise, the default value is 0. This also applies to bit 17 for Serial Hot Plug Ports. Serial Hot Plug-capable means that the PEX 8724 has detected that an I/O Expander is present. Each Transparent Downstream Port can support one Parallel Hot Plug Controller, which uses the set of on-chip Hot Plug I/O signals designated with suffixes A, B, and C. The assignment of Parallel Hot Plug Controllers, to individual Ports, is programmed in the Parallel Hot Plug Control register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3A4h[7:0, 15:8, and 23:16] for Parallel Hot Plug Controllers A, B, and C, respectively). All other Transparent Downstream Ports support a Serial Hot Plug Controller, which uses signals on an external I²C I/O Expander, for Hot Plug signaling. By default, all Transparent Downstream Ports use a Serial Hot Plug Controller, unless the Port's Power Management Hot Plug User Configuration register Serial Hot Plug Override Parallel Disable bit (offset F70h[19]) is Set. Ports that use a Serial Hot Plug Controller have the register's Port Is Serial Hot Plug Port bit (offset F70h[15]) Set.					
0	Reserved	Upstream	RsvdP	No	0
	Attention Button Present Set if the Port is Parallel or Serial Hot Plug-capable. 0 = Attention Button is not implemented 1 = Attention Button is implemented on this Hot Plug-capable Transparent Downstream Port's slot chassis	Downstream	RO	Yes	1
1	Reserved	Upstream	RsvdP	No	0
	Power Controller Present Enables or disables the Hot Plug Controller on the Hot Plug-capable Transparent Downstream Ports. Set if the Port is Parallel or Serial Hot Plug-capable. 0 = Power Controller is not implemented. The Hot Plug Controller is disabled for that slot and a power-up sequence is not executed. The slot remains in the disabled state. 1 = Power Controller is implemented for the Hot Plug-capable Transparent Downstream Port's slot. The Hot Plug Controller powers up the slot when the Manually operated Retention Latch (MRL) is closed and the Port's Slot Control register Power Controller Control bit (Downstream Ports, offset 80h[10]) is Cleared. Otherwise, if bit 2 (MRL Sensor Present) is disabled (Cleared), the MRL's position has no effect on powering up the slot.	Downstream	RO	Yes	1

**Register 13-30. 7Ch Slot Capability
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
2	Reserved	Upstream	RsvdP	No	0
	MRL Sensor Present Set if the Port is Parallel or Serial Hot Plug-capable. 0 = MRL Sensor is not implemented. MRL position is “Don’t Care” for that slot. 1 = MRL Sensor is implemented on this Hot Plug-capable Transparent Downstream Port’s slot chassis. The PEX 8724 senses whether the MRL is open or closed for a slot. The MRL should be Low for power-on for that slot.	Downstream	RO	Yes	1
3	Reserved	Upstream	RsvdP	No	0
	Attention Indicator Present Set if the Port is Parallel or Serial Hot Plug-capable. 0 = Attention Indicator is not implemented. HP_ATNLED_x# output is not functional on the slot. 1 = Attention Indicator is implemented on this Hot Plug-capable Transparent Downstream Port’s slot chassis. Controls whether the HP_ATNLED_x# output for the slot drives out Active-Low.	Downstream	RO	Yes	1

**Register 13-30. 7Ch Slot Capability
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Reserved	Upstream	RsvdP	No	0
	Power Indicator Present Set if the Port is Parallel or Serial Hot Plug-capable. 0 = Power Indicator is not implemented. HP_PWRLED_x# output is not functional on the slot. 1 = Power Indicator is implemented on the slot chassis of this Hot Plug-capable Transparent Downstream Port. Controls whether the HP_PWRLED_x# output for the slot drives out Active-Low.	Downstream	RO	Yes	1
5	Reserved	Upstream	RsvdP	No	0
	Hot Plug Surprise 0 = No device in the Downstream Port slot is removed from the system without prior notification 1 = Device in the Downstream Port slot can be removed from the system without prior notification	Downstream	RO	Yes	0
6	Reserved	Upstream	RsvdP	No	0
	Hot Plug Capable Set if the Port is Parallel or Serial Hot Plug-capable. 0 = Downstream Port slot is not capable of supporting Hot Plug operations 1 = Downstream Port slot is capable of supporting Hot Plug operations	Downstream	RO	Yes	1
14:7	Reserved	Upstream	RsvdP	No	00h
	Slot Power Limit Value The maximum power supplied by the corresponding PEX 8724 Downstream slot is determined by multiplying the value in this field (expressed in decimal; 25d = 19h) by the field [16:15] (<i>Slot Power Limit Scale</i>) value. Must be implemented if the PCI Express Capability register <i>Slot Implemented</i> bit (offset 68h[24]) is Set (default). Serial EEPROM and/or I ² C Writes to this register or a DLL Up event causes the Downstream Port to send the Set_Slot_Power_Limit Message to the device connected to it, so as to convey the Limit value to the Downstream device's Upstream Port Device Capability register <i>Captured Slot Power Limit Value</i> and <i>Captured Slot Power Limit Scale</i> fields.	Downstream	RO	Yes	19h

**Register 13-30. 7Ch Slot Capability
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
16:15	Reserved	Upstream	RsvdP	No	00b
	Slot Power Limit Scale The maximum power supplied by the corresponding PEX 8724 Downstream slot is determined by multiplying the value in this field by the field [14:7] (<i>Slot Power Limit Value</i>) value. Must be implemented if the PCI Express Capability register <i>Slot Implemented</i> bit (offset 68h[24]) is Set (default). Serial EEPROM and/or I ² C Writes to this register or a DLL Up event causes the Downstream Port to send the Set_Slot_Power_Limit Message to the device connected to it, so as to convey the Limit value to the Downstream device's Upstream Port Device Capability register <i>Captured Slot Power Limit Value</i> and <i>Captured Slot Power Limit Scale</i> fields. 00b = 1.0x 01b = 0.1x 10b = 0.01x 11b = 0.001x	Downstream	RO	Yes	00b
17	Reserved	Upstream; Downstream non-Serial Hot Plug-enabled	RsvdP	No	0
	Electromechanical Interlock Present <i>Not valid for Parallel Hot Plug Ports. Valid for Serial Hot Plug Ports that have an external I²C I/O Expander.</i> 0 = Electromechanical Interlock is not implemented on the chassis for this slot 1 = Electromechanical Interlock is implemented on the chassis for this slot	Downstream Serial Hot Plug-enabled	RO	Yes	1
18	No Command Completed Support Reserved		RsvdP	No	0

**Register 13-30. 7Ch Slot Capability
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default	
31:19	Reserved	Upstream	RsvdP	No	0-0h	
	Physical Slot Number <i>Note: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.</i> Indicates the physical Slot Number attached to this Port. If the PCI Express Capability register <i>Slot Implemented</i> bit (offset 68h[24]) is Set (default), this field must be hardware-initialized to a value that assigns a Slot Number that is unique within the chassis, regardless of the form factor associated with the slot. Must be initialized to 0h for Ports connected to devices that are integrated on the system board. Bit usage depends upon whether the Port is Serial Hot Plug-capable.	Downstream	RO	Yes	0-0h	
	Serial Hot Plug-Capable Downstream Ports					
	Bit(s)					Description/Function
	23:19					Port Numbers 0, 1, 2, 3, 8, 9, 10
	31:24					Loaded from I ² C I/O Expander
	Parallel Hot Plug-Capable Downstream Ports					
	Bit(s)					Description/Function
	23:19					Port Numbers 0, 1, 2, 3, 8, 9, 10
	25:24	I ² C_ADDR0 input binary value (input state 0, Z, or 1, is converted to binary value 00b, 01b, or 10b, respectively)				
31:26	0000_00b					

**Register 13-31. 80h Slot Status and Control
(Downstream Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: Virtual Switch mode only – To inform software of the addition of a bridge in failover while the PEX 8724 is in Virtual Switch mode, the Upstream Port Hot Plug (Device-Specific) functionality can be enabled, by Setting the Port's Power Management Hot Plug User Configuration register Upstream Hot Plug Enable bit (Upstream Port(s), offset F70h[14]). This feature enables the MRL Sensor and Presence Detect-related register bits, and generates the corresponding Hot Plug interrupts, using the Upstream Port(s) copy of this register, which is usually Reserved. The Hot Plug-related bits that can be enabled for the Upstream Port in the Slot Status and Control register are: • MRL Sensor Changed Enable (bit 2) • Presence Detect Changed Enable (bit 3) • Hot Plug Interrupt Enable (bit 5) • MRL Sensor Changed (bit 18) • Presence Detect Changed (bit 19) • MRL Sensor State (bit 21) • Presence Detect State (bit 22)					
Slot Control					
Note: To change the values of the MRL Sensor Changed Enable, Power Fault Detector Enable, and/or Attention Button Pressed Enable bits (bits [2:0], respectively), the corresponding bit(s) in the Port's Slot Capability register (Downstream Ports, offset 7Ch[2:0], respectively) must be Set first.					
0	Reserved	Upstream	RsvdP	No	0
	Attention Button Pressed Enable 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register Power State field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the Power State field bits are both Set), for an Attention Button Pressed event on this Hot Plug-capable Transparent Downstream Port	Offset 7Ch[0]=0 Downstream	RO	No	0
		Offset 7Ch[0]=1 Downstream	RW	Yes	0
1	Reserved	Upstream	RsvdP	No	0
	Power Fault Detector Enable 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register Power State field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the Power State field bits are both Set), for a Power Fault Detected event on this Hot Plug-capable Transparent Downstream Port	Offset 7Ch[1]=0 Downstream	RO	No	0
		Offset 7Ch[1]=1 Downstream	RW	Yes	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description		Ports	Type	Serial EEPROM and I ² C	Default
2	Base Mode <i>Reserved</i>	Offset F70h[14]=0	Upstream	RsvdP	No	0
	Virtual Switch Mode MRL Sensor Changed Enable Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register <i>Power State</i> field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the <i>Power State</i> field bits are both Set), for an MRL Sensor Changed event on this Upstream Port	Offset 7Ch[2]=0 and Offset F70h[14]=1	Upstream	RO	No	0
		Offset 7Ch[2]=1 and Offset F70h[14]=1	Upstream	RW	Yes	0
	Base and Virtual Switch Modes 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register <i>Power State</i> field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the <i>Power State</i> field bits are both Set), for an MRL Sensor Changed event on this Hot Plug-capable Transparent Downstream Port	Offset 7Ch[2]=0	Downstream	RO	No	0
		Offset 7Ch[2]=1	Downstream	RW	Yes	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
3	Base Mode <i>Not valid</i>	Upstream	RsvdP	No	0
	Virtual Switch Mode Presence Detect Changed Enable Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. A Presence Detect Changed event is triggered by either the Upstream Port(s)' SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), or by HP_PRSNT_x# input or Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input (from external I ² C I/O Expander) on this Upstream Port. 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register <i>Power State</i> field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the <i>Power State</i> field bits are both Set), for a Presence Detect Changed event on the Upstream Port	Upstream	RW	Yes	0
	Base and Virtual Switch Modes A Presence Detect Changed event is triggered by either the Downstream Port's SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), or by HP_PRSNT_x# input or Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input (from external I ² C I/O Expander) on this Hot Plug-capable Transparent Downstream Port. 0 = Function is disabled 1 = Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register <i>Power State</i> field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the <i>Power State</i> field bits are both Set), for a Presence Detect Changed event on the Downstream Port	Downstream	RW	Yes	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Reserved	Upstream	RsvdP	No	0
	Command Completed Interrupt Enable 0 = Function is disabled 1 = Enables software notification with an interrupt when a command is completed by the Hot Plug Controller on this Hot Plug-capable Transparent Downstream Port	Downstream	RW	Yes	0
5	Base Mode Reserved	Upstream	RsvdP	No	0
	Virtual Switch Mode Hot Plug Interrupt Enable Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. 0 = Function is disabled 1 = Enables an interrupt on enabled Hot Plug/Link State events for Upstream Port(s)	Upstream	RW	Yes	0
	Base and Virtual Switch Modes 0 = Function is disabled 1 = Enables an interrupt on enabled Hot Plug/Link State events for the Downstream Port	Downstream	RW	Yes	0
7:6	Reserved	Upstream	RsvdP	No	00b
	Attention Indicator Control Controls the Downstream Port's slot's Attention Indicator. Reads return the Hot Plug-capable Transparent Downstream Port's Attention Indicator's current state. Writing a non-zero value triggers a Command Completed event (even if the value written is the same as the existing value). Writing 00b preserves the current value and does not trigger a Command Completed event. 00b = Reserved – Writes are ignored 01b = Turns On indicator to constant On state 10b = Causes indicator to blink 11b = Turns Off indicator	Downstream	RW	Yes	11b

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
9:8	Reserved	Upstream	RsvdP	No	00b
	Power Indicator Control Controls the Hot Plug-capable Transparent Downstream Port's slot's Power Indicator. Reads return this Hot Plug-capable Transparent Downstream Port's Power Indicator's current state. Writing a non-zero value triggers a Command Completed event (even if the value written is the same as the existing value). Writing 00b preserves the current value and does not trigger a Command Completed event. 00b = Reserved – Writes are ignored 01b = Turns On indicator to constant On state 10b = Causes indicator to blink 11b = Turns Off indicator	Downstream	RW	Yes	01b (MRL closed) 11b (MRL open)
10	Reserved	Upstream	RsvdP	No	0
	Power Controller Control Controls the Hot Plug-capable Transparent Downstream Port's slot's Power Controller. 0 = Turns On the Power Controller; requires some delay to be effective 1 = Turns Off the Power Controller	Downstream	RW	Yes	0 (MRL closed) 1 (MRL open)
11	Reserved	Upstream; Downstream non-Serial Hot Plug-enabled	RsvdP	No	0
	Electromechanical Interlock Control <i>Not valid for Parallel Hot Plug Ports. Valid for Serial Hot Plug Ports that have an external I²C I/O Expander.</i> If an Electromechanical Interlock is implemented, writing 1 to this bit causes the state of the interlock to toggle. A Write of 0 to this bit has no effect. A Read of this bit always returns a value of 0.	Downstream Serial Hot Plug-enabled	RW	Yes	0
12	Not valid	Upstream	RsvdP	No	0
	Data Link Layer State Changed Enable Enables software notification with an interrupt if the Port is in the D0 Device PM state (Port's PCI Power Management Status and Control register Power State field, offset 44h[1:0], is Cleared), or with a PME Message if the Port is in the D3hot Device PM state (the Power State field bits are both Set), when the Port's Link Status register Data Link Layer Link Active bit (Downstream Ports, offset 78h[29]) is changed.	Downstream	RW	Yes	0
15:13	Reserved		RsvdP	No	000b

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Slot Status					
16	Reserved	Upstream	RsvdP	No	0
	Attention Button Pressed 1 = Hot Plug-capable Transparent Downstream Port's slot's Attention Button was pressed	Downstream	RW1C	Yes	0
17	Reserved	Upstream	RsvdP	No	0
	Power Fault Detected 1 = Hot Plug-capable Transparent Downstream Port's slot's Power Controller detected a Power Fault at the slot	Downstream	RW1C	Yes	0
18	Base Mode Reserved	Offset F70h[14]=0	Upstream	RsvdP	No
	Virtual Switch Mode MRL Sensor Changed Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. 1 = Hot Plug-capable Transparent Upstream Port's slot detected an MRL Sensor state change	Offset F70h[14]=1	Upstream	RW1C	Yes
	Base and Virtual Switch Modes 1 = Hot Plug-capable Transparent Downstream Port's slot detected an MRL Sensor state change	Downstream	RW1C	Yes	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
19	Base Mode Reserved Offset F70h[14]=0	Upstream	RsvdP	No	0
	Virtual Switch Mode Presence Detect Changed Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. A Presence Detect Changed event is triggered by either the Upstream Port's SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), or by HP_PRSNT_x# or Hot Plug-capable Transparent Downstream Port's Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input (from external I ² C I/O Expander). Write 1 to Clear. 1 = Value reported in bit 22 (<i>Presence Detect State</i>) changed	Upstream	RW1C	Yes	0
	Base and Virtual Switch Modes A Presence Detect Changed event is triggered by either the Downstream Port's SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), or by HP_PRSNT_x# or Hot Plug-capable Transparent Downstream Port's Serial Hot Plug SHP_PRSNT# (SHP_PRSNTx#) input (from external I ² C I/O Expander). Write 1 to Clear. 1 = Value reported in bit 22 (<i>Presence Detect State</i>) changed	Downstream	RW1C	Yes	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default	
20	Reserved	Upstream	RsvdP	No	0	
	Command Completed 1 = Hot Plug-capable Transparent Downstream Port's slot's Hot Plug Controller completed an issued command to: • <i>Attention Indicator Control</i> (field [7:6]) • <i>Power Indicator Control</i> (field [9:8]) • <i>Power Controller Control</i> (bit 10) • <i>Electromechanical Interlock Control</i> (bit 11) (Serial Hot Plug-enabled Ports only)	Downstream	RW1C	Yes	0	
21	Base Mode Reserved	Offset F70h[14]=0	Upstream	RsvdP	No	0
	Virtual Switch Mode MRL Sensor State Enabled for Upstream Port(s) when the Port's Power Management Hot Plug User Configuration register <i>Upstream Hot Plug Enable</i> bit (Upstream Port(s), offset F70h[14]) is Set. Reveals the Hot Plug-capable Transparent Upstream Port's MRL Sensor's current state. 0 = MRL Sensor is closed 1 = MRL Sensor is open	Offset F70h[14]=1	Upstream	RO	No	0
	Base and Virtual Switch Modes Reveals the Hot Plug-capable Transparent Downstream Port's MRL Sensor's current state. 0 = MRL Sensor is closed 1 = MRL Sensor is open		Downstream	RO	No	0

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
22	Base Mode <i>Not valid</i>	Upstream	RsvdP	No	0
	Virtual Switch Mode Presence Detect State For Upstream Port(s) that implement slots, indicates the presence of an adapter in the slot, reflected by the logical OR of the Upstream Port's SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), and, if present, the Port's HP_PRSENT_x# input (de-bounced) or Serial Hot Plug SHP_PRSENT# (SHP_PRSENTx#) input (from external I ² C I/O Expander) for the Serial Hot Plug-enabled Port. Hardwired to 1 when the Port's PCI Express Capability register <i>Slot Implemented</i> bit (offset 68h[24]) value is 0. 0 = Slot is empty, or device is not present 1 = Slot is occupied, or device is present	Upstream	RO	No	0
		Upstream	RO	No	1
	Base and Virtual Switch Modes For Downstream Ports that implement slots, indicates the presence of an adapter in the slot, reflected by the logical OR of the Downstream Port's SerDes Receiver Detect (Physical Layer Receiver Detect Status register <i>Receiver Detected on Lane x</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 200h[31:16])), and, if present, the Port's HP_PRSENT_x# input (de-bounced) or Serial Hot Plug SHP_PRSENT# (SHP_PRSENTx#) input (from external I ² C I/O Expander) for the Serial Hot Plug-enabled Port. Hardwired to 1 when the Port's PCI Express Capability register <i>Slot Implemented</i> bit (offset 68h[24]) value is 0. 0 = Slot is empty, or device is not present 1 = Slot is occupied, or device is present	Downstream	RO	No	0
		Downstream	RO	No	1

**Register 13-31. 80h Slot Status and Control
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
23	Reserved	Upstream; Downstream non-Serial Hot Plug-enabled	RsvdZ	No	0
	Electromechanical Interlock Status <i>Not valid for Parallel Hot Plug Ports. Valid for Serial Hot Plug Ports that have an external I²C I/O Expander.</i> When an Electromechanical Interlock is implemented, indicates the Electromechanical Interlock's current status. 0 = Electromechanical Interlock is disengaged 1 = Electromechanical Interlock is engaged	Downstream Serial Hot Plug-enabled	RW1C	Yes	0
24	Not valid	Upstream	RsvdP	No	0
	Data Link Layer State Changed In response to a Data Link Layer State Changed event, software must read the Port's Link Status register Data Link Layer Link Active bit (Downstream Ports, offset 78h[29]), to determine whether the Link is active before initiating Configuration Requests to the device. 1 = Value reported in the Link Status register Data Link Layer Link Active bit changed	Downstream	RW1C	Yes	0
31:25	Reserved		RsvdZ	No	0-0h

**Register 13-32. 8Ch Device Capability 2
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4:0	<i>Reserved</i>		RsvdP	No	0-0h
5	<i>Reserved</i>	Upstream	RsvdP	No	0
	ARI Forwarding Supported 0 = Alternative Routing-ID Interpretation (ARI) forwarding is not supported 1 = ARI forwarding is supported	Downstream	RO	Yes	1
6	AtomicOp Routing Supported		RO	Yes	1
10:7	<i>Reserved</i>		RsvdP	No	0h
11	LTR Mechanism Supported 0 = Latency Tolerance Reporting (LTR) mechanism is not supported 1 = LTR mechanism is supported		RO	Yes	1
17:12	<i>Reserved</i>		RsvdP	No	0-0h
18	OBFF Mechanism Supported Bit 0 Indicates whether the Optimized Buffer Flush/Fill (OBFF) mechanism is supported. 0 = OBFF is not supported 1 = OBFF is supported, using Message signaling only		RO	Yes	1
31:19	<i>Reserved</i>		RsvdP	No	0-0h

**Register 13-33. 90h Device Status and Control 2
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Device Control 2					
4:0	Reserved		RsvdP	No	0-0h
5	Reserved	Upstream	RsvdP	No	0
	ARI Forwarding Enable 0 = Disabled 1 = Enabled; Downstream Port disables its traditional Device Number field from being forced to 0 when turning a Type 1 Configuration Request into a Type 0 Configuration Request, permitting access to extended functions in an ARI device immediately below the Port	Downstream	RW	Yes	0
6	Reserved		RsvdP	No	0
7	AtomicOp Egress Blocking		RW	Yes	0
9:8	Reserved		RsvdP	No	00b
10	LTR Mechanism Enable 0 = Disables LTR mechanism 1 = Enables LTR mechanism		RW	Yes	0
12:11	Reserved		RsvdP	No	00b
14:13	OBFF Enable 00b = Disabled 01b = Enabled using Message Signaling [Variation A] 10b = Enabled using Message Signaling [Variation B] 11b = Reserved		RW	Yes	00b
15	Reserved		RsvdP	No	0
Device Status 2					
31:16	Reserved		RsvdP	No	0000h

**Register 13-34. 94h Link Capability 2
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
0	Reserved	RsvdP	No	0	
7:1	Supported Link Speeds Vector Supports 2.5, 5.0, and 8.0 GT/s Link speeds. Indicates the supported Link speed(s) of the associated Port. For each bit, a value of 1 indicates that the corresponding Link speed is supported; otherwise, the Link speed is not supported .	RO	Yes	07h	
	Bit(s)				Description/Function
	1				2.5 GT/s (Gen 1)
	2				5.0 GT/s (Gen 2)
	3				8.0 GT/s (Gen 3)
	7:4				Reserved
8	Cross-Link Supported 0 (Ports that support Link speeds of 8.0 GT/s or higher) = Associated Port does not support cross-links 0 (Ports that support only Link speeds of 2.5 or 5.0 GT/s) = Does not provide information regarding the Port’s level of cross-link support 1 = Associated Port supports cross-links Notes: <i>It is recommended that this bit be Set in any Port that supports cross-links, even though doing so is only required for Ports that also support operating at 8.0 GT/s or higher Link speeds.</i> <i>Software should use this bit when referencing fields whose definition depends upon whether the Port supports cross-links.</i>	RO	Yes	1	
31:9	Reserved	RsvdP	No	0-0h	

**Register 13-35. 98h Link Status and Control 2
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Link Control 2					
3:0	Target Link Speed Sets the Target Compliance mode speed when software is using bit 4 (<i>Enter Compliance</i>) to force a Link into Compliance mode.	Upstream	RWS	Yes	3h
	Sets an upper limit on Link operational speed, by restricting the values advertised by the Upstream component in its training sequences. Also Sets the Target Compliance mode speed when software is using bit 4 (<i>Enter Compliance</i>) to force a Link into Compliance mode. The encoding is the binary value of the bit location in the Port's Link Capability 2 register <i>Supported Link Speeds Vector</i> field (offset 94h[7:1]) that corresponds to the target Link speed. For example, a value of 3h indicates that the current Link speed is that which corresponds to bit 3 in the <i>Supported Link Speeds Vector</i> field, which is 8.0 GT/s. If a value written to this field does not correspond to a supported speed (as indicated by the Port's Link Capability register <i>Maximum Link Speed</i> field (offset 74h[3:0]), the value of this field is undefined. The default value of this field is the highest supported Link speed (as reported in the Port's <i>Maximum Link Speed</i> field); however, the default value for all Ports can be globally changed to 2h (Gen 2 maximum speed) or 1h (Gen 1 maximum speed), according to the STRAP_GEN1_GEN2 input state (Z = 2h (Gen 2), or 0 = 1h (Gen 1)). 1h = Gen 1 (2.5 GT/s) 2h = Gen 2 (5.0 GT/s) 3h = Gen 3 (8.0 GT/s)	Downstream	RWS	Yes	3h

Register 13-35. 98h Link Status and Control 2
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Enter Compliance Software is permitted to force a Link to enter Compliance mode at the speed indicated in field [3:0] (<i>Target Link Speed</i>), by Setting this bit in both components on a Link, and then initiating a Hot Reset on the Link. The default value of this bit following a Fundamental Reset is 0.		RWS	Yes	0
5	Hardware-Autonomous Speed Disable Reserved Initial transition to the highest supported common Link speed is not blocked by this bit.		RsvdP	No	0
6	Not valid	Upstream	RsvdP	Yes	0
	Selectable De-Emphasis Applicable only for Gen 2; for Gen 1, this bit should be hardwired to 0. Selects the standard de-emphasis level when the Link is operating at 5.0 GT/s. When the Link is operating at 2.5 GT/s, the Setting of this bit has no effect (de-emphasis at 2.5 GT/s is -3.5 dB). 0 = -6 dB (Link is operating at 5.0 GT/s) 1 = -3.5 dB (Link is operating at 2.5 GT/s)	Downstream	HwInit	Yes	0 (5.0 GT/s) 1 (2.5 GT/s)
9:7	Transmit Margin Intended for debug and compliance testing only.		RWS	Yes	000b
10	Enter Modified Compliance Intended for debug and compliance testing only.		RWS	Yes	0
11	Compliance SOS Applicable only when the Link is operating at 2.5 or 5.0 GT/s. 1 = LTSSM must periodically send SKIP Ordered-Sets between sequences when sending the Compliance Pattern or Modified Compliance Pattern		RWS	Yes	0

Register 13-35. 98h Link Status and Control 2
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:12	Compliance Preset/De-Emphasis Intended for debug and compliance testing only. 8.0 GT/s Data Rate Sets the Transmitter Preset in the <i>Polling.Compliance</i> substate, if the entry occurred due to bit 4 (<i>Enter Compliance</i>) being Set. The encodings are defined in the <i>PCI Express Base r3.0</i> , Section 4.2.3.2, “Encoding of Presets.” Results are undefined if a Reserved preset encoding is used when entering the <i>Polling.Compliance</i> substate in this way. 5.0 GT/s Data Rate Sets the de-emphasis level in the <i>Polling.Compliance</i> substate, if the entry occurred due to bit 4 (<i>Enter Compliance</i>) being Set. 0h = -6 dB (Link is operating at 5.0 GT/s) 1h = -3.5 dB (Link is operating at 2.5 GT/s) 2.5 GT/s Data Rate The programming of this field has no effect.		RWS	Yes	0h
Link Status 2					
16	Current De-Emphasis Level <i>Applicable only for Gen 2; for Gen 1, this bit should be hardwired to 0.</i> Reflects the de-emphasis level when the Link is operating at 5.0 GT/s. 0 = -6 dB (Link is operating at 5.0 GT/s) 1 = -3.5 dB (Link is operating at either 2.5 or 5.0 GT/s)		RO	Yes	0
17	Equalization Complete 1 = Transmitter Equalization procedure has completed		ROS	Yes	0
18	Equalization Phase 1 Successful 1 = Phase 1 of the Transmitter Equalization procedure has successfully completed		ROS	Yes	0
19	Equalization Phase 2 Successful 1 = Phase 2 of the Transmitter Equalization procedure has successfully completed		ROS	Yes	0
20	Equalization Phase 3 Successful 1 = Phase 3 of the Transmitter Equalization procedure has successfully completed		ROS	Yes	0
21	Link Equalization Request Set by hardware, to request the Link equalization process to be performed on the Link.		Differs by Silicon Revision	Yes	0
	Silicon Revision BA		RW1C		
	Silicon Revision CA		RW1CS		
31:22	Reserved		RsvdP	No	0-0h

13.11 Subsystem ID and Subsystem Vendor ID Capability Registers (Offsets A4h – FCh)

This section details the Subsystem ID and Subsystem Vendor ID Capability registers. [Table 13-11](#) defines the register map.

Table 13-11. Subsystem ID and Subsystem Vendor ID Capability Register Map (All Ports)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Reserved</i>	Next Capability Pointer (00h)	SSID/SSVID Capability ID (0Dh) A4h
Subsystem ID	Subsystem Vendor ID A8h	
<i>Reserved</i>		ACh – FCh

Register 13-36. A4h Subsystem Capability (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	SSID/SSVID Capability ID SSID/SSVID registers for the PCI-to-PCI bridge. Program to 0Dh, as required by the <i>PCI-to-PCI Bridge r1.2</i> .	RO	Yes	0Dh
15:8	Next Capability Pointer 00h = This capability is the last capability in the PEX 8724 Port's Capabilities list The PEX 8724 Extended Capabilities list starts at offset 100h.	RO	Yes	00h
31:16	<i>Reserved</i>	RsvdP	No	0000h

Register 13-37. A8h Subsystem ID and Subsystem Vendor ID (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Subsystem Vendor ID The Vendor ID (offset 00h[15:0]) identifies the manufacturer of the PEX 8724, and the Subsystem Vendor ID optionally identifies the board or system vendor. As with the Vendor ID value, the Subsystem Vendor ID value must be a valid PCI-SIG-assigned Vendor ID. The value of this field is usually identical for all PEX 8724 Transparent Ports.	RO	Yes	10B5h
31:16	Subsystem ID The Device ID (offset 00h[31:16]) identifies the PEX 8724, and optionally the Subsystem ID in combination with the Subsystem Vendor ID, uniquely identifies the board or system. The value of this field is usually identical for all PEX 8724 Ports, and is chosen or assigned only by the “owner” of the valid Vendor ID value used for the Subsystem Vendor ID. If the board or system vendor is not a PCI-SIG member, PLX can assign, free of charge, a unique Subsystem ID value, in which case the Subsystem Vendor ID remains the PLX default value, 10B5h.	RO	Yes	8724h

13.12 Device Serial Number Extended Capability Registers (Offsets 100h – 108h)

This section details the Device Serial Number Extended Capability registers. [Table 13-12](#) defines the register map.

Table 13-12. Device Serial Number Extended Capability Register Map (All Ports)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Next Capability Offset (FB4h)	Capability Version (1h)	PCI Express Extended Capability ID (0003h)
Serial Number (Lower DW)		
Serial Number (Upper DW)		

Register 13-38. 100h Device Serial Number Extended Capability Header (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0003h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	0003h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	1h
31:20	Next Capability Offset Program to FB4h, which addresses the Advanced Error Reporting Extended Capability structure.	RO	Yes	FB4h

Register 13-39. 104h Serial Number (Lower DW)
(All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	PCI Express Device Serial Number (1st DW) Lower half of a 64-bit register. Value programmed by Serial EEPROM register initialization. The Serial Number registers contain the IEEE-defined 64-bit Extended Unique Identifier (EUI-64™), of which the upper 24 bits are the Company ID value (00_0EDFh) assigned by the IEEE Registration Authority, and the lower 40 bits are the Extension ID assigned by the identified Company (PLX). The most through least significant bytes are contained within the lowest through highest Byte addresses, respectively.	RO	Yes	B5DF_0E00h

Register 13-40. 108h Serial Number (Upper DW)
(All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
31:0	PCI Express Device Serial Number (2nd DW) Upper half of a 64-bit register. Value programmed by Serial EEPROM register initialization. The Serial Number registers contain the IEEE-defined 64-bit Extended Unique Identifier (EUI-64), of which the upper 24 bits are the Company ID value (00_0EDFh) assigned by the IEEE Registration Authority, and the lower 40 bits are the Extension ID assigned by the identified Company (PLX). The most through least significant bytes are contained within the lowest through highest Byte addresses, respectively.	RO	Yes	BA_8700_10h	CA_8700_10h

13.13 Secondary PCI Express Extended Capability Registers (Offsets 10Ch – 134h)

This section details the Secondary PCI Express Extended Capability registers. Table 13-13 defines the register map.

Table 13-13. Secondary PCI Express Extended Capability Register Map (All Ports)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Next Capability Offset (148h)																Capability Version (1h)																10Ch
																PCI Express Extended Capability ID (0019h)																110h
																Link Control 3																114h
Reserved																Lane Error Status																118h
Lane 1 Equalization Control																Lane 0 Equalization Control																11Ch
Lane 3 Equalization Control																Lane 2 Equalization Control																120h
Lane 5 Equalization Control																Lane 4 Equalization Control																124h
Lane 7 Equalization Control																Lane 6 Equalization Control																128h
Lane 9 or 25 Equalization Control																Lane 8 or 24 Equalization Control																12Ch
Lane 11 or 27 Equalization Control																Lane 10 or 26 Equalization Control																130h
Lane 13 or 29 Equalization Control																Lane 12 or 28 Equalization Control																134h
Lane 15 or 31 Equalization Control																Lane 14 or 30 Equalization Control																

Register 13-41. 10Ch Secondary PCI Express Extended Capability Header (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0019h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	0019h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	1h
31:20	Next Capability Offset Program to 148h, which addresses the Virtual Channel Extended Capability structure.	RO	Yes	148h

**Register 13-42. 110h Link Control 3
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
0	Reserved if the Port's Link Capability 2 register Cross-Link Supported bit (offset 94h[8]) is Cleared.	Upstream	RsvdP	No	0
	Perform Equalization Writable if the Port's Link Capability 2 register Cross-Link Supported bit (offset 94h[8]) is programmed to a non-zero value. 1 = When 1 is written to the Port's Link Control register Retrain Link bit (offset 78h[5]), with the Port's Link Control 2 register Target Link Speed field (offset 98h[3:0]) programmed to 8.0 GT/s, the Downstream Port must perform Transmitter Equalization	Upstream/ Downstream	RW	Yes	0
1	Reserved if the Port's Link Capability 2 register Cross-Link Supported bit (offset 94h[8]) is Cleared.	Upstream	RsvdP	No	0
	Link Equalization Request Interrupt Enable Writable if the Port's Link Capability 2 register Cross-Link Supported bit (offset 94h[8]) is programmed to a non-zero value. 1 = Enables the generation of an interrupt, to indicate that the Port's Link Status 2 register Link Equalization Request bit (offset 98h[21]) has been Set	Upstream/ Downstream	RW	Yes	0
31:2	Reserved		RsvdP	No	0-0h

**Register 13-43. 114h Lane Error Status
(All Ports)**

Bit(s)	Description	Lane x	Type	Serial EEPROM and I ² C	Default
Bits [15:0] are intended for debug purposes only.					
Notes: Except for Ports that are configured as x16, the bits in this register reflect status for all Ports within the Station, rather than the individual Port associated with the Lane.					
The Lane x column is provided to indicate the Lane Number associated with each bit.					
0	Lane x Error Status 0 = No Lane-based error is detected on the Lane 1 = Lane-based error is detected on the Lane	0	RW1CS	No	0
1		1	RW1CS	No	0
2		2	RW1CS	No	0
3		3	RW1CS	No	0
4		4	RW1CS	No	0
5		5	RW1CS	No	0
6		6	RW1CS	No	0
7		7	RW1CS	No	0
8		8 or 24	RW1CS	No	0
9		9 or 25	RW1CS	No	0
10		10 or 26	RW1CS	No	0
11		11 or 27	RW1CS	No	0
12		12 or 28	RW1CS	No	0
13		13 or 29	RW1CS	No	0
14		14 or 30	RW1CS	No	0
15		15 or 31	RW1CS	No	0
31:16	Reserved		RsvdP	No	0000h

**Register 13-44. 118h Lane 0 and 1 Equalization Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 0 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

Register 13-44. 118h Lane 0 and 1 Equalization Control
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

**Register 13-44. 118h Lane 0 and 1 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 1 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

**Register 13-45. 11Ch Lane 2 and 3 Equalization Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 2 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

**Register 13-45. 11Ch Lane 2 and 3 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

**Register 13-45. 11Ch Lane 2 and 3 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 3 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

**Register 13-46. 120h Lane 4 and 5 Equalization Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 4 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

**Register 13-46. 120h Lane 4 and 5 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

**Register 13-46. 120h Lane 4 and 5 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 5 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

**Register 13-47. 124h Lane 6 and 7 Equalization Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 6 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

Register 13-47. 124h Lane 6 and 7 Equalization Control
(All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

**Register 13-47. 124h Lane 6 and 7 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 7 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

Register 13-48. 128h Lane 8 or 24 and 9 or 25 Equalization Control (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 8 or 24 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

**Register 13-48. 128h Lane 8 or 24 and 9 or 25 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

**Register 13-48. 128h Lane 8 or 24 and 9 or 25 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 9 or 25 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

Register 13-49. 12Ch Lane 10 or 26 and 11 or 27 Equalization Control (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 10 or 26 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

**Register 13-49. 12Ch Lane 10 or 26 and 11 or 27 Equalization Control
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

Register 13-49. 12Ch Lane 10 or 26 and 11 or 27 Equalization Control (All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 11 or 27 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

Register 13-50. 130h Lane 12 or 28 and 13 or 29 Equalization Control (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 12 or 28 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

Register 13-50. 130h Lane 12 or 28 and 13 or 29 Equalization Control (All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

Register 13-50. 130h Lane 12 or 28 and 13 or 29 Equalization Control (All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 13 or 29 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

Register 13-51. 134h Lane 14 or 30 and 15 or 31 Equalization Control (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 14 or 30 Equalization Control					
3:0	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
6:4	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
7	Reserved		RsvdP	No	0

Register 13-51. 134h Lane 14 or 30 and 15 or 31 Equalization Control (All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
11:8	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
14:12	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register <i>Transmit Margin</i> field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
15	Reserved		RsvdP	No	0

Register 13-51. 134h Lane 14 or 30 and 15 or 31 Equalization Control (All Ports) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Lane 15 or 31 Equalization Control					
19:16	Reserved	Upstream	RsvdP	No	0h
	Downstream Port Transmitter Preset Preset used by the Transparent Downstream Port's Transmitters during Equalization Phase 1. Can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
22:20	Reserved	Upstream	RsvdP	No	000b
	Downstream Port Receiver Preset Hint Contains the Receiver Preset Hint value received from the Downstream device during Link Equalization, which can be used to adjust the Transparent Downstream Port's Receivers upon the change to Gen 3 speed. Use of the Receiver Preset Hint is optional (per the <i>PCI Express Base r3.0</i>), and the Port does not use the Hint.	Downstream	ROS	Yes	000b
23	Reserved		RsvdP	No	0
27:24	Upstream Port Transmitter Preset For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Transmit Preset value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	Fh
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device must use this value for its Transmitter Preset upon the speed change to Gen 3 speed. For Downstream Ports, the value can be loaded using the User Lane Equalization Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BC8h).	Downstream	ROS	Yes	0h
30:28	Upstream Port Receiver Preset Hint For Upstream Port(s), this field is intended for debug and diagnostics. If cross-link (enabled by default) is disabled for this Port (bit 8 of the Port's Link Control 2 register Transmit Margin field (offset 98h[9:7]) is Cleared), this field contains the Receiver Preset Hint value captured from the Equalization TS2s that are received on the associated Lane during Link Equalization; otherwise, this field is not used.	Upstream	ROS	No	111b
	For Transparent Downstream Ports, the Port transmits this value to the Downstream device, within the equalization TS2s sent during the <i>Recovery.RcvrCfg</i> substate preceding the speed change to Gen 3 speed. The Downstream device can use this value to adjust its Receivers upon the speed change to Gen 3 speed. For Downstream Ports, contains the value sent on the associated Lane during Link Equalization.	Downstream	ROS	Yes	000b
31	Reserved		RsvdP	No	0

13.14 Power Budget Extended Capability Registers (Offsets 138h – 144h)

This section details the Power Budget Extended Capability registers. These registers work differently than the others, especially with respect to serial EEPROM Reads and Writes. *For example*, when writing to Index 5 of the **Power Budget Data** register (Upstream Port(s), offset 140h), write 5 into the **Data Select** register *Data Select* field (Upstream Port(s), offset 13Ch[7:0]), then write the value into the **Power Budget Data** register. Table 13-14 defines the register map.

Table 13-14. Power Budget Extended Capability Register Map (Upstream Port(s))

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets										
Next Capability Offset (10Ch) (Upstream)																Capability Version (1h) (Upstream)										PCI Express Extended Capability ID (0004h) (Upstream)																138h
Reserved (Downstream)																																										
Reserved (Upstream)																								Data Select (Upstream)																		13Ch
Reserved (Downstream)																																										
Power Budget Data (Upstream) Reserved (Downstream)																																140h										
Power Budget Capability (Upstream) Reserved (Downstream)																																										
																																144h										

Register 13-52. 138h Power Budget Extended Capability Header (Upstream Port(s))

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0004h, as required by the <i>PCI Express Base r3.0</i> .	Upstream	RO	Yes	0004h
	Reserved	Downstream	RsvdP	No	0000h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	Upstream	RO	Yes	1h
	Reserved	Downstream	RsvdP	No	0h
31:20	Next Capability Offset Program to 10Ch, which addresses the Secondary PCI Express Extended Capability structure.	Upstream	RO	Yes	10Ch
	Reserved	Downstream	RsvdP	No	000h

**Register 13-53. 13Ch Data Select
(Upstream Port(s))**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
7:0	Data Select Indexes the Power Budget data reported, by way of eight Power Budget Data registers, per Upstream Port, and selects the DWord of Power Budget data that appears in each Power Budget Data register. Index values start at 0, to select the first DWord of Power Budget data; subsequent DWords of Power Budget data are selected by increasing index values 1 to 7.	Upstream	RW	Yes	00h
	Reserved	Downstream	RsvdP	No	00h
31:8	Reserved		RsvdP	No	0000_00h

**Register 13-54. 140h Power Budget Data
(Upstream Port(s))**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: Eight registers, per Upstream Port, can be programmed, through the serial EEPROM, I ² C, and/or SMBus. Each non-zero register value describes the power usage for a different operating condition. Each configuration is selected by writing to the Data Select register Data Select field (Upstream Port(s), offset 13Ch[7:0]).					
7:0	Base Power Eight registers, per Upstream Port. Specifies (in Watts) the base power value in the operating condition. This value must be multiplied by the field [9:8] (Data Scale) contents, to produce the actual power consumption value.	Upstream	RO	Yes	00h
	Reserved	Downstream	RsvdP	No	00h
9:8	Data Scale Specifies the scale to apply to the Base Power value. The device power consumption is determined by multiplying the field [7:0] (Base Power) contents with the value corresponding to the encoding returned by this field. 00b = 1.0x 01b = 0.1x 10b = 0.01x 11b = 0.001x	Upstream	RO	Yes	00b
	Reserved	Downstream	RsvdP	No	00b
12:10	PM Sub-State 000b = Power Management substate of the operating condition being described	Upstream	RO	Yes	000b
	Reserved	Downstream	RsvdP	No	000b

**Register 13-54. 140h Power Budget Data
(Upstream Port(s)) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
14:13	PM State Power Management state of the operating condition being described. 00b = D0 Device PM state 11b = D3 Device PM state All other encodings are Reserved .	Upstream	RO	Yes	00b
	Reserved	Downstream	RsvdP	No	00b
17:15	Type Type of operating condition being described. 000b = PME Auxiliary 001b = Auxiliary 010b = Idle 011b = Sustained 111b = Maximum All other encodings are Reserved .	Upstream	RO	Yes	000b
	Reserved	Downstream	RsvdP	No	000b
20:18	Power Rail Power Rail of the operating condition being described. 000b = Power 12V 001b = Power 3.3V 010b = Power 1.8V 111b = Thermal All other encodings are Reserved .	Upstream	RO	Yes	000b
	Reserved	Downstream	RsvdP	No	000b
31:21	Reserved		RsvdP	No	0-0h

**Register 13-55. 144h Power Budget Capability
(Upstream Port(s))**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
0	System Allocated 1 = Power budget for the device is included within the system power budget	Upstream	HwInit	Yes	1
	Reserved	Downstream	RsvdP	No	0
31:1	Reserved		RsvdP	No	0-0h

13.15 Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)

This section details the Virtual Channel Extended Capability registers, which are duplicated for each Port, as well as the WRR Port Arbitration Table registers. [Table 13-15](#) defines the register map for one Port.

Table 13-15. Virtual Channel Extended Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset (E00h)																Capability Version (1h)				PCI Express Extended Capability ID (0002h)																148h
Port VC Capability 1																																14Ch				
Reserved																								Port VC Capability 2								150h				
Port VC Status (Reserved)																Port VC Control																154h				
VC0 Resource Capability																																158h				
VC0 Resource Control																																15Ch				
VC0 Resource Status																Reserved																160h				
Reserved																																164h – 174h				
WRR Port Arbitration Table Registers (Offsets 178h – 1BCh)																																178h ... 1BCh				

Register 13-56. 148h Virtual Channel Extended Capability Header (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0002h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	0002h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	1h
31:20	Next Capability Offset Next extended capability is the Multicast Extended Capability structure, offset E00h.	RO	No	E00h

**Register 13-57. 14Ch Port VC Capability 1
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
2:0	Extended VC Counter 000b = Port supports only one Virtual Channel, VC0 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .		RO	Yes	000b
3	Reserved		RsvdP	No	0
6:4	Low-Priority Extended VC Counter For Strict Priority arbitration, indicates the quantity of extended Virtual Channels (VCs) (those in addition to VC0) that belong to the Low-Priority VC group for this Port. 000b = For this Port, only VC0 belongs to the Low-Priority VC group 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .		RO	Yes	000b
7	Reserved		RsvdP	No	0
9:8	Reference Clock Reserved		RsvdP	No	00b
11:10	Port Arbitration Table Entry Size 00b = Port Arbitration Table entry size is 1 bit 11b = Port Arbitration Table entry size is 8 bits All other encodings are Reserved .	Upstream	RO	Yes	11b
	Reserved	Downstream	RsvdP	No	00b
31:12	Reserved		RsvdP	No	0000_0h

**Register 13-58. 150h Port VC Capability 2
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	VC Arbitration Capability Indicates the type of VC arbitration supported by the device for the LPVC (Low-Priority Extended VC) group. 00h = Round-Robin (Hardware-Fixed) Arbitration scheme is not supported 01h = <i>Reserved</i> , because the PEX 8724 supports only one VC (Port VC Capability 2 register <i>Low-Priority Extended VC Counter</i> field, offset 14Ch[6:4]=000b)	RO	No	00h
31:8	<i>Reserved</i>	RsvdP	No	0000_00h

**Register 13-59. 154h Port VC Status and Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Port VC Control				
0	Load VC Arbitration Table <i>Not supported</i> Reads always return a value of 0.	RsvdP	No	0
3:1	VC Arbitration Select Selects the Port's VC arbitration type, as per the supported arbitration type indicated by the Port VC Capability 2 register <i>VC Arbitration Capability</i> field (offset 150h[7:0]) value. 000b = Bit 0; Round-Robin (Hardware-Fixed) arbitration scheme All other encodings are <i>Reserved</i> .	RW	Yes	000b
15:4	<i>Reserved</i>	RsvdP	No	000h
Port VC Status				
16	VC Arbitration Table Status <i>Reserved</i>	RsvdP	No	0
31:17	<i>Reserved</i>	RsvdP	No	0-0h

**Register 13-60. 158h VC0 Resource Capability
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
2:0	Port Arbitration Capability <i>Note: The exception to where WRR is given to the Upstream Port(s) is described in Section 8.4.2, “Source Queuing.”</i> Bit 0 = 1 – Non-configurable Round-Robin (Hardware-Fixed) arbitration Bit 1 = 1 – Weighted Round-Robin (WRR) arbitration with 32 Phases Bit 2 = 1 – WRR arbitration with 64 Phases	Upstream	RO	No	100b
		Downstream	RO	No	001b
13:3	Reserved		RsvdP	No	0-0h
14	Advanced Packet Switching		RsvdP	No	0
15	Reject Snoop Transactions Not a PCI Express switch feature; therefore, this bit is Cleared.		RsvdP	No	0
22:16	Maximum Time Slots Reserved		RsvdP	No	000_0000b
23	Reserved		RsvdP	No	0
31:24	Port Arbitration Table Offset Offset of the Port Arbitration Table, as the quantity of DQWords from the Base address of the Virtual Channel Extended Capability structure. (Refer to Section 13.15.1 for further details.) 00h = Port Arbitration Table is not present 03h = Port Arbitration Table is located at register offset 178h All other encodings are Reserved .	Upstream	RO	No	03h
	Reserved	Downstream	RO	No	00h

**Register 13-61. 15Ch VC0 Resource Control
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
0	TC/VC Map Defines Traffic Classes [7:0], respectively, and indicates which TCs are mapped to VC0. Traffic Class 0 (TC0) must be mapped to VC0. By default, Traffic Classes [7:1] are mapped to VC0.		RO	No	1
7:1			RW	Yes	7Fh
15:8	Reserved		RsvdP	No	00h
16	Load Port Arbitration Table Software writes this bit, to load the updated WRR Port Arbitration Table value to the internal logic. Software Read always returns a value of 0.	Upstream	RW	Yes	0
	Reserved	Downstream	RsvdP	No	0
19:17	Port Arbitration Select Selects the Port's Port Arbitration type. Indicates the bit number in the VC0 Resource Capability register <i>Port Arbitration Capability</i> field (offset 158h[2:0]) that corresponds to the arbitration type. Allowed values: • 000b if the Port Arbitration Table is not present • 000b or 010b if the Port Arbitration Table is present 000b = Fair Bandwidth (Hardware-Fixed Arbitration) 010b = Weighted Round-Robin with 64 Phases Note: If software programs other values, hardware ignores the value.	Upstream	RW	Yes	010b
		Downstream	RW	Yes	000b
23:20	Reserved		RsvdP	No	0h
26:24	VC0 ID Defines the Port's VC0 ID code. 000b = VC0 (default; VC0 is the only/default VC) All other encodings are Reserved .		RO	No	000b
30:27	Reserved		RsvdP	No	0h
31	VC0 Enable 0 = Not allowed 1 = Enables the Port's VC0		RO	No	1

**Register 13-62. 160h VC0 Resource Status
(All Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	Reserved		RsvdP	No	0000h
16	Port Arbitration Table Status 0 = Hardware has finished loading values stored in the Port Arbitration Table, after software Sets the VC0 Resource Control register <i>Load Port Arbitration Table</i> bit (offset 15Ch[16]), or if the Port Arbitration Table is not implemented, then this bit is Reserved 1 = Port Arbitration Table entry was written to by software	Upstream	RO	No	0
	Reserved	Downstream	RsvdP	No	0
17	VC0 Negotiation Pending 0 = Port's VC0 negotiation is complete 1 = Port's VC0 initialization is not complete		RO	Yes	1
31:18	Reserved		RsvdP	No	0-0h

13.15.1 WRR Port Arbitration Table Registers (Offsets 178h – 1BCh)

This section details the WRR Port Arbitration Table registers. Port Arbitration Table phases are used to determine Port weighting during “Weighted Round-Robin (WRR) with 64 Phases” Port arbitration.

Table 13-16 defines the register map. The numbers along the top of the register map table indicate the 8-bit fields of each 32-bit register. There are 64 phases, and any active Port Number can go into each Port x Phase x box.

Note: *The Port Arbitration Table is used only by the Upstream Port(s) when Weighted Round-Robin with 64-Phase Port Arbitration is selected, by way of the VC0 Resource Control register Port Arbitration Select field (offset 15Ch[19:17]=010b).*

Table 13-16. WRR Port Arbitration Table Register Map
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8,
accessible through the Management Port)

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets	
Port x, Phase 3	Port x, Phase 2	Port x, Phase 1	Port x, Phase 0	178h
Port x, Phase 7	Port x, Phase 6	Port x, Phase 5	Port x, Phase 4	17Ch
Port x, Phase 11	Port x, Phase 10	Port x, Phase 9	Port x, Phase 8	180h
Port x, Phase 15	Port x, Phase 14	Port x, Phase 13	Port x, Phase 12	184h
Port x, Phase 19	Port x, Phase 18	Port x, Phase 17	Port x, Phase 16	188h
Port x, Phase 23	Port x, Phase 22	Port x, Phase 21	Port x, Phase 20	18Ch
Port x, Phase 27	Port x, Phase 26	Port x, Phase 25	Port x, Phase 24	190h
Port x, Phase 31	Port x, Phase 30	Port x, Phase 29	Port x, Phase 28	194h
Port x, Phase 35	Port x, Phase 34	Port x, Phase 33	Port x, Phase 32	198h
Port x, Phase 39	Port x, Phase 38	Port x, Phase 37	Port x, Phase 36	19Ch
Port x, Phase 43	Port x, Phase 42	Port x, Phase 41	Port x, Phase 40	1A0h
Port x, Phase 47	Port x, Phase 46	Port x, Phase 45	Port x, Phase 44	1A4h
Port x, Phase 51	Port x, Phase 50	Port x, Phase 49	Port x, Phase 48	1A8h
Port x, Phase 55	Port x, Phase 54	Port x, Phase 53	Port x, Phase 52	1ACh
Port x, Phase 59	Port x, Phase 58	Port x, Phase 57	Port x, Phase 56	1B0h
Port x, Phase 63	Port x, Phase 62	Port x, Phase 61	Port x, Phase 60	1B4h
Reserved			1B8h – 1BCh	

Register 13-63. 178h Port Arbitration Table Phases 0 to 3

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 0	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 1	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 2	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 3	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-64. 17Ch Port Arbitration Table Phases 4 to 7
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 4	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 5	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 6	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 7	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-65. 180h Port Arbitration Table Phases 8 to 11

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 8	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 9	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 10	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 11	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-66. 184h Port Arbitration Table Phases 12 to 15
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 12	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 13	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 14	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 15	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-67. 188h Port Arbitration Table Phases 16 to 19

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 16	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 17	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 18	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 19	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-68. 18Ch Port Arbitration Table Phases 20 to 23
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 20	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 21	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 22	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 23	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-69. 190h Port Arbitration Table Phases 24 to 27

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 24	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 25	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 26	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 27	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-70. 194h Port Arbitration Table Phases 28 to 31
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 28	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 29	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 30	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 31	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-71. 198h Port Arbitration Table Phases 32 to 35

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 32	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 33	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 34	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 35	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-72. 19Ch Port Arbitration Table Phases 36 to 39
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 36	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 37	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 38	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 39	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-73. 1A0h Port Arbitration Table Phases 40 to 43

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 40	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 41	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 42	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 43	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-74. 1A4h Port Arbitration Table Phases 44 to 47
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 44	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 45	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 46	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 47	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-75. 1A8h Port Arbitration Table Phases 48 to 51

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 48	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 49	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 50	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 51	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-76. 1ACh Port Arbitration Table Phases 52 to 55
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 52	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 53	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 54	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 55	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-77. 1B0h Port Arbitration Table Phases 56 to 59

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 56	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 57	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 58	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 59	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

Register 13-78. 1B4h Port Arbitration Table Phases 60 to 63
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port Arbitration Table Phase 60	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
7:5	Reserved	RsvdP	No	000b
12:8	Port Arbitration Table Phase 61	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
15:13	Reserved	RsvdP	No	000b
20:16	Port Arbitration Table Phase 62	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
23:21	Reserved	RsvdP	No	000b
28:24	Port Arbitration Table Phase 63	RW	Yes	Value is based upon the Port's Negotiated Link Width and Current Link Speed (offset 78h[25:20 and 19:16], respectively)
31:29	Reserved	RsvdP	No	000b

13.16 Device-Specific Registers (Offsets 1C0h – A74h)

This section details the Device-Specific registers located at offsets 1C0h through A74h. Device-Specific registers are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. [Table 13-17](#) defines the register map.

Other Device-Specific registers are detailed in:

- [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#)
- [Section 13.20, “Device-Specific Registers – Failover and Port Cut-Thru \(Offsets F00h – F20h\)”](#)
- [Section 13.22, “Device-Specific Registers \(Offsets F30h – FB0h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 13-17. Device-Specific Register Map (Offsets 1C0h – A74h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Reserved</i>		1C0h – 1CCh
Device-Specific Registers – Read Pacing (Offsets 1D0h – 1D8h)		1D0h ... 1D8h
Device-Specific Registers – Captured Bus and Device Numbers (Offsets 1DCh – 1E4h)		1DCh ... 1E4h
Device-Specific Registers – Read Pacing (Offset 1E8h)		1E8h
<i>Reserved</i>		1ECh – 1FCh
Device-Specific Registers – Physical Layer (Offsets 200h – 25Ch)		200h ... 25Ch
Device-Specific Registers – Serial EEPROM (Offsets 260h – 270h)		260h ... 270h
<i>Reserved</i>		274h – 28Ch
Device-Specific Registers – I2C and SMBus Slave Interfaces (Offsets 290h – 2FCh)		290h ... 2FCh

**Table 13-17. Device-Specific Register Map
(Offsets 1C0h – A74h) (Cont.)**

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Device-Specific Registers – Port Configuration and Lane Status (Offsets 300h – 46Ch)		300h ... 46Ch
Factory Test Only/Reserved		470h – 568h
Device-Specific Registers – Exerciser (Offsets 56Ch – 570h)		56Ch 570h
Factory Test Only/Reserved		574h – 5FCh
Device-Specific Registers – General-Purpose Input/Output (Offsets 600h – 68Ch)		600h ... 68Ch
Factory Test Only		690h – 69Ch
Device-Specific Registers – PORT_GOOD Selector (Offsets 6A0h – 6A8h)		6A0h ... 6A8h
Reserved		6ACh – 6FCh
Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch)		700h ... 75Ch
Device-Specific Registers – Control (Offsets 760h – 774h)		760h ... 774h
Device-Specific Registers – Error (Offsets 778h – 8FCh)		778h ... 8FCh
Reserved (Base Mode) Device-Specific Registers – Virtual Switch (Offsets 900h – 9E8h), Virtual Switch Mode		900h ... 9E8h
Device-Specific Registers – Ingress Credit Handler (Offsets 9ECh – A2Ch)		9ECh ... A2Ch
Device-Specific Registers – Virtual Switch Debug and GPIO Status and Control (Offsets A30h – A74h)		A30h ... A74h

13.16.1 Device-Specific Registers – Read Pacing (Offsets 1D0h – 1D8h)

Note: *Source Queuing and Read Pacing should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors.*

This section details the Device-Specific Read Pacing registers located at offsets 1D0h through 1D8h. [Table 13-18](#) defines the register map.

Another Device-Specific Read Pacing register is detailed in [Section 13.16.3, “Device-Specific Registers – Read Pacing \(Offset 1E8h\).”](#)

Read Pacing is described, in detail, in [Section 8.5, “Read Pacing.”](#)

Table 13-18. Device-Specific Read Pacing Register Map (Offsets 1D0h – 1D8h) (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Read Pacing Control																																1D0h
Read Pacing Threshold 1																																1D4h
Read Pacing Threshold 2																																1D8h

Register 13-79. 1D0h Read Pacing Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Read Pacing and Source Queuing should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors. Read Pacing must be enabled for Read Spreading to be enabled. (That is, for a Port to have Read Spreading enabled, the corresponding Port bits within this register, for both Read Pacing and Read Spreading, must both be Cleared.) Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Port x Read Pacing Disable 0 = Read Pacing is enabled for this Port 1 = Read Pacing is disabled for this Port	0, 1, 8, 9	RWS	Yes	1
1		2, 3, 10	RWS	Yes	1
2	Factory Test Only		RsvdP	No	0
3	Reserved		ROS	No	1
11:4	Reserved		RsvdP	No	00h
12	Port x Memory Read Outstanding Counter Reset Provides a mechanism for Clearing the Counter when an error condition occurs, with either the device that issued the Read Request, or the device that was to provide the Read Completions, where a System Level Reset will not be issued. If the Counter is not Cleared after an error condition such as this, the threshold will not be accurate. Read returns a value of 0. 0 = Read Outstanding Counter value increments, with each outstanding Read 1 = Resets Read Outstanding Counter	0, 1, 8, 9	RZ	Yes	0
13		2, 3, 10	RZ	Yes	0
14	Factory Test Only		RsvdP	No	0
15	Reserved		RZ	Yes	0

Register 13-79. 1D0h Read Pacing Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (*Cont.*)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
16	Port x Memory Read Spreading Disable 0 = Memory Read Spreading is enabled for this Port 1 = Memory Read Spreading is disabled for this Port	0, 1, 8, 9	RWS	Yes	0
17		2, 3, 10	RWS	Yes	0
18	Factory Test Only		RsvdP	No	0
19	Reserved		ROS	No	0
27:20	Reserved		RsvdP	No	00h
31:28	Maximum Read Response Time 0h = Disabled 1h = 5 ms (default) 2h = 10 ms 3h = 15 ms 4h = 100 ms 5h = 200 ms 6h = 300 ms 7h = 500 ms 8h = 1.0s 9h = 2.0s All other encodings are Reserved .		RWS	Yes	1h

Register 13-80. 1D4h Read Pacing Threshold 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
12:0	x16 Port Memory Read Outstanding Threshold Valid only for Port 0, Station 0. Specified in DWords. Default value of 800h programs the threshold to 8 KB.	RWS	Yes	800h
15:13	Reserved	RsvdP	No	000b
28:16	x8 Port Memory Read Outstanding Threshold Specified in DWords. Default value of 600h programs the threshold to 6 KB.	RWS	Yes	600h
31:29	Reserved	RsvdP	No	000b

Register 13-81. 1D8h Read Pacing Threshold 2

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Descriptions	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
12:0	x4 Port Memory Read Outstanding Threshold Specified in DWords. Default value of 400h programs the threshold to 4 KB.	RWS	Yes	400h
15:13	Reserved	RsvdP	No	000b
28:16	x2 Port Memory Read Outstanding Threshold Specified in DWords. Default value of 200h programs the threshold to 2 KB.	RWS	Yes	200h
31:29	Reserved	RsvdP	No	000b

13.16.2 Device-Specific Registers – Captured Bus and Device Numbers (Offsets 1DCh – 1E4h)

This section details the Device-Specific Captured Bus and Device Numbers register. [Table 13-19](#) defines the register map.

Table 13-19. Device-Specific Captured Bus and Device Numbers Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Captured Bus and Device Numbers		1DCh
Factory Test Only/Reserved		1E0h – 1E4h

Register 13-82. 1DCh Captured Bus and Device Numbers (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
7:0	Captured Bus Number Captured Bus Number value for the Port. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Bus Number. <i>Note: Overwriting the Captured Bus Number value is not recommended.</i>		RW	Yes	00h
12:8	Captured Device Number Captured Device Number value for the Port. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Device Number. <i>Note: Overwriting the Captured Device Number value is not recommended.</i>	Upstream	RW	Yes	0-0h
		Downstream	RO	No	0-0h
30:13	Reserved		RsvdP	No	0-0h
31	Captured BusDev Number Override 1 = Enables the Captured Bus Number and Device Number to be overridden		RW	Yes	0

13.16.3 Device-Specific Registers – Read Pacing (Offset 1E8h)

Note: Source Queuing and Read Pacing should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors.

This section details one of the Device-Specific Read Pacing registers, at offset 1E8h. [Table 13-20](#) defines the register map.

Other Device-Specific Read Pacing registers are detailed in [Section 13.16.1, “Device-Specific Registers – Read Pacing \(Offsets 1D0h – 1D8h\).”](#)

Read Pacing is described, in detail, in [Section 8.5, “Read Pacing.”](#)

Table 13-20. Device-Specific Read Pacing Register Map (Offset 1E8h) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offset
Read Pacing Watermark																																1E8h

Register 13-83. 1E8h Read Pacing Watermark (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12:0	Memory Read Outstanding Upper Watermark	ROS	No	000h
15:13	<i>Reserved</i>	RsvdP	No	000b
28:16	Memory Read Outstanding Lower Watermark	ROS	No	1FFFh
30:29	Memory Read Outstanding Counter Selection	RWS	Yes	00b
31	Memory Read Outstanding Threshold Reset	RW	Yes	0

13.16.4 Device-Specific Registers – Physical Layer (Offsets 200h – 25Ch)

This section details the Device-Specific Physical Layer (PHY) registers located at offsets 200h through 25Ch. [Table 13-21](#) defines the register map.

Other Device-Specific PHY registers are detailed in [Section 13.18.2](#), “Device-Specific Registers – Physical Layer (Offsets B80h – C8Ch).”

Table 13-21. Device-Specific PHY Register Map (Offsets 200h – 25Ch) (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Physical Layer Receiver Detect Status	Physical Layer Electrical Idle for Compliance Mask	200h
Physical Layer Receiver Not Detected Mask	Physical Layer Electrical Idle Detect Mask	204h
Port Control		208h
Physical Layer User Test Pattern, Bytes 0 through 3		20Ch
Physical Layer User Test Pattern, Bytes 4 through 7		210h
Physical Layer User Test Pattern, Bytes 8 through 11		214h
Physical Layer User Test Pattern, Bytes 12 through 15		218h
Physical Layer Command and Status		21Ch
Physical Layer Function Control		220h
Physical Layer Test 0		224h
Physical Layer Test 1	Reserved	228h
Factory Test Only		22Ch
Factory Test Only/Reserved	Physical Layer Port Command	230h
SKIP Ordered-Set Interval		234h
SerDes Quad 0 Diagnostic Data		238h
SerDes Quad 1 Diagnostic Data		23Ch
SerDes Quad 2 Diagnostic Data		240h
SerDes Quad 3 Diagnostic Data		244h
Target Link Width		248h
Factory Test Only		24Ch – 250h
Physical Layer Additional Status		254h
Physical Layer Additional Control		258h
Factory Test Only/Reserved	Physical Layer Error Injection Control	25Ch

Register 13-84. 200h Physical Layer Receiver Detect Status and Electrical Idle for Compliance Mask (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	SerDes x/Lane x	Type	Serial EEPROM and I ² C	Default
Used for specifying the pre-determined quantity of Lanes that detected a Receiver during an LTSSM <i>Detect</i> state, but never detected an exit from Electrical Idle. Because the PEX 8724 has multiple Port configurations, a Mask register is used, rather than programming a specific quantity. When multiple bits are Set, and they correspond to Lanes that belong to the same Port, any of those specified Lanes can cause entry into the LTSSM <i>Polling.Compliance</i> substate. Notes: The SerDes x/Lane x column is provided to indicate the SerDes/Lane Number associated with bits/fields that include “SerDes x” or “Lane x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, fields [23:16 and 7:0], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 00h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
Physical Layer Electrical Idle for Compliance Mask					
Allows masking that specifies which Lanes must never exit Electrical Idle, for entry to the LTSSM <i>Polling.Compliance</i> substate to occur.					
0	Electrical Idle on SerDes x Causes Entry to Compliance State When all the bits are Cleared, the LTSSM <i>Polling.Compliance</i> substate cannot be entered, due to the Electrical Idle condition. 1 = Lane must have detected a Receiver during an LTSSM <i>Detect</i> state, and must not see an exit from Electrical Idle during the <i>Polling.Active</i> substate, to cause entry to the <i>Polling.Compliance</i> substate	0	RWS	Yes	1
1		1	RWS	Yes	1
2		2	RWS	Yes	1
3		3	RWS	Yes	1
4		4	RWS	Yes	1
5		5	RWS	Yes	1
6		6	RWS	Yes	1
7		7	RWS	Yes	1
8		8 or 24	RWS	Yes	1
9		9 or 25	RWS	Yes	1
10		10 or 26	RWS	Yes	1
11		11 or 27	RWS	Yes	1
12		12 or 28	RWS	Yes	1
13		13 or 29	RWS	Yes	1
14		14 or 30	RWS	Yes	1
15		15 or 31	RWS	Yes	1

Register 13-84. 200h Physical Layer Receiver Detect Status and Electrical Idle for Compliance Mask
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	SerDes x/Lane x	Type	Serial EEPROM and I ² C	Default
Physical Layer Receiver Detect Status					
Returns the Receiver's LTSSM <i>Detect</i> state status for all Lanes within the Station.					
16	Receiver Detected on Lane x Reads back as 1 when a Receiver is detected on the Lane.	0	ROS	No	Set by SerDes
17		1	ROS	No	Set by SerDes
18		2	ROS	No	Set by SerDes
19		3	ROS	No	Set by SerDes
20		4	ROS	No	Set by SerDes
21		5	ROS	No	Set by SerDes
22		6	ROS	No	Set by SerDes
23		7	ROS	No	Set by SerDes
24		8 or 24	ROS	No	Set by SerDes
25		9 or 25	ROS	No	Set by SerDes
26		10 or 26	ROS	No	Set by SerDes
27		11 or 27	ROS	No	Set by SerDes
28		12 or 28	ROS	No	Set by SerDes
29		13 or 29	ROS	No	Set by SerDes
30		14 or 30	ROS	No	Set by SerDes
31		15 or 31	ROS	No	Set by SerDes

Register 13-85. 204h Electrical Idle Detect/Receiver Detect Mask
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	SerDes x	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
Masking Electrical Idle detect will not affect the inferred Electrical Idle detection.						
Notes: Use this register with caution.						
The SerDes x column is provided to indicate the Port Number associated with bits/fields that include “SerDes x” in their name.						
Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, and SerDes modules.						
The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, fields [23:16 and 7:0], are Factory Test Only , RsvdP, not serial EEPROM nor I ² C writable, and each has a default value of 00h.						
The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).						
Physical Layer Electrical Idle Detect Mask						
Never Detect Electrical Idle mask. Allows masking of the Electrical Idle Detect function, on a per-SerDes basis. When the bits in this register are Set, the Lane’s Electrical Idle Condition flag does not assert, regardless of the actual presence of Electrical Idle. Masking Electrical Idle detect does not affect the inferred Electrical Idle detection.						
0	SerDes x Mask Electrical Idle Detect 1 = Masks the Electrical Idle Detect for the SerDes, by Station – the corresponding Lane will never detect Electrical Idle	0	RWS	Yes	0	1
1		1	RWS	Yes	0	1
2		2	RWS	Yes	0	1
3		3	RWS	Yes	0	1
4		4	RWS	Yes	0	1
5		5	RWS	Yes	0	1
6		6	RWS	Yes	0	1
7		7	RWS	Yes	0	1
8		8 or 24	RWS	Yes	0	1
9		9 or 25	RWS	Yes	0	1
10		10 or 26	RWS	Yes	0	1
11		11 or 27	RWS	Yes	0	1
12		12 or 28	RWS	Yes	0	1
13		13 or 29	RWS	Yes	0	1
14		14 or 30	RWS	Yes	0	1
15		15 or 31	RWS	Yes	0	

Register 13-85. 204h Electrical Idle Detect/Receiver Detect Mask
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	SerDes x	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
Physical Layer Receiver Not Detected Mask						
Always Detect a Receiver mask. Allows masking of the Receiver Detect function, on a per SerDes basis. When the bits in this register are Set, the PHY functions as if the Lane detected a Receiver, regardless of the actual presence of a Receiver.						
16	SerDes x Mask Receiver Not Detected 1 = Masks the Receiver Not Detected for the SerDes, by Station – the corresponding Lane will always detect a Receiver	0	RWS	Yes	0	
17		1	RWS	Yes	0	
18		2	RWS	Yes	0	
19		3	RWS	Yes	0	
20		4	RWS	Yes	0	
21		5	RWS	Yes	0	
22		6	RWS	Yes	0	
23		7	RWS	Yes	0	
24		8 or 24	RWS	Yes	0	
25		9 or 25	RWS	Yes	0	
26		10 or 26	RWS	Yes	0	
27		11 or 27	RWS	Yes	0	
28		12 or 28	RWS	Yes	0	
29		13 or 29	RWS	Yes	0	
30		14 or 30	RWS	Yes	0	
31		15 or 31	RWS	Yes	0	

Register 13-86. 208h Port Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Used to disable or enable the LTSSM within individual Ports. The Port Control bits are intended to be used in lieu of placing the Port into the <i>Loopback.Active</i> substate as a Loopback Master. These bits enable the test patterns to be transmitted, with or without a device attached at the far end. Recommended usage is as follows: - Set the Port's <i>Disable Port x</i> and <i>Hold Port x Quiet</i> bits (bits [3:0 and 11:8], respectively). - Setting the Port's <i>Disable Port x</i> bit forces the Port into the <i>Detect.Quiet</i> substate. If no device is attached, it is not necessary to Set the Port's <i>Disable Port x</i> bit. - If 5.0 GT/s is needed, also Set the Port's <i>Port Test Pattern x Rate</i> field [25:24]. - If Set, Clear the Port's <i>Disable Port x</i> bit. - Load the UTP registers and enable UTP transmission. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include "Port x" in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Because Port 8 is configured, but not used, Set Port 8, bit 0, to 1. Additionally, Port 8, bits [19, 11, and 3], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Disable Port x While the Port is disabled, Receiver termination is disabled and the SerDes that belong to the disabled Port are placed into the L1 Link PM state.	0, 8	RWS	Yes	0
1	Note: Software should not Set a Port's <i>Disable Port x</i> bit while that Port's Link is in the L2/L3 Ready Link PM state. 0 = Enables the Port's Link Training operation.	1, 9	RWS	Yes	0
2	1 = LTSSM remains in the <i>Detect.Quiet</i> substate on the Port if it is currently in, or returns to, that substate. Unconditionally disables the Port. This is different from an LTSSM <i>Disabled</i> state, in that the Port does not attempt to enter this state. If the Port is idle, it ceases attempting to detect a Receiver. If the Port is up, it immediately returns to the <i>Detect.Quiet</i> substate and remains there. No EIOS is sent, which could force any connected device to the <i>Recovery</i> state, and then to an LTSSM <i>Detect</i> state. The Port remains disabled until its <i>Disable Port x</i> bit is Cleared.	2, 10	RWS	Yes	0
3		3	RWS	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b

Register 13-86. 208h Port Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
8	Hold Port x Quiet 0 = No effect on the LTSSM. 1 = Port remains in the <i>Detect.Quiet</i> substate once it returns there. These bits do not force the LTSSM into the <i>Detect.Quiet</i> substate. (Refer to the <i>Disable Port x</i> bits.) After entering the <i>Detect.Quiet</i> substate, Receiver termination is enabled and the Transmitters are placed into the L0 Link PM state. The Port can now transmit test patterns (UTP), with or without an attached device and without being in the <i>Loopback.Active</i> substate. <i>Note: Use these bits when it is necessary to transmit some data pattern, without first entering the Loopback.Active substate as a Loopback Master.</i>	0, 8	RWS	Yes	0
9		1, 9	RWS	Yes	0
10		2, 10	RWS	Yes	0
11		3	RWS	Yes	0
13:12	Factory Test Only		RsvdP	No	00b
15:14	Reserved		RsvdP	No	00b
16	Port x Bypass UTP Alignment Pattern Must be Set when the following conditions are met: - Link width of the Port being tested is wider than x4, and - Port being tested is a Loopback Master transmitting the User Test Pattern, and - Loopback Slave is in a different clock domain 0 = UTP Transmitter continuously transmits the UTP alignment pattern (K28.5 D3.2 D18.2 D13.2), to allow the corresponding symbol alignment logic to find the symbol boundaries before sending the sync pattern (D3.2 D18.2 D13.2 D17.1) and programmed UTP. 1 = UTP Transmitter sends one UTP alignment pattern (K28.5 D3.2 D18.2 D13.2), one sync pattern (D3.2 D18.2 D13.2 D17.1), and then the programmed UTP.	0, 8	RWS	Yes	0
17		1, 9	RWS	Yes	0
18		2, 10	RWS	Yes	0
19		3	RWS	Yes	0
21:20	Factory Test Only		RsvdP	No	00b
23:22	Reserved		RsvdP	No	00b

Register 13-86. 208h Port Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (*Cont.*)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
25:24	Port Test Pattern x Rate Select Port data rate. 00b = 2.5 GT/s 01b = 5.0 GT/s 10b = 8.0 GT/s 11b = Reserved		RWS	Yes	00b
27:26	Reserved		RsvdP	No	00b
29:28	Test Pattern x Rate Port x Select Selects the Port within the Station to which to apply the test pattern rate. 00b = Port 0, 8 01b = Port 1, 9 10b = Port 2 11b = Port 3		RWS	Yes	00b
31:30	Reserved		RsvdP	No	00b

Register 13-87. 20Ch Physical Layer User Test Pattern, Bytes 0 through 3
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
UTP Bytes 0 through 3. Used for Digital Far-End Loopback testing. A 16-byte test pattern can be written to register offsets 20Ch through 218h. When User Test Pattern (UTP) transmission is enabled, Byte 0 of register offset 20Ch is transmitted first and Byte 3 (Byte 15 of the UTP) of register offset 218h is transmitted last. (Refer to Section 17.2.3, “Digital Loopback Master Mode,” for further details.) In Gen 1 and Gen 2 modes, every byte of the UTP can be a Control or Data character. Illegal Control characters can be specified. In Gen 3 mode, the UTP generator sends a Data block (Sync Header 10b), with the UTP register data as the content. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Byte 0 of the UTP. This is the first byte transferred.	RWS	Yes	00h
15:8	Byte 1 of the UTP.	RWS	Yes	00h
23:16	Byte 2 of the UTP.	RWS	Yes	00h
31:24	Byte 3 of the UTP.	RWS	Yes	00h

Register 13-88. 210h Physical Layer User Test Pattern, Bytes 4 through 7
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
UTP Bytes 4 through 7. Used for Digital Far-End Loopback testing. A 16-byte test pattern can be written to register offsets 20Ch through 218h. When User Test Pattern (UTP) transmission is enabled, Byte 0 of register offset 20Ch is transmitted first and Byte 3 (Byte 15 of the UTP) of register offset 218h is transmitted last. (Refer to Section 17.2.3, “Digital Loopback Master Mode,” for further details.) In Gen 1 and Gen 2 modes, every byte of the UTP can be a Control or Data character. Illegal Control characters can be specified. In Gen 3 mode, the UTP generator sends a Data block (Sync Header 10b), with the UTP register data as the content. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Byte 4 of the UTP. This is the fifth byte transferred.	RWS	Yes	00h
15:8	Byte 5 of the UTP.	RWS	Yes	00h
23:16	Byte 6 of the UTP.	RWS	Yes	00h
31:24	Byte 7 of the UTP.	RWS	Yes	00h

Register 13-89. 214h Physical Layer User Test Pattern, Bytes 8 through 11
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
UTP Bytes 8 through 11. Used for Digital Far-End Loopback testing. A 16-byte test pattern can be written to register offsets 20Ch through 218h. When User Test Pattern (UTP) transmission is enabled, Byte 0 of register offset 20Ch is transmitted first and Byte 3 (Byte 15 of the UTP) of register offset 218h is transmitted last. (Refer to Section 17.2.3, “Digital Loopback Master Mode,” for further details.) In Gen 1 and Gen 2 modes, every byte of the UTP can be a Control or Data character. Illegal Control characters can be specified. In Gen 3 mode, the UTP generator sends a Data block (Sync Header 10b), with the UTP register data as the content. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Byte 8 of the UTP. This is the ninth byte transferred.	RWS	Yes	00h
15:8	Byte 9 of the UTP.	RWS	Yes	00h
23:16	Byte 10 of the UTP.	RWS	Yes	00h
31:24	Byte 11 of the UTP.	RWS	Yes	00h

Register 13-90. 218h Physical Layer User Test Pattern, Bytes 12 through 15
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
UTP Bytes 12 through 15. Used for Digital Far-End Loopback testing. A 16-byte test pattern can be written to register offsets 20Ch through 218h. When User Test Pattern (UTP) transmission is enabled, Byte 0 of register offset 20Ch is transmitted first and Byte 3 (Byte 15 of the UTP) of register offset 218h is transmitted last. (Refer to Section 17.2.3, “Digital Loopback Master Mode,” for further details.) In Gen 1 and Gen 2 modes, every byte of the UTP can be a Control or Data character. Illegal Control characters can be specified. In Gen 3 mode, the UTP generator sends a Data block (Sync Header 10b), with the UTP register data as the content. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Byte 12 of the UTP. This is the thirteenth byte transferred.	RWS	Yes	00h
15:8	Byte 13 of the UTP.	RWS	Yes	00h
23:16	Byte 14 of the UTP.	RWS	Yes	00h
31:24	Byte 15 of the UTP.	RWS	Yes	00h

Register 13-91. 21Ch Physical Layer Command and Status
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides various Command and Status bits for Physical Layer operation.				
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
3:0	Number of Ports Available in the Station Returns the quantity of enabled Ports within this Station, based upon the selected Port configuration.	ROS	No	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
4	Upstream Cross-Link Enable 0 = Disables Upstream cross-link, Upstream Port(s) cannot be connected to other Upstream Port(s) 1 = Enables Upstream cross-link, Upstream Port(s) can be connected to other Upstream Port(s)	ROS	Yes	1
5	Downstream Cross-Link Enable 0 = Disables Downstream cross-link, Downstream Ports cannot be connected to other Downstream Ports 1 = Enables Downstream cross-link, Downstream Ports can be connected to other Downstream Ports	ROS	Yes	1
6	Lane Reversal Disable 0 = Enables Lane reversal on all Ports 1 = Disables Lane reversal on all Ports	ROS	Yes	0
7	Reserved	RsvdP	No	0

Register 13-91. 21Ch Physical Layer Command and Status
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:8	LTSSM Watch Dog Timer Disable Disables the Port's LTSSM Watchdog Timer.	RWS	Yes	00h
31:16	User Test Pattern Control/Data The UTP generators send out a set of 16 bytes of User Programmable data. For Gen 1 and Gen 2 Link speeds (2.5 and 5.0 GT/s, respectively), a k-code bit can be Set for each byte. Bit 16 corresponds to Byte 0 of the User Test Pattern. ... Bit 31 corresponds to Byte 15 of the User Test Pattern. 0 = Corresponding byte of the User Test Pattern is transmitted as a Data character 1 = Corresponding byte of the User Test Pattern is transmitted as a Control character Notes: Use caution when Setting bits within this field, because UTP logic does not check the validity of Control characters. <i>In Gen 3 mode, the UTP generator sends a Data block (Sync Header 10b), with the UTP register data as the content.</i>	RWS	Yes	0000h

Register 13-92. 220h Physical Layer Function Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	SPARE2 or Port x	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
Allows for the configuration of various functions within the PHY logic.						
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name.						
Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.						
The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [27 and 19], are Factory Test Only , RsvdP, not serial EEPROM nor I ² C writable, and each has a default value of 0.						
The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).						
3:0	Configuration Fail Count Used only if the SPARE2 input is Low, to modify Link training for a non-compliant Gen 1 device, that would otherwise fail Link training if the Data Rate bits in the TS are Set. Specifies the quantity of times that Link training fails (in either the LTSSM <i>Polling</i> or <i>Configuration</i> state), after which the Port re-starts Link training (from the <i>Detect</i> state), advertising only a Gen 1 data rate. The initial value of this register is determined by the Strap Configuration register <i>Gen 1 Compatible Disable Status</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 46Ch[10]) value, which defaults to the inverse state of the SPARE2 input. If the bit value is 0 when reset de-asserts, the initial value of this field is 1h; otherwise, the initial value is 0h.	SPARE2=L	RWS	Yes	0h (offset 46Ch[10]=1) 1h (offset 46Ch[10]=0)	
		SPARE2=H	RsvdP	No	0h	
7:4	Reserved		RsvdP	No	0h	
9:8	Detect.Quiet Wait Time Select Code [1:0] Selects the amount of time to wait during the <i>Detect.Quiet</i> substate, before starting the Receiver Detect operation, when a break from Electrical Idle is detected. If Electrical Idle is detected on all Lanes, the wait time is 12 ms. 00b = 0 ms 01b = 4 ms (default) 10b = 8 ms 11b = 12 ms		RWS	Yes	01b	
11:10	Reserved		RsvdP	No	00b	

Register 13-92. 220h Physical Layer Function Control
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	SPARE2 or Port x	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
12	Elastic Buffer Sync Reset Enable 1 = Enable Elastic Buffer Reset upon entry to the <i>Recovery</i> state (if bit 14 (<i>Elastic Buffer Sync Reset on Rx Recovery Enable</i>) is Set) or entry to the <i>Configuration</i> state from the <i>Recovery</i> state for upconfigure		RWS	Yes	1	
13	Symbol-Block Align Sync Reset Enable 1 = Enable Block and Symbol Aligner Sync Reset with Elastic Buffer Sync Reset		RWS	Yes	1	
14	Elastic Buffer Sync Reset on Rx Recovery Enable 1 = Reset Elastic buffer when Receivers are detected entering the <i>Recovery</i> state		RWS	Yes	1	
15	Inferred Electrical Idle Exit Type Selects the method used to detect exit from Electrical Idle, after Electrical Idle has been inferred. 0 = Fast Method – Type 0 Exit mode is used, which uses conventional analog Electrical Idle Exit Detection circuitry 1 = Slow Method – Type 1 Exit mode is used, which uses the Symbol Framer Detection Time Select Code and Inferred Electrical Idle Exit Time Select Code Timers		RWS	Yes	0	
16	Port x Electrical Idle Inference on EIOS Receipt Enable Electrical Idle Inference on Electrical Idle Ordered-Set (EIOS) Receipt enable, for the Port. 0 = Electrical Idle inference is enabled upon EIOS receipt, if the Physical Layer Electrical Idle Detect Mask register <i>SerDes x Mask Electrical Idle Detect</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 204h[15:0]) are Set, for the SerDes associated with the Port 1 = Electrical Idle will be inferred as soon as an EIOS is received on any Lane of the Port	0, 8	RWS	Yes	0	1
17		1, 9	RWS	Yes	0	1
18		2, 10	RWS	Yes	0	1
19		3	RWS	Yes	0	1
21:20	Factory Test Only		RsvdP	No	00b	
23:22	Reserved		RsvdP	No	00b	

Register 13-92. 220h Physical Layer Function Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	SPARE2 or Port x	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
					BA	CA
24	Port x Electrical Idle Inference Disable 0 = Electrical Idle inference is enabled, if the Physical Layer Electrical Idle Detect Mask register <i>SerDes x Mask Electrical Idle Detect</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 204h[15:0]) are Set, for the SerDes associated with the Port. 1 = Port's overall Electrical Idle inference logic is disabled. Electrical Idle inference during the <i>Recovery.Speed</i> substate is not affected and will continue to operate.	0, 8	RWS	Yes	0	
25		1, 9	RWS	Yes	0	
26		2, 10	RWS	Yes	0	
27		3	RWS	Yes	0	
30:28	Factory Test Only		RsvdP	No	000b	
31	Reserved		RsvdP	No	0	

Register 13-93. 224h Physical Layer Test 0
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Provides controls to enable various PHY test modes.					
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bit 3, is Factory Test Only , RsvdP, not serial EEPROM nor I ² C writable, and has a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Port x Timer Test Mode Enable 0 = Normal PHY Timer parameters are used 1 = Millisecond scale timers in the Port’s LTSSM are reduced to microsecond scale	0, 8	RWS	Yes	0
1		1, 9	RWS	Yes	0
2		2, 10	RWS	Yes	0
3		3	RWS	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	SKIP Timer Test Mode Enable 0 = Disables Skip Timer Test mode. 1 = Enables Skip Timer Test mode. SKIP Ordered-Sets are transmitted every 256 symbol times, on all Ports, regardless of the SKIP Ordered-Set Interval register SKIP Ordered-Set Interval in Symbol Times (Gen 1 and Gen 2 Link speeds) or SKIP Ordered-Set Interval in Blocks (Gen 3 Link speeds) field (offset 234h[11:0 or 27:16], respectively) value.		RW	Yes	0
9	Ignore Compliance Receive TCB Ignore the <i>Compliance Receive</i> Training Control Bit (TCB) field in Training Sets. 1 = Causes the PHY to ignore the <i>Compliance Receive</i> TCB when it is Set in received Training Sets		RWS	Yes	0
10	Analog Loopback Enable 0 = PEX 8724 enters Digital Loopback Slave mode if an external device sends at least two consecutive TS1 Ordered-Sets that have the <i>Loopback</i> bit exclusively Set in the TS1 Training Control symbol. The PEX 8724 then loops back data through the Elastic buffer, and decoder and encoder (8b/10b for Gen 1 and Gen 2 Link speeds, or 128b/130b for Gen 3 Link speed). 1 = When operating as a Loopback Slave, the Loopback point of all Ports will be before the Elastic buffer in the Recovered Receive Clock domain.		RWS	Yes	0
11	Factory Test Only		RW	Yes	0
19:12	Factory Test Only		RWS	Yes	00h
31:20	Reserved		RsvdP	No	000h

Register 13-94. 228h Physical Layer Test 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x or SerDes Quad x/ SerDes Pair x	Type	Serial EEPROM and I ² C	Default
Provides controls to enable various PHY test modes. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. The SerDes Quad x/SerDes Pair x column is provided to indicate the SerDes quad and SerDes pair(s) associated with bits/fields that include “SerDes Quad x/SerDes Pair x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [27:24, 19, and 11], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
7:0	Reserved		RWS	No	00h
8	Port x Serial Loopback Path Enable 1 = Enables serial loopback (NES) on the Port	0, 8	RWS	Yes	0
9		1, 9	RWS	Yes	0
10		2, 10	RWS	Yes	0
11		3	RWS	Yes	0
13:12	Factory Test Only		RsvdP	No	00b
15:14	Reserved		RsvdP	No	00b
16	Port x Parallel Loopback Path Enable 1 = Enables the Parallel loopback (LB) path on the Port (analog loopback = 0 LB_PIPE, analog loopback = 1 LB_FEP)	0, 8	RWS	Yes	0
17		1, 9	RWS	Yes	0
18		2, 10	RWS	Yes	0
19		3	RWS	Yes	0
21:20	Factory Test Only		RsvdP	No	00b
23:22	Reserved		RsvdP	No	00b

Register 13-94. 228h Physical Layer Test 1
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Port x or SerDes Quad x/ SerDes Pair x	Type	Serial EEPROM and I ² C	Default
24	SerDes Quad x Pair x User Test Pattern Enable Enables the User Test Pattern generator for SerDes Pair x, by Station. 0 = Disables transmission of the 128-bit test pattern 1 = Enables transmission of the 128-bit test pattern (Physical Layer User Test Pattern, Bytes x through y registers (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 20Ch through 218h)) on SerDes Pair x, by Station, in Digital Far-End Loopback Master mode <i>Notes: UTP transmission should be enabled only when operating as a Loopback Master, or when the LTSSM has returned to the Detect.Quiet substate and the Port's Port Control register Hold Port x Quiet bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 208h[11:8]) is Set.</i> <i>The Port's Port Control register Port x Bypass UTP Alignment Pattern bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 208h[19:16]) must be Set if the Link width of the Port under test is wider than x4, the Port under test is a Loopback Master, and the Loopback Slave is in a different clock domain.</i>	SerDes Quad 0 (Lower Pair) SerDes[0, 1]	RWS	Yes	0
25		SerDes Quad 0 (Upper Pair) SerDes[2, 3]	RWS	Yes	0
26		SerDes Quad 1 (Lower Pair) SerDes[4, 5]	RWS	Yes	0
27		SerDes Quad 1 (Upper Pair) SerDes[6, 7]	RWS	Yes	0
28		SerDes Quad 2 (Lower Pair) SerDes[8, 9]/[24, 25]	RWS	Yes	0
29		SerDes Quad 2 (Upper Pair) SerDes[10, 11]/[26, 27]	RWS	Yes	0
30		SerDes Quad 3 (Lower Pair) SerDes[12, 13]/[28, 29]	RWS	Yes	0
31		SerDes Quad 3 (Upper Pair) SerDes[14, 15]/[30, 31]	RWS	Yes	0

Register 13-95. 230h Physical Layer Port Command

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides the Loopback, Scrambler Disable, and Compliance Receive commands, and Ready as Loopback Master status, for each Port. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [15:12], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
0	Port 0, 8 Loopback Command 0 = Port is not enabled to go to the <i>Loopback</i> Master state. 1 = Port attempts to enter the <i>Loopback</i> state as a Loopback Master. If this bit is Set before the <i>Configuration</i> state is reached, the <i>Configuration.Linkwidth.Start</i> to <i>Loopback</i> path is used. If this bit is Set later, the <i>Recovery.Idle</i> to <i>Loopback</i> path is used.	RWS	Yes	0
1	Port 0, 8 Scrambler Disable Command When Set, unconditionally disables the data scramblers on the Port's Lane(s), and causes the <i>Scramble Disable</i> Training Control Bit to be Set in the transmitted Training Sets. There is one bit for each Port in the associated Station. If a serial EEPROM load Sets this bit, the scrambler is disabled in a <i>Configuration.Complete</i> substate. If software Sets this bit when the Link is in the Up state, hardware disables its scrambler without executing the Link Training protocol. This scrambler disable takes effect after the Link passes through <i>Configuration</i> again. The Upstream/Downstream device scrambler will not be disabled. 0 = Port's scrambler is enabled 1 = Port's scrambler is disabled	RWS	Yes	0
2	Port 0, 8 Compliance Receive Command 0 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is not Set during the <i>Polling.Active</i> nor <i>Loopback.Entry</i> substate 1 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is Set during the <i>Polling.Active</i> or <i>Loopback.Entry</i> substate	RWS	Yes	0
3	Port 0, 8 Ready as Loopback Master Link Training and Status State Machine (LTSSM) established Loopback as a Master for the Port. 0 = Port is not in Loopback Master mode. 1 = Port has successfully transitioned to the <i>Loopback.Active</i> substate as a Loopback Master. The LTSSM remains in this substate, until bit 0 (<i>Port 0, 8 Loopback Command</i>) is Cleared. Cleared when the PEX 8724 exits the <i>Loopback.Active</i> substate.	ROS	No	0

Register 13-95. 230h Physical Layer Port Command
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	Port 1, 9 Loopback Command 0 = Port is not enabled to go to the <i>Loopback</i> Master state. 1 = Port attempts to enter the <i>Loopback</i> state as a Loopback Master. If this bit is Set before the <i>Configuration</i> state is reached, the <i>Configuration.Linkwidth.Start</i> to <i>Loopback</i> path is used. If this bit is Set later, the <i>Recovery.Idle</i> to <i>Loopback</i> path is used.	ROS	Yes	0
5	Port 1, 9 Scrambler Disable Command When Set, unconditionally disables the data scramblers on the Port's Lane(s), and causes the <i>Scramble Disable</i> Training Control Bit to be Set in the transmitted Training Sets. There is one bit for each Port in the associated station. If a serial EEPROM load Sets this bit, the scrambler is disabled in a <i>Configuration.Complete</i> substate. If software Sets this bit when the Link is in the Up state, hardware disables its scrambler without executing the Link Training protocol. This scrambler disable takes effect after the Link passes through <i>Configuration</i> again. The Upstream/Downstream device scrambler will not be disabled. 0 = Port's scrambler is enabled 1 = Port's scrambler is disabled	ROS	Yes	0
6	Port 1, 9 Compliance Receive Command 0 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is not Set during the <i>Polling.Active</i> nor <i>Loopback.Entry</i> substate 1 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is Set during the <i>Polling.Active</i> or <i>Loopback.Entry</i> substate	ROS	Yes	0
7	Port 1, 9 Ready as Loopback Master LTSSM established Loopback as a Master for the Port. 0 = Port is not in Loopback Master mode. 1 = Port has successfully transitioned to the <i>Loopback.Active</i> substate as a Loopback Master. The LTSSM remains in this substate, until bit 4 (<i>Port 1, 9 Loopback Command</i>) is Cleared. Cleared when the PEX 8724 exits the <i>Loopback.Active</i> substate.	ROS	No	0

Register 13-95. 230h Physical Layer Port Command**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Port 2, 10 Loopback Command 0 = Port is not enabled to go to the <i>Loopback</i> Master state. 1 = Port attempts to enter the <i>Loopback</i> state as a Loopback Master. If this bit is Set before the <i>Configuration</i> state is reached, the <i>Configuration.Linkwidth.Start</i> to <i>Loopback</i> path is used. If this bit is Set later, the <i>Recovery.Idle</i> to <i>Loopback</i> path is used.	ROS	Yes	0
9	Port 2, 10 Scrambler Disable Command When Set, unconditionally disables the data scramblers on the Port's Lane(s), and causes the <i>Scramble Disable</i> Training Control Bit to be Set in the transmitted Training Sets. There is one bit for each Port in the associated station. If a serial EEPROM load Sets this bit, the scrambler is disabled in a <i>Configuration.Complete</i> substate. If software Sets this bit when the Link is in the Up state, hardware disables its scrambler without executing the Link Training protocol. This scrambler disable takes effect after the Link passes through <i>Configuration</i> again. The Upstream/Downstream device scrambler will not be disabled. 0 = Port's scrambler is enabled 1 = Port's scrambler is disabled	ROS	Yes	0
10	Port 2, 10 Compliance Receive Command 0 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is not Set during the <i>Polling.Active</i> nor <i>Loopback.Entry</i> substate 1 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is Set during the <i>Polling.Active</i> or <i>Loopback.Entry</i> substate	ROS	Yes	0
11	Port 2, 10 Ready as Loopback Master LTSSM established Loopback as a Master for the Port. 0 = Port is not in Loopback Master mode. 1 = Port has successfully transitioned to the <i>Loopback.Active</i> substate as a Loopback Master. The LTSSM remains in this substate, until bit 8 (<i>Port 2, 10 Loopback Command</i>) is Cleared. Cleared when the PEX 8724 exits the <i>Loopback.Active</i> substate.	ROS	No	0

Register 13-95. 230h Physical Layer Port Command
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Port 3 Loopback Command 0 = Port is not enabled to go to the <i>Loopback</i> Master state. 1 = Port attempts to enter the <i>Loopback</i> state as a Loopback Master. If this bit is Set before the <i>Configuration</i> state is reached, the <i>Configuration.Linkwidth.Start</i> to <i>Loopback</i> path is used. If this bit is Set later, the <i>Recovery.Idle</i> to <i>Loopback</i> path is used.	ROS	Yes	0
13	Port 3 Scrambler Disable Command When Set, unconditionally disables the data scramblers on the Port's Lane(s), and causes the <i>Scramble Disable</i> Training Control Bit to be Set in the transmitted Training Sets. There is one bit for each Port in the associated station. If a serial EEPROM load Sets this bit, the scrambler is disabled in a <i>Configuration.Complete</i> substate. If software Sets this bit when the Link is in the Up state, hardware disables its scrambler without executing the Link Training protocol. This scrambler disable takes effect after the Link passes through <i>Configuration</i> again. The Upstream/Downstream device scrambler will not be disabled. 0 = Port's scrambler is enabled 1 = Port's scrambler is disabled	ROS	Yes	0
14	Port 3 Compliance Receive Command 0 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is not Set during the <i>Polling.Active</i> nor <i>Loopback.Entry</i> substate 1 = When the Port transmits TS1 Ordered-Sets, the <i>Compliance Receive</i> Training Control Bit within these Ordered-Sets is Set during the <i>Polling.Active</i> or <i>Loopback.Entry</i> substate	ROS	Yes	0
15	Port 3 Ready as Loopback Master LTSSM established Loopback as a Master for the Port. 0 = Port is not in Loopback Master mode. 1 = Port has successfully transitioned to the <i>Loopback.Active</i> substate as a Loopback Master. The LTSSM remains in this substate, until bit 12 (<i>Port 3 Loopback Command</i>) is Cleared. Cleared when the PEX 8724 exits the <i>Loopback.Active</i> substate.	ROS	No	0
23:16	Factory Test Only	ROS	No	00h
31:24	Reserved	ROS	No	00h

Register 13-96. 234h SKIP Ordered-Set Interval

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Used to adjust the distance between SKIP Ordered-Sets.					
Notes: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.</i>					
<i>The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.</i>					
<i>The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).</i>					
11:0	SKIP Ordered-Set Interval in Symbol Times Specifies the SKIP Ordered-Set interval (in symbol times). When a value of 000h is written, SKIP Ordered-Set transmission is disabled. 000h = SKIP Ordered-Set transmission is disabled 49Ch = Minimum interval (1,180 symbol times) 602h = Maximum interval (1,538 symbol times) Note: <i>A high value (such as FFFh) can cause the Link to fail.</i>	Gen 1 and Gen 2 Link speeds (2.5 and 5.0 GT/s, respectively)	RWS	Yes	49Ch
	Reserved Note: <i>For Gen 3 Link speed, refer to field [27:16] (SKIP Ordered-Set Interval in Blocks).</i>	Gen 3 Link speed (8.0 GT/s)	RsvdP	No	000h
15:12	Reserved		RsvdP	No	0h
27:16	SKIP Ordered-Set Interval in Blocks Specifies the SKIP Ordered-Set interval (in block times). When a value of 000h is written, SKIP Ordered-Set transmission is disabled. 000h = SKIP Ordered-Set transmission is disabled 173h = Minimum interval (371 block times) 177h = Maximum interval (375 block times) Note: <i>A high value (such as FFFh) can cause the Link to fail.</i>	Gen 3 Link speed (8.0 GT/s)	RWS	Yes	173h
	Reserved Note: <i>For Gen 1 and Gen 2 Link speeds, refer to field [11:0] (SKIP Ordered-Set Interval in Symbol Times).</i>	Gen 1 and Gen 2 Link speeds (2.5 and 5.0 GT/s, respectively)	RsvdP	No	000h
31:28	Reserved		RsvdP	No	0h

Register 13-97. 238h SerDes Quad 0 Diagnostic Data
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
There are four SerDes Quad x Diagnostic Data (Diagnostic Data) registers, one per SerDes quad, at offsets 238h through 244h, in each Station. The Diagnostic Data register contents reflect the performance of the SerDes that are selected by the registers' <i>SerDes Diagnostic Data Select</i> bits (field [25:24]). For example, if offset 23Ch[25:24] is programmed to 10b, the information in that Diagnostic Data register is for SerDes 2 of Quad 1 of that Station (SerDes 6 in Station 0; the Station 1 bits for that particular SerDes quad and SerDes are Factory Test Only/Reserved). Used to retrieve Diagnostic Test results for SerDes[0-3]. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	UTP Expected Data When User Test Pattern (UTP) is enabled, if an error is detected in the received UTP data, this field returns the expected data.	ROS	No	00h
15:8	UTP Actual Data When UTP is enabled, if an error is detected in the received UTP data, this field returns the actual data that was received.	ROS	No	00h
23:16	UTP Error Counter Valid only when bit [31] (<i>UTP Sync</i>) is Set. Receiver Detected flags. Returns the quantity of errors detected by the UTP Data Checker. The Error Counter saturates at 255, and is Cleared when UTP is disabled. To Clear the Counter, disable UTP mode by Clearing the SerDes Quad 0 Physical Layer Test 1 register <i>SerDes Quad x Pair x User Test Pattern Enable</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[25:24]).	ROS	No	00h
25:24	SerDes Diagnostic Data Select Used to select the SerDes (SerDes[0-3]) to which the diagnostic data in this SerDes quad pertains. Status selection code for the fields representing RO bits [23:0] of this register. The binary code represents a status selection for one of the SerDes Quad 0 Lanes. The test results for physical device Lanes [0-3] are selected with corresponding binary codes from 0-3. Note: To obtain diagnostic data on all SerDes within the quad, run a test, then cycle these bits.	RW	Yes	00b
29:26	Reserved	ROS	No	0h
30	-UTP Counter 0 = Field [23:16] (<i>UTP Error Counter</i>) is the UTP Error Counter (diagnostic data is from the UTP Data Checkers) 1 = Reserved	ROS	No	0
31	UTP Sync UTP Sync indicator. 0 = User Test Pattern sync is not detected by this Port 1 = User Test Pattern sync is detected by this Port	ROS	No	0

Register 13-98. 23Ch SerDes Quad 1 Diagnostic Data

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
There are four SerDes Quad x Diagnostic Data (Diagnostic Data) registers, one per SerDes quad, at offsets 238h through 244h, in each Station. The contents of the Diagnostic Data registers reflect the performance of the SerDes that are selected by the registers' <i>SerDes Diagnostic Data Select</i> bits (field [25:24]). <i>For example</i>, if field [25:24] of this register is programmed to 10b, the information in that Diagnostic Data register is for SerDes 2 of Quad 1 of that Station (SerDes 6 in Station 0; the Station 1 bits for that particular SerDes quad and SerDes are Factory Test Only/Reserved). Used to retrieve Diagnostic Test results for SerDes[4-7]. Notes: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes.</i> <i>The Port 8 bits are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.</i> <i>The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).</i>				
7:0	UTP Expected Data When User Test Pattern (UTP) is enabled, if an error is detected in the received UTP data, this field returns the expected data.	ROS	No	00h
15:8	UTP Actual Data When UTP is enabled, if an error is detected in the received UTP data, this field returns the actual data that was received.	ROS	No	00h
23:16	UTP Error Counter Valid only when bit [31] (<i>UTP Sync</i>) is Set. Receiver Detected flags. Returns the quantity of errors detected by the UTP Data Checker. The Error Counter saturates at 255, and is Cleared when UTP is disabled. To Clear the Counter, disable UTP mode by Clearing the SerDes Quad 1 Physical Layer Test 1 register <i>SerDes Quad x Pair x User Test Pattern Enable</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h [27:26]).	ROS	No	00h
25:24	SerDes Diagnostic Data Select Used to select the SerDes (SerDes[4-7]) to which the diagnostic data in this SerDes quad pertains. Status selection code for the fields representing RO bits [23:0] of this register. The binary code represents a status selection for one of the SerDes Quad 1 Lanes. The test results for physical device Lanes [4-7] are selected with corresponding binary codes from 0-3. Note: <i>To obtain diagnostic data on all SerDes within the quad, run a test, then cycle these bits.</i>	RW	Yes	00b
29:26	Reserved	ROS	No	0h
30	-UTP Counter 0 = Field [23:16] (<i>UTP Error Counter</i>) is the UTP Error Counter (diagnostic data is from the UTP Data Checkers) 1 = Reserved	ROS	No	0
31	UTP Sync UTP Sync indicator. 0 = User Test Pattern sync is not detected by this Port 1 = User Test Pattern sync is detected by this Port	ROS	No	0

Register 13-99. 240h SerDes Quad 2 Diagnostic Data
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
There are four SerDes Quad x Diagnostic Data (Diagnostic Data) registers, one per SerDes quad, at offsets 238h through 244h, in each Station. The contents of the Diagnostic Data registers reflect the performance of the SerDes that are selected by the registers' <i>SerDes Diagnostic Data Select</i> bits (field [25:24]). <i>For example</i>, if offset 23Ch[25:24] is programmed to 10b, the information in that Diagnostic Data register is for SerDes 2 of Quad 1 of that Station (SerDes 6 in Station 0; the Station 1 bits for that particular SerDes quad and SerDes are Factory Test Only/Reserved). Used to retrieve Diagnostic Test results for SerDes[8-11]/[24-27]. Notes: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes.</i> <i>The Port 0 bits are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.</i> <i>The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).</i>				
7:0	UTP Expected Data When User Test Pattern (UTP) is enabled, if an error is detected in the received UTP data, this field returns the expected data.	ROS	No	00h
15:8	UTP Actual Data When UTP is enabled, if an error is detected in the received UTP data, this field returns the actual data that was received.	ROS	No	00h
23:16	UTP Error Counter Valid only when bit [31] (<i>UTP Sync</i>) is Set. Receiver Detected flags. Returns the quantity of errors detected by the UTP Data Checker. The Error Counter saturates at 255, and is Cleared when UTP is disabled. To Clear the Counter, disable UTP mode by Clearing the SerDes Quad 2 Physical Layer Test 1 register <i>SerDes Quad x Pair x User Test Pattern Enable</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[29:28]).	ROS	No	00h

Register 13-99. 240h SerDes Quad 2 Diagnostic Data

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
25:24	SerDes Diagnostic Data Select Used to select the SerDes (SerDes[8-11]/[24-27]) to which the diagnostic data in this SerDes quad pertains. Status selection code for the fields representing RO bits [23:0] of this register. The binary code represents a status selection for one of the SerDes Quad 2 Lanes. The test results for physical device Lanes [8-11]/[24-27] are selected with corresponding binary codes from 0-3. <i>Note: To obtain diagnostic data on all SerDes within the quad, run a test, then cycle these bits.</i>	RW	Yes	00b
29:26	Reserved	ROS	No	0h
30	-UTP Counter 0 = Field [23:16] (UTP Error Counter) is the UTP Error Counter (diagnostic data is from the UTP Data Checkers) 1 = Reserved	ROS	No	0
31	UTP Sync UTP Sync indicator. 0 = User Test Pattern sync is not detected by this Port 1 = User Test Pattern sync is detected by this Port	ROS	No	0

Register 13-100. 244h SerDes Quad 3 Diagnostic Data
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
There are four SerDes Quad x Diagnostic Data (Diagnostic Data) registers, one per SerDes quad, at offsets 238h through 244h, in each Station. The contents of the Diagnostic Data registers reflect the performance of the SerDes that are selected by the registers' <i>SerDes Diagnostic Data Select</i> bits (field [25:24]). <i>For example</i>, if offset 23Ch[25:24] is programmed to 10b, the information in that Diagnostic Data register is for SerDes 2 of Quad 1 of that Station (SerDes 6 in Station 0; the Station 1 bits for that particular SerDes quad and SerDes are Factory Test Only/Reserved). Used to retrieve Diagnostic Test results for SerDes[12-15]/[28-31]. Notes: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes.</i> <i>The Port 0 bits are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.</i> <i>The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).</i>				
7:0	UTP Expected Data When User Test Pattern (UTP) is enabled, if an error is detected in the received UTP data, this field returns the expected data.	ROS	No	00h
15:8	UTP Actual Data When UTP is enabled, if an error is detected in the received UTP data, this field returns the actual data that was received.	ROS	No	00h
23:16	UTP Error Counter Valid only when bit [31] (<i>UTP Sync</i>) is Set. Receiver Detected flags. Returns the quantity of errors detected by the UTP Data Checker. The Error Counter saturates at 255, and is Cleared when UTP is disabled. To Clear the Counter, disable UTP mode by Clearing the SerDes Quad 3 Physical Layer Test 1 register <i>SerDes Quad x Pair x User Test Pattern Enable</i> bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[31:30]).	ROS	No	00h

Register 13-100. 244h SerDes Quad 3 Diagnostic Data

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
25:24	SerDes Diagnostic Data Select Used to select the SerDes (SerDes[12-15]/[28-31]) to which the diagnostic data in this SerDes quad pertains. Status selection code for the fields representing RO bits [23:0] of this register. The binary code represents a status selection for one of the SerDes Quad 3 Lanes. The test results for physical device Lanes [12-15]/[28-31] are selected with corresponding binary codes from 0-3. <i>Note: To obtain diagnostic data on all SerDes within the quad, run a test, then cycle these bits.</i>	RW	Yes	00b
29:26	Reserved	ROS	No	0h
30	-UTP Counter 0 = Field [23:16] (UTP Error Counter) is the UTP Error Counter (diagnostic data is from the UTP Data Checker) 1 = Reserved	ROS	No	0
31	UTP Sync UTP Sync indicator. 0 = User Test Pattern sync is not detected by this Port 1 = User Test Pattern sync is detected by this Port	ROS	No	0

Register 13-101. 248h Target Link Width
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides software with the ability to direct Link Width Up/Down configuration. The Port's <i>Target Link Width</i> field in this register is initially loaded with the Port's <i>Negotiated Link Width</i> (offset 78h[25:20]), the first time that the Port's LTSSM transitions from the <i>Configuration</i> state to the L0 Link PM state. The value remains 0 for Ports that are not enabled. Software can be used to retrain a Link to a wider or narrower width than the current width, <i>such as</i> to conserve power when maximum bandwidth is not required. Devices advertise such support, by transmitting the appropriate value for the Data Rate Identifier symbol within TS2 Ordered-Sets. If the Port's <i>Port x Upconfigure Capability Received</i> bit is Set (indicating that during the previous Link training, the Port received Upconfigure Capability notification from the connected device), software can cause the Port to: • Upconfigure the Link to a previously negotiated Link width, –or– • Downconfigure the Link to a narrower width by writing the needed Target Link Width value to this register, followed by Setting the Port's <i>Link Control</i> register <i>Retrain Link</i> bit (offset 78h[5]). If the Target Link Width is not equal to the current Link width, the LTSSM transitions from <i>Recovery</i> to <i>Configuration</i>, then renegotiates the Link width. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [31, 26:24], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
4:0	Port 0, 8 Target Link Width Provides the capability to allow software to cause Link retraining to a wider or narrower width than the current width, <i>such as</i> to conserve power when maximum bandwidth is not required. Devices advertise such support, by transmitting the appropriate value for the Data Rate Identifier symbol within TS2 Ordered-Sets. Written with the Port's Target Link width, to support Link Width Upconfiguration. The initial value of this field is the Port's <i>Negotiated Link Width</i> (offset 78h[25:20]).	RWS	No	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
6:5	Reserved	RsvdP	No	00b
7	Port 0, 8 Upconfigure Capability Received Set during Link training, if the Port received an Upconfigure Capability notification from the connected device. 0 = Device connected to the Port does not indicate that it is capable of Link Width Upconfiguration. 1 = Device connected to the Port indicates that it is capable of Link Width Upconfiguration. Software can cause the Port to either upconfigure the Link to a previously negotiated Link width, or downconfigure the Link to a narrower width, by writing the needed Target Link Width value to this register, followed by Setting the Port's <i>Link Control</i> register <i>Retrain Link</i> bit (offset 78h[5]).	ROS	No	0

Register 13-101. 248h Target Link Width**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
11:8	Port 1, 9 Target Link Width Provides the capability to allow software to cause Link retraining to a wider or narrower width than the current width, <i>such as</i> to conserve power when maximum bandwidth is not required. Devices advertise such support, by transmitting the appropriate value for the Data Rate Identifier symbol within TS2 Ordered-Sets. Written with the Port's Target Link width, to support Link Width Upconfiguration. The initial value of this field is the Port's Negotiated Link Width (offset 78h[25:20]).	RWS	No	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
14:12	Reserved	RsvdP	No	000b
15	Port 1, 9 Upconfigure Capability Received Set during Link training, if the Port received an Upconfigure Capability notification from the connected device. 0 = Device connected to the Port does not indicate that it is capable of Link Width Upconfiguration. 1 = Device connected to the Port indicates that it is capable of Link Width Upconfiguration. Software can cause the Port to either upconfigure the Link to a previously negotiated Link width, or downconfigure the Link to a narrower width, by writing the needed Target Link Width value to this register, followed by Setting the Port's Link Control register Retrain Link bit (offset 78h[5]).	ROS	No	0
18:16	Port 2, 10 Target Link Width Provides the capability to allow software to cause Link retraining to a wider or narrower width than the current width, <i>such as</i> to conserve power when maximum bandwidth is not required. Devices advertise such support, by transmitting the appropriate value for the Data Rate Identifier symbol within TS2 Ordered-Sets. Written with the Port's Target Link width, to support Link Width Upconfiguration. The initial value of this field is the Port's Negotiated Link Width (offset 78h[25:20]).	RWS	No	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
22:19	Reserved	RsvdP	No	0h
23	Port 2, 10 Upconfigure Capability Received Set during Link training, if the Port received an Upconfigure Capability notification from the connected device. 0 = Device connected to the Port does not indicate that it is capable of Link Width Upconfiguration. 1 = Device connected to the Port indicates that it is capable of Link Width Upconfiguration. Software can cause the Port to either upconfigure the Link to a previously negotiated Link width, or downconfigure the Link to a narrower width, by writing the needed Target Link Width value to this register, followed by Setting the Port's Link Control register Retrain Link bit (offset 78h[5]).	ROS	No	0

Register 13-101. 248h Target Link Width
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
26:24	Port 3 Target Link Width Provides the capability to allow software to cause Link retraining to a wider or narrower width than the current width, <i>such as</i> to conserve power when maximum bandwidth is not required. Devices advertise such support, by transmitting the appropriate value for the Data Rate Identifier symbol within TS2 Ordered-Sets. Written with the Target Link width for the Port, to support Link Width Upconfiguration. The initial value of this field is the Port's Negotiated Link Width (offset 78h[25:20]).	RWS	No	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
30:27	Reserved	RsvdP	No	0h
31	Port 3 Upconfigure Capability Received Set during Link training, if the Port received an Upconfigure Capability notification from the connected device. 0 = Device connected to the Port does not indicate that it is capable of Link Width Upconfiguration. 1 = Device connected to the Port indicates that it is capable of Link Width Upconfiguration. Software can cause the Port to either upconfigure the Link to a previously negotiated Link width, or downconfigure the Link to a narrower width, by writing the needed Target Link Width value to this register, followed by Setting the Port's Link Control register <i>Retrain Link</i> bit (offset 78h[5]).	ROS	No	0

Register 13-102. 254h Physical Layer Additional Status

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [27 and 3], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each have a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Port x Loopback Master Entry Failed 1 = Port failed to enter the Loopback state as a Loopback Master, and abandoned the attempt by returning the LTSSM to the Detect state Note: If this bit and the Port's Physical Layer Port Command register Port x Ready as Loopback Master bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 230h[3, 7, 11, and/or 15]) are both Set, the Loopback state was entered after the initial failure from the Configuration state.	0, 8	RW1C	Yes	0
1		1, 9	RW1C	Yes	0
2		2, 10	RW1C	Yes	0
3		3	RW1C	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
9:8	Link Speed Reflects the STRAP_GEN1_GEN2 input state, which indicates the Link speed.		ROS	No	STRAP_GEN1_GEN2 input state
	Field Value	STRAP_GEN1_GEN2 Input State			
	00b	0			
	01b	Z			
	10b	1			
	11b	–			Reserved
11:10	Reserved		RsvdP	No	00b

Register 13-102. 254h Physical Layer Additional Status
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
14:12	Link Speed Status Port Select Selects the Port within the Station for which to return Link speed status, in field [9:8] (<i>Link Speed</i>). 000b = Port 0 of the Station 001b = Port 1 of the Station 010b = Port 2 of the Station 011b = Port 3 of the Station (Station 0 only) All other encodings are Reserved . <i>Note:</i> If the Port Number indicated is not enabled within the Station, the associated encoding is Reserved .		RW	No	000b
15	Reserved		RsvdP	No	0
22:16	Received Modified Compliance Error Counter Returns the value received in the Modified Compliance Pattern's <i>ERR</i> field, for the Lane(s) selected by field [27:24] (<i>Received Modified Compliance Lane Select</i>).		RO	No	0-0h
23	Received Modified Compliance Pattern Lock 1 = Modified Compliance Pattern has been locked onto by the Lane(s) selected by field [27:24] (<i>Received Modified Compliance Lane Select</i>).		RO	No	0
27:24	Received Modified Compliance Lane Select Selects which Lane within the Station receives a Modified Compliance error. Status is read from the <i>Received Modified Compliance Pattern Lock</i> bit and <i>Received Modified Compliance Error Counter</i> field (bits [23 and 22:16], respectively). <i>Note:</i> If a Lane does not exist in that position, the bit is Factory Test Only, Reserved (RsvdP) , and not serial EEPROM nor I ² C writable. 0h = No Lane received an error 1h = Lane 0 within the Station received a Modified Compliance error 2h = Lane 1 within the Station received a Modified Compliance error 3h = Lane 2 within the Station received a Modified Compliance error 4h = Lane 3 within the Station received a Modified Compliance error All other encodings are Reserved .		RW	Yes	0h
31:28	Reserved		RsvdP	No	0h

Register 13-103. 258h Physical Layer Additional Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [11 and 3], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each have a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Port x External Loopback Enable 1 = Allows the Port to reach Link Up status, when receiving its own Training Sets during Link training. It is necessary to Set this bit when a Port’s Receivers are directly connected, externally, to its Transmitters.	0, 8	RWS	No	0
1		1, 9	RWS	No	0
2		2, 10	RWS	No	0
3		3	RWS	No	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x 2nd Receiver Detect Disable 1 = Prevents the Port from waiting 12 ms and Retrying the Receiver Detect operation, when Receivers are detected on a subset of Lanes after the first Receiver Detect operation. Instead, the LTSSM progresses to the <i>Polling</i> state, and operates only on the Lanes that detected Receivers.	0, 8	RWS	Yes	0
9		1, 9	RWS	Yes	0
10		2, 10	RWS	Yes	0
11		3	RWS	Yes	0
13:12	Factory Test Only		RsvdP	No	00b
15:14	Reserved		RsvdP	No	00b
16	Factory Test Only		RWS	Yes	0
17	Reserved		RsvdP	No	0
19:18	Tx Idle Minimum Time Select 00b = 20 ns (default) 01b = 30 ns 10b = 40 ns 11b = 80 ns		RWS	Yes	00b
31:20	Reserved		RsvdP	No	000h

Register 13-104. 25Ch Physical Layer Error Injection Control
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides 1-bit error injection control. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, fields [15:14 and 13:12], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 00b. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
1:0	Port 0, 8 1-Bit Error/DLLP Error Injection 00b = Disabled (default) 01b = 1-bit error every 1 μ s 10b = 1-bit error every 1 ms 11b = Error is in the next DLLP (this value self-Clears after an error is injected in a DLLP)	RW	Yes	00b
3:2	Port 0, 8 TLP Error Injection 00b = Disabled (default) 01b = Every TLP 10b = Every other TLP 11b = Every third TLP	RW	Yes	00b
5:4	Port 1, 9 1-Bit Error/DLLP Error Injection 00b = Disabled (default) 01b = 1-bit error every 1 μ s 10b = 1-bit error every 1 ms 11b = Error is in the next DLLP (this value self-Clears after an error is injected in a DLLP)	RW	Yes	00b
7:6	Port 1, 9 TLP Error Injection 00b = Disabled (default) 01b = Every TLP 10b = Every other TLP 11b = Every third TLP	RW	Yes	00b

Register 13-104. 25Ch Physical Layer Error Injection Control**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
9:8	Port 2, 10 1-Bit Error/DLLP Error Injection 00b = Disabled (default) 01b = 1-bit error every 1 μ s 10b = 1-bit error every 1 ms 11b = Error is in the next DLLP (this value self-Clears after an error is injected in a DLLP)	RW	Yes	00b
11:10	Port 2, 10 TLP Error Injection 00b = Disabled (default) 01b = Every TLP 10b = Every other TLP 11b = Every third TLP	RW	Yes	00b
13:12	Port 3 1-Bit Error/DLLP Error Injection 00b = Disabled (default) 01b = 1-bit error every 1 μ s 10b = 1-bit error every 1 ms 11b = Error is in the next DLLP (this value self-Clears after an error is injected in a DLLP)	RW	Yes	00b
15:14	Port 3 TLP Error Injection 00b = Disabled (default) 01b = Every TLP 10b = Every other TLP 11b = Every third TLP	RW	Yes	00b
23:16	Factory Test Only	RW	Yes	00h
31:24	Reserved	RsvdP	No	00h

13.16.5 Device-Specific Registers – Serial EEPROM (Offsets 260h – 270h)

This section details the Device-Specific Serial EEPROM registers. [Table 13-22](#) defines the register map.

Table 13-22. Device-Specific Serial EEPROM Register Map
(Base mode – Port 0; Virtual Switch mode – Port 0,
accessible through the Management Port)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Status Data from Serial EEPROM								Serial EEPROM Status								Serial EEPROM Control																260h
Serial EEPROM Buffer																																264h
Serial EEPROM Clock Frequency																																268h
Expansion ROM Base Address																Reserved								Serial EEPROM 3rd Address Byte								26Ch
Serial EEPROM CRC Value																																270h

Register 13-105. 260h Serial EEPROM Status and Control
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM	Default
Serial EEPROM Control				
12:0	EepBlkAddr Serial EEPROM Block Address for 32 KB.	RW	Yes	000h
15:13	EepCmd[2:0] Commands to the Serial EEPROM Controller. 000b = <i>Reserved</i> (default) 001b = Data from bits [31:24] (Status Data from Serial EEPROM register) is written to the serial EEPROM's internal Status register 010b = Write four bytes of data from the EepBuf into the memory location pointed to by field [12:0] (EepBlkAddr) 011b = Read four bytes of data from the memory location pointed to by field [12:0] (EepBlkAddr) into the EepBuf 100b = Reset Write Enable latch 101b = Data from the serial EEPROM's internal Status register is written to bits [31:24] (Status Data from Serial EEPROM register) 110b = Set Write Enable latch 111b = <i>Reserved</i> Note: For value of 001b, only bits [31, 27:26] can be written into the serial EEPROM's internal Status register.	RW	Yes	000b

Register 13-105. 260h Serial EEPROM Status and Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM	Default
Serial EEPROM Status				
17:16	EepPrsnt[1:0] Serial EEPROM Present status. 00b = Not present 01b = Serial EEPROM is present – Validation Signature is verified 10b = Reserved 11b = Serial EEPROM is present – Validation Signature is not verified	ROS	No	00b
18	EepCmdStatus Serial EEPROM Command status. 0 = Serial EEPROM Command is complete 1 = Serial EEPROM Command is not complete	ROS	No	0
19	EepCRCErr CRC error detected status. 0 = Calculated CRC value matches the stored CRC value 1 = Calculated CRC value does not match the stored CRC value	RW1CS	No	0
20	EepBlkAddr Upper Bit Serial EEPROM Block Address upper bit 13. Extends the serial EEPROM to 64 KB.	RW	Yes	0
21	EepAddrWidth Override 0 = Field [23:22] (<i>EepAddrWidth</i>) is RO 1 = Field [23:22] (<i>EepAddrWidth</i>) is software-writable	RW	Yes	0
23:22	EepAddrWidth Serial EEPROM Address width. If the addressing width cannot be determined, 00b is returned. A non-zero value is reported only if the Validation Signature (5Ah) is successfully read from the first serial EEPROM location. Usually ROS; however, this field is RW if bit 21 (<i>EepAddrWidth Override</i>) is Set. 00b = Undetermined 01b = 1 byte 10b = 2 bytes 11b = 3 bytes	Bit 21 (<i>EepAddrWidth Override</i>) = 0 ROS	No	00b
		Bit 21 (<i>EepAddrWidth Override</i>) = 1 RW	No	00b

Register 13-105. 260h Serial EEPROM Status and Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM	Default					
Status Data from Serial EEPROM ^a									
24	EepRdy Serial EEPROM RDY#. 0 = Serial EEPROM is ready to transmit data 1 = Write cycle is in-progress	RW	Yes	0					
25	EepWen Serial EEPROM Write enable. 0 = Serial EEPROM Write is disabled 1 = Serial EEPROM Write is enabled	RW	Yes	0					
27:26	EepBp[1:0] Serial EEPROM Block-Write Protect bits. Block Protection options protect the top ¼, top ½, or the entire serial EEPROM. PEX 8724 Configuration data is stored in the lower addresses; therefore, when using Block Protection, the entire serial EEPROM should be protected with BP[1:0]=11b.		RW	Yes	00b				
	BP[1:0]	Level				Array Addresses Protected, by Device Size			
						8 KB	16 KB	32 KB	64 KB
	00b	0				None	None	None	None
	01b	1 (top ¼)				1800h – 1FFFh	3000h – 3FFFh	6000h – 7FFFh	–
	10b	2 (top ½)				1000h – 1FFFh	2000h – 3FFFh	4000h – 7FFFh	–
	11b	3 (All)	0000h – 1FFFh	0000h – 3FFFh	0000h – 7FFFh	–			

Register 13-105. 260h Serial EEPROM Status and Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM	Default
30:28	EepWrStatus Serial EEPROM Write status. Value is 000b when the serial EEPROM is not in an internal Write cycle. <i>Note: The definition of this field varies among serial EEPROM manufacturers. Reads of the serial EEPROM's internal Status register can return a value of 000b or 111b, depending upon the serial EEPROM that is used.</i>	RW	No	000b
31	EepWpen Serial EEPROM Write Protect enable. Overrides the internal serial EEPROM Write Protect WP# input and enables/disables Writes to the Serial EEPROM Status register (bits [23:16] of this register): - When WP#=H or this bit is Cleared, and bit 25 (<i>EepWen</i>) is Set, the Serial EEPROM Status register is writable - When WP#=L and this bit is Set, or bit 25 (<i>EepWen</i>) is Cleared, the Serial EEPROM Status register is write-protected <i>Notes: If the internal serial EEPROM Write Protect WP# input is Low, after software Sets the <i>EepWen</i> bit to write-protect the Serial EEPROM Status register, the <i>EepWen</i> value cannot be Cleared, nor can the <i>EepBp[1:0]</i> field be Cleared to disable Block Protection, until the WP# input is High.</i> <i>Not implemented in certain serial EEPROMs. Refer to the serial EEPROM manufacturer's data sheet.</i>	RW	Yes	0

a. Within the serial EEPROM's internal **Status** register, only bits [31, 27:26] can be written.

Register 13-106. 264h Serial EEPROM Buffer**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	EepBuf Serial EEPROM RW buffer. Read/Write command to the Serial EEPROM Control register (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 260h) results in a 4-byte Read/Write from/to the serial EEPROM device.	RW	Yes	0000_0000h

Register 13-107. 268h Serial EEPROM Clock Frequency**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	EepFreq[2:0] Serial EEPROM clock (EE_SK) frequency control. 000b = 1 MHz (default) 001b = 1.98 MHz 010b = 5 MHz 011b = 9.62 MHz 100b = 12.5 MHz 101b = 15.6 MHz 110b = 17.9 MHz 111b = Reserved	RW	Yes	000b
7:3	Reserved	RsvdP	No	0-0h
10:8	EepCsStHld[2:0] CS to SCLK setup and hold timing to the serial EEPROM, between EE_CS# active and EE_SK active, and between EE_SK inactive and EE_CS# inactive. Time increases in ½ EE_SK Clock cycle increments, from a minimum of ½, to a maximum of 4. 000b = ½ EE_SK clocks (minimum time) 001b = 1 EE_SK clock (+ ½ clock) 010b = 1 ½ EE_SK clocks 011b = 2 EE_SK clocks 100b = 2 ½ EE_SK clocks 101b = 3 EE_SK clocks 110b = 3 ½ EE_SK clocks 111b = 4 EE_SK clocks (maximum time)	RW	Yes	000b
15:11	Reserved	RsvdP	No	0-0h
16	Expansion ROM Size Indicates the NT Port Expansion ROM size. 0 = 16 KB 1 = 32 KB	RW	Yes	0
31:17	Reserved	RsvdP	No	0-0h

Register 13-108. 26Ch Serial EEPROM 3rd Address Byte
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Serial EEPROM 3rd Address Byte Contains the upper Byte address when reading from or writing to a 3-address byte serial EEPROM.	RW	Yes	00h
15:8	Reserved	RsvdP	No	00h
31:16	Expansion ROM Base Address Indicates the NT Port Expansion ROM Base address within the serial EEPROM. The value is dependent upon the Serial EEPROM Clock Frequency register <i>Expansion ROM Size</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[16]) value. Maps to serial EEPROM Byte Address bits [23:8]. Address bits [7:0] = 00h so that Expansion ROM code within the serial EEPROM always begins on a 256-byte boundary. 0020h = Default Base address in serial EEPROM for a 16-KB NT Port Expansion ROM (<i>Expansion ROM Size</i> bit is Cleared) is 2000h (8 KB). The serial EEPROM size must be at least 32 KB. 0040h = Default Base address in serial EEPROM for a 32-KB NT Port Expansion ROM (<i>Expansion ROM Size</i> bit is Set) is 4000h (16 KB). The serial EEPROM size must be at least 64 KB.	RW	Yes	0020h

Register 13-109. 270h Serial EEPROM CRC Value
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	EepCRCValue Calculated serial EEPROM CRC value. There is an optional CRC checking at the end of serial EEPROM auto load. Bit 7 of the second byte (after the 5Ah Validation Signature) determines whether CRC checking is to be performed. The calculated CRC value is always placed into this register after serial EEPROM auto load. If CRC checking is enabled, and the stored 32-bit CRC value read from the end the serial EEPROM content does not match the calculated value, an Uncorrectable Internal error is triggered at Port 0 of Station 0.	RO	Yes	FFFF_FFFFh

13.16.6 Device-Specific Registers – I²C and SMBus Slave Interfaces (Offsets 290h – 2FCh)

This section details the Device-Specific I²C and SMBus Slave Interface registers. [Table 13-23](#) defines the register map.

The I²C and SMBus Slave Interfaces are described, in detail, in [Chapter 7, “I²C/SMBus Slave Interface Operation.”](#)

Table 13-23. Device-Specific I²C and SMBus Slave Interfaces Register Map
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Factory Test Only</i>		290h
I2C Configuration		294h
<i>Factory Test Only</i>		298h – 2C4h
SMBus Configuration		2C8h
<i>Reserved</i>		2CCh – 2FCh

Register 13-110. 294h I²C Configuration
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default								
6:0	Slave Address Default I²C Slave/SMBus Device address, which is dependent upon the I2C_ADDR0 input state. Other values can be programmed as well, by serial EEPROM or Memory Write. Additionally, Silicon Revision CA supports access to this field, through Configuration Requests. <table><thead><tr><th>I2C_ADDR0 Input State</th><th>I²C Slave/SMBus Device Address</th></tr></thead><tbody><tr><td>0</td><td>0111_000b</td></tr><tr><td>Z</td><td>0111_001b</td></tr><tr><td>1</td><td>0111_010b</td></tr></tbody></table> <i>Note: The I²C Slave/SMBus Device address must not be changed by an I²C/SMBus Write command.</i>	I2C_ADDR0 Input State	I ² C Slave/SMBus Device Address	0	0111_000b	Z	0111_001b	1	0111_010b	RWS	Yes	Refer to Table
I2C_ADDR0 Input State	I ² C Slave/SMBus Device Address											
0	0111_000b											
Z	0111_001b											
1	0111_010b											
9:7	Reserved	RsvdP	No	000b								
10	Factory Test Only	RWS	Yes	0								
30:11	Reserved	RWS	Yes	0-0h								
31	Factory Test Only	ROS	No	0								

Register 13-111. 2C8h SMBus Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	SMBus Enable Initially loaded from the STRAP_I2C_SMBUS_EN input state. The value can later be changed by software, serial EEPROM, I ² C, and/or SMBus. 0 = Disables SMBus for device configuration (I ² C mode is enabled) 1 = Enables SMBus without ARP for device configuration (SMBus mode is enabled)	RWS	Yes	0 (STRAP_I2C_SMBUS_EN=Z or 1) 1 (STRAP_I2C_SMBUS_EN=0)
7:1	SMBus Device Address Defined by the Address Resolution Protocol (ARP) Master, if ARP is enabled. Defaults to 0011_100b if ARP is disabled (STRAP_I2C_SMBUS_EN=0 and bit 8 (<i>ARP Disable</i>) is Set). When ARP is disabled, the serial EEPROM should simultaneously: - Set bits [15, 11, and 10 (<i>SMBus Parameter Reload</i>, <i>AR Flag</i>, and <i>AV Flag</i>, respectively), and, - Program this field to the SMBus address for this device (<i>such as</i> to 70h, to match the I²C Slave Address (I2C Configuration register <i>Slave Address</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 294h[6:0])) default value) An 8 th bit is transmitted immediately following the 7-bit address, to signal whether the instruction is Write (0) or Read (1). Therefore, when accessing the default I ² C Slave address 0111_000b , the transmitted byte value is 70h for a Write instruction and 71h for a Read instruction.	RWS	Yes	0000_000b (STRAP_I2C_SMBUS_ENSTRAP_I2C_SMBUS_EN=Z or 1) 0011_100b (STRAP_I2C_SMBUS_EN=0 and bit 8 (<i>ARP Disable</i>) is Set)
8	ARP Disable 0 = Device under test is able to respond to ARP commands 1 = Device under test is unable to respond to ARP commands	RWS	Yes	1
9	PEC Check Disable 0 = Enable PEC checking on all packets 1 = Disables Packet Error Checks (PECs) checking on all packets; packets with the wrong PECs are accepted	RWS	Yes	0
10	AV Flag Address Valid (AV) flag. Set, by default, when ARP is disabled (STRAP_I2C_SMBUS_EN=0 and bit 8 (<i>ARP Disable</i>) is Set).	RWS	Yes	1
11	AR Flag Address Resolved (AR) flag.	RWS	Yes	0

Register 13-111. 2C8h SMBus Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
13:12	UDID Address Type Unique Device Identifier (UDID) Address type. 00b = ARP is disabled (STRAP_I2C_SMBUS_EN=0 and bit 8 (ARP Disable) is Set) 10b = ARP is enabled (STRAP_I2C_SMBUS_EN=Z or 1) All other encodings are Reserved .	RWS	Yes	00b (STRAP_I2C_SMBUS_EN=0 and bit 8 (ARP Disable) is Set) 10b (STRAP_I2C_SMBUS_EN=Z or 1)
14	UDID PEC Support 1 = Sets the <i>PEC Support</i> bit in the UDID	RWS	Yes	1
15	SMBus Parameter Reload Software must Set this bit if bits [10, 8, and/or 7:1] (<i>AV Flag</i> , <i>ARP Disable</i> , and/or <i>SMBus Device Address</i> , respectively) at runtime. Effective only when bit 28 (<i>SMBus Command In-Progress</i>) is Cleared.	ROS	No	0
23:16	UDID Vendor-Specific ID Defines the MSB of the UDID Vendor-Specific ID. Bits [23:20] of this field are defined by the <i>I2C_ADDR0</i> input state, which provides the following ID values. 0 = 7000_0000h Z = B000_0000h 1 = D000_0000h	RWS	Yes	Defined by I2C_ADDR0 input state
26:24	UDID Version UDID version defined for the <i>SMBus v2.0</i> .	RWS	Yes	001b
27	Factory Test Only	RWS	Yes	0
28	SMBus Command In-Progress 0 = SMBus state machine is idle 1 = SMBus state machine is active (not idle)	ROS	No	0
29	PEC Check Failed 0 = PEC check successfully completed when receiving a packet 1 = PEC check failed when receiving a packet	RW1C	No	0
30	Unsupported SMBus Command 0 = Command received from SMBus is a supported command 1 = Command received from SMBus is an unsupported command	RW1C	No	0
31	SMBus Error Detected 0 = No error detected in STOP condition (as generated by the SMBus Master, to terminate the Data transfer) 1 = STOP detected was not on a byte boundary	RW1C	Yes	0

This section details Device-Specific Port Configuration and Lane Status registers at offsets 300h through 46Ch, including registers are related to the Management Port and Virtual Switches. [Table 13-24](#) defines the register map.

Additional information regarding programming of the Virtual Switch Table, as it relates to these registers, is provided in [Section 5.3.3.2, “Virtual Switch Table Programming Sequence.”](#)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 Offsets

Port Configuration			300h
Reserved	x1 Port Configuration		304h
Reserved	x2 Port Configuration	Reserved	308h
Factory Test Only			30Ch – 310h
Clock Enable			314h
Factory Test Only			318h – 32Ch
Station 0/1 Lane Status			330h
Factory Test Only			334h – 350h
Reserved (Base Mode) Management Port Control (Virtual Switch Mode)			354h
Reserved (Base Mode)			358h
Reserved (Virtual Switch Mode)		Virtual Switch Enable (Virtual Switch Mode)	
Reserved			35Ch
VS0 Upstream			360h
Reserved (Base Mode) VS1 Upstream (Virtual Switch Mode)			364h
Reserved (Base Mode) VS2 Upstream (Virtual Switch Mode)			368h
Reserved (Base Mode) VS3 Upstream (Virtual Switch Mode)			36Ch
Factory Test Only			370h – 37Ch

Table 13-24. Device-Specific Port Configuration and Lane Status Register Map (Ports^a) (Cont.)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
VS0 Port Vector		380h
Reserved (Base Mode) VS1 Port Vector (Virtual Switch Mode)		384h
Reserved (Base Mode) VS2 Port Vector (Virtual Switch Mode)		388h
Reserved (Base Mode) VS3 Port Vector (Virtual Switch Mode)		38Ch
Factory Test Only		390h – 39Ch
Port Reset		3A0h
Reserved	Parallel Hot Plug Control	3A4h
VSx_PERST# and NT_PERST# Control and Status		3A8h
Configuration Release		3ACh
Factory Test Only/Reserved		3B0h – 468h
Strap Configuration		46Ch

a. Certain registers are Chip-specific, others are Station-specific; all are Device-specific.

Register 13-112. 300h Port Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 lists the Port configurations for each Station. The table also indicates which Ports and SerDes/Lanes are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. In the values listed for each field, the second value is the Port Configuration Number, as listed in Table 13-5 (for example, 001b is equivalent to Port Configuration 1).				
2:0	Port Configuration for Station 0 Port Configuration Link width, per Port, for Station 0. The serial EEPROM bit values always override the STRAP_STN0_PORTCFG[1:0] input states (if the serial EEPROM values are loaded; refer to Table 13-5). Reset only by a Fundamental Reset (PEX_PERST# input assertion). 001b = 1 = x16 010b = 2 = x8x8 011b = 3 = x8x4x4 100b = 4 = x4x4x4x4 All other encodings are Reserved.	ROS	Yes	Defined by STRAP_STN0_PORTCFG[1:0] input states
5:3	Port Configuration for Station 1 Port Configuration Link width, per Port, for Station 1. The serial EEPROM bit values always override the STRAP_STN1_PORTCFG[1:0] input states (if the serial EEPROM values are loaded; refer to Table 13-5). Reset only by a Fundamental Reset (PEX_PERST# input assertion). 010b = 2 = x8 011b = 3 = x4x4 All other encodings are Reserved.	ROS	Yes	Defined by STRAP_STN1_PORTCFG[1:0] input states
31:6	Reserved	RsvdP	No	0-0h

Register 13-113. 304h x1 Port Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Forces the Port to linkup with a x1 negotiated Link width, regardless of the quantity of Lanes connected to the Port. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Ports, SerDes modules, and Lanes. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8. “X” is “Don’t Care.”				
7:0	x1 Only for Station 0 0000_0000b = None of the Ports are configured as x1 only, and x1 operation is allowed XXXX_XXX1b = Port 0 is configured as x1 only XXXX_XX1Xb = Port 1 is configured as x1 only XXXX_X1XXb = Port 2 is configured as x1 only XXXX_1XXXb = Port 3 is configured as x1 only All other encodings are Reserved.	ROS	Yes	0000_0000b
15:8	x1 Only for Station 1 0000_0000b = None of the Ports are configured as x1 only, and x1 operation is allowed XXXX_XXX1b = Port 8 is configured as x1 only XXXX_XX1Xb = Port 9 is configured as x1 only XXXX_X1XXb = Port 10 is configured as x1 only All other encodings are Reserved.	ROS	Yes	0000_0000b
31:16	Reserved	RsvdP	No	0000h

Register 13-114. 308h x2 Port Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Forces the Port to linkup with a x2 negotiated Link width, regardless of the quantity of Lanes connected to the Port. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Ports, SerDes modules, and Lanes. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8. “X” is “Don’t Care.”				
7:0	Reserved	RsvdP	No	00h
15:8	x2 Only for Station 1 0000_0000b = None of the Ports are configured as x2 only, and x2 operation is allowed XXXX_XXX1b = Port 8 is configured as x2 only XXXX_XX1Xb = Port 9 is configured as x2 only XXXX_X1XXb = Port 10 is configured as x2 only All other encodings are Reserved.	ROS	Yes	0000_0000b
31:16	Reserved	RsvdP	No	0000h

Register 13-115. 314h Clock Enable**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Ports and Stations are automatically enabled, according to the Port configuration defined by the STRAP_STNx_PORTCFGx 3-state inputs, which can be overridden by programming the Station's Port Configuration register <i>Port Configuration for Station x</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0]). An enabled Port can be selectively disabled, however, by Clearing the Port's <i>Port x Clock Enable</i> bit within this register. Port 0 must always remain enabled, and Port 8 (containing Station-specific registers) must remain enabled, if other Ports within the respective Station are enabled. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. <i>Table 13-5</i> indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8. It is not possible to enable more Ports than the maximum specified for the device.					
0	Port x Clock Enable 0 = Disables 1 = Enables	0	RWS	Yes	1
1		1	RWS	Yes	Defined by STRAP_STN0_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 0</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[2:0])
2		2	RWS	Yes	
3		3	RWS	Yes	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x Clock Enable 0 = Disables 1 = Enables	8	RWS	Yes	Defined by STRAP_STN1_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 1</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3])
9		9	RWS	Yes	
10		10	RWS	Yes	
23:11	Reserved		RsvdP	No	0-0h
24	Station 1 Root Clock Enable		RWS	Yes	Defined by STRAP_STN1_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 1</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3])
31:25	Factory Test Only/Reserved		RsvdP	No	0-0h

Register 13-116. 330h Station 0/1 Lane Status**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Lane x	Type	Serial EEPROM and I ² C	Default
Notes: The Lane x column is provided to indicate the Lane Number associated with each bit. Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Ports, SerDes modules, and Lanes.					
0	Lane x Up Status 0 = Lane is Down 1 = Lane is Up	0	ROS	Yes	1
1		1	ROS	Yes	1
2		2	ROS	Yes	1
3		3	ROS	Yes	1
4		4	ROS	Yes	1
5		5	ROS	Yes	1
6		6	ROS	Yes	1
7		7	ROS	Yes	1
8		8	ROS	Yes	1
9		9	ROS	Yes	1
10		10	ROS	Yes	1
11		11	ROS	Yes	1
12		12	ROS	Yes	1
13		13	ROS	Yes	1
14		14	ROS	Yes	1
15		15	ROS	Yes	1
23:16	Reserved		RsvdP	No	00h
24	Lane x Up Status 0 = Lane is Down 1 = Lane is Up	24	ROS	Yes	1
25		25	ROS	Yes	1
26		26	ROS	Yes	1
27		27	ROS	Yes	1
28		28	ROS	Yes	1
29		29	ROS	Yes	1
30		30	ROS	Yes	1
31		31	ROS	Yes	1

Register 13-117. 354h Management Port Control
(Virtual Switch mode – Port 0, accessible through the Management Port and Redundant Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default																								
Accessible by Management-capable Ports (Active and Redundant Management Ports), Strapping signals, and/or the I²C Bus. The Reserved register bits return zeros (0) during Reads. Writes to Reserved register bits do not affect the register. This register, in combination with the VSx Upstream and VSx Port Vector registers (offsets 360h through 36Ch, and 380h through 38Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.” Notes: Port 0 is recommended as the Management Port. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For fields [12:8 and 4:0], Station Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.																													
4:0	Active Management Port Indicates the Port Number of the Active Management Port. The value of this field is latched in, upon reset de-assertion, from the STRAP_UPSTRM_PORTSEL[2:0] input states.		RWS	Yes	Defined by STRAP_UPSTRM_PORTSEL[2:0] input states																								
	<table><thead><tr><th>Field Value</th><th>STRAP_UPSTRM_PORTSEL[2:0] Input States</th><th>Port Number</th></tr></thead><tbody><tr><td>0_0000b</td><td>000</td><td>0</td></tr><tr><td>0_0001b</td><td>00Z</td><td>1</td></tr><tr><td>0_0010b</td><td>001</td><td>2</td></tr><tr><td>0_0011b</td><td>0Z0</td><td>3</td></tr><tr><td>0_1000b</td><td>011</td><td>8</td></tr><tr><td>0_1001b</td><td>Z00</td><td>9</td></tr><tr><td>0_1010b</td><td>Z0Z</td><td>10</td></tr></tbody></table>	Field Value				STRAP_UPSTRM_PORTSEL[2:0] Input States	Port Number	0_0000b	000	0	0_0001b	00Z	1	0_0010b	001	2	0_0011b	0Z0	3	0_1000b	011	8	0_1001b	Z00	9	0_1010b	Z0Z	10	
	Field Value	STRAP_UPSTRM_PORTSEL[2:0] Input States				Port Number																							
	0_0000b	000				0																							
	0_0001b	00Z				1																							
	0_0010b	001				2																							
	0_0011b	0Z0				3																							
	0_1000b	011				8																							
	0_1001b	Z00				9																							
	0_1010b	Z0Z				10																							
All other encodings are Reserved .																													

Register 13-117. 354h Management Port Control
(Virtual Switch mode – Port 0, accessible through the Management Port and Redundant Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
5	Active Management Port Enable Enables the Active Management Port. The value of this bit is latched in, upon reset de-assertion, from the STRAP_MGMT_PORT_EN input state. 0 = Management Port is disabled (STRAP_MGMT_PORT_EN=1) 1 = Management Port is enabled (STRAP_MGMT_PORT_EN=0 or Z) If STRAP_MGMT_PORT_EN=0, the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, offset 3ACh[0]) is Set, and all Ports linkup (unless the serial EEPROM Clears the <i>Initiate Configuration</i> bit). If STRAP_MGMT_PORT_EN=Z, the <i>Initiate Configuration</i> bit value is 0, and only the Management Port links up. Other Ports linkup when the <i>Initiate Configuration</i> bit is Set by software, serial EEPROM, and/or I²C/SMBus.		RWS	Yes	Defined by STRAP_MGMT_PORT_EN input state
6	Active Management Port EEPROM Load on Hot Reset for Chip and Station Registers Enable After the Management Port receives a Hot Reset or DL_Down condition, the serial EEPROM reloads registers, as described below. 0 = Serial EEPROM reloads Management Port Port-specific registers (default) 1 = Serial EEPROM reloads: • Chip-specific registers (might affect all virtual switches), • Station-specific registers (might affect all virtual switches), and, • Management Port Port-specific registers	Management	RWS	Yes	0
	Reserved	Otherwise	RsvdP	No	0
7	Reserved		RsvdP	No	0
12:8	Redundant Management Port Indicates the Redundant Management Port's Port Number. 0_0000b = Port 0 (default) 0_0001b = Port 1 0_0010b = Port 2 0_0011b = Port 3 0_1000b = Port 8 0_1001b = Port 9 0_1010b = Port 10 All other encodings are Reserved.		RWS	Yes	0_0000b
13	Redundant Management Port Enable Enables the Redundant Management Port.		RWS	Yes	0
15:14	Reserved		RsvdP	No	00b

Register 13-117. 354h Management Port Control
(Virtual Switch mode – Port 0, accessible through the Management Port and
Redundant Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
18:16	Transparent Mode <i>Reserved</i>		RsvdP	No	000b
	NT VS Indicates the virtual switch to which the NT Port belongs. 000b = VS0 (default) 001b = VS1 010b = VS2 011b = VS3 All other encodings are <i>Reserved</i> .		RWS	Yes	000b
31:19	<i>Reserved</i>		RsvdP	No	0-0h

Register 13-118. 358h Virtual Switch Enable
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
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Indicates which virtual switches are enabled. The initial value of this register corresponds to the [STRAP_TESTMODE\[2:0\]](#) and VS_MODE input states.

Can be programmed to enable or disable virtual switches, prior to linkup, using any of the following methods:

- Serial EEPROM
- I²C, if [STRAP_I2C_SMBUS_CFG_EN#](#)=L, followed by an I²C Write that Sets the [Configuration Release](#) register *Initiate Configuration* bit (Port 0, accessible through the Management Port, offset [3ACh\[0\]](#))
- Software, followed by a Hot Reset

STRAP_TESTMODE[2:0] Input States	VS_MODE Input State	# of Enabled Virtual Switches	VSx
0ZZ, 0Z1, 010, or 01Z	0	1	VS0
	Z	2	VS0, VS1
	1	3	VS0, VS1, VS2
011, Z00, Z0Z, or Z01	0	4	VS0, VS1, VS2, VS3

All other encoding combinations are **Reserved**.

[Table 13-6](#) lists the default Upstream and Downstream Port assignments of each virtual switch.

0	VS0 Enable 0 = Disables VS0 1 = Enables VS0	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
1	VS1 Enable 0 = Disables VS1 1 = Enables VS1	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
2	VS2 Enable 0 = Disables VS2 1 = Enables VS2	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
3	VS3 Enable 0 = Disables VS3 1 = Enables VS3	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
5:4	Reserved	RsvdP	No	00b
7:6	Silicon Revision BA Reserved	RsvdP	No	00b
	Silicon Revision CA Factory Test Only	RWS	Yes	00b
31:8	Reserved	RsvdP	No	0000_00h

Register 13-119. 360h VS0 Upstream**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default																								
Can be programmed, prior to linkup, using any of the following methods:																												
- Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset																												
In Base mode, defines the VS0 Upstream Port and NT Port, and enables NT mode.																												
In Virtual Switch mode, defines the VS0 Upstream Port and NT Port, and enables NT mode. The VSx Upstream register(s) (offset(s) 360h through 36Ch), in combination with the Virtual Switch Enable and VSx Port Vector registers (offsets 354h, and 380h through 38Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.”																												
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.																												
For fields [12:8 and 4:0], Station Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.																												
Station Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.																												
4:0	VS0 Upstream Port Base Mode Identifies the VS0 Upstream Port, based upon the STRAP_UPSTRM_PORTSEL[2:0] input states.	RWS	Yes	Defined by STRAP_UPSTRM_PORTSEL[2:0] input states																								
	<table><tr><th>Field Value</th><th>STRAP_UPSTRM_PORTSEL[2:0] Input States</th><th>Port Number</th></tr><tr><td>0_0000b</td><td>000</td><td>0</td></tr><tr><td>0_0001b</td><td>00Z</td><td>1</td></tr><tr><td>0_0010b</td><td>001</td><td>2</td></tr><tr><td>0_0011b</td><td>0Z0</td><td>3</td></tr><tr><td>0_1000b</td><td>011</td><td>8</td></tr><tr><td>0_1001b</td><td>Z00</td><td>9</td></tr><tr><td>0_1010b</td><td>Z0Z</td><td>10</td></tr></table>				Field Value	STRAP_UPSTRM_PORTSEL[2:0] Input States	Port Number	0_0000b	000	0	0_0001b	00Z	1	0_0010b	001	2	0_0011b	0Z0	3	0_1000b	011	8	0_1001b	Z00	9	0_1010b	Z0Z	10
	Field Value				STRAP_UPSTRM_PORTSEL[2:0] Input States	Port Number																						
	0_0000b				000	0																						
	0_0001b				00Z	1																						
	0_0010b				001	2																						
	0_0011b				0Z0	3																						
	0_1000b				011	8																						
	0_1001b				Z00	9																						
	0_1010b				Z0Z	10																						
All other encodings are Reserved .																												
Virtual Switch Mode Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch.	RWS	Yes	00h																									
7:5	Reserved	RsvdP	No	000b																								

Register 13-119. 360h VS0 Upstream**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default																																				
12:8	NT Port The NT Port Number value is selected by the STRAP_NT_UPSTRM_PORTSEL[2:0] input states, and dependent upon whether the NT Port is enabled (bit 13, <i>NT Enable</i>, is Set). <table><thead><tr><th>Field Value</th><th>Bit 13 Value</th><th>STRAP_NT_UPSTRM_PORTSEL[2:0] Input States</th><th>Port Number</th></tr></thead><tbody><tr><td>0_0000b</td><td>1</td><td>000</td><td>0</td></tr><tr><td>0_0001b</td><td>1</td><td>00Z</td><td>1</td></tr><tr><td>0_0010b</td><td>1</td><td>001</td><td>2</td></tr><tr><td>0_0011b</td><td>1</td><td>0Z0</td><td>3</td></tr><tr><td>0_1000b</td><td>1</td><td>011</td><td>8</td></tr><tr><td>0_1001b</td><td>1</td><td>Z00</td><td>9</td></tr><tr><td>0_1010b</td><td>1</td><td>Z0Z</td><td>10</td></tr><tr><td>1_1010b</td><td>0</td><td>111</td><td>– (NT mode is disabled)</td></tr></tbody></table> All other encodings are Reserved.	Field Value	Bit 13 Value	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Port Number	0_0000b	1	000	0	0_0001b	1	00Z	1	0_0010b	1	001	2	0_0011b	1	0Z0	3	0_1000b	1	011	8	0_1001b	1	Z00	9	0_1010b	1	Z0Z	10	1_1010b	0	111	– (NT mode is disabled)	RWS	Yes	Defined by STRAP_NT_UPSTRM_P ORTSEL[2:0] input states
Field Value	Bit 13 Value	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Port Number																																					
0_0000b	1	000	0																																					
0_0001b	1	00Z	1																																					
0_0010b	1	001	2																																					
0_0011b	1	0Z0	3																																					
0_1000b	1	011	8																																					
0_1001b	1	Z00	9																																					
0_1010b	1	Z0Z	10																																					
1_1010b	0	111	– (NT mode is disabled)																																					
13	NT Enable 0 = STRAP_NT_UPSTRM_PORTSEL[2:0] = 111, NT mode is disabled 1 = STRAP_NT_UPSTRM_PORTSEL[2:0] = Port Number, NT mode is enabled	RWS	Yes	Defined by STRAP_NT_UPSTRM_P ORTSEL[2:0] input states																																				
25:14	Factory Test Only Note: When NT mode is disabled (STRAP_NT_UPSTRM_PORTSEL[2:0]=111), the value of bits [23:8] is 1A1Ah.	RWS	Yes	Refer to Note																																				
29:26	Reserved	RsvdP	No	0h																																				
31:30	Factory Test Only	RWS	Yes	00b																																				

Register 13-120. 364h VS1 Upstream
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Upstream register(s) (offset(s) 360h through 36Ch), in combination with the Virtual Switch Enable and VSx Port Vector registers (offsets 354h, and 380h through 38Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.”				
4:0	VS1 Upstream Port Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch.	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
31:5	Reserved	RsvdP	No	0-0h

Register 13-121. 368h VS2 Upstream
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Upstream register(s) (offset(s) 360h through 36Ch), in combination with the Virtual Switch Enable and VSx Port Vector registers (offsets 354h, and 380h through 38Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.”				
4:0	VS2 Upstream Port Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch.	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
31:5	Reserved	RsvdP	No	0-0h

Register 13-122. 36Ch VS3 Upstream
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register Initiate Configuration bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Upstream register(s) (offset(s) 360h through 36Ch), in combination with the Virtual Switch Enable and VSx Port Vector registers (offsets 354h, and 380h through 38Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.”				
4:0	VS3 Upstream Port Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch.	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
31:5	Reserved	RsvdP	No	0-0h

Register 13-123. 380h VS0 Port Vector**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset In Base mode, the default value is 0000_003Fh, indicating that all Ports are assigned to VS0. In Virtual Switch mode: - Defines which Upstream and Downstream Ports are active for VS0, and is initially programmed (along with other Virtual Switch Table registers; refer to the next bullet) according to the values of the STRAP_TESTMODE[2:0] and VS_MODE 3-state inputs, and/or by serial EEPROM. Immediately following PEX_PERST# input de-assertion, if the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) is Cleared (by STRAP_I2C_SMBUS_CFG_EN#=L, STRAP_MGMT_PORT_EN#=Z, and/or serial EEPROM), to allow I²C and/or Management Port software to program switch configuration prior to Setting the <i>Initiate Configuration</i> bit, the default value of this register is 0000_003Fh, indicating that all Ports are initially assigned to VS0. VSx Port Vector register(s) (offset(s) 380h through 38Ch), in combination with the Virtual Switch Enable and VSx Upstream registers (offsets 354h, and 360h through 36Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.” - Attempted access through virtual switch Upstream Ports, that are not the Management Port, is handled as NOP (Reads return a value of 0h, Writes have no effect). Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	VS0 Active Port x	0	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
1		1	RWS	Yes	
2		2	RWS	Yes	
3		3	RWS	Yes	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	VS0 Active Port x	8	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
9		9	RWS	Yes	
10		10	RWS	Yes	
31:11	Reserved		RsvdP	No	0-0h

Register 13-124. 384h VS1 Port Vector
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Defines which Upstream and Downstream Ports are associated with VS1. The initial value of this register corresponds to the STRAP_TESTMODE[2:0] and VS_MODE input states. Attempted access through virtual switch Upstream Ports, that are not the Management Port, is handled as NOP (Reads return a value of 0h, Writes have no effect). Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch. Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Port Vector register(s) (offset(s) 380h through 38Ch), in combination with the Virtual Switch Enable and VSx Upstream registers (offsets 354h, and 360h through 36Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.” Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	VS1 Active Port x	0	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
1		1	RWS	Yes	
2		2	RWS	Yes	
3		3	RWS	Yes	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	VS1 Active Port x	8	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
9		9	RWS	Yes	
10		10	RWS	Yes	
31:11	Reserved		RsvdP	No	0-0h

Register 13-125. 388h VS2 Port Vector
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Defines which Upstream and Downstream Ports are associated with VS2. The initial value of this register corresponds to the STRAP_TESTMODE[2:0] and VS_MODE input states. Attempted access through virtual switch Upstream Ports, that are not the Management Port, is handled as NOP (Reads return a value of 0h, Writes have no effect). Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch. Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Port Vector register(s) (offset(s) 380h through 38Ch), in combination with the Virtual Switch Enable and VSx Upstream registers (offsets 354h, and 360h through 36Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.” Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	VS2 Active Port x	0	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
1		1	RWS	Yes	
2		2	RWS	Yes	
3		3	RWS	Yes	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	VS2 Active Port x	8	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
9		9	RWS	Yes	
10		10	RWS	Yes	
31:11	Reserved		RsvdP	No	0-0h

Register 13-126. 38Ch VS3 Port Vector
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Defines which Upstream and Downstream Ports are associated with VS3. The initial value of this register corresponds to the STRAP_TESTMODE[2:0] and VS_MODE input states. Attempted access through virtual switch Upstream Ports, that are not the Management Port, is handled as NOP (Reads return a value of 0h, Writes have no effect). Table 13-6 lists the default Upstream and Downstream Port assignments of each virtual switch. Can be programmed, prior to linkup, using any of the following methods: - Serial EEPROM - I²C, if STRAP_I2C_SMBUS_CFG_EN#=L, followed by an I²C Write that Sets the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, accessible through the Management Port, offset 3ACh[0]) - Software, followed by a Hot Reset The VSx Port Vector register(s) (offset(s) 380h through 38Ch), in combination with the Virtual Switch Enable and VSx Upstream registers (offsets 354h, and 360h through 36Ch, respectively), comprise the Virtual Switch Table. For further details, refer to Section 5.3.3, “Virtual Switch Table.” Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	VS3 Active Port x	0	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
1		1	RWS	Yes	
2		2	RWS	Yes	
3		3	RWS	Yes	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	VS3 Active Port x	8	RWS	Yes	Based upon STRAP_TESTMODE[2:0] and VS_MODE input states
9		9	RWS	Yes	
10		10	RWS	Yes	
31:11	Reserved		RsvdP	No	0-0h

Register 13-127. 3A0h Port Reset**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Downstream Port x	Type	Serial EEPROM and I ² C	Default
Can be used in both Base mode and Virtual Switch mode; however, it is typically used for failover in Virtual Switch mode. When a <i>Reset Port x Vector</i> bit is driven with a value of 1, this register holds the Port in reset, including the Port PCI-to-PCI bridge and the hierarchy below the bridge. A value of 0 indicates to un-reset the PCI-to-PCI bridge and the hierarchy below the bridge. Notes: The Downstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. <i>Table 13-5</i> indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Reset Port x Vector 0 = Port is not reset 1 = Port is reset	0	RWS	Yes	0
1		1	RWS	Yes	0
2		2	RWS	Yes	0
3		3	RWS	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Reset Port x Vector 0 = Port is not reset 1 = Port is reset	8	RWS	Yes	0
9		9	RWS	Yes	0
10		10	RWS	Yes	0
31:11	Reserved		RsvdP	No	0-0h

Register 13-128. 3A4h Parallel Hot Plug Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Any Transparent Downstream Port can be configured as a Parallel or Serial Hot Plug Port. Hot Plug Port assignment is described in Section 10.9, “Default Parallel Hot Plug Ports.”				
0_0000b = Port 0	0_1000b = Port 8			
0_0001b = Port 1	0_1001b = Port 9			
0_0010b = Port 2	0_1010b = Port 10			
0_0011b = Port 3				
All other encodings are Reserved .				
Note: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
4:0	Parallel Hot Plug Port A Value specifies which Port is assigned Parallel Hot Plug Controller A.	RWS	Yes	TBD
6:5	Reserved	RsvdP	No	00b
7	Parallel Hot Plug Port A Enable 0 = Hot Plug Port A is not enabled 1 = Hot Plug Port A is enabled	RWS	Yes	1
12:8	Parallel Hot Plug Port B Value specifies which Port is assigned Parallel Hot Plug Controller B.	RWS	Yes	TBD
14:13	Reserved	RsvdP	No	00b
15	Parallel Hot Plug Port B Enable 0 = Hot Plug Port B is not enabled 1 = Hot Plug Port B is enabled	RWS	Yes	1
20:16	Parallel Hot Plug Port C Value specifies which Port is assigned Parallel Hot Plug Controller C.	RWS	Yes	TBD
22:21	Reserved	RsvdP	No	00b
23	Parallel Hot Plug Port C Enable 0 = Hot Plug Port C is not enabled 1 = Hot Plug Port C is enabled	RWS	Yes	1
31:24	Reserved	RsvdP	No	00h

Register 13-129. 3A8h VSx_PERST# and NT_PERST# Control and Status
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: The VSx_PERST#-related bits are valid only in Virtual Switch mode. The PEX_NT_PERST#-related bits are valid in both Base mode and Virtual Switch mode.				
0	Base Mode Reserved	RsvdP	No	0
	Virtual Switch Mode VS0_PERST# Input Value 0 = VS0_PERST# input is Low 1 = VS0_PERST# input is High	ROS	No	Reflects the VS0_PERST# input value
1	Base Mode Reserved	RsvdP	No	0
	Virtual Switch Mode VS1_PERST# Input Value 0 = VS1_PERST# input is Low 1 = VS1_PERST# input is High	ROS	No	Reflects the VS1_PERST# input value
2	Base Mode Reserved	RsvdP	No	0
	Virtual Switch Mode VS2_PERST# Input Value 0 = VS2_PERST# input is Low 1 = VS2_PERST# input is High	ROS	No	Reflects the VS2_PERST# input value
3	Base Mode Reserved	RsvdP	No	0
	Virtual Switch Mode VS3_PERST# Input Value 0 = VS3_PERST# input is Low 1 = VS3_PERST# input is High	ROS	No	Reflects the VS3_PERST# input value
7:4	Factory Test Only/Reserved	RsvdP	No	0h

Register 13-129. 3A8h VSx_PERST# and NT_PERST# Control and Status**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Base Mode <i>Reserved</i>	RsvdP	No	0
	Virtual Switch Mode VS0_PERST# Control 0 = Writing 0 Clears the VS0_PERST# reset condition 1 = Causes VS0_PERST# to receive a Fundamental Reset	RWS	Yes	0
9	Base Mode <i>Reserved</i>	RsvdP	No	0
	Virtual Switch Mode VS1_PERST# Control 0 = Writing 0 Clears the VS1_PERST# reset condition 1 = Causes VS1_PERST# Reset	RWS	Yes	0
10	Base Mode <i>Reserved</i>	RsvdP	No	0
	Virtual Switch Mode VS2_PERST# Control 0 = Writing 0 Clears the VS2_PERST# reset condition 1 = Causes VS2_PERST# Reset	RWS	Yes	0
11	Base Mode <i>Reserved</i>	RsvdP	No	0
	Virtual Switch Mode VS3_PERST# Control 0 = Writing 0 Clears the VS3_PERST# reset condition 1 = Causes VS3_PERST# Reset	RWS	Yes	0
15:12	Factory Test Only/Reserved	RsvdP	No	0h
16	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode PEX_NT_PERST# Input Value 0 = PEX_NT_PERST# input is Low 1 = PEX_NT_PERST# input is High	ROS	No	Reflects the PEX_NT_PERST# input value
23:17	Reserved	RsvdP	No	0-0h
24	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode PEX_NT_PERST# Control 0 = Writing 0 Clears the PEX_NT_PERST# reset condition 1 = Causes PEX_NT_PERST# Reset	RWS	Yes	0
31:25	Reserved	RsvdP	No	0-0h

Register 13-130. 3ACh Configuration Release**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	Initiate Configuration 0 = STRAP_I2C_SMBUS_CFG_EN#=L, –or– STRAP_MGMT_PORT_EN=Z 1 = STRAP_I2C_SMBUS_CFG_EN#=H, –or– STRAP_MGMT_PORT_EN=0 or 1, releases the hold that prevents the Links from coming Up Base Mode and Virtual Switch Mode If the STRAP_I2C_SMBUS_CFG_EN# input is pulled or tied Low to VSS (Ground), I²C is being used to configure the PEX 8724. After I²C has completed its configuration, it must write to this bit, to release the hold that is preventing the Links from coming Up. Virtual Switch Mode The STRAP_MGMT_PORT_EN 3-state input is used to control Bring-Up Options 1 and 2: • Option 1 – STRAP_MGMT_PORT_EN=Z. After the serial EEPROM (if present) is loaded, the Management Port comes up first, to configure the PEX 8724. When the Management Port has completed its configuration, Management Port software and/or I²C/SMBus must Set this bit, to release the hold that is preventing the remaining Links from coming up (assuming this hold is not de-asserted by the serial EEPROM Setting this bit). • Option 2 – STRAP_MGMT_PORT_EN=0. After the serial EEPROM (if present) is loaded, all Ports come up simultaneously (provided that the serial EEPROM does not Clear this bit).	HwInit	Yes	0 (STRAP_I2C_SMBUS_CFG_EN#=L, –or– if in Virtual Switch mode, STRAP_MGMT_PORT_EN=Z) Otherwise, 1
31:1	Reserved	RsvdP	No	0-0h

Register 13-131. 46Ch Strap Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4:0	Factory Test Only	HwInit	Yes	0-0h
5	I²C Configuration Enable Status The default value of this bit is the inverse of the STRAP_I2C_SMBUS_CFG_EN# input. The serial EEPROM/I ² C should not be used to change the value of this bit. To override the value of the STRAP_I2C_SMBUS_CFG_EN# input functionality, the serial EEPROM should program the Configuration Release register <i>Initiate Configuration</i> bit (Port 0, offset 3ACh[0]). 0 = I ² C/SMBus Configuration is disabled (STRAP_I2C_SMBUS_CFG_EN#=H) 1 = I ² C/SMBus Configuration is enabled (STRAP_I2C_SMBUS_CFG_EN#=L)	ROS	Yes	PCFG
6	SMBus Enable Status The default value of this bit is the inverse of the STRAP_I2C_SMBUS_EN input. 0 = SMBus protocol without ARP is enabled (STRAP_I2C_SMBUS_EN=Z or 1) 1 = I ² C protocol is enabled (STRAP_I2C_SMBUS_EN=0)	ROS	Yes	PCFG
7	Reserved	RsvdP	No	0
9:8	Factory Test Only	ROS	Yes	11b
10	Gen 1 Compatible Disable Status Value is dependent upon the SPARE2 input. 0 = Enable Gen 1 Compatible mode, to accommodate a connected, non-compliant Gen 1 device (SPARE2=L) 1 = Disable Gen 1 Compatible mode, for non-compliant Gen 1 device (SPARE2=H)	ROS	Yes	0 (SPARE2=L) 1 (SPARE2=H)
11	Gen 1 Mode Status Value is dependent upon the STRAP_GEN1_GEN2 3-state input. 0 = Link speed is 2.5 GT/s (STRAP_GEN1_GEN2=0) 1 = Maximum Link speed is 5.0 or 8.0 GT/s (STRAP_GEN1_GEN2=Z or 1, respectively)	ROS	Yes	PCFG

Register 13-131. 46Ch Strap Configuration**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default																		
12	Gen 2 Mode Status Value is dependent upon the STRAP_GEN1_GEN2 3-state input. 0 = Maximum Link speed is 5.0 GT/s (STRAP_GEN1_GEN2=Z) 1 = Maximum Link speed is 8.0 GT/s (STRAP_GEN1_GEN2=1)	ROS	Yes	PCFG																		
15:13	VS Mode Status Indicates the HP_PWR_GOOD_A_VS_MODE input state, primarily for the purpose of allowing software to check the board connection. The VS_MODE 3-state input is sampled and latched at PEX_PERST# and/or VSx_PERST# input de-assertion). While the serial EEPROM and/or I ² C/SMBus could change the value, Virtual Switch mode must be enabled by programming the Virtual Switch Table registers, rather than this field. (Virtual Switch Table registers are comprised of the Virtual Switch Enable , VSx Upstream , and VSx Port Vector registers (offsets 354h, 360h through 36Ch, and 380h through 38Ch, respectively).) 000b = Base mode, no virtual switches are enabled 001b = Two virtual switches are enabled (VS0, VS1) 010b = Three virtual switches are enabled (VS0, VS1, VS2) 011b = Four virtual switches are enabled (VS0, VS1, VS2, and VS3) All other encodings are Reserved .	ROS	Yes	PCFG																		
20:16	Factory Test Only	ROS	No	PCFG																		
23:21	Reserved	RsvdP	No	000b																		
28:24	Strap Test Mode Status Reflects the STRAP_TESTMODE[2:0] input states. <table><thead><tr><th>Base Mode</th><th>Virtual Switch Mode</th></tr></thead><tbody><tr><td>00000b = 000</td><td>00100b = 0ZZ</td></tr><tr><td>00001b = 00Z</td><td>00101b = 0Z1</td></tr><tr><td>00010b = 001</td><td>00110b = 010</td></tr><tr><td>00011b = 0Z0</td><td>00111b = 01Z</td></tr><tr><td></td><td>01000b = 011</td></tr><tr><td></td><td>01001b = Z00</td></tr><tr><td></td><td>01010b = Z0Z</td></tr><tr><td></td><td>01011b = Z01</td></tr></tbody></table> All other encodings are Reserved .	Base Mode	Virtual Switch Mode	00000b = 000	00100b = 0ZZ	00001b = 00Z	00101b = 0Z1	00010b = 001	00110b = 010	00011b = 0Z0	00111b = 01Z		01000b = 011		01001b = Z00		01010b = Z0Z		01011b = Z01	ROS	No	Based upon STRAP_TESTMODE[2:0] input states
Base Mode	Virtual Switch Mode																					
00000b = 000	00100b = 0ZZ																					
00001b = 00Z	00101b = 0Z1																					
00010b = 001	00110b = 010																					
00011b = 0Z0	00111b = 01Z																					
	01000b = 011																					
	01001b = Z00																					
	01010b = Z0Z																					
	01011b = Z01																					
31:29	Reserved	RsvdP	No	000b																		

13.16.8 Device-Specific Registers – Exerciser (Offsets 56Ch – 570h)

This section details the Device-Specific Exerciser registers. [Table 13-25](#) defines the register map.

Table 13-25. Device-Specific Exerciser Register Map
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets	
Exerciser DW FIFO																																	56Ch
Exerciser Control																																	570h

Register 13-132. 56Ch Exerciser DW FIFO
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	32b Write Data Write 32 bits to this register. A subsequent Write shifts over the first 32 bits, resulting in 2 x 32 bits. Up to five 32-bit Writes can be stored using this method. The first 32 bits written comprises the Most Significant DWord; the last 32 bits written comprises the Least Significant DWord. The Exerciser RAM Write Control sends the five 32-bit quantities to the on-chip RAM. Only the most recently written 32 bits can be read.	RW	No	0000_0000h

Register 13-133. 570h Exerciser Control
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
10:0	RAM Address RAM address in which to write the Exerciser DWord FIFO contents.	RW	Yes	0-0h
12:11	Reserved	RsvdP	No	00b
14:13	RAM Select 2-bit encoding that selects the RAM type. 00b = Header command 01b = Header TLP 10b = Lower Payload 11b = Upper Payload	RW	Yes	00b
15	RAM Write Enable Writing 1 creates a 1-clock strobe that writes the RAM specified by field [14:13] (<i>RAM Select</i>), at the RAM address indicated by field [10:0] (<i>RAM Address</i>), that has up to 5 DWords of data stored within Exerciser DWord FIFO.	RW1C	Yes	0

Register 13-133. 570h Exerciser Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
16	Exerciser Enable <i>Note:</i> Do not change this bit during the Exerciser program. 0 = Disables the Exerciser 1 = Enables the Exerciser	RW	Yes	0
17	Silicon Revision BA <i>Reserved</i>	RsvdP	No	0
	Silicon Revision CA <i>Not used</i>	RW	No	0
18	Tag Increment Mode 0 = Disables automatic Tag incrementation 1 = Enables automatic Tag incrementation	RW	Yes	0
26:19	Maximum NP Request Counters Defines the maximum Non-Posted Request value. 0 = 256	RW	Yes	0-0h
27	<i>Reserved</i>	RsvdP	No	0
28	Exerciser Done Status 0 = Exerciser program is still running 1 = Exerciser program has finished running	RO	Yes	0
29	Loop2 Infinite Loop Enable 0 = Loop2 is a finite loop 1 = Loop2 is an infinite loop	RW	Yes	0
30	Exerciser Reset 1 = Resets the Exerciser program	RW1C	Yes	0
31	Exerciser Run Allows the Exerciser to run or pause. Can be changed while the Exerciser is running. 0 = Pause Exerciser 1 = Run Exerciser	RW	Yes	0

13.16.9 Device-Specific Registers – General-Purpose Input/Output (Offsets 600h – 68Ch)

This section details the Device-Specific General-Purpose Input/Output (GPIO) registers. [Table 13-26](#) defines the register map.

Note: In Virtual Switch mode, each virtual switch uses its own local set of **GPIO** registers (VS Upstream Port(s), offsets [A34h](#) through [A54h](#)).

Table 13-26. Device-Specific GPIO Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
GPIO 0_9 Direction Control	600h
GPIO 10_11 Direction Control	604h
Reserved	608h – 610h
GPIO 0_11 Input De-Bounce	614h
Reserved	618h
GPIO 0_11 Input Data	61Ch
Reserved	620h
GPIO 0_11 Output Data	624h
Reserved	628h
GPIO 0_11 Interrupt Polarity	62Ch
Reserved	630h
GPIO 0_11 Interrupt Status	634h
Reserved	638h
GPIO 0_11 Interrupt Mask	63Ch
Reserved	640h – 648h
Reserved (Base Mode) Virtual Switch GPIO Update (Virtual Switch Mode)	64Ch
Reserved (Base Mode) VS0 GPIO_PG 0_11 Assignment (Virtual Switch Mode)	650h
Reserved (Base Mode) VS1 GPIO_PG 0_11 Assignment (Virtual Switch Mode)	654h
Reserved (Base Mode) VS2 GPIO_PG 0_11 Assignment (Virtual Switch Mode)	658h
Reserved (Base Mode) VS3 GPIO_PG 0_11 Assignment (Virtual Switch Mode)	65Ch
Factory Test Only/Reserved	660h – 68Ch

Register 13-134. 600h GPIO 0_9 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The GPIO 0_9 Direction Control registers control the direction, source, and destination of the PORT_GOOD[0:6]# and GPIO[7:9] signals, respectively. Notes: Register offsets 61Ch and 624h, referenced within this register, are located as follows – Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
1:0	PORT_GOOD0# Source/Destination As Input: 00b = To PORT_GOOD0# Input Data register (offset 61Ch [0]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD0# Output Data register (offset 624h [0]) 01b = PORT_GOOD0# 10b = SHP_PERST0# output 11b = Reserved	RWS	Yes	TBD
2	PORT_GOOD0# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
4:3	PORT_GOOD1# Source/Destination As Input: 00b = To PORT_GOOD1# Input Data register (offset 61Ch [1]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD1# Output Data register (offset 624h [1]) 01b = PORT_GOOD1# 10b = SHP_PERST1# output 11b = Reserved	RWS	Yes	TBD
5	PORT_GOOD1# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD

Register 13-134. 600h GPIO 0_9 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:6	PORT_GOOD2# Source/Destination As Input: 00b = To PORT_GOOD2# Input Data register (offset 61Ch[2]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD2# Output Data register (offset 624h[2]) 01b = PORT_GOOD2# 10b = SHP_PERST2# output 11b = Reserved	RWS	Yes	TBD
8	PORT_GOOD2# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
10:9	PORT_GOOD3# Source/Destination As Input: 00b = To PORT_GOOD3# Input Data register (offset 61Ch[3]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD3# Output Data register (offset 624h[3]) 01b = PORT_GOOD3# 10b = SHP_PERST3# output 11b = Reserved	RWS	Yes	TBD
11	PORT_GOOD3# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD

Register 13-134. 600h GPIO 0_9 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
13:12	PORT_GOOD4# Source/Destination As Input: 00b = To PORT_GOOD4# Input Data register (offset 61Ch[4]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD4# Output Data register (offset 624h[4]) 01b = PORT_GOOD4# 10b = SHP_PERST4# output 11b = Reserved	RWS	Yes	TBD
14	PORT_GOOD4# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
16:15	PORT_GOOD5# Source/Destination As Input: 00b = To PORT_GOOD5# Input Data register (offset 61Ch[5]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD5# Output Data register (offset 624h[5]) 01b = PORT_GOOD5# 10b = SHP_PERST5# output 11b = Reserved	RWS	Yes	TBD
17	PORT_GOOD5# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD

Register 13-134. 600h GPIO 0_9 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19:18	PORT_GOOD6# Source/Destination As Input: 00b = To PORT_GOOD6# Input Data register (offset 61Ch[6]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From PORT_GOOD6# Output Data register (offset 624h[6]) 01b = PORT_GOOD6# 10b = SHP_PERST6# output 11b = Reserved	RWS	Yes	TBD
20	PORT_GOOD6# Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
22:21	GPIO7 Source/Destination As Input: 00b = To GPIO7 Input Data register (offset 61Ch[7]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From GPIO7 Output Data register (offset 624h[7]) 01b = GPIO7 10b = SHP_PERST7# output 11b = Reserved	RWS	Yes	TBD
23	GPIO7 Direction Control 0 = Input 1 = Output	RWS	Yes	TBD

Register 13-134. 600h GPIO 0_9 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
25:24	GPIO8 Source/Destination As Input: 00b = To GPIO8 Input Data register (offset 61Ch[8]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From GPIO8 Output Data register (offset 624h[8]) 01b = GPIO8 10b = SHP_PERST8# output 11b = Reserved	RWS	Yes	TBD
26	GPIO8 Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
28:27	GPIO9 Source/Destination As Input: 00b = To GPIO9 Input Data register (offset 61Ch[9]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From GPIO9 Output Data register (offset 624h[9]) 01b = GPIO9 10b = SHP_PERST9# output 11b = Reserved	RWS	Yes	TBD
29	GPIO9 Direction Control 0 = Input 1 = Output	RWS	Yes	TBD
31:30	Reserved	RsvdP	No	00b

Register 13-135. 604h GPIO 10_11 Direction Control**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
The GPIO 10_11 Direction Control registers control the direction, source, and destination of the GPIO[10:11] signals, respectively.					
Notes: Register offsets 61Ch and 624h , referenced within this register, are located as follows – Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port.					
Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.					
1:0	GPIO10 Source/Destination As Input: 00b = To GPIO10 Input Data register (offset 61Ch [10]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From GPIO10 Output Data register (offset 624h [10]) 01b = GPIO10 10b = SHP_PERST10# output 11b = Reserved	RWS	Yes	TBD	
2	GPIO10 Direction Control 0 = Input 1 = Output	RWS	Yes	TBD	
4:3	Base Mode Reserved		RsvdP	No	00b
	Virtual Switch Mode GPIO11 Source/Destination 00b = To GPIO11 Input Data register (offset 61Ch [11]) 01b = General interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) 10b, 11b = Reserved As Output: 00b = From GPIO11 Output Data register (offset 624h [11]) 01b = GPIO11 10b = SHP_PERST11# output 11b = Reserved	Four enabled virtual switches	RWS	Yes	TBD
	Reserved	Otherwise	RsvdP	No	00b
	Base Mode Reserved		RsvdP	No	0
5	Virtual Switch Mode GPIO11 Direction Control 0 = Input 1 = Output	Four enabled virtual switches	RWS	Yes	TBD
	Reserved	Otherwise	RsvdP	No	0
	Reserved		RsvdP	No	0-0h

Register 13-136. 614h GPIO 0_11 Input De-Bounce
(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The PORT_GOOD[0:6]# and GPIO[7:11] signals are de-bounced, using the GPIO 0_11 Input De-Bounce registers, respectively. Notes: Register offsets 600h and 604h, referenced within this register, are located as follows – Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Input De-Bounce Control Controls de-bounce when PORT_GOOD0# is configured as an input (offset 600h[2] is Cleared). 0 = PORT_GOOD0# input is not de-bounced 1 = PORT_GOOD0# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
1	PORT_GOOD1# Input De-Bounce Control Controls de-bounce when PORT_GOOD1# is configured as an input (offset 600h[5] is Cleared). 0 = PORT_GOOD1# input is not de-bounced 1 = PORT_GOOD1# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
2	PORT_GOOD2# Input De-Bounce Control Controls de-bounce when PORT_GOOD2# is configured as an input (offset 600h[8] is Cleared). 0 = PORT_GOOD2# input is not de-bounced 1 = PORT_GOOD2# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
3	PORT_GOOD3# Input De-Bounce Control Controls de-bounce when PORT_GOOD3# is configured as an input (offset 600h[11] is Cleared). 0 = PORT_GOOD3# input is not de-bounced 1 = PORT_GOOD3# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0

Register 13-136. 614h GPIO 0_11 Input De-Bounce**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	PORT_GOOD4# Input De-Bounce Control Controls de-bounce when PORT_GOOD4# is configured as an input (offset 600h[14] is Cleared). 0 = PORT_GOOD4# input is not de-bounced 1 = PORT_GOOD4# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
5	PORT_GOOD5# Input De-Bounce Control Controls de-bounce when PORT_GOOD5# is configured as an input (offset 600h[17] is Cleared). 0 = PORT_GOOD5# input is not de-bounced 1 = PORT_GOOD5# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
6	PORT_GOOD6# Input De-Bounce Control Controls de-bounce when PORT_GOOD6# is configured as an input (offset 600h[20] is Cleared). 0 = PORT_GOOD6# input is not de-bounced 1 = PORT_GOOD6# input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
7	GPIO7 Input De-Bounce Control Controls de-bounce when GPIO7 is configured as an input (offset 600h[23] is Cleared). 0 = GPIO7 input is not de-bounced 1 = GPIO7 input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0

Register 13-136. 614h GPIO 0_11 Input De-Bounce**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	GPIO8 Input De-Bounce Control Controls de-bounce when GPIO8 is configured as an input (offset 600h[26] is Cleared). 0 = GPIO8 input is not de-bounced 1 = GPIO8 input is de-bounced; de-bounce time is approximately 1.3 ms		RWS	Yes	0
9	GPIO9 Input De-Bounce Control Controls de-bounce when GPIO9 is configured as an input (offset 600h[29] is Cleared). 0 = GPIO9 input is not de-bounced 1 = GPIO9 input is de-bounced; de-bounce time is approximately 1.3 ms		RWS	Yes	0
10	GPIO10 Input De-Bounce Control Controls de-bounce when GPIO10 is configured as an input (offset 604h[2] is Cleared). 0 = GPIO10 input is not de-bounced 1 = GPIO10 input is de-bounced; de-bounce time is approximately 1.3 ms		RWS	Yes	0
11	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode GPIO11 Input De-Bounce Control Controls de-bounce when GPIO11 is configured as an input (offset 604h[5], is Cleared). 0 = GPIO11 input is not de-bounced 1 = GPIO11 input is de-bounced; de-bounce time is approximately 1.3 ms	Four enabled virtual switches	RWS	Yes	0
	<i>Reserved</i>	Otherwise	RsvdP	No	0
31:12	<i>Reserved</i>		RsvdP	No	0000_0h

Register 13-137. 61Ch GPIO 0_11 Input Data**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The PORT_GOOD[0:6]# and GPIO[7:11] input values are updated into the GPIO 0_11 Input Data registers, respectively. Notes: Register offsets 600h and 604h, referenced within this register, are located as follows – Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Input Data If PORT_GOOD0# is configured as an output (offset 600h[2] is Set), Reads return a value of 0. If PORT_GOOD0# is configured as an input (offset 600h[2] is Cleared), Reads return the logic value of the voltage on PORT_GOOD0#.	ROS	No	0
1	PORT_GOOD1# Input Data If PORT_GOOD1# is configured as an output (offset 600h[5] is Set), Reads return a value of 0. If PORT_GOOD1# is configured as an input (offset 600h[5] is Cleared), Reads return the logic value of the voltage on PORT_GOOD1#.	ROS	No	0
2	PORT_GOOD2# Input Data If PORT_GOOD2# is configured as an output (offset 600h[8] is Set), Reads return a value of 0. If PORT_GOOD2# is configured as an input (offset 600h[8] is Cleared), Reads return the logic value of the voltage on PORT_GOOD2#.	ROS	No	0
3	PORT_GOOD3# Input Data If PORT_GOOD3# is configured as an output (offset 600h[11] is Set), Reads return a value of 0. If PORT_GOOD3# is configured as an input (offset 600h[11] is Cleared), Reads return the logic value of the voltage on PORT_GOOD3#.	ROS	No	0

Register 13-137. 61Ch GPIO 0_11 Input Data**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
4	PORT_GOOD4# Input Data If PORT_GOOD4# is configured as an output (offset 600h[14] is Set), Reads return a value of 0. If PORT_GOOD4# is configured as an input (offset 600h[14] is Cleared), Reads return the logic value of the voltage on PORT_GOOD4#.	ROS	No	0	
5	PORT_GOOD5# Input Data If PORT_GOOD5# is configured as an output (offset 600h[17] is Set), Reads return a value of 0. If PORT_GOOD5# is configured as an input (offset 600h[17] is Cleared), Reads return the logic value of the voltage on PORT_GOOD5#.	ROS	No	0	
6	PORT_GOOD6# Input Data If PORT_GOOD6# is configured as an output (offset 600h[20] is Set), Reads return a value of 0. If PORT_GOOD6# is configured as an input (offset 600h[20] is Cleared), Reads return the logic value of the voltage on PORT_GOOD6#.	ROS	No	0	
7	GPIO7 Input Data If GPIO7 is configured as an output (offset 600h[23] is Set), Reads return a value of 0. If GPIO7 is configured as an input (offset 600h[23] is Cleared), Reads return the logic value of the voltage on GPIO7.	ROS	No	0	
8	GPIO8 Input Data If GPIO8 is configured as an output (offset 600h[26] is Set), Reads return a value of 0. If GPIO8 is configured as an input (offset 600h[26] is Cleared), Reads return the logic value of the voltage on GPIO8.	ROS	No	0	
9	GPIO9 Input Data If GPIO9 is configured as an output (offset 600h[29] is Set), Reads return a value of 0. If GPIO9 is configured as an input (offset 600h[29] is Cleared), Reads return the logic value of the voltage on GPIO9.	ROS	No	0	
10	GPIO10 Input Data If GPIO10 is configured as an output (offset 604h[2] is Set), Reads return a value of 0. If GPIO10 is configured as an input (offset 604h[2] is Cleared), Reads return the logic value of the voltage on GPIO10.	ROS	No	0	
11	Base Mode Reserved	RsvdP	No	0	
	Virtual Switch Mode GPIO11 Input Data If GPIO11 is configured as an output (offset 604h[5] is Set), Reads return a value of 0. If GPIO11 is configured as an input (offset 604h[5] is Cleared), Reads return the logic value of the voltage on GPIO11.	Four enabled virtual switches	ROS	No	0
	Reserved	Otherwise	RsvdP	No	0
31:12	Reserved	RsvdP	No	0000_0h	

Register 13-138. 624h GPIO 0_11 Output Data**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The GPIO 0_11 Output Data registers control the PORT_GOOD[0:6]# and GPIO[7:11] outputs, respectively. Notes: Register offsets 600h and 604h, referenced within this register, are located as follows – Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Output Data If PORT_GOOD0# is configured as an output (offset 600h[2] is Set), the value written to this bit is immediately driven to the PORT_GOOD0# output. Reads return the value written.	RWS	Yes	0
1	PORT_GOOD1# Output Data If PORT_GOOD1# is configured as an output (offset 600h[5] is Set), the value written to this bit is immediately driven to the PORT_GOOD1# output. Reads return the value written.	RWS	Yes	0
2	PORT_GOOD2# Output Data If PORT_GOOD2# is configured as an output (offset 600h[8] is Set), the value written to this bit is immediately driven to the PORT_GOOD2# output. Reads return the value written.	RWS	Yes	0
3	PORT_GOOD3# Output Data If PORT_GOOD3# is configured as an output (offset 600h[11] is Set), the value written to this bit is immediately driven to the PORT_GOOD3# output. Reads return the value written.	RWS	Yes	0
4	PORT_GOOD4# Output Data If PORT_GOOD4# is configured as an output (offset 600h[14] is Set), the value written to this bit is immediately driven to the PORT_GOOD4# output. Reads return the value written.	RWS	Yes	0
5	PORT_GOOD5# Output Data If PORT_GOOD5# is configured as an output (offset 600h[17] is Set), the value written to this bit is immediately driven to the PORT_GOOD5# output. Reads return the value written.	RWS	Yes	0
6	PORT_GOOD6# Output Data If PORT_GOOD6# is configured as an output (offset 600h[20] is Set), the value written to this bit is immediately driven to the PORT_GOOD6# output. Reads return the value written.	RWS	Yes	0
7	GPIO7 Output Data If GPIO7 is configured as an output (offset 600h[23] is Set), the value written to this bit is immediately driven to the GPIO7 output. Reads return the value written.	RWS	Yes	0

Register 13-138. 624h GPIO 0_11 Output Data**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	GPIO8 Output Data If GPIO8 is configured as an output (offset 600h[26] is Set), the value written to this bit is immediately driven to the GPIO8 output. Reads return the value written.		RWS	Yes	0
9	GPIO9 Output Data If GPIO9 is configured as an output (offset 600h[29] is Set), the value written to this bit is immediately driven to the GPIO9 output. Reads return the value written.		RWS	Yes	0
10	GPIO10 Output Data If GPIO10 is configured as an output (offset 604h[2] is Set), the value written to this bit is immediately driven to the GPIO10 output. Reads return the value written.		RWS	Yes	0
11	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode GPIO11 Output Data If GPIO11 is configured as an output (offset 604h[5] is Set), the value written to this bit is immediately driven to the GPIO11 output. Reads return the value written.	Four enabled virtual switches	RWS	Yes	0
	<i>Reserved</i>	Otherwise	RsvdP	No	0
31:12	<i>Reserved</i>		RsvdP	No	0000_0h

Register 13-139. 62Ch GPIO 0_11 Interrupt Polarity**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The GPIO 0_11 Interrupt Polarity registers control the PORT_GOOD[0:6]# and GPIO[7:11] Interrupt input polarities, respectively. Notes: PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD0# Interrupt input is Active-Low 1 = PORT_GOOD0# Interrupt input is Active-High	RWS	Yes	0
1	PORT_GOOD1# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD1# Interrupt input is Active-Low 1 = PORT_GOOD1# Interrupt input is Active-High	RWS	Yes	0
2	PORT_GOOD2# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD2# Interrupt input is Active-Low 1 = PORT_GOOD2# Interrupt input is Active-High	RWS	Yes	0
3	PORT_GOOD3# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD3# Interrupt input is Active-Low 1 = PORT_GOOD3# Interrupt input is Active-High	RWS	Yes	0
4	PORT_GOOD4# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD4# Interrupt input is Active-Low 1 = PORT_GOOD4# Interrupt input is Active-High	RWS	Yes	0
5	PORT_GOOD5# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD5# Interrupt input is Active-Low 1 = PORT_GOOD5# Interrupt input is Active-High	RWS	Yes	0
6	PORT_GOOD6# Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = PORT_GOOD6# Interrupt input is Active-Low 1 = PORT_GOOD6# Interrupt input is Active-High	RWS	Yes	0
7	GPIO7 Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = GPIO7 Interrupt input is Active-Low 1 = GPIO7 Interrupt input is Active-High	RWS	Yes	0

Register 13-139. 62Ch GPIO 0_11 Interrupt Polarity**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	GPIO8 Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = GPIO8 Interrupt input is Active-Low 1 = GPIO8 Interrupt input is Active-High		RWS	Yes	0
9	GPIO9 Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = GPIO9 Interrupt input is Active-Low 1 = GPIO9 Interrupt input is Active-High		RWS	Yes	0
10	GPIO10 Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = GPIO10 Interrupt input is Active-Low 1 = GPIO10 Interrupt input is Active-High		RWS	Yes	0
11	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode GPIO11 Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High. 0 = GPIO11 Interrupt input is Active-Low 1 = GPIO11 Interrupt input is Active-High	Four enabled virtual switches	RWS	Yes	0
	<i>Reserved</i>	Otherwise	RsvdP	No	0
31:12	<i>Reserved</i>		RsvdP	No	0000_0h

Register 13-140. 634h GPIO 0_11 Interrupt Status**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The PORT_GOOD[0:6]# and GPIO[7:11] Interrupt input status values are updated into the GPIO 0_11 Interrupt Status registers, respectively. Interrupt status remains Set, as long the corresponding PORT_GOODx# or GPIOx input is asserted, and Clears on its own when the corresponding PORT_GOODx# or GPIOx input de-asserts to the inactive state. The active state of each interrupt is controlled by its respective GPIO 0_11 Interrupt Polarity register bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. offset 62Ch). Notes: The bits in this register can be masked by their respective GPIO 0_11 Interrupt Mask register bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port. offset 63Ch). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD0# interrupt is inactive 1 = PORT_GOOD0# interrupt is active	ROS	No	0
1	PORT_GOOD1# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD1# interrupt is inactive 1 = PORT_GOOD1# interrupt is active	ROS	No	0
2	PORT_GOOD2# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD2# interrupt is inactive 1 = PORT_GOOD2# interrupt is active	ROS	No	0
3	PORT_GOOD3# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD3# interrupt is inactive 1 = PORT_GOOD3# interrupt is active	ROS	No	0

Register 13-140. 634h GPIO 0_11 Interrupt Status**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	PORT_GOOD4# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD4# interrupt is inactive 1 = PORT_GOOD4# interrupt is active	ROS	No	0
5	PORT_GOOD5# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD5# interrupt is inactive 1 = PORT_GOOD5# interrupt is active	ROS	No	0
6	PORT_GOOD6# Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = PORT_GOOD6# interrupt is inactive 1 = PORT_GOOD6# interrupt is active	ROS	No	0
7	GPIO7 Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = GPIO7 interrupt is inactive 1 = GPIO7 interrupt is active	ROS	No	0
8	GPIO8 Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = GPIO8 interrupt is inactive 1 = GPIO8 interrupt is active	ROS	No	0
9	GPIO9 Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = GPIO9 interrupt is inactive 1 = GPIO9 interrupt is active	ROS	No	0
10	GPIO10 Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = GPIO10 interrupt is inactive 1 = GPIO10 interrupt is active	ROS	No	0
11	Base Mode <i>Reserved</i>	RsvdP	No	0
	Virtual Switch Mode GPIO11 Interrupt Status Indicates whether the GPIO interrupt is inactive or active. 0 = GPIO11 interrupt is inactive 1 = GPIO11 interrupt is active	Four enabled virtual switches ROS	No	0
	<i>Reserved</i>	Otherwise RsvdP	No	0
31:12	<i>Reserved</i>	RsvdP	No	0000_0h

Register 13-141. 63Ch GPIO 0_11 Interrupt Mask**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The PORT_GOOD[0:6]# and GPIO[7:11] Interrupt inputs are masked, using the GPIO 0_11 Interrupt Mask registers, respectively. Notes: The bits in this register can be used to mask their respective GPIO 0_11 Interrupt Status register bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 634h). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	PORT_GOOD0# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD0# interrupt is not masked. 1 = PORT_GOOD0# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
1	PORT_GOOD1# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD1# interrupt is not masked. 1 = PORT_GOOD1# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
2	PORT_GOOD2# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD2# interrupt is not masked. 1 = PORT_GOOD2# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
3	PORT_GOOD3# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD3# interrupt is not masked. 1 = PORT_GOOD3# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1

Register 13-141. 63Ch GPIO 0_11 Interrupt Mask**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	PORT_GOOD4# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD4# interrupt is not masked. 1 = PORT_GOOD4# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
5	PORT_GOOD5# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD5# interrupt is not masked. 1 = PORT_GOOD5# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
6	PORT_GOOD6# Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = PORT_GOOD6# interrupt is not masked. 1 = PORT_GOOD6# interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
7	GPIO7 Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = GPIO7 interrupt is not masked. 1 = GPIO7 interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1

Register 13-141. 63Ch GPIO 0_11 Interrupt Mask**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default	
8	GPIO8 Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = GPIO8 interrupt is not masked. 1 = GPIO8 interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.		RWS	Yes	1	
9	GPIO9 Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = GPIO9 interrupt is not masked. 1 = GPIO9 interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.		RWS	Yes	1	
10	GPIO10 Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = GPIO10 interrupt is not masked. 1 = GPIO10 interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.		RWS	Yes	1	
11	Base Mode <i>Reserved</i>		RsvdP	No	0	
	Virtual Switch Mode GPIO11 Interrupt Mask Indicates whether the GPIO interrupt is masked or not masked. 0 = GPIO11 interrupt is not masked. 1 = GPIO11 interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	Four enabled virtual switches	RWS	Yes	1	
	<i>Reserved</i>		Otherwise	RsvdP	No	0
31:12	<i>Reserved</i>		RsvdP	No	0000_0h	

Register 13-142. 64Ch Virtual Switch GPIO Update
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	VS GPIOs Update After the Management Port, serial EEPROM, and/or I ² C/SMBus configures the PORT_GOODx# and GPIOx signals, and assigns these individual signals to various virtual switches, by programming the VSx GPIO_PG 0_11 Assignment register(s) (Port 0, accessible through the Management Port, offset(s) 650h through 65Ch, respectively), Set this bit to cause the GPIO assignments to take effect.	RW1CS	Yes	0
31:1	Reserved	RsvdP	No	0-0h

Register 13-143. 650h VS0 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Each of bits [11:0] in this register uniquely corresponds to one of 12 PORT_GOODx#/GPIOx signals, where x in the signal name matches the bit position within this register. If bit(s) in this register are Set, the corresponding PORT_GOODx#/GPIOx signal(s) is (are) assigned to VS0. After GPIO assignment is complete (by the Management Port or I²C/SMBus Setting the Virtual Switch GPIO Update register VS GPIOs Update bit (Port 0, accessible through the Management Port, offset 64Ch[0]), the Virtual Switch GPIO_PG 0_11 Availability register Number of GPIO_PGs Available field (VS Upstream Port(s), offset A3Ch[3:0]) indicates the total quantity of PORT_GOODx#/GPIOx signals that are assigned to VS0 (same as the quantity of bits that are Set in this register). Notes: Each virtual switch uses its own local set of GPIO registers (VS Upstream Port(s), offsets A34h through A54h). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	VS0 GPIO_PG 0 Assignment 0 = PORT_GOOD0# is not assigned to VS0 1 = PORT_GOOD0# is assigned to VS0	RWS	Yes	Signal
1	VS0 GPIO_PG 1 Assignment 0 = PORT_GOOD1# is not assigned to VS0 1 = PORT_GOOD1# is assigned to VS0	RWS	Yes	Signal
2	VS0 GPIO_PG 2 Assignment 0 = PORT_GOOD2# is not assigned to VS0 1 = PORT_GOOD2# is assigned to VS0	RWS	Yes	Signal
3	VS0 GPIO_PG 3 Assignment 0 = PORT_GOOD3# is not assigned to VS0 1 = PORT_GOOD3# is assigned to VS0	RWS	Yes	Signal

Register 13-143. 650h VS0 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS0 GPIO_PG 4 Assignment 0 = PORT_GOOD4# is not assigned to VS0 1 = PORT_GOOD4# is assigned to VS0	RWS	Yes	Signal
5	VS0 GPIO_PG 5 Assignment 0 = PORT_GOOD5# is not assigned to VS0 1 = PORT_GOOD5# is assigned to VS0	RWS	Yes	Signal
6	VS0 GPIO_PG 6 Assignment 0 = PORT_GOOD6# is not assigned to VS0 1 = PORT_GOOD6# is assigned to VS0	RWS	Yes	Signal
7	VS0 GPIO_PG 7 Assignment 0 = GPIO7 is not assigned to VS0 1 = GPIO7 is assigned to VS0	RWS	Yes	Signal
8	VS0 GPIO_PG 8 Assignment 0 = GPIO8 is not assigned to VS0 1 = GPIO8 is assigned to VS0	RWS	Yes	Signal
9	VS0 GPIO_PG 9 Assignment 0 = GPIO9 is not assigned to VS0 1 = GPIO9 is assigned to VS0	RWS	Yes	Signal
10	VS0 GPIO_PG 10 Assignment 0 = GPIO10 is not assigned to VS0 1 = GPIO10 is assigned to VS0	RWS	Yes	Signal
11	VS0 GPIO_PG 11 Assignment 0 = GPIO11 is not assigned to VS0 1 = GPIO11 is assigned to VS0	RWS	Yes	Signal
	Reserved	Otherwise	No	0
31:12	Reserved	RsvdP	No	0000_0h

Register 13-144. 654h VS1 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Each of bits [11:0] in this register uniquely corresponds to one of 12 PORT_GOODx#/GPIOx signals, where x in the signal name matches the bit position within this register. If bit(s) in this register are Set, the corresponding PORT_GOODx#/GPIOx signal(s) is (are) assigned to VS0. After GPIO assignment is complete (by the Management Port or I²C/SMBus Setting the Virtual Switch GPIO Update register <i>VS GPIOs Update</i> bit (Port 0, accessible through the Management Port, offset 64Ch[0]), the Virtual Switch GPIO_PG 0_11 Availability register <i>Number of GPIO_PGs Available</i> field (VS Upstream Port(s), offset A3Ch[3:0]) indicates the total quantity of PORT_GOODx#/GPIOx signals that are assigned to VS1 (same as the quantity of bits that are Set in this register). Notes: Each virtual switch uses its own local set of GPIO registers (VS Upstream Port(s), offsets A34h through A54h). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	VS1 GPIO_PG 0 Assignment 0 = PORT_GOOD0# is not assigned to VS1 1 = PORT_GOOD0# is assigned to VS1	RWS	Yes	Signal
1	VS1 GPIO_PG 1 Assignment 0 = PORT_GOOD1# is not assigned to VS1 1 = PORT_GOOD1# is assigned to VS1	RWS	Yes	Signal
2	VS1 GPIO_PG 2 Assignment 0 = PORT_GOOD2# is not assigned to VS1 1 = PORT_GOOD2# is assigned to VS1	RWS	Yes	Signal
3	VS1 GPIO_PG 3 Assignment 0 = PORT_GOOD3# is not assigned to VS1 1 = PORT_GOOD3# is assigned to VS1	RWS	Yes	Signal
4	VS1 GPIO_PG 4 Assignment 0 = PORT_GOOD4# is not assigned to VS1 1 = PORT_GOOD4# is assigned to VS1	RWS	Yes	Signal
5	VS1 GPIO_PG 5 Assignment 0 = PORT_GOOD5# is not assigned to VS1 1 = PORT_GOOD5# is assigned to VS1	RWS	Yes	Signal
6	VS1 GPIO_PG 6 Assignment 0 = PORT_GOOD6# is not assigned to VS1 1 = PORT_GOOD6# is assigned to VS1	RWS	Yes	Signal
7	VS1 GPIO_PG 7 Assignment 0 = GPIO7 is not assigned to VS1 1 = GPIO7 is assigned to VS1	RWS	Yes	Signal

Register 13-144. 654h VS1 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	VS1 GPIO_PG 8 Assignment 0 = GPIO8 is not assigned to VS1 1 = GPIO8 is assigned to VS1		RWS	Yes	Signal
9	VS1 GPIO_PG 9 Assignment 0 = GPIO9 is not assigned to VS1 1 = GPIO9 is assigned to VS1		RWS	Yes	Signal
10	VS1 GPIO_PG 10 Assignment 0 = GPIO10 is not assigned to VS1 1 = GPIO10 is assigned to VS1		RWS	Yes	Signal
11	VS1 GPIO_PG 11 Assignment 0 = GPIO11 is not assigned to VS1 1 = GPIO11 is assigned to VS1	Four enabled virtual switches	RWS	Yes	Signal
	Reserved	Otherwise	RsvdP	No	0
31:12	Reserved		RsvdP	No	0000_0h

Register 13-145. 658h VS2 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Each of bits [11:0] in this register uniquely corresponds to one of 12 PORT_GOODx#/GPIOx signals, where x in the signal name matches the bit position within this register. If bit(s) in this register are Set, the corresponding PORT_GOODx#/GPIOx signal(s) is (are) assigned to VS0. After GPIO assignment is complete (by the Management Port or I²C/SMBus Setting the Virtual Switch GPIO Update register <i>VS GPIOs Update</i> bit (Port 0, accessible through the Management Port, offset 64Ch[0]), the Virtual Switch GPIO_PG 0_11 Availability register <i>Number of GPIO_PGs Available</i> field (VS Upstream Port(s), offset A3Ch[3:0]) indicates the total quantity of PORT_GOODx#/GPIOx signals that are assigned to VS2 (same as the quantity of bits that are Set in this register). Notes: Each virtual switch uses its own local set of GPIO registers (VS Upstream Port(s), offsets A34h through A54h). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	VS2 GPIO_PG 0 Assignment 0 = PORT_GOOD0# is not assigned to VS2 1 = PORT_GOOD0# is assigned to VS2	RWS	Yes	Signal
1	VS2 GPIO_PG 1 Assignment 0 = PORT_GOOD1# is not assigned to VS2 1 = PORT_GOOD1# is assigned to VS2	RWS	Yes	Signal
2	VS2 GPIO_PG 2 Assignment 0 = PORT_GOOD2# is not assigned to VS2 1 = PORT_GOOD2# is assigned to VS2	RWS	Yes	Signal
3	VS2 GPIO_PG 3 Assignment 0 = PORT_GOOD3# is not assigned to VS2 1 = PORT_GOOD3# is assigned to VS2	RWS	Yes	Signal
4	VS2 GPIO_PG 4 Assignment 0 = PORT_GOOD4# is not assigned to VS2 1 = PORT_GOOD4# is assigned to VS2	RWS	Yes	Signal
5	VS2 GPIO_PG 5 Assignment 0 = PORT_GOOD5# is not assigned to VS2 1 = PORT_GOOD5# is assigned to VS2	RWS	Yes	Signal
6	VS2 GPIO_PG 6 Assignment 0 = PORT_GOOD6# is not assigned to VS2 1 = PORT_GOOD6# is assigned to VS2	RWS	Yes	Signal
7	VS2 GPIO_PG 7 Assignment 0 = GPIO7 is not assigned to VS2 1 = GPIO7 is assigned to VS2	RWS	Yes	Signal

Register 13-145. 658h VS2 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	VS2 GPIO_PG 8 Assignment 0 = GPIO8 is not assigned to VS2 1 = GPIO8 is assigned to VS2		RWS	Yes	Signal
9	VS2 GPIO_PG 9 Assignment 0 = GPIO9 is not assigned to VS2 1 = GPIO9 is assigned to VS2		RWS	Yes	Signal
10	VS2 GPIO_PG 10 Assignment 0 = GPIO10 is not assigned to VS2 1 = GPIO10 is assigned to VS2		RWS	Yes	Signal
11	VS2 GPIO_PG 11 Assignment 0 = GPIO11 is not assigned to VS2 1 = GPIO11 is assigned to VS2	Four enabled virtual switches	RWS	Yes	Signal
	Reserved	Otherwise	RsvdP	No	0
31:12	Reserved		RsvdP	No	0000_0h

Register 13-146. 65Ch VS3 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Each of bits [11:0] in this register uniquely corresponds to one of 12 PORT_GOODx#/GPIOx signals, where x in the signal name matches the bit position within this register. If bit(s) in this register are Set, the corresponding PORT_GOODx#/GPIOx signal(s) is (are) assigned to VS0. After GPIO assignment is complete (by the Management Port or I²C/SMBus Setting the Virtual Switch GPIO Update register <i>VS GPIOs Update</i> bit (Port 0, accessible through the Management Port, offset 64Ch[0]), the Virtual Switch GPIO_PG 0_11 Availability register <i>Number of GPIO_PGs Available</i> field (VS Upstream Port(s), offset A3Ch[3:0]) indicates the total quantity of PORT_GOODx#/GPIOx signals that are assigned to VS3 (same as the quantity of bits that are Set in this register). Notes: Each virtual switch uses its own local set of GPIO registers (VS Upstream Port(s), offsets A34h through A54h). PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
0	VS3 GPIO_PG 0 Assignment 0 = PORT_GOOD0# is not assigned to VS3 1 = PORT_GOOD0# is assigned to VS3	RWS	Yes	Signal
1	VS3 GPIO_PG 1 Assignment 0 = PORT_GOOD1# is not assigned to VS3 1 = PORT_GOOD1# is assigned to VS3	RWS	Yes	Signal
2	VS3 GPIO_PG 2 Assignment 0 = PORT_GOOD2# is not assigned to VS3 1 = PORT_GOOD2# is assigned to VS3	RWS	Yes	Signal
3	VS3 GPIO_PG 3 Assignment 0 = PORT_GOOD3# is not assigned to VS3 1 = PORT_GOOD3# is assigned to VS3	RWS	Yes	Signal
4	VS3 GPIO_PG 4 Assignment 0 = PORT_GOOD4# is not assigned to VS3 1 = PORT_GOOD4# is assigned to VS3	RWS	Yes	Signal
5	VS3 GPIO_PG 5 Assignment 0 = PORT_GOOD5# is not assigned to VS3 1 = PORT_GOOD5# is assigned to VS3	RWS	Yes	Signal
6	VS3 GPIO_PG 6 Assignment 0 = PORT_GOOD6# is not assigned to VS3 1 = PORT_GOOD6# is assigned to VS3	RWS	Yes	Signal
7	VS3 GPIO_PG 7 Assignment 0 = GPIO7 is not assigned to VS3 1 = GPIO7 is assigned to VS3	RWS	Yes	Signal

Register 13-146. 65Ch VS3 GPIO_PG 0_11 Assignment
(Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
8	VS3 GPIO_PG 8 Assignment 0 = GPIO8 is not assigned to VS3 1 = GPIO8 is assigned to VS3		RWS	Yes	Signal
9	VS3 GPIO_PG 9 Assignment 0 = GPIO9 is not assigned to VS3 1 = GPIO9 is assigned to VS3		RWS	Yes	Signal
10	VS3 GPIO_PG 10 Assignment 0 = GPIO10 is not assigned to VS3 1 = GPIO10 is assigned to VS3		RWS	Yes	Signal
11	VS3 GPIO_PG 11 Assignment 0 = GPIO11 is not assigned to VS3 1 = GPIO11 is assigned to VS3	Four enabled virtual switches	RWS	Yes	Signal
	Reserved	Otherwise	RsvdP	No	0
31:12	Reserved		RsvdP	No	0000_0h

13.16.10 Device-Specific Registers – PORT_GOOD Selector (Offsets 6A0h – 6A8h)

This section details the Device-Specific PORT_GOOD Selector registers. [Table 13-27](#) defines the register map.

Table 13-27. Device-Specific PORT_GOOD Selector Register Map (Offsets 6A0h – 6A8h) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
PORT_GOOD Selector 0																																6A0h
PORT_GOOD Selector 1																																6A4h
Reserved																																6A8h

Register 13-147. 6A0h PORT_GOOD Selector 0**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Assigns PORT_GOODx#/GPIOx signal(s) that are enabled for PORT_GOOD output functionality (in the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and/or 604h), in which default values are globally defined, according to the STRAP_TESTMODE[2:0] input states), to designated Port(s). Has no effect on PORT_GOODx#/GPIOx signals that are enabled for Serial Hot Plug (SHP_PERSTx#) or GPIO functionality. 0_0000b = Port 0 0_1000b = Port 8 0_0001b = Port 1 0_1001b = Port 9 0_0010b = Port 2 0_1010b = Port 10 0_0011b = Port 3 All other <i>Port Up Select x</i> field encodings are Reserved. Notes: PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
4:0	Port Up Select 0 Use this field to program a different Port to be associated with PORT_GOOD0#. The default for PORT_GOOD0# is Port 0.	RWS	Yes	0_0000b
7:5	Reserved	RsvdP	No	000b
12:8	Port Up Select 1 Use this field to program a different Port to be associated with PORT_GOOD1#. The default for PORT_GOOD1# is Port 1.	RWS	Yes	0_0001b
15:13	Reserved	RsvdP	No	000b
20:16	Port Up Select 2 Use this field to program a different Port to be associated with PORT_GOOD2#. The default for PORT_GOOD2# is Port 2.	RWS	Yes	0_0010b
23:21	Reserved	RsvdP	No	000b
28:24	Port Up Select 3 Use this field to program a different Port to be associated with PORT_GOOD3#. The default for PORT_GOOD3# is Port 3.	RWS	Yes	0_0011b
31:29	Reserved	RsvdP	No	000b

Register 13-148. 6A4h PORT_GOOD Selector 1**(Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Assigns PORT_GOODx#/GPIOx signal(s) that are enabled for PORT_GOOD output functionality (in the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 600h and/or 604h), in which default values are globally defined, according to the STRAP_TESTMODE[2:0] input states), to designated Port(s). Has no effect on PORT_GOODx#/GPIOx signals that are enabled for Serial Hot Plug (SHP_PERSTx#) or GPIO functionality. 0_0000b = Port 0 0_1000b = Port 8 0_0001b = Port 1 0_1001b = Port 9 0_0010b = Port 2 0_1010b = Port 10 0_0011b = Port 3 All other <i>Port Up Select x</i> field encodings are Reserved. Notes: PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively, by default. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the PORT_GOOD Selector x register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) 6A0h and 6A4h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.				
4:0	Port Up Select 4 Use this field to program a different Port to be associated with PORT_GOOD4#. The default for PORT_GOOD4# is Port 8.	RWS	Yes	0_1000b
7:5	Reserved	RsvdP	No	000b
12:8	Port Up Select 5 Use this field to program a different Port to be associated with PORT_GOOD5#. The default for PORT_GOOD5# is Port 9.	RWS	Yes	0_1001b
15:13	Reserved	RsvdP	No	000b
20:16	Port Up Select 6 Use this field to program a different Port to be associated with PORT_GOOD6#. The default for PORT_GOOD6# is Port 10.	RWS	Yes	0_1010b
31:21	Reserved	RsvdP	No	0-0h

13.16.11 Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch)

This section details the Device-Specific Error Checking and Debug registers located at offsets 700h through 75Ch. [Table 13-28](#) defines the register map.

Other Device-Specific Error Checking and Debug registers are detailed in [Section 13.22.3](#), “Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h).”

Note: The PEX87XX Error Injection White Paper, available from the PLX PEX 8724 Toolbox, includes cross-reference information for some of the **Error Checking and Debug** registers and their associated **Error Counter** registers, located at offsets 800h through 828h.

Table 13-28. Device-Specific Error Checking and Debug Register Map (Offsets 700h – 75Ch) (Ports^a)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0														Offsets
Device-Specific Error Status 1																														700h
Device-Specific Error Mask 1																														704h
Device-Specific Error Status 2																														708h
Device-Specific Error Mask 2																														70Ch
Reserved		Device-Specific Error Status 3																												710h
Reserved		Device-Specific Error Mask 3																												714h
Reserved		Device-Specific Error Status 4																												718h
Reserved		Device-Specific Error Mask 4																												71Ch
Error Check Disable																														720h
Reserved																Gen 3 Framing Error Status														724h
Reserved																Gen 3 Framing Error Mask														728h
Reserved																Gen 3 Framing Error Disable														72Ch
Reserved																														730h – 75Ch

a. Certain registers are Port-specific, others are Station-specific or Chip-specific; all are Device-specific.

Register 13-149. 700h Device-Specific Error Status 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
ECC is checked by the egress Port, and any ECC error is reported by the egress Station (containing the egress Port), when that Port reads the Packet data from the ingress Station RAM. <i>For example</i>, when Port 8 receives a packet, the data is stored in Station 1 RAM; then, if the egress Port is in Station 0 and an ECC error is detected when the Port reads the Station 1 RAM, the error is flagged in the Port 0 register bit that reflects Station 1 error status. Notes: The bits in this register can be masked by their respective Device-Specific Error Mask 1 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 704h). All errors in this register generate an ERR_COR Message, if enabled by the following: • Device-Specific Error Mask 1 register • Correctable Error Status register Corrected Internal Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FC4h[14]) • Port's Device Status register Correctable Error Reporting Enable bit (offset 70h[0]) The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.					
0	Factory Test Only		RsvdP	No	0
1	Factory Test Only	0	RW1CS	No	0
2	Station 0 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
3	Station 1 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
4	Reserved		RsvdP	No	0
7:5	Factory Test Only		RW1CS	Yes	000b
8	Station 0 Header RAM0 Instance0 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
9	Station 1 Header RAM0 Instance0 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
10	Reserved		RsvdP	No	0
11	Station 0 Header RAM0 Instance1 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0

Register 13-149. 700h Device-Specific Error Status 1
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
12	Station 1 Header RAM0 Instance1 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
13	Reserved		RsvdP	No	0
14	Station 0 Header RAM1 Instance0 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
15	Station 1 Header RAM1 Instance0 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
16	Reserved		RsvdP	No	0
17	Station 0 Header RAM1 Instance1 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
18	Station 1 Header RAM1 Instance1 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
19	Reserved		RsvdP	No	0
22:20	Factory Test Only		RsvdP	No	000b
23	Destination Queue Link List RAM 1-Bit ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0

Register 13-149. 700h Device-Specific Error Status 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
24	Source Queue Link List RAM 1-Bit ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
25	Retry Buffer 1-Bit ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
26	Ingress Link List RAM ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
27	Source Queue Link List RAM2 ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
28	Destination Queue Data RAM ECC Error Counter Overflow Detected 0 = No overflow is detected 1 = 1-bit ECC Error Counter Overflow is detected		RW1CS	Yes	0
31:29	Reserved		RsvdP	No	000b

Register 13-150. 704h Device-Specific Error Mask 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Device-Specific Error Status 1 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 700h). The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.					
0	Factory Test Only		RWS	Yes	1
1	Factory Test Only	0	RWS	Yes	1
2	Station 0 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = Station 0 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Detected bit is masked/disabled		RWS	Yes	1
3	Station 1 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = Station 1 Egress Packet Link List RAM 1-Bit ECC Error Counter Overflow Detected bit is masked/disabled		RWS	Yes	1
4	Reserved		RsvdP	No	0
7:5	Factory Test Only		RWS	Yes	000b
8	Station 0 Header RAM0 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = Station 0 Header RAM0 Instance0 ECC Error Counter Overflow Detected bit is masked/disabled		RWS	Yes	1
9	Station 1 Header RAM0 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = Station 1 Header RAM0 Instance0 ECC Error Counter Overflow Detected bit is masked/disabled		RWS	Yes	1
10	Reserved		RsvdP	No	0
11	Station 0 Header RAM0 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = Station 0 Header RAM0 Instance1 ECC Error Counter Overflow Detected bit is masked/disabled		RWS	Yes	1

Register 13-150. 704h Device-Specific Error Mask 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
12	Station 1 Header RAM0 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM0 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
13	Reserved		RsvdP	No	0
14	Station 0 Header RAM1 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Header RAM1 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
15	Station 1 Header RAM1 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM1 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
16	Reserved		RsvdP	No	0
17	Station 0 Header RAM1 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Header RAM1 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
18	Station 1 Header RAM1 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM1 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
19	Reserved		RsvdP	No	0
22:20	Factory Test Only		RsvdP	No	000b
23	Destination Queue Link List RAM 1-Bit ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Destination Queue Link List RAM 1-Bit ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1

Register 13-150. 704h Device-Specific Error Mask 1**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
24	Source Queue Link List RAM 1-Bit ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Source Queue Link List RAM 1-Bit ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
25	Retry Buffer 1-Bit ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Retry Buffer 1-Bit ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
26	Ingress Link List RAM ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Ingress Link List RAM ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
27	Source Queue Link List RAM2 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Source Queue Link List RAM2 ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
28	Destination Queue Data RAM ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Destination Queue Data RAM ECC Error Counter Overflow Detected</i> bit is masked/disabled		RWS	Yes	1
31:29	Reserved		RsvdP	No	000b

Register 13-151. 708h Device-Specific Error Status 2

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Device-Specific Error Mask 2 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 70Ch). All errors in this register generate ERR_FATAL or ERR_NONFATAL Messages, if enabled by the following: • Device-Specific Error Mask 2 register • Uncorrectable Error Status register Uncorrectable Internal Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FB8h[22]) • Port's Device Status register Fatal Error Reporting Enable and Non-Fatal Error Reporting Enable bits (offset 70h[2:1], respectively) The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Reserved	RsvdP	No	0
1	Ingress Freelist Underrun Error Detected 0 = No error is detected 1 = Underrun error is detected	RW1CS	Yes	0
2	Station 0 Egress Packet Link List RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
3	Station 1 Egress Packet Link List RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
4	Reserved	RsvdP	No	0
7:5	Factory Test Only	RW1CS	Yes	000b
8	Station 0 Header RAM0 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
9	Station 1 Header RAM0 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
10	Reserved	RsvdP	No	0
11	Station 0 Header RAM0 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0

Register 13-151. 708h Device-Specific Error Status 2
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 1 Header RAM0 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
13	Reserved	RsvdP	No	0
14	Station 0 Header RAM1 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
15	Station 1 Header RAM1 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
16	Reserved	RsvdP	No	0
17	Station 0 Header RAM1 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
18	Station 1 Header RAM1 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
19	Reserved	RsvdP	No	0
22:20	Factory Test Only	RsvdP	No	000b
23	Destination Queue Link List RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0

Register 13-151. 708h Device-Specific Error Status 2

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Source Queue Link List RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
25	Retry Buffer 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
26	Ingress Link List RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
27	Source Queue Link List RAM2 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
28	Destination Queue Data RAM 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
29	Ingress Parity Error Detected 0 = No error is detected 1 = Ingress Parity error detected	RW1CS	Yes	0
30	Egress Parity Error Detected 0 = No error is detected 1 = Egress Parity error detected	RW1CS	Yes	0
31	Reserved	RsvdP	No	0

Register 13-152. 70Ch Device-Specific Error Mask 2
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Device-Specific Error Status 2 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 708h). The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Reserved	RsvdP	No	0
1	Ingress Freelist Underrun Error Mask 0 = No effect on reporting activity 1 = Ingress Freelist Underrun Error Detected bit is masked/disabled	RWS	Yes	1
2	Station 0 Egress Packet Link List RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = Station 0 Egress Packet Link List RAM 2-Bit Error Detected bit is masked/disabled	RWS	Yes	1
3	Station 1 Egress Packet Link List RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = Station 1 Egress Packet Link List RAM 2-Bit Error Detected bit is masked/disabled	RWS	Yes	1
4	Reserved	RsvdP	No	0
7:5	Factory Test Only	RWS	Yes	000b
8	Station 0 Header RAM0 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = Station 0 Header RAM0 Instance0 2-Bit Error Detected bit is masked/disabled	RWS	Yes	1
9	Station 1 Header RAM0 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = Station 1 Header RAM0 Instance0 2-Bit Error Detected bit is masked/disabled	RWS	Yes	1
10	Reserved	RsvdP	No	0
11	Station 0 Header RAM0 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = Station 0 Header RAM0 Instance1 2-Bit Error Detected bit is masked/disabled	RWS	Yes	1

Register 13-152. 70Ch Device-Specific Error Mask 2

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 1 Header RAM0 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM0 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
13	Reserved	RsvdP	No	0
14	Station 0 Header RAM1 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Header RAM1 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
15	Station 1 Header RAM1 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM1 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
16	Reserved	RsvdP	No	0
17	Station 0 Header RAM1 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Header RAM1 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
18	Station 1 Header RAM1 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Header RAM1 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
19	Reserved	RsvdP	No	0
22:20	Factory Test Only	RsvdP	No	000b
23	Destination Queue Link List RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Destination Queue Link List RAM 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1

Register 13-152. 70Ch Device-Specific Error Mask 2
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Source Queue Link List RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Source Queue Link List RAM 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
25	Retry Buffer 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Retry Buffer 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
26	Ingress Link List RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Ingress Link List RAM 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
27	Source Queue Link List RAM2 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Source Queue Link List RAM2 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
28	Destination Queue Data RAM 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Destination Queue Data RAM 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
29	Ingress Parity Error Mask 0 = No effect on reporting activity 1 = <i>Ingress Parity Error Detected</i> bit is masked/disabled	RWS	Yes	1
30	Egress Parity Error Mask 0 = No effect on reporting activity 1 = <i>Egress Parity Error Detected</i> bit is masked/disabled	RWS	Yes	1
31	Reserved	RsvdP	No	0

Register 13-153. 710h Device-Specific Error Status 3

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
ECC is checked by the egress Port, and any ECC error is reported by the egress Station (containing the egress Port), when that Port reads the Packet data from the ingress Station RAM. <i>For example</i>, when Port 8 receives a packet, the data is stored in Station 1 RAM; then, if the egress Port is in Station 0 and an ECC error is detected when the Port reads the Station 1 RAM, the error is flagged in the Port 0 register bit that reflects Station 1 error status. Notes: The bits in this register can be masked by their respective Device-Specific Error Mask 3 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 714h). All errors in this register generate an ERR_COR Message, if enabled by the following: • Device-Specific Error Mask 3 register • Correctable Error Status register Corrected Internal Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FC4h[14]) • Port's Device Status register Correctable Error Reporting Enable bit (offset 70h[0]) The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Station 0 Packet RAM0 Instance0 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
1	Station 1 Packet RAM0 Instance0 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
2	Reserved	RsvdP	No	0
5:3	Factory Test Only	RW1CS	Yes	000b
6	Station 0 Packet RAM0 Instance1 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
7	Station 1 Packet RAM0 Instance1 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
8	Reserved	RsvdP	No	0
11:9	Factory Test Only	RW1CS	Yes	000b

Register 13-153. 710h Device-Specific Error Status 3
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 0 Packet RAM1 Instance0 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
13	Station 1 Packet RAM1 Instance0 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
14	Reserved	RsvdP	No	0
17:15	Factory Test Only	RW1CS	Yes	000b
18	Station 0 Packet RAM1 Instance1 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
19	Station 1 Packet RAM1 Instance1 ECC Error Counter Overflow Detected 0 = No error is detected 1 = 1-bit ECC Error Counter Overflow is detected	RW1CS	Yes	0
20	Reserved	RsvdP	No	0
23:21	Factory Test Only	RsvdP	No	000b
31:24	Reserved	RsvdP	No	00h

Register 13-154. 714h Device-Specific Error Mask 3

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Device-Specific Error Status 3 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 710h). The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Station 0 Packet RAM0 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM0 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
1	Station 1 Packet RAM0 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM0 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
2	Reserved	RsvdP	No	0
5:3	Factory Test Only	RWS	Yes	000b
6	Station 0 Packet RAM0 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM0 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
7	Station 1 Packet RAM0 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM0 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
8	Reserved	RsvdP	No	0
11:9	Factory Test Only	RWS	Yes	000b

Register 13-154. 714h Device-Specific Error Mask 3
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 0 Packet RAM1 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM1 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
13	Station 1 Packet RAM1 Instance0 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM1 Instance0 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
14	Reserved	RsvdP	No	0
17:15	Factory Test Only	RWS	Yes	000b
18	Station 0 Packet RAM1 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM1 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
19	Station 1 Packet RAM1 Instance1 ECC Error Counter Overflow Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM1 Instance1 ECC Error Counter Overflow Detected</i> bit is masked/disabled	RWS	Yes	1
20	Reserved	RsvdP	No	0
23:21	Factory Test Only	RsvdP	No	000b
31:24	Reserved	RsvdP	No	00h

Register 13-155. 718h Device-Specific Error Status 4

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
ECC is checked by the egress Port, and any ECC error is reported by the egress Station (containing the egress Port), when that Port reads the Packet data from the ingress Station RAM. <i>For example</i>, when Port 8 receives a packet, the data is stored in Station 1 RAM; then, if the egress Port is in Station 0 and an ECC error is detected when the Port reads the Station 1 RAM, the error is flagged in the Port 0 register bit that reflects Station 1 error status. Notes: The bits in this register can be masked by their respective Device-Specific Error Mask 4 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 71Ch). All errors in this register generate ERR_FATAL or ERR_NONFATAL Messages, if enabled by the following: Device-Specific Error Mask 4 register Uncorrectable Error Status register Uncorrectable Internal Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FB8h[22]) - Port's Device Status register Fatal Error Reporting Enable and Non-Fatal Error Reporting Enable bits (offset 70h[2:1], respectively) The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Station 0 Packet RAM0 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
1	Station 1 Packet RAM0 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
2	Reserved	RsvdP	No	0
5:3	Factory Test Only	RW1CS	Yes	000b
6	Station 0 Packet RAM0 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
7	Station 1 Packet RAM0 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
8	Reserved	RsvdP	No	0
11:9	Factory Test Only	RW1CS	Yes	000b

Register 13-155. 718h Device-Specific Error Status 4
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 0 Packet RAM1 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
13	Station 1 Packet RAM1 Instance0 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
14	Reserved	RsvdP	No	0
17:15	Factory Test Only	RW1CS	Yes	000b
18	Station 0 Packet RAM1 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
19	Station 1 Packet RAM1 Instance1 2-Bit Error Detected 0 = No error is detected 1 = 2-bit error is detected	RW1CS	Yes	0
20	Reserved	RsvdP	No	0
23:21	Factory Test Only	RsvdP	No	000b
31:24	Reserved	RsvdP	No	00h

Register 13-156. 71Ch Device-Specific Error Mask 4

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Device-Specific Error Status 4 register bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 718h). The Stations are defined in Table 13-5. The Port 0 bits are for Station 0. The Port 8 bits are for Station 1. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h.				
0	Station 0 Packet RAM0 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM0 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
1	Station 1 Packet RAM0 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM0 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
2	Reserved	RsvdP	No	0
5:3	Factory Test Only	RWS	Yes	000b
6	Station 0 Packet RAM0 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM0 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
7	Station 1 Packet RAM0 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM0 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
8	Reserved	RsvdP	No	0
11:9	Factory Test Only	RWS	Yes	000b

Register 13-156. 71Ch Device-Specific Error Mask 4
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
12	Station 0 Packet RAM1 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM1 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
13	Station 1 Packet RAM1 Instance0 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM1 Instance0 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
14	Reserved	RsvdP	No	0
17:15	Factory Test Only	RWS	Yes	000b
18	Station 0 Packet RAM1 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 0 Packet RAM1 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
19	Station 1 Packet RAM1 Instance1 2-Bit Error Mask 0 = No effect on reporting activity 1 = <i>Station 1 Packet RAM1 Instance1 2-Bit Error Detected</i> bit is masked/disabled	RWS	Yes	1
20	Reserved	RsvdP	No	0
23:21	Factory Test Only	RsvdP	No	000b
31:24	Reserved	RsvdP	No	00h

Register 13-157. 720h Error Check Disable

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: PEX_INTA# is used only in Base mode. VSx_PEX_INTA# is used only in Virtual Switch mode. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	1-Bit ECC Error Check Disable 0 = RAM 1-Bit ECC Error Check enabled 1 = Disables RAM 1-Bit ECC Error Check	Base Mode 0 Virtual Switch Mode 0, accessible through the Management Port	RWS	Yes	0
1	2-Bit Error Check Disable 0 = RAM 2-Bit Error Check enabled 1 = Disables RAM 2-Bit Error Check	Base Mode 0 Virtual Switch Mode 0, accessible through the Management Port	RWS	Yes	0
2	Software Force Error Enable 1 = Correctable Error Status and Uncorrectable Error Status registers (offsets FC4h and FB8h, respectively) change from RW1CS to RW. Software Sets an Uncorrectable Error Status register bit when that register is Cleared, the Advanced Error Capabilities and Control register First Error Pointer field (offset FCCh[4:0]) is updated; however, the Header Log x registers (offsets FD0h through FDCh) remain unchanged.	Transparent Downstream	RWS	Yes	0
		Otherwise	RsvdP	No	0
3	Software Force Non-Posted Request Used to select software-forced errors to be associated with Posted or Non-Posted TLPs, because some errors are handled differently, depending upon the TLP type (Posted or Non-Posted). 0 = Handle software-forced errors as if the errors are associated with Posted TLPs 1 = Enables handling of errors associated with Posted TLPs as if those errors are associated with Non-Posted TLPs		RWS	Yes	0

Register 13-157. 720h Error Check Disable**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
4	Enable Ball Interrupt Output for Hot Plug or Link State Event-Triggered Interrupts 0 = Hot Plug or Link State Event Interrupt Requests send an INTx Message (and do not assert PEX_INTA# (Base mode) nor VSx_PEX_INTA# (Virtual Switch mode)) 1 = Hot Plug or Link State Event Interrupt Requests assert PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) (and do not send an INTx Message)		RWS	Yes	0
5	Transparent Mode Reserved		RsvdP	No	0
	NT Mode Enable Ball Interrupt Output for Device-Specific NT-Link Port Event-Triggered Interrupts Enables PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode), or INTx interrupt, signaling for the following NT-Link Interface events, defined in the Link Error Status Virtual and Link Error Mask Virtual registers (NT Port Virtual Interface, offsets FE0h and FE4h , respectively): - NT-Link Port Correctable error (NT Port Link Interface, offset FC4h) - NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h) - NT-Link Port Data Link Layer State change - NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message 0 = Device-Specific NT-Link Port Event Interrupt Requests send an INTx Message (and do not assert PEX_INTA# nor VSx_PEX_INTA#) 1 = Device-Specific NT-Link Port Event Interrupt Requests assert PEX_INTA# or VSx_PEX_INTA# (and do not send an INTx Message)	Base Mode 0 Virtual Switch Mode 0, accessible through the Management Port	RWS	Yes	0
	Reserved	Otherwise	RsvdP	No	0

Register 13-157. 720h Error Check Disable

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
6	Enable Ball Interrupt Output for GPIO-Generated Interrupts 0 = General-Purpose Input/Output (GPIO) Interrupt Requests send an INT _x Message (and do not assert PEX_INTA# (Base mode) nor VS _x PEX_INTA# (Virtual Switch mode)) 1 = GPIO Interrupt Requests assert PEX_INTA# (Base mode) or VS _x PEX_INTA# (Virtual Switch mode) (and do not send an INT _x Message)		RWS	Yes	0
7	Transparent Mode Reserved		RsvdP	No	0
	NT Mode Enable Ball Interrupt Output for NT-Virtual Doorbell-Generated Interrupts Enables PEX_INTA# (Base mode) or VS _x PEX_INTA# (Virtual Switch mode), or INT _x Messages, for NT-Virtual Doorbell interrupts (NT Port Virtual Interface, offsets C4Ch through C58h). 0 = NT-Virtual Doorbell Interrupt Requests send an INT _x Message (and do not assert PEX_INTA# nor VS _x PEX_INTA#) 1 = NT-Virtual Doorbell Interrupt Requests assert PEX_INTA# or VS _x PEX_INTA# (and do not send an INT _x Message)	Base Mode 0 Virtual Switch Mode 0, accessible through the Management Port	RWS	Yes	0
	Reserved	Otherwise	RsvdP	No	0

Register 13-157. 720h Error Check Disable**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
8	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode Enable Ball Interrupt Output for Management Port Doorbell-Generated Interrupts 0 = Management Port Doorbell Interrupt Requests send an INTx Message (and do not assert VSx_PEX_INTA#) 1 = Management Port Doorbell Interrupt Requests assert VSx_PEX_INTA# (and do not send an INTx Message)	0, accessible through the Management Port	RWS	Yes	0
9	Base Mode <i>Reserved</i>		RsvdP	No	0
	Virtual Switch Mode Enable Ball Interrupt Output for Management Link Status Event-Generated Interrupts 0 = Management Link Status Event Interrupt Requests send an INTx Message (and do not assert VSx_PEX_INTA#) 1 = Management Link Status Event Interrupt Requests assert VSx_PEX_INTA# (and do not send an INTx Message)	0, accessible through the Management Port	RWS	Yes	0
10	Disable Sending MSI if MSI Is Enabled after Interrupt Status Set 0 = Does not disable sending an MSI, if MSIs are enabled after an <i>Interrupt Status</i> bit is Set 1 = Disables sending an MSI, if MSIs are enabled after an <i>Interrupt Status</i> bit is Set <i>Note:</i> Must remain Cleared, for compliance to specifications governing the MSI Capability.		RWS	Yes	0
11	Egress Completion FIFO Overflow Mask <i>Note:</i> Can be used to mask bit 12 (Egress Completion FIFO Overflow Status).		RWS	Yes	1
12	Egress Completion FIFO Overflow Status <i>Note:</i> Can be masked by bit 11 (Egress Completion FIFO Overflow Mask).		RW1CS	Yes	0
31:13	<i>Reserved</i>		RsvdP	No	0-0h

Register 13-158. 724h Gen 3 Framing Error Status (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Gen 3 Framing Error Mask register bits (offset 728h). All errors in this register generate an ERR_COR Message, if enabled by the following: • Gen 3 Framing Error Mask register • Correctable Error Status register Receiver Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FC4h[0]) • Port's Device Status register Correctable Error Reporting Enable bit (offset 70h[0])				
0	Invalid Sync Header Status	RW1CS	Yes	0
1	Sync Header Not Same on All Lanes Status	RW1CS	Yes	0
2	Bad Framing Symbol in Data Block Status	RW1CS	Yes	0
3	Second Symbol of DLLP Not ACh Status 1 = Second symbol of the Data Link Layer Packet (DLLP) is not ACh	RW1CS	Yes	0
4	EDB Not following TLP Status	RW1CS	Yes	0
5	Length CRC/Parity Error Status Length Cyclic Redundancy Check (CRC)/Parity Error status.	RW1CS	Yes	0
6	End of Data Stream Not Followed by Ordered-Set Block Status	RW1CS	Yes	0
7	End of Data Stream Not in Last DWord of Data Block Status	RW1CS	Yes	0
8	Ordered-Set Block Not Followed by Data Block Status	RW1CS	Yes	0
9	FTS Not Followed by Data Block Status	RW1CS	Yes	0
10	Invalid Skip Ordered-Set Status	RW1CS	Yes	0
11	Ordered-Set Not the Same on All Lanes Status	RW1CS	Yes	0
12	Not All Subsequent Lanes Have LIDL or End of Data Stream when LIDL Status	RW1CS	Yes	0
13	Last TLP or DLLP Has No LIDL Symbols Status For x8 Gen 3 Links, if a TLP or DLLP ends on Lane 3 and is not followed by another TLP or DLLP, subsequent Lanes should have LIDL symbols. For x16 Gen 3 Links, if a TLP or DLLP ends on Lanes 3, 7, or 11 and is not followed by another TLP or DLLP, subsequent Lanes should have LIDL symbols.	RW1CS	Yes	0
14	Frame Error 14 Status When processing Symbols that are expected to be a Framing Token, receiving a Symbol or sequence of Symbols that does not match the definition of a Framing Token is a Framing error.	RW1CS	Yes	0
15	Frame Error 15 Status	RW1CS	Yes	0
31:16	Reserved	RsvdP	No	0000h

**Register 13-159. 728h Gen 3 Framing Error Mask
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Gen 3 Framing Error Status register bits (offset 724h). All errors in this register generate an ERR_COR Message, if enabled by the following: - This register Correctable Error Status register Receiver Error Status bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset FC4h[0]) - Port's Device Status register Correctable Error Reporting Enable bit (offset 70h[0])				
0	Invalid Sync Header Mask	RWS	Yes	0
1	Sync Header Not Same on All Lanes Mask	RWS	Yes	0
2	Bad Framing Symbol in Data Block Mask	RWS	Yes	0
3	Second Symbol of DLLP Not ACh Mask	RWS	Yes	0
4	EDB Not following TLP Mask	RWS	Yes	0
5	Length CRC/Parity Error Mask Length Cyclic Redundancy Check (CRC)/Parity Error mask.	RWS	Yes	0
6	End of Data Stream Not Followed by Ordered-Set Block Mask	RWS	Yes	0
7	End of Data Stream Not in Last DWord of Data Block Mask	RWS	Yes	0
8	Ordered-Set Block Not Followed by Data Block Mask	RWS	Yes	0
9	FTS Not Followed by Data Block Mask	RWS	Yes	0
10	Invalid Skip Ordered-Set Mask	RWS	Yes	0
11	Ordered-Set Not the Same on All Lanes Mask	RWS	Yes	0
12	Not All Subsequent Lanes Have LIDL or End of Data Stream when LIDL Mask	RWS	Yes	0
13	Last TLP or DLLP Has No LIDL Symbols Mask For x8 or x16 Gen 3 Links, if TLP or DLLP ends on Lane K, and not followed by another TLP or DLLP, then subsequent Lanes should have LIDL symbols.	RWS	Yes	0
14	Frame Error 14 Mask	RWS	Yes	0
15	Frame Error 15 Mask	RWS	Yes	0
31:16	Reserved	RsvdP	No	0000h

Register 13-160. 72Ch Gen 3 Framing Error Disable

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
0	Sync Header Not Same on All Lanes Disable	RWS	Yes	0
1	Invalid Block Type Disable	RWS	Yes	0
2	Not SKIP, EIOS, or EIEOS Ordered-Set in Block following EDS Disable	RWS	Yes	0
3	Any Ordered-Set Block following SDS Ordered-Set Disable	RWS	Yes	0
4	Ordered-Set Block Received Without EDS Token in Preceding Data Block Disable	RWS	Yes	0
5	Block Containing EDS Token Followed by Data Block Instead of Ordered-Set Disable	RWS	Yes	0
6	Ordered-Sets Not the Same on All Lanes Disable	RWS	Yes	1
7	TLP Length Field CRC/Parity Disable	RWS	Yes	0
8	All Four Symbols of EDB Token Not the Same Disable	RWS	Yes	0
9	EDB Token Not Immediately following a TLP Disable	RWS	Yes	0
10	TLP Length Field = 0 Disable	RWS	Yes	1
11	Received More than One STP Token in a Single Symbol Time Disable	RWS	Yes	1
12	Received More than One SDP Token in a Single Symbol Time Disable	RWS	Yes	1
13	Lanes between IDL and Next Potential Starting Lane are Not All IDL Disable	RWS	Yes	1
14	Invalid Framing Token Detected when Searching for Framing Tokens Disable	RWS	Yes	0
15	Frame Error 15 Disable	RWS	Yes	1
31:16	Reserved	RsvdP	No	0000h

13.16.12 Device-Specific Registers – Control
(Offsets 760h – 774h)

This section details the Device-Specific Control registers. [Table 13-29](#) defines the register map.

Table 13-29. Device-Specific Control Register Map
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8,
accessible through the Management Port)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Station-Based Control																												760h				
Ingress Chip Control																												764h				
Factory Test Only																	768h –											774h				

Register 13-161. 760h Station-Based Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
Notes: <i>Table 13-5</i> indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.					
The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h.					
The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Link List RAM 1-Bit Error Injection The ECC Counter for this bit is located in the Ingress PLL RAM 1-Bit ECC Counter register <i>1-Bit ECC Counter for PLL RAM Read from Ingress Block</i> field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 778h[7:0]).	RWS	Yes	0	
1	Link List RAM 2-Bit Error Injection	RWS	Yes	0	
2	Link List RAM Error Injection Field 0 = Error injection is in the <i>ECC Code</i> field 1 = Error injection is in the <i>Data</i> field	RWS	Yes	0	
4:3	Link List RAM Port Selector for Error Injection	RWS	Yes	00b	
5	Factory Test Only	RWS	Yes	0	
6	Not used	RWS	Yes	0	
7	RAM Select for ECC Injection 0 = Select Payload RAM for ECC injection 1 = Select Header RAM for ECC injection	RWS	Yes	0	
8	Header RAM 1-Bit Error Injection	RWS	Yes	0	
9	Header RAM 2-Bit Error Injection	RWS	Yes	0	
10	Header RAM Error Injection Field 0 = Error injection is in the <i>ECC Code</i> field 1 = Error injection is in the <i>Data</i> field	RWS	Yes	0	
12:11	Header RAM Port Selector for Error Injection	RWS	Yes	00b	
15:13	Factory Test Only	RWS	Yes	000b	

Register 13-161. 760h Station-Based Control**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
16	Silicon Revision BA Payload RAM 1-Bit Error Injection	RWS	Yes	0	–
	Silicon Revision CA Device-Specific Register Access by Extended Configuration Request Enable <i>Note: This bit exists within each Station, and affects all Ports within the Station.</i> Controls whether Extended Configuration Requests can access Device-Specific registers within the following offset ranges, of each Port: • 200h through AFCh • B0Ch through B6Ch • B80h through C30h Because this bit exists within the Device-Specific register range listed above, it cannot be Set by an Extended Configuration Request. Serial EEPROM, I ² C/SMBus, and Memory-Mapped access can initially Set this bit. 0 = Extended Configuration Requests cannot access Device-Specific registers (default) 1 = Extended Configuration Requests can access Device-Specific registers	RWS	Yes	–	0
19:17	Reserved	RsvdP	No	000b	
20	Source Queue Link List RAM 1-Bit Error Injection	RWS	Yes	0	
21	Source Queue Link List RAM 2-Bit Error Injection	RWS	Yes	0	
22	Source Queue Link List RAM Error Injection Field 0 = Error injection is in the <i>ECC Code</i> field 1 = Error injection is in the <i>Data</i> field	RWS	Yes	0	
23	Source Queue Link List RAM Selector for Error Injection 0 = Port-A, normal traffic 1 = Port-B, Read-Pacing traffic	RWS	Yes	0	

Register 13-161. 760h Station-Based Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
24	Disable Credit Re-Balancing 1 = No Credit re-balancing	RWS	Yes	0	
25	Use Serial EEPROM Values for Ingress Credit Initialization Allow Configuration with a Device Number that is not 0, that is accessing Downstream devices, to be forwarded. When the Device Number is 0, the Configuration terminates in a UR. 0 = Use default values for ingress credit initialization 1 = Use serial EEPROM values for ingress credit initialization (initial credit values will not be overwritten)	RWS	Yes	0	
26	INCH Credit Reserve Flag A transition from 0 to 1 indicates that I ² C has finished with all CSR to INCH Initialization registers, and credit reservation can proceed.	RWS	Yes	0	
28:27	Factory Test Only	RWS	Yes	00b	
29	No Special Treatment for Relaxed Ordering Traffic The PEX 8724 supports Relaxed Ordering for Completions. By default, if the RO attribute is Set within a Completion, that Completion can bypass Posted transactions, if Posted TLPs are blocked at the egress Port (due to insufficient Posted credits from the connected device). This behavior can be disabled by Setting this bit, in each Station. 1 = Device-Specific Relaxed Ordering Completion will not be flagged to the egress block <i>Note: When Set, affects all Ports within the Station.</i>	RWS	Yes	0	1
30	Factory Test Only	RWS	Yes	0	
31	Not used	RWS	Yes	0	

Register 13-162. 764h Ingress Chip Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
0	NT Port Expansion ROM Virtual Side 0 = Expansion ROM is located on the NT Port Link Interface 1 = Expansion ROM is located on the NT Port Virtual Interface	RWS	Yes	0
1	Reserved	RsvdP	No	0
2	LUT Format Use of this bit is described in Section 15.3.2, “Device-Specific Memory-Mapped Configuration Mechanism.”	RWS	Yes	0
16:3	Factory Test Only	RWS	Yes	0-0h
17	NT Error Message Drop 0 = If the NT Port Link Interface receives an Uncorrectable Error Message that is routed to the Root Complex, the NT Port Link Interface reports a Malformed TLP error. 1 = Do not malformed a Fatal Error Message received on the NT Port Link Interface with routing equal to 0, and instead, drop the packet and log the error into the Link Error Status Virtual register <i>Link Side Uncorrectable Error Message Drop Status</i> bit (NT Port Virtual Interface, offset FE0h[3]). If the corresponding Link Error Mask Virtual register <i>Link Side Uncorrectable Error Message Drop Mask</i> bit (NT Port Virtual Interface, offset FE4h[3]) is Set, the NT Port Virtual Interface signals an Interrupt (INTx, MSI, or PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode)) to the Virtual side Host through the Upstream Port(s), if interrupts are enabled.	RWS	Yes	0
18	Reserved	RsvdP	No	0
19	Not used	RWS	Yes	0
20	Ingress MWr32 Counter Disable 0 = Enables Ingress Memory Write 32-Bit Counter 1 = Disables Ingress Memory Write 32-Bit Counter	RWS	Yes	0
21	Ingress MWr64 Counter Disable 0 = Enables Ingress Memory Write 64-Bit Counter 1 = Disables Ingress Memory Write 64-Bit Counter	RWS	Yes	0
22	Ingress MSG Counter Disable 0 = Enables Ingress Message Counter 1 = Disables Ingress Message Counter	RWS	Yes	0
23	Ingress MRd32 Counter Disable 0 = Enables Ingress Memory Read 32-Bit Counter 1 = Disables Ingress Memory Read 32-Bit Counter	RWS	Yes	0

Register 13-162. 764h Ingress Chip Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Ingress MRd64 Counter Disable 0 = Enables Ingress Memory Read 64-Bit Counter 1 = Disables Ingress Memory Read 64-Bit Counter	RWS	Yes	0
25	Ingress Other Non-Posted Counter Disable 0 = Enables Ingress Other Non-Posted Counter 1 = Disables Ingress Other Non-Posted Counter	RWS	Yes	0
26	Ingress and Egress DLLP ACK Counter Disable 0 = Enables Ingress and Egress DLLP ACK Counter 1 = Disables Ingress and Egress DLLP ACK Counter	RWS	Yes	0
27	Ingress and Egress DLLP UpdateFC-P Counter Disable 0 = Enables Ingress and Egress DLLP UpdateFC-Posted Counter 1 = Disables Ingress and Egress DLLP UpdateFC-Posted Counter	RWS	Yes	0
28	Ingress and Egress DLLP UpdateFC-NP Counter Disable 0 = Enables Ingress and Egress DLLP UpdateFC-Non-Posted Counter 1 = Disables Ingress and Egress DLLP UpdateFC-Non-Posted Counter	RWS	Yes	0
29	Ingress and Egress DLLP UpdateFC-CPL Counter Disable 0 = Enables Ingress and Egress DLLP UpdateFC-Completion Counter 1 = Disables Ingress and Egress DLLP UpdateFC-Completion Counter	RWS	Yes	0
30	Factory Test Only	RWS	Yes	0
31	Factory Test Only	RO	Yes	0

13.16.13 Device-Specific Registers – Error (Offsets 778h – 8FCh)

This section details the Device-Specific Error registers. [Table 13-30](#) defines the register map.

Note: The PEX87XX Error Injection White Paper, available from the PLX PEX 8724 Toolbox, includes cross-reference information for some of the **Error Counter** registers and their associated **Error Checking and Debug** registers, located at offsets 700h through 71Ch.

Table 13-30. Device-Specific Error Register Map
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Reserved</i>		Ingress PLL RAM 1-Bit ECC Counter 778h
<i>Factory Test Only</i>		77Ch
Ingress Exerciser Completion Status		780h
Ingress Exerciser Completion Data		784h
<i>Reserved</i>		788h – 7FCh
Egress Station 0 Payload RAM ECC Error Counters		800h
Egress Station 1 Payload RAM ECC Error Counters		804h
<i>Reserved</i>		808h – 80Ch
<i>Reserved</i>	Egress Header RAM ECC Error Counters 1	810h
<i>Reserved</i>	Egress Header RAM ECC Error Counters 2	814h
<i>Reserved</i>	Egress Header RAM ECC Error Counters 3	818h
<i>Reserved</i>	Egress Header RAM ECC Error Counters 4	81Ch
<i>Reserved</i>	Egress PLL RAM ECC Error Counters 1	820h
Egress PLL RAM ECC Error Counters 2	<i>Reserved</i>	824h
Egress Miscellaneous RAM ECC Error Counters		828h
Error Injection		82Ch
<i>Reserved</i>		830h – 8FCh

Register 13-163. 778h Ingress PLL RAM 1-Bit ECC Counter

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	1-Bit ECC Counter for PLL RAM Read from Ingress Block A Write of 0 to bit 0 Clears the ECC Counter.	ROS	No	00h
31:8	Reserved	RsvdP	No	0000_00h

Register 13-164. 780h Ingress Exerciser Completion Status

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
0	Received a Completion 1 = New Exerciser Completion was received	RW1C	Yes	0
1	Completion Received with EP	ROS	No	0
2	Completion Received with ECRC Error	ROS	No	0
3	Reserved	RsvdP	No	0
6:4	Completion Status 1 = Received Exerciser Completion TLPs status	ROS	No	000b
31:7	Reserved	RsvdP	No	0-0h

Register 13-165. 784h Ingress Exerciser Completion Data
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
31:0	Completion Data 1 = Received Exerciser Completion data	ROS	No	0000_0000h

Register 13-166. 800h Egress Station 0 Payload RAM ECC Error Counters
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 Payload RAM0 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 0 Payload RAM0 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
23:16	Station 0 Payload RAM1 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
31:24	Station 0 Payload RAM1 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h

Register 13-167. 804h Egress Station 1 Payload RAM ECC Error Counters
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 1 Payload RAM0 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 Payload RAM0 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
23:16	Station 1 Payload RAM1 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
31:24	Station 1 Payload RAM1 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h

Register 13-168. 810h Egress Header RAM ECC Error Counters 1

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 Header RAM0 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 Header RAM0 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
31:16	Reserved	RsvdP	No	0000h

Register 13-169. 814h Egress Header RAM ECC Error Counters 2

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 Header RAM0 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 Header RAM0 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
31:16	Reserved	RsvdP	No	0000h

Register 13-170. 818h Egress Header RAM ECC Error Counters 3
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 Header RAM1 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 Header RAM1 Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
31:16	Reserved	RsvdP	No	0000h

Register 13-171. 81Ch Egress Header RAM ECC Error Counters 4
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 Header RAM1 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 Header RAM1 Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
31:16	Reserved	RsvdP	No	0000h

Register 13-172. 820h Egress PLL RAM ECC Error Counters 1
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Station 0 PLL RAM 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Station 1 PLL RAM 1-Bit ECC Error Counter Value	ROS	No	00h
31:16	Reserved	RsvdP	No	0000h

Register 13-173. 824h Egress PLL RAM ECC Error Counters 2
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Source Queue Link List RAM Instance1 1-Bit ECC Error Counter Value Reserved	RsvdP	No	00h
15:8	Reserved	RsvdP	No	00h
23:16	Destination Queue Link List RAM Instance1 1-Bit ECC Error Counter Value	ROS	No	00h
31:24	Destination Queue Link List RAM Instance2 1-Bit ECC Error Counter Value	ROS	No	00h

Register 13-174. 828h Egress Miscellaneous RAM ECC Error Counters
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Destination Queue Data RAM 1-Bit ECC Error Counter Value	ROS	No	00h
15:8	Destination Queue Link List RAM Instance0 1-Bit ECC Error Counter Value	ROS	No	00h
23:16	Source Queue Link List RAM 1-Bit ECC Error Counter Value	ROS	No	00h
31:24	Retry Buffer 1-Bit ECC Error Counter Value	ROS	No	00h

Register 13-175. 82Ch Error Injection

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
0	Destination Queue Data RAM 1-Bit ECC Error Injection Writing 1 injects one error.	RWS	Yes	0
1	Destination Queue Data RAM 2-Bit Error Injection Writing 1 injects one error.	RWS	Yes	0
2	Destination Queue Data RAM Error Injection Select 0 = Inject error in the <i>ECC Code</i> field 1 = Inject error in the <i>Data</i> field	RWS	Yes	0
3	Destination Queue Link List RAM Instance0 1-Bit ECC Error Injection Writing 1 injects one error.	RWS	Yes	0
4	Destination Queue Link List RAM Instance0 2-Bit Error Injection Writing 1 injects one error.	RWS	Yes	0
5	Destination Queue Link List RAM Instance0 Error Injection Select 0 = Inject error in the <i>ECC Code</i> field 1 = Inject error in the <i>Data</i> field	RWS	Yes	0
6	Retry Buffer 1-Bit ECC Error Injection Writing 1 injects one error.	RWS	Yes	0
7	Retry Buffer 2-Bit Error Injection Writing 1 injects one error.	RWS	Yes	0

Register 13-175. 82Ch Error Injection

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (*Cont.*)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Retry Buffer Error Injection Select 0 = Inject error in the <i>ECC Code</i> field 1 = Inject error in the <i>Data</i> field	RWS	Yes	0
9	Destination Queue Link List RAM Instance1 1-Bit ECC Error Injection Writing 1 injects one error.	RWS	Yes	0
10	Destination Queue Link List RAM Instance1 2-Bit Error Injection Writing 1 injects one error.	RWS	Yes	0
11	Destination Queue Link List RAM Instance1 Error Injection Select 0 = Inject error in the <i>ECC Code</i> field 1 = Inject error in the <i>Data</i> field	RWS	Yes	0
12	Destination Queue Link List RAM Instance2 1-Bit ECC Error Injection Writing 1 injects one error.	RWS	Yes	0
13	Destination Queue Link List RAM Instance2 2-Bit Error Injection Writing 1 injects one error.	RWS	Yes	0
14	Destination Queue Link List RAM Instance2 Error Injection Select 0 = Inject error in the <i>ECC Code</i> field 1 = Inject error in the <i>Data</i> field	RWS	Yes	0
19:15	Reserved	RsvdP	No	0-0h

Register 13-175. 82Ch Error Injection

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
20	Ingress Payload RAM0 Instance0 ECC Counter Reset Writing 1 Clears the Counter in the Egress Station x Payload RAM ECC Error Counters register <i>Station x Payload RAM0 Instance0 1-Bit ECC Error Counter Value</i> field(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 800h and 804h[7:0]). Reads always return a value of 0.	RZ	Yes	0
21	Ingress Payload RAM0 Instance1 ECC Counter Reset Writing 1 Clears the Counter in the Egress Station x Payload RAM ECC Error Counters register <i>Station x Payload RAM0 Instance1 1-Bit ECC Error Counter Value</i> field(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 800h and 804h[15:8]). Reads always return a value of 0.	RZ	Yes	0
22	Ingress Payload RAM1 Instance0 ECC Counter Reset Writing 1 Clears the Counter in the Egress Station x Payload RAM ECC Error Counters register <i>Station x Payload RAM1 Instance0 1-Bit ECC Error Counter Value</i> field(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 800h and 804h[23:16]). Reads always return a value of 0.	RZ	Yes	0
23	Ingress Payload RAM1 Instance1 ECC Counter Reset Writing 1 Clears the Counter in the Egress Station x Payload RAM ECC Error Counters register <i>Station x Payload RAM1 Instance1 1-Bit ECC Error Counter Value</i> field(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 800h and 804h[31:24]). Reads always return a value of 0.	RZ	Yes	0
24	Header RAM ECC Counter Reset Writing 1 Clears the Counter in the Egress Header RAM ECC Error Counters x register <i>Station x Header RAMx Instancex 1-Bit ECC Error Counter Value</i> field(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 810h through 81Ch). Reads always return a value of 0.	RZ	Yes	0
28:25	Factory Test Only	RZ	Yes	0h
31:29	Station Number to Reset the ECC Counter Selects the ECC Counter that corresponds to the Station Number. 000b = Station 0 (default) 001b = Station 1 All other encodings are Reserved .	RZ	Yes	000b

13.16.14 Device-Specific Registers – Virtual Switch (Offsets 900h – 9E8h), Virtual Switch Mode

Note: In Base mode, this entire structure is **Reserved**, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 0h.

This section details the Device-Specific Virtual Switch Support registers located at offsets 900h through 9E8h. These registers are implemented only in Virtual Switch mode, for use with a Management Port. Switch Link Event interrupts are sent to the Management Port (rather than to Virtual Switch Hosts). Additionally, the registers are implemented only in Port 0, accessible through the Management Port (offsets 900h through 90Ch), or VS Upstream Port(s) and Management Port (offsets 910h through 93Ch), as indicated in the registers that follow. Table 13-31 defines the register map.

Table 13-31. Device-Specific Virtual Switch Register Map (Ports^a)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Switch Link Up	900h
Switch Link Down	904h
Switch Link Event Mask	908h
Switch Link Status	90Ch
VS Upstream to Management Upstream Doorbell Request	910h
VS Upstream to Management Upstream Doorbell Mask	914h
VS Upstream to Management Upstream Scratchpad 1	918h
VS Upstream to Management Upstream Scratchpad 2	91Ch
VS Upstream to Management Upstream Scratchpad 3	920h
VS Upstream to Management Upstream Scratchpad 4	924h
Management Upstream to VS Upstream Doorbell Request	928h
Management Upstream to VS Upstream Doorbell Mask	92Ch
Management Upstream to VS Upstream Scratchpad 1	930h
Management Upstream to VS Upstream Scratchpad 2	934h
Management Upstream to VS Upstream Scratchpad 3	938h
Management Upstream to VS Upstream Scratchpad 4	93Ch
Reserved	940h – 9E8h

a. Certain registers are Port-specific, others are Chip-specific; all are Device-specific.

Register 13-176. 900h Switch Link Up
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only , and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port 0 Link Up 1 = Port 0 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
1	Port 1 Link Up 1 = Port 1 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
2	Port 2 Link Up 1 = Port 2 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
3	Port 3 Link Up 1 = Port 3 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port 8 Link Up 1 = Port 8 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
9	Port 9 Link Up 1 = Port 9 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
10	Port 10 Link Up 1 = Port 10 Link transitioned from the <i>DL_Inactive</i> state to the <i>DL_Active</i> state	0, accessible through the Management Port	RW1C	Yes	0
	Reserved	Otherwise	RsvdP	No	0
31:11	Reserved		RsvdP	No	0-0h

Register 13-177. 904h Switch Link Down
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only , and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port 0 Link Down 1 = Port 0 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
1	Port 1 Link Down 1 = Port 1 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
2	Port 2 Link Down 1 = Port 2 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
3	Port 3 Link Down 1 = Port 3 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port 8 Link Down 1 = Port 8 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
9	Port 9 Link Down 1 = Port 9 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
10	Port 10 Link Down 1 = Port 10 Link transitioned from the <i>DL_Active</i> state to the <i>DL_Inactive</i> state	0, accessible through the Management Port	RW1C	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
31:11	Reserved		RsvdP	No	0-0h

Register 13-178. 908h Switch Link Event Mask
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Switch Link Status register bits (Port 0, accessible through the Management Port, offset 90Ch). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port 0 Link Event Mask 0 = Interrupt for Port 0 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 0 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
1	Port 1 Link Event Mask 0 = Interrupt for Port 1 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 1 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
2	Port 2 Link Event Mask 0 = Interrupt for Port 2 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 2 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
3	Port 3 Link Event Mask 0 = Interrupt for Port 3 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 3 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b

Register 13-178. 908h Switch Link Event Mask**(Virtual Switch mode – Port 0, accessible through the Management Port) (Cont.)**

Bit(s)	Description	Port	Type	Serial EEPROM and I ² C	Default
8	Port 8 Link Event Mask 0 = Interrupt for Port 8 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 8 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
9	Port 9 Link Event Mask 0 = Interrupt for Port 9 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 9 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
10	Port 10 Link Event Mask 0 = Interrupt for Port 10 due to a Link Up or Link Down event is not masked 1 = Interrupt for Port 10 due to a Link Up or Link Down event is masked	0, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	0
31:11	Reserved		RsvdP	No	0-0h

Register 13-179. 90Ch Switch Link Status
(Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Port	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Switch Link Event Mask register bits (Port 0, accessible through the Management Port, offset 908h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port 0 Link Status 0 = Port 0 is in a <i>DL_Inactive</i> state 1 = Port 0 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
1	Port 1 Link Status 0 = Port 1 is in a <i>DL_Inactive</i> state 1 = Port 1 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
2	Port 2 Link Status 0 = Port 2 is in a <i>DL_Inactive</i> state 1 = Port 2 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
3	Port 3 Link Status 0 = Port 3 is in a <i>DL_Inactive</i> state 1 = Port 3 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port 8 Link Status 0 = Port 8 is in a <i>DL_Inactive</i> state 1 = Port 8 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
9	Port 9 Link Status 0 = Port 9 is in a <i>DL_Inactive</i> state 1 = Port 9 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
10	Port 10 Link Status 0 = Port 10 is in a <i>DL_Inactive</i> state 1 = Port 10 is in a <i>DL_Active</i> state	0, accessible through the Management Port	RO	Yes	PCFG
	Reserved	Otherwise	RsvdP	No	0
31:11	Reserved		RsvdP	No	0-0h

Register 13-180. 910h VS Upstream to Management Upstream Doorbell Request
(Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
<i>Note:</i> The bits in this register can be masked by their respective VS Upstream to Management Upstream Doorbell Mask register bits (VS Upstream Port(s) and Management Port, offset 914h).					
3:0	Doorbell Writing 1 to any bit within this field, in the Non-Management VS Upstream Port(s), signals an interrupt to the Management Upstream Port. The Management Port Clears the interrupt(s), by writing 1 to the bit(s) that are Set.	Non-Management VS Upstream Port	RW	Yes	0h
		Management Port	RW1C	Yes	0h
	Reserved	Otherwise	RsvdP	No	0h
31:4	Reserved		RsvdP	No	0000_000h

Register 13-181. 914h VS Upstream to Management Upstream Doorbell Mask
(Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
<i>Note:</i> The bits in this register can be used to mask their respective VS Upstream to Management Upstream Doorbell Request register bits (VS Upstream Port(s) and Management Port, offset 910h).					
3:0	Doorbell Interrupt Mask 0 = Doorbell interrupts to the Management Upstream Port are not masked 1 = Doorbell interrupts to the Management Upstream Port are masked	Non-Management VS Upstream Port	RW	Yes	Fh
		Management Port	RW1C	Yes	0h
	Reserved	Otherwise	RsvdP	No	0h
31:4	Reserved		RsvdP	No	0000_000h

Register 13-182. 918h VS Upstream to Management Upstream Scratchpad 1 (Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 1 32-bit Scratchpad 1 register.	Non-Management VS Upstream Port	RW	Yes	0000_0000h
		Management Port	ROS	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

Register 13-183. 91Ch VS Upstream to Management Upstream Scratchpad 2 (Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 2 32-bit Scratchpad 2 register.	Non-Management VS Upstream Port	RW	Yes	0000_0000h
		Management Port	ROS	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

Register 13-184. 920h VS Upstream to Management Upstream Scratchpad 3 (Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 3 32-bit Scratchpad 3 register.	Non-Management VS Upstream Port	RW	Yes	0000_0000h
		Management Port	ROS	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

Register 13-185. 924h VS Upstream to Management Upstream Scratchpad 4 (Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 4 32-bit Scratchpad 4 register.	Non-Management VS Upstream Port	RW	Yes	0000_0000h
		Management Port	ROS	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

Register 13-186. 928h Management Upstream to VS Upstream Doorbell Request
(Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: The bits in this register can be masked by their respective <i>Management Upstream to VS Upstream Doorbell Mask</i> register bits (VS Upstream Port(s) and Management Port, offset 92Ch).					
3:0	Doorbell Writing 1 to any bit within this field, in the Management Port, signals an interrupt to the VS Upstream Port(s). The Non-Management VS Upstream Port(s) Clear(s) the interrupt(s), by writing 1 to the bit(s) that are Set.	Non-Management VS Upstream Port	RW1C	Yes	0h
		Management Port	RW	Yes	0h
	Reserved	Otherwise	RsvdP	No	0h
31:4	Reserved		RsvdP	No	0000_000h

Register 13-187. 92Ch Management Upstream to VS Upstream Doorbell Mask
(Virtual Switch mode – VS Upstream Port(s) and Management Port)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: The bits in this register can be used to mask their respective <i>Management Upstream to VS Upstream Doorbell Request</i> register bits (VS Upstream Port(s) and Management Port, offset 928h).					
3:0	Doorbell Interrupt Mask 0 = Doorbell interrupts to the VS Upstream Port(s) are not masked 1 = Doorbell interrupts to the VS Upstream Port(s) are masked	Non-Management VS Upstream Port	RW1C	Yes	Fh
		Management Port	RW	Yes	0h
	Reserved	Otherwise	RsvdP	No	0h
31:4	Reserved		RsvdP	No	0000_000h

**Register 13-188. 930h Management Upstream to VS Upstream Scratchpad 1
(Virtual Switch mode – VS Upstream Port(s) and Management Port)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 1 32-bit Scratchpad 1 register.	Non-Management VS Upstream Port	ROS	Yes	0000_0000h
		Management Port	RW	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

**Register 13-189. 934h Management Upstream to VS Upstream Scratchpad 2
(Virtual Switch mode – VS Upstream Port(s) and Management Port)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 2 32-bit Scratchpad 2 register.	Non-Management VS Upstream Port	ROS	Yes	0000_0000h
		Management Port	RW	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

**Register 13-190. 938h Management Upstream to VS Upstream Scratchpad 3
(Virtual Switch mode – VS Upstream Port(s) and Management Port)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 3 32-bit Scratchpad 3 register.	Non-Management VS Upstream Port	ROS	Yes	0000_0000h
		Management Port	RW	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

**Register 13-191. 93Ch Management Upstream to VS Upstream Scratchpad 4
(Virtual Switch mode – VS Upstream Port(s) and Management Port)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 4 32-bit Scratchpad 4 register.	Non-Management VS Upstream Port	ROS	Yes	0000_0000h
		Management Port	RW	Yes	0000_0000h
	Reserved	Otherwise	RsvdP	No	0000_0000h

13.16.15 Device-Specific Registers – Ingress Credit Handler (Offsets 9ECh – A2Ch)

This section details the Device-Specific Ingress Credit Handler (INCH) registers. [Table 13-32](#) defines the register map.

Table 13-32. Device-Specific INCH Register Map (Ports^a)

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Reserved			9ECh
INCH Station Pool Values			9F0h
Factory Test Only			9F4h
INCH Reserve Pool			9F8h
Reserved		INCH Port Pool	9FCh
INCH Threshold VC0 Posted			A00h
Factory Test Only	INCH Threshold VC0 Non-Posted		A04h
INCH Threshold VC0 Completion			A08h
Reserved			A0Ch – A2Ch

a. Certain registers are Port-specific, others are Station-specific; all are Device-specific.

Register 13-192. 9F0h INCH Station Pool Values

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
INCH Reserve pool. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
8:0	Current Value of Header Pool for the Station Specifies the quantity of Common Header Credits available for sharing among the Ports within this Station.	RO	Yes	–
15:9	Reserved	RsvdP	No	0-0h
25:16	Current Value of Payload Link Pool for the Station Specifies the quantity of Common Payload Links available for sharing among the Ports within this Station. Each Payload Link can store 64 bytes.	RO	Yes	–
31:26	Reserved	RsvdP	No	0-0h

Register 13-193. 9F8h INCH Reserve Pool

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	Header Count to Remove from Station Header Pool Value of Header to remove from initial Header pool.	RW	Yes	00h
15:8	Reserved	RsvdP	No	00h
24:16	Payload Link Count to Remove from Station Payload Pool Value of Payload Links to remove from initial Payload Link pool.	RW	Yes	0-0h
31:25	Reserved	RsvdP	No	0-0h

Register 13-194. 9FCh INCH Port Pool

(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
<i>Note: Consider the INCH Port Pool register to be Reserved and only change the credit Settings, using the INCH Threshold registers (offsets A00h through A08h). Do not change the INCH Port Pool register from its default value, unless directed otherwise by PLX Technical Support.</i>				
2:0	Port Payload Pool Payload credits (other than the initial credits) for Posted/Completion TLPs that are dedicated to the Port. 000b = 0 001b = 32 010b = 64 011b = 96 100b = 128 101b = 192 110b, 111b = 256	RWS	Yes	000b
3	Unused 0 Keep value at 0. Additional bit for the Port Payload Pool.	RWS	Yes	0
6:4	Port Header Pool Combined Header credits (other than the initial credits) that are dedicated to the Port. 000b = 0 TLP 001b = 4 TLPs 010b = 8 TLPs 011b = 16 TLPs 100b = 32 TLPs 101b = 48 TLPs 110b, 111b = 64 TLPs	RWS	Yes	000b
7	Unused 1 Keep value at 0. Additional bit for the Port Header Pool.	RWS	Yes	0
31:8	Reserved	RsvdP	No	0000_00h

Register 13-195. A00h INCH Threshold VC0 Posted
(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Posted credits are used for VC0 Memory Write and Message transactions.				
Note: <i>Changing credit values from default register values must be done carefully; otherwise the PEX 8724 will not properly function.</i>				
2:0	Reserved	RsvdP	No	Upstream Payload (decimal): x16: 504 x8: 256 x4: 136
8:3	Posted Payload Credit Default advertised Posted Payload credit. Actual value is dependent upon Link width and Port configuration, and the advertised value can be 0 to 3 more Payload credits than the register value. Bit resolution is in units of 8. Each increment provides 8 Posted Payload credits (<i>for example</i> , Ah = 80 Posted Payload credits). Each credit means that 16 bytes of storage are reserved for Posted TLP Payload data.	RWS	Yes	Downstream Payload (decimal): x16: 504 x8: 256 x4: 136 Upstream: x16: C3F8h x8: 63000h x4: 3288h
15:9	Posted Header Credit Default advertised Posted Header credit. Actual value is dependent upon Link width and Port configuration. Bit resolution is 1 for 1. Each increment provides 1 Posted Header credit (<i>for example</i> , Ah = 10 Posted Header credits). Each credit means that storage is reserved for the entire Header of a Posted TLP.	RWS	Yes	Upstream Header (decimal): x16: 97 x8: 49 x4: 25 Downstream Header (decimal): x16: 124 x8: 113 x4: 57 Downstream: x16: F9F8h x8: E300h x4: 7288h

Register 13-195. A00h INCH Threshold VC0 Posted
(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the
Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
17:16	UpdateFC High-Priority Threshold for Posted Payload Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
19:18	UpdateFC High-Priority Threshold for Posted Header Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
22:20	Congested Port Weight If the effective rate Setting times the negotiated Port Link width equates to less than x1 or greater than x8, the internal Credit Allocation logic rounds to x1 or x8, respectively. 000b = Request is weighted, based upon the Port's Link width relative to the effective Link widths of the other Stations' Ports 001b = Increases the weight of a Request by 2x 010b = Increases the weight of a Request by 4x 011b = Increases the weight of a Request by 8x 100b = Port receives no credit out of the common pool, until a decongested state is reached 101b = Decreases the weight of a Request by 2x 110b = Decreases the weight of a Request by 4x 111b = Decreases the weight of a Request by 8x	RWS	Yes	000b
31:23	Reserved	RsvdP	No	0-0h

Register 13-196. A04h INCH Threshold VC0 Non-Posted
(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Non-Posted credits are used for VC0 Memory Read, I/O Read, I/O Write, Configuration Read, Configuration Write, and Atomic transactions. Note: <i>Changing credit values from default register values must be done carefully; otherwise the PEX 8724 will not properly function.</i>				
7:0	Non-Posted Payload Credit The Non-Posted Payload is stored with the Non-Posted Header.	RWS	Yes	Upstream Payload (decimal): x16: 8 x8: 8 x4: 8 Downstream Payload (decimal): x16: 8 x8: 8 x4: 8 Upstream: x16: 6108h x8: 3108h x4: 1908h
8	Reserved	RsvdP	No	0
15:9	Non-Posted Header Credit Default advertised Posted Header credit. Actual value is dependent upon Link width and Port configuration. Bit resolution is 1 for 1. Each increment provides 1 Non-Posted Header credit (<i>for example, Ah = 10 Non-Posted Header credits</i>). Each credit means that storage is reserved for the entire Header of a Non-Posted TLP.	RWS	Yes	Upstream Header (decimal): x16: 97 x8: 49 x4: 25 Downstream Header (decimal): x16: 124 x8: 62 x4: 31 Downstream: x16: 7C08h x8: 3E08h x4: 1F08h

Register 13-196. A04h INCH Threshold VC0 Non-Posted
(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the
Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
17:16	UpdateFC High-Priority Threshold for Non-Posted Payload Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
19:18	UpdateFC High-Priority Threshold for Non-Posted Header Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
22:20	<i>Not used</i>	RWS	Yes	000b
23	<i>Reserved</i>	RsvdP	No	0
29:24	<i>Factory Test Only</i>	RWS	Yes	0-0h
31:30	<i>Factory Test Only</i>	RW1C	No	00b

Register 13-197. A08h INCH Threshold VC0 Completion**(Base mode – All Ports; Virtual Switch mode – Port 0, accessible through the Management Port)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Completion credits are used for VC0 Memory Read, I/O Read, I/O Write, Configuration Read, and Configuration Write transaction Completions.				
Note: Changing credit values from default register values must be done carefully; otherwise the PEX 8724 will not properly function.				
2:0	Reserved	RsvdP	No	
8:3	Completion Payload Credit Default advertised Completion Payload credit. Actual value is dependent upon Link width and Port configuration, and the advertised value can be 0 to 3 more Payload credits than the register value. Bit resolution is in units of 8. Each increment provides 8 Completion Payload credits (<i>for example</i> , Ah = 80 Completion Payload credits). Each credit means that 16 bytes of storage are reserved for Completion TLP Payload data.	RWS	Yes	Upstream Payload (decimal): x16: 480 x8: 240 x4: 136 Downstream Payload (decimal): x16: 480 x8: 240 x4: 136 Upstream: x16: F9E0h x8: E2F0h x4: 7288h
15:9	Completion Header Credit Default advertised Completion Header credit. Actual value is dependent upon Link width and Port configuration. Bit resolution is 1 for 1. Each increment provides 1 Completion Header credit (<i>for example</i> , Ah = 10 Completion Header credits). Each credit means that storage is reserved for the entire Header of a Completion TLP.	RWS	Yes	Upstream Header (decimal): x16: 124 x8: 71 x4: 57 Downstream Header (decimal): x16: 70 x8: 36 x4: 19 Downstream: x16: 8DE0h x8: 48F0h x4: 2688h
17:16	UpdateFC High-Priority Threshold for Completion Payload Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
19:18	UpdateFC High-Priority Threshold for Completion Header Credit 00b = 75% (default) 01b = 50% 10b = 25% 11b = 100%	RWS	Yes	00b
22:20	Not used	RWS	Yes	000b
31:23	Reserved	RsvdP	No	0-0h

13.16.16 Device-Specific Registers – Virtual Switch Debug and GPIO Status and Control (Offsets A30h – A74h)

This section details the Device-Specific Virtual Switch Debug and GPIO Status and Control registers. [Table 13-33](#) defines the register map.

Table 13-33. Device-Specific Virtual Switch Debug and GPIO Status and Control Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Virtual Switch Debug		A30h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_9 Direction Control (Virtual Switch Mode)		A34h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 10_11 Direction Control (Virtual Switch Mode)		A38h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Availability (Virtual Switch Mode)		A3Ch
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Input De-Bounce (Virtual Switch Mode)		A40h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Input Data (Virtual Switch Mode)		A44h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Output Data (Virtual Switch Mode)		A48h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Interrupt Polarity (Virtual Switch Mode)		A4Ch
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Interrupt Status (Virtual Switch Mode)		A50h
<i>Reserved</i> (Base Mode) Virtual Switch GPIO_PG 0_11 Interrupt Mask (Virtual Switch Mode)		A54h
<i>Reserved</i>		A58h – A74h

Register 13-198. A30h Virtual Switch Debug (Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	Interrupt Fencing Mode Select <i>Note:</i> A Fundamental Reset is needed to recover from Fencing errors. Mode 1 (Default) - When the PEX 8724 receives a packet with a Fatal error (Malformed, DLL Protocol error) from an external device, or the device detects a Credit Overflow, Receiver Overflow, or Surprise Link Down, the switch logs the Header on the corresponding Port, sends a Fatal Error Message to the Host, then asserts FATAL_ERR# (Base mode) or VSx_FATAL_ERR# (Virtual Switch mode). - When the PEX 8724 detects an internal Fatal error (ECC failure), the switch sends a Fatal Interrupt Message to the Host and asserts FATAL_ERR# (Base mode) or VSx_FATAL_ERR# (Virtual Switch mode). In certain situations, delivery of the interrupt is not guaranteed; however, the signal is always asserted upon a Fatal event. Mode 2 (Generate Internal Reset) – Base Mode Upon Fatal error (internal or external) detection, an internal Chip Level reset is asserted (equivalent to an In-Band Reset from the Upstream Port). No Error Messages are generated, and no attempt is made to block packets in transit. Mode 3 (Block All Packet Transmission) Upon Fatal error (internal or external) detection, the Port logs the error in the Uncorrectable Error Status register (offset FB8h), then asserts FATAL_ERR# (Base mode) or VSx_FATAL_ERR# (Virtual Switch mode). This Fatal error detection blocks all the Ports from sending out TLPs. No Error Messages are generated. If a packet is already in transmission, an EDB is inserted to cancel the packet. Mode 4 (Block All Packet Transmission and Create Surprise Down) In addition to the Mode 3 actions, the PEX 8724 forces the Upstream Link to go down, thus causing a Surprise Down event on the Link, so that the Host is notified. 00b = Mode 1 (default) 01b = Mode 2 – Generate Internal Reset 10b = Mode 3 – Block All Packet Transmission 11b = Mode 4 – Block All Packet Transmission and Create Surprise Down	RWS	Yes	00b

**Register 13-198. A30h Virtual Switch Debug
(Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2	Upstream Hot Reset Control 0 = Reset all logic, except Sticky bits and Device-Specific registers 1 = Reset only the Configuration Space registers of all Ports defined by the <i>PCI Express Base r3.0</i> <i>Notes: Only a Fundamental Reset serial EEPROM load affects this bit. This bit should be Set if NT mode is enabled.</i>	RWS	Yes	0
3	Disable Serial EEPROM Load on Hot Reset 0 = Enables serial EEPROM load upon Upstream Port Hot Reset or <i>DL_Down</i> state. Port-specific registers for other virtual switches are <i>not</i> reloaded. (default) Virtual Switch mode – Chip- and Station-specific register reload from serial EEPROM (which can affect other virtual switches) can be enabled by Setting the Management Port Control register <i>Active Management Port EEPROM Load on Hot Reset for Chip and Station Registers Enable</i> bit (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[6]). 1 = Disables serial EEPROM load upon Upstream Port Hot Reset or <i>DL_Down</i> state.	RWS	Yes	0
4	Upstream Port and NT-Link Port DL_Down Reset Propagation Disable Setting this bit: - Enables the Upstream Port(s) and NT Port Link Interface to ignore a Hot Reset training sequence, - Blocks the PEX 8724 from manifesting an internal reset due to a <i>DL_Down</i> event, - Blocks the NT Port Link Interface from manifesting an internal reset due to a <i>DL_Down</i> event on the NT Port Link, and - Prevents the Downstream Ports from issuing a Hot Reset to Downstream devices when a Hot Reset or <i>DL_Down</i> event occurs on the Upstream Link	RWS	Yes	0
5	Factory Test Only	RWS	Yes	0
23:6	Reserved	RsvdP	No	0-0h

Register 13-198. A30h Virtual Switch Debug (Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode NT Virtual Interface Access Enable When the serial EEPROM is not present, the default value is 1; otherwise, the default value is 0. 0 = Retries Type 0 Configuration TLPs received on the NT Port Virtual Interface 1 = Accepts Type 0 Configuration TLPs on the NT Port Virtual Interface <i>Notes:</i> Does not affect other transaction types. Set this bit to enable Configuration accesses to the NT Port Virtual Interface.	RW	Yes	1
25	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode NT Link Interface Access Enable 0 = Retries Type 0 Configuration Requests received on the NT Port Link Interface 1 = Accepts Type 0 Configuration Requests on the NT Port Link Interface <i>Notes:</i> Does not affect other transaction types. Set this bit to enable Configuration accesses to the NT Port Link Interface.	RW	Yes	1

**Register 13-198. A30h Virtual Switch Debug
(Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
26	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode Inhibit EEPROM NT-Link Load on Hot Reset Inhibits serial EEPROM load of NT Port Link Interface registers when any one of the following conditions are met: • Upstream Port Hot Reset – Bits [3:2] (<i>Disable Serial EEPROM Load on Hot Reset</i> and <i>Upstream Hot Reset Control</i>, respectively) are Cleared • Upstream Port DL_Down state – Bits [4:2] (<i>Upstream Port and NT-Link Port DL_Down Reset Propagation Disable</i>, <i>Disable Serial EEPROM Load on Hot Reset</i>, and <i>Upstream Hot Reset Control</i>, respectively) are Cleared • NT Port Link Interface Hot Reset or DL_Down state – Bit 3 (<i>Disable Serial EEPROM Load on Hot Reset</i>) is Cleared Refer also to Section 6.8, “Serial EEPROM Loading of NT Port Link Interface Registers – NT Mode,” for further details.	RW	Yes	0
27	Transparent Mode <i>Reserved</i>	RsvdP	No	0
	NT Mode Load Only EEPROM NT-Link on Hot Reset <i>Note:</i> In Virtual Switch mode, with NT mode enabled, the default value changes to 1 in a virtual switch that implements the NT Port; the default value is 0 in a virtual switch that does not implement the NT Port. Load only serial EEPROM NT Port Link Interface register entries when any one of the following conditions are met: • Upstream Port Hot Reset – Bits [3:2] (<i>Disable Serial EEPROM Load on Hot Reset</i> and <i>Upstream Hot Reset Control</i>, respectively) are Cleared • Upstream Port DL_Down state – Bits [4:2] (<i>Upstream Port and NT-Link Port DL_Down Reset Propagation Disable</i>, <i>Disable Serial EEPROM Load on Hot Reset</i>, and <i>Upstream Hot Reset Control</i>, respectively) are Cleared • NT Port Link Interface Hot Reset or DL_Down state – Bit 3 (<i>Disable Serial EEPROM Load on Hot Reset</i>) is Cleared Refer also to Section 6.8, “Serial EEPROM Loading of NT Port Link Interface Registers – NT Mode,” for further details.	RW	Yes	Refer to Note
31:28	Reserved	RsvdP	No	0h

Register 13-199. A34h Virtual Switch GPIO_PG 0_9 Direction Control (Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Port 0, accessible through the Management Port, offset(s) 600h and/or 604h), to support the virtual switches. The main differences are as follows: - Virtual switches have no context as to which PORT_GOODx# or GPIOx signal is being used - When the PORT_GOODx# or GPIOx signal is configured as an output, value of 01b is Reserved This register, in combination with the Virtual Switch GPIO_PG 10_11 Direction Control register (VS Upstream Port(s), offset A38h), controls the 12 GPIOs assigned to the virtual switch. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Twelve bits in each of the VS Upstream Port registers located at offsets A40h through A54h correspond to the 12 GPIO 0 through 11 fields in offsets A34h and A38h, within each virtual switch. Note: Register offsets A44h and A48h, referenced within this register, are located as follows – VS Upstream Port(s).				
1:0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 0 PORT_GOODx# or GPIOx Input Data register (offset A44h[0]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 0 PORT_GOODx# or GPIOx Output Data register (offset A48h[0]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
2	VS GPIO_PG 0 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
4:3	VS GPIO_PG 1 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 1 PORT_GOODx# or GPIOx Input Data register (offset A44h[1]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 1 PORT_GOODx# or GPIOx Output Data register (offset A48h[1]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
5	VS GPIO_PG 1 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0

**Register 13-199. A34h Virtual Switch GPIO_PG 0_9 Direction Control
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:6	VS GPIO_PG 2 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 2 PORT_GOODx# or GPIOx Input Data register (offset A44h[2]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 2 PORT_GOODx# or GPIOx Output Data register (offset A48h[2]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
8	VS GPIO_PG 2 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
10:9	VS GPIO_PG 3 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 3 PORT_GOODx# or GPIOx Input Data register (offset A44h[3]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 3 PORT_GOODx# or GPIOx Output Data register (offset A48h[3]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
11	VS GPIO_PG 3 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
13:12	VS GPIO_PG 4 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 4 PORT_GOODx# or GPIOx Input Data register (offset A44h[4]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 4 PORT_GOODx# or GPIOx Output Data register (offset A48h[4]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
14	VS GPIO_PG 4 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0

Register 13-199. A34h Virtual Switch GPIO_PG 0_9 Direction Control (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
16:15	VS GPIO_PG 5 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 5 PORT_GOODx# or GPIOx Input Data register (offset A44h[5]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 5 PORT_GOODx# or GPIOx Output Data register (offset A48h[5]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
17	VS GPIO_PG 5 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
19:18	VS GPIO_PG 6 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 6 PORT_GOODx# or GPIOx Input Data register (offset A44h[6]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 6 PORT_GOODx# or GPIOx Output Data register (offset A48h[6]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
20	VS GPIO_PG 6 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
22:21	VS GPIO_PG 7 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 7 PORT_GOODx# or GPIOx Input Data register (offset A44h[7]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 7 PORT_GOODx# or GPIOx Output Data register (offset A48h[7]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
23	VS GPIO_PG 7 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0

Register 13-199. A34h Virtual Switch GPIO_PG 0_9 Direction Control
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
25:24	VS GPIO_PG 8 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 8 PORT_GOODx# or GPIOx Input Data register (offset A44h[8]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 8 PORT_GOODx# or GPIOx Output Data register (offset A48h[8]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
26	VS GPIO_PG 8 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
28:27	VS GPIO_PG 9 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 9 PORT_GOODx# or GPIOx Input Data register (offset A44h[9]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 9 PORT_GOODx# or GPIOx Output Data register (offset A48h[9]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
29	VS GPIO_PG 9 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0
31:30	Reserved	RsvdP	No	00b

Register 13-200. A38h Virtual Switch GPIO_PG 10_11 Direction Control (Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_x Direction Control register <i>Direction Control</i> bit(s) (Port 0, accessible through the Management Port, offset(s) 600h and/or 604h), to support the virtual switches. The main differences are as follows: - Virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used - When the PORT_GOODx# or GPIOx signal is configured as an output, value of 01b is Reserved This register, in combination with the Virtual Switch GPIO_PG 0_9 Direction Control register (VS Upstream Port(s), offset A34h), controls the 12 GPIOs assigned to the virtual switch. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Twelve bits in each of the VS Upstream Port registers located at offsets A40h through A54h correspond to the 12 GPIO 0 through 11 fields in offsets A34h and A38h, within each virtual switch. Note: Register offsets A44h and A48h, referenced within this register, are located in the VS Upstream Port(s).				
1:0	VS GPIO_PG 10 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 10 PORT_GOODx# or GPIOx Input Data register (offset A44h[10]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 10 PORT_GOODx# or GPIOx Output Data register (offset A48h[10]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	RWS	Yes	00b
2	VS GPIO_PG 10 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	RWS	Yes	0

Register 13-200. A38h Virtual Switch GPIO_PG 10_11 Direction Control
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
4:3	VS GPIO_PG 11 PORT_GOODx# or GPIOx Source/Destination As Input: 00b = To VS GPIO_PG 11 PORT_GOODx# or GPIOx Input Data register (offset A44h[11]) 01b = General interrupt (INTx, MSI, or VSx_PEX_INTA#) 10b, 11b = Reserved As Output: 00b = From VS GPIO_PG 11 PORT_GOODx# or GPIOx Output Data register (offset A48h[11]) 01b = PORT_GOODx# or GPIOx 10b = SHP_PERSTx# output 11b = Reserved	Four enabled virtual switches	RWS	Yes	00b
	Reserved	Otherwise	RsvdP	No	0
5	VS GPIO_PG 11 PORT_GOODx# or GPIOx Direction Control 0 = Input 1 = Output	Four enabled virtual switches	RWS	Yes	0
	Reserved	Otherwise	RsvdP	No	0
31:6	Reserved		RsvdP	No	0-0h

Register 13-201. A3Ch Virtual Switch GPIO_PG 0_11 Availability
(Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
3:0	Number of GPIO_PGs Available Indicates the quantity of PORT_GOODx# or GPIOx signals assigned to each virtual switch, up to a maximum of 11, as indicated in the VSx GPIO_PG 0_11 Assignment register(s) (Port 0, accessible through the Management Port, offset(s) 650h through 65Ch). A single GPIOx signal cannot be assigned to more than one virtual switch. 0h = 0 (default) 7h = 7 1h = 1 8h = 8 2h = 2 9h = 9 3h = 3 Ah = 10 4h = 4 Bh = 11 5h = 5 Ch = 12 6h = 6 All other encodings are Reserved . Note: Although this register is programmable, it should not be written.	RWS	Yes	0h
31:4	Reserved	RsvdP	No	0000_000h

Register 13-202. A40h Virtual Switch GPIO_PG 0_11 Input De-Bounce (Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Input De-Bounce register (Port 0, accessible through the Management Port, offset 614h), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Note: Register offsets A34h and A38h, referenced within this register, are located in the VS Upstream Port(s).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[2] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[5] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[8] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[11] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0

**Register 13-202. A40h Virtual Switch GPIO_PG 0_11 Input De-Bounce
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[14] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[17] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[20] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[23] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0

Register 13-202. A40h Virtual Switch GPIO_PG 0_11 Input De-Bounce (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[26] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A34h[29] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A38h[2] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	RWS	Yes	0
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Input De-Bounce Control Controls de-bounce when PORT_GOODx# or GPIOx is configured as an input (offset A38h[5] is Cleared). 0 = PORT_GOODx# or GPIOx input is not de-bounced 1 = PORT_GOODx# or GPIOx input is de-bounced; de-bounce time is approximately 1.3 ms	Four enabled virtual switches	Yes	0
	Reserved	Otherwise	No	0
31:12	Reserved	RsvdP	No	0000_0h

Register 13-203. A44h Virtual Switch GPIO_PG 0_11 Input Data
(Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Input Data register (Port 0, accessible through the Management Port, offset 61Ch), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Note: Register offsets A34h and A38h, referenced within this register, are located in the VS Upstream Port(s).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[2] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[2] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[5] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[5] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[8] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[8] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[11] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[11] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0

Register 13-203. A44h Virtual Switch GPIO_PG 0_11 Input Data (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[14] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[14] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[17] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[17] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[20] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[20] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[23] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[23] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0

**Register 13-203. A44h Virtual Switch GPIO_PG 0_11 Input Data
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[26] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[26] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[29] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A34h[29] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A38h[2] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A38h[2] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	ROS	No	0
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Input Data If PORT_GOODx# or GPIOx is configured as an output (offset A38h[5] is Set), Reads return a value of 0. If PORT_GOODx# or GPIOx is configured as an input (offset A38h[5] is Cleared), Reads return the logic value of the voltage on PORT_GOODx# or GPIOx.	Four enabled virtual switches ROS	No	0
	Reserved	Otherwise RsvdP	No	0
31:12	Reserved	RsvdP	No	0000_0h

Register 13-204. A48h Virtual Switch GPIO_PG 0_11 Output Data (Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Output Data register (Port 0, accessible through the Management Port, offset 624h), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Note: Register offsets A34h and A38h, referenced within this register, are located in the VS Upstream Port(s).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[2] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[5] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[8] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[11] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[14] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[17] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[20] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[23] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0

**Register 13-204. A48h Virtual Switch GPIO_PG 0_11 Output Data
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[26] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0	
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A34h[29] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0	
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A38h[2] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	RWS	Yes	0	
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Output Data If PORT_GOODx# or GPIOx is configured as an output (offset A38h[5] is Set), the value written to this bit is immediately driven to the PORT_GOODx# or GPIOx output. Reads return the value written.	Four enabled virtual switches	RWS	Yes	0
	Reserved	Otherwise	RsvdP	No	0
31:12	Reserved	RsvdP	No	0000_0h	

Register 13-205. A4Ch Virtual Switch GPIO_PG 0_11 Interrupt Polarity (Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Interrupt Polarity register (Port 0, accessible through the Management Port, offset 62Ch), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0

Register 13-205. A4Ch Virtual Switch GPIO_PG 0_11 Interrupt Polarity
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0

Register 13-205. A4Ch Virtual Switch GPIO_PG 0_11 Interrupt Polarity (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	RWS	Yes	0
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Interrupt Polarity Controls whether GPIO Interrupt input is Active-Low or Active-High for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is Active-Low 1 = GPIO Interrupt input is Active-High	Four enabled virtual switches RWS	Yes	0
	Reserved	Otherwise RsvdP	No	0
31:12	Reserved	RsvdP	No	0000_0h

Register 13-206. A50h Virtual Switch GPIO_PG 0_11 Interrupt Status
(Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Interrupt Status register (Port 0, accessible through the Management Port, offset 634h), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Interrupt status remains Set, as long the corresponding PORT_GOODx# or GPIOx signal is asserted, and Clears on its own when the corresponding PORT_GOODx# or GPIOx input de-asserts to the inactive state. The active state of each interrupt is controlled by its respective Virtual Switch GPIO_PG 0_11 Interrupt Polarity register bits (VS Upstream Port(s), offset A4Ch). Note: The bits in this register can be masked by their respective Virtual Switch GPIO_PG 0_11 Interrupt Mask register bits (VS Upstream Port(s), offset A54h).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is inactive 1 = GPIO interrupt is active	ROS	No	0
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0

Register 13-206. A50h Virtual Switch GPIO_PG 0_11 Interrupt Status (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Interrupt Status Indicates whether GPIO interrupts are inactive or active for the PORT_GOODx# or GPIOx signal. 0 = GPIO Interrupt input is inactive 1 = GPIO Interrupt input is active	ROS	No	0
	Reserved			
31:12	Reserved	RsvdP	No	0000_0h

Register 13-207. A54h Virtual Switch GPIO_PG 0_11 Interrupt Mask
(Virtual Switch mode – VS Upstream Port(s))

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Provides a virtualized copy of the GPIO 0_11 Interrupt Mask register (Port 0, accessible through the Management Port, offset 63Ch), to support the virtual switches. The main difference is that the virtual switches have no knowledge of which PORT_GOODx# or GPIOx signal is being used. A single GPIO signal cannot be assigned to more than one virtual switch. Each Virtual Switch Upstream Port can determine the quantity of GPIOs that are assigned to that virtual switch, by reading the Virtual Switch GPIO_PG 0_11 Availability register (VS Upstream Port(s), offset A3Ch). Note: The bits in this register can be used to mask their respective Virtual Switch GPIO_PG 0_11 Interrupt Status register bits (VS Upstream Port(s), offset A50h).				
0	VS GPIO_PG 0 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
1	VS GPIO_PG 1 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
2	VS GPIO_PG 2 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
3	VS GPIO_PG 3 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1

Register 13-207. A54h Virtual Switch GPIO_PG 0_11 Interrupt Mask (Virtual Switch mode – VS Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4	VS GPIO_PG 4 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
5	VS GPIO_PG 5 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
6	VS GPIO_PG 6 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
7	VS GPIO_PG 7 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1

**Register 13-207. A54h Virtual Switch GPIO_PG 0_11 Interrupt Mask
(Virtual Switch mode – VS Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	VS GPIO_PG 8 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
9	VS GPIO_PG 9 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
10	VS GPIO_PG 10 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	RWS	Yes	1
11	VS GPIO_PG 11 PORT_GOODx# or GPIOx Interrupt Mask Indicates whether GPIO interrupts are not masked or masked for the PORT_GOODx# or GPIOx signal. 0 = GPIO interrupt is not masked. 1 = GPIO interrupt is masked. When an interrupt is masked, the corresponding Interrupt Status register bit is not updated.	Four enabled virtual switches RWS	Yes	1
	Reserved	Otherwise	RsvdP	No
31:12	Reserved		RsvdP	No
				0000_0h

13.17 Latency Tolerance Reporting Extended Capability Registers (Offsets B00h – B08h)

This section details the Latency Tolerance Reporting (LTR) Extended Capability registers. [Table 13-34](#) defines the register map.

Table 13-34. LTR Extended Capability Register Map (Upstream Port(s))

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Next Capability Offset (B70h) (Upstream)												Capability Version (1h) (Upstream)				PCI Express Extended Capability ID (0018h) (Upstream)																B00h
Reserved (Downstream)																																
Max No-Snoop Latency (Upstream) Reserved (Downstream)																Max Snoop Latency (Upstream) Reserved (Downstream)																B04h
LTR Message Received Status (Upstream) Reserved (Downstream)																																B08h

Register 13-208. B00h LTR Extended Capability Header (Upstream Port(s))

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0018h, as required by the <i>PCI Express Base r3.0</i> .	Upstream	ROS	Yes	0018h
	Reserved	Downstream	RsvdP	No	0000h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	Upstream	ROS	Yes	1h
	Reserved	Downstream	RsvdP	No	0h
31:20	Next Capability Offset Program to B70h, which addresses the Vendor-Specific Extended Capability 2 structure.	Upstream	ROS	Yes	B70h
	Reserved	Downstream	RsvdP	No	000h

Register 13-209. B04h Max Snoop/No-Snoop Latency (Upstream Port(s))

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Max Snoop Latency					
9:0	Max Snoop LatencyValue Along with field [12:10] (<i>Max Snoop LatencyScale</i>), this register specifies the maximum Snoop latency that the PEX 8724 is allowed to request. Software should program this field to the platform's maximum supported latency, or less.	Upstream	RW	Yes	0-0h
	Reserved	Downstream	RsvdP	No	0-0h
12:10	Max Snoop LatencyScale Provides a scale for the value contained within field [9:0] (<i>Max Snoop LatencyValue</i>). The encoding is the same as the <i>LatencyScale</i> fields in the LTR Message. (Refer to the <i>PCI Express Base r3.0</i> , Section 6.18.) Hardware operation is undefined if software writes a Not Permitted value to this field.	Upstream	RW	Yes	000b
	Reserved	Downstream	RsvdP	No	000b
15:13	Reserved		RsvdP	No	000b
Max No-Snoop Latency					
25:16	Max No-Snoop LatencyValue Along with field [28:26] (<i>Max No-Snoop LatencyScale</i>), this register specifies the maximum No-Snoop latency that the PEX 8724 is allowed to request. Software should program this field to the platform's maximum supported latency, or less.	Upstream	RW	Yes	0-0h
	Reserved	Downstream	RsvdP	No	0-0h
28:26	Max No-Snoop LatencyScale Provides a scale for the value contained within field [25:16] (<i>Max No-Snoop LatencyValue</i>). The encoding is the same as the <i>LatencyScale</i> fields in the LTR Message. (Refer to the <i>PCI Express Base r3.0</i> , Section 6.18.) Hardware operation is undefined if software writes a Not Permitted value to this field.	Upstream	RW	Yes	000b
	Reserved	Downstream	RsvdP	No	000b
31:29	Reserved		RsvdP	No	000b

Register 13-210. B08h LTR Message Received Status (Upstream Port(s))

Bit(s)	Description	Upstream Port x	Type	Serial EEPROM and I ² C	Default
When driven with a value of 1, this register holds the Port in reset, including the Port PCI-to-PCI bridge and the hierarchy below it. A value of 0 indicates to un-reset the PCI-to-PCI bridge and the hierarchy below it. Notes: The Downstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Upstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port x LTR Initiate or Received Status bits that are used to indicate that an LTR Message was received on a specific Port. Write 1 to Clear. 0 = No LTR Message was received on the Port 1 = LTR Message was received on the Port	0	RO	Yes	0
1		1	RO	Yes	0
2		2	RO	Yes	0
3		3	RO	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x LTR Initiate or Received Status bits that are used to indicate that an LTR Message was received on a specific Port. Write 1 to Clear. 0 = No LTR Message was received on the Port 1 = LTR Message was received on the Port	8	RO	Yes	0
9		9	RO	Yes	0
10		10	RO	Yes	0
31:11	Reserved		RsvdP	No	0-0h

13.18 Device-Specific Registers (Offsets B70h – DFCh)

This section details the Device-Specific registers located at offsets B70h through DFCh. Device-Specific registers are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. Table 13-35 defines the register map.

Other Device-Specific registers are detailed in:

- Section 13.16, “Device-Specific Registers (Offsets 1C0h – A74h)”
- Section 13.20, “Device-Specific Registers – Failover and Port Cut-Thru (Offsets F00h – F20h)”
- Section 13.22, “Device-Specific Registers (Offsets F30h – FB0h)”

Note: *It is recommended that these registers not be changed from their default values.*

**Table 13-35. Device-Specific Register Map
(Offsets B70h – DFCh)**

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets																
Next Capability Offset 2 (000h)																1h																PCI Express Extended Capability ID 2 (000Bh)																B70h
Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch)																																...																
Device-Specific Registers – Physical Layer (Offsets B80h – C8Ch)																																B80h																
																																...																
																																C8Ch																
Device-Specific Registers – Requester ID Read Back (Offset C90h)																																C90h																
Reserved																																C94h –	DFCh															

13.18.1 Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch)

This section details the Device-Specific Vendor-Specific Extended Capability 2 registers. [Table 13-36](#) defines the register map.

Table 13-36. Device-Specific, Vendor-Specific Extended Capability 2 Register Map (All Ports)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Next Capability Offset 2 (000h)										Capability Version 2 (1h)			PCI Express Extended Capability ID 2 (000Bh)																B70h			
Vendor-Specific Header 2																																B74h
Hardwired Device ID																Hardwired Vendor ID																B78h
<i>Reserved</i>																				Hardwired Revision ID												B7Ch

Register 13-211. B70h Vendor-Specific Extended Capability 2 (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID 2 Program to 000Bh, to indicate that the Capability structure is the Vendor-Specific Extended Capability structure.	ROS	Yes	000Bh
19:16	Capability Version 2	ROS	Yes	1h
31:20	Next Capability Offset 2 000h = This extended capability is the last extended capability in the PEX 8724 Extended Capabilities list	ROS	Yes	000h

Register 13-212. B74h Vendor-Specific Header 2 (All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Vendor-Specific ID 2 ID Number of this Extended Capability structure.	ROS	Yes	0001h
19:16	Vendor-Specific Rev 2 Version Number of this structure.	ROS	Yes	0h
31:20	Vendor-Specific Length 2 Quantity of bytes in the entire structure.	ROS	Yes	010h

**Register 13-213. B78h PLX Hardwired Configuration ID
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Hardwired Vendor ID Always returns the PLX PCI-SIG-assigned Vendor ID value, 10B5h.	ROS	No	10B5h
31:16	Hardwired Device ID Always returns the PEX 8724 default Device ID value, 8724h.	ROS	No	8724h

**Register 13-214. B7Ch PLX Hardwired Revision ID
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
7:0	Hardwired Revision ID Always returns the PEX 8724 default Revision ID value – BAh or CAh.	ROS	No	Current Rev # (BAh)	Current Rev # (CAh)
31:8	Reserved	RsvdP	No	0000_00h	0000_00h

13.18.2 Device-Specific Registers – Physical Layer (Offsets B80h – C8Ch)

This section details the Device-Specific Physical Layer (PHY) registers located at offsets B80h through C8Ch. [Table 13-37](#) defines the register map.

Other Device-Specific PHY registers are detailed in [Section 13.16.4, “Device-Specific Registers – Physical Layer \(Offsets 200h – 25Ch\).”](#)

Table 13-37. Device-Specific PHY Register Map (Offsets B80h – C8Ch) (Ports^a)

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Factory Test Only/Reserved			SerDes Control B80h
Synchronous Advertised N_FTS			B84h
Reserved			B88h
Asynchronous Advertised N_FTS			B8Ch
Reserved			B90h
Factory Test Only/Reserved		PIPE Transmit Parameter Control	B94h
2.5 GT/s -3.5 dB Transmit Parameters			B98h
5.0 GT/s -6 dB Transmit Parameters			B9Ch
5.0 GT/s -3.5 dB Transmit Parameters			BA0h
Reserved			BA4h – BB4h
Reserved		DC Balance Control	BB8h
Trained Coefficient Status			BBCh
Port Accumulation Control			BC0h
Recovery Diagnostic			BC4h
User Lane Equalization Control			BC8h
Factory Test Only/Reserved			BCCh – BD0h
Gen 3 Coefficient Values			BD4h
Equalization Control			BD8h
Factory Test Only/Reserved		Signal Detect Level	BDCh
Gen 3 LFSR Seed			BE0h
Factory Test Only			BE4h – BE8h
TLP Round Trip Timer Control			BECh
Port Receiver Error Counter			BF0h
Factory Test Only/Reserved			BF4h – BF8h
AHB Access Control			BFCh
Reserved			C00h – C8Ch

a. Certain registers are Port-specific, others are Station-specific; all are Device-specific.

Register 13-215. B80h SerDes Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Controls the SerDes logic parameters. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
2:0	Electrical Idle Inference Time Select Valid only when the Physical Layer Electrical Idle Detect Mask register <i>SerDes x Mask Electrical Idle Detect</i> bits (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 204h[15:0]) are all Set. Selects the amount of time to wait until no SKIP Ordered-Sets are detected, for Electrical Idle to be inferred. Does not affect Electrical Idle inference during the <i>Recovery.Speed</i> substate. 000b = 4 μs 001b = 6 μs 010b = 8 μs 011b = 16 μs 100b = 32 μs 101b = 64 μs 110b = 128 μs (default) 111b = 256 μs	RWS	Yes	110b
3	Reserved	RsvdP	No	0
5:4	Recovery.Speed Electrical Idle Inference Time Select Selects the inferred Electrical Idle detect time for Link rate changes (amount of time that no SKIP, TS1, nor TS2 Ordered-Sets are detected). This inferred function is always enabled. Refer to the <i>PCI Express Base r3.0</i>, Section 4.2.4.3, Table 4-11, for the specified electrical inference times. 00b = Specified EI Inference Time 01b = Specified EI Inference Time / 2 10b = Specified EI Inference Time / 4 11b = Specified EI Inference Time / 8	RWS	Yes	00b
7:6	Reserved	RsvdP	No	00b

Register 13-215. B80h SerDes Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (*Cont.*)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Force SerDes Out Transmit Data (TD) [19:0] Force enable. 1 = All bits of the TD[19:0] inputs, of the SerDes in this Station, are forced to the state specified by bit 9 (<i>SerDes Out Data Force State</i>)	RWS	Yes	0
9	SerDes Out Data Force State TD[19:0] Force state. Specifies the state to which the TD[19:0] inputs are forced, when bit 8 (<i>Force SerDes Out</i>) is Set.	RWS	Yes	0
11:10	Reserved	RsvdP	No	00b
14:12	Tx Parameter Port Select Selects the Port to which the value written in the Tx Parameter Control registers will be applied.	RWS	Yes	000b
15	Reserved	RsvdP	No	0
21:16	Tx Margin Override	RWS	Yes	0-0h
23:22	Reserved	RsvdP	No	00b
27:24	Factory Test Only	RWS	Yes	0h
31:28	Reserved	RsvdP	No	0h

Register 13-216. B84h Synchronous Advertised N_FTS**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)**

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Advertises the Number of Fast Training Sets (N_FTS) values for synchronous clocking. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, field [31:24], is Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 00h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
7:0	Port x Synchronous Tx N_FTS	0, 8	RWS	Yes	80h
15:8	When clocking is synchronous, specifies the N_FTS field value in Training Sets transmitted by the Port, when the Port's Link Control register <i>Common Clock Configuration</i> bit (offset 78h[6]) is Set.	1, 9	RWS	Yes	80h
23:16		2, 10	RWS	Yes	80h
31:24		3	RWS	Yes	80h

Register 13-217. B8Ch Asynchronous Advertised N_FTS**(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)**

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Advertises the Number of Fast Training Sets (N_FTS) values for asynchronous clocking. Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, field [31:24], is Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 00h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
7:0	Port x Asynchronous Tx N_FTS	0, 8	RWS	Yes	A0h
15:8	When clocking is asynchronous, specifies the N_FTS field value in Training Sets transmitted by the Port, when the Port's Link Control register <i>Common Clock Configuration</i> bit (offset 78h[6]) is Cleared.	1, 9	RWS	Yes	A0h
23:16		2, 10	RWS	Yes	A0h
31:24		3	RWS	Yes	A0h

Register 13-218. B94h PIPE Transmit Parameter Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [15:12], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
2:0	Port 0, 8 Tx Margin Applied when TxMargin Override0 = 1.	RWS	Yes	000b
3	Port 0, 8 Tx Swing	RWS	Yes	0
6:4	Port 1, 9 Tx Margin Applied when TxMargin Override1 = 1.	RWS	Yes	000b
7	Port 1, 9 Tx Swing	RWS	Yes	0
10:8	Port 2, 10 Tx Margin Applied when TxMargin Override2 = 1.	RWS	Yes	000b
11	Port 2, 10 Tx Swing	RWS	Yes	0
14:12	Port 3 Tx Margin Applied when TxMargin Override3 = 1.	RWS	Yes	000b
15	Port 3 Tx Swing	RWS	Yes	0
23:16	Factory Test Only	RsvdP	No	00h
31:24	Reserved	RsvdP	No	00h

Register 13-219. B98h 2.5 GT/s -3.5 dB Transmit Parameters
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
2:0	TX_PREDRV_DLY Pre-drive delay.	RWS	Yes	000b
5:3	TXEQ_D2 Transmit equalization.	RWS	Yes	000b
7:6	TX_EQ_E Transmit equalization.	RWS	Yes	00b
10:8	TXEQ_MODE Transmit equalization mode.	RWS	Yes	000b
13:11	TX_ETLEV	RWS	Yes	000b
16:14	TX_D2LEV	RWS	Yes	000b
20:17	TX_LEV Transmit amplitude control.	RWS	Yes	0h
23:21	TX_SR_SET	RWS	Yes	000b
26:24	TX_DTLEV Transmit de-emphasis control.	RWS	Yes	000b
28:27	TX_RT_SET	RWS	Yes	00b
30:29	Reserved	RsvdP	Yes	00b
31	Transmit Parameter Override Enable	RWS	Yes	0

Register 13-220. B9Ch 5.0 GT/s -6 dB Transmit Parameters

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
2:0	TX_PREDRV_DLY Pre-drive delay.	RWS	Yes	000b
5:3	TXEQ_D2 Transmit equalization.	RWS	Yes	000b
7:6	TX_EQ_E Transmit equalization.	RWS	Yes	00b
10:8	TXEQ_MODE Transmit equalization mode.	RWS	Yes	000b
13:11	TX_ETLEV	RWS	Yes	000b
16:14	TX_D2LEV	RWS	Yes	000b
20:17	TX_LEV Transmit amplitude control.	RWS	Yes	0h
23:21	TX_SR_SET	RWS	Yes	000b
26:24	TX_DTLEV Transmit de-emphasis control.	RWS	Yes	000b
28:27	TX_RT_SET	RWS	Yes	00b
30:29	Reserved	RsvdP	Yes	00b
31	Transmit Parameter Override Enable	RWS	Yes	0

Register 13-221. BA0h 5.0 GT/s -3.5 dB Transmit Parameters
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
2:0	TX_PREDRV_DLY Pre-drive delay.	RWS	Yes	000b
5:3	TXEQ_D2 Transmit equalization.	RWS	Yes	000b
7:6	TX_EQ_E Transmit equalization.	RWS	Yes	00b
10:8	TXEQ_MODE Transmit equalization mode.	RWS	Yes	000b
13:11	TX_ETLEV	RWS	Yes	000b
16:14	TX_D2LEV	RWS	Yes	000b
20:17	TX_LEV Transmit amplitude control.	RWS	Yes	0h
23:21	TX_SR_SET	RWS	Yes	000b
26:24	TX_DTLEV Transmit de-emphasis control.	RWS	Yes	000b
28:27	TX_RT_SET	RWS	Yes	00b
30:29	Reserved	RsvdP	Yes	00b
31	Transmit Parameter Override Enable	RWS	Yes	0

Register 13-222. BB8h DC Balance Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
5:0	Port DC Balance Enable	RWS	Yes	3Fh
7:6	Reserved	RsvdP	No	00b
15:8	DC Balance Symbol 14 Threshold	RWS	Yes	20h
23:16	DC Balance Symbol 15 Threshold	RWS	Yes	10h
31:24	Reserved	RsvdP	No	00h

Register 13-223. BBCh Trained Coefficient Status

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
5:0	Pre-Cursor Coefficient Final near- or far-end pre-cursor coefficient for the selected Lane.	ROS	No	0-0h
7:6	Reserved	RsvdP	No	00b
13:8	Main Cursor Coefficient Final near or far end main-cursor coefficient for the selected Lane.	ROS	No	0-0h
15:14	Reserved	RsvdP	No	00b
21:16	Post-Cursor Coefficient Final near- or far-end post-cursor coefficient for the selected Lane.	ROS	No	0-0h
23:22	Reserved	RsvdP	No	00b
24	Near/Far End Select 1 = Coefficient fields return the final coefficients received from the far-end device. Otherwise, coefficient fields return the final coefficients transmitted by the near-end device.	RWS	Yes	0
27:25	Reserved	RsvdP	No	000b
31:28	Lane Select Selects the physical Lane for which to return the coefficients. Note: Table 13-5 lists the Lane assignments, by Port and Station, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Lanes 0 through 7 are applicable to Station 0 only, because Station 1 does not have Lanes in those Lane positions. 0h = Lane 0 of the Station (default) 8h = Lane 8 of the Station 1h = Lane 1 of the Station 9h = Lane 9 of the Station 2h = Lane 2 of the Station Ah = Lane 10 of the Station 3h = Lane 3 of the Station Bh = Lane 11 of the Station 4h = Lane 4 of the Station Ch = Lane 12 of the Station 5h = Lane 5 of the Station Dh = Lane 13 of the Station 6h = Lane 6 of the Station Eh = Lane 14 of the Station 7h = Lane 7 of the Station Fh = Lane 15 of the Station	RWS	Yes	0h

Register 13-224. BC0h Port Accumulation Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
1:0	Lock Down Mode Selects the Station's Port 0 Lock Down mode, which fixes a set of Lanes to that Port. When enabled, those Lanes cannot be assigned to any other Port, regardless of the configuration of the devices attached to the Station. Bit 0 determines the final Port configuration. Bit 1 enables/disables PIPE Reversal mode.	RWS	Yes	00b
2	Lock Down Mode Enable Enables the Station's Port 0 Lock Down mode when Autonomous Port Accumulation mode is also enabled (bit 7 (<i>Autonomous Port Accumulation Mode Enable</i>) is Set). 0 = Disables Lock Down mode 1 = Enables Lock Down mode	RWS	Yes	0
6:3	Reserved	RsvdP	No	0h
7	Autonomous Port Accumulation Mode Enable Note: Autonomous Port Accumulation is supported only on Station 0. For Station 1, this bit is RsvdZ, not serial EEPROM nor I²C writable, and collectively has a default value of 0. 0 = Disables Autonomous Port Accumulation mode 1 = Enables Autonomous Port Accumulation mode	RWS	Yes (Serial EEPROM only)	0

Register 13-224. BC0h Port Accumulation Control
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
10:8	Auto Port Configuration Returns the current Port configuration selected by the Port Accumulation logic. The returned data is valid when Autonomous Port Accumulation mode is enabled (bit 7 (<i>Autonomous Port Accumulation Mode Enable</i>) is Set) and at least one Port within the Station has reached the L0 Link PM state. Note: Refer to Table 13-5 for further details regarding these Port Configurations. 001b = 1 010b = 2 011b = 3 100b = 4 All other encodings are Reserved.	ROS	No	PCFG
11	PIPE Reverse Returns the PIPE Reversal mode selected by the Port Accumulation logic. The returned data is valid when Autonomous Port Accumulation mode is enabled (bit 7 (<i>Autonomous Port Accumulation Mode Enable</i>) is Set) and at least one Port within the Station has reached the L0 Link PM state. 0 = PIPE Reversal mode is disabled (bit 1 of field [1:0] (<i>Lock Down Mode</i>) is Cleared) 1 = PIPE Reversal mode is enabled (bit 1 of field [1:0] (<i>Lock Down Mode</i>) is Set)	ROS	No	PCFG
31:12	Reserved	RsvdP	No	0000_0h

Register 13-225. BC4h Recovery Diagnostic

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
15:0	Recovery Counter Quantity of times that the Port entered the <i>Recovery</i> state since the Counter was last Cleared.	ROS	No	PCFG
23:16	Reserved	RsvdP	No	00h
26:24	Port Select Selects the Port for which to return the Recovery Count, or the Port's Recovery Counter to Clear.	RW	Yes	000b
30:27	Reserved	RsvdP	No	0h
31	Clear Recovery Count Writing 1 to this bit Clears the Port's Recovery Counter.	RZ	Yes	0

Register 13-226. BC8h User Lane Equalization Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, field [31:24], is Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 00h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
7:0	User Lane Equalization Control Data for Port x These fields allow Memory Write commands (rather than the serial EEPROM) to load the Lane x Equalization Control register(s) (offset(s) 118h through 134h) if this register is written. When this register is used to load the Lane x Equalization Control register(s), all Lanes that belong to the same Port have the same Settings.	0, 8	RWS	Yes	00h
15:8		1, 9	RWS	Yes	00h
23:16		2, 10	RWS	Yes	00h
31:24		3	RWS	Yes	00h

Register 13-227. BD4h Gen 3 Coefficient Values

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STN _x _PORTCFG _x input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.					
The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h.					
The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
17:0	Preset 0 Coefficients Refer to Section 18.7.2 , “Tx Preshoot and De-Emphasis,” for details, including values, regarding this field.		RWS	Yes	1_0BC0h
	Bit(s)	Description			
	5:0	Preshoot			
	11:6	Main			
	17:12	De-Emphasis			
23:18	Gen 3 FS Value Gen 3 Full Swing value.	RWS	Yes	3Fh	
27:24	Preset Select Refer to Section 18.7.2 , “Tx Preshoot and De-Emphasis,” for details, including values, regarding this field.	RWS	Yes	0h	
29:28	Factory Test Only	RWS	Yes	00b	
30	Reserved	RsvdP	No	0	
31	Coefficient Write Enable Control Must be Set to enable modification of bits [23:0].	RWS	Yes	0	

Register 13-228. BD8h Equalization Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, bits [19, 11, and 3], are Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 0. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Port x Bypass Equalization Directs Downstream Ports to bypass the Transmitter Equalization phases.	0, 8	RWS	Yes	0
1		1, 9	RWS	Yes	0
2		2, 10	RWS	Yes	0
3		3	RWS	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x Set start_equalization_w_preset Directs Downstream Ports to Set the start_equalization_w_preset variable to 1 at the appropriate time.	0, 8	RWS	Yes	0
9		1, 9	RWS	Yes	0
10		2, 10	RWS	Yes	0
11		3	RWS	Yes	0
13:12	Factory Test Only		RsvdP	No	00b
15:14	Reserved		RsvdP	No	00b
16	Port x Reset EIEOS Counter Directs Ports to Set the Reset EIEOS Counter bit in the TS1 Ordered-Sets transmitted during its Equalization phase.	0, 8	RWS	Yes	0
17		1, 9	RWS	Yes	0
18		2, 10	RWS	Yes	0
19		3	RWS	Yes	0
21:20	Factory Test Only		RsvdP	No	00b
23:22	Reserved		RsvdP	No	00b

Register 13-228. BD8h Equalization Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port) (Cont.)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default																		
26:24	Receiver Evaluation Period Selects the length of time to use for the Receiver Evaluation period. <table><thead><tr><th>Silicon Revision BA</th><th>Silicon Revision CA</th></tr></thead><tbody><tr><td>000b = 1 μs</td><td>000b = 4 μs</td></tr><tr><td>001b = 2 μs</td><td>001b = 8 μs</td></tr><tr><td>010b = 4 μs</td><td>010b = 16 μs</td></tr><tr><td>011b = 8 μs</td><td>011b = 32 μs</td></tr><tr><td>100b = 26 μs (default)</td><td>100b = 64 μs (default)</td></tr><tr><td>101b = 32 μs</td><td>101b = 128 μs</td></tr><tr><td>110b = 64 μs</td><td>110b = 256 μs</td></tr><tr><td>111b = 128 μs</td><td>111b = 512 μs</td></tr></tbody></table>	Silicon Revision BA	Silicon Revision CA	000b = 1 μs	000b = 4 μs	001b = 2 μs	001b = 8 μs	010b = 4 μs	010b = 16 μs	011b = 8 μs	011b = 32 μs	100b = 26 μs (default)	100b = 64 μs (default)	101b = 32 μs	101b = 128 μs	110b = 64 μs	110b = 256 μs	111b = 128 μs	111b = 512 μs		RWS	Yes	100b
Silicon Revision BA	Silicon Revision CA																						
000b = 1 μs	000b = 4 μs																						
001b = 2 μs	001b = 8 μs																						
010b = 4 μs	010b = 16 μs																						
011b = 8 μs	011b = 32 μs																						
100b = 26 μs (default)	100b = 64 μs (default)																						
101b = 32 μs	101b = 128 μs																						
110b = 64 μs	110b = 256 μs																						
111b = 128 μs	111b = 512 μs																						
31:27	Receiver Evaluation Threshold Selects the maximum quantity of Receiver Evaluation periods per Transmitter Tuning phase.		RWS	 Yes	19h																		

Register 13-229. BDCh Signal Detect Level

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
Programs the Signal Detect Level for 2.5 and 8.0 GT/s; writes to this register do not affect the Signal Detect Level for 5.0 GT/s. The Signal Detect Level for individual Lanes and/or SerDes quads can be programmed, independently, for 2.5, 5.0 and 8.0 GT/s, through the AHB Access Control register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BFCh). Notes: <i>Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STN_x_PORTCFG_x input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.</i> <i>The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, field [14:12], is Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 000b.</i> <i>The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).</i>					
2:0	Port 0, 8 Signal Detect Level	RWS	Yes	010b	011b
3	Reserved	RsvdP	No	0	
6:4	Port 1, 9 Signal Detect Level	RWS	Yes	010b	011b
7	Reserved	RsvdP	No	0	
10:8	Port 2, 10 Signal Detect Level	RWS	Yes	010b	011b
11	Reserved	RsvdP	No	0	
14:12	Port 3 Signal Detect Level	RWS	Yes	010b	011b
22:15	Factory Test Only/Reserved	RsvdP	No	0-0h	
31:23	Reserved	RsvdP	No	0-0h	

Register 13-230. BE0h Gen 3 LFSR Seed

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Lanes [0-15]. The Port 8 bits are for Lanes [24-31].				
22:0	LFSR Seed Linear Feedback Shift register (LFSR) seed. Used as the Lane-specific scrambler seed for the Gen 3 scrambler/de-scrambler.	RWS	Yes	1D_BFBCh
23	Reserved	RsvdP	No	0
26:24	Lane Select Specifies which of eight register banks in field [22:0] (<i>LFSR Seed</i>) is to be accessed. Note: Table 13-5 lists the Lane assignments, by Port and Station, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Lanes 0 through 7 are applicable to Station 0 only, because Station 1 does not have Lanes in those Lane positions. 000b = Lanes 0, 8 of the Station (default) 001b = Lanes 1, 9 of the Station 010b = Lanes 2, 10 of the Station 011b = Lanes 3, 11 of the Station 100b = Lanes 4, 12 of the Station 101b = Lanes 5, 13 of the Station 110b = Lanes 6, 14 of the Station 111b = Lanes 7, 15 of the Station	RWS	Yes	000b
30:27	Reserved	RsvdP	No	0h
31	LFSR Seed Write Enable Control Must be Set to enable modification of field [22:0] (<i>LFSR Seed</i>).	RWS	Yes	0

Register 13-231. BECh TLP Round Trip Timer Control
(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STN _x _PORTCFG _x input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports.					
The Port 8 bits are Software Only , RsvdP, not serial EEPROM nor I ² C writable, and collectively have a default value of 0000_0000h.					
The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	TRT Enable Enable TLP Round Trip (TRT) Timer. 0 = TLP Round Trip Timer is disabled 1 = TLP Round Trip Timer is enabled	RW	Yes	0	
1	TRT Clear Always returns a value of 0 when read. 1 = Clear minimum and maximum registers	RW	Yes	0	
2	TRT Read Maximum 0 = Return minimum time in field [31:16] (<i>TRT Time</i>) 1 = Return maximum time in field [31:16] (<i>TRT Time</i>)	RW	Yes	0	
3	TRT Active 1 = TRT Timer is active, time update is pending	RO	No	0	
6:4	TRT Port Select	RW	Yes	000b	
7	Reserved	RsvdP	No	0	
14:8	TRT TLP Type and Format Select		RWS	Yes	0-0h
	Bit(s)	Description/Function			
	12:8	Non-Posted TLP Type			
	14:13	Non-Posted TLP Format			
15	TRT Ignore Format Field 1 = Detect egress TLPs, by matching only bits [12:8] (<i>Non-Posted TLP Type</i>) of field [14:8]	RW	Yes	0	
31:16	TRT Time Returns the minimum or maximum round trip time, depending upon the bit 2 (<i>TRT Read Maximum</i>) state.	RO	No	FFFFh	

Register 13-232. BF0h Port Receiver Error Counter

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. Additionally, Port 8, field [31:24], is Factory Test Only, RsvdP, not serial EEPROM nor I²C writable, and has a default value of 00h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
7:0	Port x Receiver Error Counter	0, 8	RW1C	No	00h
15:8	When read, returns the quantity of Receiver errors that the Port detected (Receiver Error Counter). The Error Counter saturates at 255.	1, 9	RW1C	No	00h
23:16		2, 10	RW1C	No	00h
31:24		3	RW1C	No	00h
	The Counter is Cleared with any Write to the corresponding byte in this register; otherwise, this field is RO.				

Register 13-233. BFCh AHB Access Control

(Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
In addition to the functions defined for each bit/field, this register can also be used to program the Signal Detect Level, per Lane or SerDes Quad, independently, for each of the three data rates (2.5, 5.0, and 8.0 GT/s), by Station. (Refer to Table 13-38.) Signal Detect Level is defined by 3-bit values (48 fields, per Station). Because the AHB Data window is 8 bits wide, a Read/Modify/Write sequence is required, to preserve the values of other bits within the register. The 3-bit field for 5.0 GT/s exists within one 8-bit window; therefore, only one Read/Modify/Write sequence is needed. However the 3-bit fields for 2.5 and 8.0 GT/s are each split between two offsets; therefore, two Read/Modify/Write sequences are required to modify a single 3-bit field. For example, to program SerDes Quad 0 to use value 3 during 8.0 GT/S, software performs the following actions: - Writes the value 000E_5800h to offset BFCh. - Reads the register (to preserve bits [7:1]). - Clears bit 0 (msb of 3-bit Signal Detect value). - ORs the field [7:0] value with 008E_5800h. - Writes the result back to offset BFCh. - Writes the value 000E_5700h to offset BFCh. - Reads the register (to preserve bits [5:0]). - Changes the value of bits [7:6] to 10b (lsb of Signal Detect value). - ORs the field [7:0] value with 008E_5700h. - Writes the result back to offset BFCh. Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and collectively have a default value of 0000_0000h. The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
7:0	AHB Data When an AHB register is being written, the data in this field is the Write data. When an AHB register is read, the response (Read) data is returned in this field.	RWS	Yes	00h
20:8	AHB Address Specifies the address of the AHB register being accessed.	RWS	Yes	0-0h
22:21	AHB Quad Select Specifies the SerDes quad to which the AHB access is directed.	RWS	Yes	00b
23	AHB -Read/+Write Specifies the AHB access type. 0 = Read 1 = Write	RWS	Yes	0
29:24	Reserved	RsvdP	No	0-0h
30	AHB Access Timeout 1 = An AHB register access did not return AHB_Ready	RW1C	No	0
31	AHB Ready 0 = AHB Controller is processing the previous access or client accesses 1 = AHB Controller is ready to accept a new access	ROS	No	1

Table 13-38. Signal Level Detect Read/Modify/Write Values for AHB Access Control Register
(Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BFCh)^a

Quad	Lane(s) ^b	Gen 1 (2.5 GT/s)						Gen 2 (5.0 GT/s)			Gen 3 (8.0 GT/s)					
		Read/Modify/Write			Read/Modify/Write			Read/Modify/Write			Read/Modify/Write			Read/Modify/Write		
		Write (bit 23 = 0)	Read [7:1] ---»	Write (bit 23 = 1) OR [7:1] OR	Write (bit 23 = 0)	Read [5:0] ---»	Write (bit 23 = 1) OR [5:0] OR	Write (bit 23 = 0)	Read [4:0] ---»	Write (bit 23 = 1) OR [4:0] OR	Write (bit 23 = 0)	Read [7:1] ---»	Write (bit 23 = 1) OR [7:1] OR	Write (bit 23 = 0)	Read [5:0] ---»	Write (bit 23 = 1) OR [5:0] OR
				[0] = msb			[7:6] = lsb			(bits [7:5])			[0] = msb			[7:6] = lsb
0	(All 4)	000E1300h		008E13__h	000E1200h		008E12__h	000E4F00h		008E4F__h	000E5800h		008E58__h	000E5700h		008E57__h
	0	00021300h		008213__h	00021200h		008212__h	00024F00h		00824F__h	00025800h		008258__h	00025700h		008257__h
	1	00041300h		008413__h	00041200h		008412__h	00044F00h		00844F__h	00045800h		008458__h	00045700h		008457__h
	2	00061300h		008613__h	00061200h		008612__h	00064F00h		00864F__h	00065800h		008658__h	00065700h		008657__h
	3	00081300h		008813__h	00081200h		008812__h	00084F00h		00884F__h	00085800h		008858__h	00085700h		008857__h
1	(All 4)	002E1300h		00AE13__h	002E1200h		00AE12__h	002E4F00h		00AE4F__h	002E5800h		00AE58__h	002E5700h		00AE57__h
	4	00221300h		00A213__h	00221200h		00A212__h	00224F00h		00A24F__h	00225800h		00A258__h	00225700h		00A257__h
	5	00241300h		00A413__h	00241200h		00A412__h	00244F00h		00A44F__h	00245800h		00A458__h	00245700h		00A457__h
	6	00261300h		00A613__h	00261200h		00A612__h	00264F00h		00A64F__h	00265800h		00A658__h	00265700h		00A657__h
	7	00281300h		00A813__h	00281200h		00A812__h	00284F00h		00A84F__h	00285800h		00A858__h	00285700h		00A857__h
2	(All 4)	004E1300h		00CE13__h	004E1200h		00CE12__h	004E4F00h		00CE4F__h	004E5800h		00CE58__h	004E5700h		00CE57__h
	8	00421300h		00C213__h	00421200h		00C212__h	00424F00h		00C24F__h	00425800h		00C258__h	00425700h		00C257__h
	9	00441300h		00C413__h	00441200h		00C412__h	00444F00h		00C44F__h	00445800h		00C458__h	00445700h		00C457__h
	10	00461300h		00C613__h	00461200h		00C612__h	00464F00h		00C64F__h	00465800h		00C658__h	00465700h		00C657__h
	11	00481300h		00C813__h	00481200h		00C812__h	00484F00h		00C84F__h	00485800h		00C858__h	00485700h		00C857__h
3	(All 4)	006E1300h		00EE13__h	006E1200h		00EE12__h	006E4F00h		00EE4F__h	006E5800h		00EE58__h	006E5700h		00EE57__h
	12	00621300h		00E213__h	00621200h		00E212__h	00624F00h		00E24F__h	00625800h		00E258__h	00625700h		00E257__h
	13	00641300h		00E413__h	00641200h		00E412__h	00644F00h		00E44F__h	00645800h		00E458__h	00645700h		00E457__h
	14	00661300h		00E613__h	00661200h		00E612__h	00664F00h		00E64F__h	00665800h		00E658__h	00665700h		00E657__h
	15	00681300h		00E813__h	00681200h		00E812__h	00684F00h		00E84F__h	00685800h		00E858__h	00685700h		00E857__h

- a. Throughout this data book, 32-bit hex values usually include an underscore character (_), separating the combined 16-bit values. In this table, however, that underscore is not used, to avoid confusion with the underscores used for placeholder values.
- b. If a SerDes module does not exist in that position, the corresponding bit(s) is (are) **Factory Test Only, Reserved** (RsvdP), and not serial EEPROM nor I²C writable.

13.18.3 Device-Specific Registers – Requester ID Read Back (Offset C90h)

This section details the Device-Specific Requester ID Read Back register. [Table 13-39](#) defines the register map.

Table 13-39. Device-Specific Requester ID Read Back Register Map (All Ports)

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offset
Reserved (Upstream)		Requester ID Read Back (Upstream)	C90h
Reserved (Downstream)			

Register 13-234. C90h Requester ID Read Back (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	Requester ID Read Back Memory Read to this location returns the Requester ID of that TLP. Determines the Requester ID values to program into the NT Port Requester ID Lookup Tables (NT Port Link Interface, offsets DB4h to DF0h , and NT Port Virtual Interface, offsets D94h to DD0h), in the Station that contains the NT Port.	Upstream	ROS	No	0000h
	Reserved	Downstream	RsvdP	No	0000h
31:16	Reserved		RsvdP	No	0000h

13.19 Multicast Extended Capability Registers (Offsets E00h – E2Ch)

This section details the Multicast Extended Capability registers. [Table 13-40](#) defines the register map. Multicast is described, in detail, in [Section 8.6, “Multicast.”](#)

Table 13-40. Multicast Extended Capability Register Map (All Ports)

31 30 29 28 27 26 25 24 23 22 21 20												19 18 17 16				15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0												Offsets
Next Capability Offset (B00h or F24h)												Capability Version (1h)				PCI Express Extended Capability ID (0012h)												E00h
Multicast Control												Multicast Extended Capability												E04h				
Multicast BAR0																								E08h				
Multicast BAR1																								E0Ch				
Multicast Receive 0																								E10h				
Multicast Receive 1																								E14h				
Multicast Block All 0																								E18h				
Multicast Block All 1																								E1Ch				
Multicast Block Untranslated 0																								E20h				
Multicast Block Untranslated 1																								E24h				
Multicast Overlay BAR0																								E28h				
Multicast Overlay BAR1																								E2Ch				

Register 13-235. E00h Multicast Extended Capability Header (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0012h, to indicate that the Extended Capability structure is the Multicast Extended Capability structure.		ROS	Yes	0012h
19:16	Capability Version		ROS	Yes	1h
31:20	Next Capability Offset Next extended capability is the LTR Extended Capability structure, offset B00h.	Upstream	ROS	Yes	B00h
	Next extended capability is the ACS Extended Capability structure, offset F24h.	Downstream	ROS	Yes	F24h

**Register 13-236. E04h Multicast Extended Capability and Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Multicast Extended Capability				
5:0	MC Max Group 00h = One Multicast Group is supported 3Fh = Maximum quantity of Multicast Groups that the component supports, encoded as <i>M</i> -1	ROS	Yes	3Fh
14:6	Reserved	RsvdP	No	0-0h
15	MC ECRC Generation Supported 0 = ECRC generation is not supported in Multicast 1 = ECRC generation is supported in Multicast	ROS	Yes	1
Multicast Control				
21:16	MC Num Group Indicates the quantity of Multicast Groups configured for use, encoded as <i>N</i> -1. The behavior of this field is undefined if its value exceeds the value indicated by field [5:0] (<i>MC Max Group</i>). Indirectly defines the upper limit of the Multicast Address range. Ignored if bit 31 (<i>MC Enable</i>) is Cleared. 00h = One Multicast Group is configured for use	RW	Yes	0-0h
30:22	Reserved	RsvdP	No	0-0h
31	MC Enable 0 = Disables Multicast 1 = Enables Multicast	RW	Yes	0

**Register 13-237. E08h Multicast BAR0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
5:0	MC Index Position The Multicast Group Number LSB location within the address. The behavior of this field is undefined if its value is less than 12 (Ch) and the Multicast Control register <i>MC Enable</i> bit (offset E04h[31]) is Set.	RW	Yes	0-0h
11:6	Reserved	RsvdP	No	0-0h
31:12	MC Base Address Multicast Lower Base Address[31:12]. Base address of the Multicast Address range. The behavior is undefined if: • Multicast Control register <i>MC Enable</i> bit (offset E04h[31]) is Set, and • Bits in this field corresponding to Address bits that contain the Multicast Group number, or Address bits less than the value indicated by field [5:0] (<i>MC Index Position</i>), are non-zero.	RW	Yes	0000_0h

**Register 13-238. E0Ch Multicast BAR1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Upper Base Address Multicast Upper Base Address[63:32]. Base address of the Multicast Address range. The behavior is undefined if: • Multicast Control register <i>MC Enable</i> bit (offset E04h[31]) is Set, and • Bits in this field corresponding to Address bits that contain the Multicast Group number, or Address bits less than the value indicated by the Multicast BAR0 register <i>MC Index Position</i> field (offset E04h[5:0]), are non-zero.	RW	Yes	0000_0000h

**Register 13-239. E10h Multicast Receive 0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Receive Multicast Receive[31:0]. Provides a bit Vector that denotes to which Multicast Groups the Port should forward Multicast TLPs. For each bit that is Set, this Port receives a copy of any Multicast TLPs that exist for the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-240. E14h Multicast Receive 1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Receive Multicast Receive[63:32]. Provides a bit Vector that denotes to which Multicast Groups the Port should forward Multicast TLPs. For each bit that is Set, this Port receives a copy of any Multicast TLPs that exist for the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-241. E18h Multicast Block All 0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Block All Multicast Block All[31:0]. Provides a bit Vector that denotes which Multicast Groups the Multicast function should block. For each bit that is Set, this Port is blocked from sending TLPs to the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-242. E1Ch Multicast Block All 1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Block All Multicast Block All[63:32]. Provides a bit Vector that denotes which Multicast Groups the Multicast function should block. For each bit that is Set, this Port is blocked from sending TLPs to the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-243. E20h Multicast Block Untranslated 0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Block Untranslated Multicast Block Untranslated[31:0]. Used to determine whether a TLP that includes an Untranslated Address should be blocked. For each bit that is Set, this Port is blocked from sending TLPs containing Untranslated addresses to the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-244. E24h Multicast Block Untranslated 1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	MC Block Untranslated Multicast Block Untranslated[63:32]. Used to determine whether a TLP that includes an Untranslated Address should be blocked. For each bit that is Set, this Port is blocked from sending TLPs containing Untranslated addresses to the associated Multicast Group. Hardware ignores bits above MC Num Group (Multicast Extended Capability register <i>MC Num Group</i> field, offset E04h[21:16]).	RW	Yes	0000_0000h

**Register 13-245. E28h Multicast Overlay BAR0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Specifies the Base address of a window in Unicast space onto which Multicast TLPs output from an egress Port are overlaid by a process of address replacement. This allows a single BAR in an endpoint attached to the PEX 8724 to be used for both Unicast and Multicast traffic. At the Upstream Port(s), this register allows the Multicast Address range, or a portion of it, to be overlaid onto Host memory.				
5:0	MC Overlay Size Less than 06h = Disables the Overlay mechanism 06h or greater = Specifies the size (in bytes) of the Overlay Address range, as a power of 2	RW	Yes	0-0h
31:6	MC Overlay BAR Multicast Overlay Lower Base Address[31:6]. Specifies the Base address of the window onto which Multicast TLPs passing through the Multicast function will be overlaid.	RW	Yes	0-0h

**Register 13-246. E2Ch Multicast Overlay BAR1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Specifies the Base address of a window in Unicast space onto which Multicast TLPs output from an egress Port are overlaid by a process of address replacement. This allows a single BAR in an endpoint attached to the PEX 8724 to be used for both Unicast and Multicast traffic. At the Upstream Port(s), this register allows the Multicast Address range, or a portion of it, to be overlaid onto Host memory.				
31:0	MC Overlay Upper Base Address Multicast Overlay Upper Base Address[63:32]. Specifies the Base address of the window onto which Multicast TLPs passing through the Multicast function will be overlaid.	RW	Yes	0000_0000h

13.20 **Device-Specific Registers – Failover and Port Cut-Thru (Offsets F00h – F20h)**

This section details Device-Specific Failover and Port Cut-Thru registers. [Table 13-41](#) defines the register map.

Other Device-Specific registers are detailed in:

- [Section 13.16, “Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#)
- [Section 13.22, “Device-Specific Registers \(Offsets F30h – FB0h\)”](#)

Table 13-41. Device-Specific Failover and Port Cut-Thru Register Map (Upstream Port(s))

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
VS Failover																																F00h
Reserved																																F04h – F1Ch
Port Cut-Thru Enable Status																																F20h

**Register 13-247. F00h VS Failover
(Upstream Port(s))**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default																								
Used for failover switching between the Upstream Port and NT Port within a virtual switch. Reflects the VS Upstream Port and the NT Port that are enabled within the virtual switch. If the virtual switch owns the NT Port, its VS <i>NTx Enable</i> bit within this register is Set. Note: In Base mode, this register maps to the VS0 Upstream register (Port 0, offset 360h), allowing either register to be used for NT Failover. In Virtual Switch mode, however, the VS0 Upstream register is accessible only to the Management Port; therefore, the VS0 Upstream register is aliased to this register (offset F00h), to provide register access to a non-Management Virtual Switch Upstream Port Host.																												
4:0	VS Upstream Port Reflects the VS Upstream Port Number, based upon the STRAP_UPSTRM_PORTSEL[2:0] input states.	RW	Yes	Defined by STRAP_UPSTRM_PORTSEL[2:0] input states																								
	<table> <tr> <th>Field Value</th> <th>STRAP_UPSTRM_PORTSEL[2:0] Input States</th> <th>Port Number</th> </tr> <tr> <td>0_0000b</td> <td>000</td> <td>0</td> </tr> <tr> <td>0_0001b</td> <td>00Z</td> <td>1</td> </tr> <tr> <td>0_0010b</td> <td>001</td> <td>2</td> </tr> <tr> <td>0_0011b</td> <td>0Z0</td> <td>3</td> </tr> <tr> <td>0_1000b</td> <td>011</td> <td>8</td> </tr> <tr> <td>0_1001b</td> <td>Z00</td> <td>9</td> </tr> <tr> <td>0_1010b</td> <td>Z0Z</td> <td>10</td> </tr> </table>				Field Value	STRAP_UPSTRM_PORTSEL[2:0] Input States	Port Number	0_0000b	000	0	0_0001b	00Z	1	0_0010b	001	2	0_0011b	0Z0	3	0_1000b	011	8	0_1001b	Z00	9	0_1010b	Z0Z	10
	Field Value				STRAP_UPSTRM_PORTSEL[2:0] Input States	Port Number																						
	0_0000b				000	0																						
	0_0001b				00Z	1																						
	0_0010b				001	2																						
	0_0011b				0Z0	3																						
	0_1000b				011	8																						
	0_1001b				Z00	9																						
	0_1010b				Z0Z	10																						
	All other encodings are Reserved .																											
7:5	Reserved	RsvdP	No	000b																								

**Register 13-247. F00h VS Failover
(Upstream Port(s)) (Cont.)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default																																				
12:8	Transparent Mode <i>Reserved</i>	RsvdP	No	0-0h																																				
	NT Mode VS NT Port Reflects the NT Port Number, based upon the STRAP_NT_UPSTRM_PORTSEL[2:0] input states, when the NT Port is enabled (bit 13, <i>VS NT Enable</i> , is Set).	RW	Yes	Defined by STRAP_NT_UPSTRM_P ORTSEL[2:0] input states																																				
	<table><tr><th>Field Value</th><th>Bit 13 Value</th><th>STRAP_NT_UPSTRM_PORTSEL[2:0] Input States</th><th>Port Number</th></tr><tr><td>0_0000b</td><td>1</td><td>000</td><td>0</td></tr><tr><td>0_0001b</td><td>1</td><td>00Z</td><td>1</td></tr><tr><td>0_0010b</td><td>1</td><td>001</td><td>2</td></tr><tr><td>0_0011b</td><td>1</td><td>0Z0</td><td>3</td></tr><tr><td>0_1000b</td><td>1</td><td>011</td><td>8</td></tr><tr><td>0_1001b</td><td>1</td><td>Z00</td><td>9</td></tr><tr><td>0_1010b</td><td>1</td><td>Z0Z</td><td>10</td></tr><tr><td>1_1010b</td><td>0</td><td>111</td><td>– (NT mode is disabled)</td></tr></table>				Field Value	Bit 13 Value	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Port Number	0_0000b	1	000	0	0_0001b	1	00Z	1	0_0010b	1	001	2	0_0011b	1	0Z0	3	0_1000b	1	011	8	0_1001b	1	Z00	9	0_1010b	1	Z0Z	10	1_1010b	0	111	– (NT mode is disabled)
	Field Value				Bit 13 Value	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Port Number																																	
	0_0000b				1	000	0																																	
	0_0001b				1	00Z	1																																	
	0_0010b				1	001	2																																	
	0_0011b				1	0Z0	3																																	
	0_1000b				1	011	8																																	
	0_1001b				1	Z00	9																																	
0_1010b	1	Z0Z	10																																					
1_1010b	0	111	– (NT mode is disabled)																																					
All other encodings are <i>Reserved</i> .																																								
13	Transparent Mode <i>Reserved</i>	RsvdP	No	0																																				
	NT Mode VS NT Enable 0 = STRAP_NT_UPSTRM_PORTSEL[2:0] = 111, NT mode is disabled 1 = STRAP_NT_UPSTRM_PORTSEL[2:0] = Port Number, NT mode is enabled	RW	Yes	Defined by STRAP_NT_UPSTRM_P ORTSEL[2:0] input states																																				
31:14	<i>Reserved</i>	RsvdP	No	0-0h																																				

Register 13-248. F20h Port Cut-Thru Enable Status (Upstream Port(s))

Bit(s)	Description	Upstream Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Downstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Upstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. For bit 8, Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port x Cut-Thru Enable Status Link Up status. 0 = Cut-Thru is disabled for this Port (Link is Down) 1 = Cut-Thru is enabled for this Port (Link is Up)	0	RO	No	Defined by STRAP_STN0_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 0</i> field (Port 0, accessible through the Management Port, offset 300h[2:0])
1		1	RO	No	
2		2	RO	No	
3		3	RO	No	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x Cut-Thru Enable Status Link Up status. 0 = Cut-Thru is disabled for this Port (Link is Down) 1 = Cut-Thru is enabled for this Port (Link is Up)	8	RO	No	Defined by STRAP_STN1_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 1</i> field (Port 0, accessible through the Management Port, offset 300h[5:3])
9		9	RO	No	
10		10	RO	No	
31:11	Reserved		RsvdP	No	0-0h

13.21 ACS Extended Capability Registers (Offsets F24h – F2Ch)

This section details the ACS Extended Capability registers. [Table 13-42](#) defines the register map.

**Table 13-42. ACS Extended Capability Register Map
(Downstream Ports)**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Reserved (Upstream)																																F24h
Next Capability Offset (B70h) (Downstream)												Capability Version (1h) (Downstream)				PCI Express Extended Capability ID (000Dh) (Downstream)																
Reserved (Upstream)																																F28h
ACS Control (Downstream)																ACS Capability (Downstream)																
Reserved (Upstream) ACS Egress Control Vector (Downstream)																																F2Ch

**Register 13-249. F24h ACS Extended Capability Header
(Downstream Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: Because this register is implemented as one physical register common to all Ports, the Upstream Port(s)' register (which is Reserved) has the same value as the Downstream Ports' registers. However, in the Upstream Port(s), the ACS Extended Capability is excluded from the Linked List of PCI Express Extended Capabilities, and therefore, the Upstream Port(s)' register is effectively hidden from system software and the non-zero value has no significant consequence.					
15:0	Reserved	Upstream	RsvdP	No	0000h
	PCI Express Extended Capability ID	Downstream	ROS	Yes	000Dh
19:16	Reserved	Upstream	RsvdP	No	0h
	Capability Version	Downstream	ROS	Yes	1h
31:20	Reserved	Upstream	RsvdP	No	000h
	Next Capability Offset Program to B70h, which addresses the Vendor-Specific Extended Capability 2 structure.	Downstream	ROS	Yes	B70h

**Register 13-250. F28h ACS Control and Capability
(Downstream Ports)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
ACS Capability					
0	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS Source Validation	Downstream	RO	Yes	1
1	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS Translation Blocking	Downstream	RO	Yes	1
2	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS P2P Request Redirect ACS Peer-to-Peer Request redirect.	Downstream	RO	Yes	1
3	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS P2P Completion Redirect ACS Peer-to-Peer Completion redirect.	Downstream	RO	Yes	1
4	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS Upstream Forwarding	Downstream	RO	Yes	1
5	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS P2P Egress Control ACS Peer-to-Peer Egress control.	Downstream	RO	Yes	1
6	<i>Reserved</i>	Upstream	RsvdP	No	0
	ACS Direct Translated P2P ACS Direct Translated Peer-to-Peer.	Downstream	RO	Yes	1
7	<i>Reserved</i>		RsvdP	No	0
12:8	<i>Reserved</i>	Upstream	RsvdP	No	00h
	Egress Control Vector Size Encodings 01h through FFh directly indicate the number of each Port's ACS Egress Control Vector register <i>ACS Peer-to-Peer Port x Control</i> bit (Downstream Ports, offset F2Ch[10:8, 3:0]). <i>Note: The ACS Egress Control Vector Size value must be adjusted for the specific Port configuration.</i>	Downstream	HwInit	Yes	18h
15:13	<i>Reserved</i>		RsvdP	No	000b

**Register 13-250. F28h ACS Control and Capability
(Downstream Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
ACS Control					
16	Reserved	Upstream	RsvdP	No	0
	ACS Source Validation Enable 0 = Disables 1 = Enables	Downstream	RW	Yes	0
17	Reserved	Upstream	RsvdP	No	0
	ACS Translation Blocking Enable 0 = Disables 1 = Enables	Downstream	RW	Yes	0
18	Reserved	Upstream	RsvdP	No	0
	ACS P2P Request Redirect Enable Enables or disables ACS Peer-to-Peer Request redirect. 0 = Disables 1 = Enables	Downstream	RW	Yes	0
19	Reserved	Upstream	RsvdP	No	0
	ACS P2P Completion Redirect Enable Enables or disables ACS Peer-to-Peer Completion redirect. 0 = Disables 1 = Enables	Downstream	RW	Yes	0
20	Reserved	Upstream	RsvdP	No	0
	ACS Upstream Forwarding Enable 0 = Disables 1 = Enables	Downstream	RW	Yes	0
21	Reserved	Upstream	RsvdP	No	0
	ACS P2P Egress Control Enable Enables or disables ACS Peer-to-Peer Egress control. 0 = Disables 1 = Enables	Downstream	RW	Yes	0
22	Reserved	Upstream	RsvdP	No	0
	ACS Direct Translated P2P Enable Enables or disables ACS Direct Translated Peer-to-Peer. 0 = Disables 1 = Enables	Downstream	RW	Yes	0
31:23	Reserved		RsvdP	No	0-0h

**Register 13-251. F2Ch ACS Egress Control Vector
(Downstream Ports)**

Bit(s)	Description	Downstream Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Upstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Downstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. The Peer-to-Peer Port x Control bits are valid when the ACS Control register ACS P2P Egress Control Enable bit (Downstream Ports, offset F28h[21]) is Set. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. Port 8 is Software Only, bit 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	ACS Peer-to-Peer Port x Control 0 = No Peer-to-Peer control 1 = ACS Peer-to-Peer control	0	RW	Yes	0
1		1	RW	Yes	0
2		2	RW	Yes	0
3		3	RW	Yes	0
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	ACS Peer-to-Peer Port x Control 0 = No Peer-to-Peer control 1 = ACS Peer-to-Peer control	8	RW	Yes	0
9		9	RW	Yes	0
10		10	RW	Yes	0
31:11	Reserved		RsvdP	No	0-0h

13.22 Device-Specific Registers (Offsets F30h – FB0h)

This section details the Device-Specific registers located at offsets F30h through FB0h. Device-Specific registers are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. [Table 13-43](#) defines the register map.

Other Device-Specific registers are detailed in:

- [Section 13.16, “Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#)
- [Section 13.20, “Device-Specific Registers – Failover and Port Cut-Thru \(Offsets F00h – F20h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 13-43. Device-Specific Register Map (Offsets F30h – FB0h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Device-Specific Registers – Egress Control (Offsets F30h – F40h)		F30h ... F40h
Device-Specific Registers – Ingress Control and Port Enable (Offsets F44h – F6Ch)		F44h ... F6Ch
Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h)		F70h ... FB0h

13.22.1 Device-Specific Registers – Egress Control (Offsets F30h – F40h)

This section details the Device-Specific Egress Control registers. [Table 13-44](#) defines the register map.

Table 13-44. Device-Specific Egress Control Register Map (All Ports)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Egress Control and Status		F30h
Reserved		F34h
Port Egress TLP Threshold		F38h
Reserved		F3Ch – F40h

**Register 13-252. F30h Egress Control and Status
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. The Port 8 bits are Software Only, RsvdP, not serial EEPROM nor I²C writable, and each has a default value of 0. For bits [12:9, 7, 5] – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).				
1:0	Reserved	RsvdP	No	00b
4:2	Factory Test Only	RWS	Yes	000b
5	Snoop Enable as Requester ID The <i>Snoop Enable</i> bit must match the incoming <i>No Snoop</i> attribute for the LUT.	RWS	Yes	0
6	Factory Test Only	RWS	Yes	0
7	Ex Tag 8 Bit 1 = 8-bit Exerciser Tag	RWS	Yes	0
8	Rdr Hdr Prefetch Disable 0 = Prefetch Headers on this Port 1 = Do not prefetch Headers on this Port	RWS	Yes	0
9	Vendor-Defined Type 0 UR 0 = Do not generate UR Vendor-Defined Type 0 Broadcast TLP in <i>DL_Down</i> state 1 = Generate UR Vendor-Defined Type 0 Broadcast TLP in <i>DL_Down</i> state	RWS	Yes	0
10	Egress Credit Timeout Enable 0 = Egress Credit Timeout mechanism is disabled. 1 = Egress Credit Timeout mechanism is enabled. The timeout period is selected in field [12:11] (<i>Egress Credit Timeout Value</i>). Status is reflected in bit 16 (<i>Egress Credit Timeout Status</i>). If the Egress Credit Timer is enabled and expires (due to lack of Flow Control credits from the connected device), the Port brings down its Link. This event generates a Surprise Down Uncorrectable error, for Transparent Downstream Ports. For Upstream Port Egress Credit Timeout, the connected Upstream device detects the Surprise Down event.	RWS	Yes	0
12:11	Egress Credit Timeout Value 00b = 1 ms 01b = 512 ms 10b = 1s 11b = Reserved	RWS	Yes	00b
15:13	Reserved	RsvdP	No	000b

Register 13-252. F30h Egress Control and Status
(All Ports) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
16	Egress Credit Timeout Status 0 = No timeout 1 = Timeout	RW1CS	No	0
18:17	Egress Credit Timeout VC&T Egress Credit timeout for Virtual Channel and Type. 00b = Posted 01b = Non-Posted 10b = Completion 11b = Reserved	ROS	No	00b
30:19	Reserved	RsvdP	No	0-0h
31	Port Activity 0 = Port is idle 1 = Port has one or more pending TLPs to transmit	ROS	No	0

Register 13-253. F38h Port Egress TLP Threshold
(All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Note: Source Queuing and Read Pacing should not be concurrently enabled. The two features are incompatible and doing so can result in Fatal errors.					
11:0	Port Lower TLP Counter When Source Scheduling is disabled due to Threshold, it is re-enabled when the Port TLP Counter goes below this Threshold value.	Upstream	RWS	Yes	003h
		Downstream	RWS	Yes	FFFh
15:12	Reserved		RsvdP	No	0h
27:16	Port Upper TLP Counter When the Port TLP Counter is greater than or equal to this value, the Source Scheduler disables TLP scheduling to this egress Port.	Upstream	RWS	Yes	006h
		Downstream	RWS	Yes	FFFh
31:28	Reserved		RsvdP	No	0h

13.22.2 Device-Specific Registers – Ingress Control and Port Enable (Offsets F44h – F6Ch)

This section details the Device-Specific Ingress Control and Port Enable registers, which include the **Negotiated Link Width** registers. [Table 13-45](#) defines the register map.

Table 13-45. Device-Specific Ingress Control and Port Enable Register Map (Ports^a)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Factory Test Only																															F44h	
Ingress Port-Based Control																															F48h	
Port Enable Status																															F4Ch	
Factory Test Only/Reserved															Negotiated Link Width for Ports 0, 1, 2, 3															F50h		
Reserved				Negotiated Link Width for Ports 8, 9, 10																											F54h	
Factory Test Only/Reserved																														F58h –	F5Ch	
Ingress Control																															F60h	
Reserved																														F64h –	F6Ch	

a. Certain registers are Port-specific, others are Station-specific or Chip-specific; all are Device-specific.

Register 13-254. F48h Ingress Port-Based Control (Base mode – All Ports; Virtual Switch mode – VS Upstream Port)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	ACK TLP Counter Timeout Sets the quantity of ingress TLP Acknowledges (ACKs) pending, which causes a high-priority ACK to be sent. 00b = 16 TLPs 01b = 8 TLPs 10b = 4 TLPs 11b = Feature is disabled	RWS	Yes	00b
31:2	Reserved	RsvdP	No	0-0h

Register 13-255. F4Ch Port Enable Status (Upstream Port(s))

Bit(s)	Description	Upstream Port x	Type	Serial EEPROM and I ² C	Default
The <i>Port x Enable Status</i> bits are qualified with the Virtual Switch Port Vector associated with the Port, which is dependent upon the Port configuration for each Station – Station 0 (bits [3:0]) and Station 1 (bits [10:8]). When a Port is enabled in the corresponding Station configuration, the bit for that Port is Set; otherwise, the bit is Cleared. Notes: The Downstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Upstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. <i>Table 13-5</i> indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations and Ports. For bit 8, Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
0	Port x Enable Status 0 = Port is disabled 1 = Port is enabled	0	RO	No	Defined by STRAP_STN0_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 0</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[2:0])
1		1	RO	No	
2		2	RO	No	
3		3	RO	No	
5:4	Factory Test Only		RsvdP	No	00b
7:6	Reserved		RsvdP	No	00b
8	Port x Enable Status 0 = Port is disabled 1 = Port is enabled	8	RO	No	Defined by STRAP_STN1_PORTCFG[1:0] input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station 1</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3])
9		9	RO	No	
10		10	RO	No	
23:11	Reserved		RsvdP	No	0-0h
28:24	Upstream Port of VS Is in Port x Indicates the virtual switch's Upstream Port Number. 0_0000b = Port 0 (default) 0_0001b = Port 1 0_0010b = Port 2 0_0011b = Port 3 0_1000b = Port 8 0_1001b = Port 9 0_1010b = Port 10 All other encodings are Reserved .		ROS	No	0_0000b
31:29	Reserved		RsvdP	No	000b

Register 13-256. F50h Negotiated Link Width for Ports 0, 1, 2, 3 (Upstream Port(s))

Bit(s)	Description	Upstream Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Downstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Upstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes.					
3:0	Negotiated Link Width and Link Speed Encoding for Port x Remainder of value divided by 5 indicates the negotiated Link width. If the Link is Down, the value is 0h.	0	RO	No	PCFG
7:4	0h = x1, or the Port is in the DL_Down state 1h = x2 2h = x4 3h = x8 4h = x16 (Port 0 only)	1	RO	No	
11:8	Quotient of division indicates the Link speed. 0h = Gen 1 (2.5 GT/s) 1h = Gen 2 (5.0 GT/s) 2h = Gen 3 (8.0 GT/s) All other encodings are Reserved .	2	RO	No	
15:12		3	RO	No	
23:16	Factory Test Only		RsvdP	No	00h
31:24	Reserved		RsvdP	No	00h

Register 13-257. F54h Negotiated Link Width for Ports 8, 9, 10 (Upstream Port(s))

Bit(s)	Description	Upstream Port x	Type	Serial EEPROM and I ² C	Default
Notes: The Downstream Port bits are Reserved, RsvdP, not serial EEPROM nor I²C writable, and have a default value of 0. The Upstream Port x column is provided to indicate the Port Number associated with bits/fields that include “Port x” in their name. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, Ports, SerDes modules, and Lanes. For field [3:0], Port 8 is Software Only, and Ports 9 and 10 are accessed through Software Only Port 8.					
3:0	Negotiated Link Width and Link Speed Encoding for Port x Remainder of value divided by 5 indicates the negotiated Link width. If the Link is Down, the value is 0h.	8	RO	No	PCFG
7:4	0h = x1, or the Port is in the DL_Down state 1h = x2 2h = x4 3h = x8 Quotient of division indicates the Link speed.	9	RO	No	
11:8	0h = Gen 1 (2.5 GT/s) 1h = Gen 2 (5.0 GT/s) 2h = Gen 3 (8.0 GT/s) All other encodings are Reserved .	10	RO	No	
23:12	Factory Test Only		RsvdP	No	000h
31:24	Reserved		RsvdP	No	00h

**Register 13-258. F60h Ingress Control
(Upstream Port(s))**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	Factory Test Only	RWS	Yes	010b
3	Route to Upstream a Downstream Memory TLP that Does Not Hit Any Downstream Port and Hits the Upstream Memory Space When inactive, such TLPs will be UR. When active, such TLPs are routed Upstream.	RWS	Yes	0
5:4	Not used	RWS	Yes	00b
6	Reserved	RsvdP	No	0
7	Not used	RsvdP	No	0
8	Drop ECRC TLPs Drop End-to-end Cyclic Redundancy Check (ECRC) TLPs. 0 = Ignores and passes the ECRC TLP, without dropping it 1 = ECRC TLP was dropped	RWS	Yes	0
9	Drop Poisoned TLPs 0 = PEX 8724 forwards (and does not modify) Poisoned TLPs 1 = Discard Poisoned TLPs Notes: If a Write Request that targets a PEX 8724 register is Poisoned, the register is not modified. If a Non-Posted TLP that targets a PEX 8724 register is Poisoned, the PEX 8724 responds with a Completion in which Status is Unsupported Request (UR), and logs this error in the Uncorrectable Error Status register Unsupported Request Error Status bit (offset FB8h [20]). The PEX 8724 never Sets the EP bit in the TLP Header. If a TLP is received with its EP bit Set (indicating Poisoned), the Port Sets its Detected Parity Error bit(s) (PCI Status register Detected Parity Error bit (offset 04h [31]) on the primary side, and/or Secondary Status register Detected Parity Error bit (offset 1Ch [31]) on the secondary side).	RWS	Yes	0
10	Factory Test Only	RWS	Yes	0
12:11	Not used	RWS	Yes	00b
14:13	Factory Test Only	RWS	Yes	00b
15	Disable NT Port Expansion ROM BAR 0 = NT Port Expansion ROM BAR requests Memory space 1 = NT Port Expansion ROM BAR always reads 0, making the Expansion ROM not present	RWS	Yes	0
23:16	Not used	RWS	Yes	00h

Register 13-258. F60h Ingress Control (Upstream Port(s)) (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
24	Limit NT Port Link Interface Memory Accesses to NT Port Link Interface 0 = NT Port Link Interface is allowed access to all PEX 8724 registers. 1 = Limit NT Port Link Interface Memory accesses that target PEX 8724 accesses to only the NT Port Link Interface Memory-Mapped registers. All Memory accesses that target registers in other PEX 8724 Ports are No Operation – Write data is ignored, Read operation returns zeros (0s).	RWS	Yes	0
25	Factory Test Only	RWS	Yes	0
26	Disable Upstream Port BAR0 and BAR1 0 = Enables all Upstream Port Base Address 0 and Base Address 1 registers (BAR0 and BAR1 , Upstream Port(s), offsets 10h and 14h , respectively) 1 = Disables all Upstream Port BAR0 and BAR1	RWS	Yes	0
27	Flag Unexpected Completion Error 0 = Flag unexpected Completion errors for Completions that hit the PEX 8724's internal virtual PCI Bus space 1 = Silently unexpected Completion errors for Completions that hit the PEX 8724's internal virtual PCI Bus space	RWS	Yes	0
28	Disable VGA BIOS Memory Access Decoding 0 = Enables decoding of PC ROM shadow addresses C_0000h to C_FFFFh (packets destined to these addresses are blocked) 1 = Disables decoding of PC ROM shadow addresses C_0000h to C_FFFFh (packets destined to these addresses are not blocked)	RWS	Yes	1
30:29	Factory Test Only	RWS	Yes	00b
31	Not used	RWS	Yes	0

13.22.3 Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h)

This section details the Device-Specific Error Checking and Debug registers located at offsets F70h through FB0h. [Table 13-46](#) defines the register map.

Other Device-Specific Error Checking and Debug registers are detailed in [Section 13.16.11](#), “Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch).”

Table 13-46. Device-Specific Error Checking and Debug Register Map (Offsets F70h – FB0h) (All Ports)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Power Management Hot Plug User Configuration																																F70h
Reserved																																F74h – FA4h
ACK Transmission Latency Limit																																FA8h
Bad TLP Count																																FACH
Bad DLLP Count																																FB0h

Register 13-259. F70h Power Management Hot Plug User Configuration (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
0	L0s Entry Idle Count Traffic Idle time to meet, to enter the ASPM L0s Link PM state. 0 = Idle condition must last 1 μ s 1 = Idle condition must last 4 μ s		RWS	Yes	0
1	Factory Test Only		RWS	Yes	0
2	Not enabled Functionality associated with this bit is enabled only on Downstream Ports.	Upstream	RWS	Yes	0
	HPC PME Turn-Off Enable 1 = PME Turn-Off Message is transmitted before the Port is turned Off on a Downstream Port	Downstream	RWS	Yes	0
4:3	Not enabled Functionality associated with this field is enabled only on Downstream Ports.	Upstream	RWS	Yes	00b
	HPC T_{pepv} Hot Plug Port time from Power Enable to Power Valid. Indicates the delay from when an HP_PWREN _x is asserted High, to when power is valid at a slot. (Refer to Section 10.8.1.2, “Slot Power-Up Sequencing when Power Controller Present Bit Is Set,” for details.) 00b = Feature is disabled, and HP_PWR_GOOD _x inputs are used for Power Valid 01b = 128 ms 10b = 256 ms 11b = 512 ms	Downstream	RWS	Yes	00b
5	Factory Test Only		RWS	Yes	0
6	Not enabled Functionality associated with this bit is enabled only on Downstream Ports.	Upstream	RWS	Yes (Serial EEPROM only)	0
	HP_PWR_GOOD_x Active-Low Enable Controls the HP_PWR_GOOD _x input polarity. (Refer to Section 10.8.1.2, “Slot Power-Up Sequencing when Power Controller Present Bit Is Set,” for details.) 0 = HP_PWR_GOOD _x inputs are Active-High 1 = HP_PWR_GOOD _x inputs are Active-Low	Downstream	RWS	Yes (Serial EEPROM only)	0
7	Factory Test Only		RWS	Yes	0

**Register 13-259. F70h Power Management Hot Plug User Configuration
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
8	DLLP Timeout Link Retrain Disable Disable Link retraining when no Data Link Layer Packets (DLLPs) are received for more than 256 μ s. 0 = Enables Link retraining when no DLLPs are received for more than 256 μ s (default) 1 = DLLP Timeout is disabled		RWS	Yes	0
9	Factory Test Only		RWS	Yes	0
10	L0s Entry Disable 0 = Enables entry into the ASPM L0s Link PM state on a Port when the state's idle conditions are met 1 = Disables entry into the ASPM L0s Link PM state on a Port when the state's idle conditions are met		RWS	Yes	0
11	Factory Test Only		RWS	Yes	0
12	Software-Controlled Hot Plug Enable 1 = Hot Plug input functionality is disabled and software controls the input functionality		RWS	Yes	0
13	Software-Controlled Power Good When bit 12 (<i>Software-Controlled Hot Plug Enable</i>) is Set, the value in this bit controls HP_PWR_GOOD_x inputs to the PEX 8724.		RWS	Yes	0
14	Upstream Hot Plug Enable 1 = Enables Presence Detect and its corresponding interrupt on the Upstream Port(s)	Upstream	RWS	Yes	0
		Downstream	RsvdP	No	0
15	Port Is Serial Hot Plug Port 1 = Indicates the Port is a Serial Hot Plug Port, using I/O Expanders		RO	No	0
16	HPC Serial Expansion Controller Disable 0 = Enables Serial Hot Plug capability on all Ports 1 = Disables Serial Hot Plug capability on all Ports	0	RWS	Yes	0
	Not used	Otherwise	RsvdP	No	0
17	40-Pin I/O Expander Scan Disable 0 = 40-pin I/O Expander scan 1 = Disables 40-pin I/O Expander scan		RWS	Yes	0
18	Serial Hot Plug INTx De-Bounce Disable 1 = Disables the 10-ms De-Bounce Counter in the Serial Hot Plug Controller, for I/O Expander Interrupt inputs		RWS	Yes	0
19	Serial Hot Plug Override Parallel Disable 0 = Defaults to Serial Hot Plug when a Port is both Parallel and Serial Hot Plug-capable 1 = Selects Parallel Hot Plug if a Port is both Parallel and Serial Hot Plug-capable		RWS	Yes	0

**Register 13-259. F70h Power Management Hot Plug User Configuration
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
20	HPC I/O Reload 1 = Parallel Hot Plug Controller/Serial Hot Plug Controller (I ² C I/O Expander) Output pin values are reloaded from field [26:21] (<i>HPC Output Reload Value</i>). After the action is complete, this bit is self-Clearing.		RWS	Yes	0
26:21	HPC Output Reload Value When bit 20 (<i>HPC I/O Reload</i>) is Set, values from this field are reloaded to the Hot Plug Controller outputs associated with the Port.		RWS	Yes	0-0h
	Bit	Description/Function			
	21	HP_PWRLED_x# or Serial Hot Plug SHP_PWRLED# (SHP_PWRLEDx#)			
	22	HP_ATNLED_x# or Serial Hot Plug SHP_ATNLED# (SHP_ATNLEDx#)			
	23	HP_PWREN_x or Serial Hot Plug SHP_PWREN (SHP_PWRENx)			
	24	HP_CLKEN_x# or Serial Hot Plug SHP_REFCLKEN# (SHP_REFCLKENx#)			
	25	HP_PERST_x# or Serial Hot Plug SHP_PERST# (SHP_PERSTx#)			
26	Serial Hot Plug SHP_INTERLOCK (SHP_INTERLOCKx)				
27	Software Present Detect State Value Presence Detect state (Port's Slot Status register <i>Presence Detect State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[22])) value is programmed from this register when bit 14 or 12 (<i>Upstream Hot Plug Enable</i> or <i>Software-Controlled Hot Plug Enable</i> , respectively) is Set.		RWS	Yes	0
28	Software MRL State Value Manually operated Retention Latch (MRL) Sensor state (Port's Slot Status register <i>MRL Sensor State</i> bit (Base mode – Downstream Ports; Virtual Switch mode – Upstream and Downstream Ports, offset 80h[21])) value is programmed from this register when bit 14 or 12 (<i>Upstream Hot Plug Enable</i> or <i>Software-Controlled Hot Plug Enable</i> , respectively) is Set.		RWS	Yes	0
29	PHY Upstream Port Control Write Enable 1 = Enables the following bits to be functional on Upstream Port(s): • Link Control register <i>Link Disable</i> bit (offset 78h[4]) • Link Control register <i>Retrain Link</i> bit (offset 78h[5]) • Link Control 2 register <i>Target Link Speed</i> field (offset 98h[3:0])	Upstream	RWS	Yes	0
	Reserved	Downstream	RsvdP	No	0
31:30	Factory Test Only		RWS	Yes	00b

**Register 13-260. FA8h ACK Transmission Latency Limit
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Register value should be valid after Link negotiation.				
11:0	ACK Transmission Latency Limit Acknowledge Control Packet (ACK) Transmission Latency Limit. Value changes, based upon the Port's Negotiated Link Width (offset 78h[25:20]), Current Link Speed (offset 78h[19:16]) after the Link is Up, and Maximum Payload Size (offset 70h[7:5]). x1 = 255d x2 = 217d x4 = 118d x8 = 107d x16 = 100d (Port 0, Station 0 only)	RWS	Yes	Defined by STRAP_STN _x _PORTCFG _x input states
15:12	Reserved	RsvdP	No	0h
23:16	Upper 8 Bits of the Replay Timer Limit If the serial EEPROM is not present, the value of this register changes based upon the negotiated Link width after the Link is Up. Value is a multiplier of the default internal timer values that are compliant to the <i>PCI Express Base r3.0</i> . Should normally remain the default value, 00h.	RWS	Yes	00h
30:24	Reserved	RsvdP	No	00h
31	ACK Transmission Latency Timer Status Indicates the written status of field [11:0] (<i>ACK Transmission Latency Limit</i>). After the register is written, either by software and/or serial EEPROM, this bit is Set and Cleared only by a Fundamental Reset.	RO	No	0

Register 13-261. FACH Bad TLP Count
(All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Bad TLP Count Counts the quantity of TLPs received with bad LCRC, or quantity of TLPs with a Sequence Number Mismatch error. The Counter saturates at FFFF_FFFFh and does not roll over to 0000_0000h.	RWS	Yes	0000_0000h

Register 13-262. FB0h Bad DLLP Count
(All Ports)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Bad DLLP Count Counts the quantity of DLLPs received with bad LCRC, or quantity of DLLPs with a Sequence Number Mismatch error. The Counter saturates at FFFF_FFFFh and does not roll over to 0000_0000h.	RWS	Yes	0000_0000h

13.23 Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)

This section details the Advanced Error Reporting Extended Capability registers. [Table 13-47](#) defines the register map.

Table 13-47. Advanced Error Reporting Extended Capability Register Map (All Ports)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Next Capability Offset (138h or 10Ch)																PCI Express Extended Capability ID (0001h)																FB4h
Uncorrectable Error Status																																FB8h
Uncorrectable Error Mask																																FBCh
Uncorrectable Error Severity																																FC0h
Reserved																Correctable Error Status																FC4h
Reserved																Correctable Error Mask																FC8h
Advanced Error Capabilities and Control																																FCCCh
Header Log 0																																FD0h
Header Log 1																																FD4h
Header Log 2																																FD8h
Header Log 3																																FDCh

Register 13-263. FB4h Advanced Error Reporting Extended Capability Header (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID		ROS	Yes	0001h
19:16	Capability Version		ROS	Yes	1h
31:20	Next Capability Offset Program to 138h, which addresses the Upstream Port Power Budget Extended Capability structure.	Upstream	ROS	Yes	138h
	Program to 10Ch, which addresses the Secondary PCI Express Extended Capability structure.	Downstream	ROS	Yes	10Ch

Register 13-264. FB8h Uncorrectable Error Status (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: The following errors are fatal to the PEX 8724, and require a Fundamental Reset (<i>PEX_PERST#</i>) to recover: • Bit 17, <i>Receiver Overflow Status</i> • Bit 18, <i>Malformed TLP Status</i> • Bit 22, <i>Uncorrectable Internal Error Status</i> If an individual error is masked (corresponding bit in the <i>Uncorrectable Error Mask</i> register (offset <i>FBCh</i>) is Set) when the error is detected, its Error Status bit is still updated; however, an error reporting Message (<i>ERR_FATAL</i> or <i>ERR_NONFATAL</i>) is not sent to the Root Complex, and the <i>Advanced Error Capabilities and Control</i> register <i>First Error Pointer</i> field (offset <i>FCCh</i>[4:0]) and <i>Header Log x</i> registers (offsets <i>FD0h</i> through <i>FDCh</i>) remain unchanged. <i>Table 13-5</i> indicates which Ports are enabled/used/Software Only, and when, based upon the <i>STRAP_STNx_PORTCFGx</i> input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. For bit 22 – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
3:0	Reserved		RsvdP	No	0h
4	Data Link Protocol Error Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
5	Reserved	Upstream	RsvdP	No	0
	Surprise Down Error Status 0 = No error is detected 1 = Error is detected	Downstream	RW1CS ^a	Yes	0
11:6	Reserved		RsvdP	No	0-0h
12	Poisoned TLP Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
13	Flow Control Protocol Error Status <i>Reserved/Not supported</i>		RsvdP	No	0
14	Completion Timeout Status <i>Not applicable to switches.</i>		RsvdP	No	0
15	Completer Abort Status		RW1CS ^a	Yes	0

**Register 13-264. FB8h Uncorrectable Error Status
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
16	Unexpected Completion Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
17	Receiver Overflow Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
18	Malformed TLP Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
19	ECRC Error Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
20	Unsupported Request Error Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
21	Reserved	Upstream	RsvdP	No	0
	ACS Violation Error Status 0 = No violation is detected 1 = Violation is detected	Downstream	RW1CS ^a	Yes	0
22	Uncorrectable Internal Error Status 0 = No error is detected 1 = Error is detected	Base Mode 0, 8 Virtual Switch Mode 0, 8, accessible through the Management Port	RW1CS ^a	Yes	0
	Reserved	Otherwise	RsvdP	No	0
23	MC Blocked TLP Status Multicast blocked TLP status. 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
24	Atomic Operation Egress Blocked Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
31:25	Reserved		RsvdP	No	0-0h

- a. When the **Error Check Disable** register **Software Force Error Enable** bit (Transparent Downstream Ports, offset 720h[2]) is Set, Type changes from RW1CS to RW.

Register 13-265. FBCh Uncorrectable Error Mask (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Uncorrectable Error Status register bits (offset FB8h). Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. For bit 22 – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
3:0	Reserved		RsvdP	No	0h
4	Data Link Protocol Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
5	Reserved	Upstream	RsvdP	No	0
	Surprise Down Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	Downstream	RWS	Yes	0
11:6	Reserved		RsvdP	No	0-0h
12	Poisoned TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
13	Flow Control Protocol Error Mask Reserved/Not supported		RsvdP	No	0
14	Completion Timeout Mask Not applicable to switches.		RsvdP	No	0
15	Completer Abort Mask		RWS	Yes	0

**Register 13-265. FBCh Uncorrectable Error Mask
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
16	Unexpected Completion Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
17	Receiver Overflow Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
18	Malformed TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
19	ECRC Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
20	Unsupported Request Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
21	Reserved	Upstream	RsvdP	No	0
	ACS Violation Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	Downstream	RWS	Yes	0
22	Uncorrectable Internal Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	Base Mode 0, 8	RWS	Yes	1
		Virtual Switch Mode 0, 8, accessible through the Management Port			
	Reserved	Otherwise	RsvdP	No	1
23	MC Blocked TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
24	Atomic Operation Egress Blocked Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error		RWS	Yes	0
31:25	Reserved		RsvdP	No	0-0h

Register 13-266. FC0h Uncorrectable Error Severity (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. For bit 22 – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
3:0	Reserved		RsvdP	No	0h
4	Data Link Protocol Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal		RWS	Yes	1
5	Surprise Down Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	Upstream	ROS	No	1
		Downstream	RWS	Yes	1
11:6	Reserved		RsvdP	No	0-0h
12	Poisoned TLP Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal		RWS	Yes	0
13	Flow Control Protocol Error Severity Reserved/Not supported		RO	No	1
14	Completion Timeout Severity Not applicable to switches. Because the Status and Mask are both Reserved for this bit, Severity can be ignored.		RsvdP	No	0
15	Completer Abort Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal		RWS	Yes	0

**Register 13-266. FC0h Uncorrectable Error Severity
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
16	Unexpected Completion Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal		RWS	Yes	0
17	Receiver Overflow Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal		RWS	Yes	1
18	Malformed TLP Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal		RWS	Yes	1
19	ECRC Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal		RWS	Yes	0
20	Unsupported Request Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal		RWS	Yes	0
21	Reserved	Upstream	RsvdP	Yes	0
	ACS Violation Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	Downstream	RWS	Yes	0
22	Uncorrectable Internal Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	Base Mode 0, 8	RWS	Yes	1
		Virtual Switch Mode 0, 8, accessible through the Management Port			
	Reserved	Otherwise	RsvdP	No	1
23	MC Blocked TLP Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal		RWS	Yes	0
24	Atomic Operation Egress Blocked Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal		RWS	Yes	0
31:25	Reserved		RsvdP	No	0-0h

Register 13-267. FC4h Correctable Error Status (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: If an individual error is masked (corresponding bit in the Correctable Error Mask register (offset FC8h) is Set) when the error is detected, its Error Status bit is still updated; however, an error reporting Message (ERR_COR) is not sent to the Root Complex. Table 13-5 indicates which Ports are enabled/used/Software Only, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. For bit 14 – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Receiver Error Status 0 = No error is detected 1 = One or more of the following Receiver errors was detected: • Code (symbol) error • Disparity error • Elastic Buffer overflow/underflow • Gen 3 Framing error		RW1CS ^a	Yes	0
5:1	Reserved		RsvdP	No	0-0h
6	Bad TLP Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
7	Bad DLLP Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
8	REPLAY NUM Rollover Status Replay Number Rollover status. 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
11:9	Reserved		RsvdP	No	000b

**Register 13-267. FC4h Correctable Error Status
(All Ports) (Cont.)**

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
12	Replay Timer Timeout Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
13	Advisory Non-Fatal Error Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
14	Corrected Internal Error Status 0 = No error is detected 1 = Error is detected	Base Mode 0, 8 Virtual Switch Mode 0, 8, accessible through the Management Port	RW1CS ^a	Yes	0
	Reserved	Otherwise	RsvdP	No	0
15	Header Log Overflow Status 0 = No error is detected 1 = Error is detected		RW1CS ^a	Yes	0
31:16	Reserved		RsvdP	No	0000h

- a. When the **Error Check Disable** register **Software Force Error Enable** bit (Transparent Downstream Ports, offset 720h[2]) is Set, Type changes from RW1CS to RW.

Register 13-268. FC8h Correctable Error Mask (All Ports)

Bit(s)	Description	Ports	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Correctable Error Status register bits (offset FC4h). Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming. The table also lists the relationship between the Stations, Station Ports, and Ports. For bit 14 – The Port 0 bits are for Ports 0, 1, 2, and 3. The Port 8 bits are for Ports 9 and 10 (accessed through Software Only Port 8).					
0	Receiver Error Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	0
5:1	Reserved		RsvdP	No	0-0h
6	Bad TLP Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	0
7	Bad DLLP Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	0
8	REPLAY NUM Rollover Mask Replay Number Rollover mask. 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	0
11:9	Reserved		RsvdP	No	000b
12	Replay Timer Timeout Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	0
13	Advisory Non-Fatal Error Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	1
14	Corrected Internal Error Mask 0 = Error reporting is not masked 1 = Error reporting is masked	Base Mode 0, 8 Virtual Switch Mode 0, 8, accessible through the Management Port	RWS	Yes	1
	Reserved	Otherwise	RsvdP	No	1
15	Header Log Overflow Mask 0 = Error reporting is not masked 1 = Error reporting is masked		RWS	Yes	1
31:16	Reserved		RsvdP	No	0000h

**Register 13-269. FCCh Advanced Error Capabilities and Control
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
4:0	First Error Pointer Identifies the bit position of the first error reported in the Uncorrectable Error Status register (offset FB8h).	ROS	No	1Fh
5	ECRC Generation Capable 0 = ECRC generation is not supported 1 = ECRC generation is supported, but must be enabled	ROS	Yes	1
6	ECRC Generation Enable 0 = ECRC generation is disabled 1 = ECRC generation is enabled	RWS	Yes	0
7	ECRC Check Capable 0 = ECRC checking is not supported 1 = ECRC checking is supported, but must be enabled	ROS	Yes	1
8	ECRC Check Enable 0 = ECRC checking is disabled 1 = ECRC checking is enabled	RWS	Yes	0
31:9	Reserved	RsvdP	No	0-0h

**Register 13-270. FD0h Header Log 0
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	TLP Header 0 First DWord Header. TLP Header associated with error.	ROS	Yes	0000_0000h

**Register 13-271. FD4h Header Log 1
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	TLP Header 1 Second DWord Header. TLP Header associated with error.	ROS	Yes	0000_0000h

**Register 13-272. FD8h Header Log 2
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	TLP Header 2 Third DWord Header. TLP Header associated with error.	ROS	Yes	0000_0000h

**Register 13-273. FDCh Header Log 3
(All Ports)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	TLP Header 3 Fourth DWord Header. TLP Header associated with error.	ROS	Yes	0000_0000h

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14.1 Introduction

Note: Check the latest design guides, application notes and errata list for Non-Transparent (NT) usage.

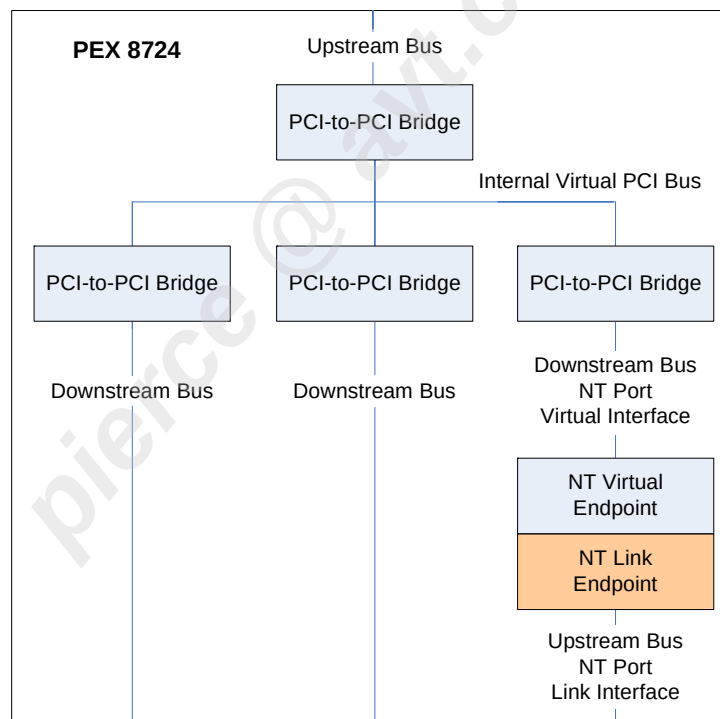
A Non-Transparent bridge (NT bridge, or NTB) consists of two back-to-back PCI Express endpoints, a Virtual side endpoint, and a Link side endpoint. In the PEX 8724, the Virtual side endpoint shows up to software behind two PCI-to-PCI bridges on the switch (a switch is a hierarchy of PCI-to-PCI bridges), while the Link side endpoint is all that the Link side Host “finds,” as illustrated in [Figure 14-1](#).

The NTB is used to connect two independent address/Host domains, and can be used to implement High-Availability systems or Intelligent I/O modules using PCI Express technology. NT bridges allow systems to isolate Address spaces, by appearing as an endpoint to each Host.

The NTB exposes a Type 0 Configuration Space register (CSR) Header to each side – Virtual and Link – and forwards transactions from one domain to the other, using Address and Requester ID translation through Memory windows. The NTB also includes **Doorbell** registers, for transmitting interrupts from one side of the bridge to the other, and **Scratchpad** registers, accessible from both domains for inter-Host communication. The PEX 8724, with a single Port, supports the [Intelligent Adapter Mode](#) system model.

[Figure 14-1](#) illustrates an external connection between two virtual switches, through an NT Port (AC-coupling capacitors are required, but are not shown).

Figure 14-1. NT Bridge – NT Mode



14.1.1 NT Device Type Identification

An NT bridge identifies itself as **PCI Class Code** “other,” **068000h**, with a Type 0 Header. A Type 0 Header is used by PCI endpoints. In contrast, a Transparent PCI-to-PCI bridge identifies itself as a **PCI Class Code 060400h**, with a Type 1 Header (PCI-to-PCI bridge).

The **PCI Express Capability** register includes a **Device/Port Type** field (offset **68h[23:20]**). In this register, a Transparent bridge/switch Port identifies itself as an *Upstream Port* or *Downstream Port*, while an NTB/NT Port identifies itself as a *PCI Express endpoint*.

14.1.2 Enabling Non-Transparent Bridging

NT mode is enabled if the **STRAP_NT_UPSTRM_PORTSEL[2:0]** 3-state inputs are not all pulled or tied High to **VDD18**. Valid values on these 3-state inputs are 0, Z (floating), and 1, and when used together, provide 27 possible combinations, of which values within the ranges of 0 through 3, and 8 through 10 represent corresponding Port Numbers. The selected value determines which Port is assigned to be the NT Port. The Port Numbers listed above do not exist in all Port configurations, and the programmed value must correspond to a valid Port Number; otherwise, that value is **Reserved**. **Table 14-1** lists the 27 combinations for the three 3-state inputs, and their equivalent binary and decimal values (indicating register values and Port Numbers), in incrementing order of the 3-state encoding.

Alternately, I²C, software, and/or an optional serial EEPROM (if present) can enable NT, by Setting the **VS0 Upstream** register **NT Enable** bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset **360h[13]**). The NT Port is determined by the register’s **NT Port** field [12:8].

Table 14-1. 3-State-Encoded STRAP_NT_UPSTRM_PORTSEL[2:0] Values

STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Binary Value	Port Number/ Decimal Value ^a
000	0_0000b	0
00Z	0_0001b	1
001	0_0010b	2
0Z0	0_0011b	3
0ZZ, 0Z1, 010, 01Z	Reserved	Reserved
011	0_1000b	8
Z00	0_1001b	9
Z0Z	0_1010b	10
Z01, ZZ0, ZZZ, ZZ1, Z10, Z1Z, Z11, 100, 10Z, 101, 1Z0, 1ZZ, 1Z1, 110, 11Z	Reserved	Reserved
111	1_1010b	Disabled

- a. **Table 13-5, “Port Configurations”** indicates which Ports are enabled/used/**Software Only**, and when, based upon the **STRAP_STNx_PORTCFGx** input states and/or serial EEPROM programming.

14.1.3 NT Port Features

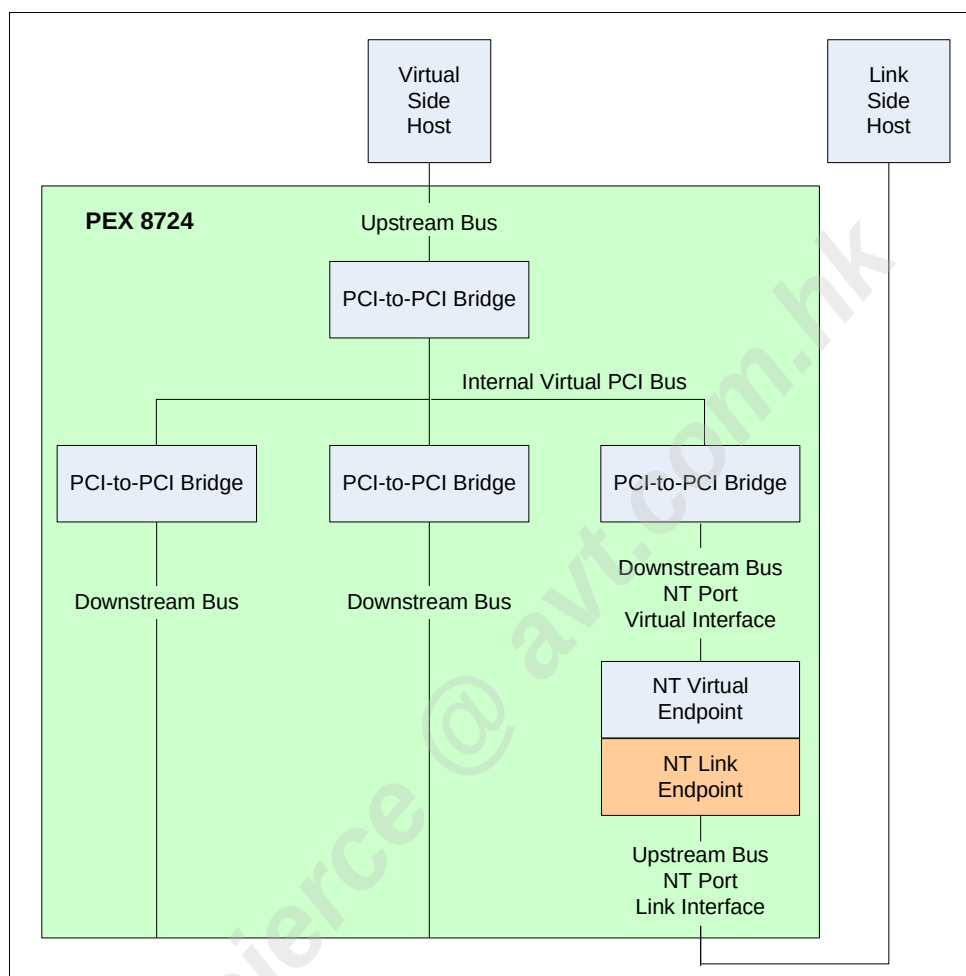
- Maps PEX 8724 Configuration registers into either 32- or 64-bit Memory space
- Base Address registers (BARs)
 - Implements four 32-bit, two 32-bit and one 64-bit, or two 64-bit BARs
 - Supports BAR Size programming, through the **BARx Setup** register(s)
 - Allows BARs to be individually disabled, including Memory-Mapped BARs
- Supports [Direct Address Translation](#)
 - 32-to-32-bit address conversion
 - 32-to-64-bit address conversion
 - 64-to-32-bit address conversion
 - 64-to-64-bit address conversion
 - Requester ID (Bus Number, Device Number, and Function Number) conversion across the NT bridge
- 16 **Doorbell** registers
- 8 **Scratchpad** registers
- Supports Requester ID and Completion ID translation
- NT Port Link Interface *DL_Active* state change generates interrupt to Virtual side Host
- Supports Cursor mechanism
- Supports Expansion ROM on the NT Port Virtual or Link Interface
- Supports End-to-end Cyclic Redundancy Check (ECRC)
- Provides ability to Clear *No Snoop* Transaction Layer Packet (TLP) attribute (if enabled)
- Programmable Upstream Port(s) and NT Port for the enabling of High-Availability systems (Failover and Redundant systems)
- Ability to bring down the NT Port Link side, when the NT Port Virtual side domain is Down
- Supports the Fencing mechanism
- Signals Device-Specific interrupt to the Virtual side Host when the NT Port Link Interface detects TLP errors
- Signals Device-Specific interrupt to the Virtual side Host when the NT Port Link Interface receives Error Messages (Safety bit-controllable)
- Ability to disable NT Port Link Interface Hot Reset from resetting NT Port Link Interface registers
- Supports Configuration Space access control
- NT Port Link Interface Reset input ([PEX_NT_PERST#](#))
- Programmable **Captured Bus and Device Numbers** register

14.1.4 Intelligent Adapter Mode

The diagram provided in [Figure 14-2](#) has two Type 0 CSR Headers in the NTB. The one nearer the internal virtual PCI Bus is referred to as the *NT Virtual endpoint*, and is accessed by way of the *NT Port Virtual Interface*. The one nearer the PCI Express Link is referred to as the *NT Link endpoint*, and is accessed by way of the *NT Port Link Interface*.

The Intelligent Adapter model includes the NT Link endpoint that is connected to the System domain. The Link side Host manages only the NT Link endpoint Type 0 function. The Virtual side Host manages all PEX 8724 Transparent Port Type 1 and NT Virtual endpoint Type 0 functions. Cross-domain traffic is routed through an Address Translation mechanism. (Refer to [Section 14.1.11](#).)

Figure 14-2. Intelligent Adapter Software Model – Base Mode



14.1.5 NT Port Reset

The section discusses NT mode exceptions and enhancements to Transparent mode PCI Express (standard) reset behavior.

In discussing reset, it is useful to consider three different types of registers:

- **Setup** registers (**BARx Setup**, **Memory BARx Address Translation**, and Requester ID LUT) for transaction forwarding and address translation
- **Doorbell** registers, for interrupt-based synchronization between Host domains, and
- **Scratchpad** registers, for information exchange

14.1.5.1 Fundamental Reset (PERST#)

The Fundamental Reset input, **PEX_PERST#**, resets all PEX 8724 states, including all NT Port states, to an initial power-up value. This includes resetting all Sticky bits and all **Setup**, **Doorbell**, and **Scratchpad** registers.

VSx_PERST# provides a similar Fundamental Reset, but only to its specific virtual switch, and does not reset states nor registers that are used by other virtual switches.

PEX_NT_PERST# resets the NT Port Link Interface.

14.1.5.2 Virtual Side Hot Reset

The Virtual side of the NTB sees a Hot Reset, if any of the following occurs:

- Transparent Upstream Port goes Down or receives an in-band reset (and reset propagation is not blocked)
- Transparent Upstream Port(s)' PCI-to-PCI bridge Sets the **Bridge Control** register *Secondary Bus Reset* bit (offset **3Ch**[22])
- PCI-to-PCI bridge above the Virtual side endpoint Sets the *Secondary Bus Reset* bit

A Virtual side Hot Reset resets all the NT Port Virtual Interface states. Hot Reset does not propagate through the NT Port; therefore, the NT Port Link Interface is **not** reset by the Virtual side Hot Reset.

The PEX 8724 supports options that allow the Virtual side Hot Reset conditions to have no effect:

- Reset, from either Hot Reset being received at the PEX 8724 Upstream Port, or from an Upstream Port DL_Down, both of which usually reset all Ports, and can be blocked by Setting the **Virtual Switch Debug** register *Upstream Port and NT-Link Port DL_Down Reset Propagation Disable* bit (Upstream Port(s), offset **A30h**[4])
- Hot Reset can be blocked at the NT Port Virtual Interface, by Setting the **Link Control** register *Link Disable* bit (offset **78h**[4]) in the PCI-to-PCI bridge that is logically connected to the NT Port Virtual Interface

The NT Port Virtual Interface **BARx Setup** and **Doorbell** registers are reset, while the NT Port Link Interface **BARx Setup** and **Doorbell** registers are unaffected. The **Scratchpad** registers on both sides are also unaffected.

14.1.5.3 Link Side Hot Reset

The Link side of the NTB sees a Hot Reset, if any of the following occurs:

- NT Link goes down
- NT Link receives an in-band reset

When the Link side receives a Hot Reset, the NT Port Link Interface registers are reset.

The PEX 8724 supports an option to bring down the NT Port Link, which resets the NT Link side logic, by Setting the Port's **Port Control** register *Disable Port x* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 208h[3:0]). The identified Port (one hot) goes to, and remains in, the *Detect.Quiet* substate, until the bit is Cleared. No EIOS is sent, which should force any connected device to the *Recovery*, and then *Detect*, state.

The NT Port Link Interface **Setup** and **Doorbell** registers are reset, while the NT Port Virtual Interface **Setup** and **Doorbell** registers are unaffected. The **Scratchpad** registers on both interfaces are also unaffected.

14.1.5.4 Reset Propagation

Reset propagation, during a Hot Reset or using the **Bridge Control** register *Secondary Bus Reset* mechanism, is limited to Transparent Downstream Ports. In the NTB, this reset cannot be propagated across the bridge (across the NT Port). (Refer to [Section 5.1, “Reset,”](#) for details regarding PEX 8724 Transparent mode reset behavior.)

14.1.6 NT Port Memory-Mapped Base Address Registers

The NT Port Virtual and Link Interfaces individually claim 256 KB of memory, using **BAR0** and **BAR1**. The 256-KB space contains the Configuration Space registers for all PEX 8724 Ports. **BAR0** and **BAR1** can be programmed as one of the following:

- 32-bit BAR (**BAR0** is used, **BAR1** is **Reserved**; default mode)
- 64-bit BAR, by programming the Configuration **BAR0/1 Setup** register
(NT Port Virtual Interface, offset 3_E0D0h or NT Port Link Interface, offset 3_F0E4h)
- **BAR0** and **BAR1** can be completely disabled

Figure 14-3 provides a memory-mapped view of the PEX 8724 Configuration Space registers. This view is the same from the Upstream Port(s), NT Port Virtual Interface, or NT Port Link Interface, although the Base address can be different in each case. Each Port has a 4-KB Memory space for Configuration registers. Starting at **BAR0**, Port 0 is at offset 0000h, Port 1 is at offset 1000h, and so forth, with 4-KB increments, per Port. There are also NT function registers at the top of the space, with the NT Port Virtual Interface at 3_E000h and NT Port Link Interface at 3_F000h. **Reserved** spaces should **not** be used.

14.1.7 Doorbell Registers

Doorbell registers are used to signal interrupts from one side of the NTB to the other. This section describes a typical set of Doorbell Control registers.

A 16-bit software-controlled Interrupt Request register and associated 16-bit Mask register are implemented for the NT Port Virtual and Link Interfaces. The Doorbell mechanisms consist of the following registers:

- **Virtual Interface IRQ Set** (offset C4Ch)
- **Virtual Interface IRQ Clear** (offset C50h)
- **Virtual Interface IRQ Mask Set** (offset C54h)
- **Virtual Interface IRQ Mask Clear** (offset C58h)
- **Link Interface IRQ Set** (offset C5Ch)
- **Link Interface IRQ Clear** (offset C60h)
- **Link Interface IRQ Mask Set** (offset C64h)
- **Link Interface IRQ Mask Clear** (offset C68h)

Note: *The NT Port Virtual and Link Interfaces have access to each of the eight registers listed above. These registers perform the same function, regardless of which interface is used to access the registers.*

Each NT Port interface has its own register (*that is*, there are two of each register). The offsets are all located in NT 4-KB space. *For example*, offset C4Ch can be 3_EC4Ch (NT Port Virtual Interface) and/or 3_FC4Ch (NT Port Link Interface).

The Virtual Interface IRQ is for interrupts that exit the NT Port Virtual Interface. An interrupt is asserted on the NT Port Virtual Interface when one or more of the **Virtual Interface IRQ Set** register bits are Set by the NT Port Link Interface and their corresponding **Virtual Interface IRQ Mask Set** register bits are Cleared. An interrupt is de-asserted on the NT Port Virtual Interface when one or more of the **Virtual Interface IRQ Clear** register bits are Set from the NT Port Virtual Interface or their corresponding **Virtual Interface IRQ Mask Clear** register bits are Cleared. The interrupt is de-asserted when all Set bits are masked or Cleared.

The Link Interface IRQ is for interrupts that exit the NT Port Link Interface. An interrupt is asserted on the NT Port Link Interface when one or more of the **Link Interface IRQ Set** register bits are Set by the NT Port Virtual Interface and their corresponding **Link Interface IRQ Mask Set** register bits are Cleared. An interrupt is de-asserted on the NT Port Link Interface when one or more of the **Link Interface IRQ Clear** register bits are Set from the NT Port Link Interface or their corresponding **Link Interface IRQ Mask Clear** register bits are Cleared. The interrupt is de-asserted when all Set bits are masked or Cleared.

Because Memory Requests can access both sets of NT-Virtual and NT-Link Doorbell registers, software in either domain can generate Doorbell interrupts to both domains.

Internally, the **Set IRQ** and **Clear IRQ** registers are the same register. One location is used to Set bits and the other is used to Clear bits. The status can be read from either register.

In a PCI Express switch, interrupt state transitions (from Setting to Clearing, or vice versa) result in packets being transmitted Upstream on the appropriate side of the bridge, when INTx are enabled (**PCI Command** register *Interrupt Disable* bit (NT Port, offset 04h[10]) is Cleared). Standard PCI Express Capability structures allow these interrupts to be configured as INTx or MSIs. When MSIs are enabled (**MSI Control** register *MSI Enable* bit (NT Port, offset 48h[16]) is Set), packets are transmitted only when interrupts transition from Clear IRQ to Set IRQ.

NT-Virtual Doorbell interrupts can optionally use the **PEX_INTA#** (Base mode) or **VSx_PEX_INTA#** (Virtual Switch mode) output for interrupt signaling, instead of the INTx signaling mechanism or MSI mechanism (both mutually exclusive), by Setting the **Error Check Disable** register *Enable Ball Interrupt Output for NT-Virtual Doorbell-Generated Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[7]).

NT-Virtual Doorbell interrupts are de-asserted when the NT Port goes to the *DL_Down* state.

14.1.8 Scratchpad Registers

Scratchpad registers are readable from and writable to both sides of the NTB, providing a generic means for inter-Host communication. A block of eight registers are provided, accessible in Memory space from the NT Port Virtual and Link Interfaces. These registers pass Control and Status information between Virtual and Link Interface devices, or they can be generic RW registers. Reading from or writing to **Scratchpad** registers does not cause interrupts to assert – **Doorbell** registers are used for that purpose. **Scratchpad** registers are reset only by a Fundamental Reset (**PEX_PERST#**, Serial Hot Plug **SHP_PERST#** (SHP_PERSTx#), VSx_PERST#, and/or **PEX_NT_PERST#**).

Some non-active registers can also be used for **Scratchpad** registers, to extend the number available. The **Physical Layer User Test Pattern, Bytes x through y** registers (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 20Ch through 218h) provide four additional registers, per Port.

14.1.9 NT Base Address Registers

There are two sets of NT Base Address registers (BARs) – one for each NT Port Virtual and Link Interface. Each BAR has its own **Setup** and **Address Translation** register:

- **BARx Setup** registers enable/disable the BAR and define the window size and type. Program the **BARx Setup** registers prior to allowing configuration software to assign a resource for these BARs. (Refer to [Section 14.1.9.1](#) for further details.)
- **Memory BARx Address Translation** registers allow for an address change on the upper bits (up to the size of the space). Program the **Memory BARx Address Translation** registers, before generating traffic across the NT Port. This programming is typically performed by information downloaded from I²C, software, and/or an optional serial EEPROM (if present) on the destination side. The source side does not need to know what the Address Translation is.
- The address could change size. *For example*, the NT Port allows a 32-bit device to communicate to a 64-bit device, and vice versa. (The same is true when the addresses are the same size, as well – a 32-bit device can communicate with a 32-bit device, and a 64-bit can communicate with a 64-bit device.)

14.1.9.1 NT BARx Setup Registers

PCI defines BARs for a PCI device to claim Address space. For the PEX 8724, **BAR0** and **BAR1** provide Memory-Mapped access to the CSRs, while **BAR2**, **BAR3**, **BAR4**, and **BAR5** provide programmable window sizes to the other side of the NTB.

The **BARx Setup** registers are used to program each BAR's window size. [Table 14-2](#) briefly describes each NT Port BAR. **BAR2**, **BAR3**, **BAR4**, and **BAR5** can be configured for accessing the Address space across the NT Port Virtual and Link Interfaces.

Each **BARx Setup** register defines the memory window size that is to be assigned by a system enumerator (*that is*, BIOS or firmware). The smallest size allowed by PCI rules is 1 MB. Furthermore, the **BARx Setup** registers define a space as a 32- or 64-bit Memory space, and whether the space is I/O, Non-Prefetchable Memory, or Prefetchable Memory space.

For example, if the window size must be 1 MB in a Prefetchable Memory space, the **BARx Setup** register in 32-bit Address space is FFF0_0008h. The “FFF” value on bits [31:20] provides the size, 1 MB in this case. The 0_000 on bits [19:4] are **Reserved**. Bit 3 denotes whether the space is prefetchable – a value of 1 indicates that it is prefetchable (*that is*, cacheable). Bits [2:1] allow a choice of 32- or 64-bit Address space – 00b indicates 32-bit.

Note: Only bits [2:1] of the **BAR2** and **BAR4 Setup** registers are programmable – if programmed for 64-bit Address space, **BAR2** and **BAR3** work together as **BAR2/3**, and/or **BAR4** and **BAR5** work together as **BAR4/5**, respectively.

In most cases, the **BARx Setup** registers are programmed using an optional serial EEPROM (if present), before the BIOS or firmware allocates the resources (because resource enumeration is done before the system software can access these devices). I²C can also be used to program the **BARx Setup** registers, as long as it finishes before PCI Express enumeration completes.

Table 14-2. NT Port Virtual and Link Interface BARs

BAR	NT Port Virtual Interface Description	NT Port Link Interface Description
BAR0	All Port Configuration registers are mapped into Memory space, using BAR0 and BAR1. The Virtual side Host, connected to the Transparent Upstream Port(s), can use the Transparent Upstream Port(s) BAR0/1 or NT Port Virtual Interface BAR0/1 to access the Port Configuration registers. The NT Port Virtual Interface BAR0/1 Setup register controls the BAR0 and BAR1 implementation, as follows: - Disables BAR0 and BAR1 - Enables BAR0 and disables BAR1 (BAR0 is a 32-bit BAR) - Enables BAR0 and BAR1 (BAR0/1 is a 64-bit BAR) BAR0 and BAR1 claim 256-KB Memory space to the system.	All Port Configuration registers are mapped into Memory space, using BAR0 and BAR1. The Link side Host, connected to the NT Port, can use BAR0/1 to access the Port Configuration registers. The NT Port Link Interface BAR0/1 Setup register controls the BAR0 and BAR1 implementation, as follows: - Disables BAR0 and BAR1 - Enables BAR0 and disables BAR1 (BAR0 is a 32-bit BAR) - Enables BAR0 and BAR1 (BAR0/1 is a 64-bit BAR) BAR0 and BAR1 claim 256-KB Memory space to the system.
BAR1	Configured by the NT Port Virtual Interface BAR0/1 Setup register. BAR1 is implemented as an upper 32-bit address of the NT Port Virtual Interface memory-mapped 64-bit BAR; otherwise, it is Reserved .	Configured by the NT Port Link Interface BAR0/1 Setup register. BAR1 is implemented as an upper 32-bit address of the NT Port Link Interface memory-mapped 64-bit BAR; otherwise, it is Reserved .
BAR2	Configured by the NT Port Virtual Interface Memory BAR2 Setup register. BAR2 can be implemented as a 32-bit BAR or lower half of a 64-bit BAR, by combining it with BAR3 (BAR2/3). BAR2 uses Direct Address Translation.	Configured by the NT Port Link Interface Memory BAR2 Setup register. BAR2 can be implemented as a 32-bit BAR or lower half of a 64-bit BAR, by combining it with BAR3 (BAR2/3). BAR2 uses Direct Address Translation.
BAR3	Configured by the NT Port Virtual Interface Memory BAR2/3 Setup register. BAR3 can be implemented as a 32-bit BAR or upper half of a 64-bit BAR, by combining it with BAR2 (BAR2/3). BAR3 uses Direct Address Translation.	Configured by the NT Port Link Interface Memory BAR2/3 Setup register. BAR3 can be implemented as a 32-bit BAR or upper half of a 64-bit BAR, by combining it with BAR2 (BAR2/3). BAR3 uses Direct Address Translation.
BAR4	Configured by the NT Port Virtual Interface Memory BAR4 Setup register. BAR4 can be implemented as a 32-bit BAR or lower half of a 64-bit BAR, by combining it with BAR5 (BAR4/5). BAR4 uses Direct Address Translation.	Configured by the NT Port Link Interface Memory BAR4 Setup register. BAR4 can be implemented as a 32-bit BAR or lower half of a 64-bit BAR, by combining it with BAR5 (BAR4/5). BAR4 uses Direct Address Translation.
BAR5	Configured by the NT Port Virtual Interface Memory BAR4/5 Setup register. BAR5 can be implemented as a 32-bit BAR or upper half of a 64-bit BAR, by combining it with BAR4 (BAR4/5). BAR5 uses Direct Address Translation.	Configured by the NT Port Link Interface Memory BAR4/5 Setup register. BAR5 can be implemented as a 32-bit BAR or upper half of a 64-bit BAR, by combining it with BAR4 (BAR4/5). BAR5 uses Direct Address Translation.

14.1.10 TLP Routing

To understand what structures are required in an NTB, a brief overview of PCI routing is provided.

Memory and I/O Requests are routed by comparing an address against the PCI-to-PCI bridge **Base** and **Limit** registers. Any Memory or I/O Requests, that fall within the Base and Limit, route out the Downstream side of that one PCI-to-PCI bridge. All Downstream devices must be mapped in contiguous address regions, such that a single Address range (between the Base and Limit, inclusive) within each space is sufficient.

Completions and Configuration Requests are routed, using a Bus Number that is within the range of the PCI-to-PCI bridge's **Secondary Bus Number** and **Subordinate Bus Number** values (offsets 19h and 1Ah, respectively, within the **Bus Number** register (offset 18h)). PCI uses a tree of Bus Numbers, with the Upstream Port(s) having the lowest Bus Number. Each PCI-to-PCI bridge has:

- An Upstream Primary Bus Number,
- A Downstream Secondary Bus Number, and
- A range of other Downstream buses, up to a Subordinate Bus Number.

Similar to the Base and Limit comparison for Memory and I/O Requests, any Completions or Configuration Requests, that fall within the Secondary and Subordinate Bus Number range, route out the Downstream side of that one PCI-to-PCI bridge.

TLPs that target the PEX 8724, however, are treated a bit differently. Memory Requests must hit the BAR space. Completions or Configuration Requests must exactly match the Bus Number of a PCI-to-PCI bridge within the PEX 8724.

Upstream routing is simply the negation of Downstream and switch routing. If a packet does not route Downstream or hit the PEX 8724, the packet must route Upstream.

14.1.11 Address Translation

In multi-domain systems, each Host domain has its own Address space and Bus Numbering scheme, independent from that of other Host domain(s). Hence, any transaction crossing the inter-domain boundary, by way of an NTB or other means, must support address, as well as Bus Number, translations.

Before a transaction (PCI Express packet) can go through the NTB (either from the Virtual to Link side, or from the Link to Virtual side) in a multi-domain system, one or more sets of Memory resources must be assigned to the NTB. To request this resource from the system enumerator (BIOS or firmware), the NTB must be programmed with the **BARx Setup** register(s). (Refer to [Section 14.1.9.1](#).) These registers request the window space size, memory type, 32- or 64-bit Address space, and prefetchable or non-prefetchable area, using one 32-bit register for 32-bit Address space or the two 32-bit registers (**BAR2/3** or **BAR4/5**) for 64-bit Address space. In return, the system enumerator assigns resources to the NTB in **BAR0** through **BAR5**. Any transactions that target **BAR2** through **BAR5** on the NT Port Link Interface result in a transaction across the NTB, to the secondary address domain.

The NTB must be enumerated to accommodate the resources assigned to the NT endpoint, to allow packets to logically traverse the bridge. Its Device Number (on the Downstream Internal Virtual PCI Bus) value is the NT Port Number. Device enumeration minimally includes the **PCI Command**, **Bus Number**, and **Memory Base and Limit** and/or **Prefetchable Memory Base and Limit** registers (offsets 04h, 18h, and 20h, and/or 24h). The Internal NT Virtual Bus (connecting the NT PCI-to-PCI secondary interface and NT endpoint, as illustrated in [Figure 14-2](#)) can be assigned any available Bus Number within the Upstream Port's range of Subordinate Bus Numbers.

In addition to the BARs, re-maps for the other side (Virtual or Link) must be programmed. The **Memory BARx Address Translation** registers provide memory translation, with a one-to-one correspondence between **BAR2** through **BAR5**. This is fairly straightforward:

address{Host1_abc} changes to address{Host2_abc}

where *abc* represents a lower Address bit offset from the BAR. As previously mentioned, the Host1 and Host2 Address spaces can independently be a 32- or 64-bit Address space; the address translation accounts for any alteration.

The Requester ID Lookup Table (RID-LUT) is a little more complicated, because it processes more information. The TLP's Requester ID is used to select an entry in the RID-LUT to translate. The Requester ID is composed of the following:

- 8-bit Bus Number,
- 5-bit Device Number, and
- 3-bit Tag

The Bus Number must change as it crosses Host domains, and there must be a mechanism to change it back (for Completions); therefore, the egress TLP needs a way to route back to the initial Requester. That way is the RID-LUT, where the egress TLP is sent with an index to the RID-LUT for the return trip Bus Number recovery. Because there is an association of initial Requester ID into the RID-LUT, a Requester can be prevented from crossing the NTB if it does not have an entry enabled in the RID-LUT, adding security to the NTB, by limiting the devices that can generate transactions across the NTB.

Users of legacy PLX NT devices used to have difficulty determining the Requester's Requester ID. This issue has been addressed in the PEX 8724, with the [Requester ID Read Back](#) register (NT Port Link Interface, offset [C90h](#)). Any Requester can read this register and the Completion is its own Requester ID, on the lower 16 bits:

- [15:8] is the Bus Number
- [7:3] is the Device Number, and
- [2:0] is the Function Number

The Address Translation registers and RID-LUT can be changed during runtime, as long as they are not being used by pending transactions.

The NT Port Virtual and Link Interfaces support [Direct Address Translation](#), described in the next section.

14.1.11.1 Direct Address Translation

The **BAR_x Setup** registers define a mask that splits the address into an upper *Base* field and a lower *Offset* field. Translation then consists of replacing, under the maskable portion of the **BAR_x Setup** register, the Address Base register bits with the corresponding Address Translation register bits. Accordingly, the Address Translation register value must be a multiple of the corresponding BAR size. [Figure 14-4](#) illustrates Direct Address Translation.

The device(s) on the originating Host domain can communicate to a single device or multiple devices mapped to consecutive Memory Address space on the Target Host domain, by using the Direct Address Translation mechanism. [Figure 14-5](#) illustrates the entire Address map, claimed by the NT Port, mapped into the single target device. [Figure 14-6](#) illustrates the entire Address map claimed by the NT Port, mapped into multiple target devices. Multiple devices must be in contiguous Memory ranges.

Figure 14-4. Direct Address Translation

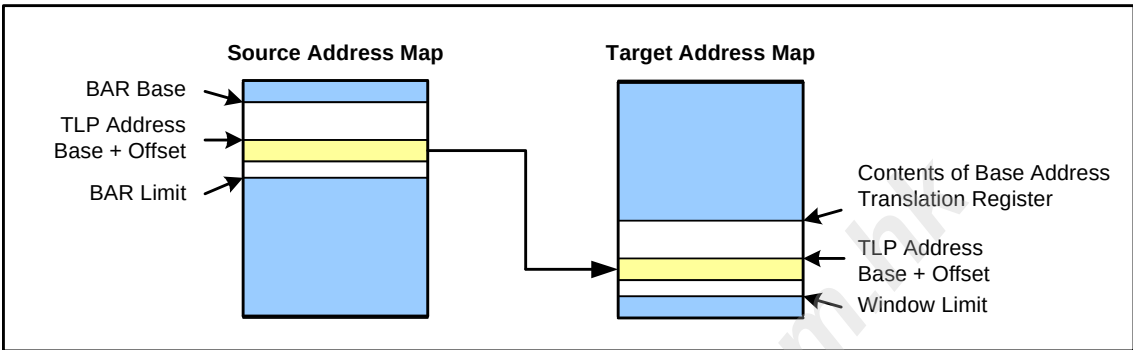


Figure 14-5. NT Port Mapped into Single Target Device

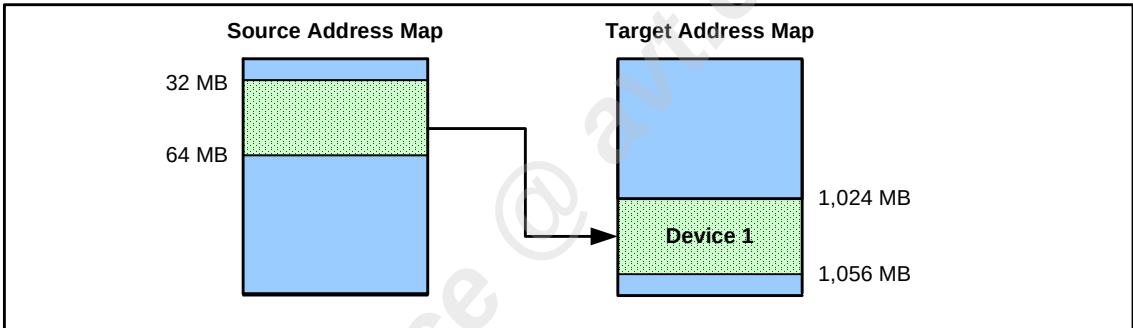
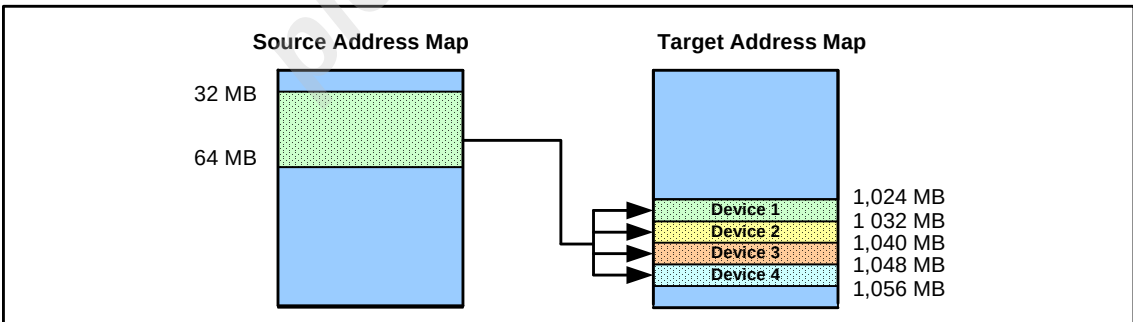


Figure 14-6. NT Port Mapped into Multiple Target Devices



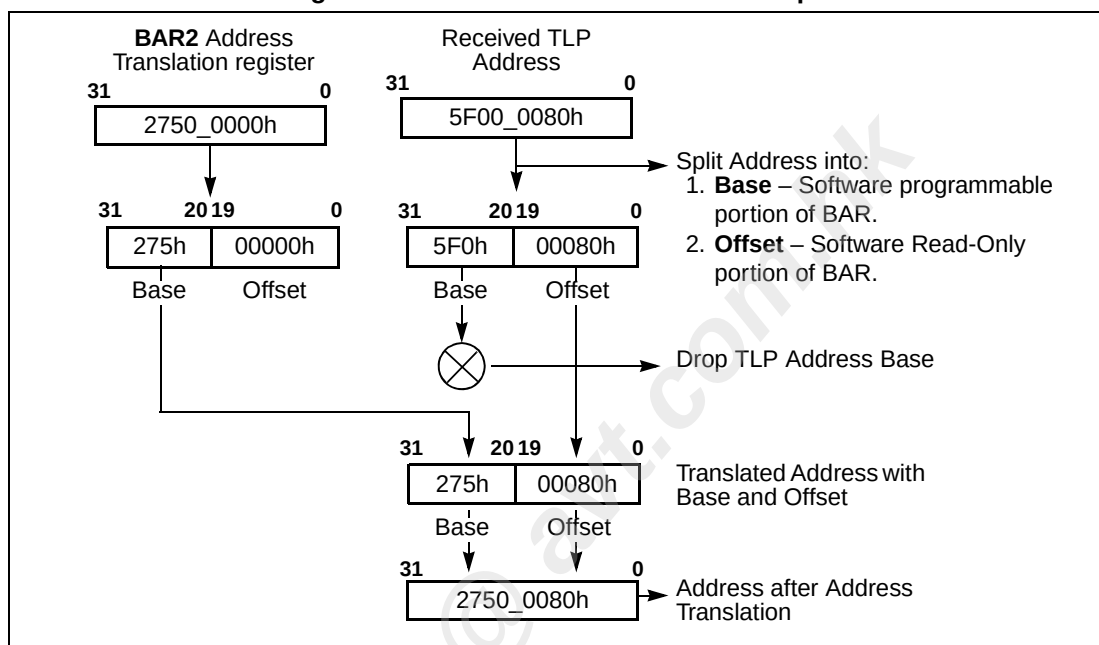
Address Translation Example

Assume the following initial conditions:

1. NT Port Virtual Interface **BAR2** claims 1-MB Memory space, by Setting bits [31:20] (**BAR2 Setup** register = FFF0_0000h).
2. Enumeration software on the Virtual side assigns the 5F00_0000h address value to NT Port Virtual Interface **BAR2** (an endpoint from the software point of view) and makes the PCI-to-PCI bridges Upstream of that endpoint have a Base and Limit that include that **BAR2** space.
3. Device driver software programs the **BAR2** Address Translation register to 2750_0000h.

The PEX 8724 receives a transaction to the NT Port Virtual Interface, with address 5F00_0080h. The received transaction address is hitting the NT Port Virtual Interface **BAR2**. The PEX 8724 claims the transaction and executes the address translation described in Figure 14-7.

Figure 14-7. Address Translation Example



14.2 Requester ID Translation

Some Messages, and Configuration and Completion transactions, are Bus Number-routed instead of address-routed. Of these, the NT Port forwards only the Completion transaction between the two Host domains. PCI Express switches and bridges use the Requester ID (defined in the Completion TLP Header) to route these packets.

The Requester ID consists of the following:

- Requester's PCI Bus Number
- Device Number
- Function Number

The Completer ID consists of the following:

- Completer's PCI Bus Number
- Device Number
- Function Number

Note: The PCI Bus Number is unique for each Host domain.

Figure 14-8 illustrates the Memory Request TLP Header format. Figure 14-9 illustrates the Completion TLP Header format.

Figure 14-8. Memory Request TLP Header Format

Bytes 0-3	Byte 0								Byte 1								Byte 2								Byte 3							
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
	R	Fmtx1		Type				R	TC		R				TD	EP	Attr		R	Length												
	Requester ID																Tag				Last DW BE				1st DW BE							
	Address[63:32]																															
Bytes 4-7																																
Bytes 8-11																																
Bytes 12-15	Address[31:0]																														R	

Figure 14-9. Completion TLP Header Format

	Byte 0								Byte 1								Byte 2								Byte 3							
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Bytes 0-3	R	Fmt		Type					R	TC			R			TD	EP	Attr		R		Length										
Bytes 4-7	Completer ID																Completer Status		BCM	Byte Count												
Bytes 8-11	Requester ID																Tag						R	Lower Address								

14.2.1 Transaction Sequence

To implement a memory sequence from a Requester in Host1 space to a Target in Host2 space:

1. Hardware inserts its Requester ID information into the Memory Read TLP that it generates on the initiating Host1 domain.
2. Switches and bridges between the transaction initiator and NT Port route this Memory Read TLP, based upon the Memory address.
3. NT Port replaces the Memory Read TLP Requester ID with its own Requester ID, and conducts the address translation. The NT Port's Requester ID consists of the Host2 Bus Number of the NT Port and an index to the RID-LUT, as well as three original bits. This modified TLP is sent to the target Host (Host2) domain.
4. Switches and bridges between the NT Port and target device route this Memory Read TLP, based upon the address and using Host2 Address space.
5. A Host2 target is found, and a Completion is generated. When the target device generates the Completion TLP, it copies the Memory Read TLP Requester ID (which was the NT Port Bus Number with RID-LUT index) into the corresponding Completion's *TLP Requester ID* field and inserts its ID into the *TLP Completer ID* field.
6. Switches and bridges between the target device and NT Port route the Completion TLP, based upon the Host2 Bus Numbering scheme and Bus Number in the *Requester ID* field, back to the NT Port.
7. NT Port restores the original Requester ID value from the RID-LUT pointed to by the RID-LUT index, replaces the Completer ID with the NT Port's Host1 Completer ID (using the Host1 Bus Number), and forwards the modified Completion TLP to the original Requester in the Host1 domain.
8. Switches and bridges between the NT Port and Requester route the Completion TLP, based upon the Bus Number in the *Requester ID* field.
9. Requester accepts and processes the Completion TLP.

Transactions originating in both the Virtual- and Link-side Host domains use the same mechanism and have the same hardware resources. Therefore, any transactions use the same processes.

Both directions have their own unique RID-LUT with 32 entries, as detailed in:

- [Section 15.13.2, “NT Port Virtual Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry \(Addresses D94h – DD0h\)”](#)
- [Section 16.13.2, “NT Port Link Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry \(Addresses DB4h – DF0h\)”](#)

This data structure supports up to 32 devices, each of which can simultaneously transmit Requests through the associated NT Port. Because the Function Number is not used in the RID-LUT association, a separate RID-LUT entry is not required for each requesting or phantom function device.

The RID-LUT must be configured before transmitting Requests through the NT Port. This Requester registration process, which cannot be accomplished by a peer, is an effective security and protection mechanism.

When a Request is received and routed to the NT Port, its Requester ID is translated – Bus Number and Device Number, but not Function Number. The 3-bit Function Number remains the same on both sides of the NTB. The remainder of the Requester ID is translated, based upon the RID-LUT. The received Memory Request TLP Requester ID is associated into the RID-LUT, to find a single matching entry. The address of this matching entry (RxIndex) is substituted into the 5-bit *Device Number* field of the Memory Request's TLP *Requester ID* field. The NT Port's Bus Number is substituted into the 8-bit *Bus Number* field.

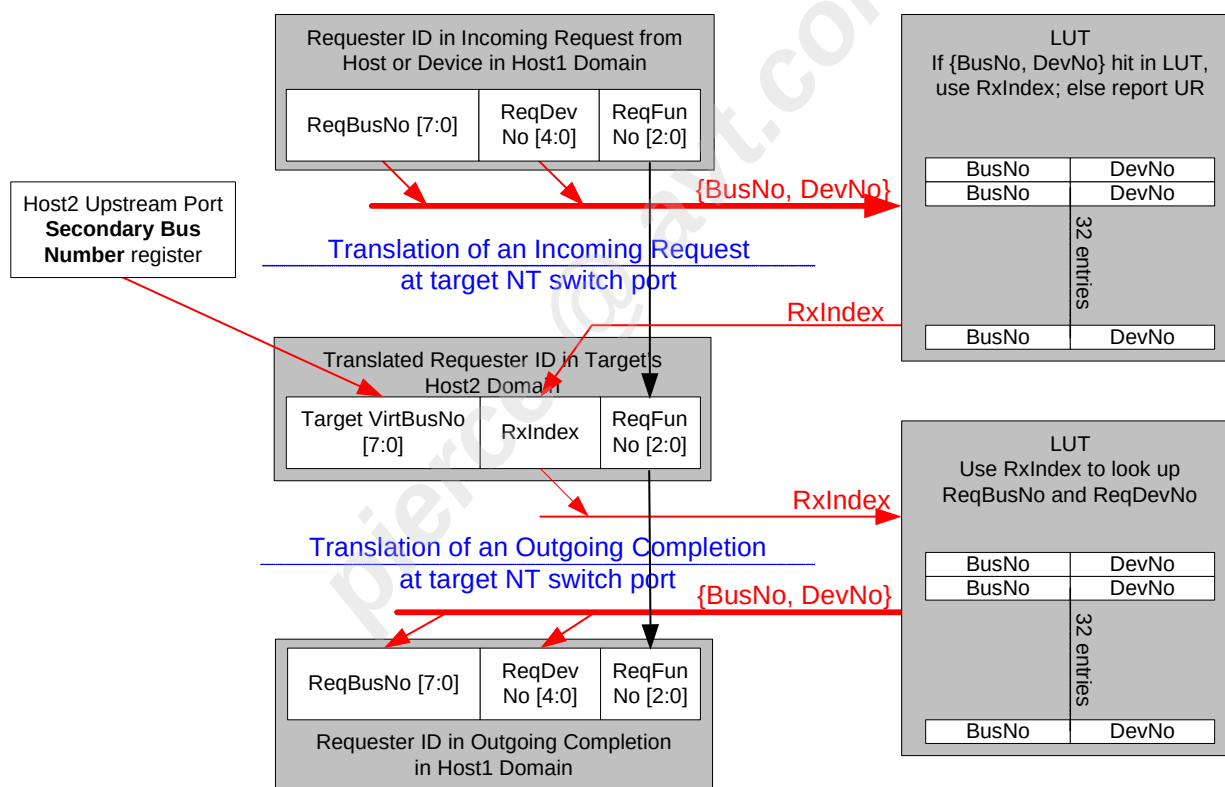
If no match is found, or the matched entry is not enabled, the Request receives an Unsupported Request (UR). If a match is found, and matched entry is enabled, the translated Memory Request TLP is address-translated and forwarded into the Virtual side Host domain.

The NT Port's Bus Number is sufficient to route the Completion from the Completer back to the NT Port in the Completer's domain, because the NT Port is the only possible Requester on that bus segment (because the NT Port is an endpoint). Elsewhere in the PCI Express hierarchy, the Bus Number is sufficient to route the Completion back into the switch that contains the NT Port.

The inverse translation occurs, when a Completion passes through the NTB. The RxIndex is retrieved from the *Device Number* field of the received Completion TLP *Requester ID* Header field and used to look up the 32-entry RID-LUT. The Completion TLP *Requester ID* values for the *Bus Number* and *Device Number* fields are replaced by the decoded RID-LUT-entry Bus Number and Device Number values, if the entry is valid; otherwise, an Unexpected Completion is reported.

The Completion TLP Completer ID is replaced by the NT Port Bus Number, Device Number, and Function Number values before forwarding the Completion TLP to the Link side Host domain. (Refer to Figure 14-10.)

Figure 14-10. Requester ID Translation for Request Originating in Host1 Domain



14.3 NT Port Power Management Handling

In NT mode, the PEX 8724 supports the ASPM L0s, ASPM L1, PCI L1, and L2/L3 Ready Link PM states (ASPM L0s must be enabled through the optional serial EEPROM and/or I²C).

14.3.1 Active State Power Management

The NT Port Link Interface endpoint supports the ASPM L0s and L1 Link PM states. The NT Port Virtual Interface endpoint implements the Configuration Space registers for ASPM support. However, the NT Port Virtual Interface endpoint does not enter into the low-power states, because there is no physical Link associated with the endpoint.

Note: *The NT Port Virtual Interface does not have a Link. Therefore, it never enters any Link PM states. From the Upstream scoreboard point of view, the NT Port Virtual Interface emulates an endpoint device in the DL_Down condition.*

14.3.1.1 ASPM L0s Link Power Management State Transition – NT Port Link Interface

The L0s Link PM state is allowable only in Active State Link Power Management (ASPM). Each device reports its support to ASPM L0s Link PM state entry through bit 10 of the **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10])^a. Software enables ASPM L0s Link PM state entry, by writing 1 to bit 0 of the **Link Control** register *Active State Power Management (ASPM)* field (NT Port Link Interface, offset 78h[1:0]). The ASPM L0s Link PM state is optimized for short entry and exit latencies, while providing substantial power savings. If a device has the ASPM L0s Link PM state enabled, the device's unused Transmitter Links must be brought into the ASPM L0s Link PM state.

Link Layer Timers are **not** affected by a transition to the L0 Link PM state. When in the ASPM L0s Link PM state, Links return to the L0 Link PM state, regardless of whether there are any TLP traffic situations to transmit Flow Control (FC) updates, which occurs every 30 μ s minimum.

Tx L0s Link Power Management State Entry Conditions – NT Port Link Interface

The conditions for the NT Port Link Interface to enter the ASPM L0s Link PM state are similar to an endpoint's ASPM L0s Link PM state entry conditions:

1. No pending TLPs (including Retry TLPs) or credits available to transmit pending TLPs.
2. No pending DLLPs in the DLL.
3. Enabled for entering into the ASPM L0s Link PM state:
 - Bit 10 of the **Link Capability** register *Active State Power Management (ASPM) Support* field (offset 74h[11:10]) is Set
 - Bit 0 of the **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]) is Set

The PM module directs the NT Port Link Interface to enter the ASPM L0s Link PM state when Conditions 1 and 2 are met for a specified period of time and Condition 3 is satisfied. The idle time can be selected, as either 1 μ s (default) or 4 μ s, in the **Power Management Hot Plug User Configuration** register *L0s Entry Idle Count* bit (NT Port Link Interface, offset F70h[0]).

*a. The ASPM L0s Link PM state is **not** supported, by default. To enable the ASPM L0s Link PM state, use the serial EEPROM and/or I²C to write 1 to bit 0 of the Active State Power Management (ASPM) field.*

L0s Link Power Management State Exit Conditions – NT Port Link Interface

Components from either end of a PCI Express Link can initiate exit from the ASPM L0s Link PM state. The following conditions must be met for the NT Port Link Interface to exit the ASPM L0s Link PM state:

- If there are pending DLLPs or TLPs to transmit
- Link enters retrain

14.3.1.2 L1 Link Power Management State Entry – NT Port Link Interface

The L1 Link PM state is a deeper power-saving state than the ASPM L0s Link PM state, with both Tx and Rx Lanes settling into Electrical Idle after L1 Link PM state negotiation is complete.

PCI L1 Link PM state entry is always initiated by an Upstream Port or Downstream device. PCI L1 Link PM state entry is also possible when an Upstream Port is placed into a state other than the D0 Device PM state. This transition is required. Downstream Ports never initiate PCI L1 Link PM state entry.

The ASPM L1 Link PM state is optional, and software-controlled by writing 1 to bit 1 of the **Link Control** register *Active State Power Management (ASPM)* field (offset 78h[1:0]). Conditions for entering the ASPM L1 Link PM state are implementation-specific.

A Port cannot directly transition from the L0s to L1 Link PM state while in ASPM – the Port transitions from L0s -> L0 -> L1, instead. Flow Control Update Timers are suspended after the Port enters the L1 Link PM state.

PCI L1 Link PM state entry protocol uses a PM_Enter_L1 DLLP. In contrast, ASPM L1 Link PM state entry protocol uses a PM_Active_State_Request_L1 DLLP. Downstream Ports never NAK the PM_Enter_L1 DLLP received from a Downstream device; however, Downstream Ports can NAK a PM_Active_State_Request_L1 DLLP, if there is a packet to transmit.

The NT Port Link Interface requests PCI L1 Link PM state entry, when the following six conditions are met:

- Software places the NT Port Link Interface into the D3hot Device PM state.
- NT Port Link Interface suspends scheduling of new TLPs, after it has sent the Completion of the TLP that enabled the D3hot Device PM state.
- NT Port Link Interface waits until all previously transmitted TLPs are acknowledged. During this time, the NT Port Link Interface receives TLPs and DLLPs, and transmits ACK/NAK and FC DLLPs.
- NT Port Link Interface waits for a sufficient quantity of FC credits, for all TLP packet types.
- NT Port Link Interface waits for all Retry TLPs to complete.
- After the NT Port Link Interface transmits all pending TLPs, it transmits a PM_Enter_L1 DLLP Upstream, with no more than a 4-symbol time gap.

When the NT Port receives a PM_Request_Ack DLLP, it directs the Link to enter the PCI L1 Link PM state.

When in the PCI L1 Link PM state, if an Electrical Idle exit is detected, or a TLP must be transmitted, the NT Port exits the PCI L1 Link PM state, transmits the pending TLP(s), if required, then re-enters the PCI L1 Link PM state after a 16-μs delay. The NT Port must detect a continuous 16 μs of the PHY L0 Link PM state for the next state entry.

L1 Link Power Management State Entry Negotiation Interruption – NT Port Link Interface

Per the *PCI Express Base r3.0*, when L1 Link PM state entry negotiation is interrupted, the state machine on both ends of the PCI Express Link should return to an Idle state. A Port can be in any of the states described in the sections that follow, when L1 Link PM state entry negotiation is interrupted, usually as the result of a trip through the *Recovery* state.

What occurs after L1 Link PM state entry negotiation is interrupted on the NT Port Link Interface is dependent upon the following:

- **Started requesting L1 Link PM state entry, using PM_Request DLLPs** – When the Tx or Rx *Recovery* state occurs, the Port stops receiving Requests, returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **Waiting for a PM_Request_Ack DLLP** – When the Tx or Rx *Recovery* state occurs, the Port returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **After receiving a PM_Request_Ack DLLP, directs the PHY to the L1 Link PM state, and the PHY has not reached the L1 Link PM state and the PHY enters the Tx or Rx *Recovery* state** – The Port returns to an Idle state, then waits 10 μ s before making another L1 Link PM State Entry Request, –or– the Tx Lane should enter the ASPM L0s Link PM state.
- **After the PHY transmits an EIOS and settles into the Tx L1 Link PM state, the Rx has not yet reached the L1 Link PM state, and an Rx *Recovery* state occurs or the Port's Link Control register *Retrain Link* bit (offset 78h[5]) is Set** – The Port directs the PHY Tx to exit the L1 Link PM state.

14.3.2 L2/L3 Ready Link Power Management State Entry – NT Port Link Interface

The L2/L3 Ready and L3 Link PM states are the staging points for removing main power. Support for the L2/L3 Ready Link PM state transition protocol is required.

The L2/L3 Ready Link PM state is related to the Device PM state transitions – it is the state that a given Link enters when the platform is preparing to enter its system sleep or shutdown state. Following completion of the L2/L3 Ready Link PM state transition protocol, the Link enters the L2/L3 Ready Link PM state. After main power is removed, the Link settles into the L3 Link PM state. The time for the L2/L3 Ready Link PM state entry transition is indicated by completion of the PME_Turn_Off/PME_TO_Ack handshake sequence. Any actions on the part of the Downstream component necessary to ready itself for power loss must be completed before initiating transition to the L2/L3 Ready Link PM state. After all preparations for loss of power and clock are complete, the Downstream component initiates L2/L3 Ready Link PM state entry, by transmitting the PM_Enter_L23 DLLP Upstream.

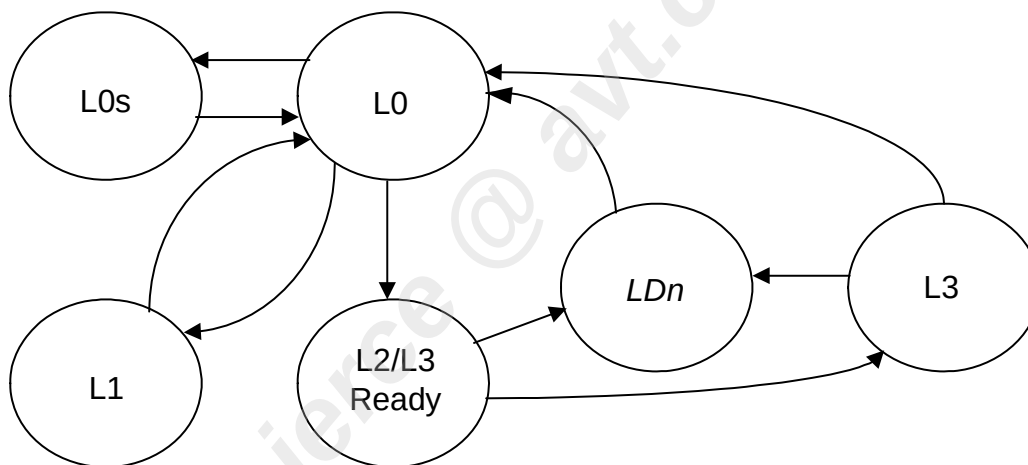
Exit from the L2/L3 Ready Link PM state, back to the L0 Link PM state, can be initiated only by an Upstream transaction that targets the Downstream component (Upstream Electrical Idle exit) in the same manner that an Upstream transaction would trigger transition from the L1 to L0 Link PM states.

When the NT Port Link Interface receives a PME_Turn_Off Message, it behaves the same way as an endpoint. The NT Port finishes the pending TLPS and ACK/NAK DLLPs, transmits a PME_TO_Ack and subsequently transmits L2/L3 Ready Link PM State Request DLLPs, then enters the L2/L3 Ready Link PM state.

When the NT Port Link Interface is in the L2/L3 Ready Link PM state, Rx Electrical Idle exit causes the Link to transition to the *LDn* state, then subsequently enter the *Detect* state.

Figure 14-11 illustrates the Link PM state transitions.

Figure 14-11. Link Power Management State Transitions



14.3.3 PCI-PM and PME Turn Off Support

When NT mode is enabled, the NT Port Link Interface behaves like any other endpoint in the D3hot Device PM state. After entering the D3hot Device PM state, the NT Port Link Interface requests PCI L1 Link PM state entry and finally settles into the PCI L1 Link PM state. Only Configuration accesses and Messages to the NT Port Link Interface are supported in the D3hot Device PM state. The Root Complex transmits PME_Turn_Off Messages when the NT Host decides to turn Off the main power and Reference Clock. The NT Port Link Interface indicates its readiness to lose power, by transmitting a PME_TO_Ack Message to the Upstream device. The PME_TO_Ack Message is transmitted when there are no pending TLPs to transmit Upstream, toward the NT Port Link Interface. The Port requests the L2/L3 Ready Link PM state, by transmitting PM_Enter_L23 DLLPs to the Upstream device after transmitting a PME_TO_Ack TLP. The Port settles into the L3 Link PM state when the Power Controller removes the main power and Reference Clock.

When the PME_Turn_Off Message is received on the Transparent Upstream Port(s), the Port broadcasts this Message to all Downstream devices, including the NT Port Virtual Interface. After the PME_TO_Ack Message is received from all Downstream devices and the NT Port Virtual Interface, the Transparent Upstream Port(s) transmit(s) an aggregated PME_TO_Ack Message to the Upstream component after it finishes transmitting all pending TLPs to the Upstream component. When NT mode is enabled, the Transparent Downstream Ports allow the attached devices to enter the PCI L1 Link PM state. The NT Port Virtual Interface never enters the PCI L1 Link PM state.

14.3.4 Message Generation

The NT Port Link Interface never generates PM_PME Messages. The NT Port Virtual Interface never receives Set_Slot_Power_Limit Messages nor generates PM_PME Messages.

14.4 NT Port Expansion ROM

The NT Port Link Interface supports Expansion ROM, by default. Expansion ROM support can be moved from the NT Port Link Interface to the NT Port Virtual Interface, by Setting the **Ingress Chip Control** register *NT Port Expansion ROM Virtual Side* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[0]).

The NT Port supports 16- or 32-KB-sized Expansion ROM, based upon the **Serial EEPROM Clock Frequency** register *Expansion ROM Size* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[16]) value.

Notes: *The NT Port Expansion ROM can be enabled in either the NT Port Virtual or Link Interface, but not both concurrently. By default, the NT Port Expansion ROM is enabled in the NT Port Link Interface (**Ingress Chip Control** register NT Port Expansion ROM Virtual Side bit is Cleared).*

*The NT Port Expansion ROM can be disabled, by Setting the **Ingress Control** register *Disable NT Port Expansion ROM BAR* bit (Upstream Port(s), offset F60h[15]).*

14.5 NT Port Interrupts

The NT Port Virtual and Link Interfaces can both generate interrupts in response to specific events. The NT Port must not receive any INTx Message Requests. If the NT Port receives an INTx Message Request, the Request is handled as a Malformed TLP error.

14.5.1 NT Port Virtual Interface Interrupts

The NT Port Virtual Interface generates interrupts to the Virtual side Host for the following reasons (all are masked, by default, and must not be masked to be enabled):

- Doorbell interrupts
- NT Port Link Interface detected an Correctable TLP error
- NT Port Link Interface detected an Uncorrectable TLP error (option to signal Fatal and/or Non-Fatal)
- NT Port Link Interface *DL_Active* state change
- NT Port Link Interface received an Uncorrectable Error Message

NT-Virtual Doorbell and NT-Link Error and Event interrupts can use the **PEX_INTA#** (Base mode) or **VSx_PEX_INTA#** (Virtual Switch mode), or INTx signaling mechanism or MSI mechanism (all mutually exclusive), as follows:

- **PEX_INTA#** or **VSx_PEX_INTA#** output can be enabled for NT-Virtual Doorbell interrupts, by Setting the **Error Check Disable** register *Enable Ball Interrupt Output for NT-Virtual Doorbell-Generated Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[7])
- **PEX_INTA#** or **VSx_PEX_INTA#** output can be enabled for NT-Link Error and Event interrupts, by Setting the **Error Check Disable** register *Enable Ball Interrupt Output for Device-Specific NT-Link Port Event-Triggered Interrupts* bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 720h[5])

Refer to [Section 14.1.7](#) for Doorbell interrupt details.

The NT Port Virtual Interface de-asserts INTx, or **PEX_INTA#** (Base mode) or **VSx_PEX_INTA#** (Virtual Switch mode), interrupts in response to one or more of the following conditions:

- **PCI Command** register *Interrupt Disable* bit (NT Port Virtual Interface, offset 04h[10]) is Set
- Corresponding *Interrupt Mask* bit is Set
- NT Port Link goes down (DL_Down condition) or the NT Port Link Interface receives a Hot Reset
- Software Clears the *Interrupt Status* bit, –or– Sets the *Doorbell Interrupt Request Clear* bit

For tracking purposes, an NT Port Virtual Interface-generated interrupt is treated like an event that is external to the PCI-to-PCI bridge. If software asserts a **Secondary Bus Reset** from this PCI-to-PCI bridge, the PCI-to-PCI bridge de-asserts the NT Port Virtual Interface interrupt.

When the NT Port Link Interface detects Correctable TLP errors, the NT Port Virtual Interface signals the interrupt to the Virtual side Host, if the interrupt signaling is enabled and not masked (**Link Error Status Virtual** register *Link Side Correctable Error Status* bit is Set, and **Link Error Mask Virtual** register *Link Side Correctable Error Mask* bit is Cleared (NT Port Virtual Interface, offsets FE0h[0] and FE4h[0], respectively)).

When the NT Port Link Interface detects Uncorrectable TLP errors, the NT Port Virtual Interface signals the interrupt to the Virtual side Host, if the interrupt signaling is enabled and not masked (**Link Error Status Virtual** register *Link Side Uncorrectable Error Status* bit is Set, and **Link Error Mask Virtual** register *Link Side Uncorrectable Error Mask* bit is Cleared (NT Port Virtual Interface, offsets FE0h[1] and FE4h[1], respectively)).

An NT Port Link Interface *DL_Active* state change occurs upon detection of an NT Port Link Interface *DL_Down* state rise edge and fall edge. This signals the interrupt to the Virtual side Host, if the interrupt signaling is enabled and not masked (**Link Error Status Virtual** register *Link Side DL Active Change Status* bit is Set, and **Link Error Mask Virtual** register *Link Side DL Active Change Mask* bit is Cleared (NT Port Virtual Interface, offsets FE0h[2] and FE4h[2], respectively)).

When the NT Port Link Interface receives an Uncorrectable Error Message, the NT Port Virtual Interface signals the interrupt to the Virtual side Host, if the interrupt signaling is enabled and not masked (**Link Error Status Virtual** register *Link Side Uncorrectable Error Message Drop Status* bit is Set, and **Link Error Mask Virtual** register *Link Side Uncorrectable Error Message Drop Mask* bit is Cleared (NT Port Virtual Interface, offsets FE0h[3] and FE4h[3], respectively)).

14.5.2 NT Port Link Interface Interrupts

The NT Port Link Interface generates interrupts to the Link side Host for NT-Link Doorbell interrupts detected at the NT Port Link Interface ingress Port (interrupts are masked, by default). The NT Port Link Interface should not detect any Device-Specific errors (because these errors are reported to the Upstream Port Host to handle).

NT-Link Doorbell interrupts and Device-Specific errors can use the INTx signaling mechanism or MSI mechanism (both mutually exclusive). Refer to [Section 14.1.7](#) for Doorbell interrupt details.

The NT Port Link Interface de-asserts INTx interrupts in response to one or more of the following conditions:

- **PCI Command** register *Interrupt Disable* bit (NT Port Link Interface, offset 04h[10]) is Set
- Corresponding *Interrupt Mask* bit is Set
- NT Port Link goes Down (DL_Down condition) or the NT Port Link Interface receives a Hot Reset
- Software Clears the *Interrupt Status* bit, or Sets the *Doorbell Interrupt Request Clear* bit

14.6 NT Port Error Handling

The NT Port Virtual Interface endpoint logs TLP errors, for TLPs that travel from the NT Port Virtual-to-Link Interface. The PEX 8724 signals Error Messages to the Virtual side Host. The PEX 8724 provides an option to communicate this error condition to the Link side Host, by signaling an interrupt.

The NT Port Link Interface endpoint logs TLP errors, for TLPs that travel from the NT Port Link-to-Virtual Interface. The PEX 8724 signals Error Messages to the Link side Host.

When the PEX 8724 receives a TLP and before it transmits the TLP through the NT Port, the following is performed, in the sequence listed:

1. TLP integrity check.
2. Address decode.
3. Address translation.
4. Requester ID translation.
5. ECRC re-generation.

If the PEX 8724 detects an ECRC error, it corrupts the re-generated ECRC before transmitting the TLP through the NT Port. The PEX 8724 also provides options for dropping error-detected poisoned or ECRC TLPs (**Ingress Control** register *Drop Poisoned TLPs* and *Drop ECRC TLPs* bits (Upstream Port(s), offset F60h[9:8], respectively)).

The PEX 8724 does not generate the ECRC for a TLP that passes through the NT Port, if the received TLP does not have its *TD* bit Set.

The PEX 8724 drops all TLPs traveling from the NT Port Virtual Interface to the NT Port Link Interface, if the internal RAM Fatal Error-Correcting Code (ECC) error is detected, until the PEX 8724 receives a Hot Reset from a Virtual side Host.

14.6.1 NT Port Link Interface Error Handling

If the NT Port Link Interface receives an Uncorrectable Error Message, it reports a Malformed TLP error, by default. However, if the **Ingress Chip Control** register *NT Error Message Drop* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[17]) is Set, the NT Port Link Interface drops the Message, and logs the error into the **Link Error Status Virtual** register *Link Side Uncorrectable Error Message Drop Status* bit (NT Port Virtual Interface, offset FE0h[3]). If the corresponding **Link Error Mask Virtual** register *Link Side Uncorrectable Error Message Drop Mask* bit (NT Port Virtual Interface, offset FE4h[3]) is Set, the NT Port Virtual Interface signals an interrupt (INTx, MSI, or **PEX_INTA#** (Base mode) or **VSx_PEX_INTA#** (Virtual Switch mode)) to the Virtual side Host through the Upstream Port(s), if interrupts are enabled.

14.6.2 NT Virtual Side Error Handling

For the NT Port Virtual Interface to generate an Error Message, the following error-forwarding *Enable* bits must be Set in both the NT PCI-to-PCI bridge and Upstream Port(s):

- **PCI Command** register (offset 04h)
 - *SERR# Enable* (bit 8)
- **Bridge Control** register (offset 3Ch)
 - *SERR# Enable* (bit 17)
- **Device Control** register (offset 70h)
 - *Correctable Error Reporting Enable* (bit 0)
 - *Non-Fatal Error Reporting Enable* (bit 1)
 - *Fatal Error Reporting Enable* (bit 2)

When the NT Port Virtual Interface generates an Error Message, the NT PCI-to-PCI bridge also logs the error status in the following NT Port registers, while sending the Message in the Upstream direction:

- **Secondary Status** register (offset 1Ch)
 - *Received System Error* (bit 30)
- **PCI Status** register (offset 04h)
 - *Signaled System Error* (bit 30)

14.7 Cursor Mechanism

A software application can use the [Device-Specific Cursor Mechanism](#) to access the NT Port Configuration Space registers. The registers that support the Device-Specific Cursor Mechanism are the [Configuration Address Window](#) and [Configuration Data Window](#) registers (offsets [F8h](#) and [FCh](#), respectively). A software application can also:

- Select the Configuration Register offset, by using the [Configuration Address Window](#) register
- Perform Read accesses to the [Configuration Data Window](#) register, to read from the selected Configuration register
- Perform Write accesses to the [Configuration Data Window](#) register, to write to the selected Configuration register

Configuration transactions have access to this Device-Specific Cursor Mechanism, if NT mode is enabled.

For details regarding the [Configuration Address Window](#) and [Configuration Data Window](#) registers, refer to:

- [Section 15.9, “NT Port Virtual Interface Vendor-Specific Capability 3 Registers \(Offsets C8h – FCh\)”](#)
- [Section 16.9, “NT Port Link Interface Vendor-Specific Capability 3 Registers \(Offsets C8h – FCh\)”](#)

14.8 Port Programmability

The PEX 8724 supports the capability of programming the VS0 Upstream Port and NT Port, by way of the [VS0 Upstream](#) register [VS0 Upstream Port](#) and [NT Port](#) fields (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset [360h\[4:0 and 12:8\]](#), respectively). This register is updated, based upon the external [STRAP_NT_UPSTRM_PORTSEL\[2:0\]](#) input states, by default. I²C, software, and/or an optional serial EEPROM (if present) can be used to override the external Strap values.

A software application can change the VS0 Upstream Port and NT Port locations to another Port Number during runtime, or as part of a failover sequence. It is recommended that the PEX 8724 be in an Idle state (no traffic) when changing the VS0 Upstream Port and NT Port Numbers during runtime. During a failover sequence, application software should be able to handle all spurious TLPs that it receives as a result of the failover process.

14.9 NT Failover

To protect against a failing Host taking the entire system down, a backup Host can be in place, ready to take over. One method is to have a secondary Host behind an NTB and use an NT failover sequence.

This section describes four of the major failover scenarios:

- [Active-Passive NT Failover](#)
- [Active-Active NT Failover](#)
- [NT Failover for Virtual Switch Operation](#)
- [Failover Completion](#)

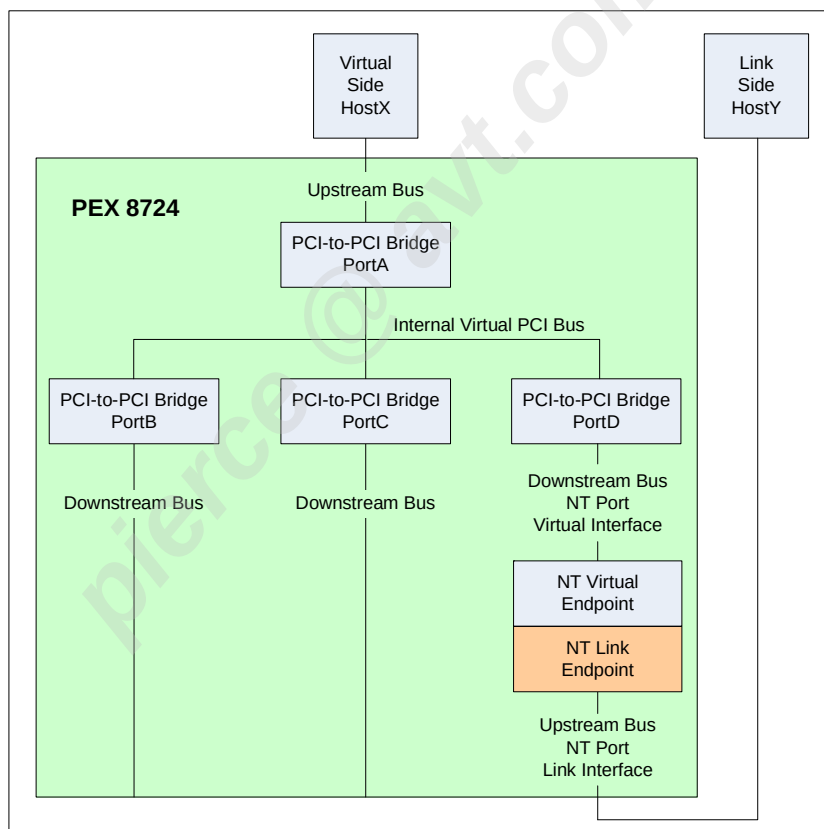
Additional failover information related to NT mode is discussed in [Section 12.5.3, “Virtual Switch Host to NT Host Failover.”](#)

14.9.1 Active-Passive NT Failover

In an active-passive Host scenario, the passive Host is waiting for the active Host to fail. The PEX 8724 might be initially configured as follows – the active HostX is the Virtual side Host connected to PortA, and the passive HostY is the Link side Host connected to PortD. In this case (refer to [Figure 14-12](#)):

- HostX owns all the PEX 8724’s Downstream Ports and the NTB’s NT Port Virtual Interface endpoint
- HostY owns only the NTB’s Link side

Figure 14-12. Example – Active-Passive NT Failover Host Scenario

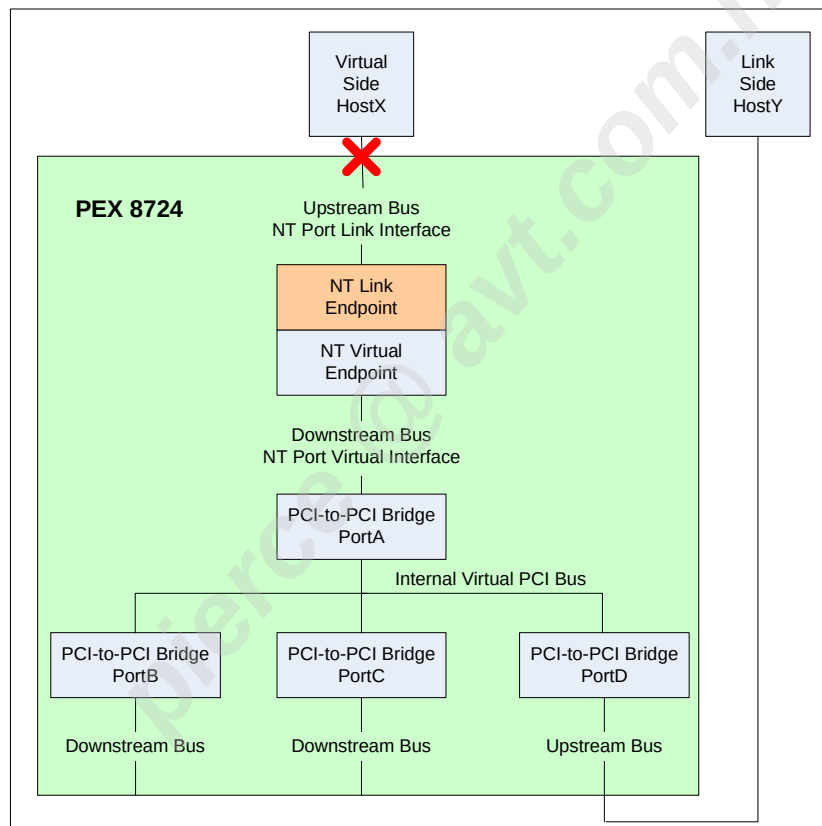


When the Virtual side Host (HostX) fails, and that failure is detected (by means beyond the scope of this data book) by the Link side Host (HostY), HostY initiates the sequence listed below.

1. Configure PortA to be the NT Port and PortD to be the Upstream Port.
 - This is a single CSR Write to the **VS Failover** register (NT Port Link Interface, offset **F00h**)
 - The register's **VS Upstream Port** field (field [4:0]) indicates the Upstream Port, and should be changed from the Port Number of PortA to the Port Number of PortD
 - The register's **VS NT Port** field (field [12:8]) indicates the NT Port, and should be changed from the Port Number of PortD to the Port Number of PortA
 - The register's **VS NT Enable** bit (bit 13) enables NT mode, and should remain Set
2. Reset the PEX 8724 through an Upstream Port's **Bridge Control** register **Secondary Bus Reset** bit (offset **3Ch[22]**).
3. Re-enumerate the hierarchy and start operation. The Downstream Bus Numbers change as a result of the re-enumeration.

Figure 14-13 illustrates how the PEX 8724 looks after this failover sequence. When HostX failure can be resolved, the red X goes away, and HostX becomes the passive standby backup processor, and the process is ready to repeat.

Figure 14-13. Example – End Result of Active-Passive NT Failover Host Scenario



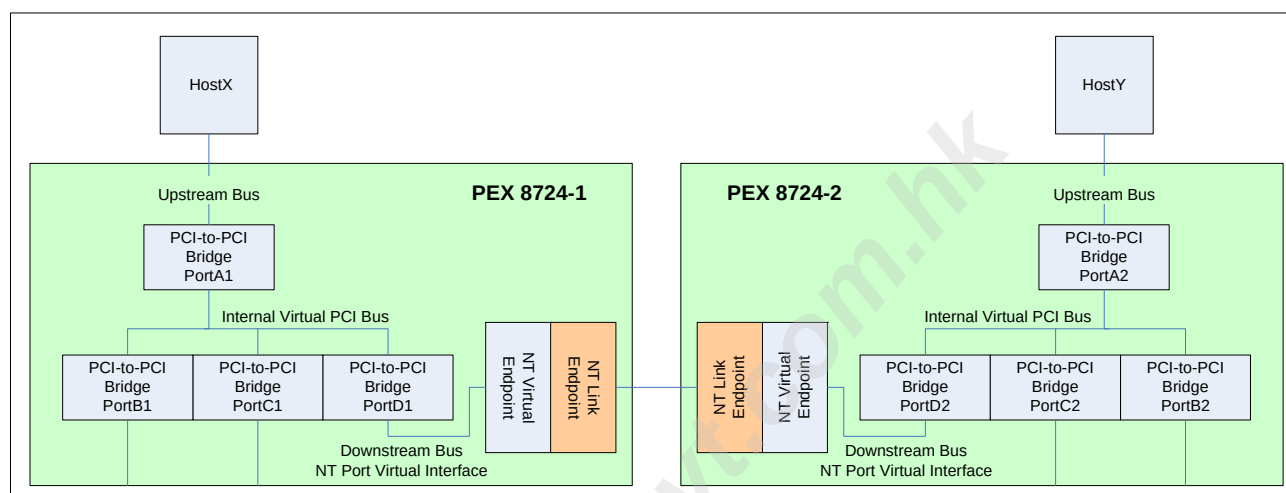
14.9.2 Active-Active NT Failover

In an active-active Host scenario, both Hosts are busy with their own hierarchies and ready to take over, should the other Host fail. A symmetric arrangement is the easiest to understand, but is not required. In [Figure 14-14](#), either Host can start the failover sequence. Initially:

- HostX owns all the PEX 8724-1 Downstream Ports and the NTB-1 NT Port Virtual Interface endpoint
- HostY owns all the PEX 8724-2 Downstream Ports and the NTB-2 NT Port Virtual Interface endpoint

Note that neither Host owns the Link between the two PEX 8724 switches nor the Link side endpoints. The default values for these endpoints use a Captured Bus Number of 0, which must be taken into consideration when programming the RID-LUT.

Figure 14-14. Example – Active-Active NT Failover Host Scenario

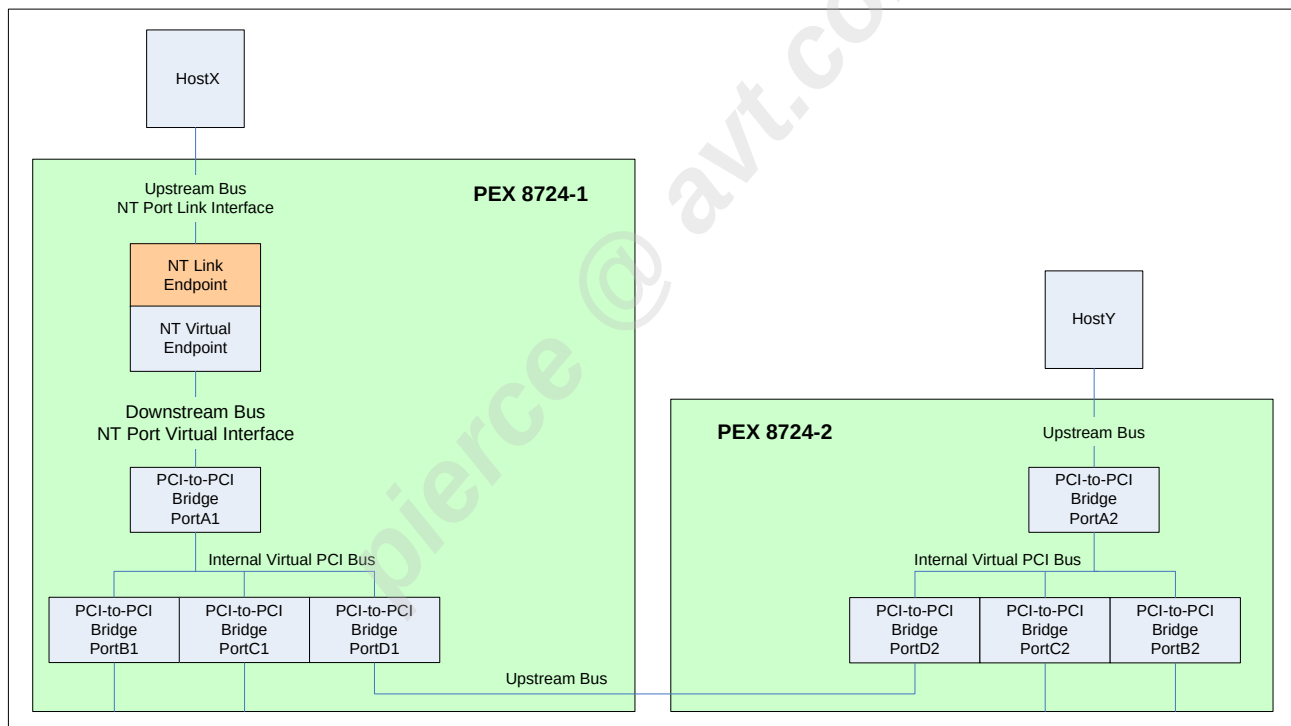


When HostY has determined that HostX has failed (by means beyond the scope of this data book), HostY initiates the sequence listed below.

1. HostY configures the PEX 8724-2 PortD2 to be a Transparent Downstream Port, by Clearing the **VS Failover** register *VS NT Enable* bit (Upstream Port(s), offset **F00h**[13]).
2. HostY configures the PEX 8724-1 PortD1 to be the Upstream Port, and PortA1 to be the NT Port, and NT-enabled. This is a single CSR Write to the **VS Failover** register (NT Port Link Interface, on PEX 8724-1, offset **F00h**).
 - The register's *VS Upstream Port* field (field [4:0]) indicates the Upstream Port, and should be changed from A to D
 - The register's *VS NT Port* field (field [12:8]) indicates the NT Port, and should be changed from D to A
 - The register's *VS NT Enable* bit (bit 13) enables NT mode, and should remain Set
3. HostY resets the entire PEX 8724-1 hierarchy. It may also need to reset the PEX 8724-2 hierarchy, unless space in the Bus Numbering of PCI-to-PCI Bridge PortD2 was reserved to inherit many new buses. The reset is Set through either the PEX 8724-1 Upstream Port's, or PEX 8724-2 PortD2's, **Bridge Control** register *Secondary Bus Reset* bit (offset **3Ch**[22]).
4. Re-enumerate the hierarchy and start operation.

Post failover, the system resembles the example illustrated in [Figure 14-15](#). From the PEX 8724-1 perspective, it looks just like an active-passive failover. When HostX is repaired, it can negotiate with HostY when to resume control of the PEX 8724-1 hierarchy.

Figure 14-15. Example – End Result of Active-Active NT Failover Host Scenario



To return to failover protection, reverse the previous steps. Steps 4 through 6 can occur in parallel, because the two Hosts have independent scheduling.

1. HostX negotiates with HostY when to resume control of the PEX 8724-1 hierarchy, and thereby resume failover protection.
2. HostY resets the PEX 8724-2 PortD2 Downstream hierarchy, by Setting and then Clearing the PEX 8724-2 PortD2's **Bridge Control** register *Secondary Bus Reset* bit (offset 3Ch[22]).
3. HostX returns the PEX 8724-1 to its initial configuration, by programming PortA1 to be Upstream, PortD1 to be NT, and NT-enabled. This is a single CSR Write to the PEX 8724-1's **VS Failover** register (NT Port Link Interface, offset F00h).
 - The register's *VS Upstream Port* field (field [4:0]) indicates the Upstream Port, and should be changed from D to A
 - The register's *VS NT Port* field (field [12:8]) indicates the NT Port, and should be changed from A to D
 - The register's *VS NT Enable* bit (bit 13) enables NT mode, and should remain Set
4. HostX resets the entire PEX 8724-1 hierarchy, by Setting and then Clearing the PEX 8724-1 PortA1's **Bridge Control** register *Secondary Bus Reset* bit (offset 3Ch[22]).
5. HostY adds the NT Port back into its hierarchy, by Setting the PEX 8724-2's **VS Failover** register *VS NT Enable* bit (NT Port Link Interface, offset F00h[13]) (the NT Port assignment has not changed, only the enable has changed).
6. Both HostX and HostY re-enumerate their hierarchies and begin operation.

14.9.3 NT Failover for Virtual Switch Operation

In a single virtual switch system (Base mode), by definition, an NT Port and Upstream Port exist in the same virtual switch hierarchy, because there is only one hierarchy. In a single virtual switch, an NT Port can automatically promote itself to be the Upstream Port, and it can assign a new NT Port to replace itself. This same functionality is available with multiple roots and multiple virtual switch hierarchies, with the one restriction being that the NT Port must belong to the same virtual switch as the Upstream Port it is trying to replace in the failover operation.

The first step for NT failover with multiple virtual switches, then, is to have the NT Port belong to the same virtual switch hierarchy as the failed Upstream Port (this process is described later in this section). The Management Port is the only Port that can control virtual switch assignment. The **Management Port Control** register *NT VS* field (Port 0, accessible through the Management Port and Redundant Management Port, offset 354h[18:16]) is used for assigning the virtual switch for the NT Port. If the NT Port must migrate to another virtual switch hierarchy, the Management Port modifies the Virtual Switch Number for the NT Port.

The Management Port also assigns which other Port(s) belong to which virtual switch, in the **VSx Port Vector** register(s) (Port 0, accessible through the Management Port, offset(s) 380h through 38Ch), and controls which Ports are the Upstream Ports, per virtual switch. There are physically four separate **VSx Upstream** registers that control the Upstream and NT Port assignments, one per virtual switch. The Management Port sees all four registers.

After the NT and Upstream Port are in the same virtual switch, the NT failover process is identical to single virtual switch (Base mode) NT failover. The NT Port does not need to know to which virtual switch it belongs, or whether multiple virtual switches are enabled. Hardware forces the NT Port to see only the one **VSx Upstream** register that matches its virtual switch register. The hardware automatically links the NT register to the correct **VSx Upstream** register, because a Read on the **VSx Upstream** register sees the change that the NT Port makes on the **NT Upstream** register in NT Link space. The NT Port reads from and writes to the **VS Failover** register *VS Upstream Port* field (NT Port Link Interface, offset F00h[4:0]), and hardware points it to the single correct **VSx Upstream** register.

The VS0 Upstream Port ID, NT Port ID, and NT Enable are all programmable, by any privileged source within that virtual switch, whether it is the Link side NT Host or Virtual side Upstream Host. (Refer to the **VS0 Upstream** register *VS0 Upstream Port*, *NT Port*, and *NT Enable* fields/bits (Port 0, accessible through the Management Port, offset 360h[4:0, 12:8, and 13], respectively), for further details.)

The NT Port, as the first step for failover, writes to the **VS Failover** register (NT Port Link Interface, offset F00h), to simultaneously promote itself to be the Upstream Port while downgrading the (broken) Upstream Port to a Downstream Port or the new (replacement) NT Port. The new Upstream Port will commonly be the previous NT Port ID, which by definition is in the same virtual switch hierarchy, but hardware does not care which Port it is, as long as it is located within the same virtual switch. (Refer to Section 14.9 for subsequent failover steps.)

14.9.4 Failover Completion

When the Management Port knows that traffic is quiesced for planned or unplanned events, it can re-program the owner of the Downstream Port(s). (The method used for determining the new Port assignments to the virtual switch is beyond the scope of this data book.) The Management Port performs two Memory Writes to the **VSx Port Vector** register(s) (Port 0, accessible through the Management Port, offset(s) [380h](#) through [38Ch](#)) – one to disable the virtual switch that failed, and one to enable the new virtual switch.

The Host that received the new Ports receives a Hot Plug Message, informing it of the new Ports. Alternately, the Management Host can message the applicable Host regarding the new Ports.

Depending upon the system, it may not be necessary to re-enumerate the recently acquired Ports. If bus numbering control is available, each Downstream Port can be assigned unique Bus Number ranges (across all virtual switches), thus allowing any Host to own any Downstream Port without creating a Bus Number conflict.

If bus numbering control is not available, re-enumeration is required. It is possible that the entire virtual switch requires re-enumeration, from the Root Port down, because the Primary, Secondary, and Subordinate Bus Numbers of each PCI-to-PCI bridge might need to be changed.

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Chapter 15 NT Port Virtual Interface Registers – NT Mode

15.1 Introduction

Note: Check the latest design guides, application notes and errata list for Non-Transparent (NT) usage.

In NT mode, the NT Port includes two sets of Configuration, Capability, Control, and Status registers, to support the Virtual and Link Interfaces. This chapter defines the NT Port Virtual Interface registers. Other registers are defined in:

- [Chapter 13, “Transparent Port Registers”](#)
- [Chapter 16, “NT Port Link Interface Registers – NT Mode”](#)

All PEX 8724 registers can be accessed by Configuration or Memory Requests.

For further details regarding register names and descriptions, refer to the following specifications:

- *PCI r3.0*
- *PCI Power Mgmt. r1.2*
- *PCI Express Base r3.0*

15.2 NT Port Virtual Interface Type 0 Register Map

Table 15-1 defines the NT Port Virtual Interface Type 0 register mapping.

Table 15-1. NT Port Virtual Interface Type 0 Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16															15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0															Offsets	
NT Port Virtual Interface PCI-Compatible Type 0 Configuration Header Registers (Offsets 00h – 3Ch)																														00h	
																														...	
																														34h	
																														...	
																														3Ch	
NT Port Virtual Interface PCI Power Management Capability Registers (Offsets 40h – 44h)															Next Capability Pointer (48h)	Capability ID (01h)	40h														
																														44h	
NT Port Virtual Interface MSI Capability Registers (Offsets 48h – 64h)															Next Capability Pointer (68h)	Capability ID (05h)	48h														
																														...	
																														64h	
NT Port Virtual Interface PCI Express Capability Registers (Offsets 68h – A0h)															Next Capability Pointer (C8h)	Capability ID (10h)	68h														
																														...	
																														A0h	
Reserved																														A4h –	C4h
NT Port Virtual Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh)															Next Capability Pointer (00h)	Capability ID 3 (09h)	C8h														
																														...	
																														FCh	
Next Capability Offset (FB4h)										1h					PCI Express Extended Capability ID (0003h)										100h						
Device Serial Number Extended Capability Registers (Offsets 100h – 108h) ^a																														...	
																														108h	
Reserved																														10Ch –	144h
Next Capability Offset 2 (C34h)										1h					PCI Express Extended Capability ID (0002h)										148h						
NT Port Virtual Interface Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)																														...	
																														1BCh	
NT Port Virtual Interface Device-Specific Registers (Offsets 1C0h – A74h)																														1C0h	
																														...	
																														A74h	
Reserved																														A78h –	B6Ch

Table 15-1. NT Port Virtual Interface Type 0 Register Map (Cont.)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets																
Next Capability Offset 2 (000h)																1h																PCI Express Extended Capability ID 2 (000Bh)																B70h
NT Port Virtual Interface Device-Specific Registers (Offsets B70h – C88h)																																...																
Next Capability Offset 4 (B70h)																1h																PCI Express Extended Capability ID 4 (000Bh)																C34h
NT Port Virtual Interface Device-Specific Registers (Offsets B70h – C88h)																																...																
																																C88h																
NT Port Virtual Interface NT Bridging-Specific Registers (Offsets C8Ch – DFCh)																																C8Ch																
																																...																
																																DFCh																
Multicast Extended Capability Registers (Offsets E00h – E2Ch) ^b																																E00h																
																																...																
																																E2Ch																
Reserved																																E30h –	FB0h															
Next Capability Offset (148h)																1h																PCI Express Extended Capability ID (0001h)																FB4h
NT Port Virtual Interface Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)																																...																
																																FDCh																
NT Port Virtual Interface Device-Specific Registers – Link Error (Offsets FE0h – FFCh)																																FE0h																
																																...																
																																FFCh																

- a. Register offsets *100h* through *108h* are detailed in Section 13.12, “Device Serial Number Extended Capability Registers (Offsets *100h* – *108h*).”
- b. Register offsets *E00h* through *E2Ch* are detailed in Section 13.19, “Multicast Extended Capability Registers (Offsets *E00h* – *E2Ch*).”

15.3 Register Access

The NT Port Virtual Interface implements a 4-KB Configuration Space. The lower 256 bytes (offsets 00h through FFh) comprise the PCI-compatible Configuration Space, and the upper 960 DWords (offsets 100h through FFFh) comprise the PCI Express Extended Configuration Space. The PEX 8724 supports three mechanisms for accessing the NT Port Virtual Interface registers:

- [PCI Express Base r3.0 Configuration Mechanism](#)
- [Device-Specific Memory-Mapped Configuration Mechanism](#)
- [Device-Specific Cursor Mechanism](#)

15.3.1 PCI Express Base r3.0 Configuration Mechanism

The *PCI Express Base r3.0* Configuration mechanism is divided into two mechanisms:

- [PCI r3.0-Compatible Configuration Mechanism](#) – Provides Conventional PCI access to the first 256 bytes (the bytes at offsets 00h through FFh) of the NT Port Virtual Interface Configuration Register space
- [PCI Express Enhanced Configuration Access Mechanism](#) – Provides access to the entire 4-KB Configuration Space

Both are described in the sections that follow.

15.3.1.1 PCI r3.0-Compatible Configuration Mechanism

The *PCI r3.0-Compatible* Configuration mechanism provides standard access to the NT Port Virtual Interface's first 256 bytes (the bytes at offsets 00h through FFh) of the PCI Express Configuration Space. (Refer to [Figure 15-1](#).)

The mechanism uses PCI Type 0 and Type 1 Configuration transactions to access the PEX 8724 Configuration registers. All Ports capture the Bus Number and Device Number assigned by the Upstream device on the PCI Express Link attached to the PEX 8724 Upstream Port, as required by the *PCI Express Base r3.0*.

The PEX 8724 decodes all Type 1 Configuration accesses received on its Upstream Port, when any of the following conditions are met:

- If the Bus Number specified in the Configuration access is the Bus Number of the PEX 8724 internal virtual PCI Bus, the PEX 8724 automatically converts the Type 1 Configuration access into the appropriate Type 0 Configuration access for the specified device.
 - If the specified device corresponds to the NT Port Virtual Interface (or to the PCI-to-PCI bridge in one of the Transparent Downstream Ports), the PEX 8724 processes the Read or Write Request
 - If the specified Device Number does not correspond to any of the Downstream Port Device Numbers nor NT Port Number, the PEX 8724 responds with an Unsupported Request (UR)

This mechanism uses the same Request format as the [PCI Express Enhanced Configuration Access Mechanism](#). For PCI-compatible Configuration Requests, the *Extended Register Address* field must be all zeros (0).

Because the mechanism is limited to the first 256 bytes of the NT Port Virtual Interface Configuration register space, one of the following must be used to access beyond Byte FFh:

- [PCI Express Enhanced Configuration Access Mechanism](#)
- [Device-Specific Memory-Mapped Configuration Mechanism](#)
- [Device-Specific Cursor Mechanism](#)

15.3.1.2 PCI Express Enhanced Configuration Access Mechanism

The PCI Express Enhanced Configuration Access mechanism (ECAM) uses a flat, Root Complex Memory-Mapped Address space to access the device Configuration registers. In this case, the Memory address determines the Configuration register accessed, and the Memory data returns the addressed register's contents. The Root Complex converts the Memory transaction into a Configuration transaction.

The mechanism can be used to access PEX 8724 registers that are defined by Specifications. This mechanism typically **cannot** access the Device-Specific registers (which are instead accessible by the Device-Specific Memory-Mapped Configuration Mechanism) in the following offset ranges, per Port:

- 200h through AFCh
- B0Ch through B6Ch
- B80h through C30h

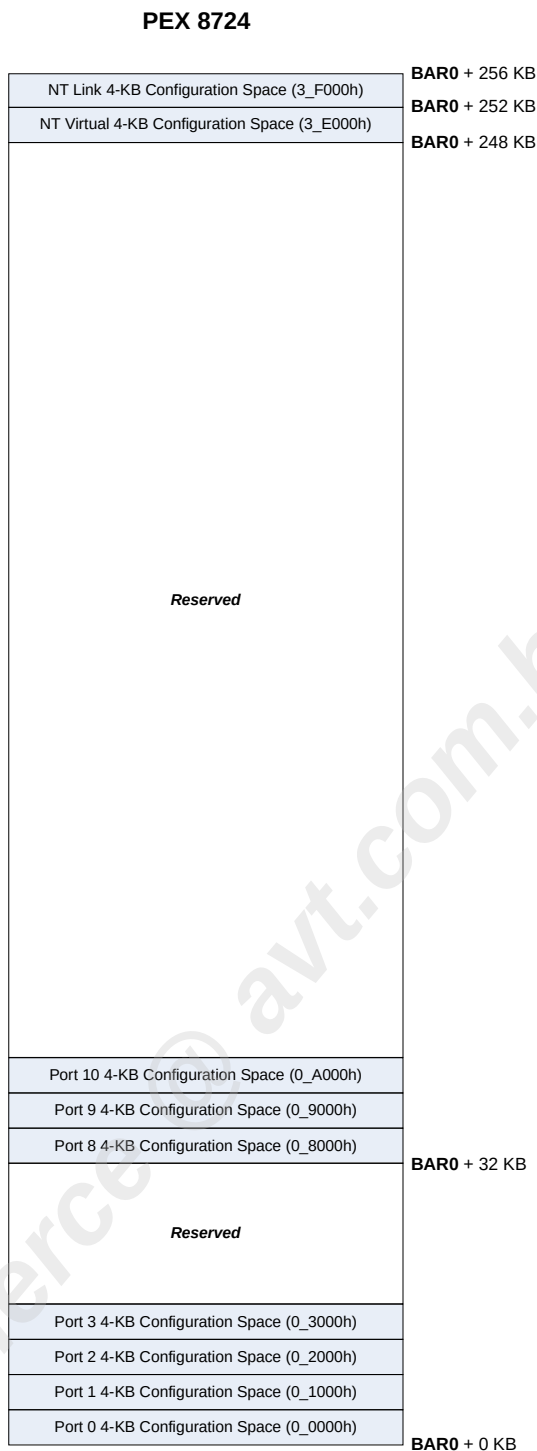
Silicon Revision CA provides an option to enable ECAM access to all PEX 8724 registers, however, by Setting the **Station-Based Control** register *Device-Specific Register Access by Extended Configuration Request Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 760h[16]), in the Station(s) that have an Upstream Port.

15.3.2 Device-Specific Memory-Mapped Configuration Mechanism

The Device-Specific Memory-Mapped Configuration mechanism provides a method to access the registers for all Ports within a single 256-KB Memory map, as illustrated in [Figure 15-1](#). This Memory map is identical for Upstream Port(s) **BAR0/1**, NT Port Virtual Interface **BAR0/1**, and NT Port Link Interface **BAR0/1**. The registers of each Port are located within a 4-KB range.

When the NT Port is enabled at Fundamental Reset, the NT Port Virtual and Link Interface registers use the *PCI r3.0* Type 0 Configuration Space Header. The NT PCI-to-PCI bridge (between the NT Port Virtual Interface and internal virtual PCI Bus) registers use the *PCI r3.0* Type 1 Configuration Space Header, and are mapped to the 4-KB Address space of the Port Number that is assigned as the NT Port (indicated in the **VS0 Upstream** register *NT Port* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 360h[12:8])).

To use this mechanism, use the [PCI r3.0-Compatible Configuration Mechanism](#) to program the NT Port Virtual Interface **Base Address 0** and **Base Address 1** registers (**BAR0** and **BAR1**, offsets 10h and 14h, respectively). After the NT Port Virtual Interface Memory-Mapped register Base address is Set, the Configuration Space registers are accessed, using Memory Reads and Writes to the 4-KB range, starting at offset 248 KB (3_E000h, NT Port Virtual Interface) and offset 252 KB (3_F000h, NT Port Link Interface).

Figure 15-1. NT Mode Configuration Register Mapping to Memory-Mapped BAR

Note: [Table 13-5](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

15.3.3 Device-Specific Cursor Mechanism

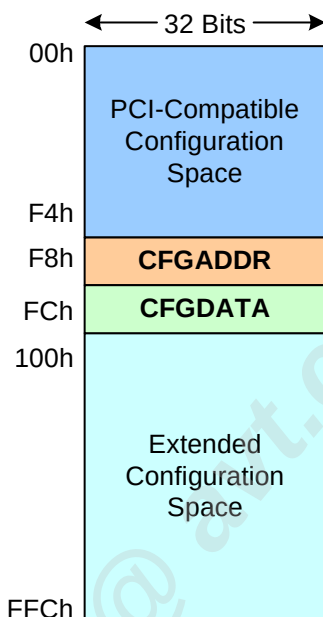
The Device-Specific Cursor mechanism is provided for use in development systems that can only generate *PCI r3.0* Configuration cycles (*that is*, the system cannot use either the [Device-Specific Memory-Mapped Configuration Mechanism](#), nor generate Extended Configuration Requests to access the Extended Configuration Space).

In [Figure 15-2](#), the software uses the **Configuration Address Window** (CFGADDR) register (offset **F8h**) to point to the NT Port Virtual or Link Interface Configuration Space registers, including the PCI Express Extended Configuration Space registers (offsets 100h through FFFh).

Software uses the **Configuration Data Window** (CFGDATA) register (offset **FCh**) to write to or read from the selected Configuration Space registers.

Refer to [Section 15.9](#), “NT Port Virtual Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh),” for the register descriptions.

Figure 15-2. Configuration Space View



15.4 Register Descriptions

The remainder of this chapter details the NT Port Virtual Interface registers, including:

- Bit/field names
- Description of register functions in the NT Port Virtual and Link Interfaces
- Type (*such as* RW or HwInit; refer to [Table 13-4, “Register Types, Grouped by User Accessibility,”](#) for Type descriptions)
- Whether the power-on/reset value can be modified, by way of the PEX 8724 serial EEPROM and/or I²C/SMBus Initialization feature
- Default power-on/reset value

15.5 NT Port Virtual Interface PCI-Compatible Type 0 Configuration Header Registers (Offsets 00h – 3Ch)

This section details the NT Port Virtual Interface PCI-Compatible Type 0 Configuration Header registers. Table 15-2 defines the register map.

Table 15-2. NT Port Virtual Interface PCI-Compatible Type 0 Configuration Header Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Device ID																Vendor ID																00h				
PCI Status																PCI Command																04h				
PCI Class Code																		PCI Revision ID														08h				
PCI BIST (Not Supported)				PCI Header Type												Master Latency Timer (Not Supported)										Cache Line Size										0Ch
Base Address 0																																10h				
Base Address 1																																14h				
Base Address 2																																18h				
Base Address 3																																1Ch				
Base Address 4																																20h				
Base Address 5																																24h				
Reserved																																28h				
Subsystem ID																Subsystem Vendor ID																2Ch				
NT Port Expansion ROM Base Address																																30h				
Reserved																		Capability Pointer (40h)														34h				
Reserved																																38h				
Max_Lat (Reserved)								Min_Gnt (Reserved)								PCI Interrupt Pin								PCI Interrupt Line								3Ch				

Register 15-1. 00h PCI Configuration ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Vendor ID Identifies the device manufacturer. Defaults to the PCI-SIG-issued Vendor ID of PLX (10B5h), if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	10B5h
31:16	Device ID Identifies the particular device. Defaults to the PLX part number for the PEX 8724, if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	87B0h

Register 15-2. 04h PCI Command/Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Command				
0	I/O Access Enable The PEX 8724 does not claim I/O resources, nor does it forward I/O transactions through the NT Port. The value of this register is “Don’t Care.”	RW	Yes	0
1	Memory Access Enable 0 = PEX 8724 ignores Memory Space Requests on the NT Port Virtual Interface 1 = PEX 8724 accepts Memory Space Requests received on the NT Port Virtual Interface	RW	Yes	0
2	Bus Master Enable Controls PEX 8724 forwarding of Memory Requests Upstream. Does not affect Message forwarding nor Completions. 0 = PEX 8724 handles Memory Requests received on the NT Port Link Interface as Unsupported Requests (URs); for Non-Posted Requests, the PEX 8724 returns a Completion with UR Completion status 1 = PEX 8724 forwards Memory Requests in the Upstream direction	RW	Yes	0
3	Special Cycle Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
4	Memory Write and Invalidate Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
5	VGA Palette Snoop <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
6	Parity Error Response Enable Controls bit 24 (<i>Master Data Parity Error Detected</i>).	RW	Yes	0
7	IDSEL Stepping/Wait Cycle Control <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
8	SERR# Enable Controls bit 30 (<i>Signaled System Error</i>). 1 = Enables reporting of Fatal and Non-Fatal errors detected by the NT Port Virtual Interface to the Root Complex	RW	Yes	0
9	Fast Back-to-Back Transactions Enable <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
10	Interrupt Disable 0 = NT Port Virtual Interface is enabled to generate INTx Interrupt Messages 1 = NT Port Virtual Interface is prevented from generating INTx Interrupt Messages	RW	Yes	0
15:11	Reserved	RsvdP	No	0-0h

Register 15-2. 04h PCI Command/Status (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Status				
18:16	Reserved	RsvdP	No	000b
19	Interrupt Status 0 = No pending INTx Interrupt Message 1 = INTx interrupt is pending internal to the NT Port Virtual Interface, –or– PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) (if enabled) is asserted	RO	No	0
20	Capability List Capability function is supported. Set, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	1
21	66 MHz Capable Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
22	Reserved	RsvdP	No	0
23	Fast Back-to-Back Transactions Capable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
24	Master Data Parity Error Detected If bit 6 (<i>Parity Error Response Enable</i>) is Set, the NT Port Virtual Interface Sets this bit when the NT Port: - Forwards the poisoned Transaction Layer Packet (TLP) Write Request from the NT Port Link-to-Virtual Interface, –or– - Receives a Completion marked as poisoned on the NT Port Virtual Interface If the <i>Parity Error Response Enable</i> bit is Cleared, the PEX 8724 never Sets this bit. This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
26:25	DEVSEL# Timing Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00b
27	Signaled Target Abort The NT Port Virtual Interface Sets this bit when any of the following conditions are met: - NT Port Virtual Interface receives a Completion (from a Transparent Port) that has a Completion status of Completer Abort (CA), –or– - NT Port Virtual Interface receives a Memory Request targeting a PEX 8724 register, and the Payload Length (indicated within the Memory Request Header) is greater than 1 DWord - NT Port Virtual Interface receives a Memory Request targeting a PEX 8724 register address within a non-existent Port - NT Port Virtual Interface receives a Memory Write Request targeting enabled NT Port Expansion ROM Address space (NT Port Expansion ROM Base Address Register (BAR), offset 30h) Note: When Set during a forwarded Completion, the Uncorrectable Error Status register <i>Completer Abort Status</i> bit (offset FB8h[15]) is not Set.	RW1C	Yes	0

Register 15-2. 04h PCI Command/Status (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
28	Received Target Abort Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
29	Received Master Abort Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
30	Signaled System Error If bit 8 (<i>SERR# Enable</i>) is Set, the NT Port Virtual Interface Sets this bit when transmitting an ERR_FATAL or ERR_NONFATAL Message to the Upstream Port.	RW1C	Yes	0
31	Detected Parity Error This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility. 1 = NT Port Virtual Interface received a Poisoned TLP, regardless of the bit 6 (<i>Parity Error Response Enable</i>) state	RW1C	Yes	0

Register 15-3. 08h PCI Class Code and Revision ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
PCI Revision ID					
7:0	Revision ID Unless overwritten by the serial EEPROM, returns the Silicon Revision (BAh or CAh), the PLX-assigned Revision ID for this version of the PEX 8724. The PEX 8724 Serial EEPROM register Initialization capability is used to replace the PLX Revision ID with another Revision ID.	RO	Yes	BAh	CAh
PCI Class Code				068000h	
15:8	Register-Level Programming Interface	RO	Yes	00h	
23:16	Sub-Class Code Other bridge devices.	RO	Yes	80h	
31:24	Base Class Code Bridge devices.	RO	Yes	06h	

Register 15-4. 0Ch Miscellaneous Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Cache Line Size				
7:0	Cache Line Size System Cache Line Size. Implemented as a RW field for Conventional PCI compatibility purposes and does not impact PEX 8724 functionality.	RW	Yes	00h
Master Latency Timer				
15:8	Master Latency Timer <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00h
PCI Header Type				
22:16	Configuration Layout Type Type 0 Configuration Header for the NT Port.	RO	No	00h
23	Multi-Function Device 0 = Single-function device 1 = Indicates multiple (up to eight) functions (logical devices), each containing its own, individually addressable Configuration Space, 256 DWords in size	RO	No	0
PCI BIST				
31:24	PCI BIST <i>Not supported</i> Built-In Self-Test (BIST) Pass or Fail.	RsvdP	No	00h

**Register 15-5. 10h Base Address 0
(NT Port Virtual Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: By default, NT Port Virtual Interface BAR0 is enabled and BAR1 is disabled, to provide a 32-bit BAR0 for register access. BAR1 can be enabled (by serial EEPROM and/or I ² C/SMBus), to provide a 64-bit BAR0/1 , by programming the NT Port Virtual Interface BAR0/1 Setup register BAR0/1 Enable field (NT Port Virtual Interface, offset D0h[1:0]) to 11b (which enables both BAR0 and BAR1).				
0	Memory Space Indicator When enabled, the Base Address register maps PEX 8724 Port Configuration registers into Memory space. Note: Hardwired to 0.	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are Reserved .	ROS	Yes	00b
3	Prefetchable 0 = Base Address register maps the PEX 8724 Port Configuration registers into Non-Prefetchable Memory space	ROS	Yes	0
17:4	Reserved	RsvdP	No	0-0h
31:18	Base Address 0 256-KB-aligned Base address used for Memory-Mapped access to the 256-KB block of all PEX 8724 registers (4 KB, per Port).	RW	Yes	0-0h

**Register 15-6. 14h Base Address 1
(NT Port Virtual Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Upper 32-Bit Address for Memory-Mapped BAR For 64-bit addressing (BAR0/1), Base Address 1 (BAR1) extends Base Address 0 (BAR0) to provide the upper 32 Address bits when the Base Address 0 register Memory Map Type field (offset 10h[2:1]) is programmed to 10b.	RW	Yes	0000_0000h
	RO when the Base Address 0 (BAR0) register is not enabled as a 64-bit BAR (Memory Map Type field (offset 10h[2:1]) is not programmed to 10b).	RO	Yes	0000_0000h

Register 15-7. 18h Base Address 2
(NT Port Virtual Interface Memory Space)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	Memory Space Indicator 0 = Implemented as a Memory BAR	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are <i>Reserved</i> .	ROS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	ROS	Yes	0
19:4	<i>Reserved</i>	RsvdP	No	0_000h
31:20	Base Address 2 Resolution is 1 MB.	RW	Yes	000h

Register 15-8. 1Ch Base Address 3
(NT Port Virtual Interface Memory Space)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Note: This register has RW privilege if BAR2/3 is configured as a 64-bit BAR (Base Address 2 register Memory Map Type field (NT Port Virtual Interface, offset 18h[2:1]), is programmed to 10b).					
0	Memory Space Indicator BAR3 can be used as an independent 32-bit only BAR, or as the upper 32 bits of 64-bit BAR2/3 . 0 = Memory BAR – only value supported	Offset 18h[2:1]=00b	RO	No	0
		Offset 18h[2:1]=10b	RW	Yes	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space All other encodings are Reserved .	Offset 18h[2:1]=00b	RO	No	00b
		Offset 18h[2:1]=10b	RW	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	Offset 18h[2:1]=00b	ROS	No	0
		Offset 18h[2:1]=10b	RW	Yes	0
19:4	Reserved	Offset 18h[2:1]=00b	RsvdP	No	0_000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset 18h[2:1]=10b	RW	Yes	0_000h
31:20	Base Address 3	RW	Yes	000h	

**Register 15-9. 20h Base Address 4
(NT Port Virtual Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	Memory Space Indicator 0 = Memory BAR – only value supported	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are Reserved .	ROS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	ROS	Yes	0
19:4	Reserved	RsvdP	No	0_000h
31:20	Base Address 4	RW	Yes	000h

**Register 15-10. 24h Base Address 5
(NT Port Virtual Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Note: This register has RW privilege if BAR4/5 is configured as a 64-bit BAR (Base Address 4 register Memory Map Type field (NT Port Virtual Interface, offset 20h[2:1]), is programmed to 10b).					
0	Memory Space Indicator BAR5 can be used as an independent 32-bit only BAR, or as the upper 32 bits of 64-bit BAR4/5 . 0 = Memory BAR – only value supported	Offset 20h[2:1]=00b	RO	No	0
		Offset 20h[2:1]=10b	RW	Yes	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space All other encodings are Reserved .	Offset 20h[2:1]=00b	RO	No	00b
		Offset 20h[2:1]=10b	RW	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	Offset 20h[2:1]=00b	ROS	Yes	0
		Offset 20h[2:1]=10b	RW	Yes	0
19:4	Reserved	Offset 20h[2:1]=00b	RsvdP	No	0_000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset 20h[2:1]=10b	RW	Yes	0_000h
31:20	Base Address 5	RW	Yes	000h	

Register 15-11. 2Ch Subsystem ID and Subsystem Vendor ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Subsystem Vendor ID				
15:0	Subsystem Vendor ID Identifies the device manufacturer. Defaults to the PCI-SIG-issued Vendor ID of PLX (10B5h), if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	10B5h
Subsystem ID				
31:16	Subsystem ID Identifies the particular device. Defaults to the PLX part number for the PEX 8724, if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	87B0h

Register 15-12. 30h NT Port Expansion ROM Base Address

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Note: Expansion ROM can be enabled in either the NT Port Virtual or Link Interface, but not both concurrently. Expansion ROM is enabled, by default, in the NT Port Link Interface (Ingress Chip Control register NT Port Expansion ROM Virtual Side bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[0]) is Cleared). Expansion ROM can be disabled, by Setting the Ingress Control register Disable NT Port Expansion ROM BAR bit (Base mode – All Ports; Virtual Switch mode – VS Upstream Port, offset F60h[15]).					
0	Expansion ROM Enable 0 = NT Port Virtual Interface Expansion ROM is disabled	Upstream Port(s), offset F60h[15]=1	RsvdP	No	0
	1 = NT Port Virtual Interface Expansion ROM is enabled, and NT Port Link Interface Expansion ROM is disabled	Upstream Port(s), offset F60h[15]=0	RW	Yes	0
13:1	Reserved		RsvdP	No	0-0h
31:14	Expansion ROM Base Address If the Serial EEPROM Clock Frequency register Expansion ROM Size bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[16]) value is 0, the NT Port Expansion ROM size is 16 KB (default value is FFFF_C001h). Bit 14 is RW.	Upstream Port(s), offset F60h[15]=1	RsvdP	No	0-0h
	If the Expansion ROM Size bit value is 1, the NT Port Expansion ROM size is 32 KB (default value is FFFF_8001h). Bit 14 is RO.	Upstream Port(s), offset F60h[15]=0	RW	Yes	0-0h

Register 15-13. 34h Capability Pointer

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability Pointer Default 40h points to the PCI Power Management Capability structure.	RO	Yes	40h
31:8	Reserved	RsvdP	No	0000_00h

Register 15-14. 3Ch PCI Interrupt

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	PCI Interrupt Line The Interrupt Line Routing value communicates interrupt line routing information. Values within this register are programmed by system software and are system architecture-specific. The value is used by device drivers and operating systems.	RW	Yes	00h
15:8	PCI Interrupt Pin Identifies the Conventional PCI Interrupt Message(s) the device (or device function) uses. Only value 00h or 01h is allowed in the PEX 8724. 00h = Device does not use Conventional PCI Interrupt Message(s) 01h, 02h, 03h, and 04h = Maps to Conventional PCI Interrupt Messages for INTA#, INTB#, INTC#, and INTD#, respectively	RO	Yes	01h
23:16	Min_Gnt Reserved Minimum Grant. Does not apply to PCI Express.	RsvdP	No	00h
31:24	Max_Lat Reserved Maximum Latency. Does not apply to PCI Express.	RsvdP	No	00h

15.6 NT Port Virtual Interface PCI Power Management Capability Registers (Offsets 40h – 44h)

This section details the NT Port Virtual Interface PCI Power Management Capability registers. Table 15-3 defines the register map.

Table 15-3. NT Port Virtual Interface PCI Power Management Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0														Offsets					
PCI Power Management Capability																Next Capability Pointer (48h)								Capability ID (01h)								40h			
PCI Power Management Data								PCI Power Management Control/Status Bridge Extensions <i>(Reserved)</i>								PCI Power Management Status and Control																44h			

Register 15-15. 40h PCI Power Management Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability ID Default = 01h – only value allowed.	RO	Yes	01h
15:8	Next Capability Pointer Default 48h points to the MSI Capability structure.	RO	Yes	48h
18:16	Version Default = 011b – only value allowed.	RO	Yes	011b
19	PME Clock Power Management Event (PME) clock. Does not apply to PCI Express. Returns a value of 0.	RsvdP	No	0
20	Reserved	RsvdP	No	0
21	Device-Specific Initialization 0 = Device-Specific Initialization is <i>not</i> required	RO	Yes	0
24:22	AUX Current The PEX 8724 does <i>not support</i> PME generation from the D3cold Device PM state; therefore, the serial EEPROM value for this field should be 000b.	RO	Yes	000b
25	D1 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D1 Device PM state	RsvdP	No	0
26	D2 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D2 Device PM state	RsvdP	No	0
31:27	PME Support The default value is applied to bits [31, 30, and 27] only. PME Messages are disabled, by default.	RO	Yes	0000_0b

Register 15-16. 44h PCI Power Management Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Status and Control				
1:0	Power State Used to determine the Port's current Device PM state, and to Set the Port into a new Device PM state. 00b = D0 01b = D1 – <i>Not supported</i> 10b = D2 – <i>Not supported</i> 11b = D3hot If software attempts to write an unsupported state to this field, the Write operation completes normally; however, the data is discarded and no state change occurs.	RW	Yes	00b
2	Reserved	RsvdP	No	0
3	No Soft Reset 1 = Devices transitioning from the D3hot to D0 Device PM state, because of Power State commands, do not perform an internal reset	ROS	Yes	1
7:4	Reserved	RsvdP	No	0h
8	PME Enable Tied to 0, because the PEX 8724 does <i>not</i> generate PME in PCI Express mode.	RsvdP	No	0
12:9	Data Select Initially writable by serial EEPROM and/or I ² C only ^a . This Configuration Space register (CSR) access privilege changes to RW after a Serial EEPROM and/or I ² C Write occurs to this register. Selects the Power Number at a particular state (for field [14:13] (<i>Data Scale</i>)), which is reflected in the PCI Power Management Data register <i>Data</i> field (offset 44h[31:24]). 0h = D0 Device PM state power consumed 3h = D3hot Device PM state power consumed 4h = D0 Device PM state power dissipated 7h = D3hot Device PM state power dissipated All other encodings are Reserved .	RO	Yes	0h
14:13	Data Scale Writable by serial EEPROM and/or I ² C only ^a . Indicates the scaling factor to be used when interpreting the PCI Power Management Data register <i>Data</i> field (offset 44h[31:24]) value. The value and meaning of the <i>Data Scale</i> field varies, depending upon which data value is selected by field [12:9] (<i>Data Select</i>). There are four internal <i>Data Scale</i> fields (one each, per <i>Data Select</i> values 0h, 3h, 4h, and 7h). For other <i>Data Select</i> values, the <i>Data Scale</i> value returned is 0h.	RO	Yes	00b
15	PME Status 0 = PME is not being generated by the NT Port	RsvdP	No	0

Register 15-16. 44h PCI Power Management Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Control/Status Bridge Extensions				
21:16	Reserved	RsvdP	No	0-0h
22	B2/B3 Support Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
23	Bus Power/Clock Control Enable Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
PCI Power Management Data				
31:24	Data Writable by serial EEPROM and/or I ² C only ^a . There are four supported <i>Data Select</i> values (0h, 3h, 4h, and 7h). For other <i>Data Select</i> values, the <i>Data</i> value returned is 00h. Selected by the PCI Power Management Status and Control register <i>Data Select</i> field (offset 44h[12:9]).	RO	Yes	00h

- a. With no serial EEPROM nor previous I²C programming, Reads return a value of 00h for the **PCI Power Management Status and Control** register *Data Scale* and **Data** register *Data* fields (for all *Data Selects*).

15.7 NT Port Virtual Interface MSI Capability Registers (Offsets 48h – 64h)

The registers detailed in [Section 13.9, “Message Signaled Interrupt Capability Registers \(Offsets 48h – 64h\),”](#) are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-4](#) (register map), and [Register 15-17](#) through [Register 15-19](#).

Table 15-4. NT Port Virtual Interface MSI Capability Register Map^a

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
MSI Control																Next Capability Pointer (68h)								Capability ID (05h)								48h
MSI Address																																4Ch
MSI Upper Address																																50h
Reserved																MSI Data																54h
MSI Mask																																58h
MSI Status																																5Ch
Reserved																																60h – 64h

- a. Offsets 54h, 58h, and 5Ch change to 50h, 54h, and 58h, respectively, when the **MSI Control** register **MSI 64-Bit Address Capable** bit (offset 48h[23]) is Cleared.

Register 15-17. 48h MSI Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
MSI Capability Header				
7:0	Capability ID Program to 05h, as required by the <i>PCI r3.0</i> .	RO	Yes	05h
15:8	Next Capability Pointer Program to 68h, to point to the PCI Express Capability structure.	RO	Yes	68h
MSI Control				
16	MSI Enable 0 = MSIs for the NT Port Virtual Interface are disabled 1 = MSIs for the NT Port Virtual Interface are enabled, and INTx Interrupt Messages and PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output assertion are disabled	RW	Yes	0
19:17	Multiple Message Capable 000b = NT Port Virtual Interface can request only one MSI Vector 001b = NT Port Virtual Interface can request two MSI Vectors 010b = NT Port Virtual Interface can request four MSI Vectors 011b = NT Port Virtual Interface can request eight MSI Vectors All other encodings are Reserved .	RO	Yes	011b
22:20	Multiple Message Enable 000b = NT Port Virtual Interface is allocated one MSI Vector, by default. 001b = NT Port Virtual Interface is allocated two MSI Vectors. 010b = NT Port Virtual Interface is allocated four MSI Vectors. 011b = NT Port Virtual Interface is allocated eight MSI Vectors. Up to six MSI Vectors are used; the upper two MSI Vectors (having the highest values of Message Data bits [2:0]) are not used. All other encodings are Reserved . Note: This field should not be programmed with a value larger than that of field [19:17] (Multiple Message Capable). If the value of this field is larger than that of field [19:17], the Multiple Message Capable value takes effect.	RW	Yes	000b
23	MSI 64-Bit Address Capable 0 = PEX 8724 is capable of generating MSI 32-bit addresses (MSI Address register, offset 4Ch, is the Message address) 1 = PEX 8724 is capable of generating MSI 64-bit addresses (MSI Address register, offset 4Ch, is the lower 32 bits of the Message address, and MSI Upper Address register, offset 50h, is the upper 32 bits of the Message address)	RO	Yes	1
24	Per Vector Masking Capable 0 = PEX 8724 does not have Per Vector Masking capability 1 = PEX 8724 has Per Vector Masking capability	RO	Yes	1
31:25	Reserved	RsvdP	No	0-0h

Register 15-18. 58h MSI Mask

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
NT Port Virtual Interface interrupt sources are grouped into four categories – Link State-/PM-triggered interrupt events, Device-Specific NT-Link Port events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts. The quantity of MSI Vectors that are generated is determined by the MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (NT Port Virtual Interface, offset 48h[19:17 and 22:20], respectively): • If one MSI Vector is enabled (default), all four interrupt events are combined into a single Vector • If two MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Link State-/PM-triggered interrupt events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts – Vector[1] Device-Specific NT-Link Port events • If four MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Link State-/PM-triggered interrupt events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts – Vector[1] Device-Specific NT-Link Port events – Vector[2] GPIO-generated interrupts – Vector[3] NT-Virtual Doorbell-generated interrupts Notes: The offset for this register changes from 58h, to 54h, when the MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be used to mask their respective MSI Status register bits (offset 5Ch).					
0	MSI Mask for Link State Events MSI mask for Power Management- or Link State-triggered interrupt events.	Offset 48h[22:20]≥010b	RW	Yes	0
	MSI Mask for Shared Interrupt Sources MSI mask for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	Offset 48h[22:20]≤001b	RW	Yes	0
1	MSI Mask for Device-Specific NT-Link Port Events MSI mask for Device-Specific NT-Link Port event-generated interrupts. Enables MSIs for the following NT-Link Port events, defined in the Link Error Status Virtual and Link Error Mask Virtual registers (NT Port Virtual Interface, offsets FE0h and FE4h, respectively): • NT-Link Port Correctable error (NT Port Link Interface, offset FC4h) • NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h) • NT-Link Port Data Link Layer State change • NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message	Offset 48h[22:20]≥001b	RW	Yes	0
	Reserved	Offset 48h[22:20]=000b	RsvdP	No	0

Register 15-18. 58h MSI Mask (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
2	MSI Mask for GPIO-Generated Interrupts	Offset 48h[22:20]≥010b	RW	Yes	0
	Reserved	Offset 48h[22:20]≤001b	RsvdP	No	0
3	MSI Mask for NT-Virtual Doorbell-Generated Interrupts Refer to NT Port registers located at offsets C4Ch through C58h.	Offset 48h[22:20]≥010b	RW	Yes	0
	Reserved	Offset 48h[22:20]≤001b	RsvdP	No	0
31:4	Reserved		RsvdP	No	0000_000h

Register 15-19. 5Ch MSI Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
NT Port Virtual Interface interrupt sources are grouped into four categories – Link State-/PM-triggered interrupt events, Device-Specific NT-Link Port events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts. The quantity of MSI Vectors that are generated is determined by the MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (NT Port Virtual Interface, offset 48h[19:17 and 22:20], respectively): • If one MSI Vector is enabled (default), all four interrupt events are combined into a single Vector • If two MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Link State-/PM-triggered interrupt events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts – Vector[1] Device-Specific NT-Link Port events • If four MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Link State-/PM-triggered interrupt events, GPIO-generated interrupts, and NT-Virtual Doorbell-generated interrupts – Vector[1] Device-Specific NT-Link Port events – Vector[2] GPIO-generated interrupts – Vector[3] NT-Virtual Doorbell-generated interrupts Notes: The offset for this register changes from 5Ch, to 58h, when the MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be masked by their respective MSI Mask register bits (offset 58h).					
0	MSI Pending Status for Link State Events MSI pending status for Power Management- or Link State-triggered interrupt events.	Offset 48h[22:20]≥010b	RO	No	0
	MSI Pending Status for Shared Interrupt Sources MSI pending status for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	Offset 48h[22:20]≤001b	RO	No	0
1	MSI Pending Status for Device-Specific Error Triggered Event Indicates the MSI pending status for the Device-Specific NT-Link Port event-generated interrupts defined in the Link Error Status Virtual and Link Error Mask Virtual registers (NT Port Virtual Interface, offsets FE0h and FE4h, respectively): • NT-Link Port Correctable error (NT Port Link Interface, offset FC4h) • NT-Link Port Uncorrectable error (NT Port Link Interface, offset FB8h) • NT-Link Port Data Link Layer State change • NT-Link Port received (and consequently dropped) a Fatal or Non-Fatal Error Message	Offset 48h[22:20]≥001b	RO	No	0
	Reserved	Offset 48h[22:20]=000b	RsvdP	No	0

Register 15-19. 5Ch MSI Status (Cont.)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
2	MSI Pending Status for GPIO-Generated Interrupts	Offset 48h[22:20]≥010b	RO	No	0
	Reserved	Offset 48h[22:20]≤001b	RsvdP	No	0
3	MSI Pending Status for NT-Virtual Doorbell-Generated Interrupts Refer to NT Port registers located at offsets C4Ch through C58h.	Offset 48h[22:20]≥010b	RO	No	0
	Reserved	Offset 48h[22:20]≤001b	RsvdP	No	0
31:4	Reserved		RsvdP	No	0000_000h

15.8 NT Port Virtual Interface PCI Express Capability Registers (Offsets 68h – A0h)

The registers detailed in [Section 13.10, “PCI Express Capability Registers \(Offsets 68h – A0h\),”](#) are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-5](#) (register map; offsets 7Ch and 80h are **Reserved**), and [Register 15-20](#) through [Register 15-28](#).

Table 15-5. NT Port Virtual Interface PCI Express Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
PCI Express Capability	Next Capability Pointer (C8h)	Capability ID (10h)
Device Capability		6Ch
Device Status	Not Supported/Reserved	Device Control
Link Capability		74h
Link Status	Reserved	Link Control
Reserved		7Ch – 88h
Device Capability 2		8Ch
Device Status 2 (Reserved)	Device Control 2	
Reserved		Link Capability 2
Link Status 2	Link Control 2	
Reserved		9Ch – A0h

Register 15-20. 68h PCI Express Capability List and Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Express Capability List				
7:0	Capability ID Program to 10h, by default, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	10h
15:8	Next Capability Pointer Program to C8h, to point to the Vendor-Specific Capability 3 structure.	RO	Yes	C8h
PCI Express Capability				
19:16	Capability Version The NT Port Virtual Interface programs this field to 2h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	2h
23:20	Device/Port Type Default = PCI Express endpoint device.	RO	Yes	0h
24	Slot Implemented <i>Not valid for PCI Express endpoint devices</i>	RsvdP	No	0
29:25	Interrupt Message Number The serial EEPROM writes 00_000b, because the Base Message and MSI Messages are the same.	RO	Yes	00_000b
31:30	Reserved	RsvdP	No	00b

Register 15-21. 6Ch Device Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	Maximum Payload Size Supported <i>Note:</i> The default Maximum Payload Size supported is based upon the STRAP_STNx_PORTCFGx 3-state inputs. If the serial EEPROM or I²C/SMBus changes the Port configuration (from the strapped value), it must also program the value of this field, accordingly (with the value dependent upon the Station having the most enabled Ports). The value for all Ports is determined by the Station having the most enabled Ports: - One or two enabled Ports = 100b - Three enabled Ports = 011b - Four enabled Ports = 010b 000b = NT Port Virtual Interface supports a 128-byte maximum payload 001b = NT Port Virtual Interface supports a 256-byte maximum payload 010b = NT Port Virtual Interface supports a 512-byte maximum payload 011b = NT Port Virtual Interface supports a 1,024-byte maximum payload 100b = NT Port Virtual Interface supports a 2,048-byte maximum payload No other encodings are supported.	HwInit	Yes	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
4:3	Phantom Functions Supported <i>Not supported</i>	RO	Yes	00b
5	Extended Tag Field Supported 0 = Maximum <i>Tag</i> field is 5 bits 1 = Maximum <i>Tag</i> field is 8 bits	RO	Yes	0
8:6	Endpoint L0s Acceptable Latency 111b = No Limit	RO	Yes	111b
11:9	Endpoint L1 Acceptable Latency 111b = No Limit	RO	Yes	111b
14:12	Reserved	RsvdP	No	000b
15	Role-Based Error Reporting	RO	Yes	1

Register 15-21. 6Ch Device Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
17:16	Reserved	RsvdP	No	00b
25:18	Captured Slot Power Limit Value For the NT Port Virtual Interface, the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [27:26] (<i>Captured Slot Power Limit Scale</i>).	RO	Yes	00h
27:26	Captured Slot Power Limit Scale For the NT Port Virtual Interface, the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [25:18] (<i>Captured Slot Power Limit Value</i>). 00b = 1.0 01b = 0.1 10b = 0.01 11b = 0.001	RO	Yes	00b
31:28	Reserved	RsvdP	No	0h

Register 15-22. 70h Device Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Control				
0	Correctable Error Reporting Enable 0 = Disables 1 = Enables the NT Port Virtual Interface to report Correctable errors to the Virtual side Host	RW	Yes	0
1	Non-Fatal Error Reporting Enable 0 = Disables 1 = Enables the NT Port Virtual Interface to report Non-Fatal errors to the Virtual side Host	RW	Yes	0
2	Fatal Error Reporting Enable 0 = Disables 1 = Enables the NT Port Virtual Interface to report Fatal errors to the Virtual side Host	RW	Yes	0
3	Unsupported Request Reporting Enable 0 = Disables 1 = Enables the NT Port Virtual Interface to report UR errors to the Virtual side Host	RW	Yes	0
4	Enable Relaxed Ordering <i>Not supported</i>	RW	Yes	1
7:5	Maximum Payload Size The NT Port Virtual Interface power-on/reset value is 000b, to support a Maximum Payload Size of 128 bytes. Software can change this field to configure the NT Port Virtual Interface to support other Payload sizes; however, software cannot change this field to a value larger than that indicated by the Device Capability register <i>Maximum Payload Size Supported</i> field (offset 6Ch[2:0]), for the NT Port Virtual and Link Interfaces. (Requester and Completer domains must possess the same Maximum Payload Size.) 000b = NT Port Virtual Interface supports a 128-byte maximum payload 001b = NT Port Virtual Interface supports a 256-byte maximum payload 010b = NT Port Virtual Interface supports a 512-byte maximum payload 011b = NT Port Virtual Interface supports a 1,024-byte maximum payload 100b = NT Port Virtual Interface supports a 2,048-byte maximum payload No other encodings are supported. Note: Software must halt all transactions through the NT Port before changing this field.	RW	Yes	000b

Register 15-22. 70h Device Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Extended Tag Field Enable <i>Not supported</i>	RsvdP	No	0
9	Phantom Functions Enable <i>Not supported</i>	RsvdP	No	0
10	AUX Power PM Enable <i>Not supported</i>	RsvdP	No	0
11	Enable No Snoop <i>Not supported</i>	RW	Yes	1
14:12	Maximum Read Request Size <i>Not supported</i>	RsvdP	No	000b
15	Reserved	RsvdP	No	0
Device Status				
16	Correctable Error Detected 0 = NT Port Virtual Interface did not detect a Correctable error 1 = NT Port Virtual Interface detected a Correctable error, regardless of the bit 0 (<i>Correctable Error Reporting Enable</i>) state	RW1C	Yes	0
17	Non-Fatal Error Detected 0 = NT Port Virtual Interface did not detect a Non-Fatal error 1 = NT Port Virtual Interface detected a Non-Fatal error, regardless of the bit 1 (<i>Non-Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
18	Fatal Error Detected 0 = NT Port Virtual Interface did not detect a Fatal error 1 = NT Port Virtual Interface detected a Fatal error, regardless of the bit 2 (<i>Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
19	Unsupported Request Detected 0 = NT Port Virtual Interface did not detect a UR 1 = NT Port Virtual Interface detected a UR, regardless of the bit 3 (<i>Unsupported Request Reporting Enable</i>) state	RW1C	Yes	0
20	AUX Power Detected <i>Not supported</i>	RsvdP	No	0
21	Transactions Pending <i>Not supported</i> Because the NT Port is a bridging device, it does not track Completion for the corresponding Non-Posted transactions. Therefore, the NT Port Virtual Interface does not implement this bit.	RsvdP	No	0
31:22	Reserved	RsvdP	No	0-0h

Register 15-23. 74h Link Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
Note: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STN _x _PORTCFG _x input states and/or serial EEPROM programming. The table also lists the relationship between the Ports and Lanes (Link widths).					
3:0	Maximum Link Speed Indicates the NT Port Virtual Interface’s maximum Link speed. The encoding is the binary value of the bit location in the Link Capability 2 register <i>Supported Link Speeds Vector</i> field (NT Port Virtual Interface, offset 94h[7:1]) that corresponds to the maximum Link speed. <i>For example</i> , the default value 3h indicates that the maximum Link speed is that which corresponds to bit 3 in the <i>Supported Link Speeds Vector</i> field, which is 8.0 GT/s. The default value for all Ports can be globally changed to 2h (Gen 2 maximum speed) or 1h (Gen 1 maximum speed), according to the STRAP_GEN1_GEN2 input state (Z = 2h (Gen 2), or 0 = 1h (Gen 1)). 1h = Gen 1 (2.5 GT/s) 2h = Gen 2 (5.0 GT/s) 3h = Gen 3 (8.0 GT/s)	RO	Yes	3h	
9:4	Maximum Link Width The PEX 8724 maximum Link width is x16 = 01_0000b. Actual maximum Link width is defined by the STRAP_STN _x _PORTCFG _x 3-state inputs. 00_0000b = Reserved 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are not supported .	Refer to Default value	No	Defined by STRAP_STN _x _PORTCFG _x input states, or programmed by serial EEPROM value for the Port Configuration -related registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 300h through 308h)	
11:10	Active State Power Management (ASPM) Support Active State Link PM support. Indicates the level of ASPM supported by the NT Port Virtual Interface. 00b = Reserved 01b = ASPM L0s Link PM state entry is supported 10b = ASPM L1 Link PM state is supported (Silicon Revision CA only) 11b = ASPM L0s and L1 Link PM states are supported	RO	Yes	11b	10b

Register 15-23. 74h Link Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
14:12	L0s Exit Latency Indicates the ASPM L0s Link PM state exit latency for the Port's PCI Express Link, when the state is supported and enabled. The value reported indicates the length of time this Port requires to complete the ASPM L0s to L0 Link PM state transition. If the ASPM L0s Link PM state is not supported, the value of this field is undefined; however, for the recommended value, refer to the Implementation Note in the <i>PCI Express Base r3.0</i>, Section 5.4.1.1, "Potential Issues with Legacy Software When L0s is Not Supported." 000b = Less than 64 ns 001b = 64 ns to less than 128 ns 010b = 128 ns to less than 256 ns 011b = 256 ns to less than 512 ns 100b = 512 ns to less than 1 μs 101b = 1 μs to less than 2 μs 110b = 2 to 4 μs (default) 111b = More than 4 μs	RO	No	110b	
17:15	L1 Exit Latency Indicates the ASPM L1 Link PM state exit latency for the Port's PCI Express Link. The value reported indicates the length of time this Port requires to complete the ASPM L1 to L0 Link PM state transition. If the L0 Link PM state is not supported, the value of this field is undefined. 000b = Less than 1 μs 001b = 1 μs to less than 2 μs (5.0 GT/s) 010b = 2 μs to less than 4 μs (default, 2.5 GT/s) 011b = 4 μs to less than 8 μs 100b = 8 μs to less than 16 μs 101b = 16 μs to less than 32 μs 110b = 32 to 64 μs 111b = More than 64 μs Note: The NT Port Virtual Interface never enters the ASPM L1 Link PM state, because there is no physical Link attached to it.	RO	Yes	010b	
18	Clock Power Management	RO	Yes	0	
21:19	Reserved	RsvdP	No	000b	
22	ASPM Optionality Compliance	HwInit	Yes	1	
23	Reserved	RsvdP	No	0	

Register 15-23. 74h Link Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision																									
				BA	CA																								
31:24	NT Port Number The NT Port Number value is determined by the STRAP_NT_UPSTRM_PORTSEL[2:0] input states, combined with the VS0 Upstream register <i>NT Port</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 360h[12:8]) value.	Refer to Default value	No	Defined by STRAP_NT_UPSTRM_PORTSEL [2:0] input states																									
	<table> <tr> <th>STRAP_NT_UPSTRM_PORTSEL[2:0] Input States</th> <th>Offset 360h[12:8] Value</th> <th>Port Number</th> </tr> <tr> <td>000</td> <td>0_0000b</td> <td>0</td> </tr> <tr> <td>00Z</td> <td>0_0001b</td> <td>1</td> </tr> <tr> <td>001</td> <td>0_0010b</td> <td>2</td> </tr> <tr> <td>0Z0</td> <td>0_0011b</td> <td>3</td> </tr> <tr> <td>011</td> <td>0_1000b</td> <td>8</td> </tr> <tr> <td>Z00</td> <td>0_1001b</td> <td>9</td> </tr> <tr> <td>Z0Z</td> <td>0_1010b</td> <td>10</td> </tr> </table>					STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Offset 360h[12:8] Value	Port Number	000	0_0000b	0	00Z	0_0001b	1	001	0_0010b	2	0Z0	0_0011b	3	011	0_1000b	8	Z00	0_1001b	9	Z0Z	0_1010b	10
	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States					Offset 360h[12:8] Value	Port Number																						
	000					0_0000b	0																						
	00Z					0_0001b	1																						
	001					0_0010b	2																						
	0Z0					0_0011b	3																						
	011					0_1000b	8																						
	Z00					0_1001b	9																						
	Z0Z					0_1010b	10																						
All other encodings are Reserved .																													

Register 15-24. 78h Link Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Control				
1:0	Active State Power Management (ASPM) Control The NT Port Virtual Interface ignores this register value, because no external Port connection exists.	RW	Yes	00b
2	Reserved	RsvdP	No	0
3	Read Request Return Parameter Control Read Request Return Parameter “R” control. Read Completion Boundary (RCB).	RO	Yes	0
4	Link Disable Reserved for the NT Port Virtual Interface.	RsvdP	No	0
5	Retrain Link Reserved for the NT Port Virtual Interface.	RsvdP	No	0
6	Common Clock Configuration The NT Port Virtual Interface ignores this register value, because no external Port connection exists.	RW	Yes	0
7	Extended Sync The NT Port Virtual Interface ignores this register value, because no external Port connection exists.	RW	Yes	0
15:8	Reserved	RsvdP	No	00h

Register 15-24. 78h Link Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Status				
19:16	Current Link Speed Indicates the negotiated Link speed of the Port's PCI Express Link. The encoding is the binary value of the bit location in the Link Capability 2 register <i>Supported Link Speeds Vector</i> field (NT Port Virtual Interface, offset 94h[7:1]) that corresponds to the current Link speed. <i>For example</i> , a value of 1h indicates that the current Link speed is that which corresponds to bit 1 in the <i>Supported Link Speeds Vector</i> field, which is 2.5 GT/s. If the Link is not up, the value of this field is undefined.	RO	No	1h
25:20	Negotiated Link Width Reports the Link status of the NT Port Link Interface. Dependent upon the configuration of the physical Ports. Link width is determined by the negotiated value with the attached Lane/Port. If the Link is not up, the value of this field is undefined. 00_0000b = Link is Down (default) 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are not supported .	RO	No	00_0000b
26	Reserved	RsvdP	No	0
27	Link Training Reserved for the NT Port Virtual Interface. Always read as 0.	RsvdP	No	0
28	Slot Clock Configuration Because there is no external connection to the NT Port Virtual Interface, this bit is always Cleared, which indicates that the PEX 8724 uses an independent clock.	HwInit	Yes	0
31:29	Reserved	RsvdP	No	000b

Register 15-25. 8Ch Device Capability 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
6:0	Reserved	RsvdP	No	0-0h
7	AtomicOp 32-Bit Completer Supported 1 = Supports Atomic 32-bit Completer	ROS	Yes	1
8	AtomicOp 64-Bit Completer Supported 1 = Supports Atomic 64-bit Completer	ROS	Yes	1
9	AtomicOp 128-Bit Completer Supported 1 = Supports Atomic 128-bit Completer	ROS	Yes	1
31:10	Reserved	RsvdP	No	0-0h

Register 15-26. 90h Device Status and Control 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Control 2				
5:0	<i>Reserved</i>	RsvdP	No	0-0h
6	AtomicOp Requester Enable	RW	Yes	0
15:7	<i>Reserved</i>	RsvdP	No	0-0h
Device Status 2				
31:16	<i>Reserved</i>	RsvdP	No	0000h

Register 15-27. 94h Link Capability 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	<i>Reserved</i>	RsvdP	No	0
7:1	Supported Link Speeds Vector Supports 2.5, 5.0, and 8.0 GT/s Link speeds. Indicates the supported Link speed(s) of the associated Port. For each bit, a value of 1 indicates that the corresponding Link speed is supported; otherwise, the Link speed is <i>not supported</i> .		RO	07h
	Bit(s)	Description/Function		
	1	2.5 GT/s		
	2	5.0 GT/s		
	3	8.0 GT/s		
	7:4	<i>Reserved</i>		
31:8	<i>Reserved</i>	RsvdP	No	0000_00h

Register 15-28. 98h Link Status and Control 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Control 2				
Bits in this register have no effect and are provided for compliance testing only.				
3:0	Target Link Speed	RWS	Yes	3h
4	Enter Compliance	RWS	Yes	0
5	Hardware Autonomous Speed Disable <i>Reserved</i>	RsvdP	No	0
6	Selectable De-Emphasis <i>Reserved</i>	RsvdP	Yes	0
9:7	Transmit Margin	RWS	Yes	000b
10	Enter Modified Compliance	RWS	Yes	0
11	Compliance SOS	RWS	Yes	0
15:12	Compliance Preset/De-Emphasis	RWS	Yes	0h
Link Status 2				
16	Current De-Emphasis Level	RO	Yes	0
31:17	<i>Reserved</i>	RsvdP	No	0-0h

15.9 NT Port Virtual Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh)

This section details the NT Port Virtual Interface Vendor-Specific Capability 3 registers, which include the **Memory BARx Setup** registers and **Configuration Address** and **Data Window** registers. Table 15-6 defines the register map used by the NT Port Virtual Interface.

The Cursor Mechanism registers at offsets F8h and FCh provide a means for accessing PCI Express Extended Configuration Space registers (offsets 100h through FFFh) within the NT Port Virtual and Link Interfaces, when only standard PCI Configuration transactions (that do not support the *Extended Register Number* field within the Completion Request Header) are available. The Cursor Mechanism can generally access only those registers that are defined by the *PCI Express Base r3.0*, and not the Device-Specific registers. However, if Port 0 is the NT Port, the Cursor Mechanism in the NT Port Virtual Interface registers can also access the Device-Specific registers.

Table 15-6. NT Port Virtual Interface Vendor-Specific Capability 3 Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Reserved																Next Capability Pointer (00h)								Capability ID 3 (09h)								C8h
Reserved																																CCh
NT Port Virtual Interface BAR0/1 Setup																																D0h
NT Port Virtual Interface Memory BAR2 Setup																																D4h
NT Port Virtual Interface Memory BAR2/3 Setup																																D8h
NT Port Virtual Interface Memory BAR4 Setup																																DCh
NT Port Virtual Interface Memory BAR4/5 Setup																																E0h
Reserved																														E4h –		F4h
Configuration Address Window																Reserved																F8h
Configuration Data Window																																FCh

Register 15-29. C8h Vendor-Specific Capability 3

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability ID 3	ROS	Yes	09h
15:8	Next Capability Pointer 00h = This capability is the last capability in the Linked List	ROS	Yes	00h
31:16	<i>Reserved</i>	RsvdP	No	0000h

Register 15-30. D0h NT Port Virtual Interface BAR0/1 Setup

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	BAR0/1 Enable 00b = Disables the NT Port Virtual Interface BAR0 and BAR1 01b = <i>Reserved</i> 10b = Enables the NT Port Virtual Interface BAR0 and disables BAR1 (BAR0 is a 32-bit BAR) 11b = Enables the NT Port Virtual Interface BAR0 and BAR1 (BAR0/1 is a 64-bit BAR)	RWS	Yes	10b
2	BAR0 Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	RWS	Yes	0
31:3	<i>Reserved</i>	RsvdP	No	0-0h

Register 15-31. D4h NT Port Virtual Interface Memory BAR2 Setup

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
0	Type Selector	RsvdP	No	0	
2:1	BAR2 Type 00b = BAR2 is implemented as a 32-bit Memory BAR 10b = BAR2/3 is implemented as a 64-bit Memory BAR No other encodings are allowed.	RWS	Yes	00b	
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	RWS	Yes	0	
19:4	Reserved	RsvdP	No	0_000h	
30:20	BAR2 Size Specifies the Address Range size requested by BAR2 . 0 = Corresponding BAR2 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR2 bits are RWS bits Note: If <i>BAR[30:n]</i> is the Base field (<i>BAR</i> size is 2 ⁿ), <i>BAR[30:n]</i> should have all ones (1).	RWS	Yes	0-0h	
31	BAR2 Enable 0 = BAR2 is disabled, all BAR2 bits read 0 1 = BAR2 is enabled	Field [2:1] (<i>BAR2 Type</i>) = 00b	RWS	Yes	0
	BAR2 Size Included with field [30:20] when this BAR is used as a 64-bit BAR.	Field [2:1] (<i>BAR2 Type</i>) = 10b	RWS	Yes	0

Register 15-32. D8h NT Port Virtual Interface Memory BAR2/3 Setup

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
Note: This register has RWS privilege if BAR2/3 is configured as a 64-bit BAR (<i>NT Port Virtual Interface Memory BAR2 Setup register BAR2 Type field (NT Port Virtual Interface, offset D4h[2:1]), is programmed to 10b).</i>					
0	Type Selector		RsvdP	No	0
2:1	BAR3 Type 00b = Selects 32-bit Memory BAR (BAR3) No other encodings are allowed.		ROS	No	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable		RWS	Yes	0
19:4	Reserved	Offset D4h[2:1] (<i>BAR2 Type</i>) = 00b	RsvdP	No	0_000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset D4h[2:1] (<i>BAR2 Type</i>) = 10b	RWS	Yes	0_000h
30:20	BAR3 Size Specifies the Address Range size requested by BAR3 . 0 = Corresponding BAR3 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR3 bits are RWS bits Note: If <i>BAR[30:n]</i> is the Base field (BAR size is 2 ⁿ), <i>BAR[30:n]</i> should have all ones (1).		RWS	Yes	0-0h
31	BAR3 Enable 32-Bit BAR 0 = BAR3 is disabled 1 = BAR3 is enabled as a 32-bit BAR	Offset D4h[2:1] (<i>BAR2 Type</i>) = 00b	RWS	Yes	0
	64-Bit BAR 0 = BAR2/3 is disabled, all BAR2/3 bits read 0 1 = BAR2/3 is enabled as a 64-bit BAR	Offset D4h[2:1] (<i>BAR2 Type</i>) = 10b	RWS	Yes	0

Register 15-33. DCh NT Port Virtual Interface Memory BAR4 Setup

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
0	Type Selector	RsvdP	No	0	
2:1	BAR4 Type 00b = BAR4 is implemented as a 32-bit Memory BAR (BAR4) 10b = BAR4/5 is implemented as a 64-bit Memory BAR (BAR4/5) No other encodings are allowed.	RWS	Yes	00b	
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	RWS	Yes	0	
19:4	Reserved	RsvdP	No	0_000h	
30:20	BAR4 Size Specifies the Address Range size requested by BAR4 . 0 = Corresponding BAR4 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR4 bits are RWS bits Note: If BAR[30:n] is the Base field (BAR size is 2 ⁿ), BAR[30:n] should have all ones (1).	RWS	Yes	0-0h	
31	BAR4 Enable 0 = BAR4 is disabled, all BAR4 bits read 0 1 = BAR4 is enabled	Field [2:1] (BAR4 Type) = 00b	RWS	Yes	0
	BAR4 Size Included with field [30:20] when this BAR is used as a 64-bit BAR.	Field [2:1] (BAR4 Type) = 10b	RWS	Yes	0

Register 15-34. E0h NT Port Virtual Interface Memory BAR4/5 Setup

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
Note: This register has RWS privilege if BAR4/5 is configured as a 64-bit BAR (NT Port Virtual Interface Memory BAR4 Setup register BAR4 Type field (NT Port Virtual Interface, offset DCh[2:1]), is programmed to 10b).					
0	Type Selector		RsvdP	No	0
2:1	BAR5 Type 00b = Selects 32-bit Memory BAR (BAR5) No other encodings are allowed.		ROS	No	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable		RWS	Yes	0
19:4	Reserved	Offset DCh[2:1] (BAR4 Type) = 00b	RsvdP	No	0_000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset DCh[2:1] (BAR4 Type) = 10b	RWS	Yes	0_000h
30:20	BAR5 Size Specifies the Address Range size requested by BAR5 . 0 = Corresponding BAR5 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR5 bits are RWS bits Note: If BAR[30:n] is the Base field (BAR size is 2 ⁿ), BAR[30:n] should have all ones (1).		RWS	Yes	0-0h
31	BAR5 Enable 32-Bit BAR 0 = BAR5 is disabled 1 = BAR5 is enabled as a 32-bit BAR	Offset DCh[2:1] (BAR4 Type) = 00b	RWS	Yes	0
	64-Bit BAR 0 = BAR4/5 is disabled, all BAR4/5 bits read 0 1 = BAR4/5 is enabled as a 64-bit BAR	Offset DCh[2:1] (BAR4 Type) = 10b	RWS	Yes	0

**Register 15-35. F8h Configuration Address Window
(Device-Specific Cursor Mechanism)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	<i>Reserved</i>	RsvdP	No	0000h
25:16	Register Offset	RW	Yes	0-0h
30:26	<i>Reserved</i>	RsvdP	No	0-0h
31	Interface Select 0 = Access is to the NT Port Link Interface Type 0 Configuration Space register 1 = Access is to the NT Port Virtual Interface Type 0 Configuration Space register	RW	Yes	0

**Register 15-36. FCh Configuration Data Window
(Device-Specific Cursor Mechanism)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Register Data Software selects a register by writing into the NT Port Virtual Interface Configuration Address window, then reads from or writes to that register using this register.	RW	Yes	0000_0000h

15.10 NT Port Virtual Interface Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)

The registers detailed in [Section 13.15, “Virtual Channel Extended Capability Registers \(Offsets 148h – 1BCh\),”](#) are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-7](#) (register map; offsets 178h through 1B4h are **Reserved**), and [Register 15-37](#) through [Register 15-40](#).

Table 15-7. NT Port Virtual Interface Virtual Channel Extended Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset 2 (C34h)																Capability Version (1h)				PCI Express Extended Capability ID (0002h)																148h
Port VC Capability 1																																14Ch				
Port VC Capability 2																																150h				
Port VC Status <i>(Reserved)</i>																Port VC Control																154h				
VC0 Resource Capability																																158h				
VC0 Resource Control																																15Ch				
VC0 Resource Status																<i>Reserved</i>																160h				
<i>Reserved</i>																																164h – 1BCh				

Register 15-37. 148h Virtual Channel Extended Capability Header

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0002h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	0002h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	1h
31:20	Next Capability Offset 2 Next extended capability is the Vendor-Specific Extended Capability 2 structure, offset C34h.	RO	Yes	C34h

Register 15-38. 14Ch Port VC Capability 1

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	Extended VC Counter 000b = NT Port Virtual Interface supports only one Virtual Channel, VC0 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .	RO	Yes	000b
3	Reserved	RsvdP	No	0
6:4	Low-Priority Extended VC Counter For Strict Priority arbitration, indicates the quantity of extended Virtual Channels (VCs) (those in addition to VC0) that belong to the Low-Priority VC group for the NT Port Virtual Interface. 000b = For the NT Port Virtual Interface, only VC0 belongs to the Low-Priority VC group 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .	RO	Yes	000b
7	Reserved	RsvdP	No	0
9:8	Reference Clock Reserved	RsvdP	No	00b
11:10	Port Arbitration Table Entry Size	RO	Yes	00b
31:12	Reserved	RsvdP	No	0000_0h

Register 15-39. 158h VC0 Resource Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	Port Arbitration Capability Bit 0 = 1 – Non-configurable Round-Robin (Hardware-Fixed) arbitration Bit 1 = 1 – Weighted Round-Robin (WRR) arbitration with 64 Phases	RO	No	00b
13:2	Reserved	RsvdP	No	0-0h
14	Advanced Packet Switching	RsvdP	No	0
15	Reject Snoop Transactions Not a PCI Express switch feature; therefore, this bit is Cleared.	RsvdP	No	0
22:16	Maximum Time Slots Reserved	RsvdP	No	000_0000b
23	Reserved	RsvdP	No	0
31:24	Port Arbitration Table Offset	RO	No	00h

Register 15-40. 160h VC0 Resource Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Reserved	RsvdP	No	0000h
16	Port Arbitration Table Status Not implemented	RO	No	0
17	VC0 Negotiation Pending 0 = VC0 negotiation completed (NT Port Virtual Interface Link to the internal virtual PCI Bus is up) 1 = VC0 initialization is not complete for the NT Port Link Interface	RO	Yes	0
31:18	Reserved	RsvdP	No	0-0h

15.11 NT Port Virtual Interface Device-Specific Registers (Offsets 1C0h – A74h)

The registers detailed in [Section 13.16, “Device-Specific Registers \(Offsets 1C0h – A74h\),”](#) are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. These registers are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-8](#) (register map; offsets 1D0h through 1D8h, 200h through 270h, 290h through 46Ch, 56Ch and 570h, 600h through 68Ch, 6A0h through 6A8h, 760h through 8FCh, 900h through 93Ch, 9F0h through A08h, and A30h through A54h are **Reserved**) and [Table 15-9](#), and [Register 15-41](#).

Other NT Port Virtual Interface Device-Specific registers are detailed in:

- [Section 15.12, “NT Port Virtual Interface Device-Specific Registers \(Offsets B70h – C88h\)”](#)
- [Section 15.15, “NT Port Virtual Interface Device-Specific Registers – Link Error \(Offsets FE0h – FFCh\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 15-8. NT Port Virtual Interface Device-Specific Register Map (Offsets 1C0h – A74h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Reserved</i>		1C0h – 1D8h
NT Port Virtual Interface Device-Specific Registers – Captured Bus Number (Offsets 1DCh – 1E4h)		1DCh
		...
		1E4h
<i>Factory Test Only/Reserved</i>		1E8h – A74h

15.11.1 NT Port Virtual Interface Device-Specific Registers – Captured Bus Number (Offsets 1DCh – 1E4h)

The **Captured Bus Number** register detailed in [Section 13.16.2, “Device-Specific Registers – Captured Bus and Device Numbers \(Offsets 1DCh – 1E4h\),”](#) is also applicable to the NT Port Virtual Interface, except as defined in [Table 15-9](#) (register map; offset 1E4h is **Reserved**, rather than **Factory Test Only**), and [Register 15-41](#).

Note: *It is recommended that this register not be changed from its default value.*

Table 15-9. NT Port Virtual Interface Device-Specific Captured Bus Number Register Map

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
NT Captured Bus Number																																1DCh
Reserved																																1E0h – 1E4h

Register 15-41. 1DCh NT Captured Bus Number

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Captured Bus Number Captured Bus Number value for the NT Port Virtual Interface. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Bus Number. Note: Overwriting the Captured Bus Number value is not recommended.	RW	Yes	00h
12:8	Captured Device Number Captured Device Number value for the NT Port Virtual Interface. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Device Number. Note: Overwriting the Captured Device Number value is not recommended.	RW	Yes	0-0h
30:13	Reserved	RsvdP	No	0-0h
31	Captured BusDev Number Override 1 = Enables the Captured Bus Number and Device Number (fields [7:0 and 12:8], respectively) to be overridden	RW	Yes	0

15.12 NT Port Virtual Interface Device-Specific Registers (Offsets B70h – C88h)

The registers detailed in [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#) (for offsets B70h through C88h), are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. These registers are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-10](#) (register map; offsets B80h through C30h are **Reserved**, and offsets C34h through C88h are associated with NT Port-specific registers) and [Table 15-11](#), and [Register 15-42](#) through [Register 15-63](#).

Other NT Port Virtual Interface Device-Specific registers are detailed in:

- [Section 15.11, “NT Port Virtual Interface Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 15.15, “NT Port Virtual Interface Device-Specific Registers – Link Error \(Offsets FE0h – FFCh\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 15-10. NT Port Virtual Interface Device-Specific Register Map (Offsets B70h – C88h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset 2 (000h)																1h				PCI Express Extended Capability ID 2 (000Bh)																B70h
Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch) ^a																																...	B7Ch			
Reserved																																B80h –	C30h			
Next Capability Offset 4 (B70h)																1h				PCI Express Extended Capability ID 4 (000Bh)																C34h
NT Port Virtual Interface Device-Specific Registers – Vendor-Specific Extended Capability 4 (Offsets C34h – C88h)																																...	C88h			

- a. Register offsets B70h through B7Ch are detailed in [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#)

This section details the NT Port Virtual Interface Device-Specific, Vendor-Specific Extended Capability 4 registers, which include the **Memory BARx Address Translation**, **Doorbell**, and **Scratchpad** registers. Table 15-11 defines the register map used by the NT Port Virtual Interface.

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 Offsets

Next Capability Offset 4 (B70h)	Capability Version 4 (1h)	PCI Express Extended Capability ID 4 (000Bh)	C34h
Vendor-Specific Header 4			C38h
Memory BAR2 Address Translation Lower			C3Ch
Memory BAR3 Address Translation Upper			C40h
Memory BAR4 Address Translation Lower			C44h
Memory BAR5 Address Translation Upper			C48h
Reserved		Virtual Interface IRQ Set	C4Ch
Reserved		Virtual Interface IRQ Clear	C50h
Reserved		Virtual Interface IRQ Mask Set	C54h
Reserved		Virtual Interface IRQ Mask Clear	C58h
Reserved		Link Interface IRQ Set	C5Ch
Reserved		Link Interface IRQ Clear	C60h
Reserved		Link Interface IRQ Mask Set	C64h
Reserved		Link Interface IRQ Mask Clear	C68h
NT Port SCRATCH0			C6Ch
NT Port SCRATCH1			C70h
NT Port SCRATCH2			C74h
NT Port SCRATCH3			C78h
NT Port SCRATCH4			C7Ch
NT Port SCRATCH5			C80h
NT Port SCRATCH6			C84h
NT Port SCRATCH7			C88h

Register 15-42. C34h Vendor-Specific Extended Capability 4

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID 4 Program to 000Bh, to indicate that the Extended Capability structure is the Vendor-Specific Extended Capability structure.	ROS	Yes	000Bh
19:16	Capability Version 4	ROS	Yes	1h
31:20	Next Capability Offset 4 Program to B70h, which addresses the Vendor-Specific Extended Capability 4 structure.	ROS	Yes	B70h

Register 15-43. C38h Vendor-Specific Header 4

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Vendor-Specific ID 4 ID Number of this Extended Capability structure.	ROS	Yes	0003h
19:16	Vendor-Specific Rev 4 Version Number of this structure.	ROS	Yes	0h
31:20	Vendor-Specific Length 4 Quantity of bytes in the entire structure.	ROS	Yes	078h

Register 15-44. C3Ch Memory BAR2 Address Translation Lower

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19:0	Reserved	RsvdP	No	0_0000h
31:20	NT Port Virtual-to-Link Interface BAR2 Base Translation Address NT Port Virtual-to-Link Interface Base Translation address when BAR2 is enabled (NT Port Virtual Interface Memory BAR2 Setup register BAR2 Enable bit (NT Port Virtual Interface, offset D4h[31]) is Set).	RW	Yes	000h

Register 15-45. C40h Memory BAR3 Address Translation Upper

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
19:0	Reserved	Offset D8h[31]=0	RsvdZ	No	0_0000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:0]) are used as the upper 32 bits.	Offset D8h[31]=1	RW	Yes	0_0000h
31:20	NT Port Virtual-to-Link Interface BAR3 Base Translation Address NT Port Virtual-to-Link Interface Base Translation address when BAR3 is enabled (NT Port Virtual Interface Memory BAR2/3 Setup register BAR3 Enable bit (NT Port Virtual Interface, offset D8h[31]) is Set).		RW	Yes	000h

Register 15-46. C44h Memory BAR4 Address Translation Lower

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19:0	Reserved	RsvdP	No	0_0000h
31:20	NT Port Virtual-to-Link Interface BAR4 Base Translation Address NT Port Virtual-to-Link Interface Base Translation address when BAR4 is enabled (NT Port Virtual Interface Memory BAR4 Setup register BAR4 Enable bit (NT Port Virtual Interface, offset DCh[31]) is Set).	RW	Yes	000h

Register 15-47. C48h Memory BAR5 Address Translation Upper

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
19:0	Reserved	Offset E0h[31]=0	RsvdZ	No	0_0000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:0]) are used as the upper 32 bits.	Offset E0h[31]=1	RW	Yes	0_0000h
31:20	NT Port Virtual-to-Link Interface BAR5 Base Translation Address NT Port Virtual-to-Link Interface Base Translation address when BAR5 is enabled (NT Port Virtual Interface Memory BAR4/5 Setup register BAR5 Enable bit (NT Port Virtual Interface, offset E0h[31]) is Set).		RW	Yes	000h

Register 15-48. C4Ch Virtual Interface IRQ Set

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Virtual Interface IRQ Mask Set register bits (offset C54h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	SET_IRQ Set Virtual IRQ. Controls the state of the NT-Virtual Doorbell Interrupt Request. Reading returns the status of the bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Sets the corresponding Interrupt Request. The Virtual Interface interrupt is asserted when the following conditions are met: - This register (offset C4Ch or C50h) value is non-zero, and, - Corresponding Virtual Interface IRQ Mask Set or Virtual Interface IRQ Mask Clear register (offset C54h or C58h, respectively) <i>Interrupt Mask</i> bit is not Set, and, - Interrupts (either INTx or MSIs) are enabled	RW1S	Yes	0000h
31:16	Reserved	RsvdP	No	0000h

Register 15-49. C50h Virtual Interface IRQ Clear

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Virtual Interface IRQ Mask Clear register bits (offset C58h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	CLR_IRQ Clear Virtual IRQ. Controls the state of the NT-Virtual Doorbell Interrupt Request. Reading returns the status of the bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Clears the corresponding Interrupt Request. The Virtual Interface interrupt is de-asserted when the following conditions are met: - This register (offset C50h or C4Ch) value is zero (0), –or– Virtual Interface IRQ Mask Set or Virtual Interface IRQ Mask Clear register (offset C54h or C58h, respectively) masks all its Set or Clear register (offset C50h or C4Ch) <i>Set</i> bits, and - INTx interrupts are enabled	RW1C	Yes	0000h
31:16	Reserved	RsvdP	No	0000h

Register 15-50. C54h Virtual Interface IRQ Mask Set

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Virtual Interface IRQ Set register bits (offset C4Ch). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	SET_IRQM Virtual Interface interrupt IRQ Mask Set. Reading returns the state of the <i>Interrupt Mask</i> bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Clears the corresponding <i>Interrupt Mask</i> bit. 0 = Corresponding Virtual Interface IRQ Set register (offset C4Ch) <i>Interrupt Request</i> bit is not masked 1 = Corresponding Virtual Interface IRQ Set register (offset C4Ch) <i>Interrupt Request</i> bit is masked/disabled	RW1S	Yes	FFFFh
31:16	Reserved	RsvdP	No	0000h

Register 15-51. C58h Virtual Interface IRQ Mask Clear

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Virtual Interface IRQ Clear register bits (offset C50h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	CLR_IRQM Clear Virtual IRQ Mask. Controls the state of the Virtual Interface <i>Interrupt Request</i> bits. Reading returns the state of the <i>Interrupt Mask</i> bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Clears the corresponding <i>Interrupt Mask</i> bit. 0 = Corresponding Virtual Interface IRQ Clear register (offset C50h) <i>Interrupt Request</i> bit is not masked 1 = Corresponding Virtual Interface IRQ Clear register (offset C50h) <i>Interrupt Request</i> bit is masked/disabled	RW1C	Yes	FFFFh
31:16	Reserved	RsvdP	No	0000h

Register 15-52. C5Ch Link Interface IRQ Set

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Link Interface IRQ Mask Set register bits (offset C64h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	SET_IRQ Set Link IRQ. Controls the state of the NT-Link Doorbell Interrupt Request. Reading returns the status of the bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Sets the corresponding Interrupt Request. The Link Interface interrupt is asserted when the following conditions are met: - This register (offset C5Ch or C60h) value is non-zero, and, - Corresponding Link Interface IRQ Mask Set or Link Interface IRQ Mask Clear register (offset C64h or C68h, respectively) Interrupt Mask bit is not Set, and, - Interrupts (either INTx or MSIs) are enabled	RW1S	Yes	0000h
31:16	Reserved	RsvdP	No	0000h

Register 15-53. C60h Link Interface IRQ Clear

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be masked by their respective Link Interface IRQ Mask Clear register bits (offset C68h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	CLR_IRQ Clear Link IRQ. Controls the state of the NT-Link Doorbell Interrupt Request. Reading returns the status of the bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Clears the corresponding Interrupt Request. The Link Interface interrupt is de-asserted when the following conditions are met: - This register (offset C60h or C5Ch) value is zero (0), –or– Link Interface IRQ Mask Set or Link Interface IRQ Mask Clear register (offset C64h or C68h, respectively) masks all its Set or Clear register (offset C60h or C5Ch) Set bits, and - INTx interrupts are enabled	RW1C	Yes	0000h
31:16	Reserved	RsvdP	No	0000h

Register 15-54. C64h Link Interface IRQ Mask Set

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Link Interface IRQ Set register bits (offset C5Ch). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	SET_IRQM Link Interface Interrupt IRQ Mask Set. Reading returns the state of the <i>Interrupt Mask</i> bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Sets the corresponding <i>Interrupt Mask</i> bit. 0 = Corresponding Link Interface IRQ Set register (offset C5Ch) <i>Interrupt Request</i> bit is not masked 1 = Corresponding Link Interface IRQ Set register (offset C5Ch) <i>Interrupt Request</i> bit is masked/disabled	RW1S	Yes	FFFFh
31:16	Reserved	RsvdP	No	0000h

Register 15-55. C68h Link Interface IRQ Mask Clear

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Notes: The bits in this register can be used to mask their respective Link Interface IRQ Clear register bits (offset C60h). The NT Port Virtual and Link Interfaces both have access to the eight Doorbell registers. These registers perform the same function, regardless of which interface is used to access the registers.				
15:0	CLR_IRQM Link Interface Interrupt IRQ Mask Clear. Reading returns the state of the <i>Interrupt Mask</i> bits. Writing 0 to a bit in the register has no effect. Writing 1 to a bit in the register Clears the corresponding <i>Interrupt Mask</i> bit. 0 = Corresponding Link Interface IRQ Clear register (offset C60h) <i>Interrupt Request</i> bit is not masked 1 = Corresponding Link Interface IRQ Clear register (offset C60h) <i>Interrupt Request</i> bit is masked/disabled	RW1C	Yes	FFFFh
31:16	Reserved	RsvdP	No	0000h

Register 15-56. C6Ch NT Port SCRATCH0

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 0 32-bit Scratchpad 0 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-57. C70h NT Port SCRATCH1

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 1 32-bit Scratchpad 1 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-58. C74h NT Port SCRATCH2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 2 32-bit Scratchpad 2 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-59. C78h NT Port SCRATCH3

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 3 32-bit Scratchpad 3 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-60. C7Ch NT Port SCRATCH4

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 4 32-bit Scratchpad 4 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-61. C80h NT Port SCRATCH5

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 5 32-bit Scratchpad 5 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-62. C84h NT Port SCRATCH6

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 6 32-bit Scratchpad 6 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

Register 15-63. C88h NT Port SCRATCH7

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Scratchpad 7 32-bit Scratchpad 7 register. Reset only by a Fundamental Reset (PEX_PERST# input assertion).	RW	Yes	0000_0000h

15.13 NT Port Virtual Interface NT Bridging-Specific Registers (Offsets C8Ch – DFCh)

Table 15-12 defines the register map of the NT Port Virtual Interface NT Bridging-Specific registers.

Table 15-12. NT Port Virtual Interface NT Bridging-Specific Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
NT Port Virtual Interface NT Bridging-Specific Registers – NT Port ID (Offsets C8Ch – C94h)		C8Ch ... C94h
Reserved		C98h – D90h
NT Port Virtual Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry (Addresses D94h – DD0h)		D94h ... DD0h
NT Port Virtual Interface NT Bridging-Specific Registers – Lookup Table Enable Link (Offsets DD4h – DDCh)		DD4h ... DDCh
Reserved		DE0h – DFCh

15.13.1 NT Port Virtual Interface NT Bridging-Specific Registers – NT Port ID (Offsets C8Ch – C94h)

This section details the NT Bridging-Specific NT Port ID, Requester ID Read Back registers for the NT Port Virtual Interface. [Table 15-13](#) defines the register map.

Table 15-13. NT Port Virtual Interface NT Bridging-Specific NT Port ID Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
NT Port ID		C8Ch
Reserved		C90h
Factory Test Only		C94h

Register 15-64. C8Ch NT Port ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	NT Port ID 0 = NT Port 1 = Reserved	ROS	No	0
30:1	Reserved	RsvdP	No	0-0h
31	Virtual Side Identifier	ROS	No	0

15.13.2 NT Port Virtual Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry (Addresses D94h – DD0h)

This section details the NT Port Virtual Interface NT Bridging-Specific Requester ID Translation Lookup Table (LUT) Entry registers. The NT Port uses these registers for Requester ID translation when it forwards:

- Memory Requests from the NT Port Virtual-to-Link Interface, –or–
- Completion TLPs from the NT Port Link-to-Virtual Interface

If the application must send traffic through the NT Port Virtual Interface, program the registers listed in this group with the corresponding Requester's Requester ID, then Set the *LUT Entry_n Enable* and *LUT Entry_m Enable* bits (bits 0 and 16, respectively) for each LUT entry, as needed.

Table 15-14 defines the register and address locations, as they relate to Register 15-65.

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The Requester ID LUT can be one of two formats, selected by the **Ingress Chip Control** register *LUT Format* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[2]):

1. **LUT Format bit is Cleared** – LUT entry is not qualified by the TLP Requester ID Function Number. The LUT format is:
 - Two entries per 32-bit register
 - Each 16-bit entry is defined as follows:
 - Bits [15:8] = Bus Number
 - Bits [7:3] = Device Number
 - Bit 2 = *Not used*
 - Bit 1 = No Snoop Control
 - Bit 0 = Enable
2. **LUT Format bit is Set** – LUT entry can be qualified by the TLP Requester ID Function Number. The Table format is:
 - Two entries per 32-bit register
 - Each 16 bit-entry is defined as follows:
 - Bits [15:8] = Bus Number
 - Bits [7:3] = Device Number
 - bits [2:0] = Function Number
 - Separate 32-bit Requester ID Enable register (**Lookup Table Enable Virtual** register, offset DD4h), in which each bit is the *Enable* bit for the corresponding LUT entry
 - Separate 32-bit No Snoop register (**Lookup Table No Snoop Enable Virtual** register, offset DD8h), in which each bit is the *No Snoop Enable* bit for the corresponding LUT entry
 - A separate 32-bit Function Number Enable register (**Lookup Table Function Enable Virtual** register, offset DDCh), in which each bit is the *Enable* bit that specifies whether the Function Number is included in the decode

Table 15-14. NT Port Virtual Interface NT Bridging-Specific Requester ID Translation Lookup Table Entry_ *n_m* Register Locations

ADDR Location	Lookup Table Entry_ <i>n_m</i>	ADDR Location	Lookup Table Entry_ <i>n_m</i>
D94h	0_1	DB4h	16_17
D98h	2_3	DB8h	18_19
D9Ch	4_5	DBCh	20_21
DA0h	6_7	DC0h	22_23
DA4h	8_9	DC4h	24_25
DA8h	10_11	DC8h	26_27
DACH	12_13	DCCCh	28_29
DB0h	14_15	DD0h	30_31

Register 15-65. D94h – DD0h NT Port Virtual Interface Requester ID Translation
LUT Entry_{n_m} (where $n_m = 0_1$ through 30_31)

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
0	LUT Entry_n Enable 0 = Disables 1 = Enables		RW	Yes	0
1	LUT Entry_n No Snoop Enable If Set, the NT Port Clears the TLP <i>No Snoop</i> attribute bit for the Memory Request, then goes from the NT Port Virtual-to-Link Interface, and re-calculates the ECRC. If the original TLP has an ECRC error, the NT Port corrupts the re-calculated ECRC before transmitting to the other Host domain. The NT Port sets the <i>No Snoop</i> attribute bit when it forwards the Completion TLP from the NT Port Link-to-Virtual Interface if this bit is Set for the corresponding Requester ID entry. This ECRC rule applies to Completion TLPs as well. 0 = Disables 1 = Enables		RW	Yes	0
2	LUT Entry_n Function Number Bit 2		RW	Yes	0
7:3	Requester ID on Link Side	Device Number LUT Entry _n Requester Device Number.	RW	Yes	0000_0b
15:8		Bus Number LUT Entry _n Requester Bus Number.	RW	Yes	00h
16	LUT Entry_m Enable 0 = Disables 1 = Enables		RW	Yes	0
17	LUT Entry_m No Snoop Enable If Set, the NT Port Clears the TLP <i>No Snoop</i> attribute bit for the Memory Request, then goes from the NT Port Virtual-to-Link Interface, and re-calculates the ECRC. If the original TLP has an ECRC error, the NT Port corrupts the re-calculated ECRC before transmitting to the other Host domain. The NT Port sets the <i>No Snoop</i> attribute bit when it forwards the Completion TLP from the NT Port Link-to-Virtual Interface if this bit is Set for the corresponding Requester ID entry. This ECRC rule applies to Completion TLPs as well. 0 = Disables 1 = Enables		RW	Yes	0
18	LUT Entry_n Function Number Bit 2		RW	Yes	0
23:19	Requester ID on Link Side	Device Number LUT Entry _m Requester Device Number.	RW	Yes	0000_0b
31:24		Bus Number LUT Entry _m Requester Bus Number.	RW	Yes	00h

15.13.3 NT Port Virtual Interface NT Bridging-Specific Registers – Lookup Table Enable Link (Offsets DD4h – DDCh)

This section details the NT Port Virtual Interface NT Bridging-Specific Lookup table Enable Link registers. [Table 15-15](#) defines the register map. Information describing when these registers are enabled, and how they are used, is provided in [Section 15.13.2](#).

Table 15-15. NT Port Virtual Interface NT Bridging-Specific Lookup Table Enable Link Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Lookup Table Enable Virtual		DD4h
Lookup Table No Snoop Enable Virtual		DD8h
Lookup Table Function Enable Virtual		DDCh

Register 15-66. DD4h Lookup Table Enable Virtual

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table Enable Enable bit per Lookup Table.	RWS	Yes	0000_0000h

Register 15-67. DD8h Lookup Table No Snoop Enable Virtual

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table No Snoop Enable No Snoop Enable bit per Lookup Table.	RWS	Yes	0000_0000h

Register 15-68. DDCh Lookup Table Function Enable Virtual

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table Function Check Enable Function Check Enable bit per Lookup Table.	RWS	Yes	0000_0000h

15.14 NT Port Virtual Interface Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)

The registers detailed in [Section 13.23, “Advanced Error Reporting Extended Capability Registers \(Offsets FB4h – FDCh\),”](#) are also applicable to the NT Port Virtual Interface, except as defined in [Table 15-16](#) (register map), and [Register 15-69](#) through [Register 15-74](#).

Table 15-16. NT Port Virtual Interface Advanced Error Reporting Extended Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20																19 18 17 16				15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Next Capability Offset (148h)												Capability Version (1h)				PCI Express Extended Capability ID (0001h)																FB4h				
Reserved												Uncorrectable Error Status																FB8h								
Reserved												Uncorrectable Error Mask																FBCh								
Reserved												Uncorrectable Error Severity																FC0h								
Reserved												Correctable Error Status																FC4h								
Reserved												Correctable Error Mask																FC8h								
Advanced Error Capabilities and Control																				FCCh																
Header Log 0																				FD0h																
Header Log 1																				FD4h																
Header Log 2																				FD8h																
Header Log 3																				FDCh																

Register 15-69. FB4h Advanced Error Reporting Extended Capability Header

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID	ROS	Yes	0001h
19:16	Capability Version	ROS	Yes	1h
31:20	Next Capability Offset Program to 148h, which addresses the Virtual Channel Extended Capability structure.	ROS	Yes	148h

Register 15-70. FB8h Uncorrectable Error Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Virtual Interface: - Data Link Protocol error - Surprise Down error Note: If an individual error is masked (corresponding bit in the Uncorrectable Error Mask register (offset FBCh) is Set) when the error is detected, its Error Status bit is still updated; however, an error reporting Message (ERR_FATAL or ERR_NONFATAL) is not sent to the Root Complex, and the Advanced Error Capabilities and Control register First Error Pointer field (offset FCCh[4:0]) and Header Log x registers (offsets FD0h through FDCh) remain unchanged.				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
5	Surprise Down Error Status Reserved	RsvdP	No	0
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
13	Flow Control Protocol Error Status Reserved/Not supported	RsvdP	No	0
14	Completion Timeout Status Not applicable to switches.	RsvdP	No	0
15	Completer Abort Status	RW1CS ^a	Yes	0
16	Unexpected Completion Status 0 = No error is detected 1 = Error is detected Note: Unexpected Completion is never reported by the internal NT bridge, because this error is detected and reported by the PEX 8724 ingress Ports.	RW1CS ^a	Yes	0
17	Receiver Overflow Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
18	Malformed TLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
19	ECRC Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0

Register 15-70. FB8h Uncorrectable Error Status (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
20	Unsupported Request Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
21	Reserved	RsvdP	No	0
22	Uncorrectable Internal Error Status Reserved	RsvdP	No	0
23	MC Blocked TLP Status Multicast blocked TLP status. 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
31:24	Reserved	RsvdP	No	00h

- a. When the **Error Check Disable** register *Software Force Error Enable* bit (offset 720h[2]) is Set, Type changes from RW1CS to RW.

Register 15-71. FBCh Uncorrectable Error Mask

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Virtual Interface: • Data Link Protocol error • Surprise Down error Note: The bits in this register can be used to mask their respective Uncorrectable Error Status register bits (offset FB8h).				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
5	Surprise Down Error Mask Reserved	RsvdP	No	0
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
13	Flow Control Protocol Error Mask Reserved/Not supported	RsvdP	No	0
14	Completion Timeout Mask Not applicable to switches.	RWS	Yes	0
15	Completer Abort Mask	RWS	Yes	0

Register 15-71. FBCh Uncorrectable Error Mask (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
16	Unexpected Completion Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
17	Receiver Overflow Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
18	Malformed TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
19	ECRC Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
20	Unsupported Request Error Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
21	Reserved	RsvdP	No	0
22	Uncorrectable Internal Error Mask Reserved	RsvdP	No	0
23	MC Blocked TLP Mask 0 = No mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
31:24	Reserved	RsvdP	No	00h

Register 15-72. FC0h Uncorrectable Error Severity

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Virtual Interface: • Data Link Protocol error • Surprise Down error				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
5	Surprise Down Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	ROS	No	1
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
13	Flow Control Protocol Error Severity	ROS	No	1
14	Completion Timeout Severity <i>Not applicable to switches.</i> Because the Status and Mask are both Reserved for this bit, Severity can be ignored.	RsvdP	No	0
15	Completer Abort Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0

Register 15-72. FC0h Uncorrectable Error Severity (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
16	Unexpected Completion Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
17	Receiver Overflow Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
18	Malformed TLP Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
19	ECRC Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
20	Unsupported Request Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
21	Reserved	RsvdP	No	0
22	Uncorrectable Internal Error Severity Reserved	RsvdP	No	0
23	MC Blocked TLP Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	0
31:24	Reserved	RsvdP	No	00h

Register 15-73. FC4h Correctable Error Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Virtual Interface: • Receiver error • Bad TLP error • Bad Data Link Layer Packet (DLLP) error • Replay Number Rollover error • Replay Timer Timeout error Note: If an individual error is masked (corresponding bit in the Correctable Error Mask register (offset FC8h) is Set) when the error is detected, its Error Status bit is still updated; however, an error reporting Message (ERR_COR) is not sent to the Root Complex.				
0	Receiver Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
5:1	Reserved	RsvdP	No	0-0h
6	Bad TLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
7	Bad DLLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
8	REPLAY NUM Rollover Status Replay Number Rollover status. 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
11:9	Reserved	RsvdP	No	000b
12	Replay Timer Timeout Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
13	Advisory Non-Fatal Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
14	Corrected Internal Error Status Reserved	RsvdP	No	0
15	Header Log Overflow Status Reserved	RsvdP	No	0
31:16	Reserved	RsvdP	No	0000h

a. When the **Error Check Disable** register **Software Force Error Enable** bit (offset **720h[2]**) is Set, Type changes from RW1CS to RW.

Register 15-74. FC8h Correctable Error Mask

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Virtual Interface: • Receiver error • Bad TLP error • Bad DLLP error • Replay Number Rollover error • Replay Timer Timeout error Note: The bits in this register can be used to mask their respective <i>Correctable Error Status</i> register bits (offset <i>FC4h</i>).				
0	Receiver Error Mask 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	0
5:1	Reserved	RsvdP	No	0-0h
6	Bad TLP Mask 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	0
7	Bad DLLP Mask 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	0
8	REPLAY NUM Rollover Mask Replay Number Rollover mask. 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	0
11:9	Reserved	RsvdP	No	000b
12	Replay Timer Timeout Mask 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	0
13	Advisory Non-Fatal Error Mask 0 = Error reporting is not masked 1 = Error reporting is masked	RWS	Yes	1
14	Corrected Internal Error Mask Reserved	RsvdP	No	0
15	Header Log Overflow Mask Reserved	RsvdP	No	0
31:16	Reserved	RsvdP	No	0000h

15.15 NT Port Virtual Interface Device-Specific Registers – Link Error (Offsets FE0h – FFCh)

This section details the NT Port Virtual Interface Device-Specific Link Error registers. These Device-Specific registers are unique to the NT Port Virtual Interface and not referenced in the *PCI Express Base r3.0*. [Table 15-17](#) defines the register map.

Other NT Port Virtual Interface Device-Specific registers are detailed in:

- [Section 15.11, “NT Port Virtual Interface Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 15.12, “NT Port Virtual Interface Device-Specific Registers \(Offsets B70h – C88h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 15-17. NT Port Virtual Interface Device-Specific Register Map – Link Error (Offsets FE0h – FFCh)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Link Error Status Virtual		FE0h
Link Error Mask Virtual		FE4h
Reserved		FE8h – FFCh

**Register 15-75. FE0h Link Error Status Virtual
(NT Port Virtual Interface)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: The bits in this register can be masked by their respective bits in the Link Error Mask Virtual register (NT Port Virtual Interface, offset FE4h).				
0	Link Side Correctable Error Status 1 = NT Port Link Interface detected a Correctable TLP error, and signaled the interrupt to the Virtual side Host	RW1CS	Yes	0
1	Link Side Uncorrectable Error Status 1 = NT Port Link Interface detected an Uncorrectable TLP error, and signaled the interrupt to the Virtual side Host	RW1CS	Yes	0
2	Link Side DL Active Change Status 1 = NT Port Link Interface <i>DL_Active</i> state change occurred upon detection of an NT Port Link Interface <i>DL_Down</i> state rise edge and fall edge	RW1CS	Yes	0
3	Link Side Uncorrectable Error Message Drop Status 1 = NT Port Link Interface received an Uncorrectable Error Message, and signaled the interrupt to the Virtual side Host	RW1CS	Yes	0
31:4	Reserved	RsvdP	No	0000_000h

**Register 15-76. FE4h Link Error Mask Virtual
(NT Port Virtual Interface)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: The bits in this register can be used to mask their respective bits in the Link Error Status Virtual register (NT Port Virtual Interface, offset FE0h).				
0	Link Side Correctable Error Mask 0 = No effect on reporting activity 1 = Link Side Correctable Error Status bit is masked/disabled	RWS	Yes	1
1	Link Side Uncorrectable Error Mask 0 = No effect on reporting activity 1 = Link Side Uncorrectable Error Status bit is masked/disabled	RWS	Yes	1
2	Link Side DL Active Change Mask 0 = No effect on reporting activity 1 = Link Side DL Active Change Status bit is masked/disabled	RWS	Yes	1
3	Link Side Uncorrectable Error Message Drop Mask 0 = No effect on reporting activity 1 = Link Side Uncorrectable Error Message Drop Status bit is masked/disabled	RWS	Yes	1
31:4	Reserved	RsvdP	No	0000_000h



Chapter 16 NT Port Link Interface Registers – NT Mode

16.1 Introduction

Note: Check the latest design guides, application notes and errata list for Non-Transparent (NT) usage.

In NT mode, the NT Port includes two sets of Configuration, Capability, Control, and Status registers, to support the Link and Virtual Interfaces. This chapter defines the NT Port Link Interface registers. Other registers are defined in:

- [Chapter 13, “Transparent Port Registers”](#)
- [Chapter 15, “NT Port Virtual Interface Registers – NT Mode”](#)

All PEX 8724 registers can be accessed by Configuration or Memory Requests.

For further details regarding register names and descriptions, refer to the following specifications:

- *PCI r3.0*
- *PCI Power Mgmt. r1.2*
- *PCI Express Base r3.0*

16.2 NT Port Link Interface Type 0 Register Map

Table 16-1 defines the NT Port Link Interface Type 0 register mapping.

Table 16-1. NT Port Link Interface Type 0 Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets																
NT Port Link Interface PCI-Compatible Type 0 Configuration Header Registers (Offsets 00h – 3Ch)																																00h																
																																...																
																																34h																
																																...																
																																3Ch																
																Next Capability Pointer (48h)																Capability ID (01h)																40h
NT Port Link Interface PCI Power Management Capability Registers (Offsets 40h – 44h)																																44h																
																Next Capability Pointer (68h)																Capability ID (05h)																48h
NT Port Link Interface MSI Capability Registers (Offsets 48h – 64h)																																...																
																																64h																
																Next Capability Pointer (C8h)																Capability ID (10h)																68h
NT Port Link Interface PCI Express Capability Registers (Offsets 68h – A0h)																																...																
																																A0h																
Reserved																																A4h –	C4h															
																Next Capability Pointer (00h)																Capability ID 3 (09h)																C8h
NT Port Link Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh)																																...																
																																FCh																
Next Capability Offset (FB4h)								1h								PCI Express Extended Capability ID (0003h)																100h																
Device Serial Number Extended Capability Registers (Offsets 100h – 108h) ^a																																...																
																																108h																
Next Capability Offset (148h)								1h								PCI Express Extended Capability ID (0019h)																10Ch																
Secondary PCI Express Extended Capability Registers (Offsets 10Ch – 134h) ^b																																...																
																																134h																
Next Capability Offset (10Ch)								1h								PCI Express Extended Capability ID (0004h)																138h																
Power Budget Extended Capability Registers (Offsets 138h – 144h) ^c																																...																
																																144h																
Next Capability Offset (C34h)								1h								PCI Express Extended Capability ID (0002h)																148h																
NT Port Link Interface Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)																																...																
																																1BCh																
NT Port Link Interface Device-Specific Registers (Offsets 1C0h – A74h)																																1C0h																
																																...																
																																A74h																
Reserved																																A78h –	B6Ch															

Table 16-1. NT Port Link Interface Type 0 Register Map (Cont.)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset 2 (000h)																1h				PCI Express Extended Capability ID 2 (000Bh)																B70h
NT Port Link Interface Device-Specific Registers (Offsets B70h – C88h)																																...				
Next Capability Offset 4 (B70h)																1h				PCI Express Extended Capability ID 4 (000Bh)																C34h
NT Port Link Interface Device-Specific Registers (Offsets B70h – C88h)																																...				
																																C88h				
NT Port Link Interface NT Bridging-Specific Registers (Offsets C8Ch – EFCh)																																C8Ch				
																																...				
																																EFCh				
NT Port Link Interface Device-Specific Registers (Offsets F00h – FB0h)																																F00h				
																																...				
																																FB0h				
Next Capability Offset (138h)																1h				PCI Express Extended Capability ID (0001h)																FB4h
NT Port Link Interface Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)																																...				
																																FDCh				
Reserved																																FE0h –	FFCh			

- Register offsets 100h through 108h are detailed in Section 13.12, “Device Serial Number Extended Capability Registers (Offsets 100h – 108h).”
- Register offsets 10Ch through 134h are detailed in Section 13.13, “Secondary PCI Express Extended Capability Registers (Offsets 10Ch – 134h).”
- Register offsets 138h through 144h are detailed in Section 13.14, “Power Budget Extended Capability Registers (Offsets 138h – 144h).”

16.3 Register Access

The NT Port Link Interface implements a 4-KB Configuration Space. The lower 256 bytes (offsets 00h through FFh) comprise the PCI-compatible Configuration Space, and the upper 960 DWords (offsets 100h through FFFh) comprise the PCI Express Extended Configuration Space. The PEX 8724 supports three mechanisms for accessing the NT Port Link Interface registers:

- [PCI Express Base r3.0 Configuration Mechanism](#)
- [Device-Specific Memory-Mapped Configuration Mechanism](#)
- [Device-Specific Cursor Mechanism](#)

16.3.1 *PCI Express Base r3.0* Configuration Mechanism

The *PCI Express Base r3.0* Configuration mechanism is divided into two mechanisms:

- **PCI r3.0-Compatible Configuration Mechanism** – Provides Conventional PCI access to the first 256 bytes (the bytes at offsets 00h through FFh) of the NT Port Link Interface Configuration Register space
- **PCI Express Enhanced Configuration Access Mechanism** – Provides access to the entire 4-KB Configuration Space

Both are described in the sections that follow.

The PEX 8724 decodes Type 0 Configuration transactions received on its NT Port Link Interface. The PEX 8724 reads from or writes to the NT Port Link Interface register, as specified in the original Type 0 Configuration access.

16.3.1.1 *PCI r3.0-Compatible* Configuration Mechanism

The *PCI r3.0-Compatible* Configuration Space consists of the first 256 bytes of the NT Port Link Interface Configuration Space. (Refer to [Figure 16-1](#).) The *PCI r3.0-Compatible* Configuration mechanism provides standard access to the NT Port Link Interface's first 256 bytes (the bytes at offsets 00h through FFh) of the PCI Express Configuration Space.

This mechanism uses the same Request format as the [PCI Express Enhanced Configuration Access Mechanism](#). For PCI-compatible Configuration Requests, the *Extended Register Address* field must be all zeros (0).

Because the mechanism is limited to the first 256 bytes of the NT Port Link Interface Configuration Register space, one of the following must be used to access beyond Byte FFh:

- [PCI Express Enhanced Configuration Access Mechanism](#)
- [Device-Specific Memory-Mapped Configuration Mechanism](#)
- [Device-Specific Cursor Mechanism](#)

16.3.1.2 PCI Express Enhanced Configuration Access Mechanism

The PCI Express Enhanced Configuration Access mechanism (ECAM) uses a flat, Root Complex Memory-Mapped Address space to access the device Configuration registers. In this case, the Memory address determines the Configuration register accessed, and Memory data returns the addressed register contents. The Root Complex converts the Memory transaction into a Configuration transaction before transmitting this access to the Downstream devices.

This mechanism is used to access the NT Port Link Interface Type 0 registers:

- NT Port Link Interface PCI-Compatible Type 0 Configuration Header Registers (Offsets 00h – 3Ch)
- NT Port Link Interface PCI Power Management Capability Registers (Offsets 40h – 44h)
- NT Port Link Interface MSI Capability Registers (Offsets 48h – 64h)
- NT Port Link Interface PCI Express Capability Registers (Offsets 68h – A0h)
- NT Port Link Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh)
- Device Serial Number Extended Capability Registers (Offsets 100h – 108h)
- Power Budget Extended Capability Registers (Offsets 138h – 144h)
- NT Port Link Interface Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)
- Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch)
- NT Port Link Interface Device-Specific Registers – Vendor-Specific Extended Capability 4 (Offsets C34h – C88h)
- NT Port Link Interface Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)

This mechanism typically **cannot** access the Device-Specific registers (which are instead accessible by the Device-Specific Memory-Mapped Configuration Mechanism) in the following offset ranges, per Port:

- 200h through AFCh
- B0Ch through B6Ch
- B80h through C30h

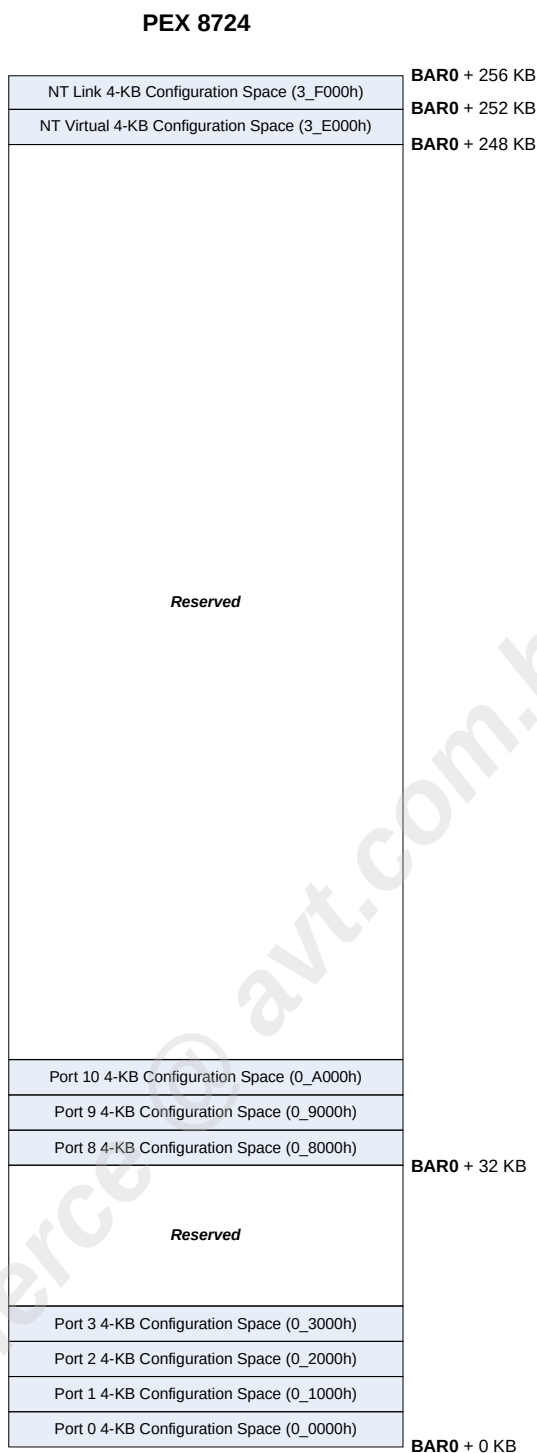
Silicon Revision CA provides an option to enable ECAM access to all PEX 8724 registers, however, by Setting the **Station-Based Control** register *Device-Specific Register Access by Extended Configuration Request Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 760h[16]), in the Station that has the NT Port.

16.3.2 Device-Specific Memory-Mapped Configuration Mechanism

The Device-Specific Memory-Mapped Configuration mechanism provides a method to access the registers for all Ports within a single 256-KB Memory map, as illustrated in [Figure 16-1](#). This Memory map is identical for Upstream Port(s) **BAR0/1**, NT Port Virtual Interface **BAR0/1**, and NT Port Link Interface **BAR0/1**. The registers of each Port are located within a 4-KB range.

When the NT Port is enabled at Fundamental Reset, the NT Port Virtual and Link Interface registers use the *PCI r3.0* Type 0 Configuration Space Header. The NT PCI-to-PCI bridge (between the NT Port Virtual Interface and internal virtual PCI Bus) registers use the *PCI r3.0* Type 1 Configuration Space Header, and are mapped to the 4-KB Address space of the Port Number that is assigned as the NT Port (indicated in the **VS0 Upstream** register *NT Port* field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset **360h[12:8]**)).

To use this mechanism, use the [PCI r3.0-Compatible Configuration Mechanism](#) to program the Upstream Port(s) **Base Address 0** and **Base Address 1** registers (**BAR0** and **BAR1**, offsets **10h** and **14h**, respectively). After the NT Port Link Interface Memory-Mapped register Base address is Set, the Configuration Space registers are accessed, using Memory Reads and Writes to the 4-KB range, starting at offset 248 KB (3_E000h, NT Port Virtual Interface) and offset 252 KB (3_F000h, NT Port Link Interface).

Figure 16-1. NT Mode Configuration Register Mapping to Memory-Mapped BAR

Note: [Table 13-5](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

16.3.3 Device-Specific Cursor Mechanism

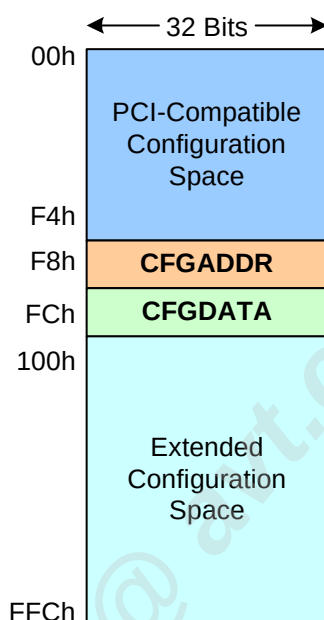
The Device-Specific Cursor mechanism is provided for use in development systems that can only generate *PCI r3.0* Configuration cycles (*that is*, the system cannot use either the [Device-Specific Memory-Mapped Configuration Mechanism](#), nor generate Extended Configuration Requests to access the Extended Configuration Space).

In [Figure 16-2](#), the software uses the **Configuration Address Window** (CFGADDR) register (offset **F8h**) to select the NT Port Virtual or Link Interface Configuration Space registers, including the PCI Express Extended Configuration Space registers (offsets 100h through FFFh).

Software uses the **Configuration Data Window** (CFGDATA) register (offset **FCh**) to write to or read from the selected Configuration Space registers.

Refer to [Section 16.9](#), “NT Port Link Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh),” for the register descriptions.

Figure 16-2. Configuration Space View



16.4 Register Descriptions

The remainder of this chapter details the NT Port Link Interface registers, including:

- Bit/field names
- Description of register functions in the NT Port Virtual and Link Interfaces
- Type (*such as* RW or HwInit; refer to [Table 13-4, “Register Types, Grouped by User Accessibility,”](#) for Type descriptions)
- Whether the power-on/reset value can be modified, by way of the PEX 8724 serial EEPROM and/or I²C/SMBus Initialization feature
- Default power-on/reset value

16.5 NT Port Link Interface PCI-Compatible Type 0 Configuration Header Registers (Offsets 00h – 3Ch)

This section details the NT Port Link Interface PCI-Compatible Type 0 Configuration Header registers. Table 16-2 defines the register map.

Table 16-2. NT Port Link Interface PCI-Compatible Type 0 Configuration Header Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Device ID																Vendor ID																00h
PCI Status																PCI Command																04h
PCI Class Code																PCI Revision ID																08h
PCI BIST (<i>Not Supported</i>)				PCI Header Type				Master Latency Timer (<i>Not Supported</i>)				Cache Line Size				0Ch																
Base Address 0																10h																
Base Address 1																14h																
Base Address 2																18h																
Base Address 3																1Ch																
Base Address 4																20h																
Base Address 5																24h																
Reserved																28h																
Subsystem ID								Subsystem Vendor ID								2Ch																
NT Port Expansion ROM Base Address																30h																
Reserved												Capability Pointer (40h)				34h																
Reserved																38h																
Max_Lat (<i>Reserved</i>)				Min_Gnt (<i>Reserved</i>)				PCI Interrupt Pin				PCI Interrupt Line				3Ch																

Register 16-1. 00h PCI Configuration ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Vendor ID Identifies the device manufacturer. Defaults to the PCI-SIG-issued Vendor ID of PLX (10B5h), if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	10B5h
31:16	Device ID Identifies the particular device. Defaults to the PLX part number for the PEX 8724, if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	87A0h

Register 16-2. 04h PCI Command/Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Command				
0	I/O Access Enable The NT Port Link Interface ignores the value of this register, because it does not support I/O resources.	RW	Yes	0
1	Memory Access Enable 0 = PEX 8724 ignores Memory Space Requests received on the NT Port Link Interface 1 = PEX 8724 accepts Memory Space Requests received on the NT Port Link Interface	RW	Yes	0
2	Bus Master Enable Controls PEX 8724 forwarding of Memory Requests Upstream. Does not affect Message forwarding nor Completions. 0 = PEX 8724 handles Memory Requests received on the NT Port Virtual Interface as Unsupported Requests (URs); for Non-Posted Requests, the PEX 8724 returns a Completion with UR Completion status 1 = PEX 8724 forwards Memory Requests from the NT Port Virtual-to-Link Interface	RW	Yes	0
3	Special Cycle Enable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
4	Memory Write and Invalidate Enable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
5	VGA Palette Snoop Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
6	Parity Error Response Enable Controls bit 24 (<i>Master Data Parity Error Detected</i>).	RW	Yes	0
7	IDSEL Stepping/Wait Cycle Control Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
8	SERR# Enable Controls bit 30 (<i>Signaled System Error</i>). 1 = Enables reporting of Fatal and Non-Fatal errors detected by the NT Port Link Interface to the Root Complex	RW	Yes	0
9	Fast Back-to-Back Transactions Enable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
10	Interrupt Disable 0 = NT Port Link Interface is enabled to generate INTx Interrupt Messages 1 = NT Port Link Interface is prevented from generating INTx Interrupt Messages	RW	Yes	0
15:11	Reserved	RsvdP	No	0-0h

Register 16-2. 04h PCI Command/Status (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Status				
18:16	Reserved	RsvdP	No	000b
19	Interrupt Status 0 = No pending INTx Interrupt Message 1 = INTx interrupt is pending internal to the NT Port Link Interface, –or– PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) (if enabled) is asserted	RO	No	0
20	Capability List Capability function is supported. Set, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	1
21	66 MHz Capable Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
22	Reserved	RsvdP	No	0
23	Fast Back-to-Back Transactions Capable Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
24	Master Data Parity Error Detected If bit 6 (<i>Parity Error Response Enable</i>) is Set, the NT Port Link Interface Sets this bit when the NT Port: - Forwards the poisoned Transaction Layer Packet (TLP) Write Request from the NT Port Virtual-to-Link Interface, –or– - Receives a Completion marked as poisoned on the NT Port Link Interface If the <i>Parity Error Response Enable</i> bit is Cleared, the PEX 8724 never Sets this bit. This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
26:25	DEVSEL# Timing Not supported Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00b
27	Signaled Target Abort The NT Port Link Interface Sets this bit when any of the following conditions are met: - NT Port Link Interface receives a Completion (from a Transparent Port) that has a Completion status of Completer Abort (CA), –or– - NT Port Link Interface receives a Memory Request targeting a PEX 8724 register, and the Payload Length (indicated within the Memory Request Header) is greater than 1 DWord - NT Port Link Interface receives a Memory Request targeting a PEX 8724 register address within a non-existent Port - NT Port Link Interface receives a Memory Write Request targeting enabled NT Port Expansion ROM Address space (NT Port Expansion ROM Base Address register (BAR), offset 30h) Note: When Set during a forwarded Completion, the Uncorrectable Error Status register <i>Completer Abort Status</i> bit (offset FB8h[15]) is not updated, because the NT Port does not log the corresponding Requests that it forwards.	RW1C	Yes	0

Register 16-2. 04h PCI Command/Status (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
28	Received Target Abort Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
29	Received Master Abort Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	0
30	Signaled System Error If bit 8 (<i>SERR# Enable</i>) is Set, the NT Port Link Interface Sets this bit when transmitting an ERR_FATAL or ERR_NONFATAL Message to its Upstream device. This error is natively reported by the Device Status register <i>Fatal Error Detected</i> and <i>Non-Fatal Error Detected</i> bits (offset 70h[18:17], respectively), which are mapped to this bit for Conventional PCI backward compatibility.	RW1C	Yes	0
31	Detected Parity Error This error is natively reported by the Uncorrectable Error Status register <i>Poisoned TLP Status</i> bit (offset FB8h[12]), which is mapped to this bit for Conventional PCI backward compatibility. 1 = NT Port Link Interface received a Poisoned TLP, regardless of the bit 6 (<i>Parity Error Response Enable</i>) state	RW1C	Yes	0

Register 16-3. 08h PCI Class Code and Revision ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
PCI Revision ID					
7:0	Revision ID Unless overwritten by the serial EEPROM, returns the Silicon Revision (BAh or CAh), the PLX-assigned Revision ID for this version of the PEX 8724. The PEX 8724 Serial EEPROM register Initialization capability is used to replace the PLX Revision ID with another Revision ID.	RO	Yes	BAh	CAh
PCI Class Code				068000h	
15:8	Register-Level Programming Interface Cleared, as required by the <i>PCI r3.0</i> for other bridge devices.	RO	Yes	00h	
23:16	Sub-Class Code Other bridge devices.	RO	Yes	80h	
31:24	Base Class Code Bridge devices.	RO	Yes	06h	

Register 16-4. 0Ch Miscellaneous Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Cache Line Size				
7:0	Cache Line Size System Cache Line Size. Implemented as a RW field for Conventional PCI compatibility purposes and does not impact PEX 8724 functionality.	RW	Yes	00h
Master Latency Timer				
15:8	Master Latency Timer <i>Not supported</i> Cleared, as required by the <i>PCI Express Base r3.0</i> .	RsvdP	No	00h
PCI Header Type				
22:16	Configuration Layout Type Type 0 Configuration Header for the NT Port.	RO	Yes	00h
23	Multi-Function Device 0 = Single-function device 1 = Indicates multiple (up to eight) functions (logical devices), each containing its own, individually addressable Configuration Space, 256 DWords in size	RO	Yes	0
PCI BIST				
31:24	PCI BIST <i>Not supported</i> Built-In Self-Test (BIST) Pass or Fail.	RsvdP	No	00h

**Register 16-5. 10h Base Address 0
(NT Port Link Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Note: By default, NT Port Link Interface BAR0 is enabled and BAR1 is disabled, to provide a 32-bit BAR0 for register access. BAR1 can be enabled (by serial EEPROM and/or I ² C/SMBus), to provide a 64-bit BAR0/1 , by programming the NT Port Link Interface BAR0/1 Setup register BAR0/1 Enable field (NT Port Link Interface, offset E4h[1:0]) to 11b (which enables both BAR0 and BAR1).				
0	Memory Space Indicator When enabled, the Base Address register maps PEX 8724 Port Configuration registers into Memory space. Note: Hardwired to 0.	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are Reserved .	ROS	Yes	00b
3	Prefetchable 0 = Base Address register maps the PEX 8724 Port Configuration registers into Non-Prefetchable Memory space	ROS	Yes	0
17:4	Reserved	RsvdP	No	0-0h
31:18	Base Address 0 256-KB-aligned Base address used for Memory-Mapped access to the 256-KB block of all PEX 8724 registers (4 KB, per Port).	RW	Yes	0-0h

**Register 16-6. 14h Base Address 1
(NT Port Link Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Upper 32-Bit Address for Memory-Mapped BAR For 64-bit addressing (BAR0/1), Base Address 1 (BAR1) extends Base Address 0 (BAR0) to provide the upper 32 Address bits when the Base Address 0 register Memory Map Type field (offset 10h[2:1]) is programmed to 10b.	RW	Yes	0000_0000h
	Read-Only when the Base Address 0 (BAR0) register is not enabled as a 64-bit BAR (Memory Map Type field (offset 10h[2:1]) is not programmed to 10b).	RO	Yes	0000_0000h

Register 16-7. 18h Base Address 2
(NT Port Link Interface Memory Space)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	Memory Space Indicator 0 = Implemented as a Memory BAR	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are Reserved .	ROS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	ROS	Yes	0
19:4	Reserved	RsvdP	No	0_000h
31:20	Base Address 2 Resolution is 1 MB.	RW	Yes	000h

Register 16-8. 1Ch Base Address 3
(NT Port Link Interface Memory Space)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Note: This register has RW privilege if BAR2/3 is configured as a 64-bit BAR (Base Address 2 register Memory Map Type field (NT Port Link Interface, offset 18h[2:1]), is programmed to 10b).					
0	Memory Space Indicator BAR3 can be used as an independent 32-bit only BAR, or as the upper 32 bits of 64-bit BAR2/3 . 0 = Implemented as a Memory BAR in 32-Bit mode	Offset 18h[2:1]=00b	RO	No	0
		Offset 18h[2:1]=10b	RW	Yes	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space All other encodings are Reserved .	Offset 18h[2:1]=00b	RO	No	00b
		Offset 18h[2:1]=10b	RW	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	Offset 18h[2:1]=00b	ROS	No	0
		Offset 18h[2:1]=10b	RW	Yes	0
19:4	Reserved	Offset 18h[2:1]=00b	RsvdP	No	0_000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset 18h[2:1]=10b	RW	Yes	0_000h
31:20	Base Address 3	RW	Yes	000h	

**Register 16-9. 20h Base Address 4
(NT Port Link Interface Memory Space)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	Memory Space Indicator 0 = Implemented as a Memory BAR; otherwise, <i>Reserved</i>	RO	No	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space 10b = Base Address register is 64 bits wide and can be mapped anywhere in the 64-bit Address space All other encodings are <i>Reserved</i> .	ROS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	ROS	Yes	0
19:4	<i>Reserved</i>	RsvdP	No	0_000h
31:20	Base Address 4	RW	Yes	000h

**Register 16-10. 24h Base Address 5
(NT Port Link Interface Memory Space)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
Note: This register has RW privilege if BAR4/5 is configured as a 64-bit BAR (Base Address 4 register Memory Map Type field (NT Port Link Interface, offset 20h[2:1]), is programmed to 10b).					
0	Memory Space Indicator BAR5 can be used as an independent 32-bit only BAR, or as the upper 32 bits of 64-bit BAR4/5 . 0 = Implemented as a Memory BAR in 32-Bit mode	Offset 20h[2:1]=00b	RO	No	0
		Offset 20h[2:1]=10b	RW	Yes	0
2:1	Memory Map Type 00b = Base Address register is 32 bits wide and can be mapped anywhere in the 32-bit Memory space All other encodings are Reserved .	Offset 20h[2:1]=00b	RO	No	00b
		Offset 20h[2:1]=10b	RW	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	Offset 20h[2:1]=00b	ROS	No	0
		Offset 20h[2:1]=10b	RW	Yes	0
19:4	Reserved	Offset 20h[2:1]=00b	RsvdP	No	0_000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset 20h[2:1]=10b	RW	Yes	0_000h
31:20	Base Address 5		RW	Yes	000h

Register 16-11. 2Ch Subsystem ID and Subsystem Vendor ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Subsystem Vendor ID				
15:0	Subsystem Vendor ID Identifies the device manufacturer. Defaults to the PCI-SIG-issued Vendor ID of PLX (10B5h), if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	10B5h
Subsystem ID				
31:16	Subsystem ID Identifies the particular device. Defaults to the PLX part number for the PEX 8724, if not overwritten by serial EEPROM and/or I ² C.	RO	Yes	87A0h

Register 16-12. 30h NT Port Expansion ROM Base Address

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
Note: Expansion ROM can be enabled in either the NT Port Link or Virtual Interface, but not both concurrently. Expansion ROM is enabled, by default, in the NT Port Link Interface (Ingress Chip Control register NT Port Expansion ROM Virtual Side bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 764h[0]) is Cleared). Expansion ROM can be disabled, by Setting the Ingress Control register Disable NT Port Expansion ROM BAR bit (Base mode – All Ports; Virtual Switch mode – VS Upstream Port, offset F60h[15]).					
0	Expansion ROM Enable 0 = NT Port Link Interface Expansion ROM is disabled	Upstream Port(s), offset F60h[15]=1	RsvdP	No	0
	1 = NT Port Link Interface Expansion ROM is enabled, and NT Port Virtual Interface Expansion ROM is disabled	Upstream Port(s), offset F60h[15]=0	RO	Yes	0
13:1	Reserved		RsvdP	No	0-0h
31:14	Expansion ROM Base Address If the Serial EEPROM Clock Frequency register Expansion ROM Size bit (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 268h[16]) value is 0, the NT Port Expansion ROM size is 16 KB (default value is FFFF_C001h). Bit 14 is RW.	Upstream Port(s), offset F60h[15]=1	RsvdP	No	0-0h
	If the Expansion ROM Size bit value is 1, the NT Port Expansion ROM size is 32 KB (default value is FFFF_8001h). Bit 14 is RO.	Upstream Port(s), offset F60h[15]=0	RW	Yes	0-0h

Register 16-13. 34h Capability Pointer

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability Pointer Default 40h points to the PCI Power Management Capability structure.	RO	Yes	40h
31:8	Reserved	RsvdP	No	0000_00h

Register 16-14. 3Ch PCI Interrupt

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	PCI Interrupt Line The Interrupt Line Routing Value communicates interrupt line routing information. Values within this register are programmed by system software and are system architecture-specific. The value is used by device drivers and operating systems.	RW	Yes	00h
15:8	PCI Interrupt Pin Identifies the Conventional PCI Interrupt Message(s) the device (or device function) uses. Only value 00h or 01h is allowed in the PEX 8724. 00h = Device does not use Conventional PCI Interrupt Message(s) 01h, 02h, 03h, and 04h = Maps to Conventional PCI Interrupt Messages for INTA#, INTB#, INTC#, and INTD#, respectively	RO	Yes	01h
23:16	Min_Gnt Minimum Grant. Reserved Does not apply to PCI Express.	RsvdP	No	00h
31:24	Max_Lat Maximum Latency. Reserved Does not apply to PCI Express.	RsvdP	No	00h

16.6 NT Port Link Interface PCI Power Management Capability Registers (Offsets 40h – 44h)

This section details the NT Port Link Interface PCI Power Management Capability registers. [Table 16-3](#) defines the register map.

Table 16-3. NT Port Link Interface PCI Power Management Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
PCI Power Management Capability																Next Capability Pointer (48h)								Capability ID (01h)								40h
PCI Power Management Data								PCI Power Management Control/Status Bridge Extensions <i>(Reserved)</i>								PCI Power Management Status and Control																44h

Register 16-15. 40h PCI Power Management Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability ID Default = 01h – only value allowed.	RO	Yes	01h
15:8	Next Capability Pointer Default 48h points to the MSI Capability structure.	RO	Yes	48h
18:16	Version Default = 011b – only value allowed.	RO	Yes	011b
19	PME Clock Power Management Event (PME) clock. Does not apply to PCI Express. Returns a value of 0.	RsvdP	No	0
20	Reserved	RsvdP	No	0
21	Device-Specific Initialization 0 = Device-Specific Initialization is <i>not</i> required	RO	Yes	0
24:22	AUX Current The PEX 8724 does <i>not support</i> PME generation from the D3cold Device PM state; therefore, the serial EEPROM value for this field should be 000b.	RO	Yes	000b
25	D1 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D1 Device PM state	RsvdP	No	0
26	D2 Support <i>Not supported</i> 0 = PEX 8724 does <i>not support</i> the D2 Device PM state	RsvdP	No	0
31:27	PME Support The default value is applied to bits [31, 30, and 27] only. PME Messages are disabled, by default.	RO	Yes	0000_0b

Register 16-16. 44h PCI Power Management Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Status and Control				
1:0	Power State Used to determine the Port's current Device PM state, and to Set the Port into a new Device PM state. 00b = D0 01b = D1 – <i>Not supported</i> 10b = D2 – <i>Not supported</i> 11b = D3hot If software attempts to write an unsupported state to this field, the Write operation completes normally; however, the data is discarded and no state change occurs.	RW	Yes	00b
2	Reserved	RsvdP	No	0
3	No Soft Reset 1 = Devices transitioning from the D3hot to D0 Device PM state, because of Power State commands, do not perform an internal reset	ROS	Yes	1
7:4	Reserved	RsvdP	No	0h
8	PME Enable Default value of 0 indicates that PME generation is disabled.	RsvdP	No	0
12:9	Data Select Initially writable by serial EEPROM and/or I ² C only ^a . This Configuration Space register (CSR) access privilege changes to RW after a Serial EEPROM and/or I ² C Write occurs to this register. Selects the Power Number at a particular state (for field [14:13] (<i>Data Scale</i>) and PCI Power Management Data register <i>Data</i> field (offset 44h[31:24])). 0h = D0 Device PM state power consumed 3h = D3hot Device PM state power consumed 4h = D0 Device PM state power dissipated 7h = D3hot Device PM state power dissipated All other encodings are Reserved .	RO	Yes	0h
14:13	Data Scale Writable by serial EEPROM and/or I ² C only ^a . Indicates the scaling factor to be used when interpreting the PCI Power Management Data register <i>Data</i> field (offset 44h[31:24]) value. The value and meaning of the <i>Data Scale</i> field varies, depending upon which data value is selected by field [12:9] (<i>Data Select</i>). There are four internal <i>Data Scale</i> fields (one each, per <i>Data Select</i> values 0h, 3h, 4h, and 7h). For other <i>Data Select</i> values, the <i>Data Scale</i> value returned is 0h.	RO	Yes	00b
15	PME Status 0 = PME is not being generated by the NT Port	RsvdP	No	0

Register 16-16. 44h PCI Power Management Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Power Management Control/Status Bridge Extensions				
21:16	Reserved	RsvdP	No	0-0h
22	B2/B3 Support Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
23	Bus Power/Clock Control Enable Reserved Cleared, as required by the <i>PCI Power Mgmt. r1.2</i> .	RsvdP	No	0
PCI Power Management Data				
31:24	Data Writable by serial EEPROM and/or I ² C only ^a . There are four supported <i>Data Select</i> values (0h, 3h, 4h, and 7h). For other <i>Data Select</i> values, the <i>Data</i> value returned is 00h. Selected by the PCI Power Management Status and Control register <i>Data Select</i> field (offset 44h[12:9]).	RO	Yes	00h

- a. With no serial EEPROM nor previous I²C programming, Reads return a value of 00h for the **PCI Power Management Status and Control** register *Data Scale* and **PCI Power Management Data** register *Data* fields (for all *Data Selects*).

16.7 NT Port Link Interface MSI Capability Registers (Offsets 48h – 64h)

The registers detailed in [Section 13.9, “Message Signaled Interrupt Capability Registers \(Offsets 48h – 64h\),”](#) are also applicable to the NT Port Link Interface, except as defined in [Table 16-4](#) (register map), and [Register 16-17](#) through [Register 16-19](#).

Table 16-4. NT Port Link Interface MSI Capability Register Map^a

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
MSI Control																Next Capability Pointer (68h)										Capability ID (05h)										48h
MSI Address																																4Ch				
MSI Upper Address																																50h				
Reserved																MSI Data																54h				
MSI Mask																																58h				
MSI Status																																5Ch				
Reserved																																60h – 64h				

- a. Offsets 54h, 58h, and 5Ch change to 50h, 54h, and 58h, respectively, when the **MSI Control** register **MSI 64-Bit Address Capable** bit (offset 48h[23]) is Cleared.

Register 16-17. 48h MSI Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
MSI Capability Header				
7:0	Capability ID Program to 05h, as required by the <i>PCI r3.0</i> .	RO	Yes	05h
15:8	Next Capability Pointer Program to 68h, to point to the PCI Express Capability structure.	RO	Yes	68h
MSI Control				
16	MSI Enable 0 = MSIs for the NT Port Link Interface are disabled 1 = MSIs for the NT Port Link Interface are enabled, and INTx Interrupt Messages and PEX_INTA# (Base mode) or VSx_PEX_INTA# (Virtual Switch mode) output assertion are disabled	RW	Yes	0
19:17	Multiple Message Capable 000b = NT Port Link Interface can request only one MSI Vector 001b = NT Port Link Interface can request two MSI Vectors 010b = NT Port Link Interface can request four MSI Vectors All other encodings are Reserved .	RO	Yes	010b
22:20	Multiple Message Enable 000b = NT Port Link Interface is allocated one MSI Vector, by default 001b = NT Port Link Interface is allocated two MSI Vectors 010b = NT Port Link Interface is allocated four MSI Vectors All other encodings are Reserved . Note: This field should not be programmed with a value larger than that of field [19:17] (Multiple Message Capable). If the value of this field is larger than that of field [19:17], the Multiple Message Capable value takes effect.	RW	Yes	000b
23	MSI 64-Bit Address Capable 0 = PEX 8724 is capable of generating MSI 32-bit addresses (MSI Address register, offset 4Ch, is the Message address) 1 = PEX 8724 is capable of generating MSI 64-bit addresses (MSI Address register, offset 4Ch, is the lower 32 bits of the Message address, and MSI Upper Address register, offset 50h, is the upper 32 bits of the Message address)	RO	Yes	1
24	Per Vector Masking Capable 0 = PEX 8724 does not have Per Vector Masking capability 1 = PEX 8724 has Per Vector Masking capability	RO	Yes	1
31:25	Reserved	RsvdP	No	0-0h

Register 16-18. 58h MSI Mask

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
NT Port Link Interface interrupt sources are grouped into two categories – Link State-/PM-triggered interrupt events and NT-Link Doorbell-generated interrupts. The quantity of MSI Vectors that are generated is determined by the MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (NT Port Link Interface, offset 48h[19:17 and 22:20], respectively): • If one MSI Vector is enabled (default), both interrupt events are combined into a single Vector • If two MSI Vectors are allocated, the individual Vectors indicate: – Vector[0] Link State-/PM-triggered interrupt events – Vector[1] NT-Link Doorbell-generated interrupts Notes: The offset for this register changes from 58h, to 54h, when the MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be used to mask their respective MSI Status register bits (offset 5Ch).					
0	MSI Mask for Link State Events MSI mask for Power Management- or Link State-triggered interrupt events.	Offset 48h[22:20]=001b	RW	Yes	0
	MSI Mask for Shared Interrupt Sources MSI mask for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	Offset 48h[22:20]=000b	RW	Yes	0
2:1	Reserved		RsvdP	No	00b
3	MSI Mask for NT-Link Doorbell-Generated Interrupts Refer to NT Port registers located at offsets C5Ch through C68h.	Offset 48h[22:20]=001b	RW	Yes	0
	Reserved	Offset 48h[22:20]=000b	RsvdP	No	0
31:4	Reserved		RsvdP	No	0000_000h

Register 16-19. 5Ch MSI Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
NT Port Link Interface interrupt sources are grouped into two categories – Link State-/PM-triggered interrupt events and NT-Link Doorbell-generated interrupts. The quantity of MSI Vectors that are generated is determined by the MSI Control register <i>Multiple Message Capable</i> and <i>Multiple Message Enable</i> fields (NT Port Link Interface, offset 48h[19:17 and 22:20], respectively): - If one MSI Vector is enabled (default), both interrupt events are combined into a single Vector - If two MSI Vectors are allocated, the individual Vectors indicate: - Vector[0] Link State-/PM-triggered interrupt events - Vector[1] NT-Link Doorbell-generated interrupts Notes: The offset for this register changes from 5Ch, to 58h, when the MSI Control register <i>MSI 64-Bit Address Capable</i> bit (offset 48h[23]) is Cleared. The bits in this register can be masked by their respective MSI Mask register bits (offset 58h).					
0	MSI Pending Status for Link State Events MSI pending status for Power Management- or Link State-triggered interrupt events.	Offset 48h[22:20]=001b	RO	No	0
	MSI Pending Status for Shared Interrupt Sources MSI pending status for all interrupt sources when the MSI Control register <i>Multiple Message Enable</i> field indicates that the Host has allocated one or two Vectors.	Offset 48h[22:20]=000b	RO	No	0
2:1	Reserved		RsvdP	No	00b
3	MSI Pending Status for NT-Link Doorbell-Generated Interrupts Refer to NT Port registers located at offsets C5Ch through C68h.	Offset 48h[22:20]=001b	RO	No	0
	Reserved	Offset 48h[22:20]=000b	RsvdP	No	0
31:4	Reserved		RsvdP	No	0000_000h

16.8 NT Port Link Interface PCI Express Capability Registers (Offsets 68h – A0h)

The registers detailed in [Section 13.10, “PCI Express Capability Registers \(Offsets 68h – A0h\),”](#) are also applicable to the NT Port Link Interface, except as defined in [Table 16-5](#) (register map; offsets 7Ch and 80h are **Reserved**), and [Register 16-20](#) through [Register 16-27](#).

Table 16-5. NT Port Link Interface PCI Express Capability Register Map

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0	Offsets
PCI Express Capability		Next Capability Pointer (C8h)	Capability ID (10h)	68h
Device Capability				6Ch
Device Status		Not Supported/Reserved	Device Control	70h
Link Capability				74h
Link Status		Reserved	Link Control	78h
Reserved				7Ch – 88h
Device Capability 2				8Ch
Device Status 2 (Reserved)		Device Control 2		90h
Link Capability 2				94h
Link Status 2		Link Control 2		98h
Reserved				9Ch – A0h

Register 16-20. 68h PCI Express Capability List and Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
PCI Express Capability List				
7:0	Capability ID Program to 10h, by default.	RO	Yes	10h
15:8	Next Capability Pointer Program to C8h, to point to the Vendor-Specific Capability 3 structure.	RO	Yes	C8h
PCI Express Capability				
19:16	Capability Version The PEX 8724 NT Port Link Interface programs this field to 2h, as required by the <i>PCI Express Base r3.0</i> .	RO	Yes	2h
23:20	Device/Port Type Default = PCI Express endpoint device.	RO	No	0h
24	Slot Implemented <i>Not valid for PCI Express endpoint devices</i>	RsvdP	No	0
29:25	Interrupt Message Number The serial EEPROM writes 00_000b, because the Base Message and MSI Messages are the same.	RO	Yes	00_000b
31:30	Reserved	RsvdP	No	00b

Register 16-21. 6Ch Device Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	Maximum Payload Size Supported <i>Note:</i> The default Maximum Payload Size supported is based upon the STRAP_STNx_PORTCFGx 3-state inputs. If the serial EEPROM or I²C/SMBus changes the Port configuration (from the strapped value), it must also program the value of this field, accordingly (with the value dependent upon the Station having the most enabled Ports). The value for all Ports is determined by the Station having the most enabled Ports: - One or two enabled Ports = 100b - Three enabled Ports = 011b - Four enabled Ports = 010b 000b = NT Port Link Interface supports a 128-byte maximum payload 001b = NT Port Link Interface supports a 256-byte maximum payload 010b = NT Port Link Interface supports a 512-byte maximum payload 011b = NT Port Link Interface supports a 1,024-byte maximum payload 100b = NT Port Link Interface supports a 2,048-byte maximum payload No other encodings are supported.	HwInit	Yes	Defined by STRAP_STNx_PORTCFGx input states, or programmed by serial EEPROM value for the Port Configuration register <i>Port Configuration for Station x</i> field(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 300h[5:3 and/or 2:0])
4:3	Phantom Functions Supported <i>Not supported</i>	RO	Yes	00b
5	Extended Tag Field Supported 0 = Maximum <i>Tag</i> field is 5 bits 1 = Maximum <i>Tag</i> field is 8 bits	RO	Yes	0
8:6	Endpoint L0s Acceptable Latency 111b = No Limit	RO	Yes	111b
11:9	Endpoint L1 Acceptable Latency 111b = No Limit	RO	Yes	111b
14:12	Reserved	RsvdP	No	000b
15	Role-Based Error Reporting	RO	Yes	1

Register 16-21. 6Ch Device Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
17:16	Reserved	RsvdP	No	00b
25:18	Captured Slot Power Limit Value For the NT Port Link Interface, the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [27:26] (<i>Captured Slot Power Limit Scale</i>).	RO	Yes	00h
27:26	Captured Slot Power Limit Scale For the NT Port Link Interface, the upper limit on power supplied by the slot is determined by multiplying the value in this field by the value in field [25:18] (<i>Captured Slot Power Limit Value</i>). 00b = 1.0 01b = 0.1 10b = 0.01 11b = 0.001	RO	Yes	00b
31:28	Reserved	RsvdP	No	0h

Register 16-22. 70h Device Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Control				
0	Correctable Error Reporting Enable 0 = Disables 1 = Enables the NT Port Link Interface to report Correctable errors to the Link side Host	RW	Yes	0
1	Non-Fatal Error Reporting Enable 0 = Disables 1 = Enables the NT Port Link Interface to report Non-Fatal errors to the Link side Host	RW	Yes	0
2	Fatal Error Reporting Enable 0 = Disables 1 = Enables the NT Port Link Interface to report Fatal errors to the Link side Host	RW	Yes	0
3	Unsupported Request Reporting Enable 0 = Disables 1 = Enables the NT Port Link Interface to report UR errors as Error Messages with a programmed uncorrectable error severity	RW	Yes	0
4	Enable Relaxed Ordering <i>Not supported</i>	RW	Yes	1
7:5	Maximum Payload Size The NT Port Link Interface power-on/reset value is 000b, to support a Maximum Payload Size of 128 bytes. Software can change this field to configure the NT Port Link Interface to support other Payload sizes; however, software cannot change this field to a value larger than that indicated by the Device Capability register <i>Maximum Payload Size Supported</i> field (offset 6Ch[2:0]), for the NT Port Virtual and Link Interfaces. (Requester and Completer domains must possess the same Maximum Payload Size.) 000b = NT Port Link Interface supports a 128-byte maximum payload 001b = NT Port Link Interface supports a 256-byte maximum payload 010b = NT Port Link Interface supports a 512-byte maximum payload 011b = NT Port Link Interface supports a 1,024-byte maximum payload 100b = NT Port Link Interface supports a 2,048-byte maximum payload No other encodings are supported. Note: Software must halt all transactions through the NT Port before changing this field.	RW	Yes	000b

Register 16-22. 70h Device Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
8	Extended Tag Field Enable <i>Not supported</i>	RsvdP	No	0
9	Phantom Functions Enable <i>Not supported</i>	RsvdP	No	0
10	AUX Power PM Enable <i>Not supported</i>	RsvdP	No	0
11	Enable No Snoop <i>Not supported</i>	RW	Yes	1
14:12	Maximum Read Request Size <i>Not supported</i>	RsvdP	No	000b
15	Reserved	RsvdP	No	0
Device Status				
16	Correctable Error Detected 0 = NT Port Link Interface did not detect a Correctable error 1 = NT Port Link Interface detected a Correctable error, regardless of the bit 0 (<i>Correctable Error Reporting Enable</i>) state.	RW1C	Yes	0
17	Non-Fatal Error Detected 0 = NT Port Link Interface did not detect a Non-Fatal error 1 = NT Port Link Interface detected a Non-Fatal error, regardless of the bit 1 (<i>Non-Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
18	Fatal Error Detected 0 = NT Port Link Interface did not detect a Fatal error 1 = NT Port Link Interface detected a Fatal error, regardless of the bit 2 (<i>Fatal Error Reporting Enable</i>) state	RW1C	Yes	0
19	Unsupported Request Detected 0 = NT Port Link Interface did not detect a UR 1 = NT Port Link Interface detected a UR, regardless of the bit 3 (<i>Unsupported Request Reporting Enable</i>) state	RW1C	Yes	0
20	AUX Power Detected <i>Not supported</i>	RsvdP	No	0
21	Transactions Pending <i>Not supported</i> Because the NT Port is a bridging device, it does not track Completion for the corresponding Non-Posted transactions. Therefore, the NT Port Link Interface does not implement this bit.	RsvdP	No	0
31:22	Reserved	RsvdP	No	0-0h

Register 16-23. 74h Link Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
Note: Table 13-5 indicates which Ports are enabled/used/ Software Only , and when, based upon the STRAP_STN _x _PORTCFG _x input states and/or serial EEPROM programming. The table also lists the relationship between the Ports and Lanes (Link widths).					
3:0	Maximum Link Speed Indicates the NT Port Link Interface’s maximum Link speed. The encoding is the binary value of the bit location in the Link Capability 2 register <i>Supported Link Speeds Vector</i> field (NT Port Link Interface, offset 94h[7:1]) that corresponds to the maximum Link speed. <i>For example</i> , the default value 3h indicates that the maximum Link speed is that which corresponds to bit 3 in the <i>Supported Link Speeds Vector</i> field, which is 8.0 GT/s. The default value for all Ports can be globally changed to 2h (Gen 2 maximum speed) or 1h (Gen 1 maximum speed), according to the STRAP_GEN1_GEN2 input state (Z = 2h (Gen 2), or 0 = 1h (Gen 1)). 1h = Gen 1 (2.5 GT/s) 2h = Gen 2 (5.0 GT/s) 3h = Gen 3 (8.0 GT/s)	RO	Yes	3h	
9:4	Maximum Link Width The PEX 8724 maximum Link width is x16 = 01_0000b. Actual maximum Link width is defined by the STRAP_STN _x _PORTCFG _x 3-state inputs. 00_0000b = Reserved 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are not supported .	Refer to Default value	No	Defined by STRAP_STN _x _PORTCFG _x input states, or programmed by serial EEPROM value for the Port Configuration -related registers (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offsets 300h through 308h)	
11:10	Active State Power Management (ASPM) Support Active State Link PM support. Indicates the level of ASPM supported by the NT Port Link Interface. 00b = Reserved 01b = ASPM L0s Link PM state entry is supported 10b = ASPM L1 Link PM state is supported (Silicon Revision CA only) 11b = ASPM L0s and L1 Link PM states are supported	RO	Yes	11b	10b

Register 16-23. 74h Link Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
14:12	L0s Exit Latency Indicates the ASPM L0s Link PM state exit latency for the Port's PCI Express Link, when the state is supported and enabled. The value reported indicates the length of time this Port requires to complete the ASPM L0s to L0 Link PM state transition. If the ASPM L0s Link PM state is not supported, the value of this field is undefined; however, for the recommended value, refer to the Implementation Note in the <i>PCI Express Base r3.0</i>, Section 5.4.1.1, "Potential Issues with Legacy Software When L0s is Not Supported." 000b = Less than 64 ns 001b = 64 ns to less than 128 ns 010b = 128 ns to less than 256 ns 011b = 256 ns to less than 512 ns 100b = 512 ns to less than 1 μs 101b = 1 μs to less than 2 μs 110b = 2 to 4 μs (default) 111b = More than 4 μs	RO	No	110b	
17:15	L1 Exit Latency Indicates the ASPM L1 Link PM state exit latency for the Port's PCI Express Link. The value reported indicates the length of time this Port requires to complete the ASPM L1 to L0 Link PM state transition. If the L0 Link PM state is not supported, the value of this field is undefined. 000b = Less than 1 μs 001b = 1 μs to less than 2 μs (5.0 GT/s) 010b = 2 μs to less than 4 μs (default, 2.5 GT/s) 011b = 4 μs to less than 8 μs 100b = 8 μs to less than 16 μs 101b = 16 μs to less than 32 μs 110b = 32 to 64 μs 111b = More than 64 μs	RO	Yes	010b	
18	Clock Power Management	RO	Yes	0	
21:19	Reserved	RsvdP	No	000b	
22	ASPM Optionality Compliance	HwInit	Yes	1	
23	Reserved	RsvdP	No	0	

Register 16-23. 74h Link Capability (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision		
				BA	CA	
31:24	NT Port Number The NT Port Number value is determined by the STRAP_NT_UPSTRM_PORTSEL[2:0] input states, combined with the VS0 Upstream register <i>NT Port</i> field (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 360h[12:8]) value.			Refer to Default value	No	Defined by STRAP_NT_UPSTRM_PORTSEL [2:0] input states
	STRAP_NT_UPSTRM_PORTSEL[2:0] Input States	Offset 360h[12:8] Value	Port Number			
	000	0_0000b	0			
	00Z	0_0001b	1			
	001	0_0010b	2			
	0Z0	0_0011b	3			
	011	0_1000b	8			
	Z00	0_1001b	9			
	Z0Z	0_1010b	10			
	All other encodings are Reserved .					

Register 16-24. 78h Link Status and Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Control				
1:0	Active State Power Management (ASPM) Control 00b = Disable ^a 01b = Enables only ASPM L0s Link PM state entry 10b = Enables only ASPM L1 Link PM state entry 11b = Enables both ASPM L0s and L1 Link PM state entries	RW	Yes	00b
2	Reserved	RsvdP	No	0
3	Read Request Return Parameter Control Read Request Return Parameter “R” control. Read Completion Boundary (RCB).	RO	Yes	0
4	Reserved Offset F70h[11]=0	RsvdP	No	0
	Link Disable Functionality is enabled when the Power Management Hot Plug User Configuration register <i>PHY Upstream Port Control Write Enable</i> bit (NT Port Link Interface, offset F70h[11]) is Set. 1 = Places the NT Port Link Interface’s Link into the <i>Disabled</i> Link Training state	Offset F70h[11]=1 RW	Yes	0
5	Reserved Always returns a value of 0 when read; however, software is allowed to write this register.	Offset F70h[11]=0 RsvdP	No	0
	Retrain Link Functionality is enabled when the Power Management Hot Plug User Configuration register <i>PHY Upstream Port Control Write Enable</i> bit (NT Port Link Interface, offset F70h[11]) is Set. Always returns a value of 0 when read; however, software is allowed to write this register. Writing 1 to this bit causes the NT Port Link Interface to initiate retraining of its PCI Express Link.	Offset F70h[11]=1 RZ	Yes	0
6	Common Clock Configuration 0 = NT Port Link Interface and the device at the other end of the Port’s PCI Express Link uses an asynchronous Reference Clock source 1 = NT Port Link Interface and the device at the other end of the Port’s PCI Express Link uses a common Reference Clock source (constant phase relationship)	RW	Yes	0
7	Extended Sync Setting this bit causes the NT Port Link Interface to transmit: • 4,096 FTS Ordered-Sets in the ASPM L0s Link PM state, • Followed by a single SKIP Ordered-Set prior to entering the L0 Link PM state, • Finally, transmission of 1,024 TS1 Ordered-Sets in the Recovery state.	RW	Yes	0
15:8	Reserved	RsvdP	No	00h

Register 16-24. 78h Link Status and Control (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Status				
19:16	Current Link Speed Indicates the negotiated Link speed of the Port's PCI Express Link. The encoding is the binary value of the bit location in the Link Capability 2 register Supported Link Speeds Vector field (NT Port Link Interface, offset 94h[7:1]) that corresponds to the current Link speed. <i>For example</i> , a value of 1h indicates that the current Link speed is that which corresponds to bit 1 in the Supported Link Speeds Vector field, which is 2.5 GT/s. If the Link is not up, the value of this field is undefined.	RO	No	1h
25:20	Negotiated Link Width Dependent upon the configuration of the physical Ports. Link width is determined by the negotiated value with the attached Lane/Port. If the Link is not up, the value of this field is undefined. 00_0000b = Link is Down (default) 00_0001b = x1 00_0010b = x2 00_0100b = x4 00_1000b = x8 01_0000b = x16 (Port 0, Station 0 only) All other encodings are not supported .	RO	No	00_0000b
26	Reserved	RsvdP	No	0
27	Link Training Reserved for the NT Port Link Interface.	RsvdP	No	0
28	Slot Clock Configuration Set by the Upstream Port(s) or NT Port Link Interface, but not both. 0 = PEX 8724 uses an independent clock 1 = PEX 8724 uses the same physical Reference Clock that the platform provides on the connector	HwInit	Yes	0
31:29	Reserved	RsvdP	No	000b

- a. The Port Receiver must be capable of entering the ASPM L0s Link PM state, regardless of whether the state is disabled.

Register 16-25. 8Ch Device Capability 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
6:0	<i>Reserved</i>	RsvdP	No	0-0h
7	AtomicOp 32-Bit Completer Supported 1 = Supports Atomic 32-bit Completer	ROS	Yes	1
8	AtomicOp 64-Bit Completer Supported 1 = Supports Atomic 64-bit Completer	ROS	Yes	1
9	AtomicOp 128-Bit Completer Supported 1 = Supports Atomic 128-bit Completer	ROS	Yes	1
31:10	<i>Reserved</i>	RsvdP	No	0000_01h

Register 16-26. 90h Device Status and Control 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Device Control 2				
5:0	<i>Reserved</i>	RsvdP	No	0-0h
6	AtomicOp Requester Enable	RW	Yes	0
15:7	<i>Reserved</i>	RsvdP	No	0-0h
Device Status 2				
31:16	<i>Reserved</i>	RsvdP	No	0000h

Register 16-27. 98h Link Status and Control 2

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Control 2				
3:0	Target Link Speed Sets the Target Compliance mode speed when software is using bit 4 (<i>Enter Compliance</i>) to force a Link into Compliance mode.	RWS	Yes	3h
4	Enter Compliance Software is permitted to force a Link to enter Compliance mode at the speed indicated in field [3:0] (<i>Target Link Speed</i>), by Setting this bit in both components on a Link, and then initiating a Hot Reset on the Link. The default value of this bit following a Fundamental Reset is 0.	RWS	Yes	0
5	Hardware Autonomous Speed Disable <i>Reserved</i> Initial transition to the highest supported common Link speed is not blocked by this bit.	RsvdP	No	0
6	Selectable De-Emphasis <i>Reserved</i>	RsvdP	Yes	0
9:7	Transmit Margin Intended for debug and compliance testing only.	RWS	Yes	000b
10	Enter Modified Compliance Intended for debug and compliance testing only.	RWS	Yes	0
11	Compliance SOS <i>Applicable only when the Link is operating at 2.5 or 5.0 GT/s.</i> 1 = Link Training and Status State Machine (LTSSM) must periodically send SKIP Ordered-Sets between sequences when sending the Compliance Pattern or Modified Compliance Pattern	RWS	Yes	0
15:12	Compliance Preset/De-Emphasis Intended for debug and compliance testing only. 8.0 GT/s Data Rate Sets the Transmitter Preset in the <i>Polling.Compliance</i> substate, if the entry occurred due to bit 4 (<i>Enter Compliance</i>) being Set. The encodings are defined in the <i>PCI Express Base r3.0</i> , Section 4.2.3.2, “Encoding of Presets.” Results are undefined if a <i>Reserved</i> preset encoding is used when entering the <i>Polling.Compliance</i> substate in this way. 5.0 GT/s Data Rate Sets the de-emphasis level in the <i>Polling.Compliance</i> substate, if the entry occurred due to bit 4 (<i>Enter Compliance</i>) being Set. 0h = -6 dB (Link is operating at 5.0 GT/s) 1h = -3.5 dB (Link is operating at 2.5 GT/s) 2.5 GT/s Data Rate The programming of this field has no effect.	RWS	Yes	0h

Register 16-27. 98h Link Status and Control 2 (Cont.)

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
Link Status 2				
16	Current De-Emphasis Level <i>Applicable only for Gen 2; for Gen 1, this bit should be hardwired to 0.</i> Reflects the de-emphasis level when the Link is operating at 5.0 GT/s. 0 = -6 dB (Link is operating at 5.0 GT/s) 1 = Undefined	RO	No	0
17	Equalization Complete 1 = Transmitter Equalization procedure has completed	ROS	Yes	0
18	Equalization Phase 1 Successful 1 = Phase 1 of the Transmitter Equalization procedure has successfully completed	ROS	Yes	0
19	Equalization Phase 2 Successful 1 = Phase 2 of the Transmitter Equalization procedure has successfully completed	ROS	Yes	0
20	Equalization Phase 3 Successful 1 = Phase 3 of the Transmitter Equalization procedure has successfully completed	ROS	Yes	0
21	Link Equalization Request Set by hardware, to request the Link equalization process to be performed on the Link.	Differs by Silicon Revision	Yes	0
	Silicon Revision BA	RW1C		
	Silicon Revision CA	RW1CS		
31:22	Reserved	RsvdP	No	0-0h

16.9 NT Port Link Interface Vendor-Specific Capability 3 Registers (Offsets C8h – FCh)

This section details the NT Port Link Interface Vendor-Specific Capability 3 registers. [Table 16-6](#) defines the register map used by the NT Port Link Interface.

The Cursor Mechanism registers at offsets [F8h](#) and [FCh](#) provide a means for accessing PCI Express Extended Configuration Space registers (offsets 100h through FFFh) within the NT Port Virtual and Link Interfaces, when only standard PCI Configuration transactions (that do not support the *Extended Register Number* field within the Completion Request Header) are available.

Table 16-6. NT Port Link Interface Vendor-Specific Capability 3 Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets	
Reserved	Next Capability Pointer (00h)	Capability ID 3 (09h)	C8h
Reserved			CCh – E0h
NT Port Link Interface BAR0/1 Setup			E4h
NT Port Link Interface Memory BAR2 Setup			E8h
NT Port Link Interface Memory BAR2/3 Setup			ECh
NT Port Link Interface Memory BAR4 Setup			F0h
NT Port Link Interface Memory BAR4/5 Setup			F4h
Configuration Address Window	Reserved		F8h
Configuration Data Window			FCh

Register 16-28. C8h Vendor-Specific Capability 3

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Capability ID 3	ROS	Yes	09h
15:8	Next Capability Pointer 00h = This capability is the last capability in the Linked List	ROS	Yes	00h
31:16	<i>Reserved</i>	RsvdP	No	0000h

Register 16-29. E4h NT Port Link Interface BAR0/1 Setup

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	BAR0/1 Enable 00b = Disables the NT Port Link Interface BAR0 and BAR1 01b = <i>Reserved</i> 10b = Enables the NT Port Link Interface BAR0 and disables BAR1 (BAR0 is a 32-bit BAR) 11b = Enables the NT Port Link Interface BAR0 and BAR1 (BAR0/1 is a 64-bit BAR)	RWS	No	10b
2	BAR0 Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	RWS	No	0
31:3	<i>Reserved</i>	RsvdP	No	0-0h

Register 16-30. E8h NT Port Link Interface Memory BAR2 Setup

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default	
0	Type Selector	RsvdP	No	0	
2:1	BAR2 Type 00b = BAR2 is implemented as a 32-bit Memory BAR 10b = BAR2/3 is implemented as a 64-bit Memory BAR No other encodings are allowed.	RWS	Yes	00b	
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	RWS	Yes	0	
19:4	Reserved	RsvdP	No	0_000h	
30:20	BAR2 Size Specifies the Address Range size requested by BAR2 . 0 = Corresponding BAR2 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR2 bits are RWS bits	RWS	Yes	0-0h	
31	BAR2 Enable 0 = BAR2 is disabled, all BAR2 bits read 0 1 = BAR2 is enabled	Field [2:1] (<i>BAR2 Type</i>) = 00b	RWS	Yes	0
	BAR2 Size Included with field [30:20] when this BAR is used as a 64-bit BAR.	Field [2:1] (<i>BAR2 Type</i>) = 10b	RWS	Yes	0

Register 16-31. ECh NT Port Link Interface Memory BAR2/3 Setup

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
Note: This register has RWS privilege if BAR2/3 is configured as a 64-bit BAR (NT Port Link Interface Memory BAR2 Setup register BAR2 Type field (NT Port Link Interface, offset E8h[2:1]), is programmed to 10b).					
0	Type Selector	Offset E8h[2:1]=00b	RsvdP	No	0
		Offset E8h[2:1]=10b	RWS	Yes	0
2:1	BAR3 Type 00b = BAR3 is implemented as a 32-bit Memory BAR No other encodings are allowed.	Offset E8h[2:1]=00b	RsvdP	No	00b
		Offset E8h[2:1]=10b	RWS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable		RWS	Yes	0
19:4	Reserved	Offset E8h[2:1]=00b	RsvdP	No	0_000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset E8h[2:1]=10b	RWS	Yes	0_000h
30:20	BAR3 Size Specifies the Address Range size requested by BAR3 . 0 = Corresponding BAR3 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR3 bits are RWS bits		RWS	Yes	0-0h
31	BAR3 Enable 32-Bit BAR 0 = BAR3 is disabled 1 = BAR3 is enabled as a 32-bit BAR	Offset E8h[2:1]=00b	RWS	Yes	0
	64-Bit BAR 0 = BAR2/3 is disabled, all BAR2/3 bits read 0 1 = BAR2/3 is enabled as a 64-bit BAR	Offset E8h[2:1]=10b	RWS	Yes	0

Register 16-32. F0h NT Port Link Interface Memory BAR4 Setup

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
0	Type Selector		RsvdP	No	0
2:1	BAR4 Type 00b = BAR4 is implemented as a 32-bit Memory BAR (BAR4) 10b = BAR4/5 is implemented as a 64-bit Memory BAR (BAR4/5) No other encodings are allowed.		RWS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable		RWS	Yes	0
19:4	Reserved		RsvdP	No	0_000h
30:20	BAR4 Size Specifies the Address Range size requested by BAR4 . 0 = Corresponding BAR4 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR4 bits are RWS bits		RWS	Yes	0-0h
31	BAR4 Enable 0 = BAR4 is disabled, all BAR4 bits read 0 1 = BAR4 is enabled	Field [2:1] (<i>BAR4 Type</i>) = 00b	RWS	Yes	0
	BAR4 Size Included with field [30:20] when this BAR is used as a 64-bit BAR.	Field [2:1] (<i>BAR4 Type</i>) = 10b	RWS	Yes	0

Register 16-33. F4h NT Port Link Interface Memory BAR4/5 Setup

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
Note: This register has RWS privilege if BAR4/5 is configured as a 64-bit BAR (NT Port Link Interface Memory BAR4 Setup register BAR4 Type field (NT Port Link Interface, offset F0h[2:1]), is programmed to 10b).					
0	Type Selector	Offset F0h[2:1]=00b	RsvdP	No	0
		Offset F0h[2:1]=10b	RWS	Yes	0
2:1	BAR5 Type 00b = BAR5 is implemented as a 32-bit Memory BAR No other encodings are allowed.	Offset F0h[2:1]=00b	RsvdP	No	00b
		Offset F0h[2:1]=10b	RWS	Yes	00b
3	Prefetchable 0 = Non-Prefetchable 1 = Prefetchable	Offset F0h[2:1]=00b	RsvdP	No	0
		Offset F0h[2:1]=10b	RWS	Yes	0
19:4	Reserved	Offset F0h[2:1]=00b	RsvdP	No	0_000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:4]) are used as the upper 32 bits.	Offset F0h[2:1]=10b	RWS	Yes	0_000h
30:20	BAR5 Size Specifies the Address Range size requested by BAR5 . 0 = Corresponding BAR5 bits are RO bits that always return a value of 0, and Writes are ignored 1 = Corresponding BAR5 bits are RWS bits		RWS	Yes	0-0h
31	BAR5 Enable 32-Bit BAR 0 = BAR5 is disabled 1 = BAR5 is enabled as a 32-bit BAR	Offset F0h[2:1]=00b	RWS	Yes	0
	64-Bit BAR 0 = BAR4/5 is disabled, all BAR4/5 bits read 0 1 = BAR4/5 is enabled as a 64-bit BAR	Offset F0h[2:1]=10b	RWS	Yes	0

**Register 16-34. F8h Configuration Address Window
(Device-Specific Cursor Mechanism)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	<i>Reserved</i>	RsvdP	No	0000h
25:16	Register Offset	RW	Yes	0-0h
30:26	<i>Reserved</i>	RsvdP	No	0-0h
31	Interface Select 0 = Access is to the NT Port Link Interface Type 0 Configuration Space register 1 = Access is to the NT Port Virtual Interface Type 0 Configuration Space register	RW	Yes	0

**Register 16-35. FCh Configuration Data Window
(Device-Specific Cursor Mechanism)**

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Register Data Software selects a register by writing into the NT Port Link Interface Configuration Address window, then reads from or writes to that register using this register.	RW	Yes	0000_0000h

16.10 NT Port Link Interface Virtual Channel Extended Capability Registers (Offsets 148h – 1BCh)

The registers detailed in [Section 13.15, “Virtual Channel Extended Capability Registers \(Offsets 148h – 1BCh\),”](#) are also applicable to the NT Port Link Interface, except as defined in [Table 16-7](#) (register map; offsets 178h through 1B4h are **Reserved**), and [Register 16-36](#) through [Register 16-40](#).

Table 16-7. NT Port Link Interface Virtual Channel Extended Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets
Next Capability Offset (C34h)												Capability Version (1h)				PCI Express Extended Capability ID (0002h)																148h
Reserved														Port VC Capability 1												14Ch						
Port VC Capability 2																												150h				
Port VC Status (Reserved)												Port VC Control														154h						
Reserved												VC0 Resource Capability														158h						
VC0 Resource Control																												15Ch				
VC0 Resource Status												Reserved														160h						
Reserved																												164h – 1BCh				

Register 16-36. 148h Virtual Channel Extended Capability Header

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID Program to 0002h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	0002h
19:16	Capability Version Program to 1h, as required by the <i>PCI Express Base r3.0</i> .	RO	No	1h
31:20	Next Capability Offset Next extended capability is the Vendor-Specific Extended Capability 2 structure, offset C34h.	RO	No	C34h

Register 16-37. 14Ch Port VC Capability 1

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
2:0	Extended VC Counter 000b = NT Port Link Interface supports only one Virtual Channel, VC0 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .	RO	Yes	000b
3	Reserved	RsvdP	No	0
6:4	Low-Priority Extended VC Counter For Strict Priority arbitration, indicates the quantity of extended Virtual Channels (VCs) (those in addition to VC0) that belong to the Low-Priority VC group for the NT Port Link Interface. 000b = For the NT Port Link Interface, only VC0 belongs to the Low-Priority VC group 001b = Reserved , because the PEX 8724 supports only one VC All other encodings are Reserved .	RO	Yes	000b
7	Reserved	RsvdP	No	0
9:8	Reference Clock Reserved	RsvdP	No	00b
11:10	Port Arbitration Table Entry Size Reserved	RsvdP	No	00b
31:12	Reserved	RsvdP	No	0000_0h

Register 16-38. 158h VC0 Resource Capability

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
1:0	Port Arbitration Capability Bit 0 = 1 – Non-configurable Round-Robin (Hardware-Fixed) arbitration Bit 1 = 1 – Weighted Round-Robin (WRR) arbitration with 64 Phases	RO	No	00b
13:2	Reserved	RsvdP	No	0-0h
14	Advanced Packet Switching Reserved	RsvdP	No	0
15	Reject Snoop Transactions Not a PCI Express switch feature; therefore, this bit is Cleared. Reserved	RsvdP	No	0
22:16	Maximum Time Slots Not supported	RsvdP	No	000_0000b
23	Reserved	RsvdP	No	0
31:24	Port Arbitration Table Offset Reserved	RsvdP	No	00h

Register 16-39. 15Ch VC0 Resource Control

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	TC/VC Map Defines Traffic Classes [7:0], respectively, and indicates which TCs are mapped into VC0. Traffic Class 0 (TC0) must be mapped to VC0. By default, Traffic Classes [7:1] are mapped to VC0.	RO	No	1
7:1		RW	Yes	7Fh
15:8	Reserved	RsvdP	No	00h
16	Load Port Arbitration Table Hardware writable and software readable.	RW	Yes	0
19:17	Port Arbitration Select Selects the Port Arbitration type for the NT Port Link Interface. Indicates the bit number in the VC0 Resource Capability register <i>Port Arbitration Capability</i> field (offset 158h[1:0]) that corresponds to the arbitration type. 0 = Round-Robin (Hardware-Fixed) arbitration scheme	RW	Yes	000b
23:20	Reserved	RsvdP	No	0h
26:24	VC ID Defines the NT Port Link Interface VC0 ID code. 000b = VC0 (default; VC0 is the only/default VC) All other encodings are Reserved .	RO	No	000b
30:27	Reserved	RsvdP	No	0h
31	VC Enable 0 = Not allowed 1 = Enables the NT Port Link Interface VC0	RO	No	1

Register 16-40. 160h VC0 Resource Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Reserved	RsvdP	No	0000h
16	Port Arbitration Table Status <i>Not implemented</i>	RO	No	0
17	VC0 Negotiation Pending 0 = VC0 negotiation completed 1 = VC0 initialization is not complete for the NT Port Link Interface	RO	Yes	1
31:18	Reserved	RsvdP	No	0-0h

16.11 NT Port Link Interface Device-Specific Registers (Offsets 1C0h – A74h)

The registers detailed in [Section 13.16, “Device-Specific Registers \(Offsets 1C0h – A74h\),”](#) are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. These registers are also applicable to the NT Port Link Interface, except as defined in [Table 16-8](#) (register map; offsets 1D0h through 1D8h, 200h through 270h, 290h through 46Ch, 56Ch and 570h, 600h through 68Ch, 6A0h through 6A8h, 760h through 8FCh, 900h through 93Ch, 9ECh through A2Ch, and A30h through A54h, are **Reserved**) through [Table 16-10](#), and [Register 16-41](#).

Other NT Port Link Interface Device-Specific registers are detailed in:

- [Section 16.12, “NT Port Link Interface Device-Specific Registers \(Offsets B70h – C88h\)”](#)
- [Section 16.14, “NT Port Link Interface Device-Specific Registers \(Offsets F00h – FB0h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 16-8. NT Port Link Interface Device-Specific Register Map (Offsets 1C0h – A74h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
<i>Reserved</i>		1C0h – 1D8h
NT Port Link Interface Device-Specific Registers – Captured Bus Number (Offsets 1DC <h>h</h> – 1E4h)		1DC <h>h</h>
		...
		1E4h
<i>Reserved</i>		1E8h – 6FCh
NT Port Link Interface Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch)		700h
		...
		75Ch
<i>Reserved</i>		760h – A74h

16.11.1 NT Port Link Interface Device-Specific Registers – Captured Bus Number (Offsets 1DCh – 1E4h)

The **Captured Bus Number** register detailed in Section 13.16.2, “Device-Specific Registers – Captured Bus and Device Numbers (Offsets 1DCh – 1E4h),” is also applicable to the NT Port Link Interface, except as defined in Table 16-9 (register map; offset 1E4h is *Reserved*, rather than *Factory Test Only*), and Register 16-41.

Note: It is recommended that this register not be changed from its default value.

Table 16-9. NT Port Link Interface Device-Specific Captured Bus Number Register Map

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
NT Captured Bus Number																															1DCh	
Reserved																															1E0h – 1E4h	

Register 16-41. 1DCh NT Captured Bus Number

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
7:0	Captured Bus Number Captured Bus Number value for the NT Port Link Interface. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Bus Number. Note: Overwriting the Captured Bus Number value is not recommended.	RW	Yes	00h
12:8	Captured Device Number Captured Device Number value for the NT Port Link Interface. The value of this field can be overwritten, if bit 31 (<i>Captured BusDev Number Override</i>) is Set prior to changing the Captured Device Number. Note: Overwriting the Captured Device Number value is not recommended.	RW	Yes	0-0h
30:13	Reserved	RsvdP	No	0-0h
31	Captured BusDev Number Override 1 = Enables the Captured Bus Number and Device Number (fields [7:0 and 12:8], respectively) to be overridden	RW	Yes	0

16.11.2 NT Port Link Interface Device-Specific Registers – Error Checking and Debug (Offsets 700h – 75Ch)

The registers detailed in [Section 13.16.11, “Device-Specific Registers – Error Checking and Debug \(Offsets 700h – 75Ch\)”](#), are also applicable to the NT Port Link Interface, except as defined in [Table 16-10](#) (register map; offsets 700h through 720h, and 72Ch, are **Reserved** in the NT Port Link Interface).

Note: *It is recommended that these registers not be changed from their default values.*

Table 16-10. NT Port Link Interface Device-Specific Error Checking and Debug Register Map (Offsets 700h – 75Ch)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Reserved																700h – 720h																
Reserved																Gen 3 Framing Error Status																724h
Reserved																Gen 3 Framing Error Mask																728h
Reserved																72Ch – 75Ch																

16.12 NT Port Link Interface Device-Specific Registers (Offsets B70h – C88h)

The registers detailed in [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#) (for offsets B70h through C88h), are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. These registers are also applicable to the NT Port Link Interface, except as defined in [Table 16-11](#) (register map; offsets B80h through C30h are **Reserved**, and offsets C34h through C88h are associated with NT Port-specific registers) and [Table 16-12](#), and [Register 16-42](#) through [Register 16-45](#).

Other NT Port Link Interface Device-Specific registers are detailed in:

- [Section 16.11, “NT Port Link Interface Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 16.14, “NT Port Link Interface Device-Specific Registers \(Offsets F00h – FB0h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 16-11. NT Port Link Interface Device-Specific Register Map (Offsets B70h – C88h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16																15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0																Offsets				
Next Capability Offset 2 (000h)																1h				PCI Express Extended Capability ID 2 (000Bh)																B70h
Device-Specific Registers – Vendor-Specific Extended Capability 2 (Offsets B70h – B7Ch) ^a																																...	B7Ch			
Reserved																																B80h –	C30h			
Next Capability Offset 4 (B70h)																1h				PCI Express Extended Capability ID 4 (000Bh)																C34h
NT Port Link Interface Device-Specific Registers – Vendor-Specific Extended Capability 4 (Offsets C34h – C88h)																																...	C88h			

- a. Register offsets B70h through B7Ch are detailed in [Section 13.18, “Device-Specific Registers \(Offsets B70h – DFCh\)”](#)

Register 16-42. C3Ch Memory BAR2 Address Translation Lower

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19:0	Reserved	RsvdP	No	0_0000h
31:20	NT Port Link-to-Virtual Interface BAR2 Base Translation Address Base Translation address when BAR2 is enabled (NT Port Link Interface Memory BAR2 Setup register BAR2 Enable bit (NT Port Link Interface, offset E8h [31]) is Set).	RW	Yes	000h

Register 16-43. C40h Memory BAR3 Address Translation Upper

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
19:0	Reserved	Offset ECh[31]=0	RsvdP	No	0_0000h
	When BAR2/3 are used as a 64-bit BAR, bits [31:0] (including bits [19:0]) are used as the upper 32 bits.	Offset ECh[31]=1	RW	Yes	0_0000h
31:20	NT Port Link-to-Virtual Interface BAR3 Base Translation Address Base Translation address when BAR3 is enabled (NT Port Link Interface Memory BAR2/3 Setup register <i>BAR3 Enable</i> bit (NT Port Link Interface, offset ECh[31]) is Set).		RW	Yes	000h

Register 16-44. C44h Memory BAR4 Address Translation Lower

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
19:0	Reserved	RsvdP	No	0_0000h
31:20	NT Port Link-to-Virtual Interface BAR4 Base Translation Address Base Translation address when BAR4 is enabled (NT Port Link Interface Memory BAR4 Setup register BAR4 Enable bit (NT Port Link Interface, offset F0h [31]) is Set).	RW	Yes	000h

Register 16-45. C48h Memory BAR5 Address Translation Upper

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
19:0	Reserved	Offset F4h [31]=0	RsvdP	No	0_0000h
	When BAR4/5 are used as a 64-bit BAR, bits [31:0] (including bits [19:0]) are used as the upper 32 bits.	Offset F4h [31]=1	RW	Yes	0_0000h
31:20	NT Port Link-to-Virtual Interface BAR5 Base Translation Address Base Translation address when BAR5 is enabled (NT Port Link Interface Memory BAR4/5 Setup register BAR5 Enable bit (NT Port Link Interface, offset F4h [31]) is Set).		RW	Yes	000h

16.13 NT Port Link Interface NT Bridging-Specific Registers (Offsets C8Ch – EFCh)

Table 16-13 defines the register map of the NT Port Link Interface NT Bridging-Specific registers.

Table 16-13. NT Port Link Interface NT Bridging-Specific Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
NT Port Link Interface NT Bridging-Specific Registers – NT Port ID and Requester ID Read Back (Offsets C8Ch – C94h)		C8Ch ... C94h
Reserved		C98h – DB0h
NT Port Link Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry (Addresses DB4h – DF0h)		DB4h ... DF0h
NT Port Link Interface NT Bridging-Specific Registers – Lookup Table Enable Link (Offsets DF4h – DFCh)		DF4h ... DFCh
Reserved		E00h – EFCh

16.13.1 NT Port Link Interface NT Bridging-Specific Registers – NT Port ID and Requester ID Read Back (Offsets C8Ch – C94h)

The registers detailed in Section 15.13.1, “NT Port Virtual Interface NT Bridging-Specific Registers – NT Port ID (Offsets C8Ch – C94h),” are also applicable to the NT Port Link Interface, except as defined in Table 16-14 (register map; offset C90h is not *Reserved*), and Register 16-46 and Register 16-47.

Table 16-14. NT Port Link Interface NT Bridging-Specific NT Port ID and Requester ID Read Back Register Map

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
NT Port ID																												C8Ch				
Reserved																Requester ID Read Back												C90h				
Factory Test Only																												C94h				

Register 16-46. C8Ch NT Port ID

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
0	NT Port ID 0 = NT Port 1 = <i>Reserved</i>	ROS	No	0
30:1	<i>Reserved</i>	RsvdP	No	0-0h
31	Link Side Identifier	ROS	No	0

Register 16-47. C90h Requester ID Read Back

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	Requester ID Read Back Memory Read to this location returns the Requester ID of the last TLP.	RO	No	0000h
31:16	<i>Reserved</i>	RsvdP	No	0000h

16.13.2 NT Port Link Interface NT Bridging-Specific Registers – Requester ID Translation Lookup Table Entry (Addresses DB4h – DF0h)

This section details the NT Port Link Interface NT Bridging-Specific Requester ID Translation Lookup Table (LUT) Entry registers. The NT Port uses these registers for Requester ID translation when it forwards:

- Memory Requests from the NT Port Link-to-Virtual Interface, –or–
- Completion TLPs from the NT Port Virtual-to-Link Interface

If the application must send traffic through the NT Port Link Interface, program the registers listed in this group with the corresponding Requester's Requester ID, then Set the *LUT Entry_n Enable* and *LUT Entry_m Enable* bits (bits 0 and 16, respectively) for each LUT entry, as needed.

Table 16-15 defines the register and address locations, as they relate to Register 16-48.

Table 16-15. NT Port Link Interface NT Bridging-Specific Requester ID Translation LUT Entry_n_m Register Locations

ADDR Location	Lookup Table Entry_n_m	ADDR Location	Lookup Table Entry_n_m
DB4h	0_1	DD4h	16_17
DB8h	2_3	DD8h	18_19
DBCh	4_5	DDCh	20_21
DC0h	6_7	DE0h	22_23
DC4h	8_9	DE4h	24_25
DC8h	10_11	DE8h	26_27
DCCh	12_13	DECh	28_29
DD0h	14_15	DF0h	30_31

**Register 16-48. DB4h – DF0h NT Port Link Interface Requester ID Translation
LUT Entry $_n_m$ (where $_n_m = 0_1$ through 30_31)**

Bit(s)	Description		Type	Serial EEPROM and I ² C	Default
0	LUT Entry $_n$ Enable 0 = Disables 1 = Enables		RW	Yes	0
1	LUT Entry $_n$ No Snoop Enable If Set, the NT Port Clears the TLP <i>No Snoop</i> attribute bit for the Memory Request, then goes from the NT Port Link-to-Virtual Interface, and re-calculates the ECRC. If the original TLP has an ECRC error, the NT Port corrupts the re-calculated ECRC before transmitting to the other Host domain. The NT Port sets the <i>No Snoop</i> attribute bit when it forwards the Completion TLP from the NT Port Virtual-to-Link Interface if this bit is Set for the corresponding Requester ID entry. This ECRC rule applies to Completion TLPs as well. 0 = Disables 1 = Enables		RW	Yes	0
2	LUT Entry $_n$ Function Number Bit 2		RW	Yes	0
7:3	Requester ID on Link Side	Device Number LUT Entry $_n$ Requester Device Number.	RW	Yes	0000_0b
15:8		Bus Number LUT Entry $_n$ Requester Bus Number.	RW	Yes	00h
16	LUT Entry $_m$ Enable 0 = Disables 1 = Enables		RW	Yes	0
17	LUT Entry $_m$ No Snoop Enable If Set, the NT Port Clears the TLP <i>No Snoop</i> attribute bit for the Memory Request, then goes from the NT Port Link-to-Virtual Interface, and re-calculates the ECRC. If the original TLP has an ECRC error, the NT Port corrupts the re-calculated ECRC before transmitting to the other Host domain. The NT Port sets the <i>No Snoop</i> attribute bit when it forwards the Completion TLP from the NT Port Virtual-to-Link Interface if this bit is Set for the corresponding Requester ID entry. This ECRC rule applies to Completion TLPs as well. 0 = Disables 1 = Enables		RW	Yes	0
18	LUT Entry $_n$ Function Number Bit 2		RW	Yes	0
23:19	Requester ID on Link Side	Device Number LUT Entry $_m$ Requester Device Number.	RW	Yes	0000_0b
31:24		Bus Number LUT Entry $_m$ Requester Bus Number.	RW	Yes	00h

16.13.3 NT Port Link Interface NT Bridging-Specific Registers – Lookup Table Enable Link (Offsets DF4h – DFCh)

This section details the NT Port Link Interface NT Bridging-Specific Lookup Table Enable Link registers. [Table 16-16](#) defines the register map. Information describing when these registers are enabled, and how they are used, is provided in [Section 16.13.2](#).

Table 16-16. NT Port Link Interface NT Bridging-Specific Lookup Table Enable Link Register Map

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
Lookup Table Enable Link																																DF4h
Lookup Table No Snoop Enable Link																																DF8h
Lookup Table Function Enable Link																																DFCh

Register 16-49. DF4h Lookup Table Enable Link

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table Enable Enable bit per Lookup Table.	RWS	Yes	0000_0000h

Register 16-50. DF8h Lookup Table No Snoop Enable Link

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table No Snoop Enable No Snoop Enable bit per Lookup Table.	RWS	Yes	0000_0000h

Register 16-51. DFCh Lookup Table Function Enable Link

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
31:0	Lookup Table Function Check Enable Function Check Enable bit per Lookup Table.	RWS	Yes	0000_0000h

16.14 NT Port Link Interface Device-Specific Registers (Offsets F00h – FB0h)

The registers detailed in [Section 13.20, “Device-Specific Registers – Failover and Port Cut-Thru \(Offsets F00h – F20h\),”](#) and [Section 13.22, “Device-Specific Registers \(Offsets F30h – FB0h\),”](#) are unique to the PEX 8724 and not referenced in the *PCI Express Base r3.0*. These registers are also applicable to the NT Port Link Interface, except as defined in [Table 16-17](#) (register map; offsets F20h, F30h, F38h, F44h, and F4Ch through F54h are **Reserved**) through [Table 16-19](#), and [Register 16-52](#).

Other NT Port Link Interface Device-Specific registers are detailed in:

- [Section 16.11, “NT Port Link Interface Device-Specific Registers \(Offsets 1C0h – A74h\)”](#)
- [Section 16.12, “NT Port Link Interface Device-Specific Registers \(Offsets B70h – C88h\)”](#)

Note: *It is recommended that these registers not be changed from their default values.*

Table 16-17. NT Port Link Interface Device-Specific Register Map (Offsets F00h – FB0h)

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
NT Port Link Interface Device-Specific Registers – Failover (Offsets F00h – F20h)		F00h ... F20h
Reserved		F24h – F6Ch
NT Port Link Interface Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h)		F70h ... FB0h

16.14.1 NT Port Link Interface Device-Specific Registers – Failover (Offsets F00h – F20h)

The **VS Failover** register detailed in [Section 13.20, “Device-Specific Registers – Failover and Port Cut-Thru \(Offsets F00h – F20h\),”](#) is also applicable to the NT Port Link Interface, as defined in [Table 16-18](#) (register map; offset F20h is **Reserved** in the NT Port Link Interface).

Table 16-18. NT Port Link Interface Device-Specific Failover Register Map

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offsets
VS Failover																															F00h	
Reserved																															F04h – F20h	

16.14.2 NT Port Link Interface Device-Specific Registers – Error Checking and Debug (Offsets F70h – FB0h)

The registers detailed in [Section 13.22.3, “Device-Specific Registers – Error Checking and Debug \(Offsets F70h – FB0h\)”](#) are also applicable to the NT Port Link Interface, except as defined in [Table 16-19](#) (register map), and [Register 16-52](#).

Other NT Port Link Interface Device-Specific Error Checking and Debug registers are detailed in [Section 16.11.2, “NT Port Link Interface Device-Specific Registers – Error Checking and Debug \(Offsets 700h – 75Ch\)”](#).

Table 16-19. NT Port Link Interface Device-Specific Error Checking and Debug Register Map

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Power Management Hot Plug User Configuration		F70h
Reserved		F74h – FA4h
ACK Transmission Latency Limit		FA8h
Bad TLP Count		FACh
Bad DLLP Count		FB0h

Register 16-52. F70h Power Management Hot Plug User Configuration

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default, by Silicon Revision	
				BA	CA
0	L0s Entry Idle Count Traffic Idle time to meet, to enter the ASPM L0s Link PM state. 0 = Idle condition must last 1 μ s 1 = Idle condition must last 4 μ s	RWS	Yes	0	
1	Factory Test Only	RWS	Yes	0	
6:2	Factory Test Only	RWS	Yes	00h	02h
7	Factory Test Only	RWS	Yes	0	
8	DLLP Timeout Link Retrain Disable Disable Link retraining when no Data Link Layer Packets (DLLPs) are received for more than 256 μ s. 0 = Enables Link retraining when no DLLPs are received for more than 256 μ s (default) 1 = DLLP Timeout is disabled	RWS	Yes	0	
9	Factory Test Only	RWS	Yes	0	
10	L0s Entry Disable 0 = Enables entry into the ASPM L0s Link PM state on the NT Port Link Interface when the state's idle conditions are met 1 = Disables entry into the ASPM L0s Link PM state on the NT Port Link Interface when the state's idle conditions are met	RWS	Yes	0	
11	PHY Upstream Port Control Write Enable Enables the Link Control register Retrain Link and Link Disable bits (offset 78h[5:4], respectively) to be functional on the NT Port Link Interface. 0 = Disables 1 = Enables	RW	Yes	0	
31:12	Reserved	RsvdP	No	0000_0h	

16.15 NT Port Link Interface Advanced Error Reporting Extended Capability Registers (Offsets FB4h – FDCh)

The registers detailed in [Section 15.14, “NT Port Virtual Interface Advanced Error Reporting Extended Capability Registers \(Offsets FB4h – FDCh\),”](#) are also applicable to the NT Port Link Interface, except as defined in [Table 16-20](#) (register map), and [Register 16-53](#) through [Register 16-56](#).

Table 16-20. NT Port Link Interface Advanced Error Reporting Extended Capability Register Map

31 30 29 28 27 26 25 24 23 22 21 20	19 18 17 16	15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0	Offsets
Next Capability Offset (138h)	Capability Version (1h)	PCI Express Extended Capability ID (0001h)	FB4h
Uncorrectable Error Status			FB8h
Uncorrectable Error Mask			FBCh
Uncorrectable Error Severity			FC0h
Reserved		Correctable Error Status	FC4h
Reserved		Correctable Error Mask	FC8h
Advanced Error Capabilities and Control			FCCh
Header Log 0			FD0h
Header Log 1			FD4h
Header Log 2			FD8h
Header Log 3			FDCh

Register 16-53. FB4h Advanced Error Reporting Extended Capability Header

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
15:0	PCI Express Extended Capability ID	ROS	Yes	0001h
19:16	Capability Version	ROS	Yes	1h
31:20	Next Capability Offset Program to 138h, which addresses the Power Budget Extended Capability structure structure.	ROS	Yes	138h

Register 16-54. FB8h Uncorrectable Error Status

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Link Interface: - Data Link Protocol error - Surprise Down error Note: If an individual error is Masked (corresponding bit in the Uncorrectable Error Mask register (offset FBCh) is Set) when the error is detected, its Error Status bit is still updated; however, an error reporting Message (ERR_FATAL or ERR_NONFATAL) is not sent to the Root Complex, and the Advanced Error Capabilities and Control register First Error Pointer field (offset FCCh[4:0]) and Header Log x registers (offsets FD0h through FDCh) remain unchanged.				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
5	Surprise Down Error Status Reserved	RsvdP	No	0
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
13	Flow Control Protocol Error Status Reserved/Not supported	RsvdP	No	0
14	Completion Timeout Status Not applicable to switches.	RsvdP	No	0
15	Completer Abort Status	RW1CS ^a	Yes	0
16	Unexpected Completion Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
17	Receiver Overflow Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
18	Malformed TLP Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
19	ECRC Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
20	Unsupported Request Error Status 0 = No error is detected 1 = Error is detected	RW1CS ^a	Yes	0
31:21	Reserved	RsvdP	No	0-0h

- a. When the **Error Check Disable** register **Software Force Error Enable** bit (offset **720h[2]**) is Set, Type changes from RW1CS to RW.

Register 16-55. FBCh Uncorrectable Error Mask

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Link Interface: • Data Link Protocol error • Surprise Down error <i>Note: The bits in this register can be used to mask their respective Uncorrectable Error Status register bits (offset FB8h).</i>				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
5	Surprise Down Error Mask Reserved	RsvdP	No	0
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
13	Flow Control Protocol Error Mask Reserved/Not supported	RsvdP	No	0
14	Completion Timeout Mask Not applicable to switches.	RWS	Yes	0
15	Completer Abort Mask	RWS	Yes	0
16	Unexpected Completion Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
17	Receiver Overflow Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
18	Malformed TLP Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
19	ECRC Error Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
20	Unsupported Request Error Mask 0 = No Mask is Set 1 = Masks error reporting, first error update, and Header logging for this error	RWS	Yes	0
31:21	Reserved	RsvdP	No	0-0h

Register 16-56. FC0h Uncorrectable Error Severity

Bit(s)	Description	Type	Serial EEPROM and I ² C	Default
The following PCI Express errors are not valid for the NT Port Link Interface: • Data Link Protocol error • Surprise Down error				
3:0	Reserved	RsvdP	No	0h
4	Data Link Protocol Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
5	Surprise Down Error Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	ROS	Yes	1
11:6	Reserved	RsvdP	No	0-0h
12	Poisoned TLP Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
13	Flow Control Protocol Error Severity	ROS	Yes	1
14	Completion Timeout Severity <i>Not applicable to switches.</i> Because the Status and Mask are both Reserved for this bit, Severity can be ignored.	RsvdP	No	0
15	Completer Abort Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
16	Unexpected Completion Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
17	Receiver Overflow Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
18	Malformed TLP Severity 0 = Error is reported as non-fatal 1 = Error is reported as fatal	RWS	Yes	1
19	ECRC Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
20	Unsupported Request Error Severity 0 = Error is handled as an Advisory Non-Fatal error, and reported as a Correctable error 1 = Error is reported as fatal	RWS	Yes	0
31:21	Reserved	RsvdP	No	0-0h



Chapter 17 Test and Debug

17.1 Introduction

This chapter describes the following test- and debug-related information:

- [Physical Layer Loopback Operation](#)
- [Built-In Self-Test](#)
- [Using the SerDes Quad x Diagnostic Data Registers](#)
- [PHY Testability Features](#)
- [JTAG Interface](#)
- [Port Good Status LEDs](#)
- [Thermal Sensor Diode](#)

17.2 Physical Layer Loopback Operation

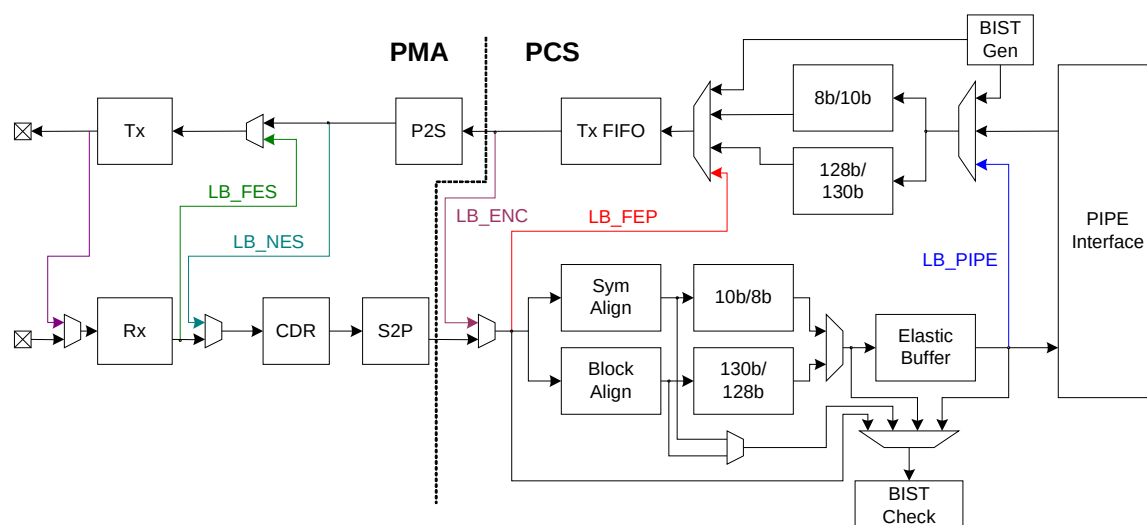
17.2.1 Overview

Physical Layer (PHY) Loopback functions are used to test the SerDes in the PEX 8724, connections between devices, and SerDes of external devices, as well as various PEX 8724 and external digital logic. The PEX 8724 supports five types of Loopback operations, as described in [Table 17-1](#). Additional information regarding each type is provided in the sections that follow.

[Figure 17-1](#) illustrates the Physical Media Attachment Layer (PMA) and Physical Coding Sublayer (PCS) Loopback paths. [Table 17-2](#) provides additional Loopback information.

Table 17-1. Loopback Operations

Operation	Description
Internal Loopback Mode (LB_NES)	This mode connects the SerDes serial Tx output to the serial Rx input.
Digital Loopback Master Mode	This mode also depends upon an external device to loopback the transmitted data. This method is best used with an external device that includes at least its Elastic buffer in the Loopback data path. The PEX 8724 provides a user-definable Test Pattern generator and checker that inserts SKIP Ordered-Sets at the proper intervals.
Analog Loopback Slave Mode (LB_FEP)	If the PEX 8724 registers are programmed to enable Loopback (as described in Section 17.2.4), the PEX 8724 enters this mode when an external device transmits Training Sets with the <i>Loopback Training Control Bit Set</i> and the Port's Physical Layer Test 0 register <i>Analog Loopback Enable</i> bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 224h[10]) is Set. The PEX 8724 automatically performs the necessary sequence of AHB register Writes, to enable this Loopback data path.
Digital Loopback Slave Mode (LB_PIPE)	The PEX 8724 enters this mode when the following conditions are met: - Port's Physical Layer Test 0 register <i>Analog Loopback Enable</i> bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 224h[10]) is Cleared, and - External device transmits Training Sets with the <i>Loopback Training Control Bit Set</i>, –or– - Port's Physical Layer Test 1 register <i>Port x Parallel Loopback Path Enable</i> bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[19:16]) is Set This loopback path includes the Elastic buffer and 8b/10b (Gen 1 and Gen 2 Link speeds) or 128b/130b (Gen 3 Link speed) encode/decode.

Figure 17-1. PMA and PCS Loopback Paths**Table 17-2. Loopback**

Loopback Name	Loopback Description	AHB Register/PIPE Control	Loopback Direction	Notes
LB_NES	Near-End Serial Loopback	Lane R3[6] = 1 (nes_lb_ena)	Tx -> Rx	Only runs up to 5 GT/s
LB_FES	Far-End Serial Loopback	Lane R1[6] = 1 (fes_lb_ena)	Rx-> Tx	Only runs up to 5 GT/s
LB_ENC	Encoder Loopback	Lane R0[2] = 1 (rx_src) Lane R0[1] = 0 (rx_clk_src)	Tx -> Rx	Requires reset.
LB_FEP	Far-End Parallel Loopback	Lane R1[6:4] = 110b (dmux_txbsel[2:0]) Lane R3[7] = 1 (rxclk_lb_ena) Lane R2[7] = 1 (cktrans_en)	Rx- > Tx	Requires reset and sequencing. Tx FIFO Write clock becomes recovered byte block.
LB_PIPE	PIPE-compliant loopback. Used when the PEX 8724 is a PCI Express Loopback Slave.	Lane R0[7:6] = 01b (dmux_txasel[1:0]) or TxDetRxLpbk && PowerDown = P0	Rx -> Tx	Automatically selected when LTSSM enters LB_ActSlv state.

17.2.2 Internal Loopback Mode (LB_NES)

SerDes BIST is intended to be overlapped with the serial EEPROM load operation. To achieve this overlap, the Port's **Physical Layer Test 0** register **Factory Test Only** bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 224h[11]) is written early in the serial EEPROM load operation. After the bit is Set, the SerDes are placed into Loopback mode. The BIST is run for 15 ms, if an error is detected on any of the SerDes, the BIST_ERROR pin associated with the Station that includes the SerDes in error is asserted. The continuing serial EEPROM register load has no effect on SerDes BIST.

17.2.3 Digital Loopback Master Mode

The only difference between Analog and Digital Loopback Master modes is that the external device is assumed to have some of its digital logic in the Loopback data path. Because this includes the Elastic buffer, SKIP Ordered-Sets must be included in the test data pattern.

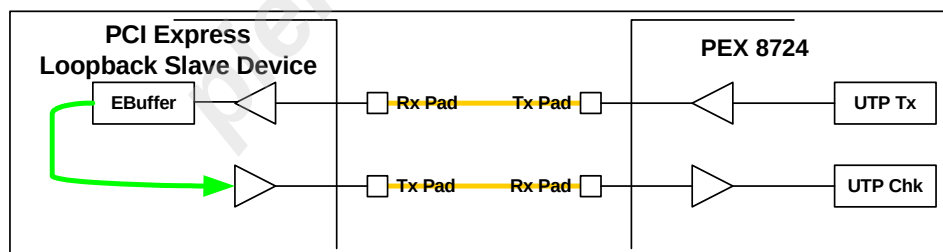
Figure 17-2 illustrates a Far-End Digital Loopback Master connection and data path.

The PEX 8724 provides the User Data Pattern Transmitter for Digital Far-End Loopback testing. After the Loopback Master mode is established, Configuration Writes are used to fill the Test Data Pattern registers. One or more **Physical Layer Test 1** register *SerDes Quad x Pair x User Test Pattern Enable* bits (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[31:24]) is Set, which starts the transmission of the User Data Pattern on all Lanes. If the Port's *Enable by Port* bit is also Set, the test pattern is transmitted on all Lanes of the corresponding Port, regardless of its Link width. If the *Enable by Port* bit is Cleared, the test pattern is transmitted only on the Lanes of the corresponding SerDes quad. SKIP Ordered-Sets are inserted at the interval determined by the **SKIP Ordered-Set Interval** register *SKIP Ordered-Set Interval in Blocks* or *SKIP Ordered-Set Interval in Symbol Times* field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 234h[27:16 or 11:0], respectively) value, at the nearest data pattern boundary:

- **Gen 3 (8.0 GT/s)** – Field [27:16]. Default interval is 371 block times.
- **Gen 1 and Gen 2 (2.5 and 5.0 GT/s, respectively)** – Field [11:0]. Default interval is 1,180 symbol times.

The Test Pattern checker ignores SKIP Ordered-Sets that are returned by the Loopback Slave, because the quantity of SKIP symbols received can be different than the quantity transmitted. All other data is compared to the transmitted data. Any errors are logged in the **SerDes Quad x Diagnostic Data** register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 238h through 244h) that corresponds to the SerDes quad being tested.

Figure 17-2. Digital Far-End Loopback



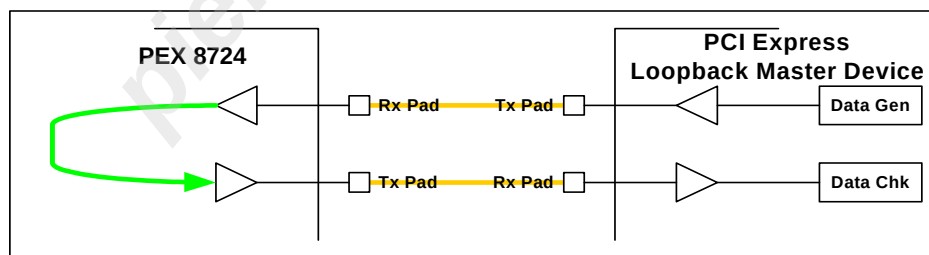
17.2.4 Analog Loopback Slave Mode (LB_FEP)

The PEX 8724 becomes an Analog Loopback Slave (as illustrated in Figure 17-3) if it receives Training Sets with the *Loopback* Training Control Bit Set, after the Port's register bits are programmed as follows, in the sequence listed:

1. **Port Control** register *Disable Port x* bit(s) (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 208h[3:0]) – Set to disable Port.
2. **Port Control** register *Hold Port x Quiet* bit(s) (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 208h[11:8]) – Set to enable Port Quiet.
3. **Port Control** register *Test Pattern x Rate Port x Select* field (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 208h[29:28]) – Select Port.
4. **Port Control** register *Disable Port x* bit(s) (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 208h[3:0]) – Clear to enable Port.
5. **Port Control** register *Port Test Pattern x Rate* field (Base mode – Ports 0 and 8; Virtual Switch mode – Ports 0 and 8, accessible through the Management Port, offset 208h[25:24]) – Select data rate (00b = 2.5GT/s, 01b = 5.0 GT/s, 10b = 8GT/s).
6. **Physical Layer Test 0** register *Analog Loopback Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 224h[10]) – Set to enable Analog Loopback.
7. **Physical Layer Test 1** register *Port x Parallel Loopback Path Enable* bit(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[19:16]) – Set to enable Parallel Loopback Path.
8. **Physical Layer Electrical Idle Detect Mask** register *SerDes x Mask Electrical Idle Detect* bit(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 204h[15:0]) – Clear to enable Electrical Idle Detection.
9. **Physical Layer Function Control** register *Port x Electrical Idle Inference Disable* bit(s) (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 220h[27:24]) – Set to disable Electrical Idle Inference.

While an Analog Loopback Slave, the PEX 8724 includes only the SerDes in the Loopback data path. The Loopback Master must provide the test data pattern and data pattern checking. It is not necessary for the Loopback Master to include SKIP Ordered-Sets in the data pattern.

Figure 17-3. Analog Loopback Slave Mode



17.2.5 Digital Loopback Slave Mode (LB_PIPE)

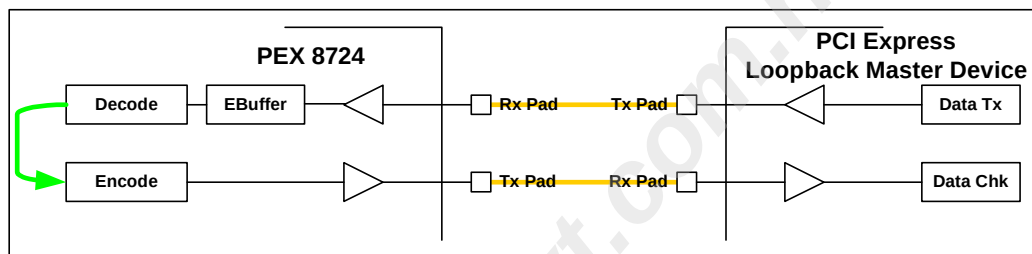
The PEX 8724 becomes a Digital Loopback Slave if it receives Training Sets with the *Loopback Training Control Bit Set*, –or– the Port's **Physical Layer Test 1** register *Port x Parallel Loopback Path Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 228h[19:16]) is Set, while the Port's **Physical Layer Test 0** register *Analog Loopback Enable* bit (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset 224h[10]) is Cleared.

When a PEX 8724 Port is a Digital Loopback Slave:

- **Gen 1 and Gen 2 Link speeds (2.5 and 5.0 GT/s, respectively)** – Loopback data path includes the Elastic buffer and symbol alignment logic
- **Gen 3 Link speed (8.0 GT/s)** – Loopback data path includes the Elastic buffer and block aligner, as well as the 128b/130b encoder and decoder

The Loopback Master must provide the test data pattern and data pattern checker (such as a PEX 8724 User Test Pattern). The Loopback Master must also transmit SKIP Ordered-Sets with the data pattern. The PEX 8724 can return more or fewer SKIP symbols than the switch received from the Master. Therefore, the Master's data pattern checker must make provisions for this when decoding for errors.

Figure 17-4. Digital Loopback Slave Mode



Note: 8b/10b (Gen 1 and Gen 2 Link speeds) or 128b/130b (Gen 3 Link speed) encode/decode.

17.3 Built-In Self-Test

Each Lane contains a Built-In Self-Test (BIST) generator and checker.

17.4 Using the SerDes Quad x Diagnostic Data Registers

Each SerDes quad has its own Diagnostic Data register, per Station. The **SerDes Quad x Diagnostic Data** register (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offsets 238h through 244h) contents reflect the performance of the SerDes selected by the registers' *SerDes Diagnostic Data Select* field [25:24]. This control is specific to this register (which reports results of UTP tests).

When field [25:24] is Cleared, the information in that Diagnostic Data register is for the first SerDes within that SerDes quad. When field [25:24] is programmed to 01b, the information in that Diagnostic Data register is for the second SerDes within that SerDes quad, as illustrated in Table 17-3. Following this pattern, a value of 10b indicates the third SerDes within that SerDes quad, and a value of 11b indicates the fourth SerDes within that SerDes quad. If a SerDes module does not exist in that position, the corresponding bit(s) is (are) **Factory Test Only, Reserved** (RsvdP), and not serial EEPROM nor I²C writable.

Table 17-3. SerDes Quad x Diagnostic Data Register Contents When SerDes Diagnostic Data Select Field [25:24]=01b (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port), by Station Port

Register Offset	Register	Port 0	Port 8
238h	SerDes Quad 0 Diagnostic Data	SerDes 1	<i>Factory Test Only</i>
23Ch	SerDes Quad 1 Diagnostic Data	SerDes 5	<i>Factory Test Only</i>
240h	SerDes Quad 2 Diagnostic Data	SerDes 9	SerDes 25
244h	SerDes Quad 3 Diagnostic Data	SerDes 13	SerDes 29

17.5 PHY Testability Features

The PEX 8724 includes several Configuration bits to ease PHY testability. Features include:

- Full support of the standard and modified Compliance patterns
- Register controllability of the common block and Lane-specific inputs of the SerDes

Table 17-4 describes the Configuration bits.

Table 17-4. Configuration Bits to Ease PHY Testability

Register Bit(s) ^a	Description
<i>SerDes x Mask Electrical Idle Detect</i> Physical Layer Electrical Idle Detect Mask register (offset 204h[15:0])	Never Detect Electrical Idle Mask. When any one of these bits is Set, the Lane's Electrical Idle condition flag does not assert, regardless of the actual presence of Electrical Idle.
<i>SerDes x Mask Receiver Not Detected</i> Physical Layer Receiver Not Detected Mask register (offset 204h[31:16])	Always Detect a Receiver Mask. When any one of these bits is Set, the PHY functions as if the Lane detected a Receiver, regardless of the actual presence of a Receiver.
<i>Disable Port x</i> Port Control register (offset 208h[3:0])	Unconditional Port Disable. When Set, the LTSSM remains in the <i>Detect.Quiet</i> substate on the Port if it is currently in, or returns to, that substate. Unconditionally disables the Port. This is different from the LTSSM <i>Disabled</i> state, in that the Port does not attempt to enter this state. If the Port is idle, it ceases attempting to detect a Receiver. If the Port is up, it immediately returns to the <i>Detect.Quiet</i> substate and remains there. No Electrical Idle Ordered-Set (EIOS) is sent, which could force any connected device to the <i>Recovery</i> state, and then to an LTSSM <i>Detect</i> state. The Port remains disabled until its <i>Disable Port x</i> bit is Cleared. While the Port is disabled, Receiver termination is disabled and the SerDes that belong to the disabled Port are placed into the L1 Link PM state.
<i>Hold Port x Quiet</i> Port Control register (offset 208h[11:8])	Port Quiet. When Set, the LTSSM remains in the <i>Detect.Quiet</i> substate if it is currently in, or returns to, that substate. These bits do not force the LTSSM to the <i>Detect.Quiet</i> substate. (Refer to the <i>Disable Port x</i> bits.) After entering the <i>Detect.Quiet</i> substate, Receiver termination is enabled and the Transmitters are placed into the L0 Link PM state. This Port can now transmit test patterns (UTP), with or without an attached device and without being in the <i>Loopback.Active</i> substate.
<i>Port x Bypass UTP Alignment Pattern</i> Port Control register (offset 208h[19:16])	Bypass Alignment Pattern. When Cleared, the UTP Transmitter continuously transmits the UTP alignment pattern (K28.5 D3.2 D18.2 D13.2), to allow the corresponding symbol alignment logic to find the symbol boundaries before sending the sync pattern (D3.2 D18.2 D13.2 D17.1) and programmed UTP. When Set, the UTP Transmitter sends one UTP alignment pattern (K28.5 D3.2 D18.2 D13.2), one sync pattern (D3.2 D18.2 D13.2 D17.1), and then the programmed UTP.

Table 17-4. Configuration Bits to Ease PHY Testability (Cont.)

Register Bit(s) ^a	Description
<i>Port Test Pattern x Rate</i> Port Control register (offset 208h[25:24])	Test Pattern Rate. Select 2.5, 5.0, or 8.0 GT/s.
<i>User Test Pattern</i> Physical Layer User Test Pattern, Bytes x through y registers (offsets 20Ch through 218h)	A 16-byte test pattern can be written to these four registers. When UTP transmission is enabled, Byte 0 of register offset 20Ch is transmitted first and Byte 3 (Byte 15 of the UTP) of register offset 218h is transmitted last. (Refer to Section 17.2.3 for further details.) Every byte of the UTP can be a Control or Data character. Illegal Control characters can be specified.
<i>Port x Scrambler Disable Command</i> Physical Layer Port Command register (offset 230h[1, 5, 9, and/or 13])	Scramble Disable. Unconditionally disables the data scramblers on the Port's Lane(s), and causes the <i>Scramble Disable</i> Training Control Bit to be Set in transmitted Training Sets. There is one bit, per Port, per Station.
Physical Layer Error Injection Control register (offset 25Ch)	1-Bit Error Injection. Provides Port-specific controls for 1-bit error injection from the PEX 8724 Transmitters. Far-end devices can use this feature to test their error-handling capability.
<i>x1 Only for Station x</i> x1 Port Configuration register (offset 304h[15:0])	x1 Only. Forces the Port to linkup with a x1 negotiated Link width, regardless of the quantity of Lanes connected to the Port.
<i>x2 Only for Station x</i> x2 Port Configuration register (offset 308h[15:8])	x2 Only. Forces the Port to linkup with a x2 negotiated Link width, regardless of the quantity of Lanes connected to the Port.
<i>Port x Receiver Error Counter</i> Port Receiver Error Counter register (offset BF0h[31:0])	Receiver Error Counter. Contains four 8-bit fields that, when read, return the quantity of Receiver errors that the Port detected. The Error Counter saturates at 255. The Counter is Cleared with any Write to the corresponding byte in this register; otherwise, this register is RO.

- a. All register bits listed in this table are located, as follows:
 Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port.

17.6 JTAG Interface

The PEX 8724 provides a Joint Test Action Group (JTAG) Boundary Scan interface, which is used to debug board connectivity for each signal.

17.6.1 *IEEE 1149.1* and *IEEE 1149.6* Test Access Port

The *IEEE Standard 1149.1* Test Access Port (TAP), commonly called the *JTAG Debug Port*, is an architectural standard described in the *IEEE Standard 1149.1-1990*. The *IEEE Standard 1149.6-2003* defines extensions to *1149.1* to support PCI Express SerDes testing. These standards describe methods for accessing internal device facilities, using a four- or five-signal interface.

The JTAG Debug Port, originally designed to support scan-based board testing, is enhanced to support the attachment of debug tools. The enhancements, which comply with the *IEEE Standard 1149.1-1994 Specifications for Vendor-Specific Extensions*, are compatible with standard JTAG hardware for boundary-scan system testing.

- **JTAG Signals** – JTAG Debug Port implements the four required JTAG signals – [JTAG_TCK](#), [JTAG_TDI](#), [JTAG_TDO](#), [JTAG_TMS](#) – and optional [JTAG_TRST#](#) signal
- **Clock Requirements** – JTAG_TCK signal frequency ranges from 0 to 15 MHz
- **JTAG Reset Requirements** – Refer to [Section 17.6.4](#)

17.6.2 JTAG Instructions

The JTAG Debug Port provides the *IEEE Standard 1149.1-1990* BYPASS, EXTEST, SAMPLE, PRELOAD, CLAMP, and IDCODE instructions. *IEEE Standard 1149.6-2003* EXTEST_PULSE and EXTEST_TRAIN instructions are also supported. [Table 17-5](#) lists the JTAG instructions, along with their input codes.

The PEX 8724 returns the JTAG IDCODE values listed in [Table 17-6](#).

Table 17-5. JTAG Instructions

Instruction	Input Code	Comments
BYPASS	7EFFh	<i>IEEE Standard 1149.1-1990</i>
EXTEST	7FDFh	
SAMPLE	7FCFh	
PRELOAD	7FCFh	
EXTEST_PULSE	7F9Fh	<i>IEEE Standard 1149.6-2003</i>
EXTEST_TRAIN	7F5Fh	
CLAMP	7FEFh	<i>IEEE Standard 1149.1-1990</i>
IDCODE	1FFFh	

Table 17-6. JTAG IDCODE Values

Units	Version, by Silicon Revision		Part Number	PLX Manufacturer Identity	Least Significant Bit
	BA	CA			
Bits	0001b	0010b	1000_0111_0010_0100b	001_1100_1101b	1
Hex	1h	2h	8724h	1CDh	1h

17.6.3 JTAG Boundary Scan

Boundary Scan Description Language (BSDL), IEEE Standard 1149.1-1994, is a supplement to the IEEE Standard 1149.1-1990 and IEEE Standard 1149.1a-1993, *IEEE Standard Test Access Port and Boundary-Scan Architecture*. BSDL, a subset of the IEEE 1076-1993 *Standard VHSIC Hardware Description Language (VHDL)*, allows a rigorous description of testability features in components which comply with the standard. This standard is used by automated test pattern generation tools for package interconnect tests and Electronic Design Automation (EDA) tools for synthesized test logic and verification. BSDL supports robust extensions that can be used for internal test generation and to write software for hardware debug and diagnostics.

The primary components of BSDL include the logical Port description, physical ball map, instruction set, and **Boundary** register description.

The logical Port description assigns symbolic names to the device's signal balls. Each ball includes a logical type of *in*, *out*, *in out*, *buffer*, or *linkage* that defines the logical direction of signal flow.

The physical ball map correlates the device's logical Ports to the physical balls of a specific package. A BSDL description can include several physical ball maps, and maps are provided with a unique name.

Instruction set statements describe the bit patterns that must be shifted into the **Instruction** register to place the device in the various test modes defined by the standard. Instruction set statements also support descriptions of instructions that are unique to the PEX 8724.

The **Boundary** register description lists each cell or shift stage of the **Boundary** register. Each cell has a unique number – the cell numbered 0 is the closest to the [JTAG Test Data Output \(JTAG_TDO\)](#), and the cell with the highest number is closest to the [JTAG Test Data Input \(JTAG_TDI\)](#). Each cell includes additional information, *such as*:

- Cell type
- Logical Port associated with the cell
- Logical function of the cell
- Safe value
- Control cell number
- Disable value
- Result value

17.6.4 JTAG Reset Input – JTAG_TRST#

The [JTAG_TRST#](#) input is the asynchronous reset input to the JTAG TAP Controller logic. JTAG_TRST# input assertion places the PEX 8724's TAP Controller into the *Test-Logic-Reset* state, which selects the PEX 8724 standard logic path (core-to-I/O), as required for typical functionality of the switch. JTAG_TRST# input de-assertion enables the TAP Controller for test and debug functionality, *such as* boundary scan.

- If JTAG functionality is required, the JTAG_TRST# input should be transitioned from Low-to-High once during PEX 8724 power-up, along with [PEX_PERST#](#) input de-assertion. In any JTAG state, the JTAG TAP Controller can be returned to the *Test-Logic-Reset* (initial) state, by holding the [JTAG_TMS](#) input High while clocking the [JTAG_TCK](#) input at least five times.
- If the design is not going to use the JTAG TAP Controller, an external pull-down resistor is recommended on the JTAG_TRST# input, because its internal pull-down resistor is weak.

17.7 Port Good Status LEDs

The PEX 8724 provides Port Good outputs, PORT_GOOD[6:0]# (PORT_GOODx#), that can be used to control external circuitry, *such as* LEDs, to provide visual indication that the PHY of that Port's Link is trained to at least x1 width. These outputs can:

- Default to the PORT_GOOD output function, when the STRAP_TESTMODE[2:0] 3-state inputs select the PORT_GOOD output function (*that is*, Test modes 0, 4, or 8, and additionally for PORT_GOOD0# only, in Test modes 3, 7, and 11; refer to [Section 3.4.4.1, “STRAP_TESTMODE\[2:0\] 3-State Input Configurations,”](#) for details), –or–
- Be programmed as a general-purpose I/O, to assume the PORT_GOOD output function (refer to the **GPIO 0_x Direction Control** register *Direction Control* bit(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) [600h](#) and/or [604h](#)))

The PORT_GOODx#/GPIOx I/Os operate from the [VDD18](#) power supply, and the maximum input voltage is 2.5V. Because the LED forward voltage drop is typically greater than 1.8V, the LED voltage source will likely be higher than 2.5V. In this case, if the LED is connected directly to the PORT_GOODx# input, the input voltage level (while the LED is turned Off) would exceed the specification. Therefore, a PORT_GOODx# signal typically cannot drive an LED directly; however, it can be used to gate a transistor that switches the LED On and Off.

Notes: PORT_GOOD[6:0]# correspond to Ports 10, 9, 8, 3, 2, 1, and 0, respectively. PORT_GOODx# signals can be re-assigned to any Port, however, by programming the **PORT_GOOD Selector x** register(s) (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset(s) [6A0h](#) and [6A4h](#)).

The GPIO11 signal exists only in Virtual Switch mode, when four virtual switches are enabled. This signal does not exist in Base mode.

[Table 4-1, “Port Configurations,”](#) indicates which Ports are enabled/used/**Software Only**, and when, based upon the STRAP_STNx_PORTCFGx input states and/or serial EEPROM programming.

[Table 17-7](#) describes the LED On/Off patterns when connected to the PORT_GOODx# signals.

Table 17-7. PORT_GOODx# LED On/Off Patterns, by State

State	LED Pattern
Link is Down	Off
Link is Up, 2.5 GT/s, any negotiated Link width	Blinking, 512 ms On, 512 ms Off (1 Hz)
Link is Up, 5.0 GT/s, any negotiated Link width	Blinking, 256 ms On, 256 ms Off (2 Hz)
Link is Up, 8.0 GT/s, any negotiated Link width	On

Software can determine:

- Which Lanes have completed PHY linkup, by performing a Memory Read of the **Station 0/1 Lane Status** register *Lane x Up Status* bits (Base mode – Port 0; Virtual Switch mode – Port 0, accessible through the Management Port, offset 330h[31:24 and 15:0]).
- Whether the Link for each Port has trained, by reading either the Port's **Link Status** register *Data Link Layer Link Active* bit (Downstream Ports, offset 78h[29]), –or– **VC0 Resource Status** register *VC0 Negotiation Pending* bit (offset 160h[17]). If the *Data Link Layer Link Active* bit is Set, or *VC0 Negotiation Pending* bit is Cleared, the Link has completed Flow Control (FC) initialization.

The **Link Status** register can be read by either a PCI Express Configuration Request or Memory Read. The **VC0 Resource Status** register can be read by either a PCI Express Enhanced Configuration access or Memory Read.

- The negotiated Link width of each Port, by reading the Port's **Link Status** register *Negotiated Link Width* field (offset 78h[25:20]). This register can be read by either a Configuration Request or Memory Read.

17.8 Thermal Sensor Diode

The PEX 8724 includes a thermal-sensing diode, to facilitate the measurement of on-die device junction temperature. On the package level, there are two external inputs – [THERMAL_DIODEn](#) and [THERMAL_DIODEp](#), that connect to anode and cathode terminals of a diode-connected transistor (PNP) implemented on the die.

At the board level, a temperature sensor chip, *such as* the Maxim MAX6657^a, can be used.

For further details, refer to the manufacturer's data sheet for the device being used.

Note: *The PEX 8724 thermal diode produces an offset in the external temperature sensor that results in a temperature reading that is 24°C above the approximate die temperature. Consequently, a temperature sensor chip that measures a maximum temperature of 128°C might not be sufficient, because the maximum temperature reported is effectively reduced to 104°C (128 - 24 = 104). Some sensor ICs support higher temperatures, such as 145°C; however, these devices have not been tested with the PEX 8724 thermal diode.*

a. *The MAX6657, MAX6658, and MAX6659 are ±1°C, SMBus-Compatible Remote/Local Temperature Sensors with Overtemperature Alarms,” by Maxim Integrated Products.*

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Chapter 18 Electrical Specifications

18.1 Introduction

This chapter provides the PEX 8724 electrical specifications:

- [Power-Up/Power-Down Sequence](#)
- [Absolute Maximum Ratings](#)
- [Power Characteristics](#)
- [Power Consumption Estimates](#)
- [I/O Interface Signal Groupings](#)
- [Transmit Characteristics](#)
- [Receive Characteristics](#)

18.2 Power-Up/Power-Down Sequence

The PEX 8724 does not have power sequencing requirements. The power rails can be powered up and powered down, in any sequence.

18.3 Absolute Maximum Ratings

Warning: *Maximum limits indicate the temperatures and voltages above which permanent damage can occur. Proper operation at these conditions is not guaranteed, and continuous operation of the PEX 8724 at these limits is not recommended.*

Table 18-1. Absolute Maximum Rating (All Voltages Referenced to VSS System Ground)

Item	Symbol	Absolute Maximum Rating	Units
I/O Interface Supply Voltage	VDD18	1.98	V
PLL Supply Voltage	VDD18A	1.98	V
Core (Logic) Supply Voltage	VDD09	0.99	V
SerDes Analog Supply Voltage	VDD09A	0.99 ^a	V
Input Voltage (1.8V Interface)	V _I	2.75	V
Operating Ambient Temperature (Commercial)	T _A	0 to +70	°C
Storage Temperature	T _{STG}	-65 to +150	°C

a. Most users tie VDD09A to VDD09.

18.4 Power Characteristics

Table 18-2. Operating Condition Power Supply Rails

Symbol	Parameter	Voltage and Ranges	Min	Typ	Max	Units
VDD09	Digital Core Supply	0.9V -5%, +5.55%	0.855	0.9	0.95	V
VDD09A	Analog SerDes Supply	0.9V -5%, +5.55%	0.855	0.9	0.95	V
VDD18 ^a	I/O Supply	1.8V -5%, +10%	1.71	1.8	1.98	V
VDD18A	Phase-Locked Loop (PLL) Supply	1.8V -5%, +10%	1.71	1.8	1.98	V

a. The PEX 8724 I/Os are 2.5V-tolerant.

18.5 Power Consumption Estimates

Table 18-3. Power Consumption Estimates (W)

# of Lanes/ Ports	Link Speed (GT/s)	T _j (°C)	Process ^{a, b, c, d}	Power Rails			Power (W)				
				VDD09 (V)	VDD09A (V)	VDD18A (V)	Digital (VDD09)	SerDes Analog (VDD09A)	PLL (VDD18A)	I/O (VDD18)	Total
24/6	8.0	25	Typical	0.9	0.9	1.8	2.814	2.238	0.437	0.018	5.508
	5.0						2.658	2.002	0.437	0.018	5.114
	2.5						2.579	1.936	0.437	0.018	4.97
	8.0	80	Worst Case	0.95	0.95	1.98	6.427	2.738	0.491	0.022	9.678
	5.0						6.116	2.448	0.491	0.022	9.077
	2.5						5.961	2.368	0.491	0.022	8.841
	8.0	110	Worst Case	0.95	0.95	1.98	8.266	2.817	0.495	0.022	11.599
	5.0						7.955	2.519	0.495	0.022	10.99
	2.5						7.799	2.436	0.495	0.022	10.752

- a. Typical power based upon 35% (8.0 GT/s), 17.5% (5.0 GT/s), or 8.75% (2.5 GT/s) traffic.
- b. Thermal simulation undertaken at approximately 90% worst-case power estimate value.
- c. Maximum power based upon 85% (8.0 GT/s), 42.5% (5.0 GT/s), or 21.25% (2.5 GT/s) traffic.
- d. Maximum power is at 110°C Junction temperature and Fast/Fast (FF) process corner silicon.

18.6 I/O Interface Signal Groupings

Table 18-4. Signal Group PCI Express Analog Interface

Signal Group	Signal Type	Signals	Notes
(a)	PCI Express Output (Transmit)	PEX_PETp/nx	Refer to Table 18-6 and Table 18-7
(b)	PCI Express Input (Receive)	PEX_PERp/nx	Refer to Table 18-6 and Table 18-12
(c)	PCI Express Differential Clock Input	PEX_REFCLKp/n	Refer to Table 18-6
(d)	SerDes External Bias Resistor	REXT_Ax, REXT_Bx, REXT_CMU	3.00K Ω , $\pm 1\%$, and refer to Table 18-6

Table 18-5. Signal Group Digital Interface

Group	Type	Signals ^a	Note
(e)	Digital Input	STRAP_RESERVED16, THERMAL_DIODEn/p	Refer to Table 18-6
(f)	Digital Input with Internal Pull-Up Resistor	PEX_NT_PERST#, PEX_PERST#, STRAP_PLL_BYPASS#, STRAP_PROBE_MODE#, STRAP_SERDES_MODE_EN#	
(g)	Digital Input with Internal Pull-Down Resistor	JTAG_TCK, JTAG_TDI, JTAG_TMS, JTAG_TRST#	
(h)	Bidirectional with Internal Pull-Up Resistor (8 mA Drive)	EE_CS#, EE_DO, EE_SK, GPIO[11:9], HP_ATNLED_x#, HP_BUTTON_x#, HP_CLKEN_x#, HP_MRL_x#, HP_PERST_x#, HP_PRSTNT_x#, HP_PWRFLT_x#, HP_PWRLED_x#, PORT_GOOD[6:0]#, SHPC_INT#, SPARE[2:0], STRAP_I2C_SMBUS_CFG_EN#, VS0_FATAL_ERR#, VS1_FATAL_ERR#, VS3_FATAL_ERR#, VSx_PERST#, VSx_PEX_INTA#	
(j)	Bidirectional with Internal Pull-Down Resistor (8 mA Drive)	GPIO[8:7], HP_PWREN_x, HP_PWR_GOOD_x, SPARE3, VS2_FATAL_ERR#	
(k)	Bidirectional Tri-Statable Output with Limited Slew Rate Input	I2C_SCL0, I2C_SCL1, I2C_SDA0, I2C_SDA1, PEX_INTA#	
(m)	Bidirectional Tri-Statable Output	EE_DI, FATAL_ERR#, JTAG_TDO	
(n)	Bidirectional with 3-State Input	I2C_ADDR0, STRAP_GEN1_GEN2, STRAP_I2C_SMBUS_EN, STRAP_MGMT_PORT_EN, STRAP_NT_UPSTRM_PORTSEL[2:0], STRAP_STN0_PORTCFG[1:0], STRAP_STN1_PORTCFG[1:0], STRAP_TESTMODE[2:0], STRAP_UPSTRM_PORTSEL[2:0], VS_MODE	

a. For STRAP_RESERVED-related signals, refer to [Table 3-7](#), “Strapping Signals,” for pull/tie requirements.

Table 18-6. Analog and Digital Interfaces (All Signal Groups) – DC Electrical Characteristics

Symbol	Signal Group	Parameter	Min	Typ	Max	Unit	Conditions
I_{OL}	(h) (j) (k) (m)	Output Low Current	7.0	12.0	17.6	mA	@ $V_{OLmax} = 0.45V$
I_{OH}	(h) (j) (m)	Output High Current	6.0	13.5	24.5	mA	@ $V_{OHmin} = 1.35V$
V_{IL}	(e) (f) (g) (h) (j) (k)	Input Low Voltage	-0.3		0.63	V	
V_{IH}	(e) (f) (g) (h) (j) (k)	Input High Voltage	1.17		2.75	V	
V_T	(e) (g) (h) (j) (k)	Threshold Point	0.78	0.84	0.89	V	
C_{PIN}	(a) (b) (c) (d) (e) (f) (g) (h) (j) (k)	Signal Capacitance			5	pF	
$I_{LEAKAGE}$	(m)	Tri-State Leakage			± 10	μA	
	(e) (f)	Input Leakage			± 10	μA	
R_{PU}	(f) (h)	Pull-Up Impedance	38K	57K	92K	Ω	
R_{PD}	(g) (j)	Pull-Down Impedance	37K	53K	99K	Ω	

18.7 Transmit Characteristics

18.7.1 PCI Express Transmitter AC and DC Characteristics

Table 18-7. PCI Express Transmitter (Signal Group a) – AC and DC Characteristics

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
UI	Unit Interval	399.88 (min) 400.12 (max)	199.94 (min) 200.06 (max)	124.9625 (min) 125.0375 (max)	ps	The specified UI is equivalent to a tolerance of ± 300 ppm for each REFCLK source. Period does not account for Spread-Spectrum Clock (SSC)-induced variations. Refer to Note 1.
BW _{TX-PLL}	Tx PLL Bandwidth for 2.5 GT/s	1.5 (min) 22 (max)	Not specified	Not specified	MHz	Refer to Note 6.
BW _{TX-PKG-PLL1}	Tx PLL Bandwidth corresponding to PKG _{TX-PLL1}	Not specified	8 (min) 16 (max)	2 (min) 4 (max)	MHz	Second Order PLL Jitter Transfer Bounding function. Refer to Note 6.
BW _{TX-PKG-PLL2}	Tx PLL Bandwidth corresponding to PKG _{TX-PLL2}	Not specified	5 (min) 16 (max)	2 (min) 5 (max)	MHz	Second Order PLL Jitter Transfer Bounding function. Refer to Note 6.
PKG _{TX-PLL1}	Tx PLL Peaking	Not specified	3.0 (max)	2.0 (max)	dB	PLL Bandwidth = 8 MHz (min) at 5 GT/s or Bandwidth = 4 MHz (max) at 8 GT/s. Refer to Notes 6 and 8.
PKG _{TX-PLL2}	Tx PLL Peaking	Not specified	1.0 (max)	1.0 (max)	dB	PLL Bandwidth = 5 MHz (min) at 5 GT/s or Bandwidth = 5 MHz (max) at 8 GT/s. Refer to Note 8.
V _{TX-DIFF-PP}	Differential Peak-to-Peak Tx Voltage Swing	0.8 (min) 1.2 (max)	0.8 (min) 1.2 (max)	Refer to Table 18-9	VPP	Measured with compliance test load. Defined as $2 \times V_{TX-D+} - V_{TX-D-} $
V _{TX-DIFF-PP-LOW}	Low Power Differential Peak-to-Peak Tx Voltage Swing	0.4 (min) 1.2 (max)	0.4 (min) 1.2 (max)	Refer to Table 18-9	VPP	Measured with compliance test load. Defined as $2 \times V_{TX-D+} - V_{TX-D-} $. Refer to Note 9.
V _{TX-DE-RATIO-3.5dB}	Tx De-Emphasis Level Ratio	3.0 (min) 4.0 (max)	3.0 (min) 4.0 (max)	Refer to Table 18-9	dB	
V _{TX-DE-RATIO-6dB}	Tx De-Emphasis Level Ratio	–	5.5 (min) 6.5 (max)	Refer to Table 18-9	dB	
T _{MIN-PULSE}	Instantaneous Lone Pulse Width	Not specified	0.9 (min)	Refer to Table 18-9	UI	Measured relative to rising/falling pulse. Refer to Notes 2 and 10.

Table 18-7. PCI Express Transmitter (Signal Group a) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
T_{TX-EYE}	Tx Eye Width (including all jitter sources)	0.75 (min)	0.75 (min)	Refer to Table 18-9	UI	Does not include SSC nor REFCLK jitter. Includes R_j at 10^{-12} . Refer to Notes 2, 3, 4, and 10. Note: 2.5 and 5.0 GT/s use different jitter determination methods.
$T_{TX-EYE-MEDIAN-to-MAX-JITTER}$	Maximum Time between the Jitter Median and Maximum Deviation from the Median	0.125 (max)	Not specified	Not specified	UI	Measured differentially at zero crossing points, after applying the 2.5 GT/s Clock Recovery function. Refer to Note 2.
$T_{TX-HF-DJ-DD}$	Tx Deterministic Jitter > 1.5 MHz	Not specified	0.15 (max)	Refer to Table 18-9	UI	Deterministic jitter only. Refer to Notes 2 and 10.
$T_{TX-LF-RMS}$	Tx RMS Jitter < 1.5 MHz	Not specified	3.0	Refer to Table 18-9	ps RMS	Total energy measured over a 10-kHz to 1.5-MHz range.
$T_{TX-RISE-FALL}$	Tx Rise and Fall Time	0.125 (min)	0.15 (min)	Not specified	UI	Measured differentially from 20 to 80% of swing.
$T_{RF-MISMATCH}$	Tx Rise/Fall Mismatch	Not specified	0.1 (max)	Not specified	UI	Measured from 20 to 80% differentially. Refer to Note 2.
$RL_{TX-DIFF}$	TX Differential Return Loss (Package + Silicon)	10 (min)	10 (min) for 0.05 to 1.25 GHz 8 (min) for 1.25 to 2.5 GHz	10 (min) for 0.05 to 1.25 GHz 8 (min) for 1.25 to 2.5 GHz 4 (min) for 2.5 to 4 GHz	dB	
RL_{TX-CM}	TX Common Mode Return Loss (Package + Silicon)	6 (min) for 0.05 to 2.5 GHz	6 (min) for 0.05 to 2.5 GHz	6 (min) for 0.05 to 2.5 GHz 3 (min) for 2.5 GHz	dB	
$Z_{TX-DIFF-DC}$	DC Differential Tx Impedance	80 (min) 120 (max)	120 (max)	120 (max)	Ω	Low impedance defined during signaling. Parameter is captured for 5.0 GHz by $RL_{TX-DIFF}$.

Table 18-7. PCI Express Transmitter (Signal Group a) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
$V_{TX-CM-AC-PP}$	Tx AC Peak-to-Peak Common Mode Voltage (5.0 GT/s)	Not specified	150 (max)	150 (max)	mVPP	For 8 GT/s, no more than 50 mVPP within the 0.03- to 500-MHz range. Refer to Notes 5 and 12.
$V_{TX-CM-AC-P}$	Tx AC Peak Common Mode Voltage (2.5 GT/s)	20	Not specified	Not specified	mV	Refer to Note 5.
$I_{TX-SHORT}$	Tx Short Circuit Current Limit	90 (max)	90 (max)	90 (max)	mA	Total single-ended current the Transmitter can supply when shorted to its Ground. Refer to Note 13.
$V_{TX-DC-CM}$	Tx DC Common Mode Voltage	0 (min) 3.6 (max)	0 (min) 3.6 (max)	0 (min) 3.6 (max)	V	Allowed DC common mode voltage at a Transmitter signal, under any conditions. Refer to Note 13.
$V_{TX-CM-DC-ACTIVE-IDLE-DELTA}$	Absolute Delta of DC Common Mode Voltage during L0 Link PM state and Electrical Idle	0 (min) 100 (max)	0 (min) 100 (max)	0 (min) 100 (max)	mV	$ V_{TX-CM-DC} [\text{during L0}] - V_{TX-CM-Idle-DC} [\text{during Electrical Idle}] \leq 100 \text{ mV}$ $V_{TX-CM-DC} = DC_{(avg)} \text{ of } V_{TX-D+} + V_{TX-D-} / 2 [L0]$ $V_{TX-CM-Idle-DC} = DC_{(avg)} \text{ of } V_{TX-D+} + V_{TX-D-} / 2 [\text{Electrical Idle}]$
$V_{TX-CM-DC-LINE-DELTA}$	Absolute Delta of DC Common Mode Voltage between D+ and D-	0 (min) 25 (max)	0 (min) 25 (max)	0 (min) 25 (max)	mV	$ V_{TX-CM-DC-D+} - V_{TX-CM-DC-D-} \leq 25 \text{ mV}$ $V_{TX-CM-DC-D+} = DC_{(avg)} \text{ of } V_{TX-D+} $ $V_{TX-CM-DC-D-} = DC_{(avg)} \text{ of } V_{TX-D-} $
$V_{TX-IDLE-DIFF-AC-p}$	Electrical Idle Differential Peak Output Voltage	0 (min) 20 (max)	0 (min) 20 (max)	0 (min) 20 (max)	mV	$V_{TX-IDLE-DIFFp} = V_{TX-Idle-D+} - V_{TX-Idle-D-} \leq 20 \text{ mV}$ Voltage must be high-pass filtered, to remove any DC component and HF noise. The bandpass is constructed from two first-order filters, the high- and low-pass 3-dB bandwidths are 10 kHz and 1.25 GHz, respectively.
$V_{TX-IDLE-DIFF-DC}$	DC Electrical Idle Differential Peak Output Voltage	Not specified	0 (min) 5 (max)	0 (min) 5 (max)	mV	$V_{TX-IDLE-DIFF-DC} = V_{TX-Idle-D+} - V_{TX-Idle-D-} \leq 5 \text{ mV}$ Voltage must be high-pass filtered, to remove any AC component. The low pass filter is first-order with a 3-dB bandwidth of 10 kHz.

Table 18-7. PCI Express Transmitter (Signal Group a) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
$V_{TX-RCV-DETECT}$	Amount of Voltage Change Allowed during Receiver Detection	600 (max)	600 (max)	600 (max)	mV	Total amount of voltage change in a positive direction that a Transmitter can apply, to sense whether a Low-Impedance Receiver is present. <i>Note:</i> Receivers display substantially different impedance for $V_{IN} < 0$ versus $V_{IN} > 0$.
$T_{TX-IDLE-MIN}$	Minimum Time Spent in Electrical Idle	20 (min)	20 (min)	20 (min)	ns	Minimum time a Transmitter must be in Electrical Idle.
$T_{TX-IDLE-SET-TO-IDLE}$	Maximum Time to Transition to a Valid Electrical Idle after Sending an Electrical Idle Ordered-Set	8 (max)	8 (max)	8 (max)	ns	After sending the required Electrical Idle Ordered-Set (EIOS), the Transmitter must meet all Electrical Idle specifications within this time. This is measured from the end of the last UI of the last EIOS to the Transmitter in Electrical Idle.
$T_{TX-IDLE-TO-DIFF-DATA}$	Maximum Time to Transition to Valid Differential Signaling after Leaving Electrical Idle	8 (max)	8 (max)	8 (max)	ns	Maximum time to transition to valid differential signaling, after leaving Electrical Idle. This is considered a de-bounce time to the Tx.
$T_{CROSSLINK}$	Cross-Link Random Timeout	1.0 (max)	1.0 (max)	1.0 (max)	ms	Random timeout that helps resolve potential conflicts in the cross-link configuration.
$L_{TX-SKEW}$	Lane-to-Lane Output Skew	500 ps + 2 UI (max)	500 ps + 4 UI (max)	500 ps + 6 UI (max)	ps	Static skew between any two Lanes within a single Transmitter.
C_{TX}	AC-Coupling Capacitor	75 (min) 265 (max)	75 (min) 265 (max)	176 (min) 265 (max)	nF	All Transmitters shall be AC-coupled. The AC coupling is required either within the media, or within the transmitting component itself. Refer to Note 14.

Notes:

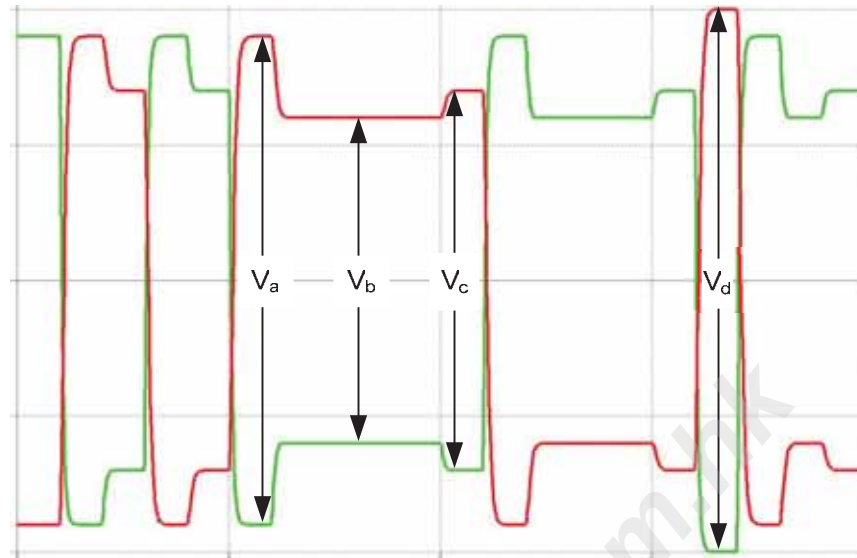
1. SSC permits a ± 0 , $\pm 5,000$ ppm modulation of the clock frequency, at a modulation rate not to exceed 33 kHz.
2. Measurements at 5.0 GT/s require an oscilloscope with a bandwidth of ≥ 12.5 GHz, or equivalent, while measurements made at 2.5 GT/s require an oscilloscope with at least 6.2 GHz bandwidth. Measurement at 5.0 GT/s must de-convolve effects of a compliance test board, to yield an effective measurement at the Tx signals. 2.5 GT/s can be measured within 200 mils of the Tx device's pins, although de-convolution is recommended.
3. Transmitter jitter is measured by driving the Transmitter under test with a low jitter "ideal" clock and connecting the device under test (DUT) to a reference load.
4. Transmitter raw jitter data must be convolved with a filtering function that represents the worst case Clock and Data Recovery (CDR) circuit tracking bandwidth. 2.5 and 5.0 GT/s use different filter functions. After the convolution process is applied, the center of the resulting eye must be determined and then used as a reference point for obtaining eye voltage and margins.
5. $V_{TX-AC-CM-PP}$ and $V_{TX-AC-CM-P}$ are defined in the PCI Express Base r3.0, Section 4.3.4.9. Measurement is made over at least 10^6 UI.
6. The Tx PLL bandwidth must lie between the minimum and maximum ranges listed in [Table 18-7](#). PLL peaking must lie below the values listed in [Table 18-7](#). The PLL bandwidth extends from zero (0), up to the value(s) specified in [Table 18-7](#).
7. Measurements are made for both common mode and differential return loss. The DUT must be powered up and DC isolated, and its data+/data- outputs must be in the low-Z state at a static value
8. A single combination of PLL bandwidth and peaking is specified for 2.5 GT/s implementations. For 5.0 GT/s, two combinations of PLL bandwidth and peaking are specified, to permit designers to make a tradeoff between the two parameters. If the PLL's minimum bandwidth is ≥ 8 MHz, then up to 3.0 dB of peaking is permitted. If the PLL's minimum bandwidth is relaxed to ≥ 5.0 MHz, then a tighter peaking value of 1.0 dB must be met. In both cases, the maximum PLL bandwidth is 16 MHz.
9. Low swing output, defined by $V_{TX-DIFF-PP-LOW}$, must be implemented with no de-emphasis.
10. For 5.0 GT/s, de-emphasis timing jitter must be removed. This parameter is measured by accumulating a record length of 106 UI while the DUT outputs a compliance pattern. $T_{MIN-PULSE}$ is defined to be nominally 1 UI-wide, and is bordered on both sides by pulses of the opposite polarity.
11. Root Complex Tx de-emphasis is configured from an Upstream controller. Downstream Tx de-emphasis is Set by way of a command, issued at 2.5 GT/s. For details, refer to the appropriate location in the PCI Express Base r3.0, Section 4.2.
12. Tx common mode (CM) noise for 8.0 GT/s is measured at TP1, without de-embedding the breakout channel. The parameter captures device CM noise only, and is not intended to capture system CM noise. CM noise is measured by applying an LPF with a -3 dB corner frequency at 4 GHz to the raw data. No more than 50 mVPP of the 150 mVPP total can lie within the 0.03- to 500-MHz range.
13. $I_{TX-SHORT}$ and $V_{TX-DC-CM}$ stipulate the maximum current/voltage levels that a Transmitter can generate, and therefore define, the worst case transients that a Receiver must tolerate.
14. All Transmitters supporting 8.0 GT/s must implement the 180 to 265 nF C_{TX} value. Transmitters operating at 2.5 or 5.0 GT/s only can implement over a range of 75 to 265 nF.

18.7.2 Tx Preshoot and De-Emphasis

The equalization ratios implemented by the Transmitter can be calculated from the PIPE tap-coefficients. These ratios are computed, based upon the voltage levels defined in [Figure 18-1](#).

[Table 18-8](#) defines the Tx preshoot and de-emphasis presets.

Figure 18-1. Definition of Tx Voltage Levels and Equalization Ratios



where

- De-Emphasis = $20 \log_{10} (V_b / V_a)$
- Preshoot = $20 \log_{10} (V_c / V_b)$
- Boost = $20 \log_{10} (V_d / V_b)$

Table 18-8. Tx Preshoot and De-Emphasis Presets

Preset Select ^a (Decimal)	Preshoot (dB)	De-Emphasis (dB)	De-Emphasis ^b (Decimal)	Main ^c (Decimal)	Preshoot ^d (Decimal)	Total
0	0	-6.0	16	47	0	63
1	0	-3.5	11	52	0	63
2	0	-4.5	13	50	0	63
3	0	-2.5	8	55	0	63
4	0	0	0	63	0	63
5	2.0	0	0	57	6	63
6	2.5	0	0	55	8	63
7	3.5	-6.0	13	44	6	63
8	3.5	-3.5	8	47	8	63
9	3.5	0	0	52	11	63
10	0	-9.5	21	42	0	63
11 to 15	Reserved					

- Gen 3 Coefficient Values** register **Preset Select** field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BD4h[27:24]).
- Gen 3 Coefficient Values** register **Preset 0 Coefficients – De-Emphasis** field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BD4h[17:12]).
- Gen 3 Coefficient Values** register **Preset 0 Coefficients – Main** field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BD4h[11:6]).
- Gen 3 Coefficient Values** register **Preset 0 Coefficients – Preshoot** field (Base mode – Port 0 and/or 8; Virtual Switch mode – Port 0 and/or 8, accessible through the Management Port, offset BD4h[5:0]).

18.7.3 Tx Voltage and Jitter

Table 18-9. Tx Voltage and Jitter Parameters

Symbol	Parameter	Value	Units	Notes
$V_{TX-FS-NO-EQ}$	Full Swing Tx Voltage with no TxEq	800 (min) 1,300 (max)	mVPP	Refer to Note 1.
$V_{TX-RS-NO-EQ}$	Reduced Swing Tx Voltage with no TxEq	1,300 (max)	mVPP	Refer to Note 1.
$V_{TX-EIEOS-FS}$	Min Swing during Electrical Idle Exit Ordered-Set (EIEOS) for Full Swing	250 (min)	mVPP	Refer to Note 2.
$V_{TX-EIEOS-RS}$	Min Swing During EIEOS for Reduced Swing	232 (min)	mVPP	Refer to Note 2.
T_{TX-UTJ}	Tx Uncorrelated Total Jitter	31.25 (max)	psPP at 10^{-12}	
$T_{TX-UDJDD}$	Tx Uncorrelated Deterministic Jitter	12 (max)	psPP	
$T_{TX-UPW-TJ}$	Total Uncorrelated PWJ	24 (max)	psPP at 10^{-12}	Refer to Notes 3 and 4.
$T_{TX-UPW-DJDD}$	Deterministic DjDD Uncorrelated PWJ	10 (max)	psPP	Refer to Notes 3 and 4.
T_{TX-DDJ}	Data-Dependent Jitter	18 (max)	psPP	Refer to Notes 4 and 5.
ps_{21TX}	Pseudo Package Loss	-3.0 (min)	dB	PP ratio of 64 ones/64 zeros pattern versus 0101 pattern. No Tx equalization. Refer to Note 6.
$V_{TX-BOOST-FS}$	Tx Boost Ratio for Full Swing	8.0 (min)	dB	
$V_{TX-BOOST-RS}$	Tx Boost Ratio for Reduced Swing	2.5 (min)	dB	
$EQ_{TX-COEFF-RES}$	Tx Coefficient Resolution	1/63	–	

Notes:

1. Voltage measurements for $V_{TX-FS-NO-EQ}$ and $V_{TX-RS-NO-EQ}$ are made using the 64-zeros/64-ones pattern in the compliance pattern.
2. Voltage limits comprehend both full swing and reduced swing modes. The Tx must reject any changes that would violate this specification. The maximum level is covered in the $V_{TX-FS-NO-EQ}$ measurement which represents the maximum peak voltage that the Tx can drive. The $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ voltage limits are imposed, to guarantee the EIEOS threshold of 175 mVPP at the Rx signal. This parameter is measured, using the actual EIEOS pattern that is part of the compliance pattern, and then removing the ISI contribution of the breakout channel. The Transmitter must advertise a value for LF during TS1 at 8.0 GT/s that ensures that these parameters are met.
3. PWJ parameters shall be measured after DDJ separation.
4. Measured with optimized preset value, after de-embedding to the Tx signal.
5. The 18 ps number takes measurement errors into account.
6. The -3.0 dB number takes measurement errors into account. For some Tx package/driver combinations, $ps21_{TX}$ can be greater than 0 dB.

18.8 Receive Characteristics

The Receiver circuit includes programmable equalization, to further compensate for the low-pass FR4 loss characteristics of the channel.

18.8.1 Receive Equalization

Figure 18-2 illustrates an example frequency response of the attenuator (ATT), which displays nearly 24 dB of programmable attenuation. Also, the ATT is capable of providing up to 9 dB of analog boost at high frequencies, for channel compensation.

The Rx Attenuator can be programmed, using the registers listed in Table 18-10.

Figure 18-2. ATT Frequency Response for Different ATT Settings

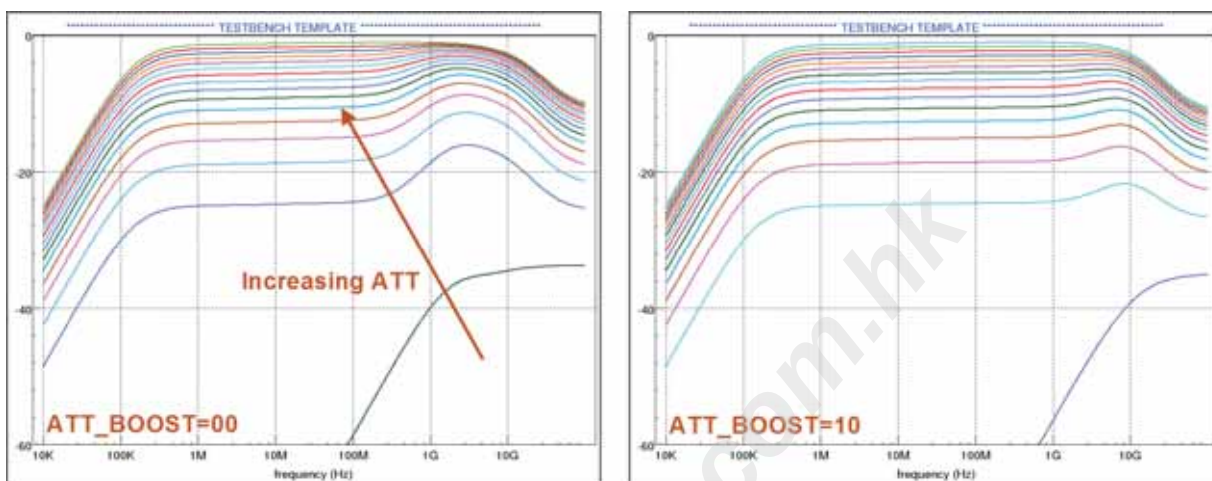


Table 18-10. Rx Attenuator Control

Register	Description
rx_att[3:0]	Rx attenuation value (15 minimum, 0 maximum).
rx_att_boost[1:0]	Rx attenuator analog boost value (2 minimum, 0 maximum, 3 invalid).

Figure 18-3. Increasing Boost

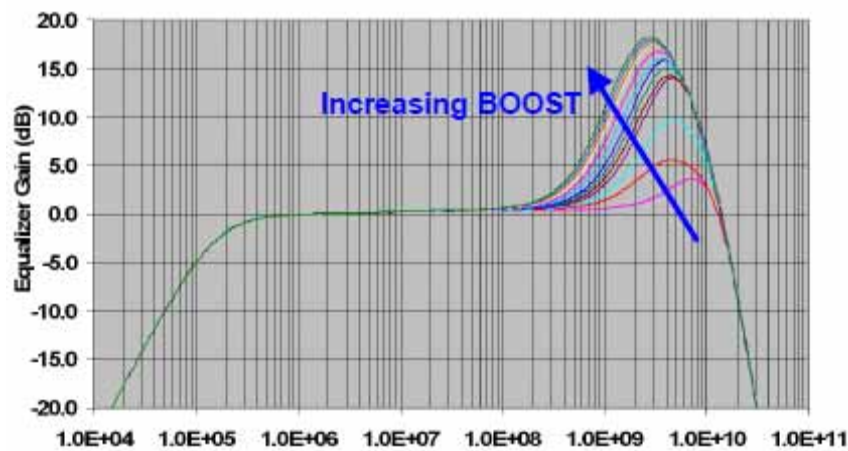


Table 18-11. Boost Settings and Gain

Boost Setting	Gain @ 4 GHz (dB)
0	0.75
1	1.31
2	1.74
3	2.46
4	4.69
5	5.46
6	7.84
7	8.18
8	10.10
9	11.78
10	12.98
11	15.20
12	15.61
13	17.25
14	17.25
15	17.25

18.8.2 PCI Express Receiver AC and DC Characteristics

Table 18-12. PCI Express Receiver (Signal Group b) – AC and DC Characteristics

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
UI	Unit Interval	399.88 (min) 400.12 (max)	199.94 (min) 200.06 (max)	124.9625 (min) 125.0375 (max)	ps	UI does not account for variations caused by SSC.
$V_{RX-DIFF-PP-CC}$	Differential Rx Peak-to-Peak Voltage for Common REFCLK Rx Architecture	0.175 (min) 1.2 (max)	0.120 (min) 1.2 (max)	Refer to Table 18-13 and Table 18-14	V	
$V_{RX-DIFF-PP-DC}$	Differential Rx Peak-to-Peak Voltage for Data Clocked Rx Architecture	0.175 (min) 1.2 (max)	0.100 (min) 1.2 (max)	Refer to Table 18-13 and Table 18-14	V	
T_{RX-EYE}	Receiver Eye Time Opening	0.40 (min)	–	Refer to Table 18-13 and Table 18-14	UI	Minimum eye time at Rx signals to yield a 10^{-12} Bit Error Rate. Receiver eye margins are defined into a $2 \times 50\Omega$ reference load.
$T_{RX-TJ-CC}$	Maximum Rx Inherent Timing Error for Common REFCLK Rx Architecture	–	0.40 (max)	Refer to Table 18-13 and Table 18-14	UI	Refer to Note 1.
$T_{RX-TJ-DC}$	Maximum Rx Inherent Timing Error for Data Clocked Rx Architecture	–	0.34 (max)	Refer to Table 18-13 and Table 18-14	UI	Refer to Note 1.
$T_{RX-DJ-DD-CC}$	Maximum Rx Inherent Deterministic Timing Error for Common REFCLK Rx Architecture	–	0.30 (max)	Refer to Table 18-13 and Table 18-14	UI	Refer to Note 1.
$T_{RX-DJ-DD-DC}$	Maximum Rx Inherent Deterministic Timing Error for Data Clocked Rx Architecture	–	0.24 (max)	Refer to Table 18-13 and Table 18-14	UI	Refer to Note 1.

Table 18-12. PCI Express Receiver (Signal Group b) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
$T_{RX-EYE-MEDIAN-to-MAX-JITTER}$	Maximum Time Delta between the Median and Deviation from the Median	0.3 (max)	Not specified	Not specified	UI	Only specified for 2.5 GT/s.
$T_{RX-MIN-PULSE}$	Minimum Width Pulse at Rx	Not specified	0.6 (min)	Not specified	UI	Measured to account for worst Tj at 10^{-12} Bit Error Rate.
$V_{RX-MAX-MIN-RATIO}$	Minimum/Maximum Pulse Voltage on Consecutive UI	Not specified	5 (max)	Not specified	–	Rx eye must simultaneously meet V_{RX-EYE} limits.
BW_{RX-PLL}	Maximum Rx PLL Bandwidth for 2.5 GT/s	1.5 (min) 22 (max)	Not specified	Not specified	MHz	Refer to Note 2.
$BW_{RX-PKG-PLL1}$	Rx PLL Bandwidth corresponding to $PKG_{RX-PLL1}$	Not specified	8 (min) 16 (max)	2 (min) 4 (max)	MHz	Second Order PLL Jitter Transfer Bounding function. Refer to Note 2.
$BW_{RX-PKG-PLL2}$	Rx PLL Bandwidth corresponding to $PKG_{RX-PLL2}$	Not specified	5 (min) 16 (max)	2 (min) 5 (max)	MHz	Second Order PLL Jitter Transfer Bounding function. Refer to Note 2.
$PKG_{RX-PLL1}$	Rx PLL Peaking Limit for PLL1	Not specified	3.0 (max)	2.0 (max)	dB	PLL Bandwidth = 8 MHz (min) at 5 GT/s or Bandwidth = 4 MHz (max) at 8 GT/s. Refer to Note 2.
$PKG_{RX-PLL2}$	Rx PLL Peaking Limit for PLL2	Not specified	1.0 (max)	1.0 (max)	dB	PLL Bandwidth = 5 MHz (min) at 5 GT/s or Bandwidth = 5 MHz (max) at 8 GT/s. Refer to Note 2.
$RL_{RX-DIFF}$	Rx Differential Return Loss (Package + Silicon)	10 (min)	10 (min) for 0.05 to 1.25 GHz 8 (min) for 1.25 to 2.5 GHz	10 (min) for 0.05 to 1.25 GHz 8 (min) for 1.25 to 2.5 GHz 5 (min) for 2.5 to 4.0 GHz	dB	Refer to Note 3.

Table 18-12. PCI Express Receiver (Signal Group b) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
RL_{RX-CM}	Common Mode Return Loss	6 (min)	6 (min)	6 (min) for 0.05 to 1.25 GHz 8 (min) for 2.5 to 4.0 GHz	dB	Refer to Note 3.
Z_{RX-DC}	Rx DC Single Ended Impedance	40 (min) 60 (max)	40 (min) 60 (max)	Not specified	Ω	DC impedance limits are needed to guarantee Receiver detect. Refer to Note 4.
$T_{RX-GND-FLOAT}$	Rx Termination Ground Float Time	Not specified	Not specified	500	μs	Time allowed to float Rx internal Ground in 2.5 or 5.0 to 8.0 GT/s configuration change. Refer to Note 7.
$Z_{RX-DIFF-DC}$	DC Differential Rx Impedance	80 (min) 120 (max)	Not specified	Not specified	Ω	For 5.0 and 8.0 GT/s, covered under the $RL_{RX-DIFF}$ parameter. Refer to Note 4.
$V_{RX-CM-AC-P}$	Rx AC Common Mode Voltage	150 (max)	150 (max)	75 mV (max) (EH <100 mVPP) 125 mV (max) (EH $\geq$ 100 mVPP)	mVP	Measured at Rx signals, into a pair of 50 Ω terminations into Ground. Refer to Note 5.
$Z_{RX-HIGH-IMP-DC-POS}$	DC Input Common Mode Input Impedance for Voltage >0 during Reset or Power-Down	10K (min)	10K (min)	10K (min)	Ω	Rx DC common mode impedance with the Rx terminations not powered, measured over the range 0 to 200 mV (with respect to Ground). Refer to Note 6.
$Z_{RX-HIGH-IMP-DC-NEG}$	DC Input Common Mode Input Impedance for Voltage <0 during Reset or Power-Down	1.0K (min)	1.0K (min)	1.0K (min)	Ω	Rx DC common mode impedance with the Rx terminations not powered, measured over the range -150 to 0 mV (with respect to Ground). Refer to Note 6.

Table 18-12. PCI Express Receiver (Signal Group b) – AC and DC Characteristics (Cont.)

Symbol	Parameter	Link Speed			Units	Comments
		2.5 GT/s	5.0 GT/s	8.0 GT/s		
$V_{RX-IDLE-DET-DIFFp-p}$	Electrical Idle Detect Threshold	65 (min) 175 (max)	65 (min) 175 (max)	65 (min) 175 (max)	mV	$V_{RX-IDLE-DET-DIFFp-p} = \frac{1}{2} \times V_{RX-D+} - V_{RX-D-} $ Measured at the Receiver's package pins.
$T_{RX-IDLE-DET-DIFF-ENTERTIME}$	Unexpected Electrical Idle Enter Idle Detect Threshold Integration Time	10 (max)	10 (max)	10 (max)	ms	An un-expected Electrical Idle ($V_{RX-DIFFp-p} < V_{RX-IDLE-DET-DIFFp-p}$) must be recognized no longer than $T_{RX-IDLE-DET-DIFF-ENTERTIME}$ to signal an unexpected idle condition.
$L_{RX-SKEW}$	Total Lane-to-Lane Skew	20 (max)	8 (max)	6 (max)	ns	Across all Lanes on a Port. $L_{RX-SKEW}$ comprehends Lane-to-Lane variations due to channel and repeater delay differences.

Notes:

- The four inherent timing error parameters are defined for the convenience of Rx designers, and they are measured during Receiver tolerancing.
- Two combinations of PLL bandwidth and peaking are specified at 5.0 GT/s, to permit designers to make trade-offs between the two parameters. If the PLL's minimum bandwidth is ≥ 8 MHz, then up to 3.0 dB of peaking is permitted. If the PLL's minimum bandwidth is relaxed to ≥ 5.0 MHz, then a tighter peaking value of 1.0 dB must be met.

A PLL bandwidth extends from zero up to the value(s) defined as the minimum or maximum in [Table 18-12](#). For 2.5 GT/s, a single PLL bandwidth and peaking value of 1.5 to 22 MHz and 3.0 dB are defined.
- Measurements must be made for both common mode and differential return loss. In both cases, the DUT must be powered up and DC-isolated, and its D+/D- inputs must be in the low-Z state.
- The Rx DC single-ended impedance must be present when the Receiver terminations are first enabled, to ensure that the Receiver Detect properly occurs. Compensation of this impedance can start immediately, and the Rx single-ended impedance (constrained by RL_{RX-CM} to $50\Omega \pm 20\%$) must be within the specified range by the time Detect is entered.
- Common mode peak voltage is defined by the expression:
$$\max\{|(V_{d+} - V_{d-}) - V_{CMDC}|\}$$
- $Z_{RX-HIGH-IMP-DC-NEG}$ and $Z_{RX-HIGH-IMP-DC-POS}$ are defined, respectively, for negative and positive voltages at the input of the Receiver. Transmitter designers must comprehend the large difference between >0 and <0 Rx impedances, when designing Receiver detect circuits.
- Defines the time for the Receiver's input pads to settle to new common mode on 2.5 or 5.0 GT/s transition to 8.0 GT/s.

Table 18-13. Stressed Voltage Eye Parameters

Symbol	Parameter	Limits at 8.0 GT/s	Units	Comments
$V_{RX-LAUNCH-8G}$	Generator Launch Voltage	800	mVPP	Measured at TP1. Refer to the <i>PCI Express Base r3.0</i> , Figure 4-65. $V_{RX-LAUNCH-8G}$ can be adjusted, if necessary, to yield the proper EH, as long as the outside eye voltage at TP2 does not exceed 1,300 mVPP.
$T_{RX-UI-8G}$	Unit Interval	125.00	ps	Nominal value is sufficient for Rx tolerancing. Value does not account for SSC.
$V_{RX-SV-8G}$	Eye Height at TP2P	25 (-20 dB channel) 50 (-12 dB channel) 200 (-3 dB channel)	mVPP	Eye height at BER = 10^{-12} . Refer to Notes 1, 2.
$T_{RX-SV-8G}$	Eye Width at TP2P	0.3 to 0.35	UI	Eye width at BER = 10^{-12} . Refer to Note 2.
$V_{RX-SV-DIFF-8G}$	Differential Mode Interference	14 or greater	mVPP	Adjusted to Set EH. Frequency = 2.10 GHz. Refer to Note 3.
$V_{RX-SV-CM-8G}$	Rx AC Common Mode Voltage at TP2P	150 (EH < 100 mVPP) 250 (EH ≥ 100 mVPP)	mVPP	Defined for a single tone at 120 MHz. Refer to Note 3.
$T_{RX-SV-SJ-8G}$	Sinusoidal Jitter at 100 MHz	0.1	UI PP	Fixed at 100 MHz. Refer to Note 4.
$T_{RX-SV-RJ-8G}$	Random Jitter	2.0	ps RMS	Rj is spectrally flat before filtering. Refer to Notes 4, 5
$V_{RX-MAX-SE-SW}$	Maximum Single-Ended Swing	±300	mVP	Refer to Note 6.

Notes:

- $V_{RX-SV-8G}$ is tested at three different voltages, to ensure that the Rx DUT is capable of equalizing over a range of channel loss profiles. The test also guarantees that the Rx is capable of operating over a sufficient dynamic range of eye heights. “SV” in the parameter names refers to “stressed voltage”.
- $V_{RX-ST-8G}$ and $T_{RX-ST-8G}$ are referenced to TP2P, and are obtained after post processing data captured at TP2.

 $V_{RX-ST-8G}$ and $T_{RX-ST-8G}$ include the effects of applying the behavioral Rx model and Rx behavioral equalization.
- $V_{RX-SV-DIFF-8G}$ measurement is made at TP2, without post processing. $V_{RX-SV-CM-8G}$ can be made at either TP1 or TP2. $V_{RX-SV-DIFF-8G}$ voltage might need to be adjusted over a wide range for the different loss calibration channels.
- $T_{RX-SV-SJ-8G}$ and $T_{RX-SV-RJ-8G}$ measurements are made at TP1, without post processing.
- Rj is applied over the following range. The low frequency limit can be between 1.5 and 10 MHz, and the upper limit is 1.0 GHz. Refer to the *PCI Express Base r3.0*, Figure 4-74, for details.
- $V_{RX-MAX-SE-SW}$ sets the maximum outer, single-ended eye voltage limit in the presence of differential and CM noise applied to the Rx, as observed at TP2 relative to ground with no behavioral RxEq post processing.

Table 18-14. Stressed Jitter Eye Parameters

Symbol	Parameter	Limits at 8.0 GT/s	Units	Comments
$V_{RX-LAUNCH-8G}$	Generator Launch Voltage	800 (nominal)	mVPP	Measured at TP1. Refer to the <i>PCI Express Base r3.0</i> , Figure 4-65. Refer to Note 1.
$T_{RX-UI-8G}$	Unit Interval	125.00	ps	Nominal value is sufficient for Rx tolerancing. Value does not account for SSC.
$V_{RX-ST-8G}$	Eye height at TP2P	25 (min) 35 (max)	mVPP	At BER = 10^{-12} . Refer to Note 2.
$T_{RX-ST-8G}$	Eye width at TP2P	0.30	UI	At BER = 10^{-12} . Refer to Note 2.
$T_{RX-ST-SJ-8G}$	Sinusoidal Jitter	0.1 to 1.0	UI PP	Measured at TP1. Refer to the <i>PCI Express Base r3.0</i> , Figure 4-74. Refer to Note 3.
$T_{RX-ST-RJ-8G}$	Random Jitter	3.0	ps RMS	Rj spectrally flat before filtering. Measured at TP1. Refer to Note 4.

Notes:

- $V_{RX-LAUNCH-8G}$ can be adjusted to meet $V_{RX-ST-8G}$, as long as the outside eye voltage at TP2 does not exceed 1,300 mVPP.
- $V_{RX-ST-8G}$ and $T_{RX-ST-8G}$ are referenced to TP2P, and are obtained after post processing data captured at TP2.

 $V_{RX-ST-8G}$ and $T_{RX-ST-8G}$ include the effects of applying the behavioral Rx model and Rx behavioral equalization.
- $T_{RX-ST-SJ-8G}$ can be measured at either TP1 or TP2.
- While the nominal value is specified at 3.0 ps RMS, it can be adjusted to meet the 0.3 UI value for $T_{RX-ST-8G}$. Rj is measured at TP1, to prevent data-channel interaction from adversely affecting Rj calibration accuracy.

Rj is applied over the following range. The low-frequency limit can be between 1.5 and 10 MHz, and the upper limit is 1.0 GHz.

19.1 Thermal Characteristics

Table 19-1. Sample Thermal Data, Commercial Temperature, with Heat Sink^{a, b, c, d}

Air Flow Velocity (m/s)	Θ_{JA} (°C/W) JEDEC 4-Layer Board	Θ_{JA} (°C/W) JEDEC 8-Layer Board	Ψ_{JB} (°C/W) JEDEC 4-Layer Board	Ψ_{JB} (°C/W) JEDEC 8-Layer Board	Θ_{JC} (°C/W) JEDEC 4-Layer Board	Θ_{JC} (°C/W) JEDEC 8-Layer Board	Θ_{JB} (°C/W) JEDEC 4-Layer Board	Θ_{JB} (°C/W) JEDEC 8-Layer Board	Heat Sink
0.00	9.37	5.76	2.51	1.9	0.65	0.6	3.35	2.16	Aavid Thermalloy, 374024B00035G
1.00	5.52	3.9	1.8	1.54					
2.00	4.83	3.52	1.7	1.47					

- Thermal data is based upon simulation.
- For all conditions, ambient temperature is assumed to be 70°C.
- The Maximum Operating Junction Temperature is 110°C.
The Maximum Junction Temperature for Reliability is 110°C.
- For all conditions, maximum power is assumed to be at 8.0 GT/s, as listed in [Table 18-3, “Power Consumption Estimates \(W\).”](#)

19.2 General Package Specifications

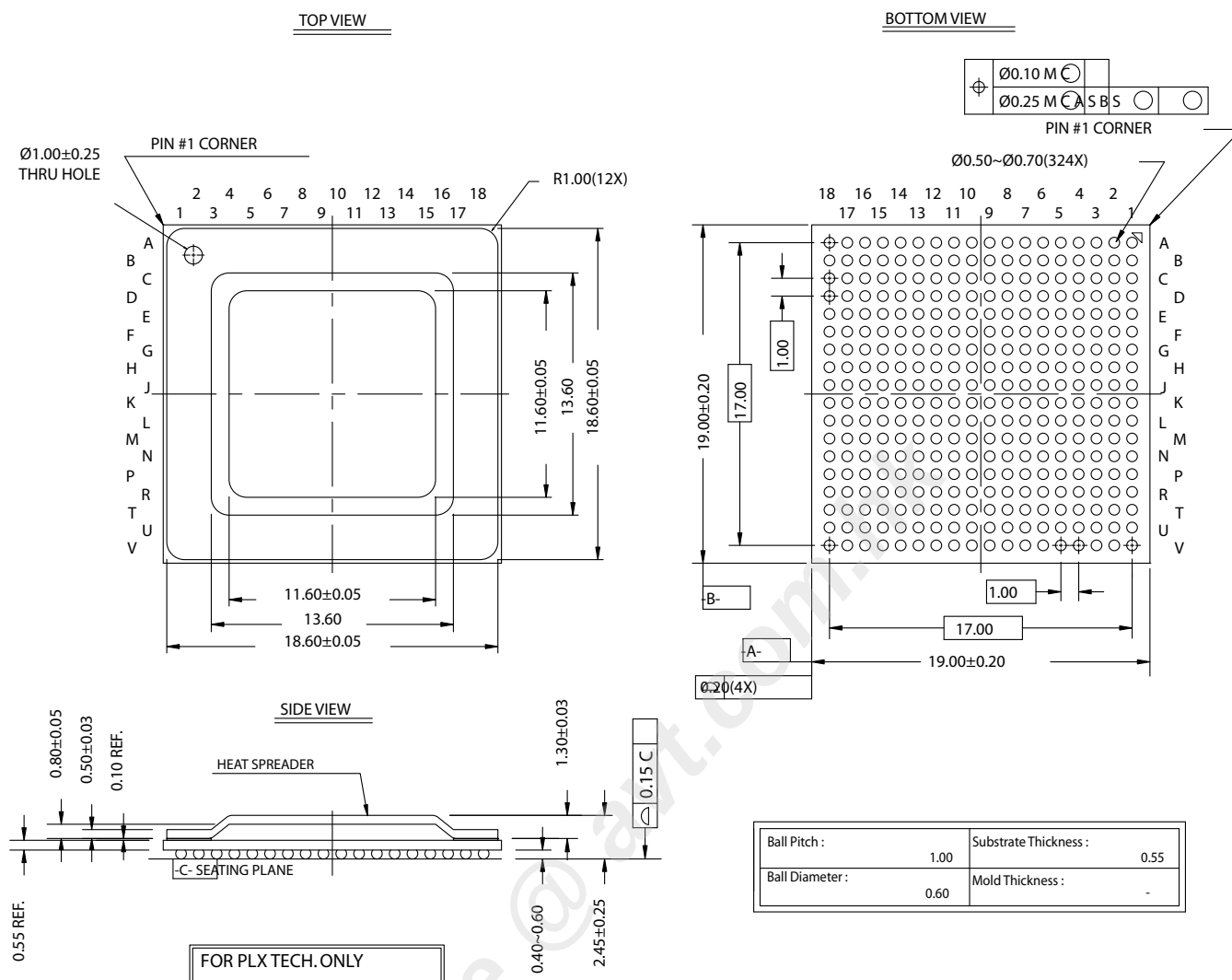
Table 19-2 lists general package specifications. (Refer also to Figure 19-1.)

Table 19-2. General Package Specifications

Parameter	Specification
Package Type	Flip-Chip Ball Grid Array with Heat Spreader (HFC-BGA)
Ball Quantity	324
Package Dimensions	19 x 19 mm ² (approximately 2.45 ±0.25-mm high)
Ball Matrix Pattern	18 x 18
Ball Pitch	1.0 mm
Ball Diameter	0.60 ±0.1 mm
Ball Spacing	0.40 mm
Ball Pad Diameter	0.58 mm
Solder Mask Opening on Ball Pad	0.48 mm

19.3 Mechanical Dimensions

Figure 19-1. Mechanical Dimensions – 19 x 19 mm² Package



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Appendix A General Information

A.1 Product Ordering Information

Contact your local [PLX Sales Representative](#) for ordering information.

Table A-1. Product Ordering Information

Part Numbers	Description
PEX8724-CA80BC G	PEX 8724 24-Lane, 6-Port PCI Express Gen 3 Multi-Root Switch Lead-Free HFC-BGA Package (19 x 19 mm ² , 324-ball)
where	PEX – PCI Express Product Family 8724 – Part Number CA – Silicon Revision 80 – Signaling Rate (8.0 GT/s) B – Flip-Chip Ball Grid Array C – Commercial Temperature G – Lead-free (RoHS 6/6 and Green compliant)
PEX8724-CA RDK	PEX 8724 Rapid Development Kit with x8 Edge Connector
x1 Adapter	PCI Express x8 to x1 Adapter
x4 Adapter	PCI Express x8 to x4 Adapter

A.2 United States and International Representatives and Distributors

PLX Technology, Inc., representatives and distributors are listed at www.plxtech.com.

A.3 Technical Support

PLX Technology, Inc., technical support information is listed at www.plxtech.com/support, or call 800 759-3735 (domestic only) or 408 774-9060.

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